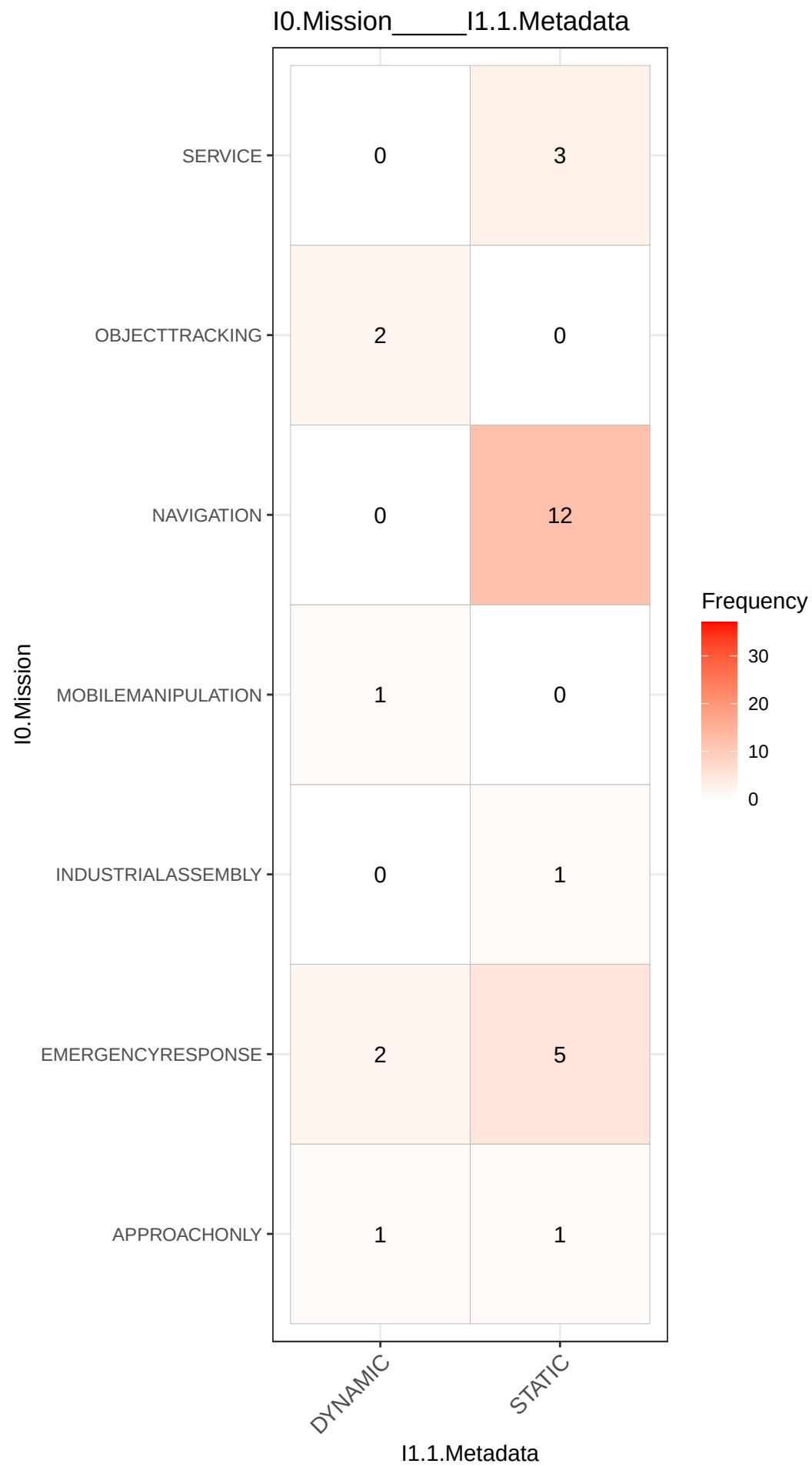
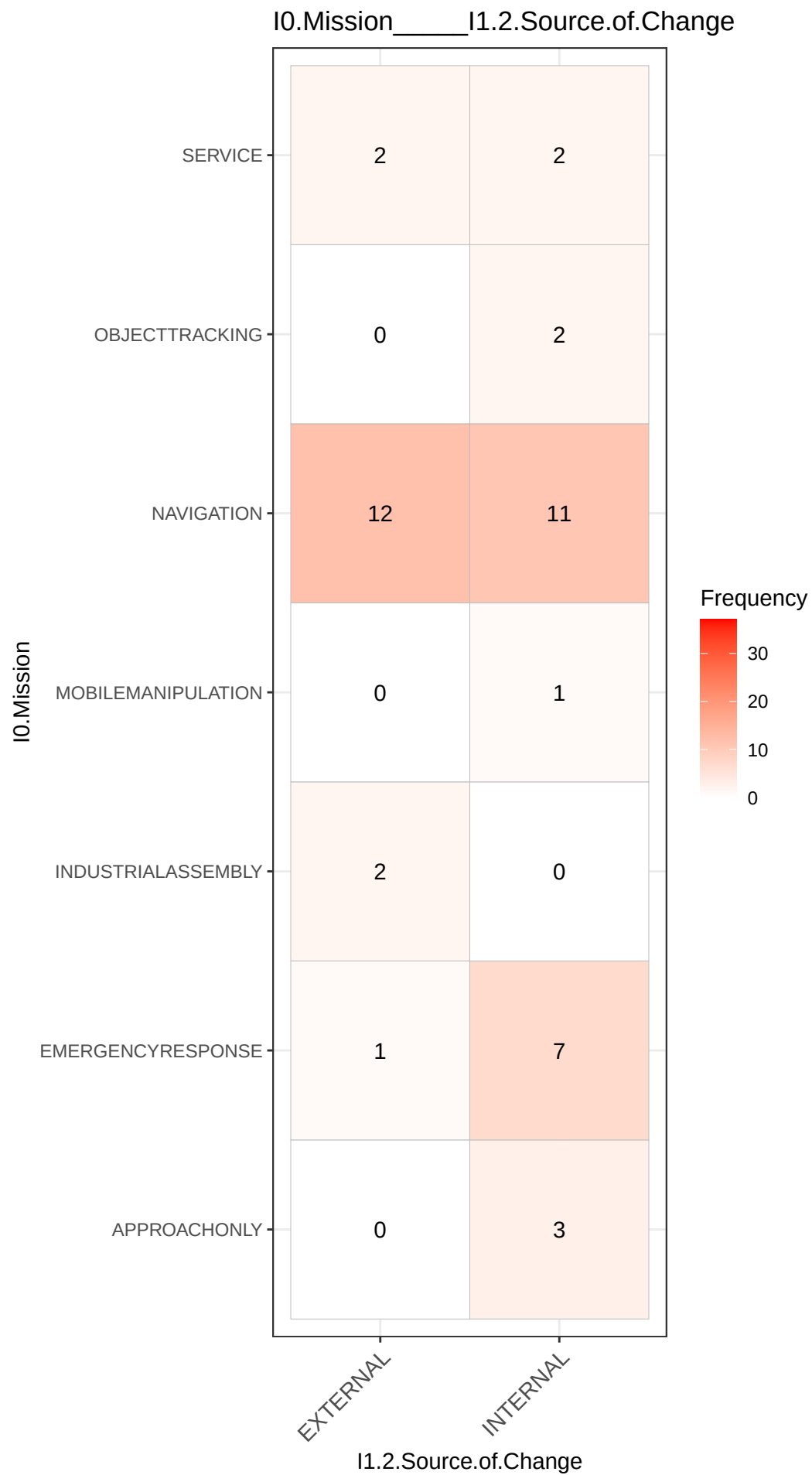
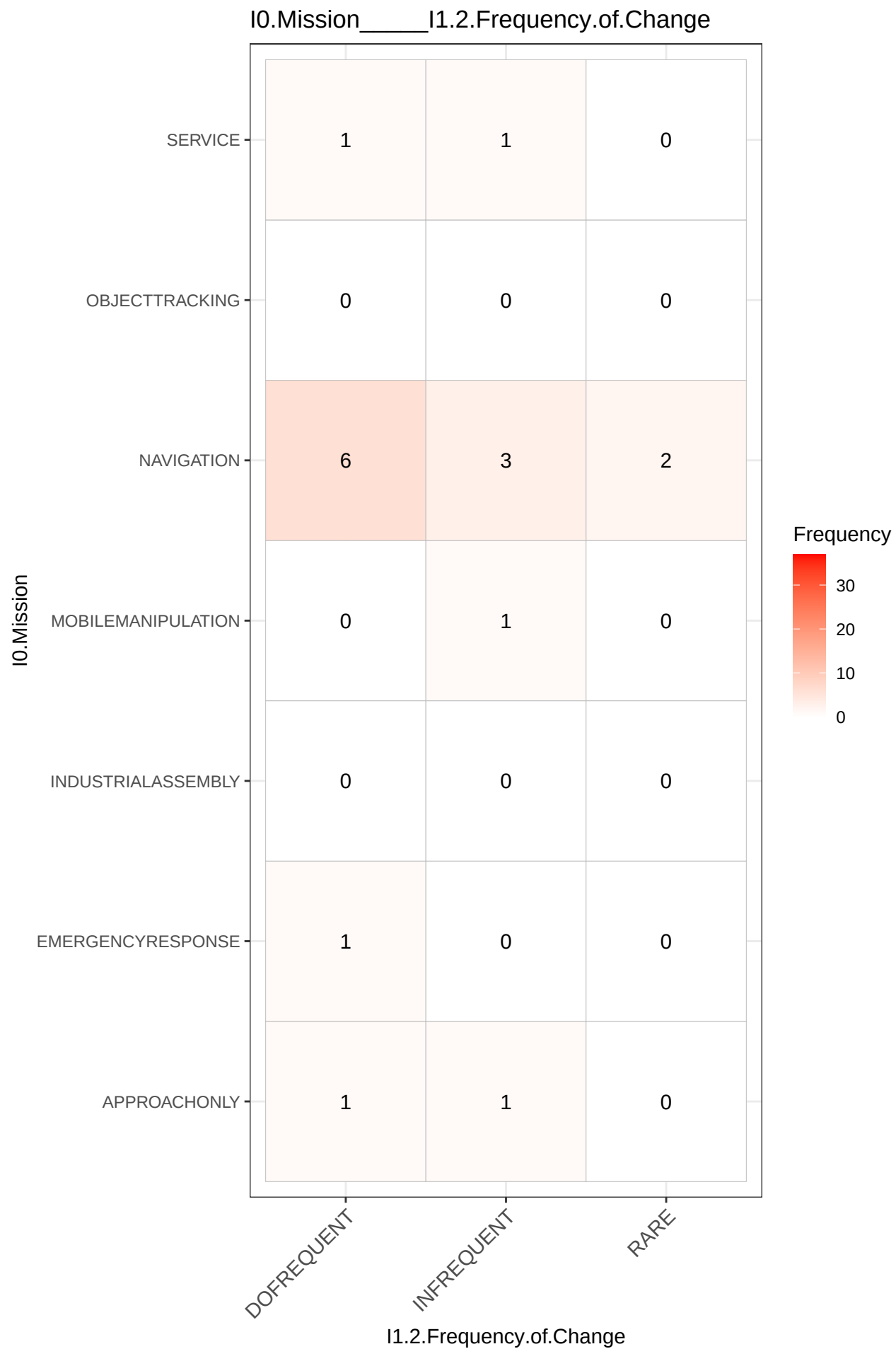


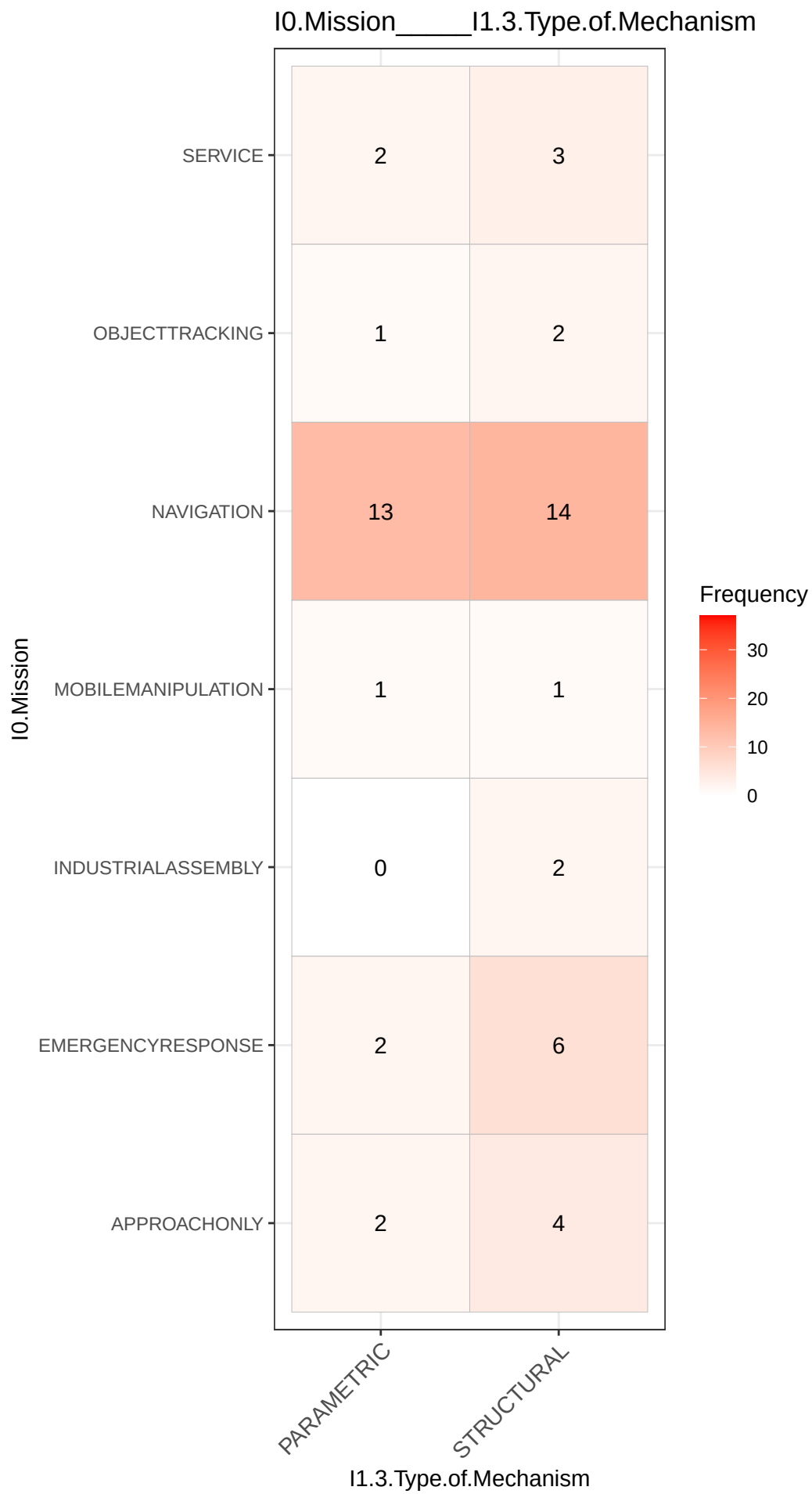


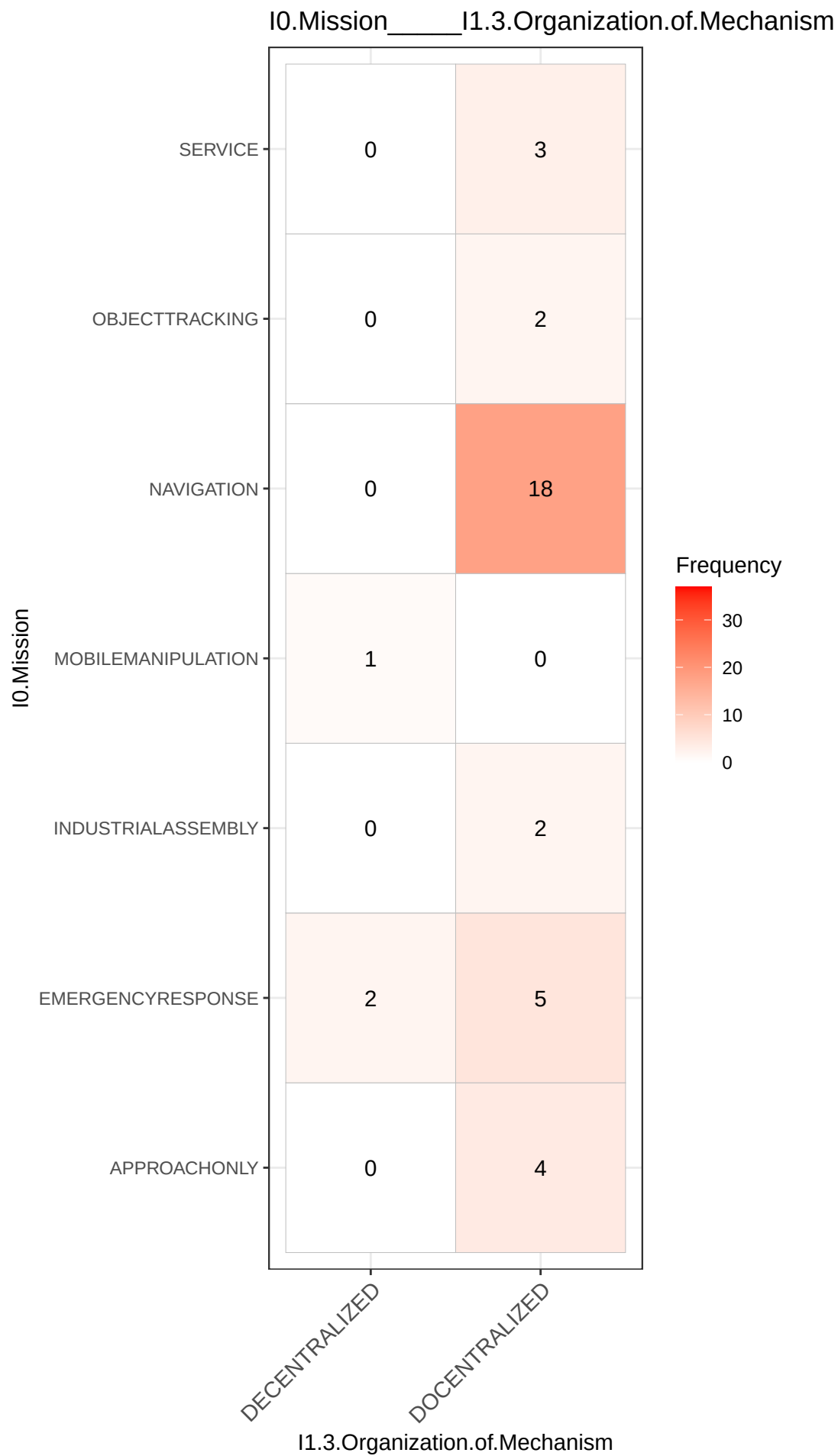


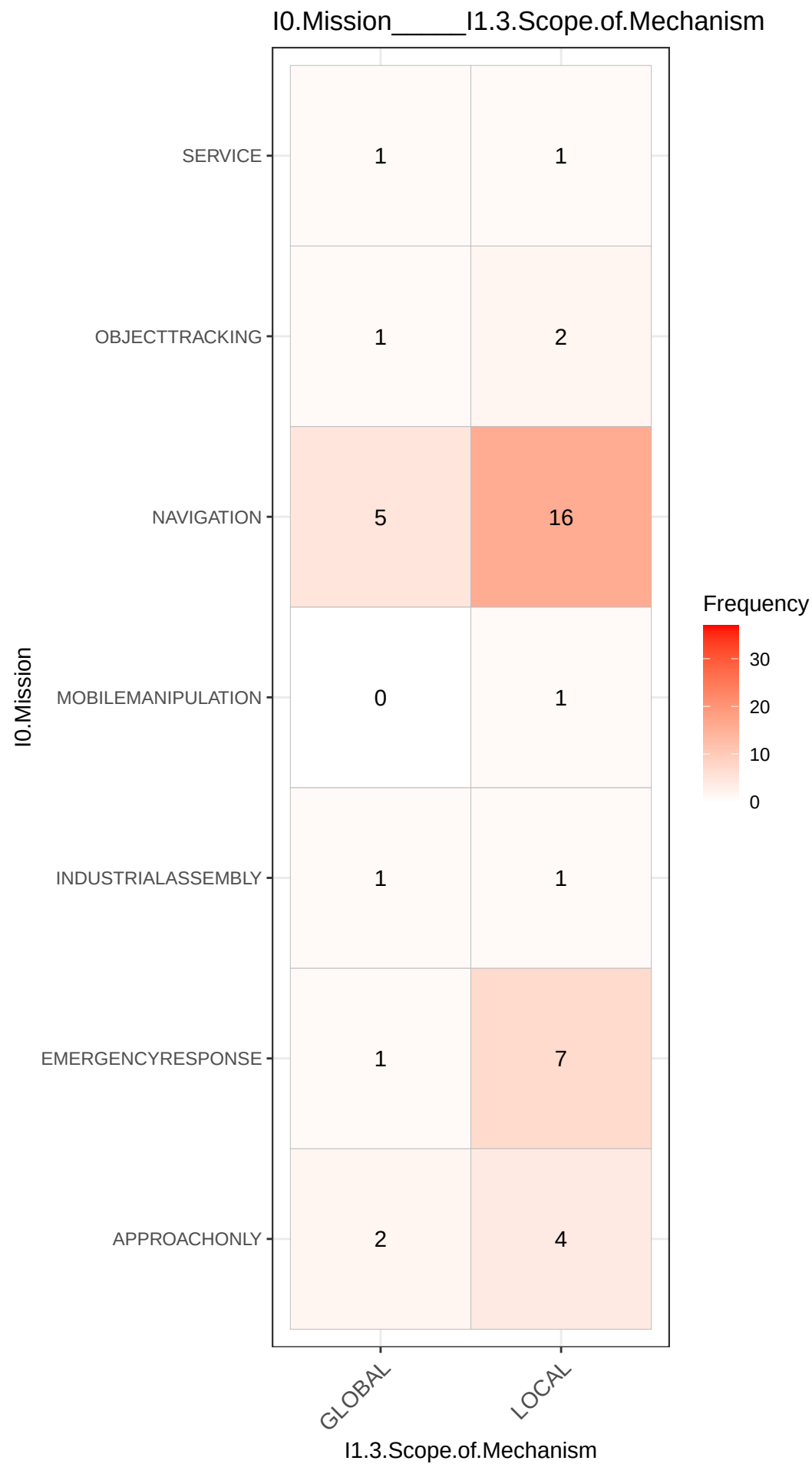


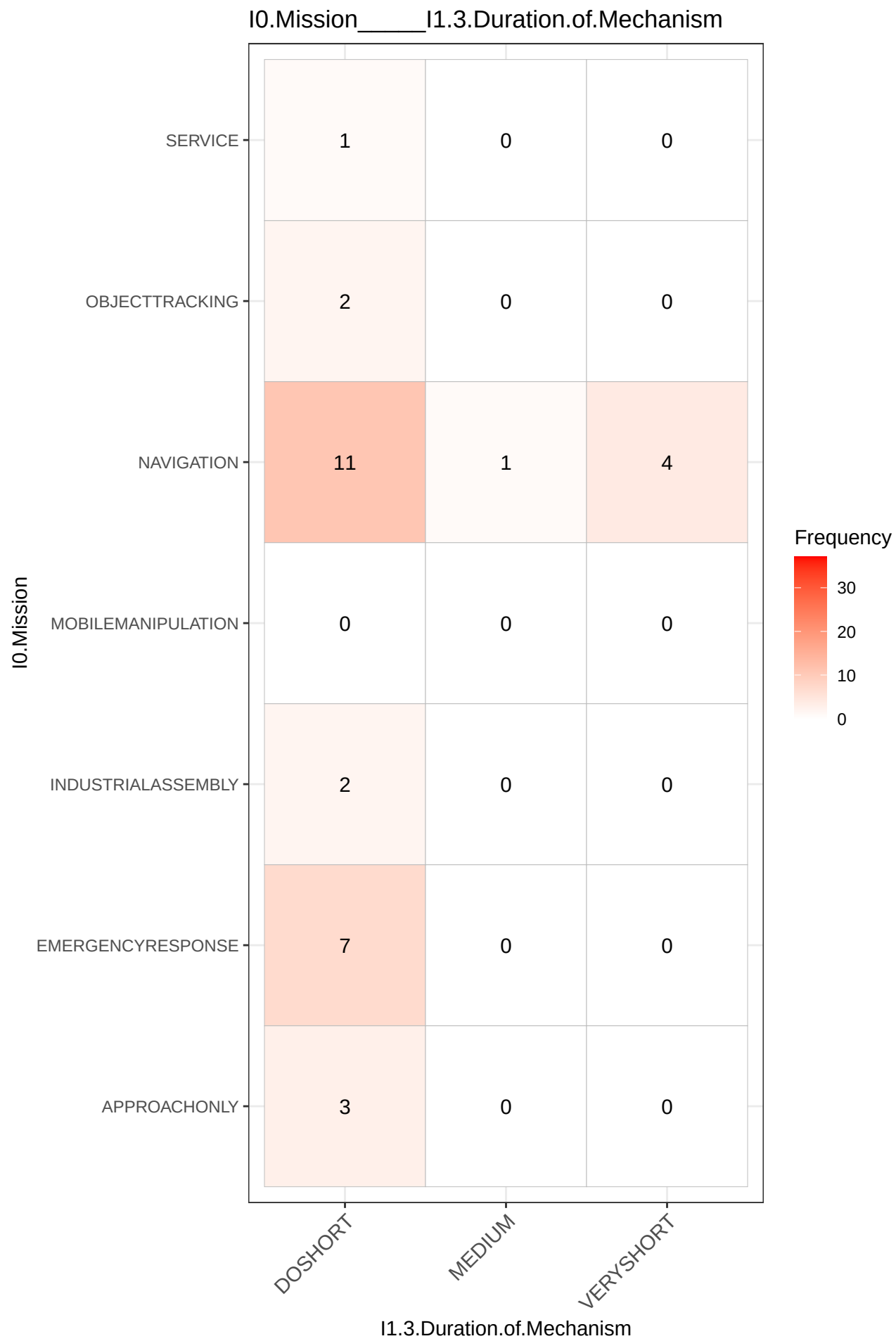


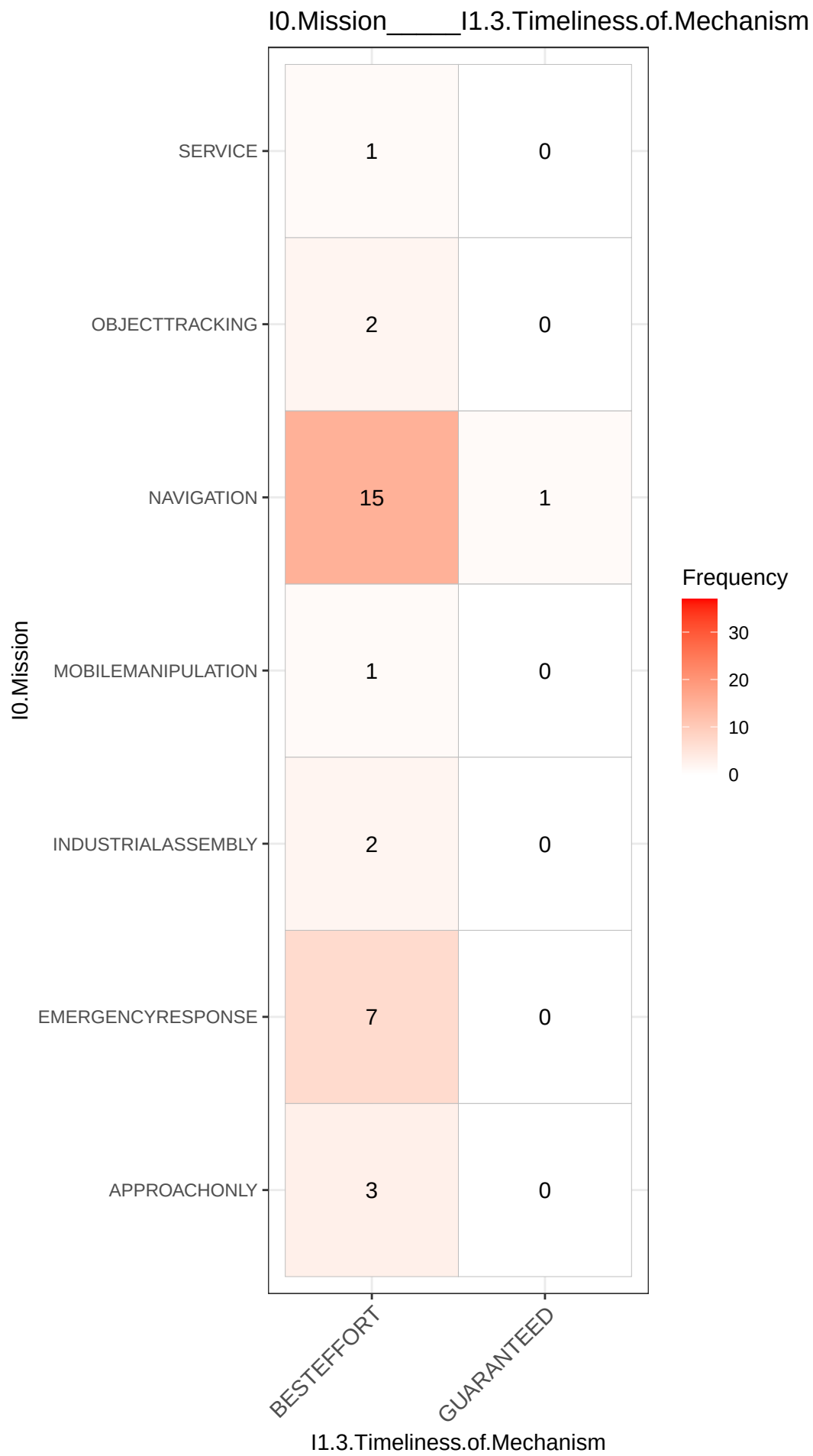


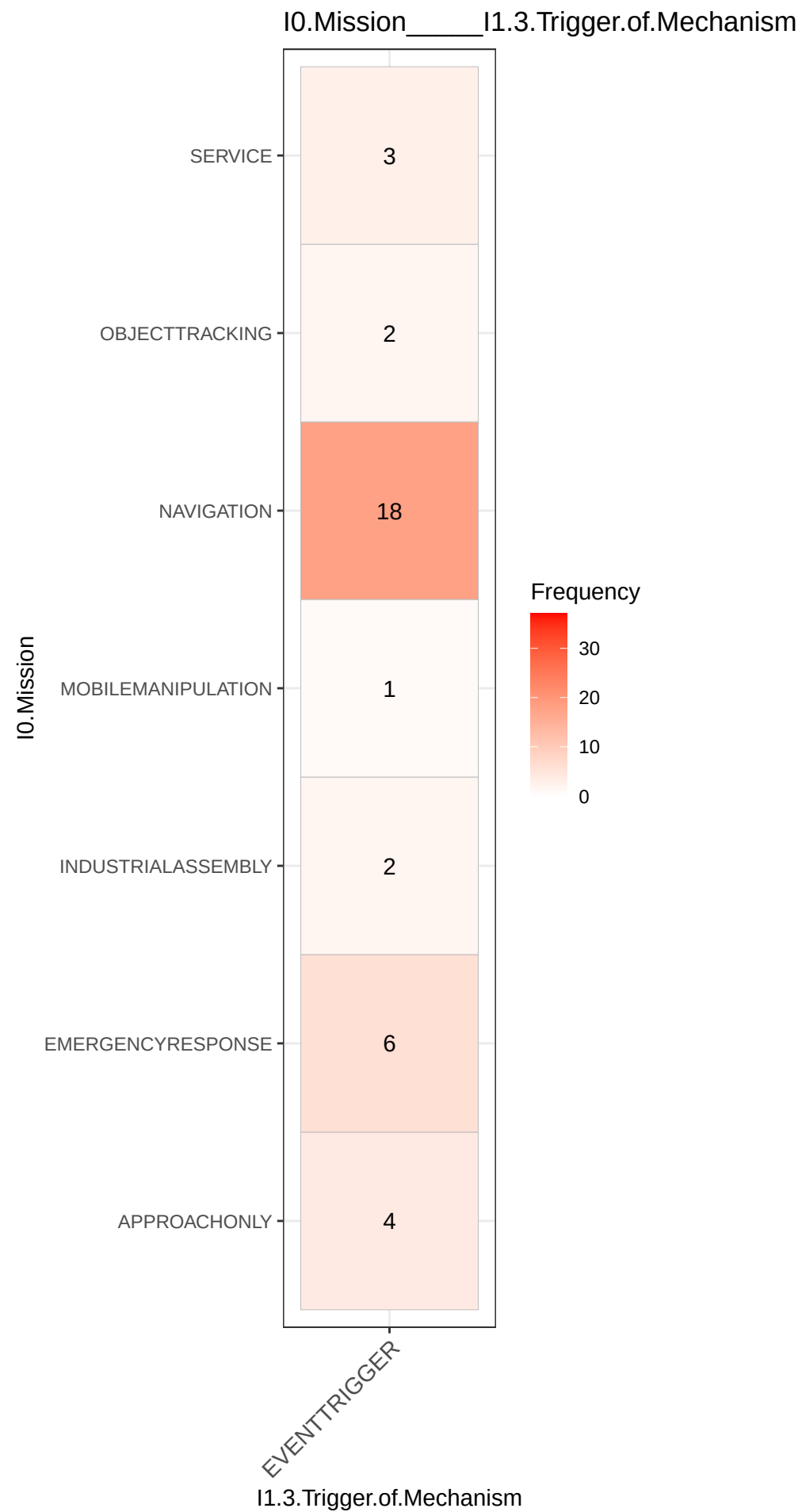


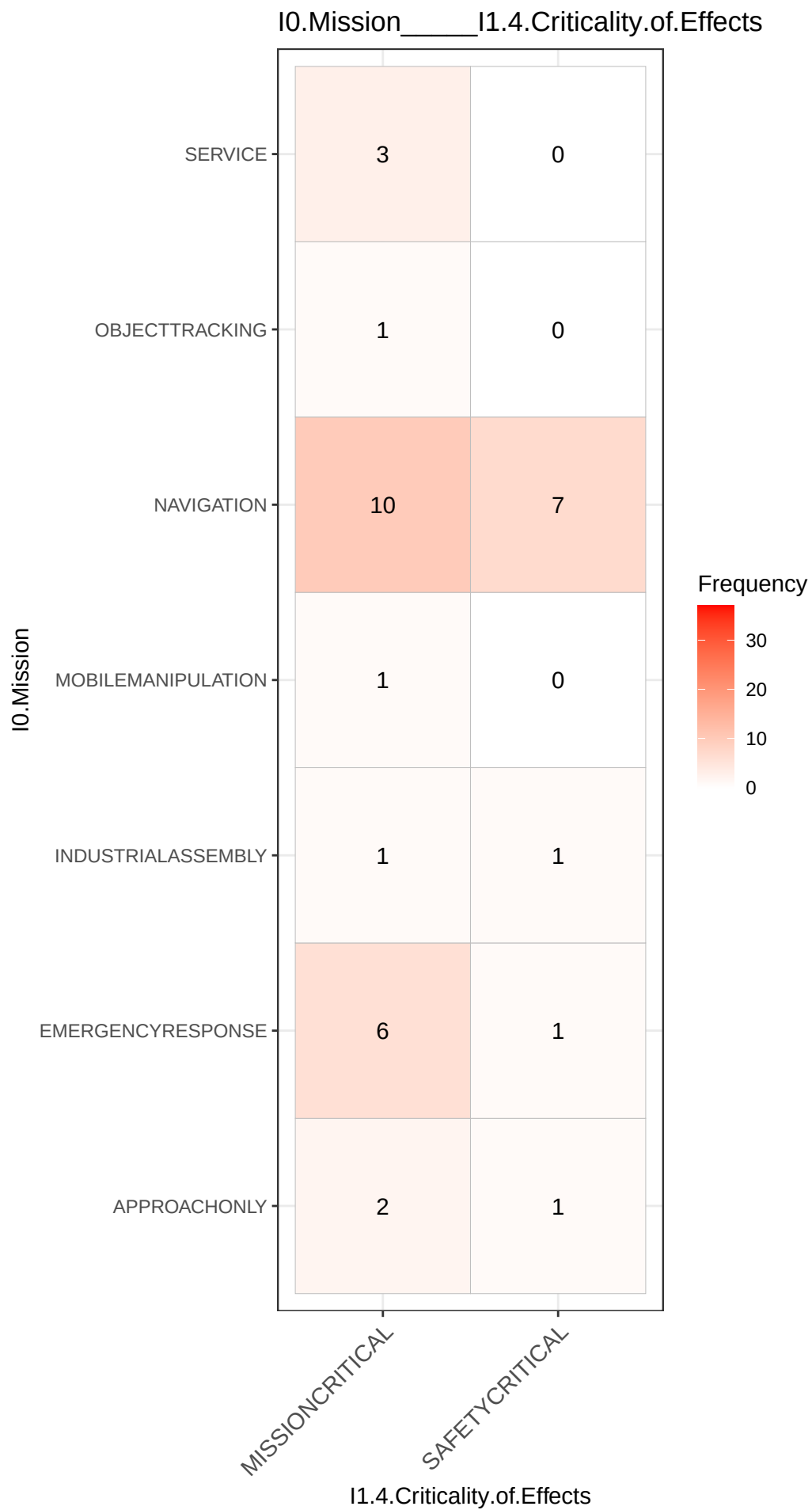


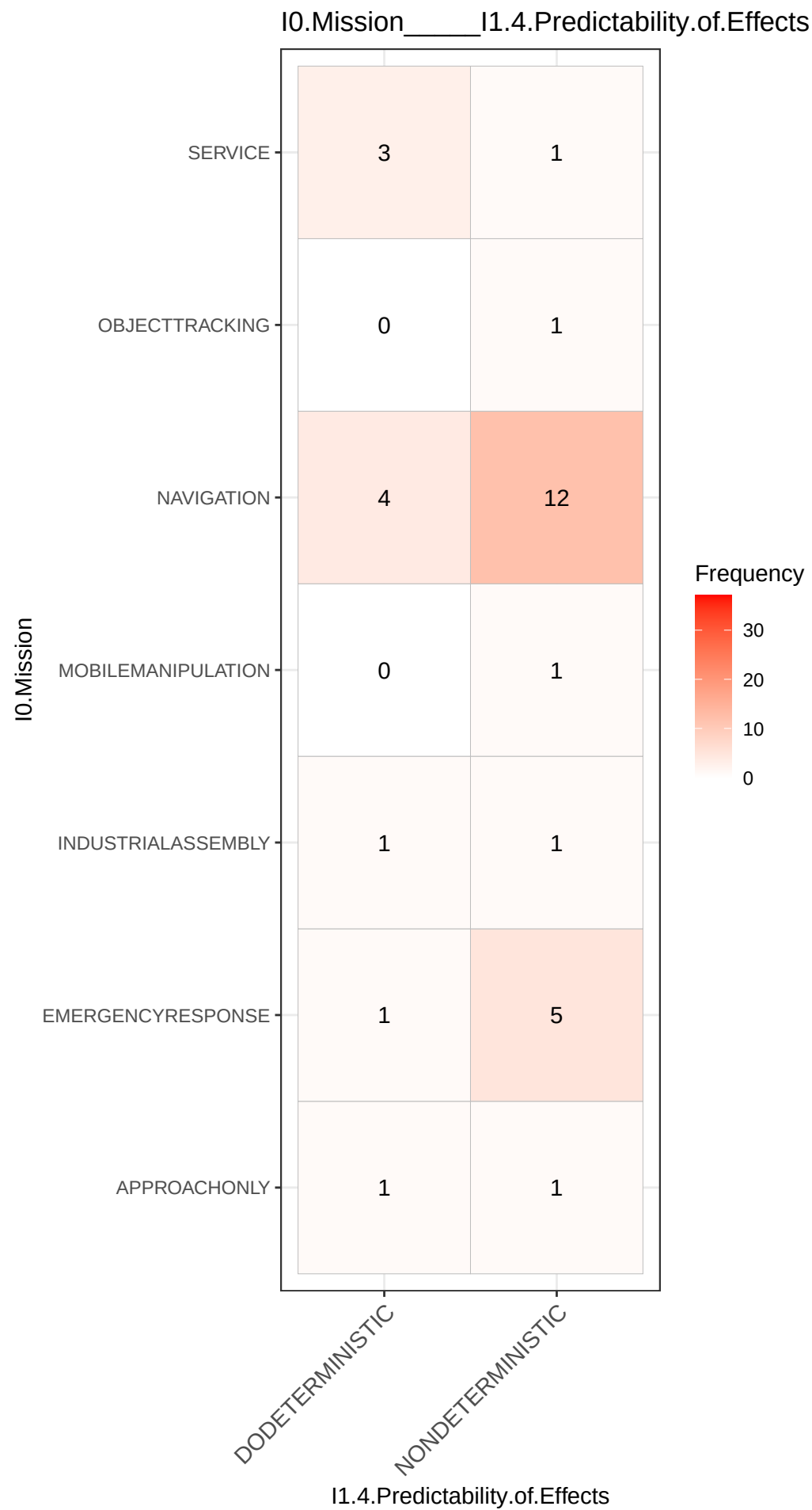


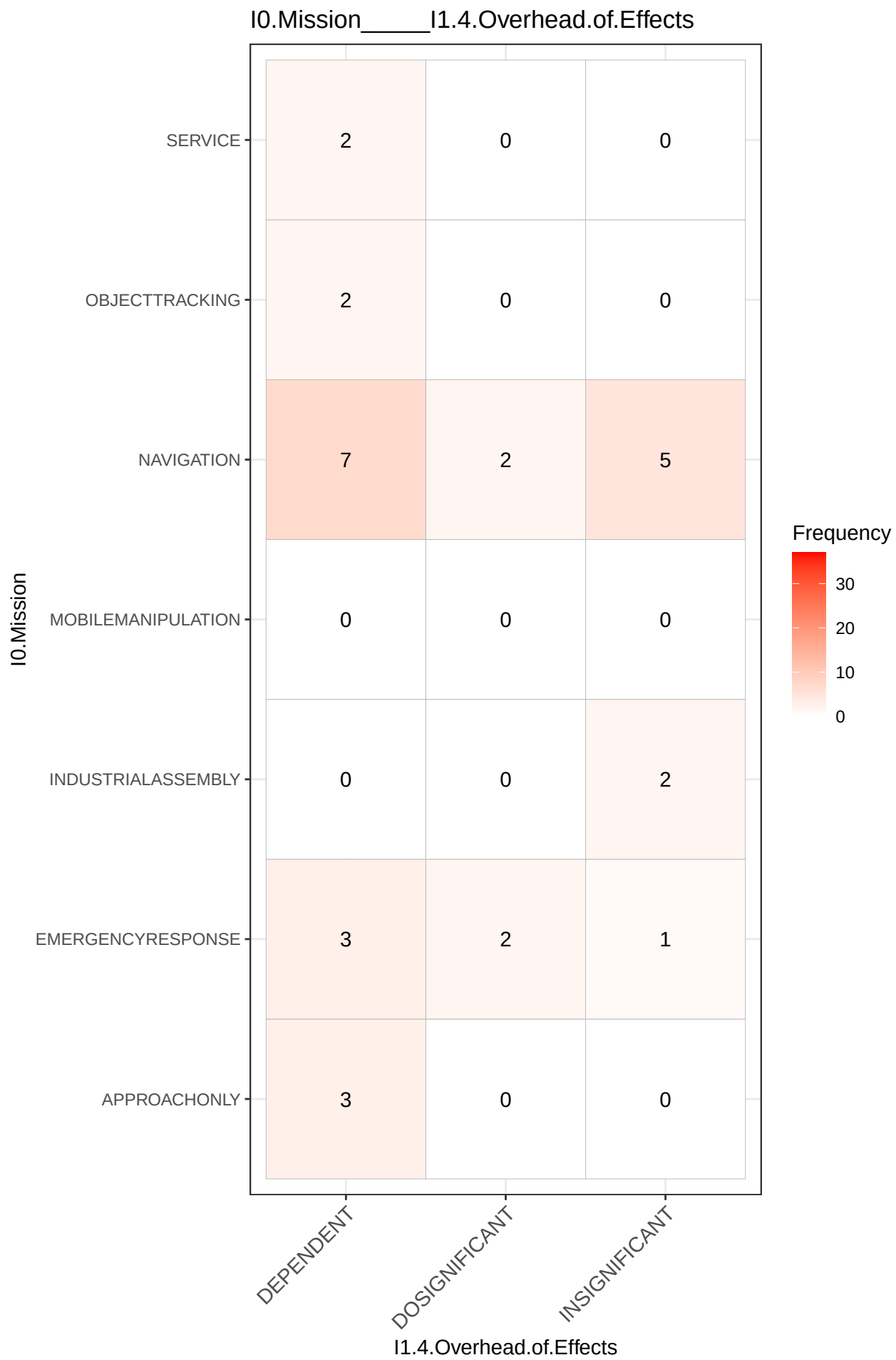


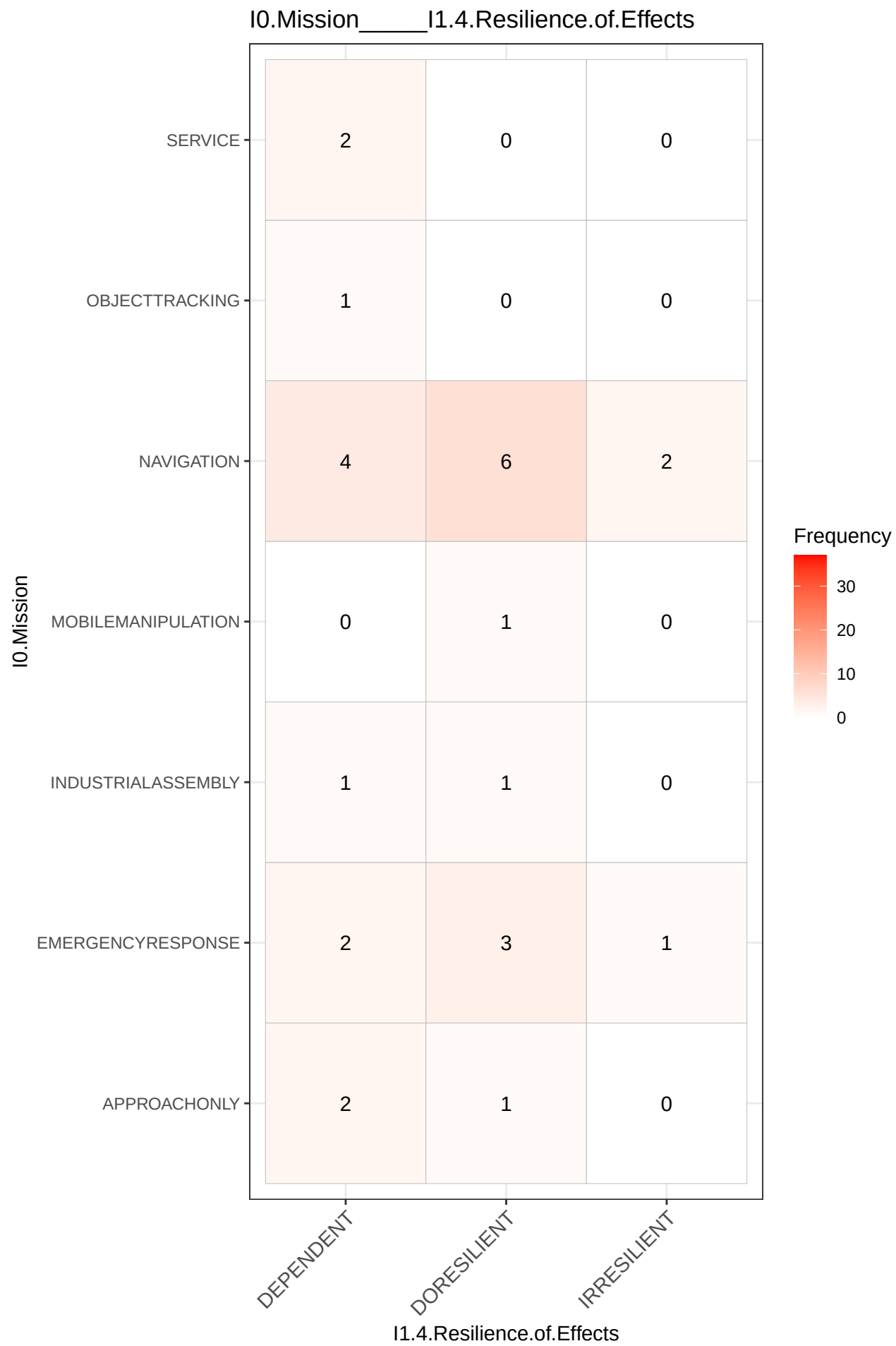




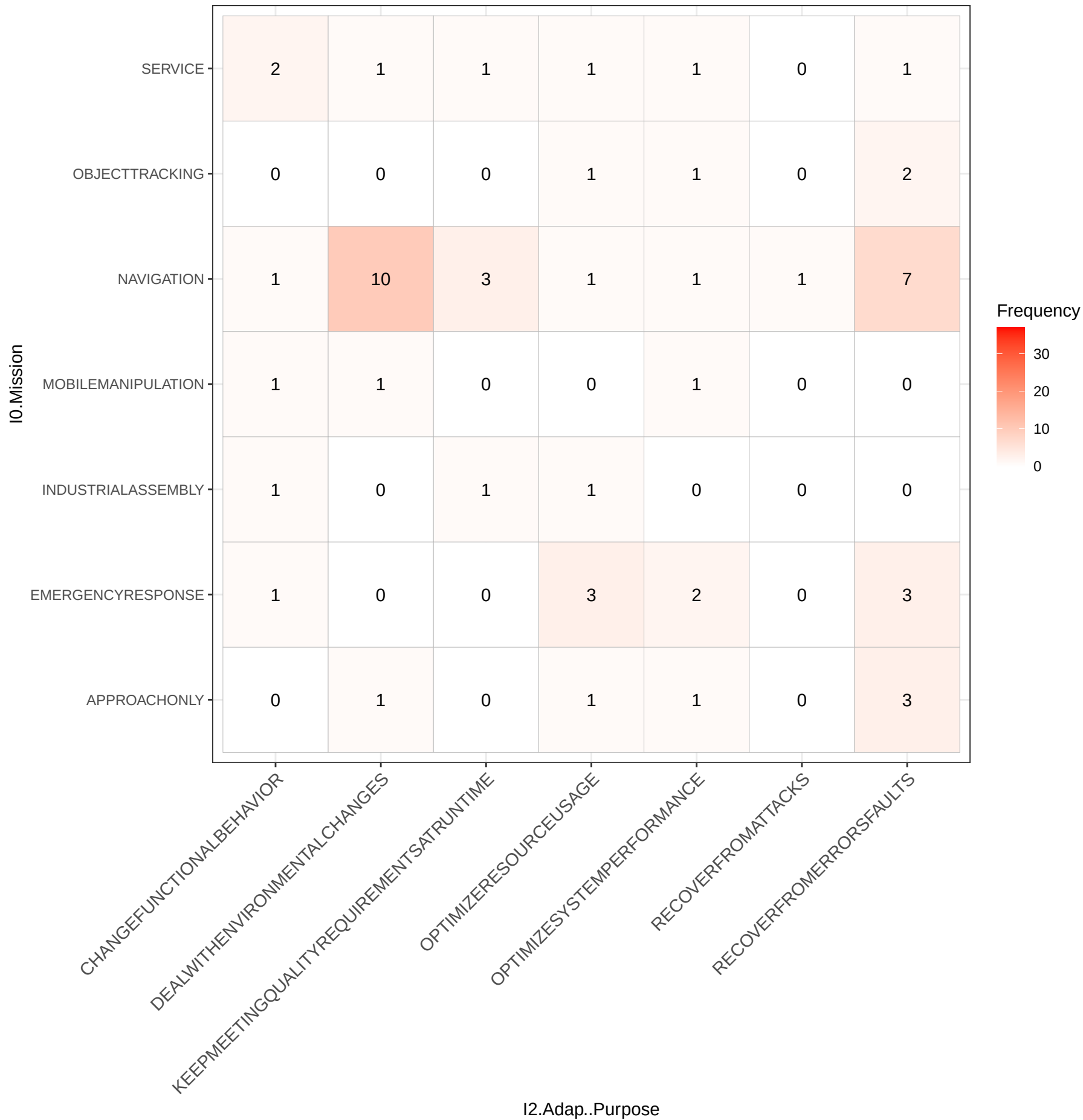


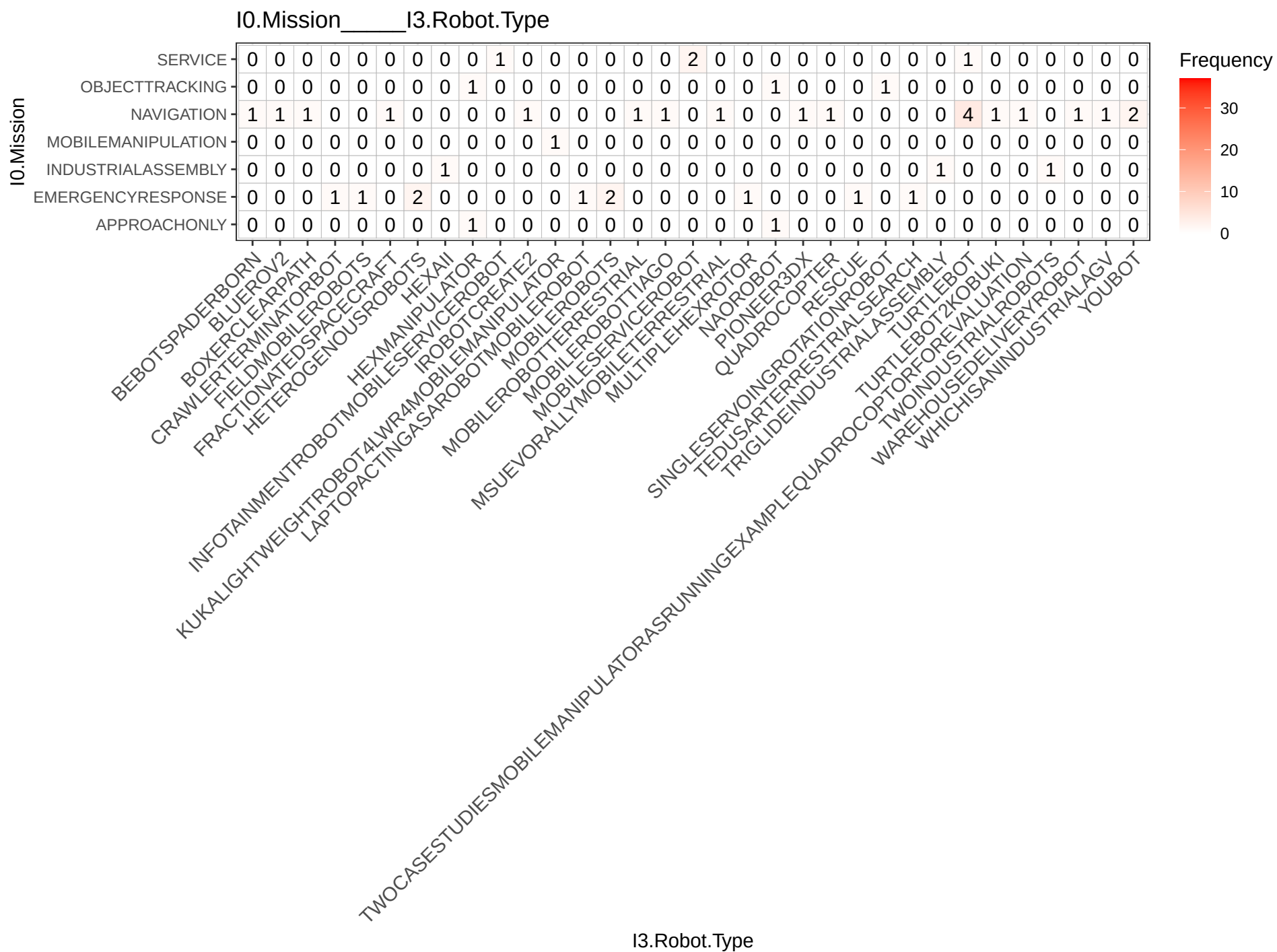


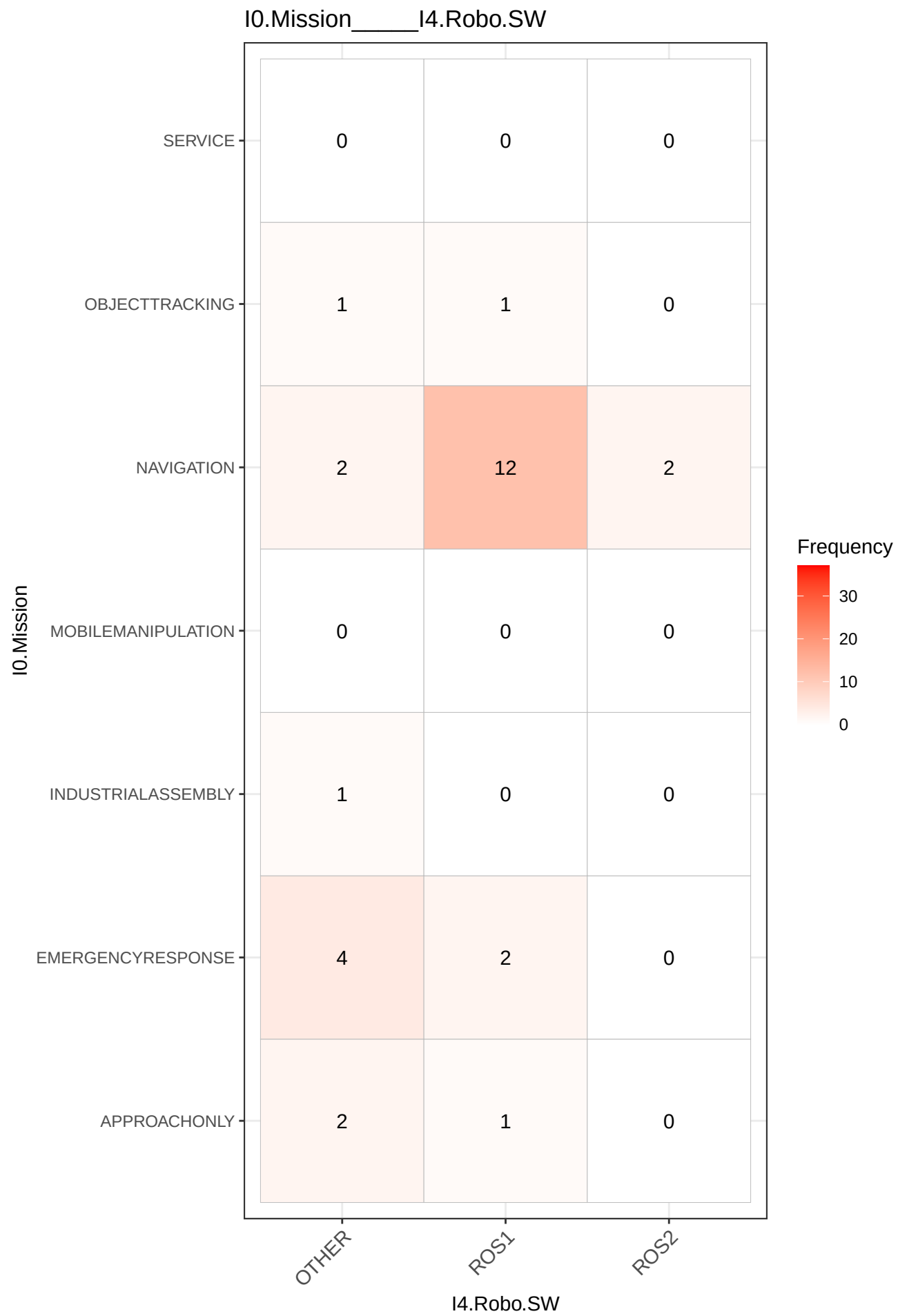


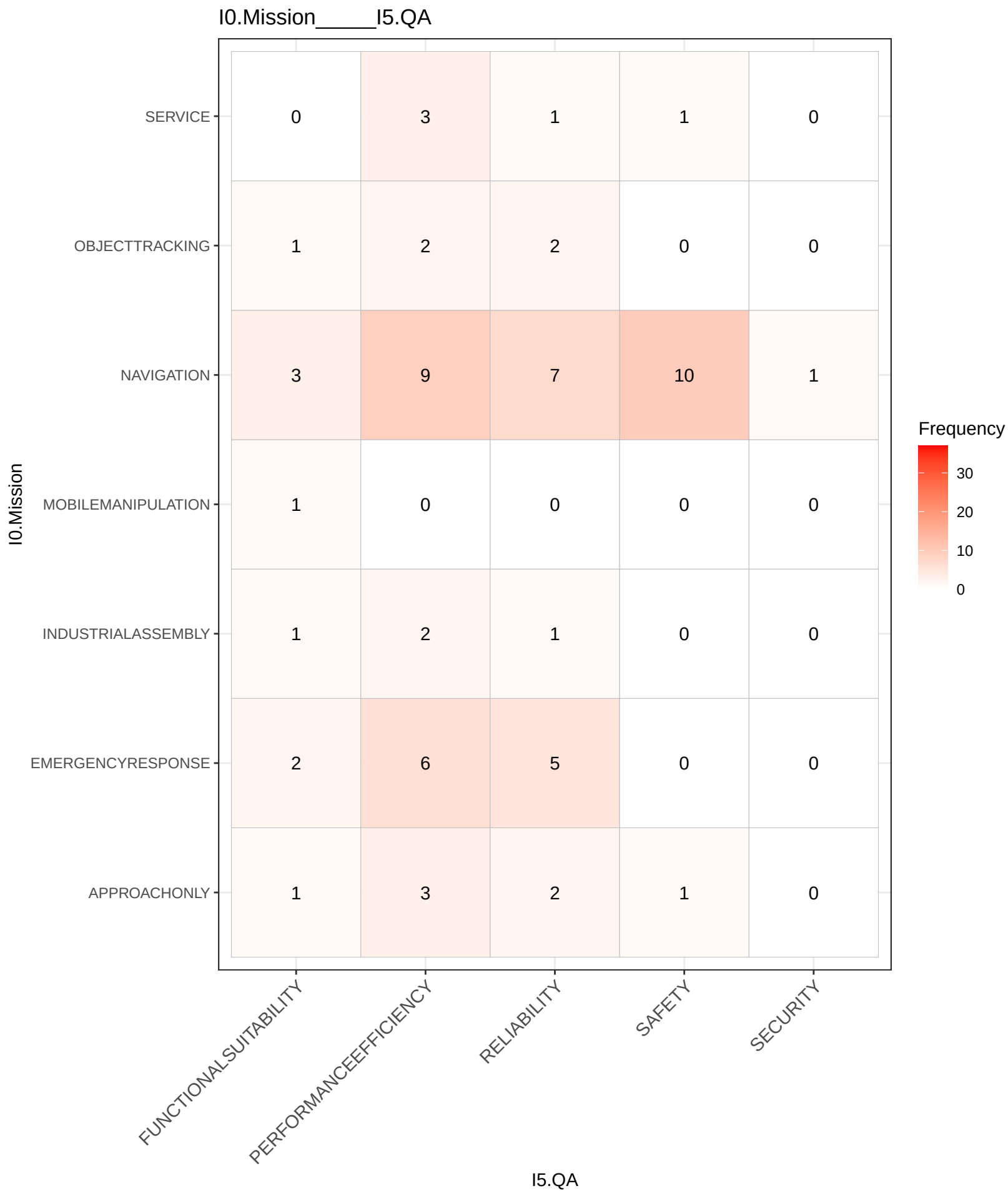


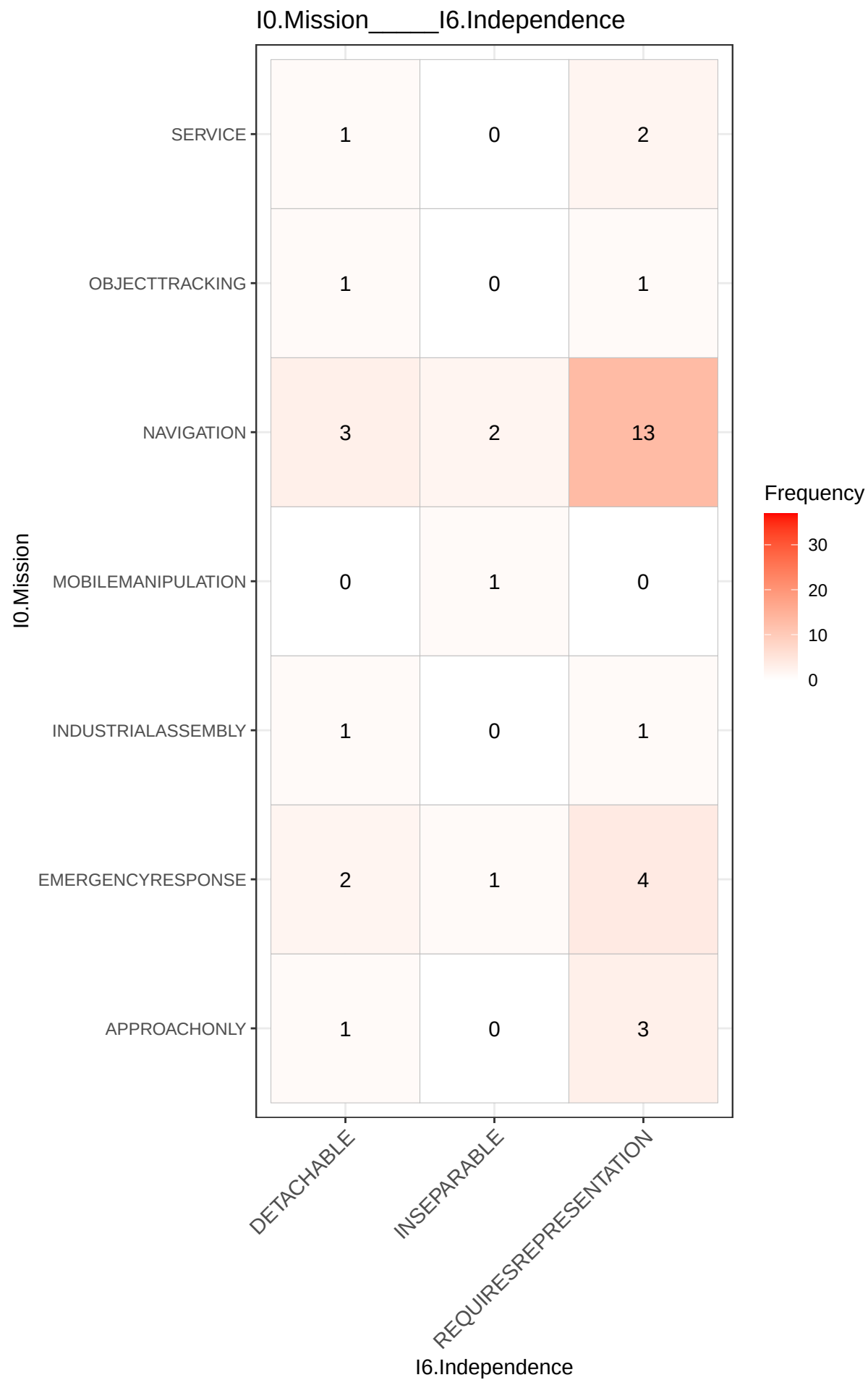
I0.Mission_____I2.Adap..Purpose



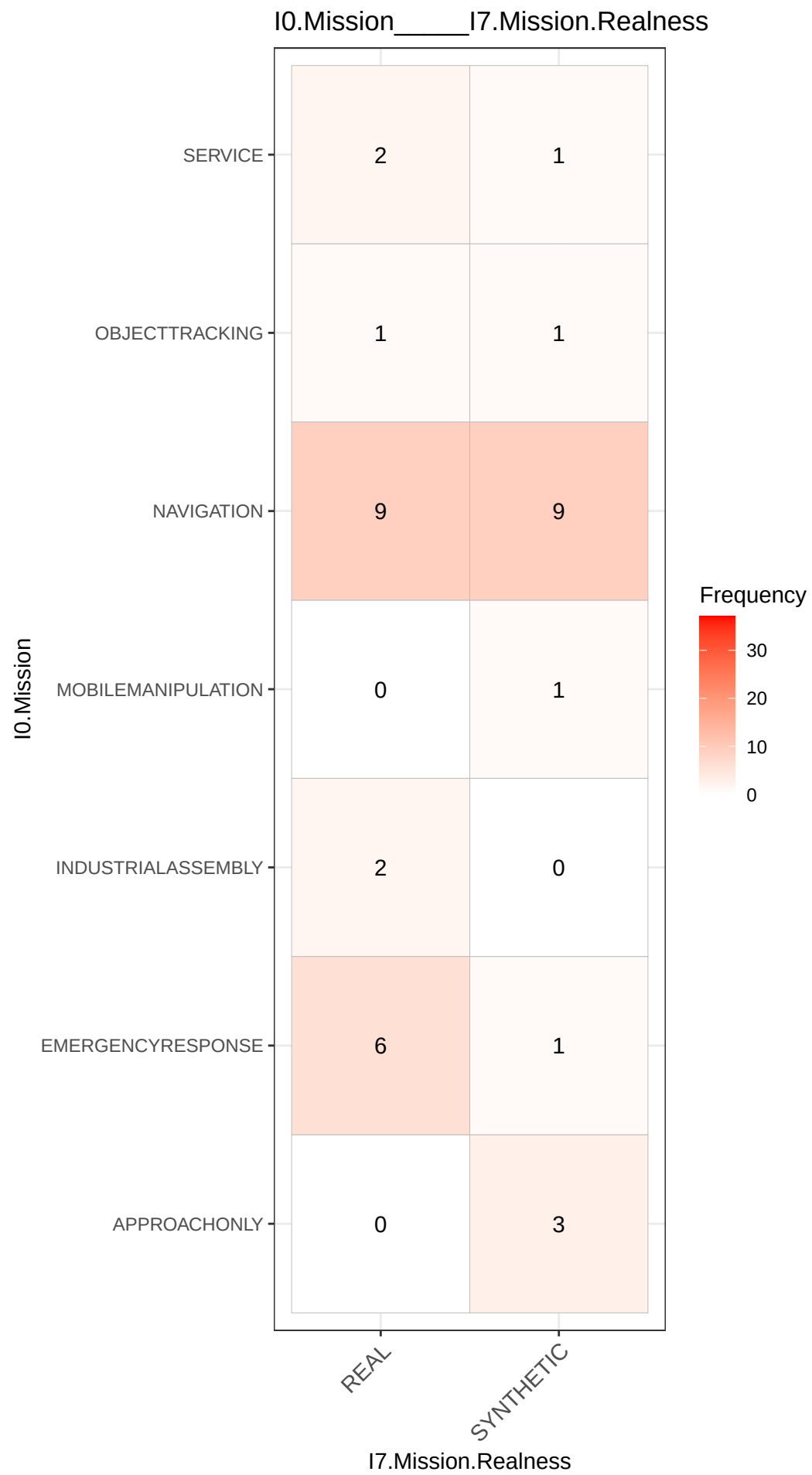


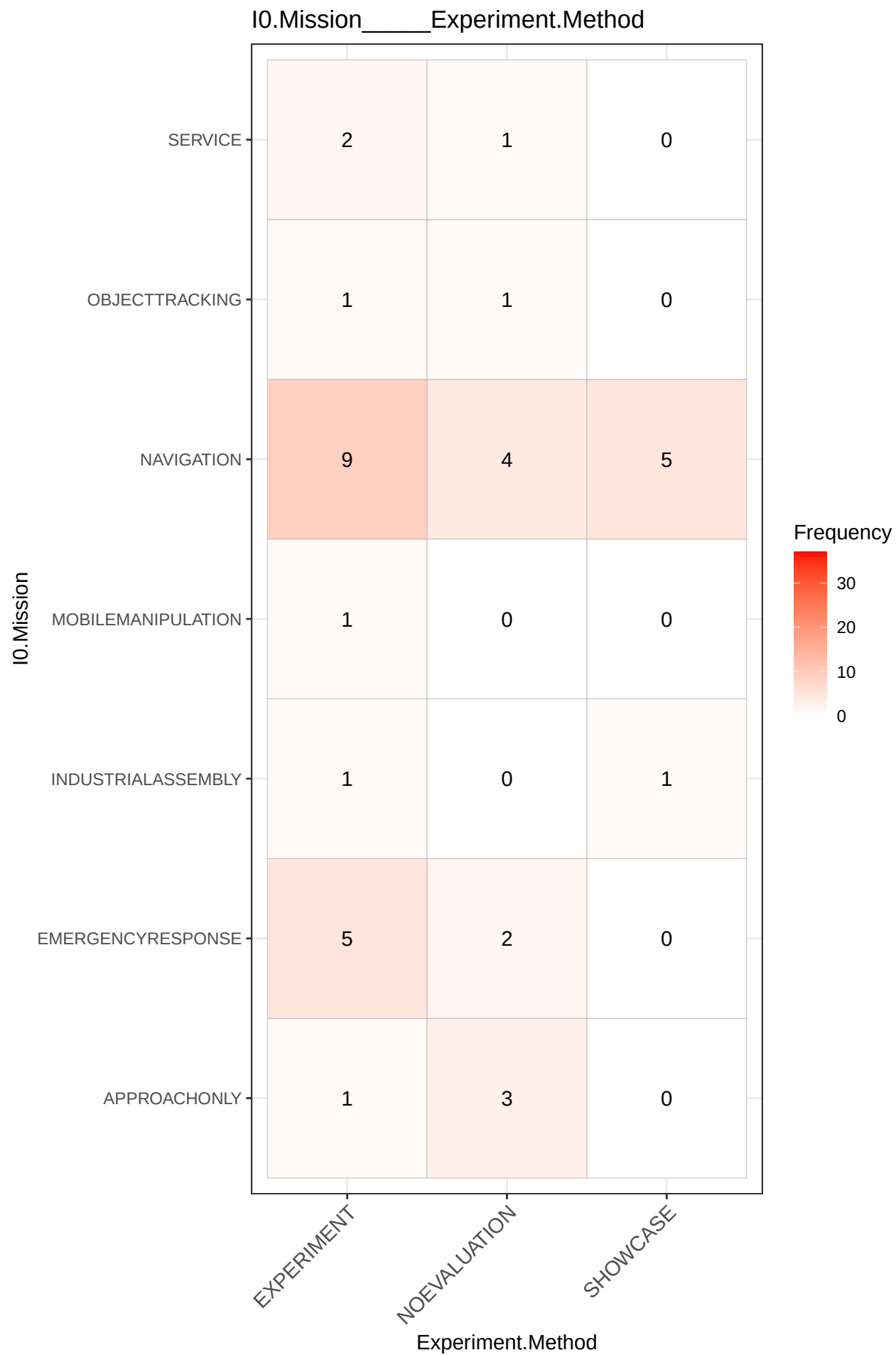




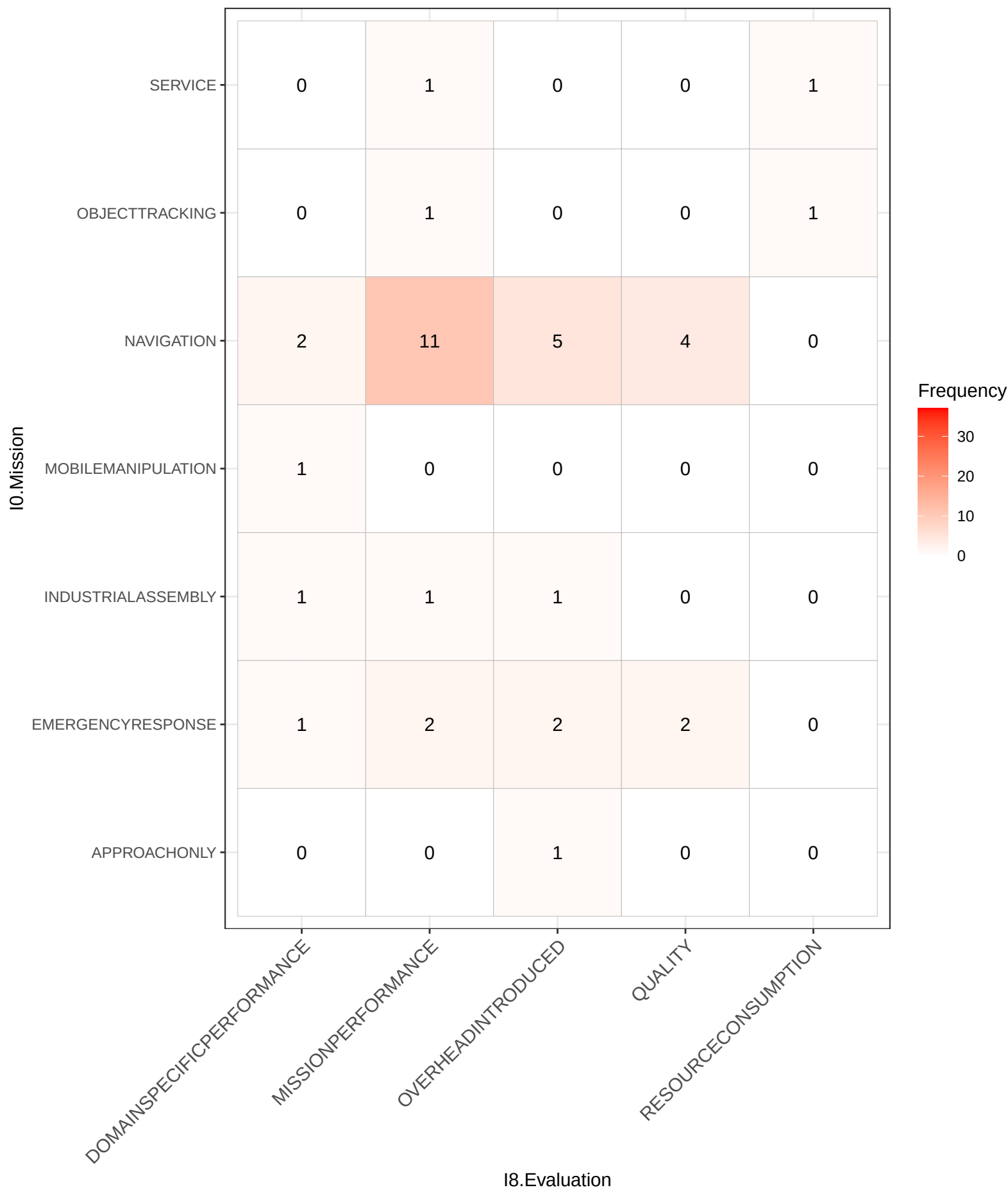






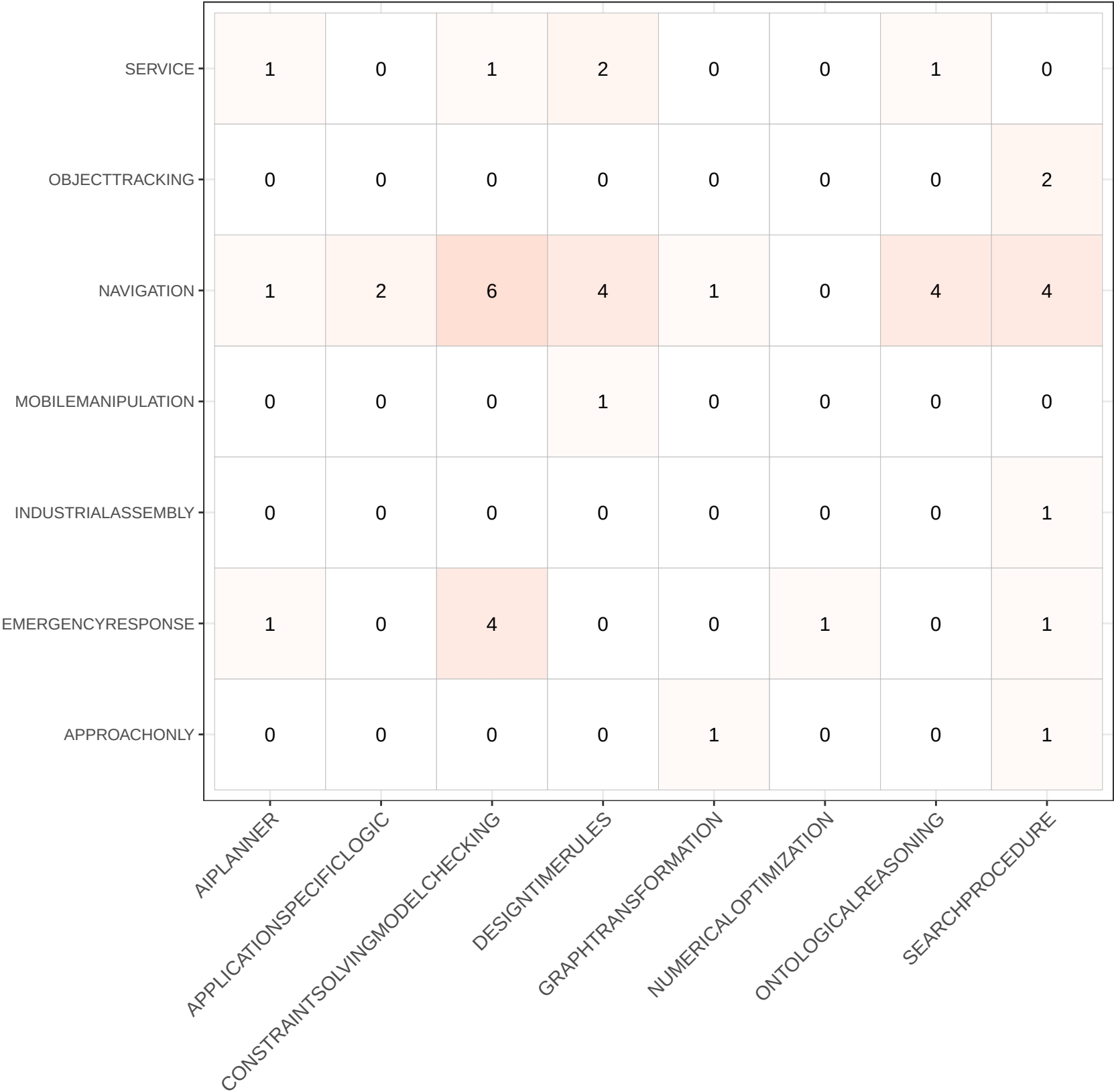


10.Mission_____18.Evaluation



IO.Mission_____I9.Adap..Logic

IO.Mission



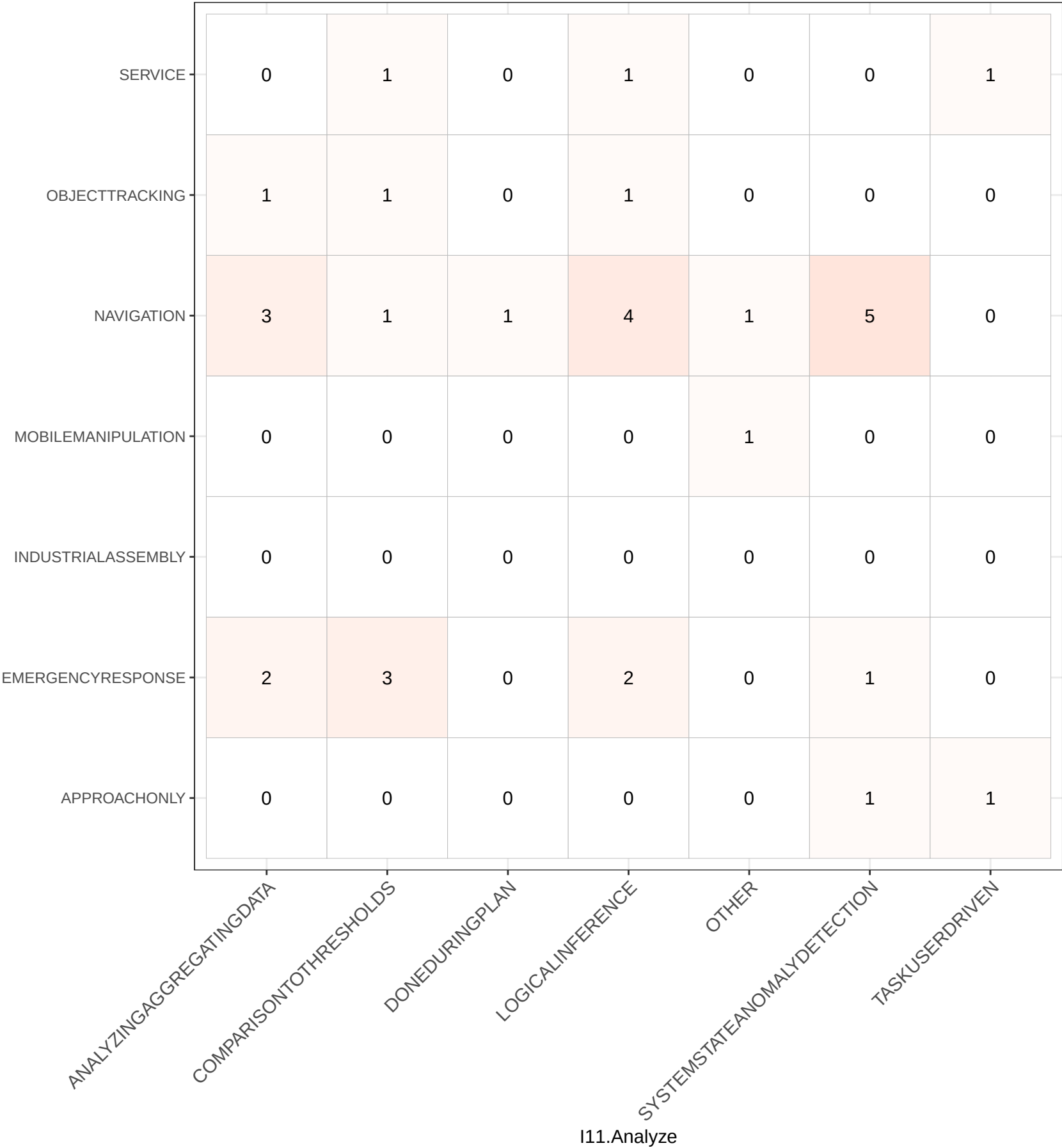
Frequency





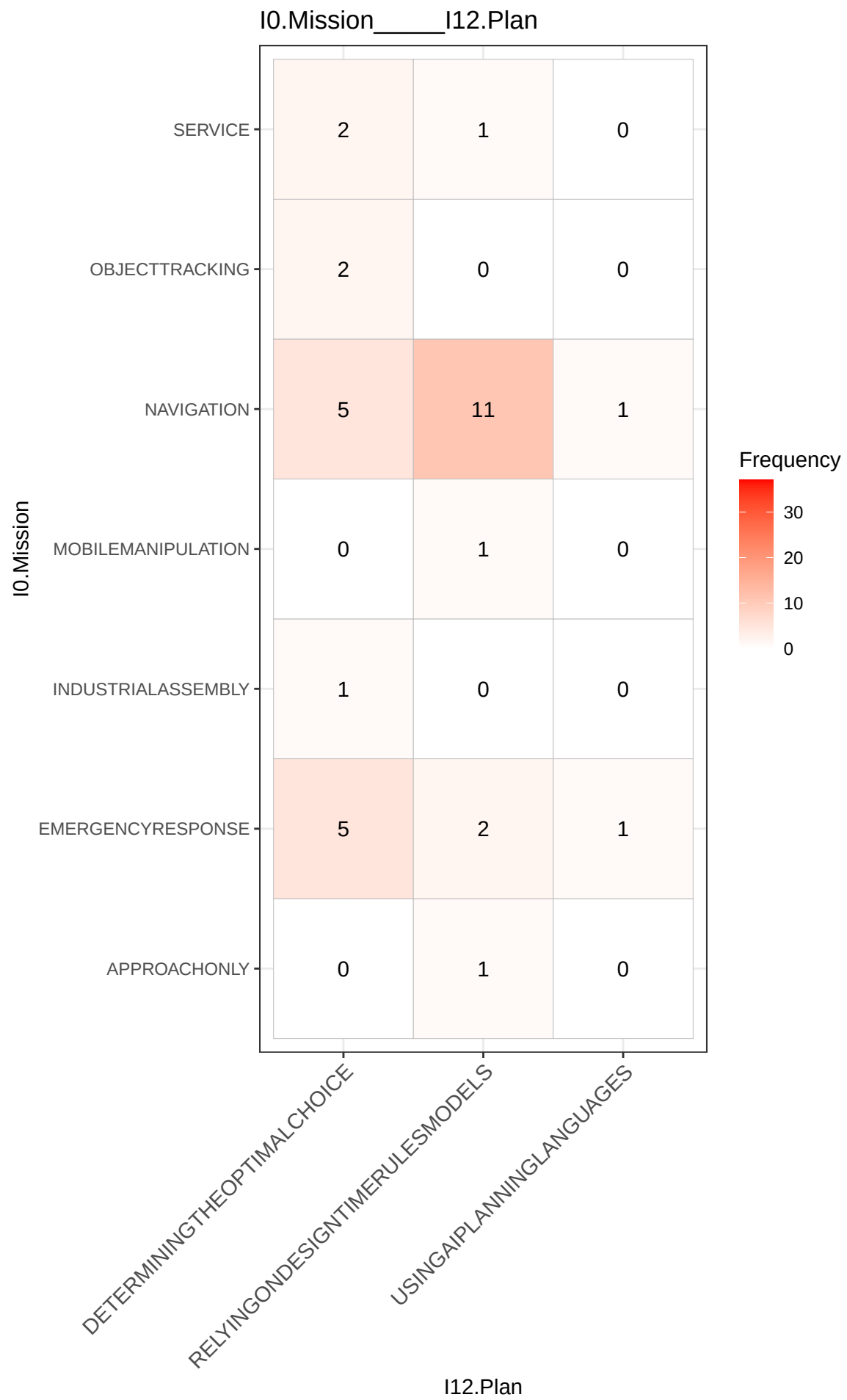
I0.Mission_____I11.Analyze

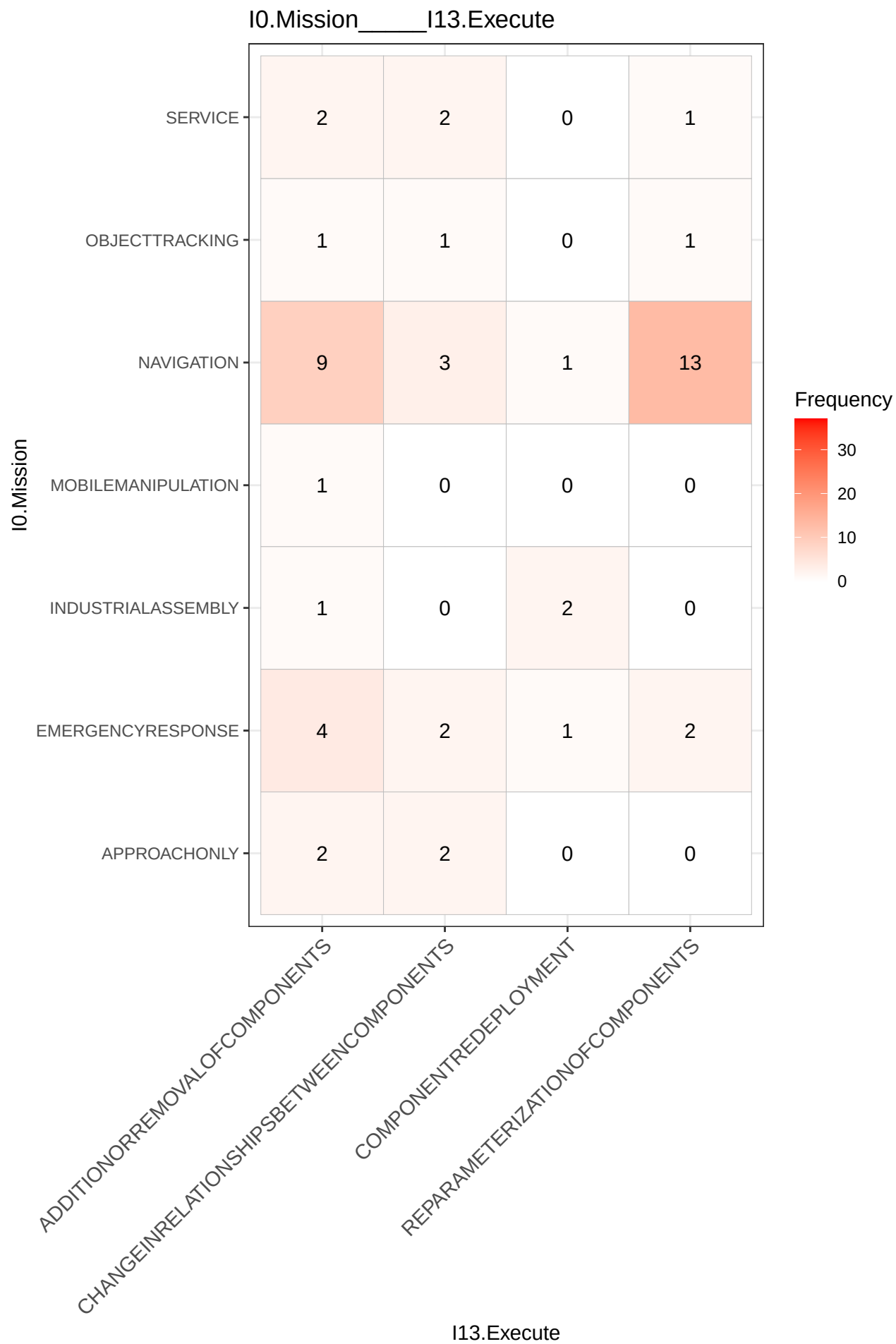
I0.Mission



Frequency

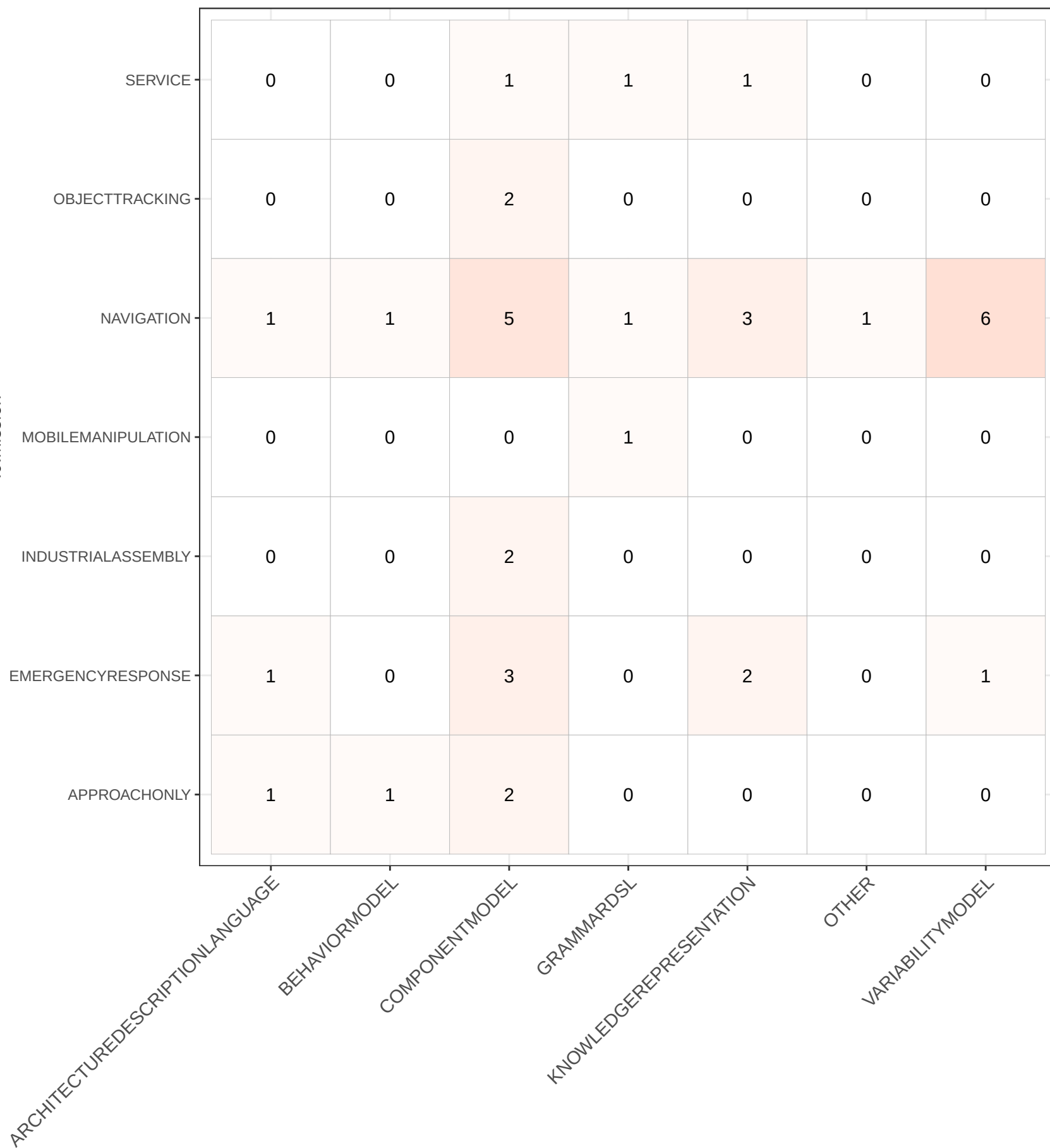






I0.Mission_____I14.Knowledge

I0.Mission



Frequency



I1.1.Metadata_____I1.2.Source.of.Change

I1.1.Metadata

STATIC

13

14

DYNAMIC

0

6

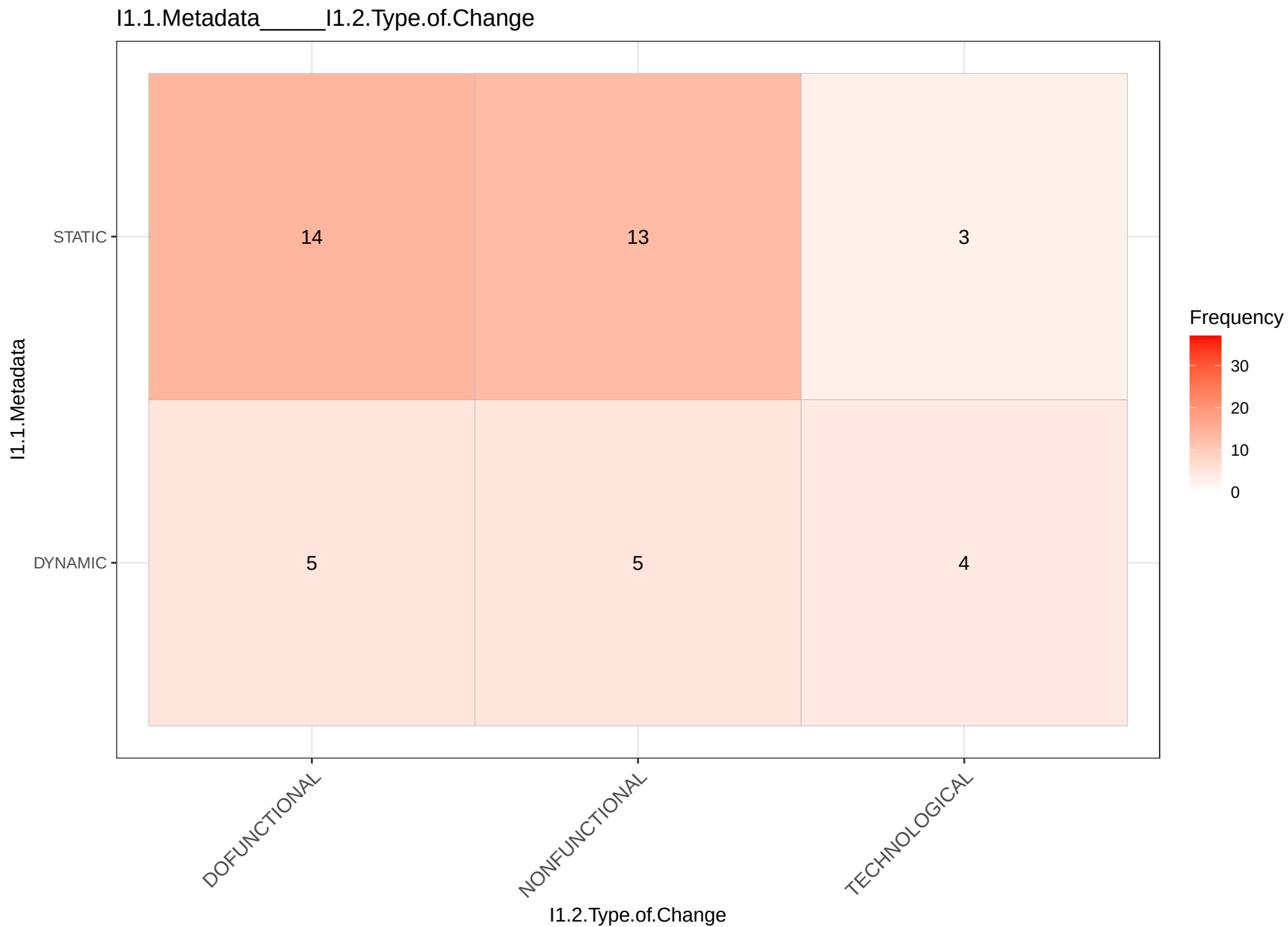
EXTERNAL

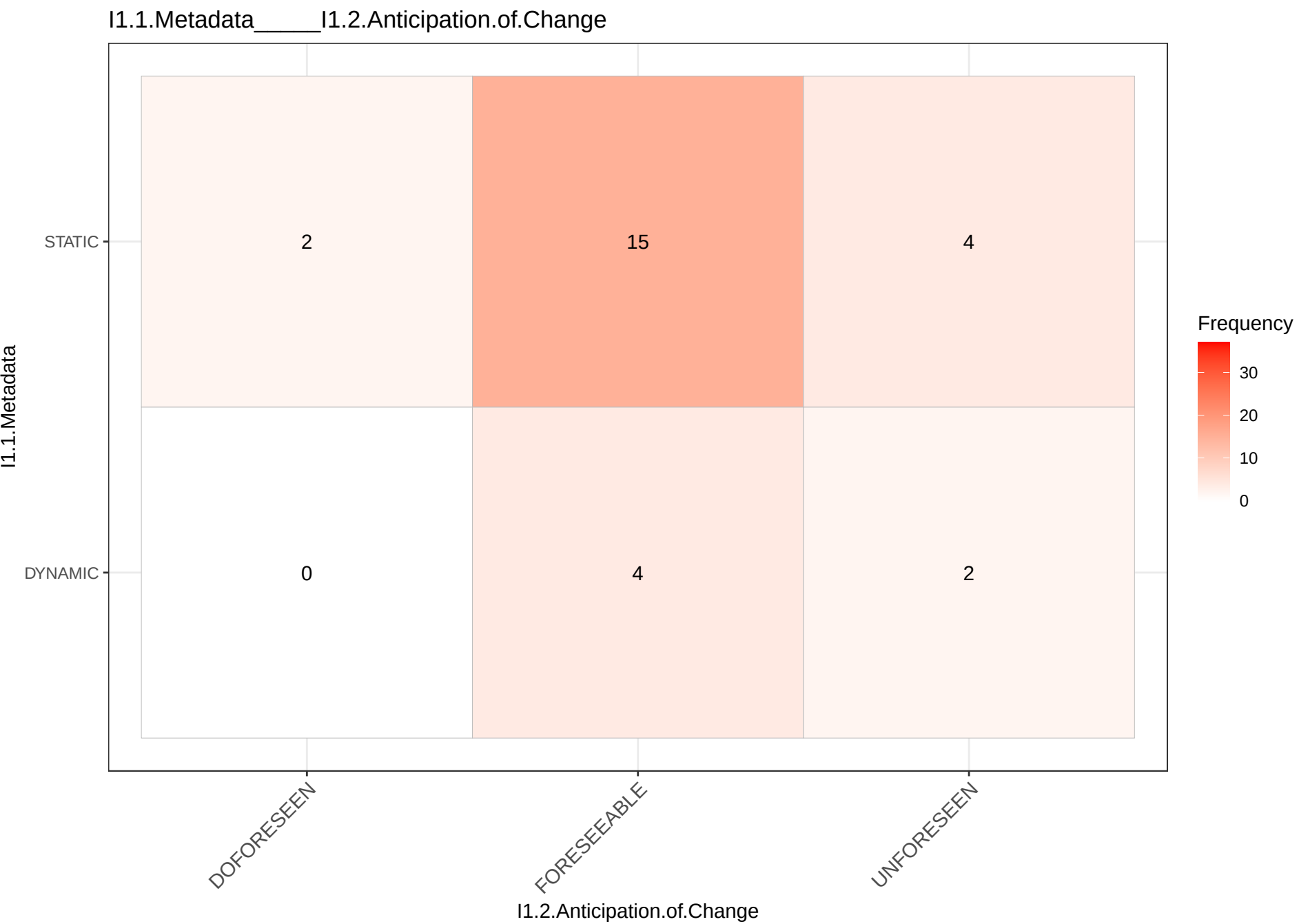
INTERNAL

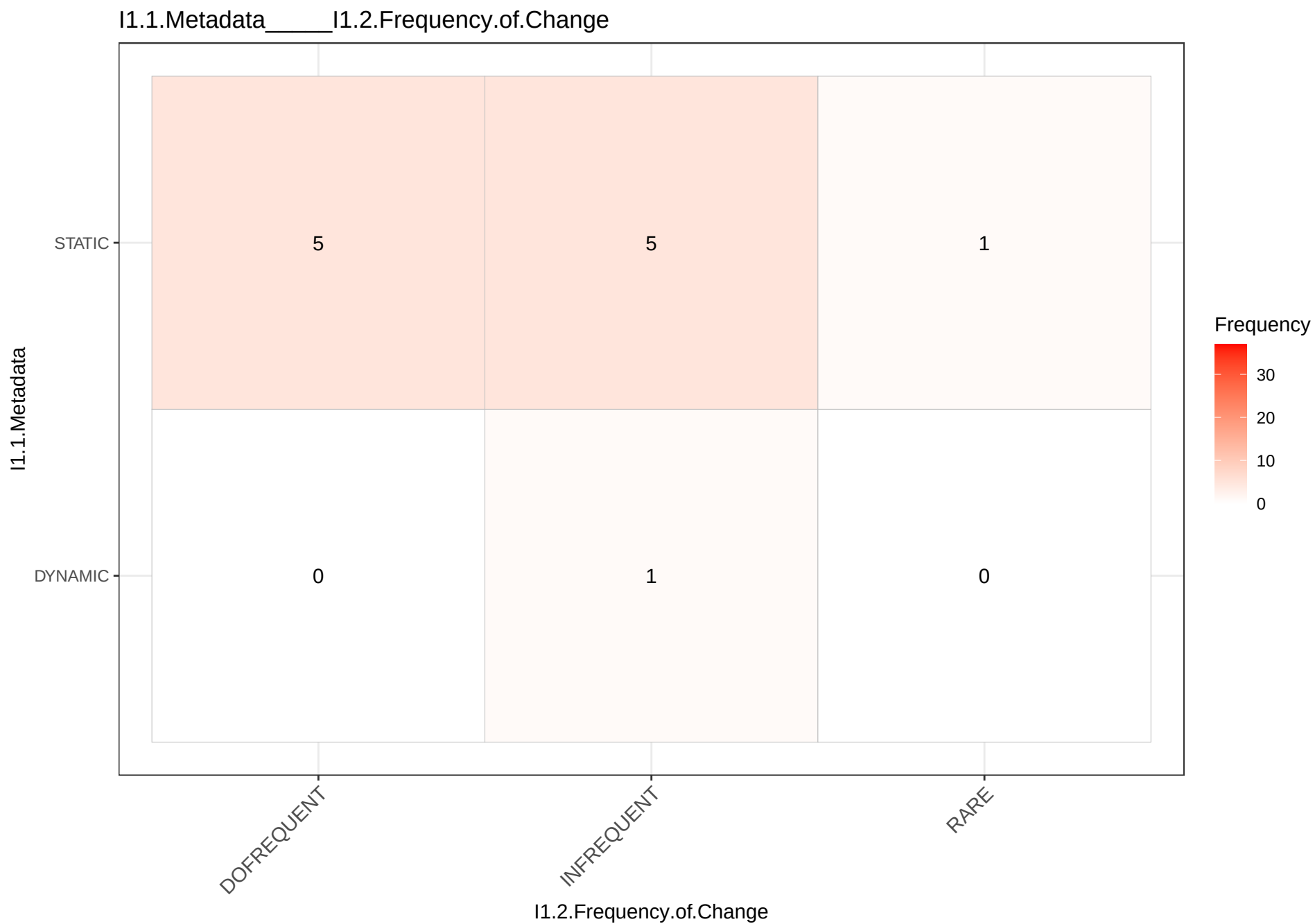
I1.2.Source.of.Change

Frequency

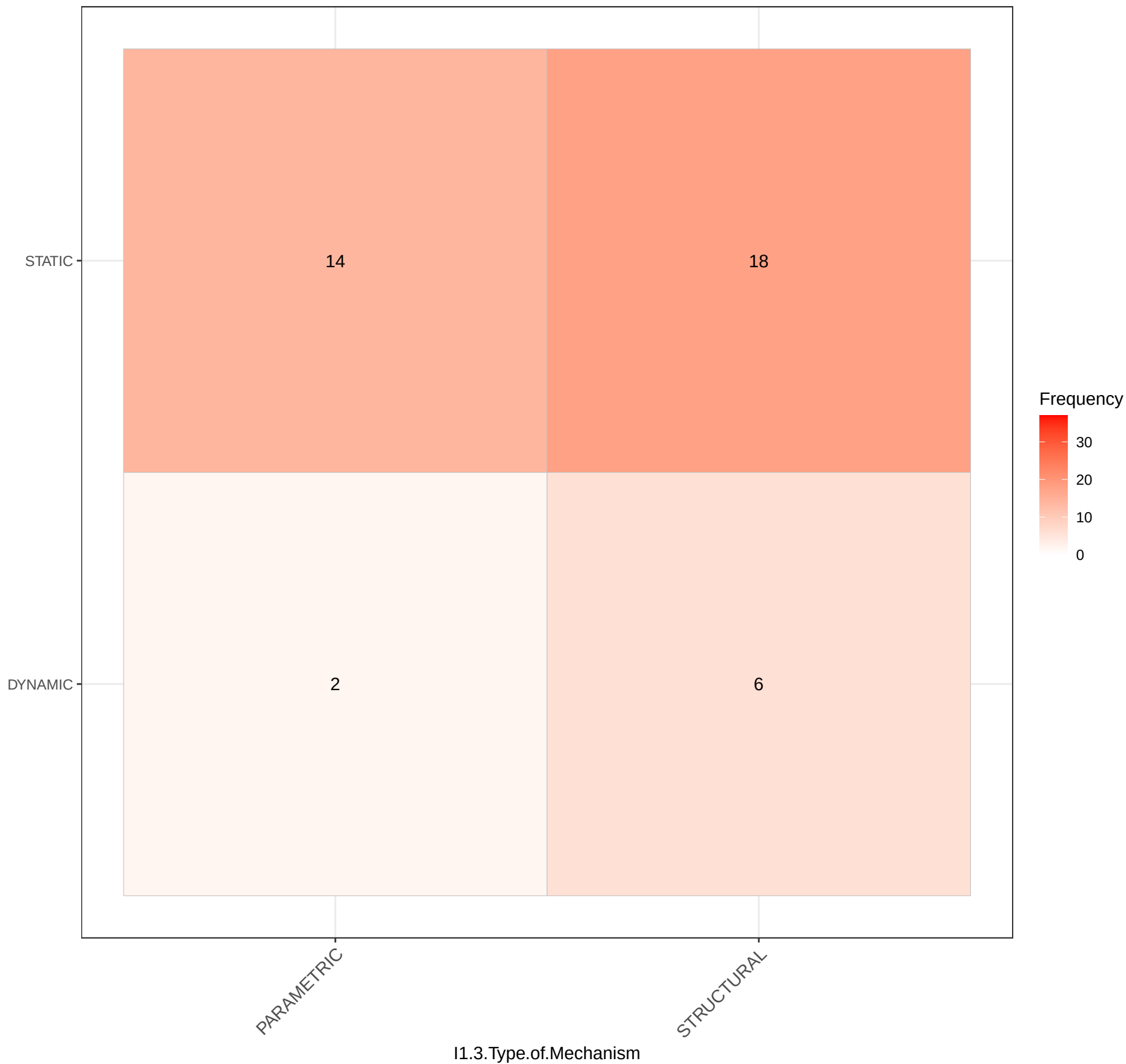








I1.1.Metadata_____I1.3.Type.of.Mechanism



STATIC

2

20

DYNAMIC

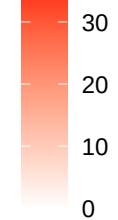
1

5

DECENTRALIZED

DOCENTRALIZED

Frequency



I1.1.Metadata_____I1.3.Scope.of.Mechanism

I1.1.Metadata

STATIC

4

19

DYNAMIC

1

6

GLOBAL

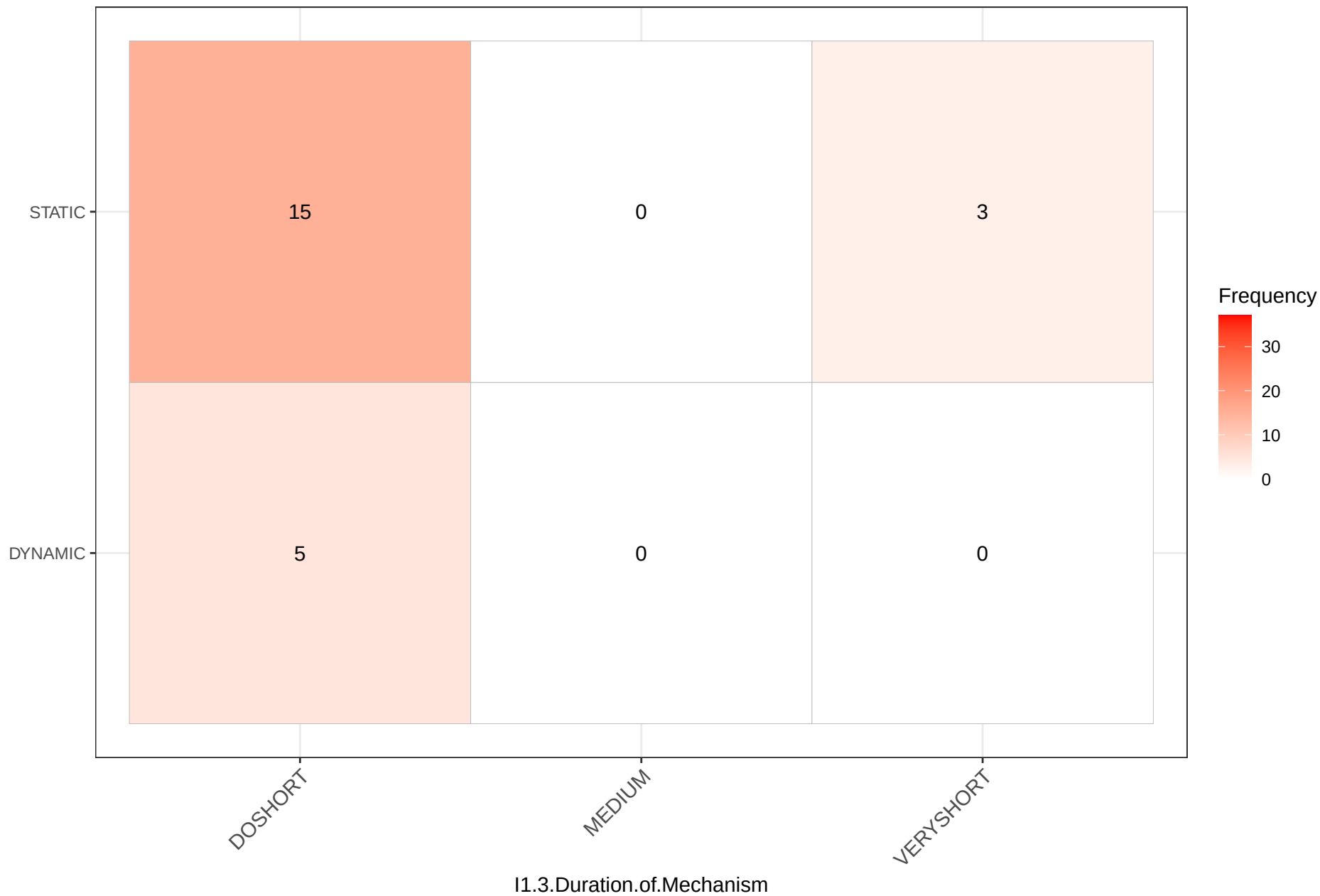
LOCAL

I1.3.Scope.of.Mechanism

Frequency



I1.1.Metadata_____I1.3.Duration.of.Mechanism



I1.1.Metadata_____I1.3.Timeliness.of.Mechanism

I1.1.Metadata

STATIC

18

0

DYNAMIC

6

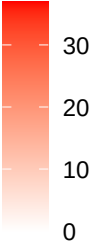
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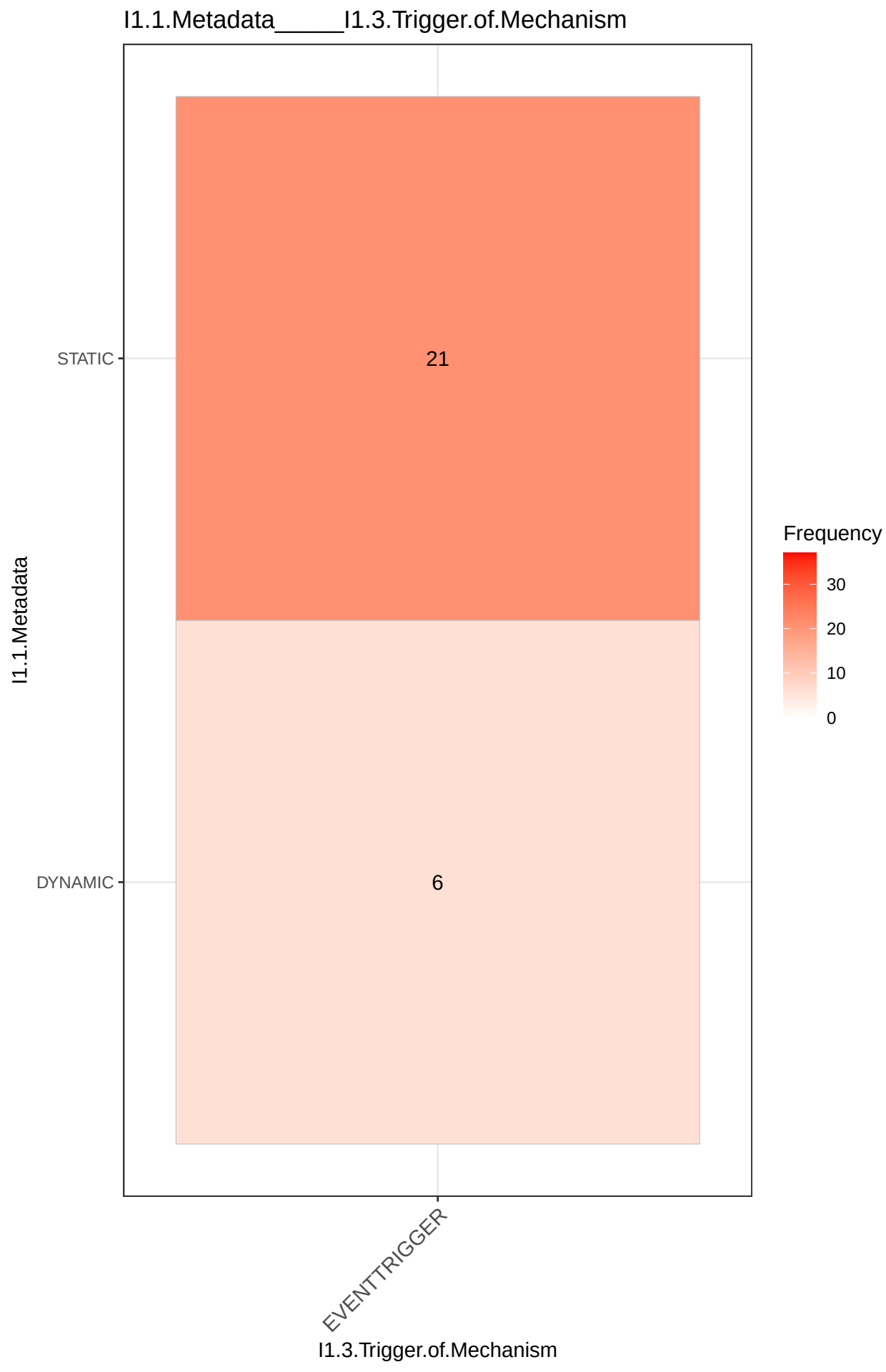
BESTEFFORT

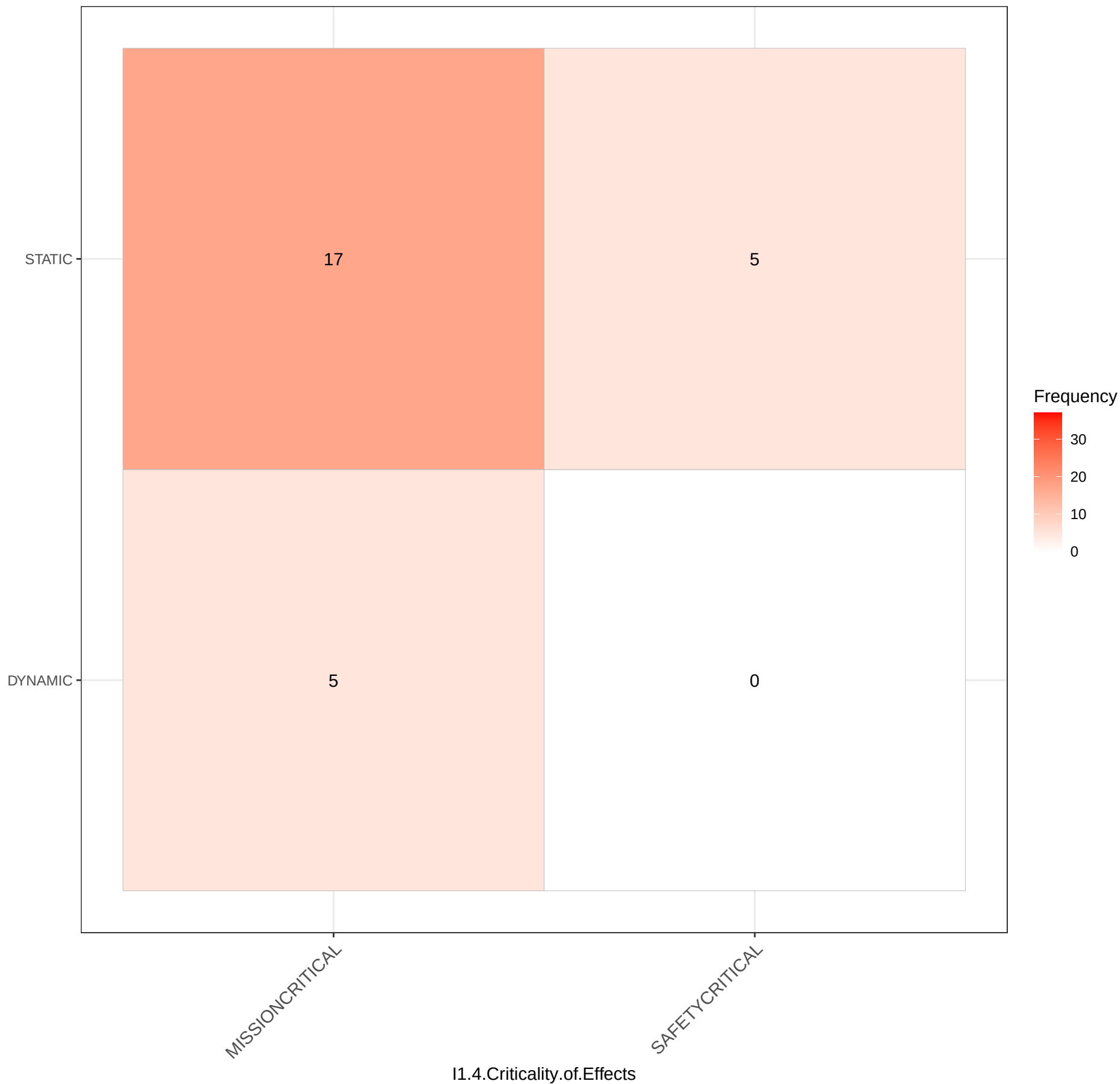
GUARANTEED

I1.3.Timeliness.of.Mechanism

Frequency







I1.1.Metadata_____I1.4.Predictability.of.Effects

I1.1.Metadata

STATIC

6

15

DYNAMIC

0

4

DODETERMINISTIC

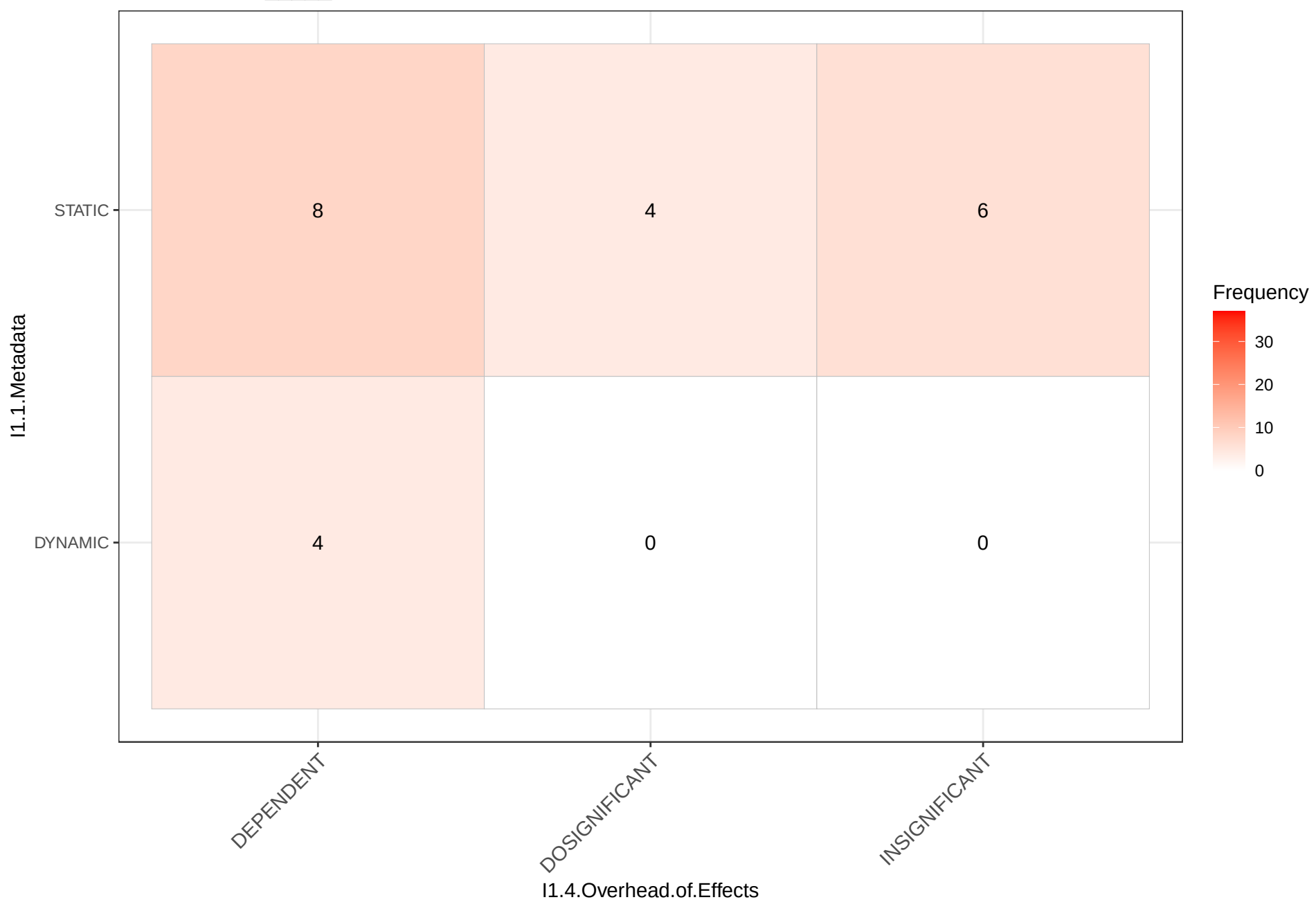
NONDETERMINISTIC

I1.4.Predictability.of.Effects

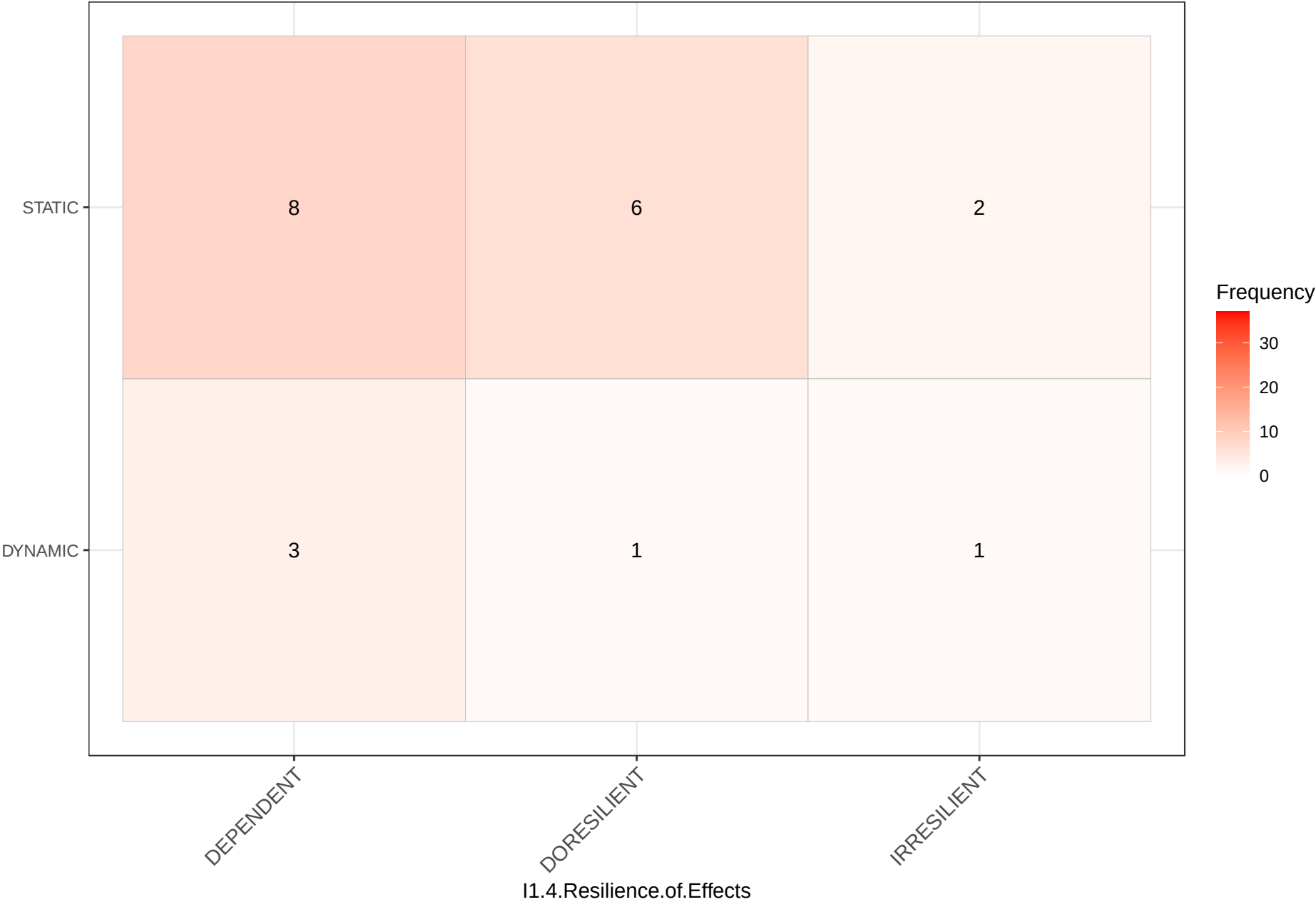
Frequency

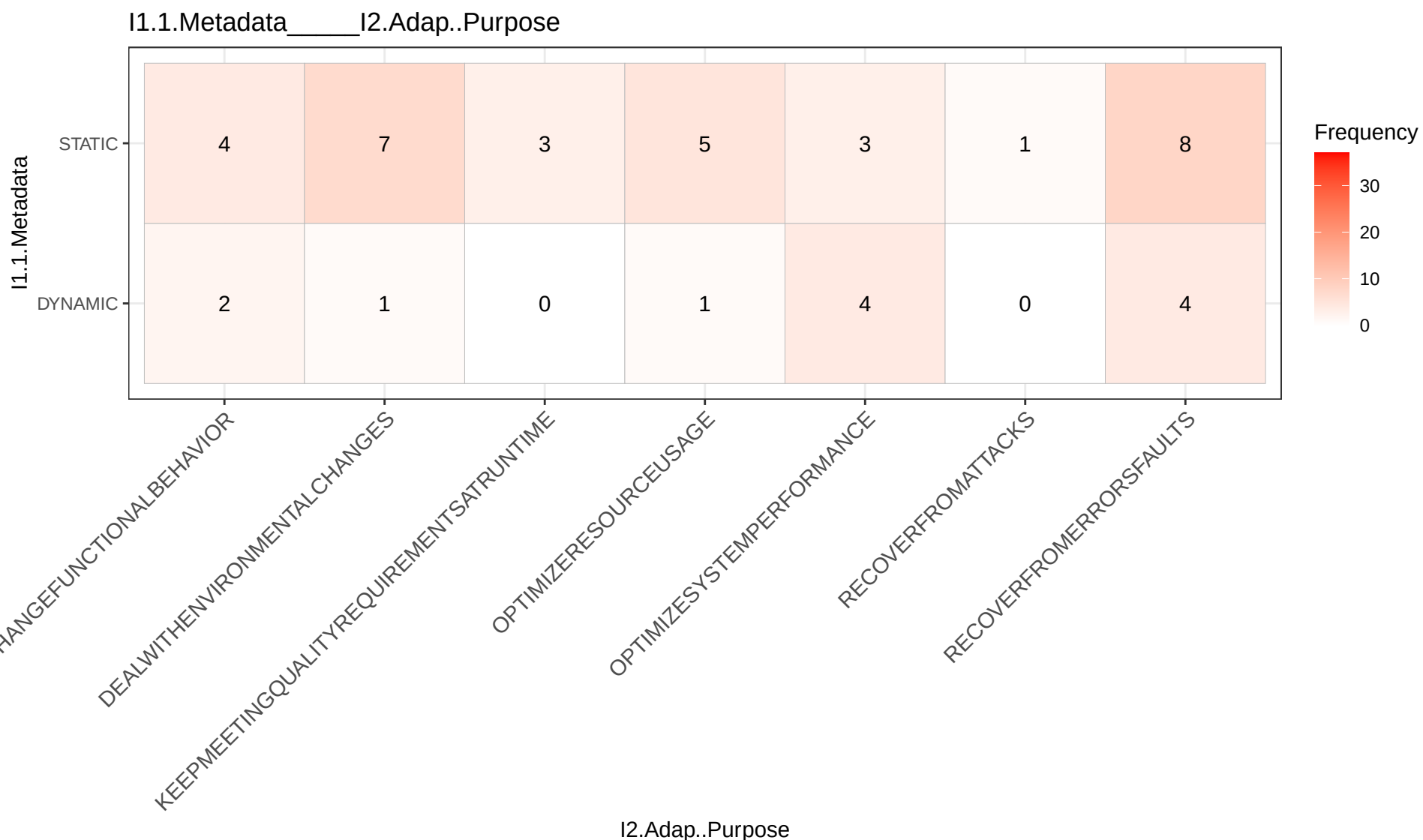


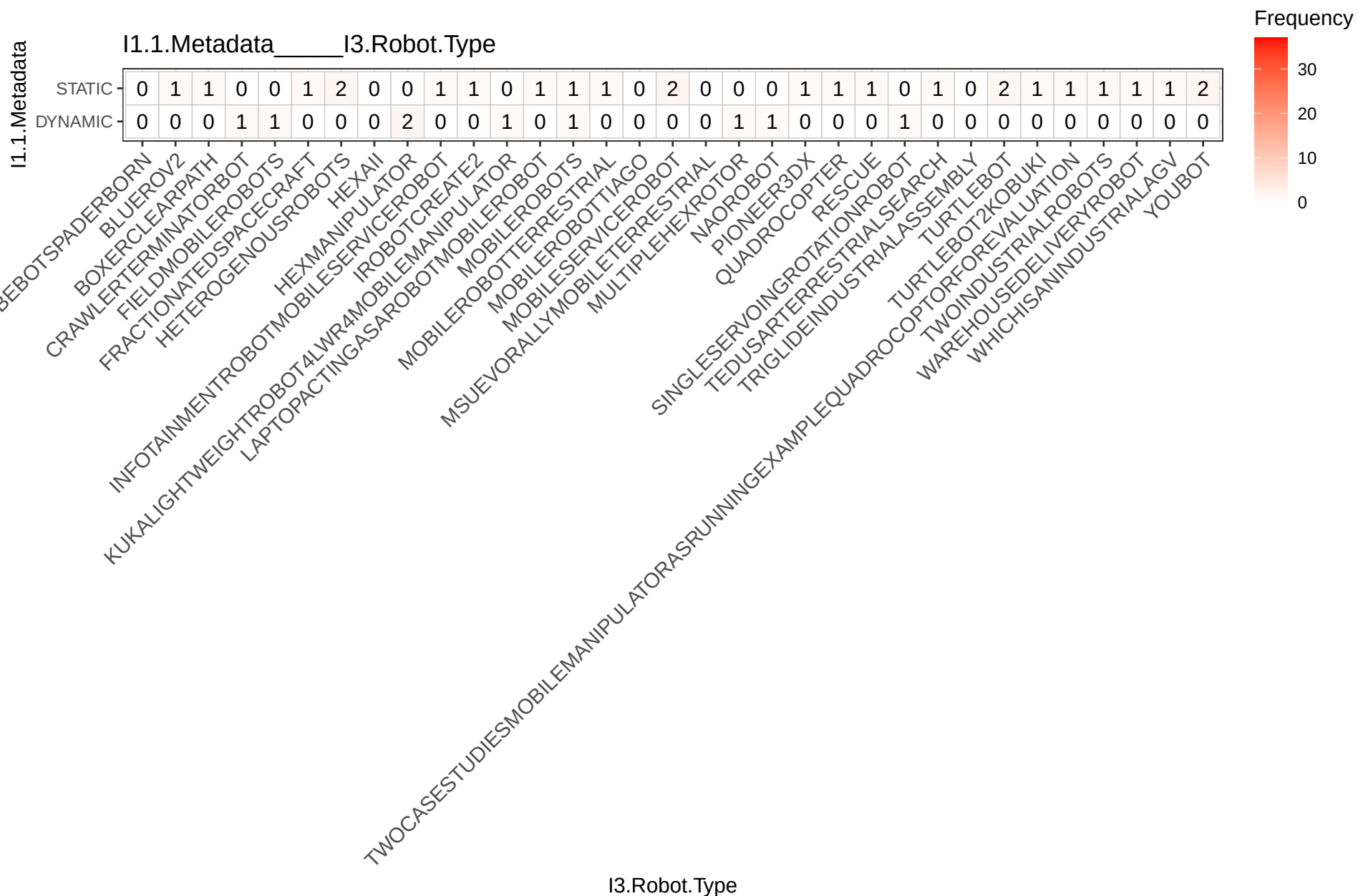
I1.1.Metadata_____I1.4.Overhead.of.Effects

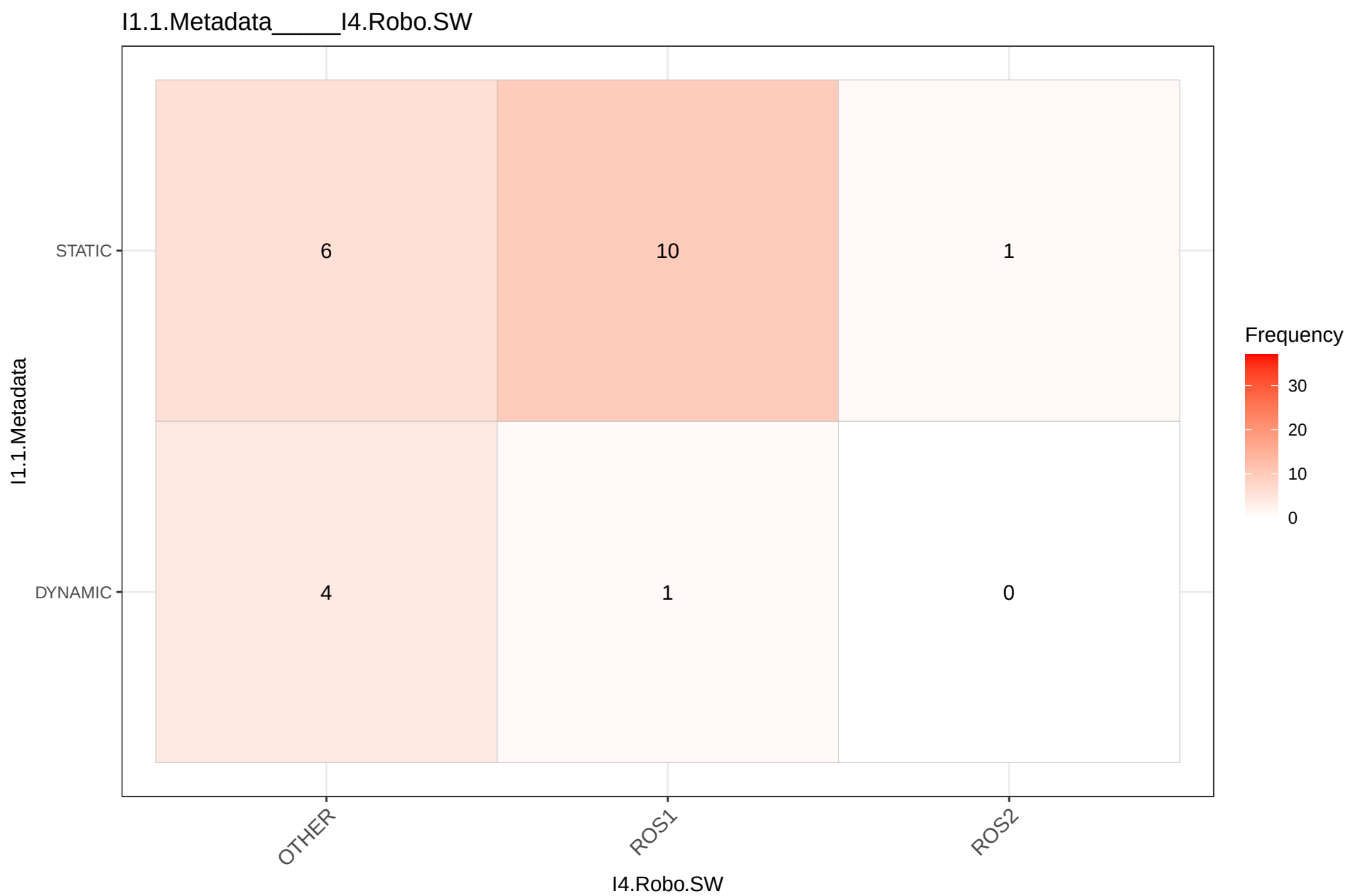


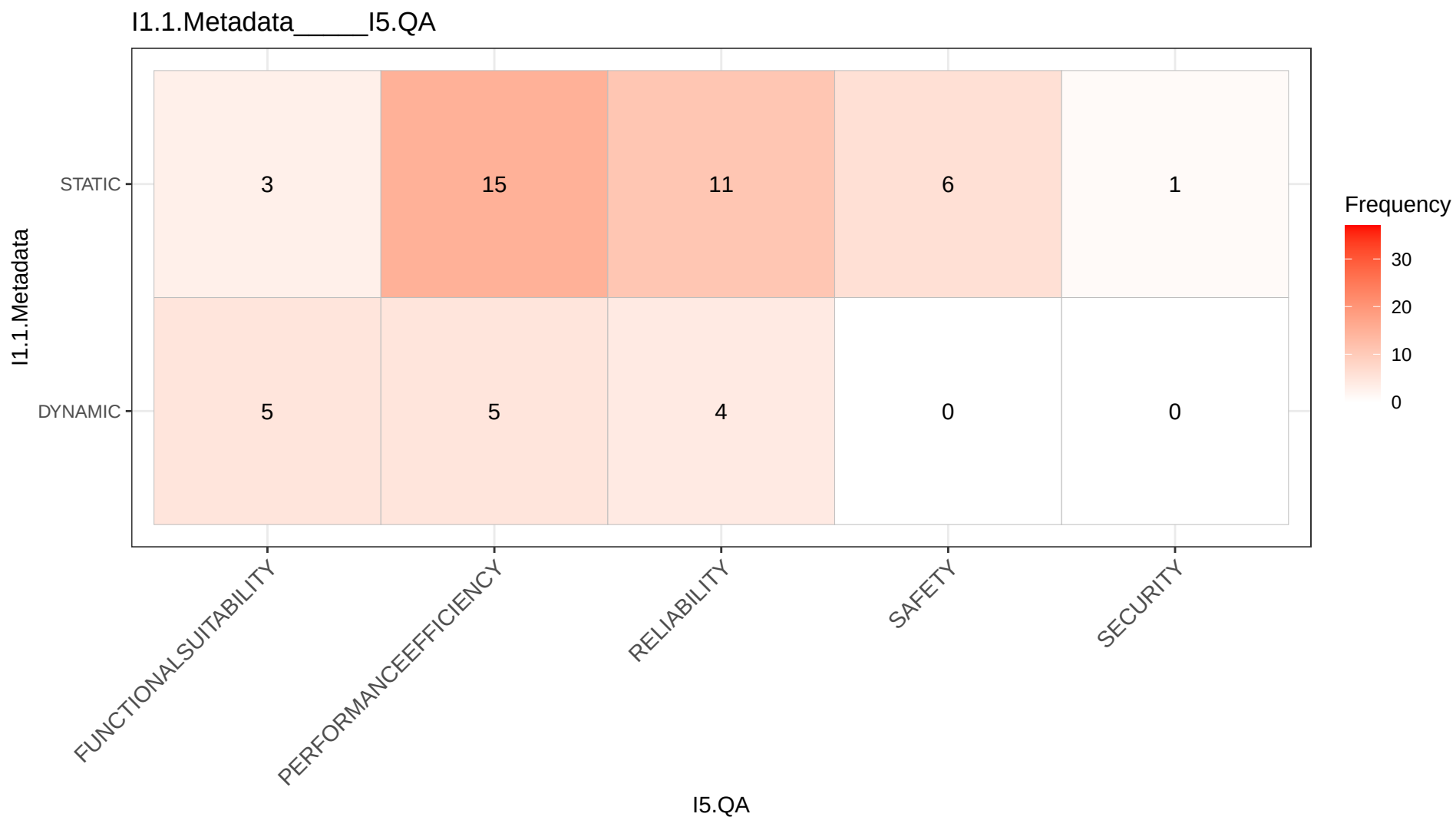
I1.1.Metadata_____I1.4.Resilience.of.Effects



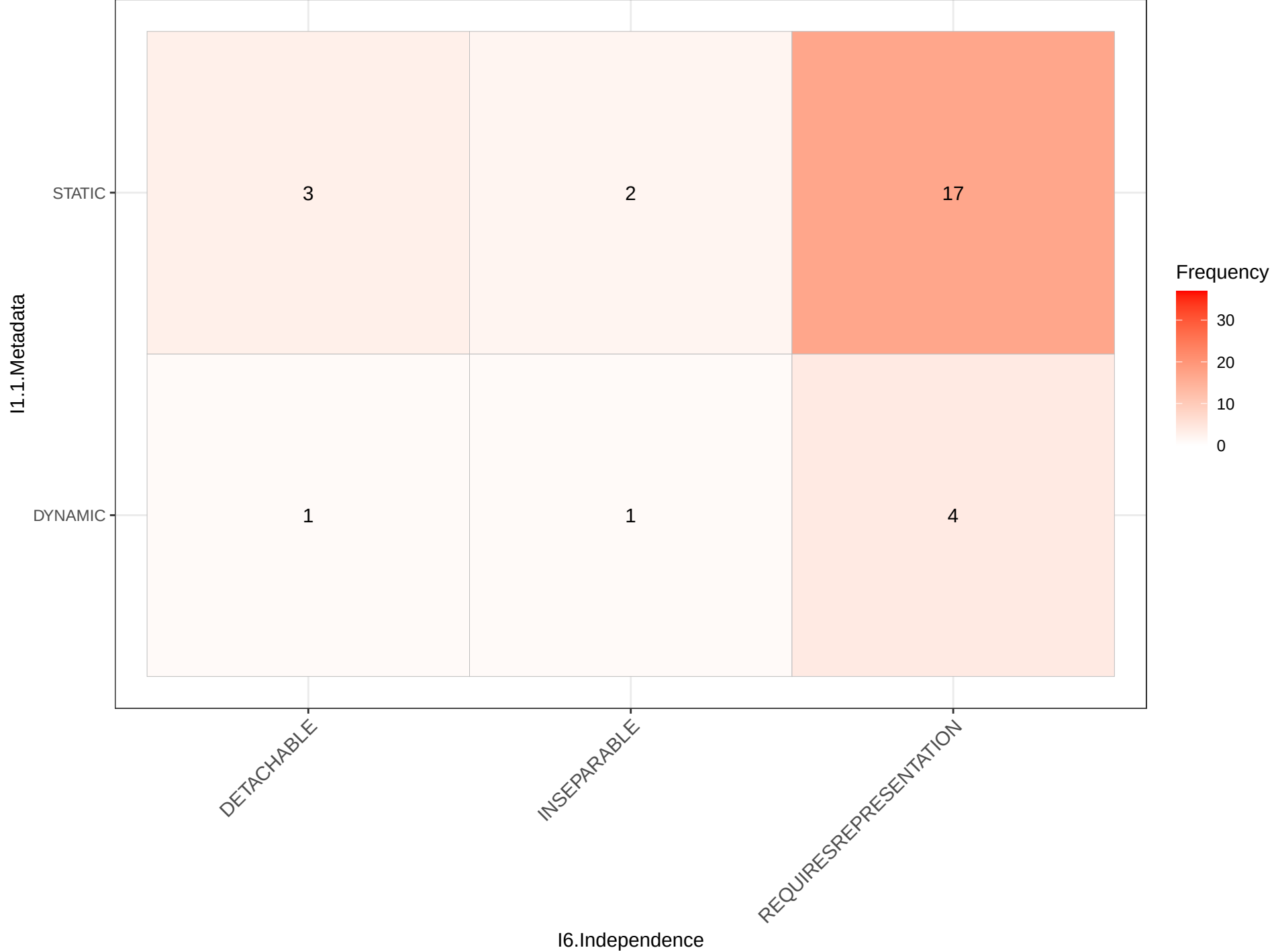








I1.1.Metadata_____I6.Independence



I1.1.Metadata_____I7.Deployment.Realness

I1.1.Metadata

STATIC

9

7

DYNAMIC

4

2

REAL

SIMULATED

I7.Deployment.Realness

Frequency



30

20

10

0

I1.1.Metadata_____I7.Mission.Realness

I1.1.Metadata

STATIC

15

7

DYNAMIC

3

3

REAL

SYNTHETIC

I7.Mission.Realness

Frequency



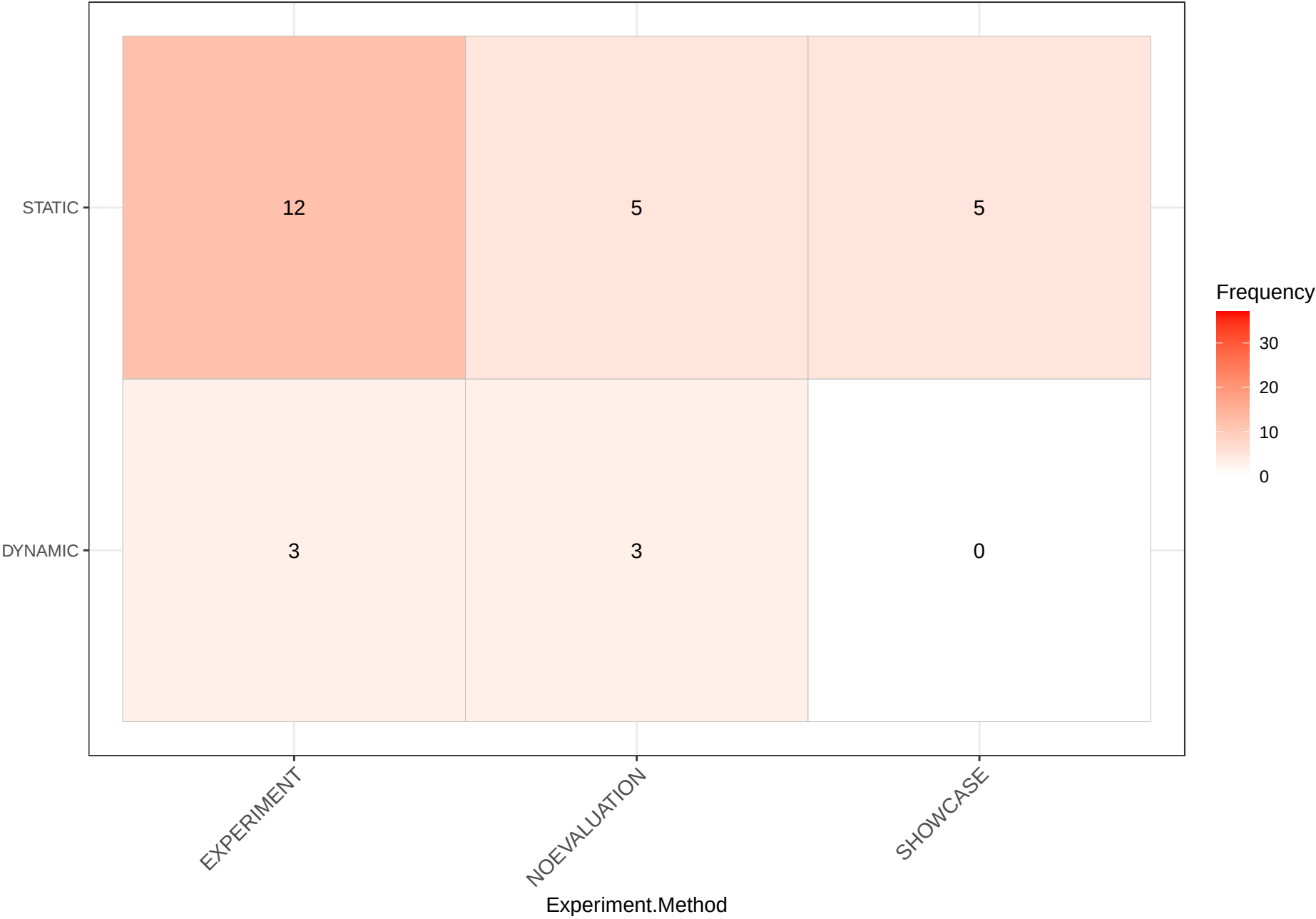
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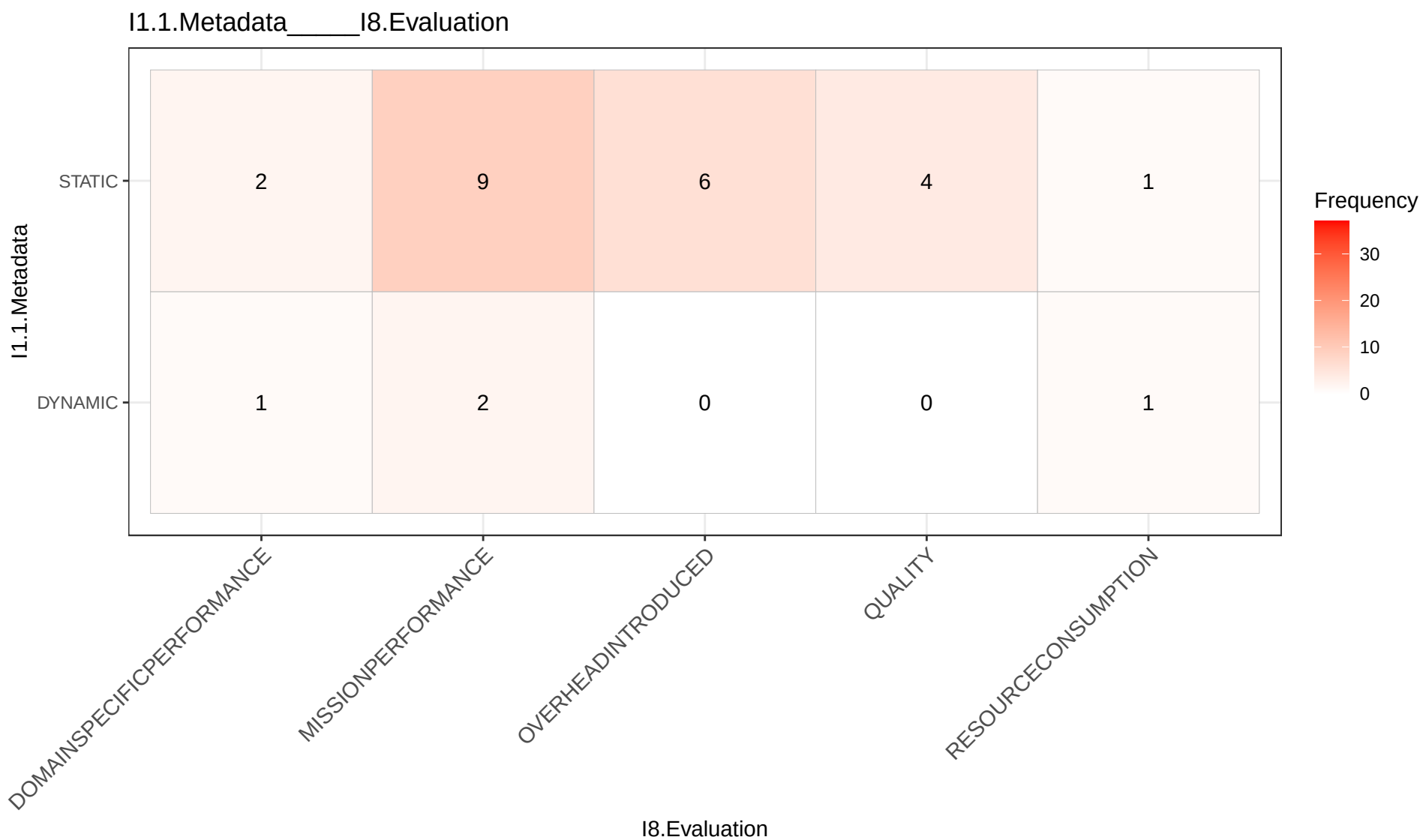
20

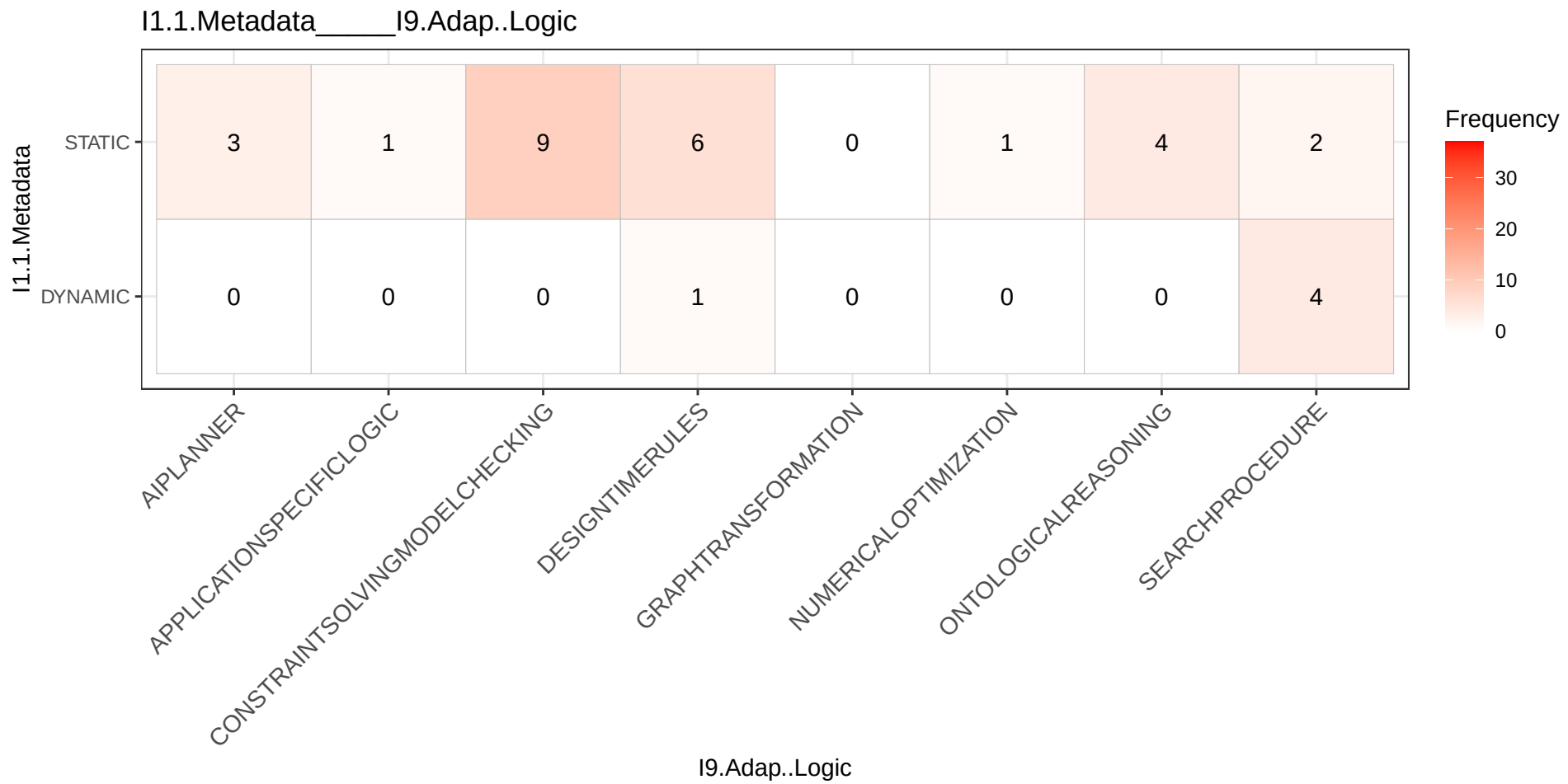
10

0

I1.1.Metadata_____Experiment.Method







I1.1.Metadata_____I10.Monitor

I1.1.Metadata

STATIC

11

16

3

DYNAMIC

0

5

1

ENVIRONMENT

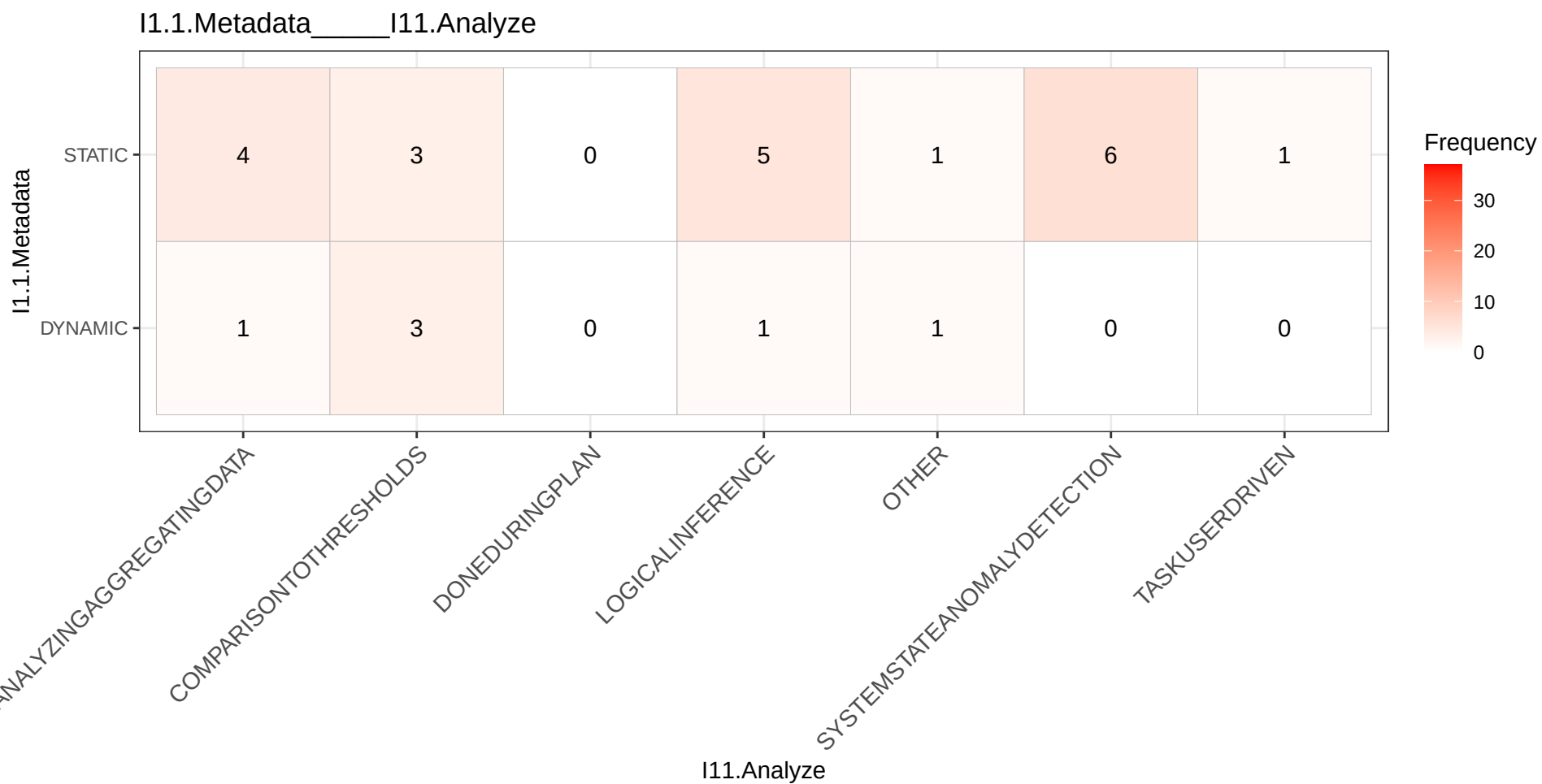
MANAGEDSYSTEM

MISSION

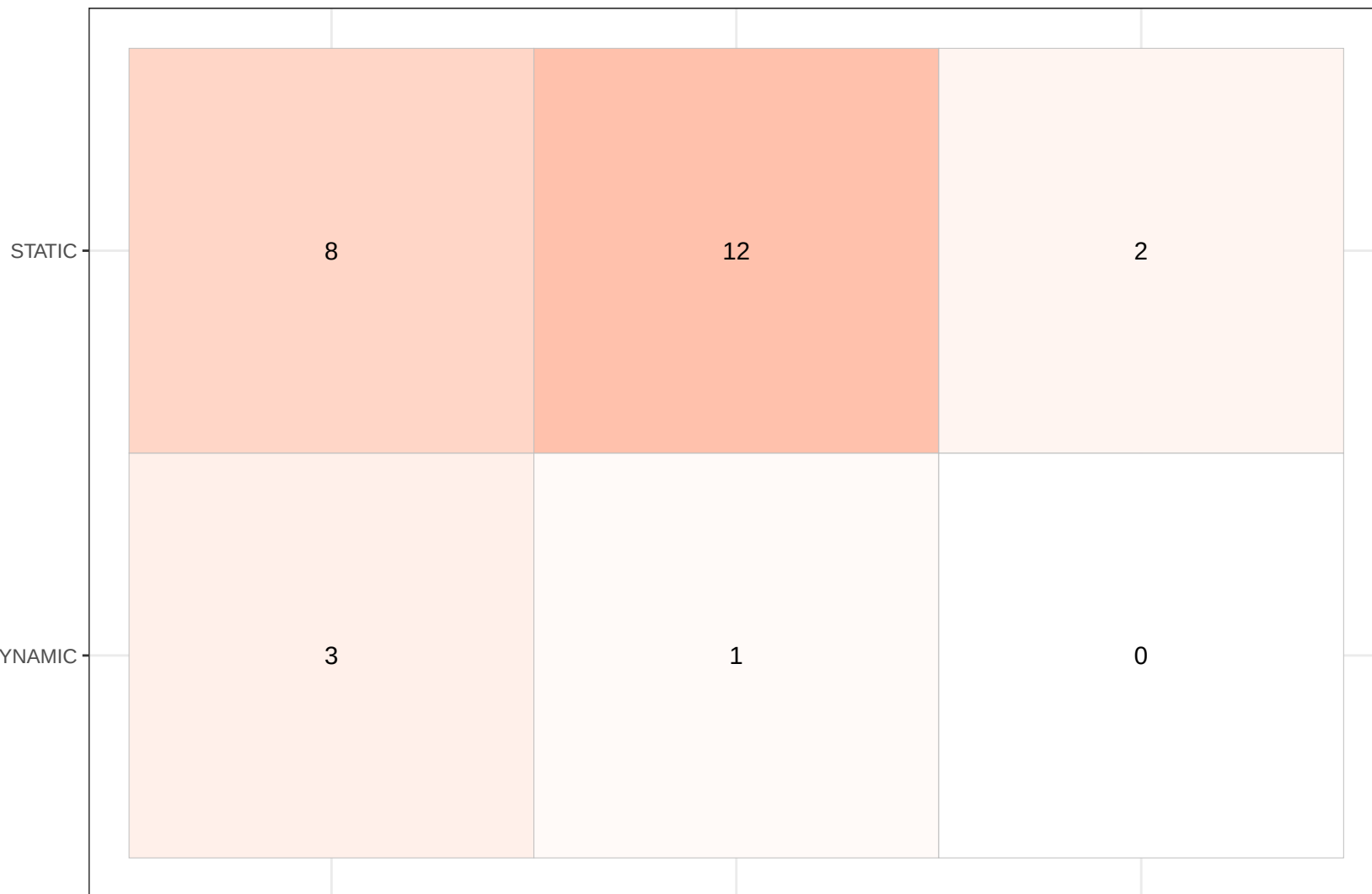
I10.Monitor

Frequency



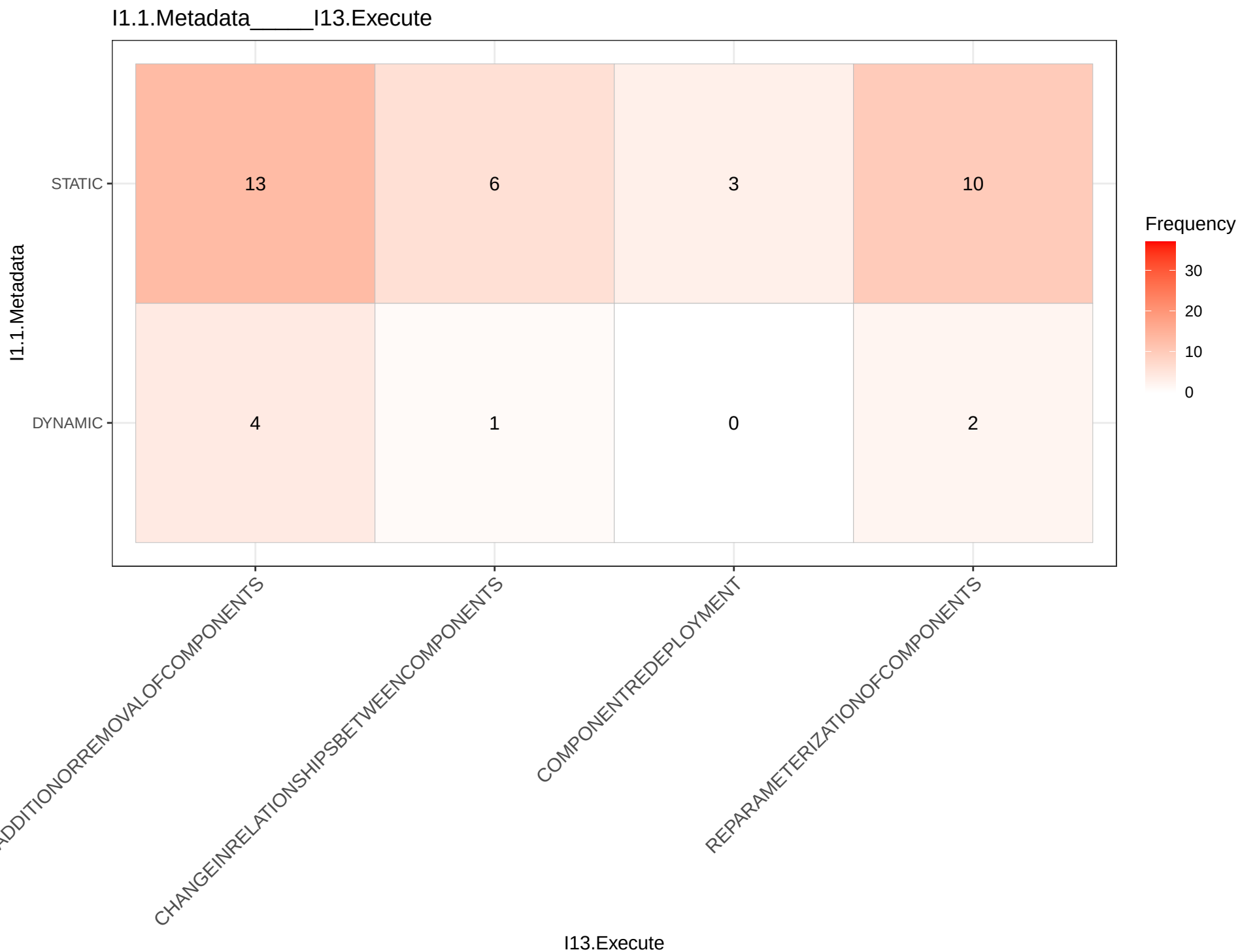


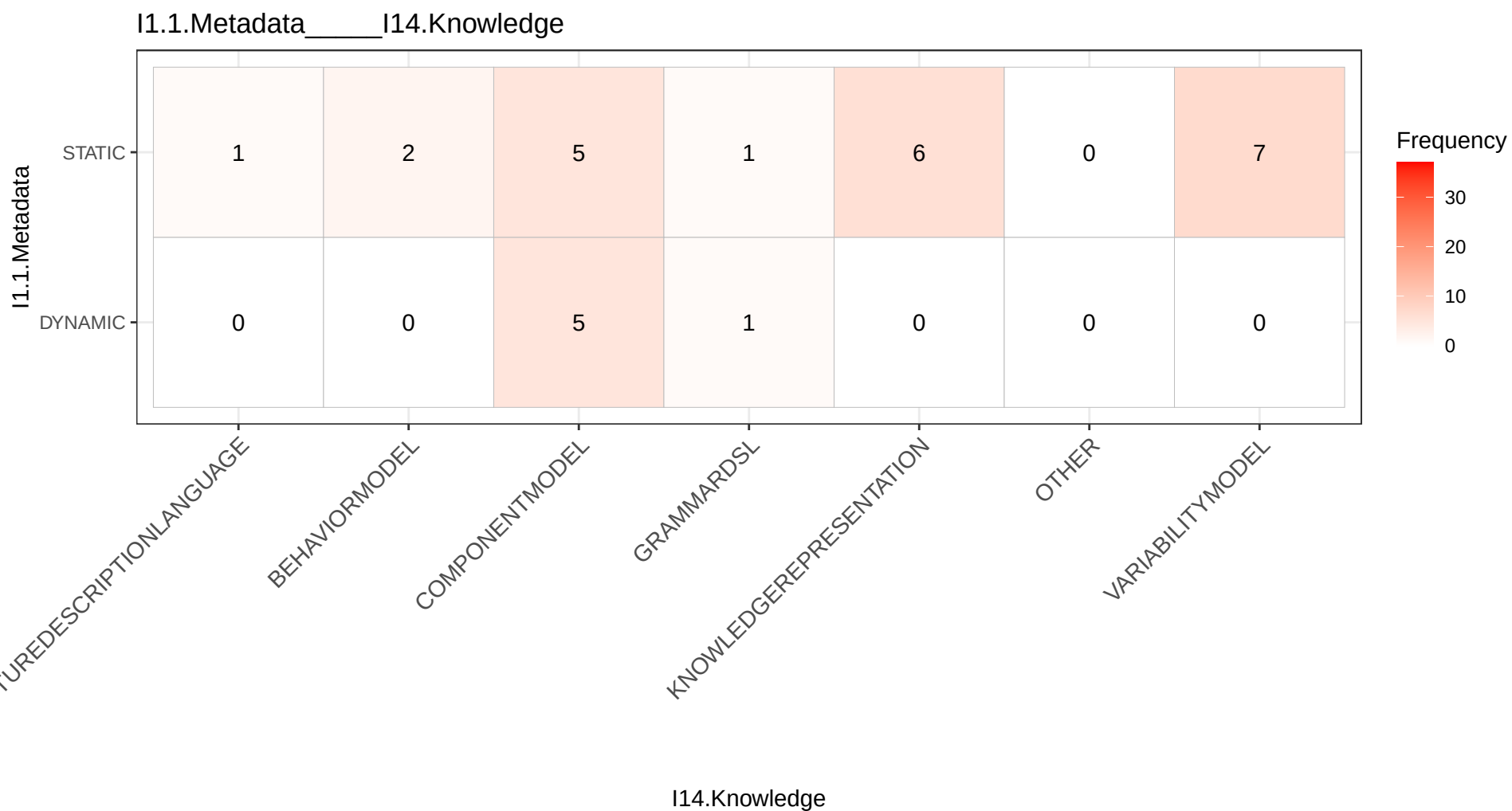
I1.1.Metadata_____I12.Plan

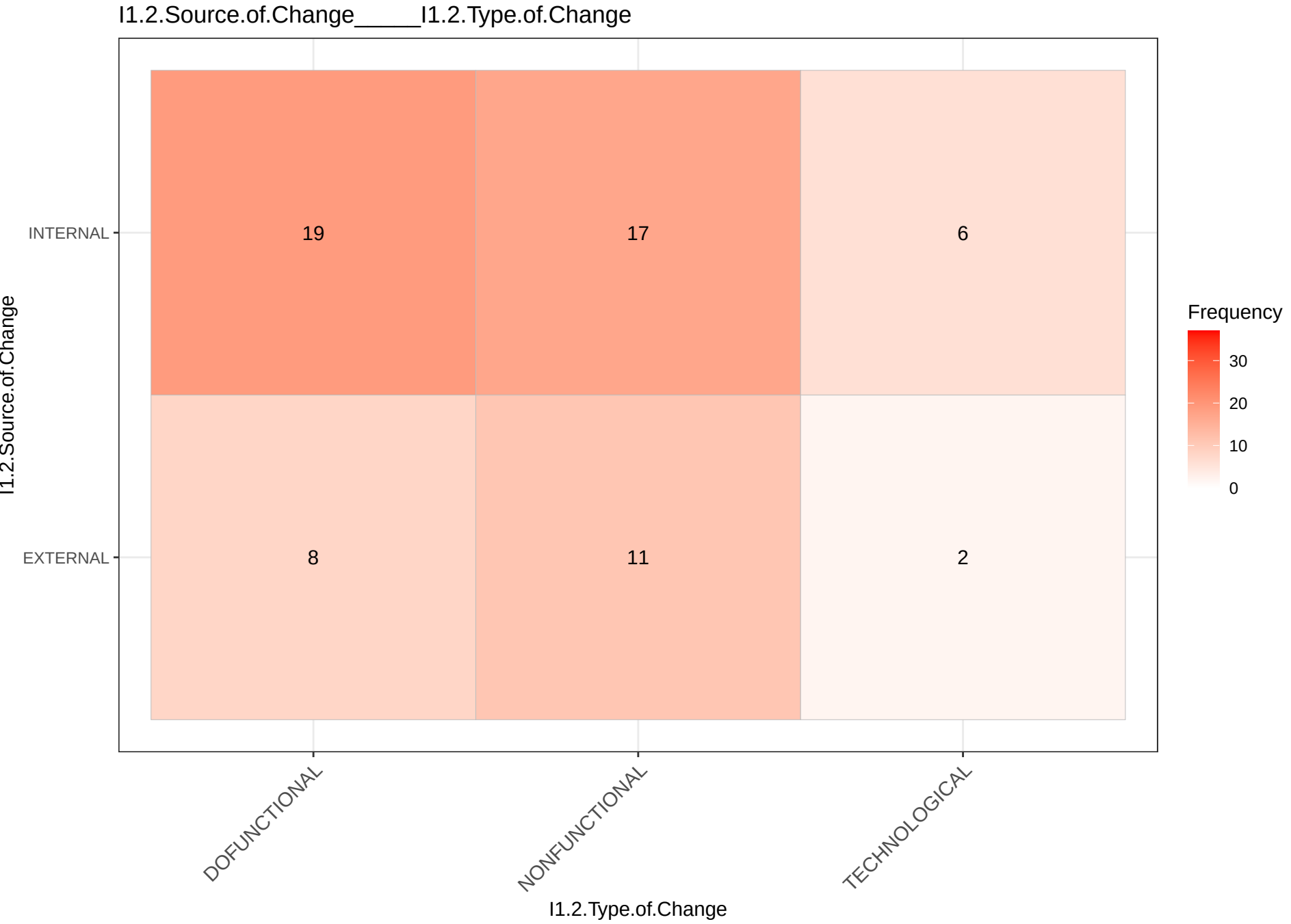


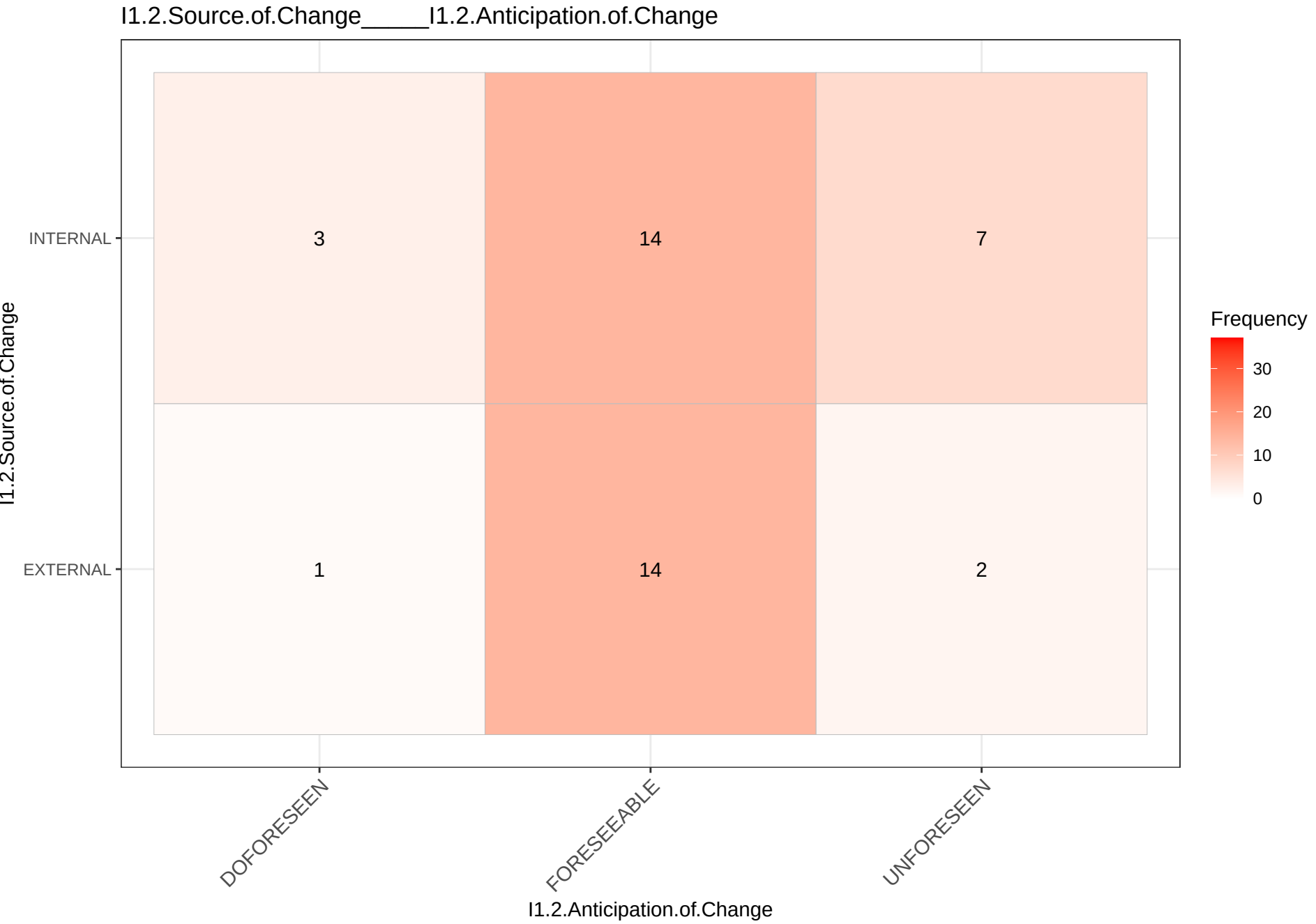
Frequency

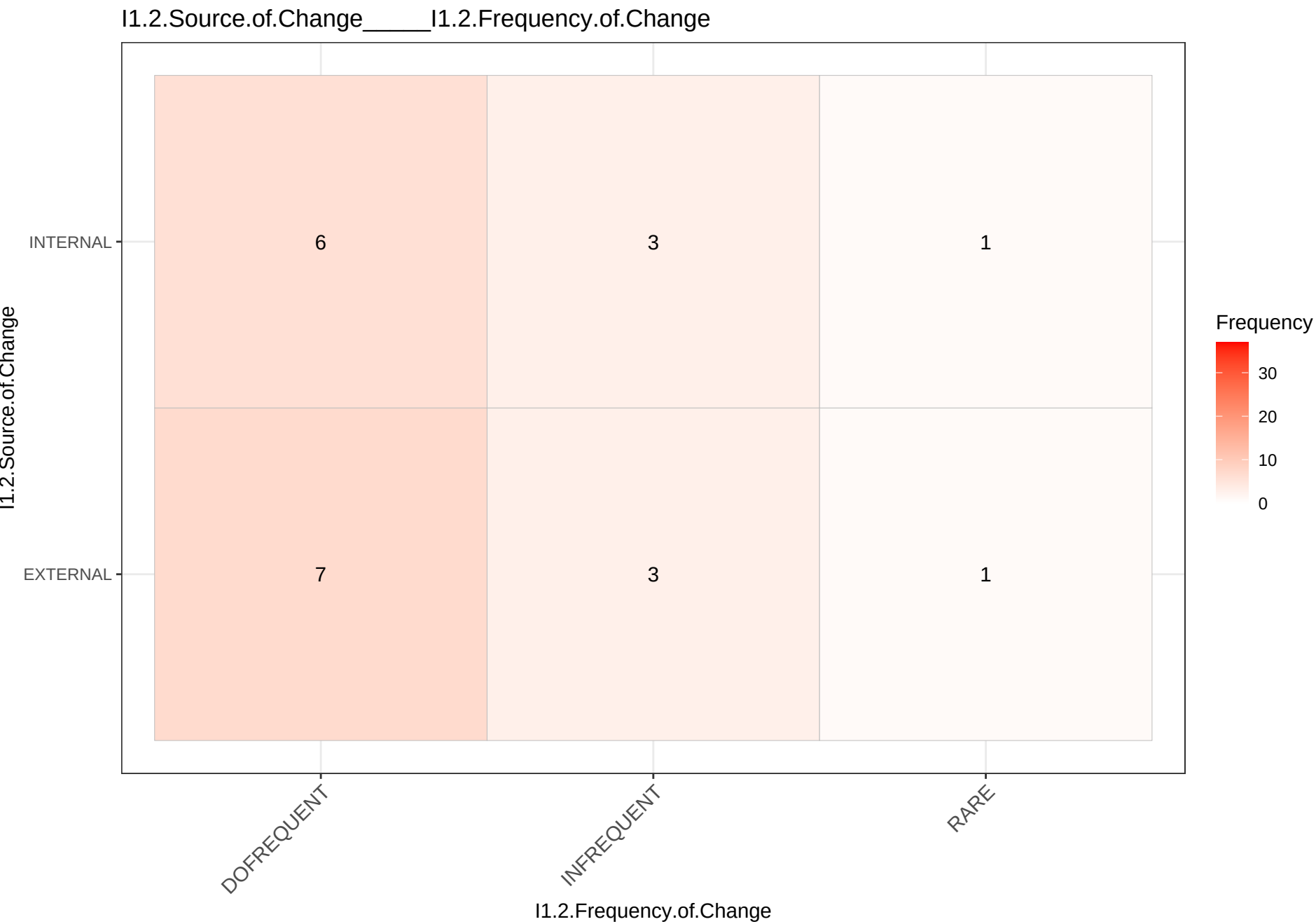












I1.2.Source.of.Change_____I1.3.Type.of.Mechanism

I1.2.Source.of.Change

INTERNAL

16

22

EXTERNAL

12

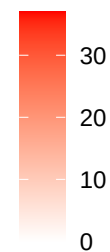
15

PARAMETRIC

STRUCTURAL

I1.3.Type.of.Mechanism

Frequency



I1.2.Source.of.Change_____I1.3.Organization.of.Mechanism

I1.2.Source.of.Change

INTERNAL

3

23

EXTERNAL

0

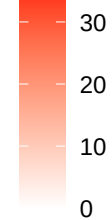
17

DECENTRALIZED

DOCENTRALIZED

I1.3.Organization.of.Mechanism

Frequency



I1.2.Source.of.Change_____I1.3.Scope.of.Mechanism

I1.2.Source.of.Change

INTERNAL

8

24

EXTERNAL

7

13

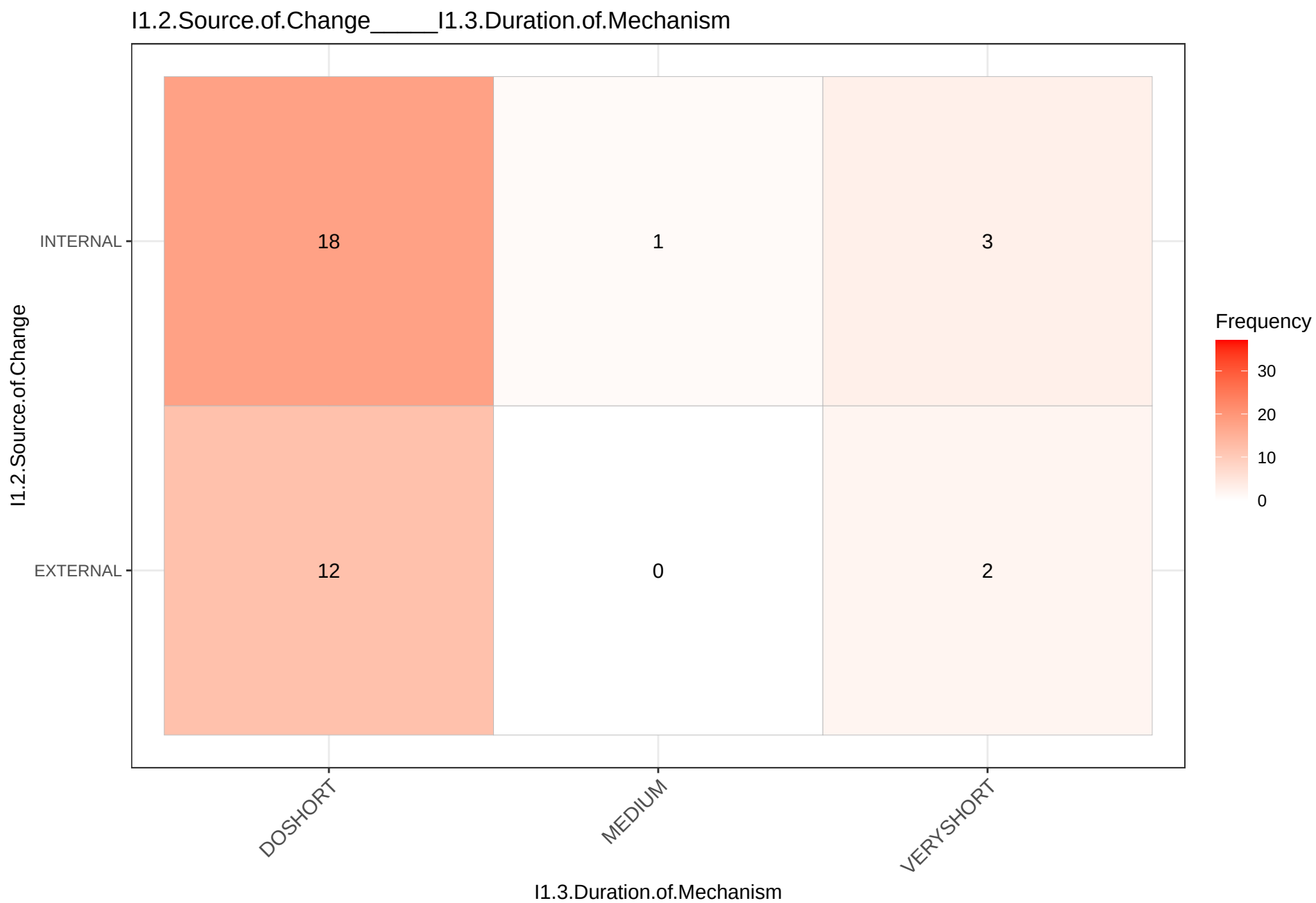
GLOBAL

LOCAL

I1.3.Scope.of.Mechanism

Frequency





I1.2.Source.of.Change_____I1.3.Timeliness.of.Mechanism

I1.2.Source.of.Change

INTERNAL

23

0

EXTERNAL

14

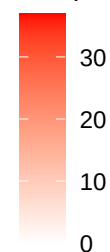
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BESTEFFORT

GUARANTEED

I1.3.Timeliness.of.Mechanism

Frequency



I1.2.Source.of.Change_____I1.4.Criticality.of.Effects

I1.2.Source.of.Change

INTERNAL

17

7

EXTERNAL

11

6

MISSIONCRITICAL

SAFETYCRITICAL

I1.4.Criticality.of.Effects

Frequency



30

20

10

0

I1.2.Source.of.Change_____I1.4.Predictability.of.Effects

I1.2.Source.of.Change

INTERNAL

6

15

EXTERNAL

5

12

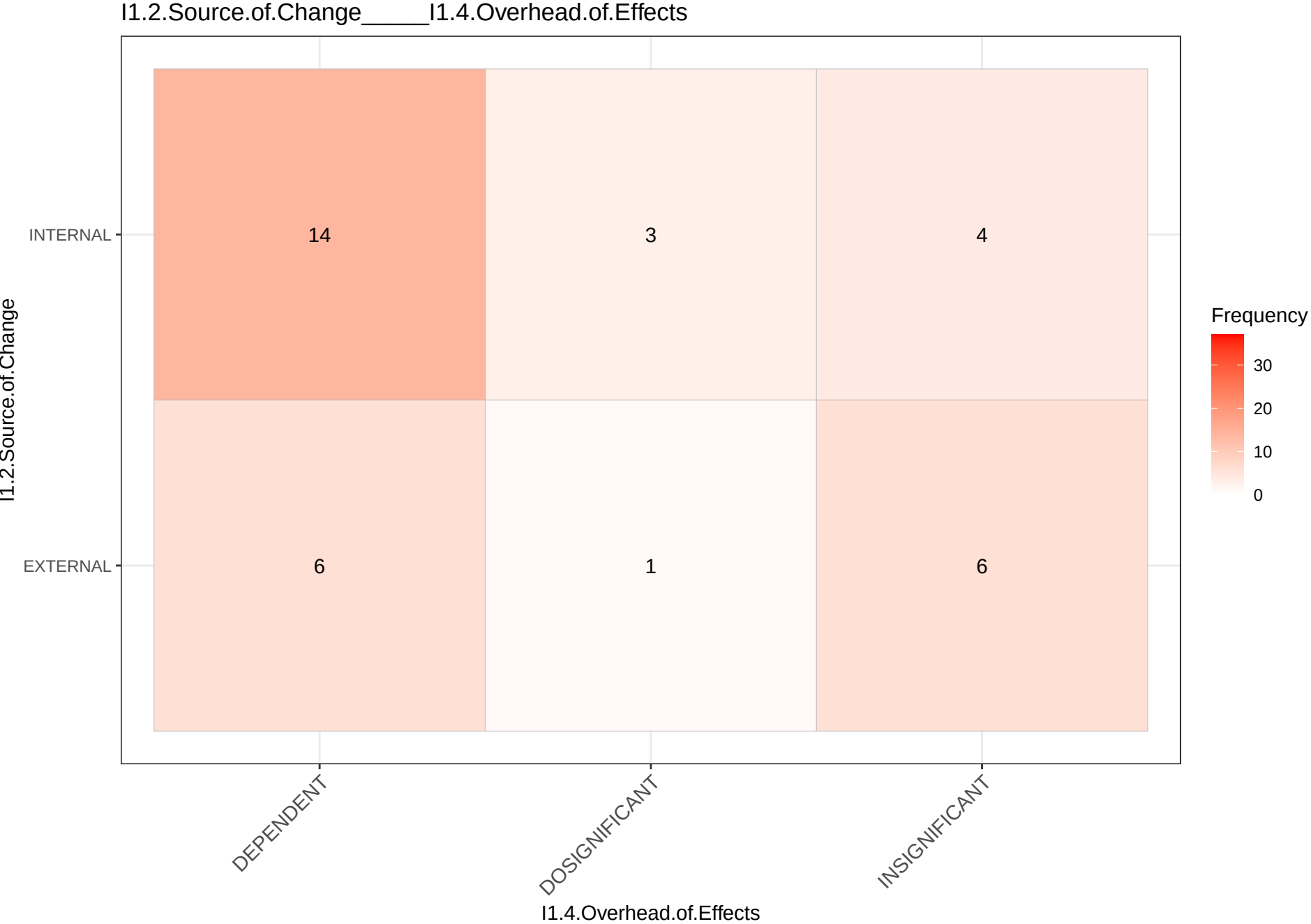
DODETERMINISTIC

NONDETERMINISTIC

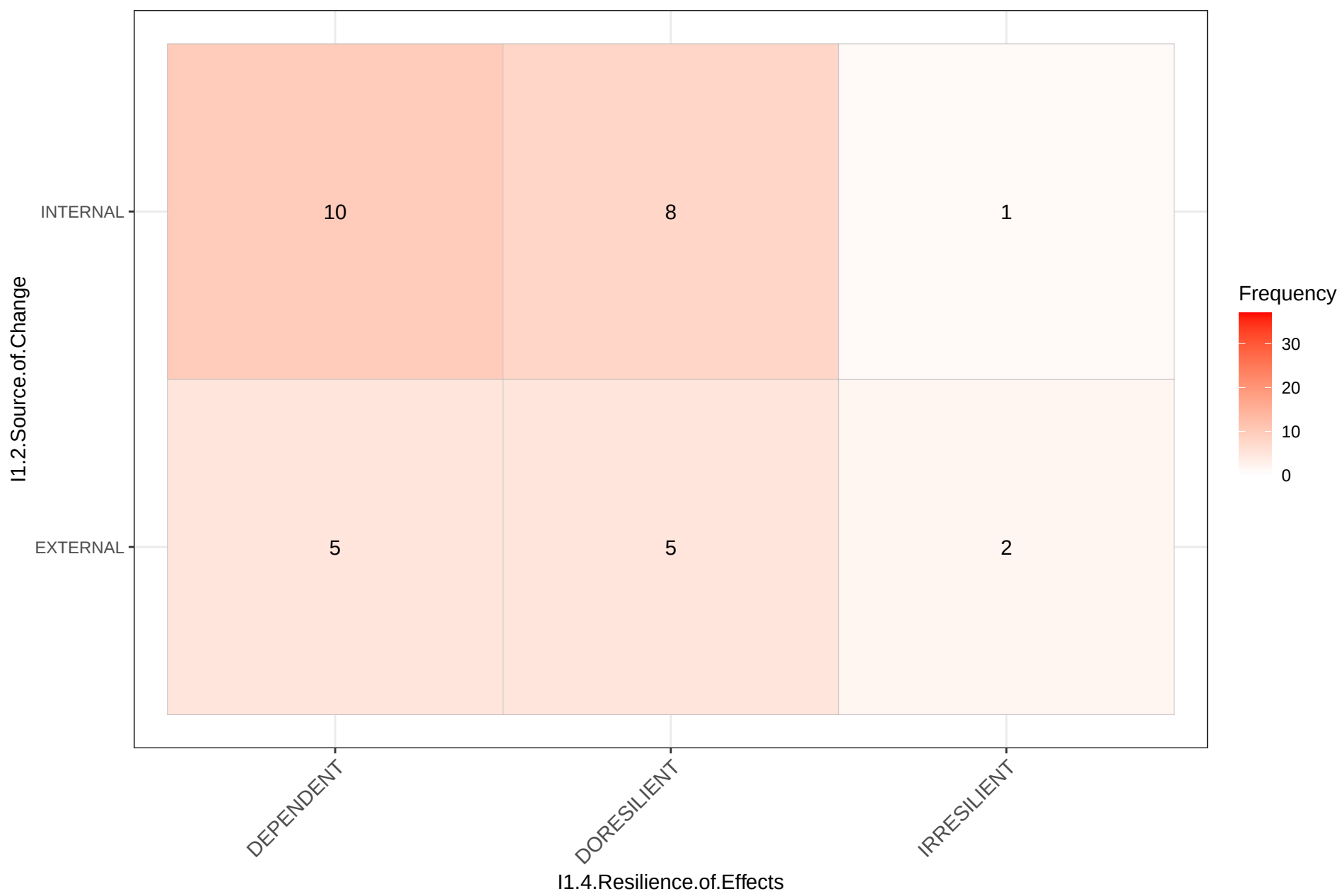
I1.4.Predictability.of.Effects

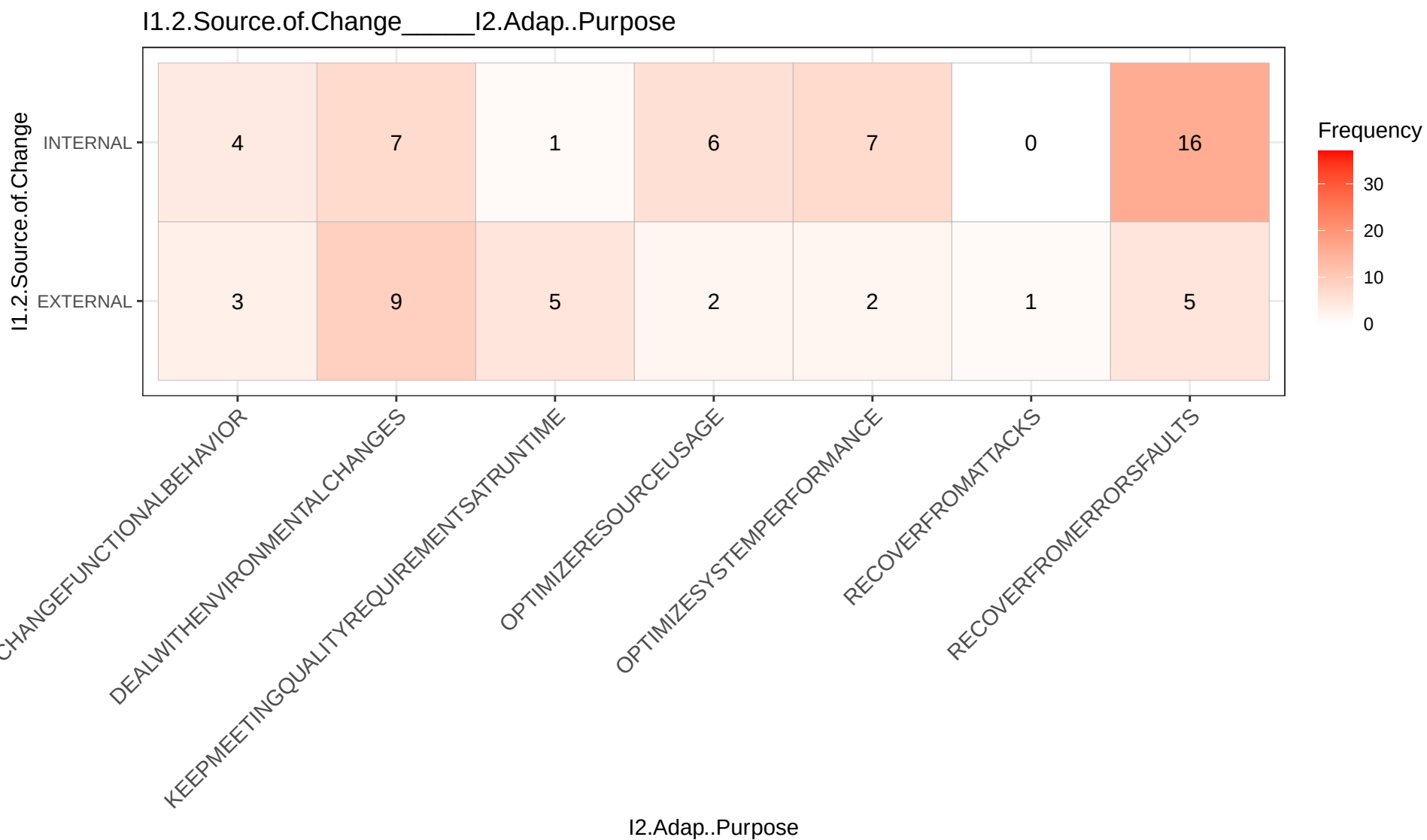
Frequency





I1.2.Source.of.Change_____I1.4.Resilience.of.Effects





1.2.Source.of.Change

Frequency

30

20

10

0

I1.2.Source.of.Change	I3.Robot.Type
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
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16	16
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94	94
95	95
96	96
97	97
98	98
99	99
100	100

INTERNAL

EXTERNAL

l1.2.S

BEBOTSPADERBORN
BLUEROV2
BOXERCLEARPATH
CRAWLERTERMINATORBOT
FIELDMOBILEROBOTS
FRACTIONATEDSPACECRAFT
HETEROGENOUSROBOTS
HEXAII
HEXMANIPULATOR
IROBOTSERVICEROBOT
INFOENTAINMENTROBOT
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULAT2
LAPTOPACTINGASAROBOTMOBILEROBOT
MSUEVORALLYMOBILETERRESTRIAL
MULTIPLEHEXROT
NAOROBOT
PIONEER3DX
QUADROCOPTER
RESCUE
SINGLESERVOINGROTATIONROBOT
TEDUSARTERRESTRIALSEARCH
TRIGLIDEINDUSTRIALASSEMBLY
TURTLEBOT
TWOINDUSTRIALROBOTS
WAREHOUSEDELIVERYROBOT
WHICHISANINDUSTRIALAGV
YOUNGBOY

13.Robot.Type

I1.2.Source.of.Change_____I4.Robo.SW

I1.2.Source.of.Change

INTERNAL

EXTERNAL

OTHER

ROS1

ROS2

I4.Robo.SW

Frequency



8

12

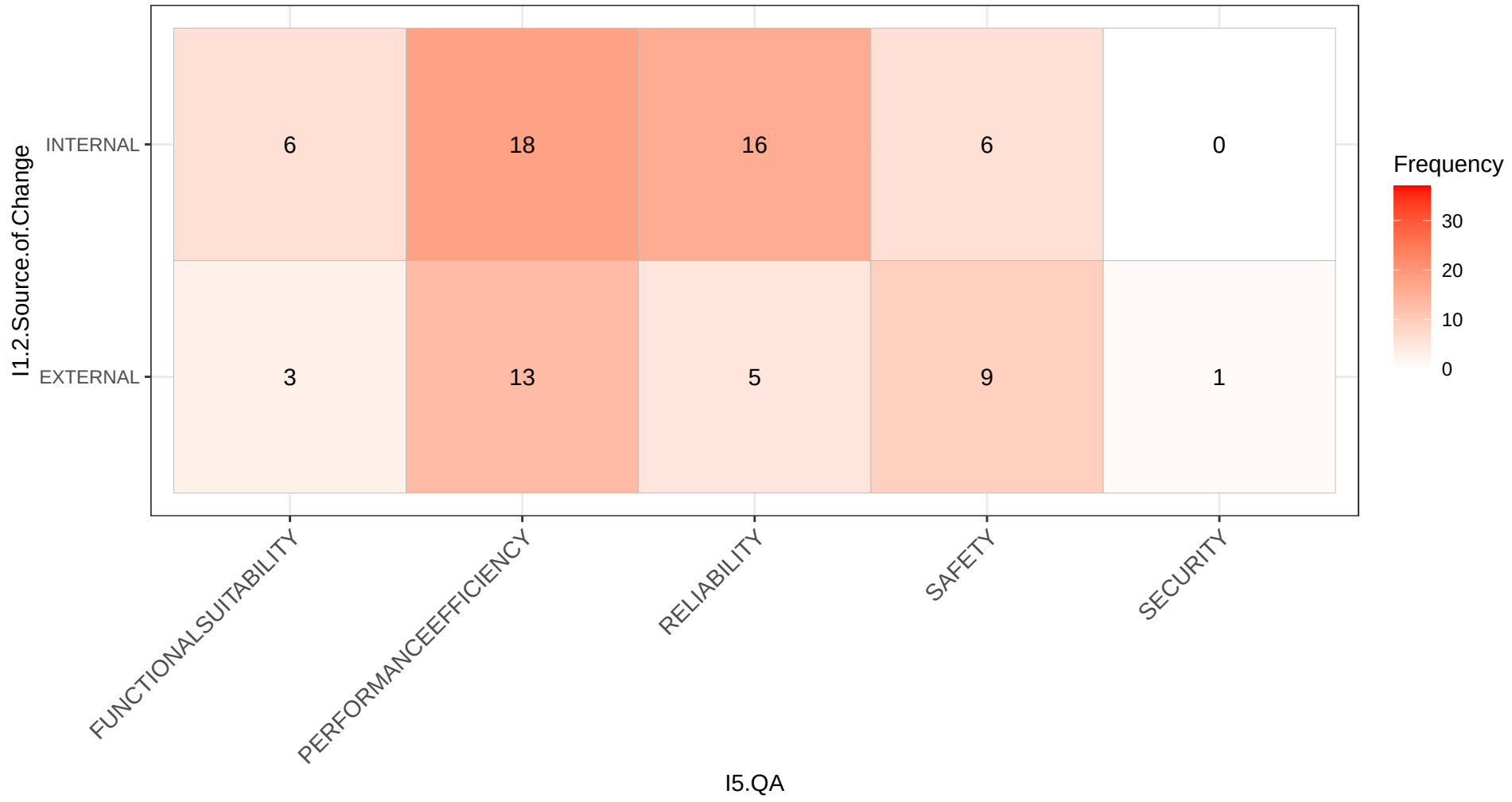
2

3

8

2

I1.2.Source.of.Change_____I5.QA



I1.2.Source.of.Change_____I6.Independence

I1.2.Source.of.Change

INTERNAL

EXTERNAL

DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

Frequency



30

20

10

0

8

2

16

3

1

13

I1.2.Source.of.Change_____I7.Deployment.Realness

I1.2.Source.of.Change

INTERNAL

12

9

EXTERNAL

5

9

REAL

SIMULATED

I7.Deployment.Realness

Frequency



30

20

10

0

I1.2.Source.of.Change_____I7.Mission.Realness

I1.2.Source.of.Change

INTERNAL

13

13

EXTERNAL

11

6

REAL

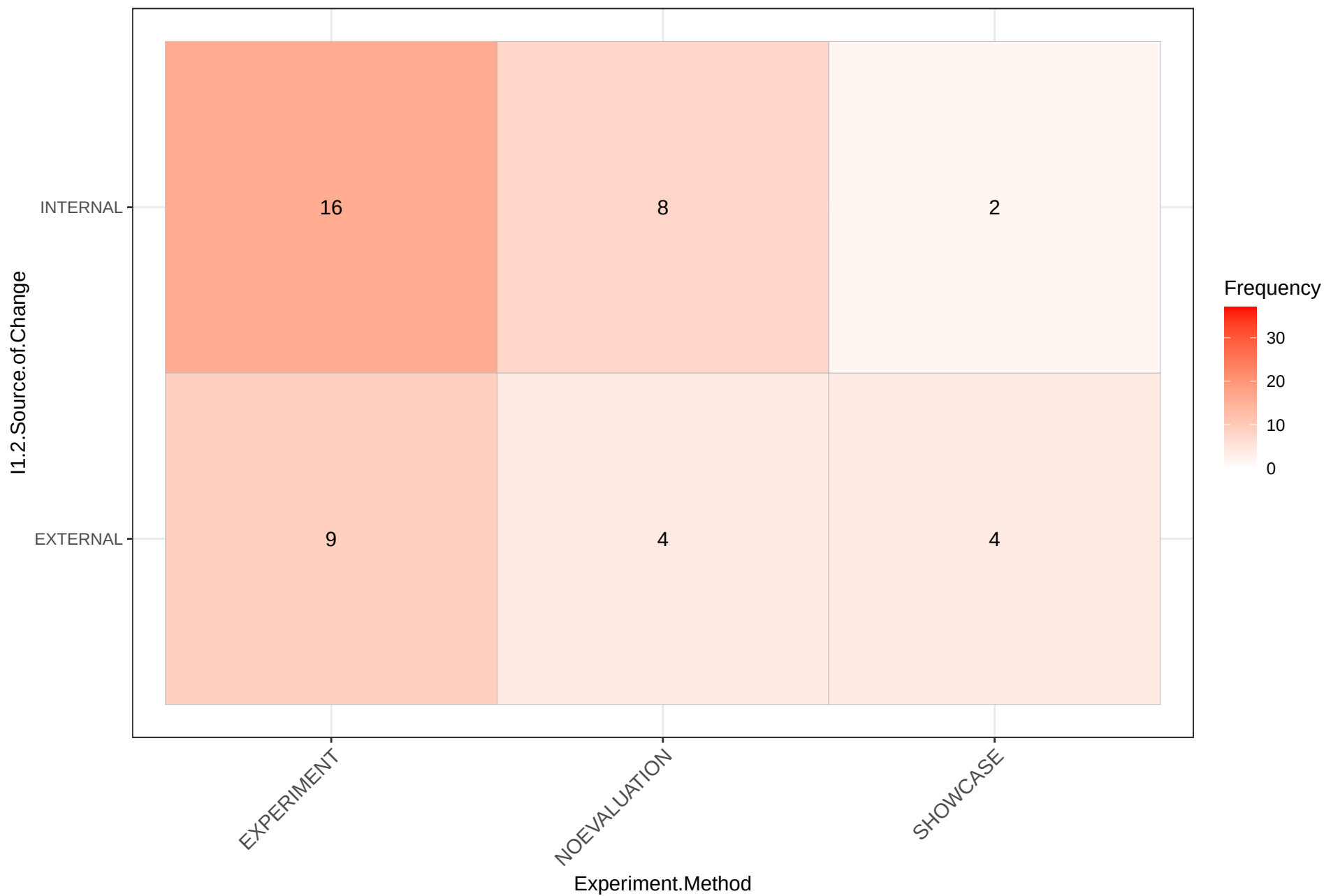
SYNTHETIC

I7.Mission.Realness

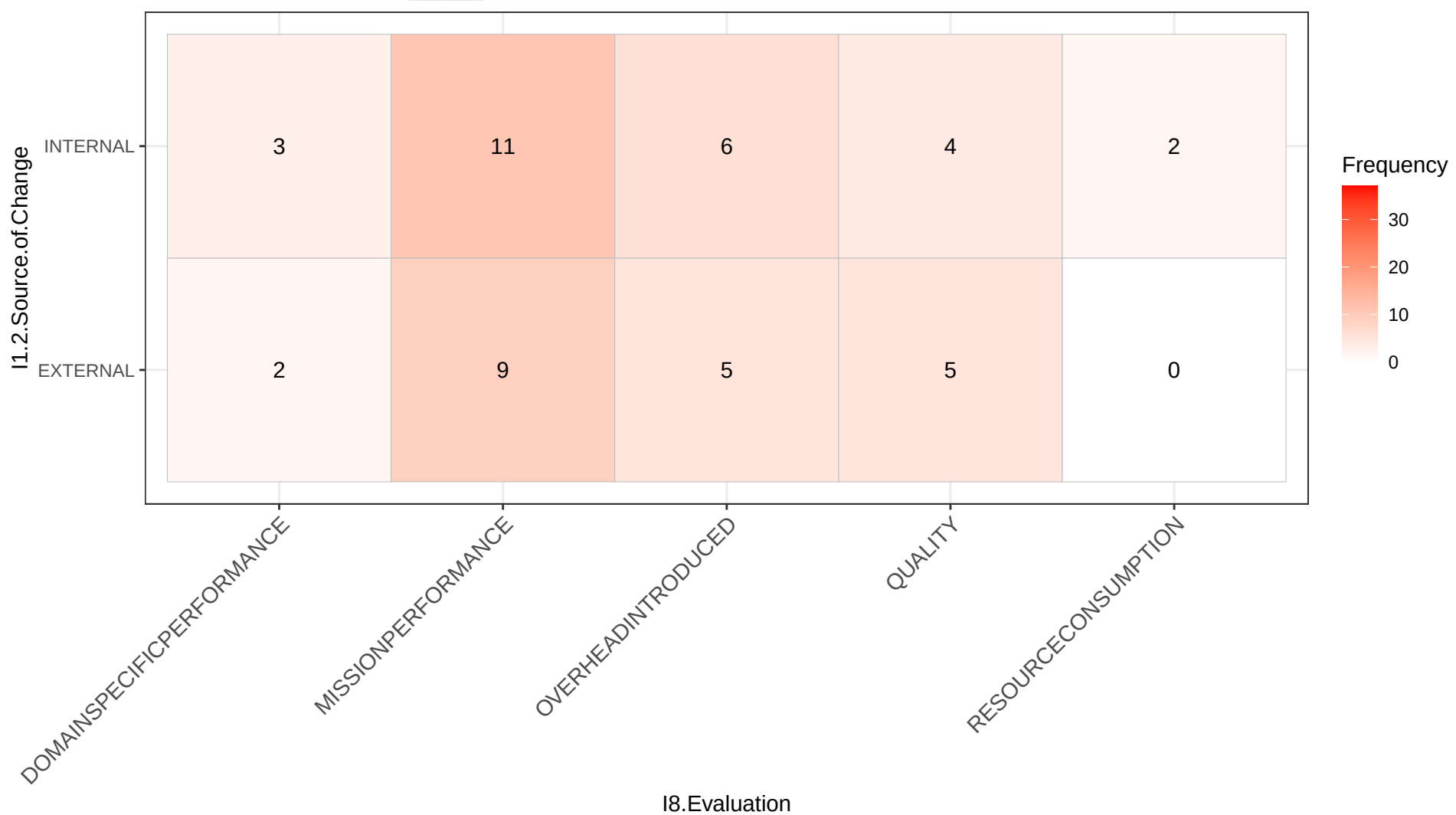
Frequency

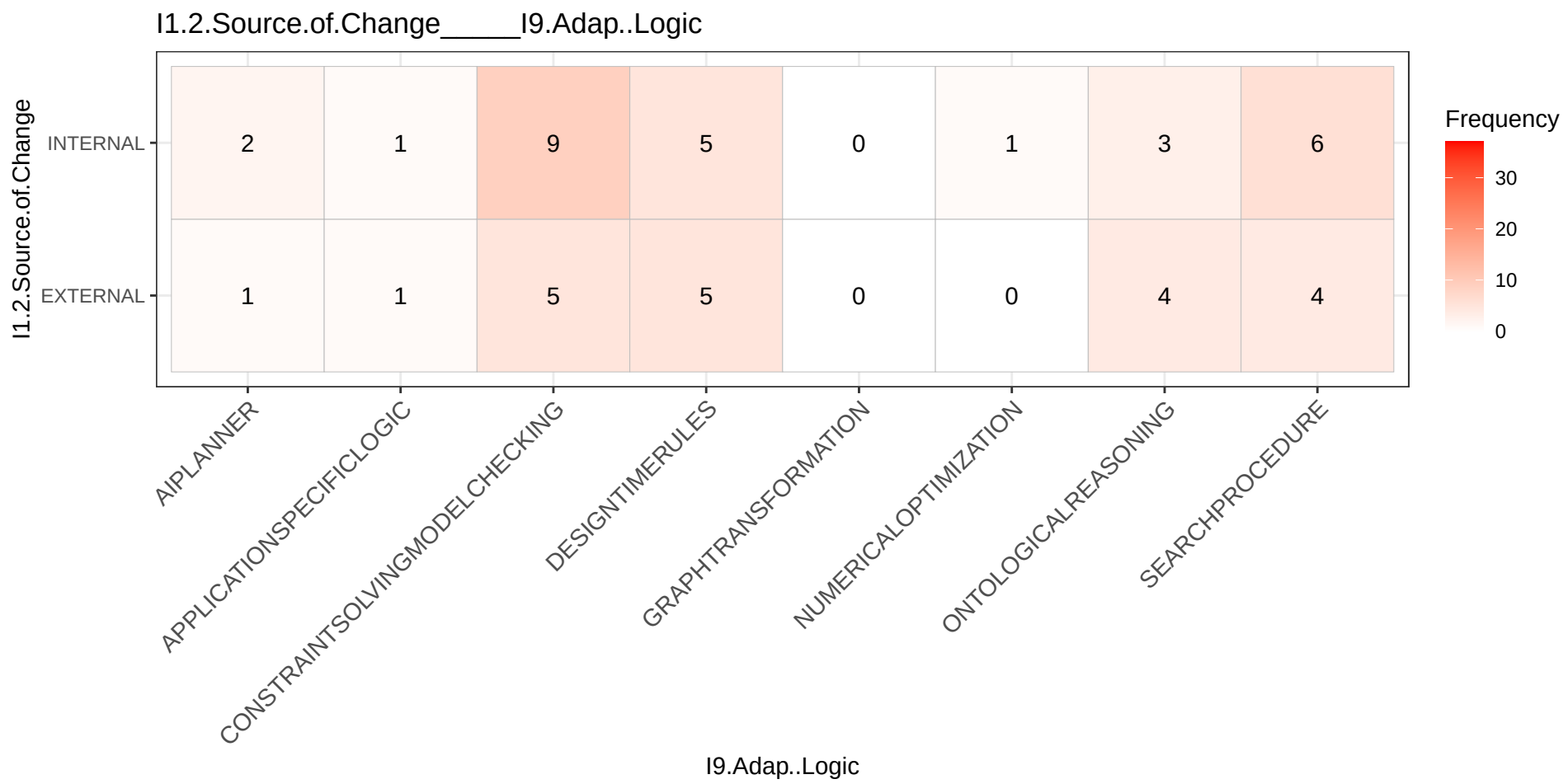


I1.2.Source.of.Change_____Experiment.Method

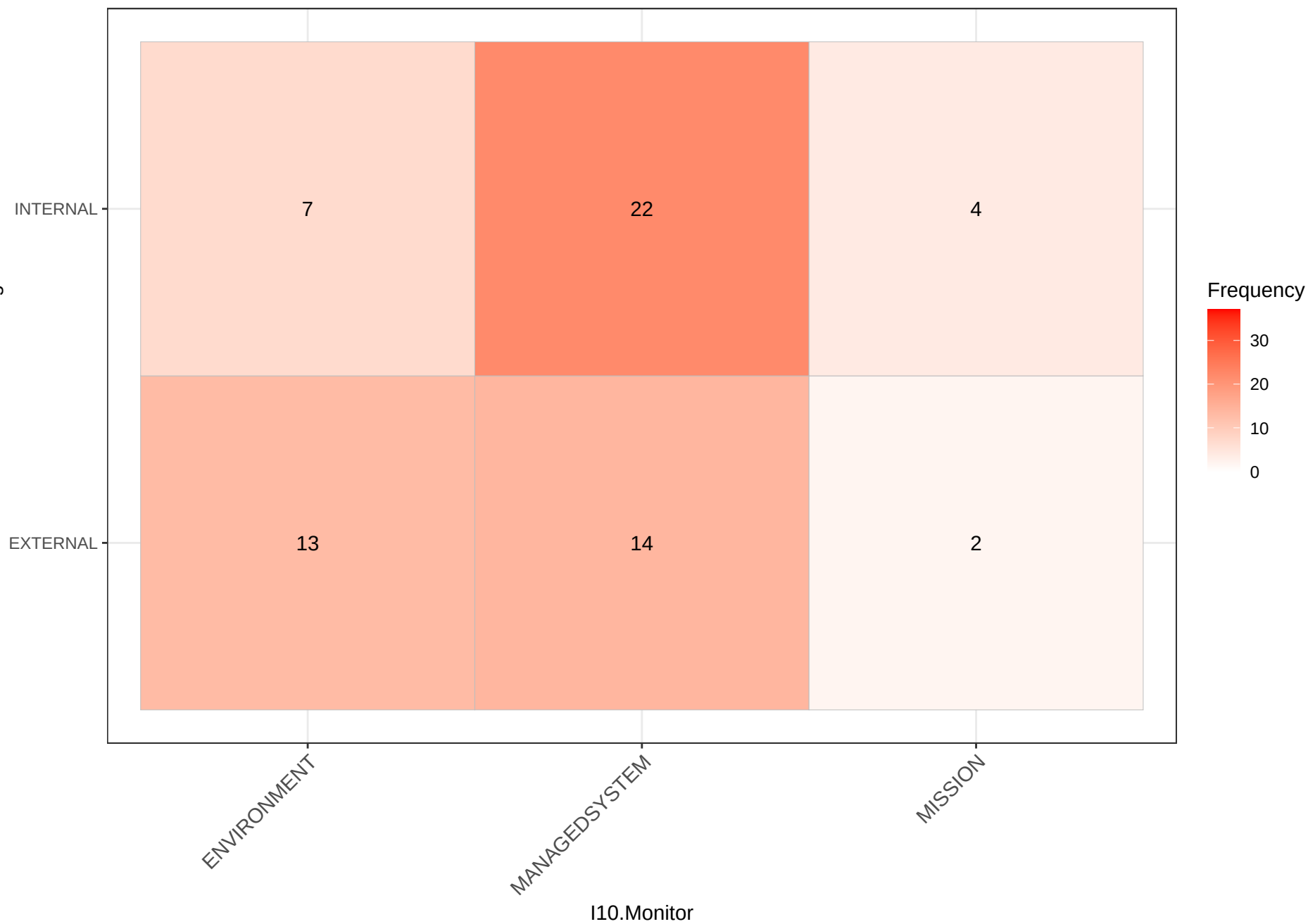


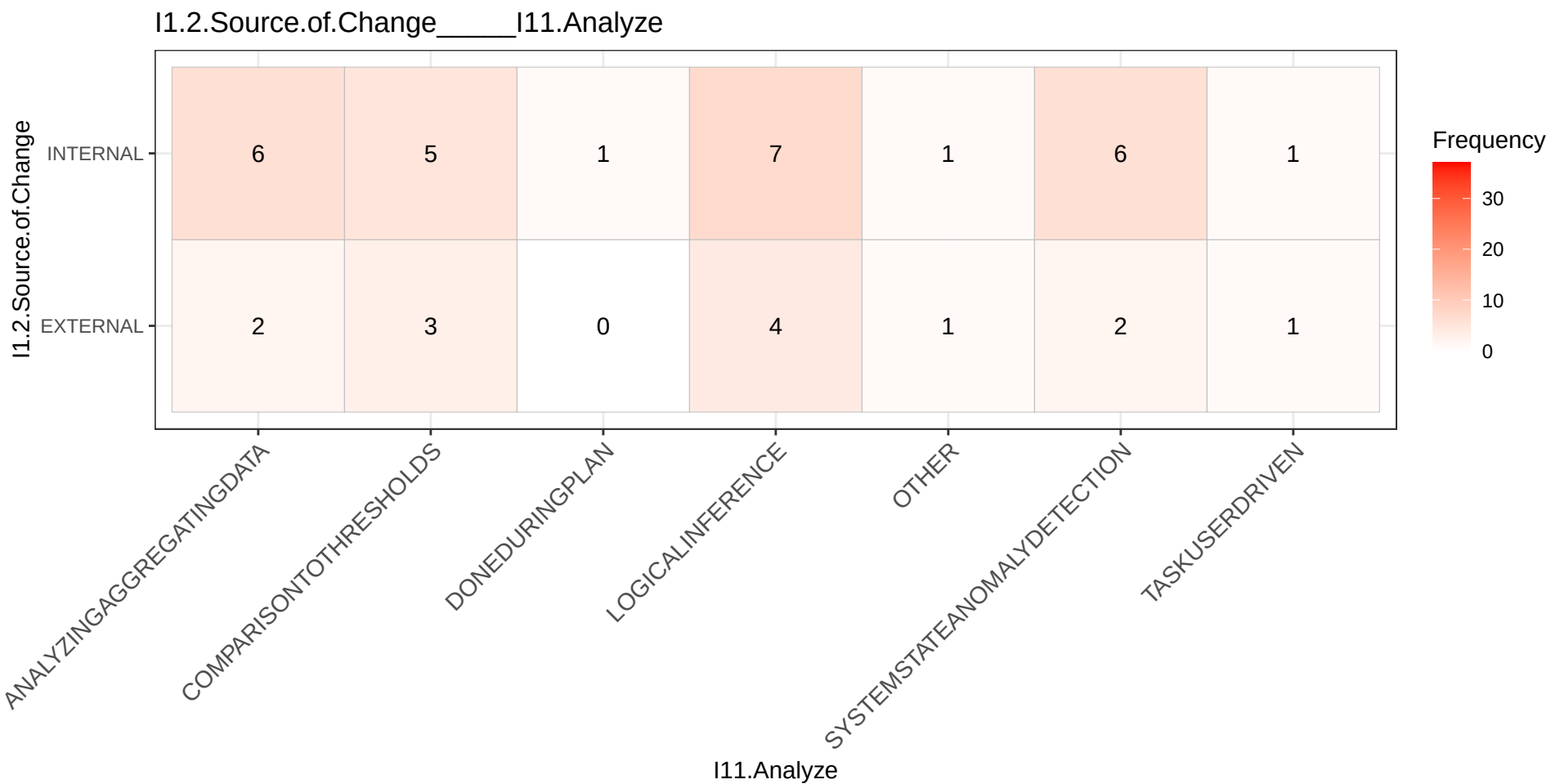
I1.2.Source.of.Change_____I8.Evaluation





I1.2.Source.of.Change_____I10.Monitor





I1.2.Source.of.Change _____ I12.Plan

I1.2.Source.of.Change

INTERNAL

12

10

2

EXTERNAL

7

8

0

DETERMININGTHEOPTIMALCHOICE

RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

I12.Plan

Frequency

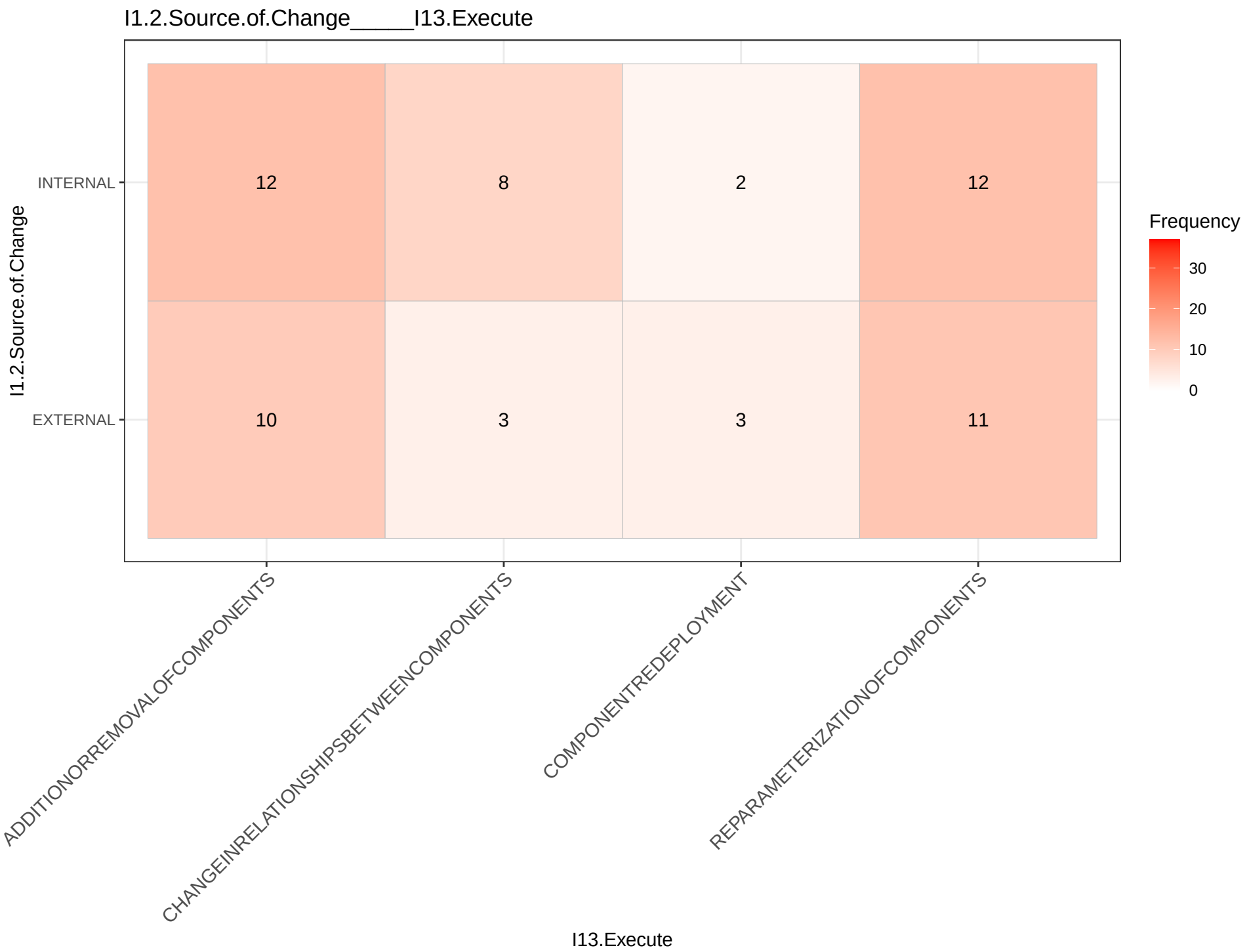


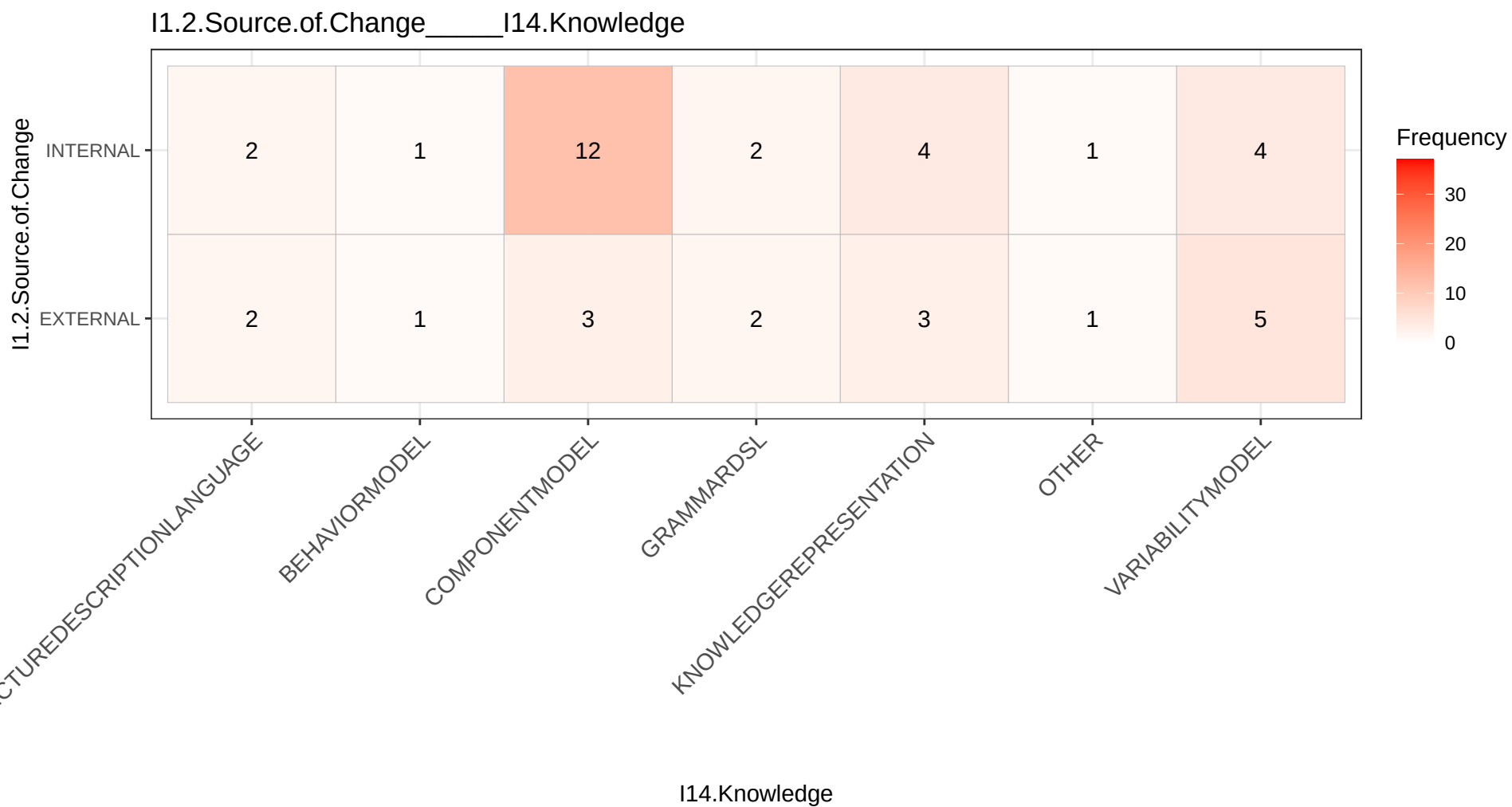
30

20

10

0





I1.2.Type.of.Change_____I1.2.Anticipation.of.Change

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

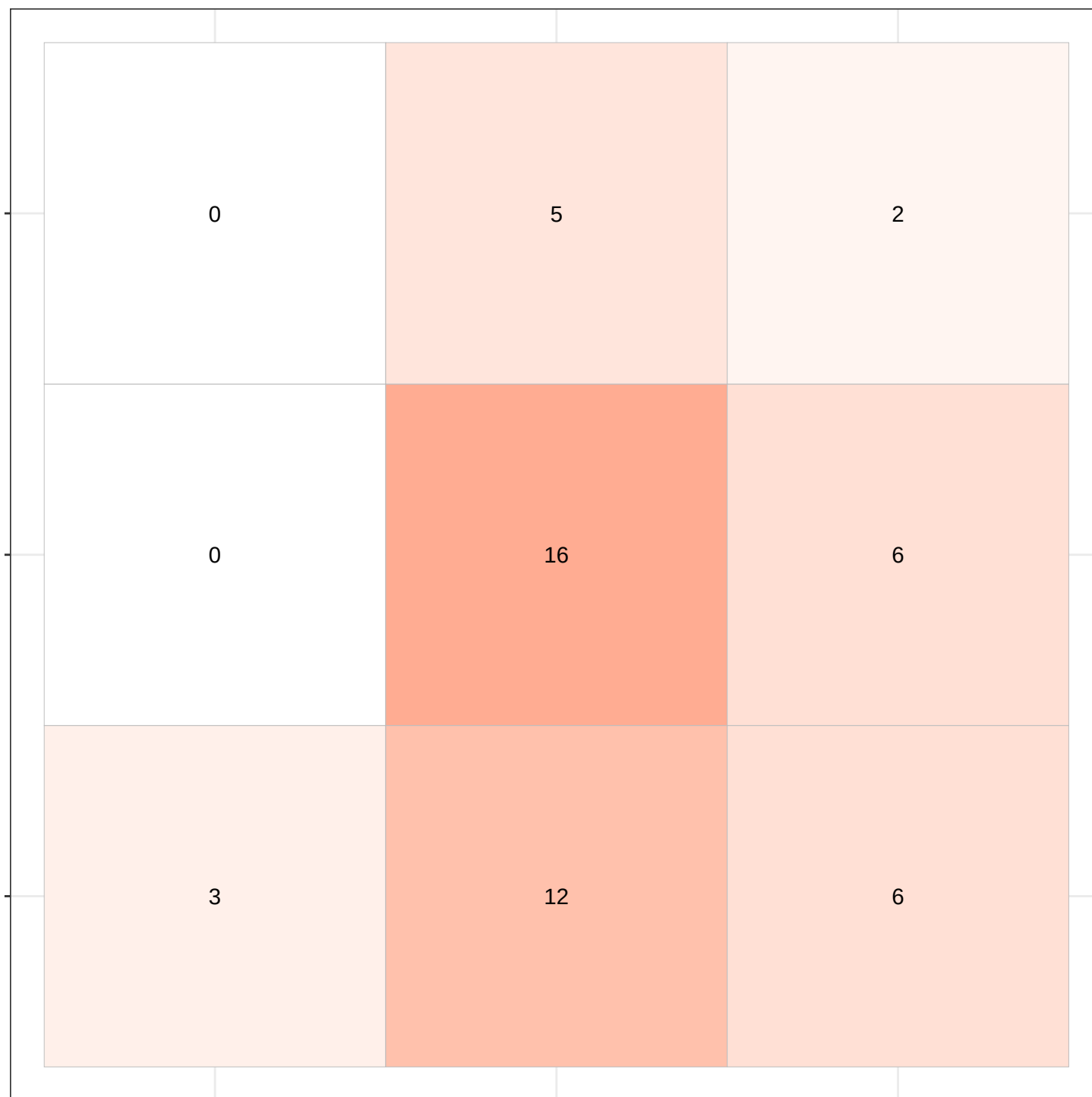
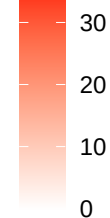
DOFORESEEN

FORESEEABLE

UNFORESEEN

I1.2.Anticipation.of.Change

Frequency



I1.2.Type.of.Change_____I1.2.Frequency.of.Change

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

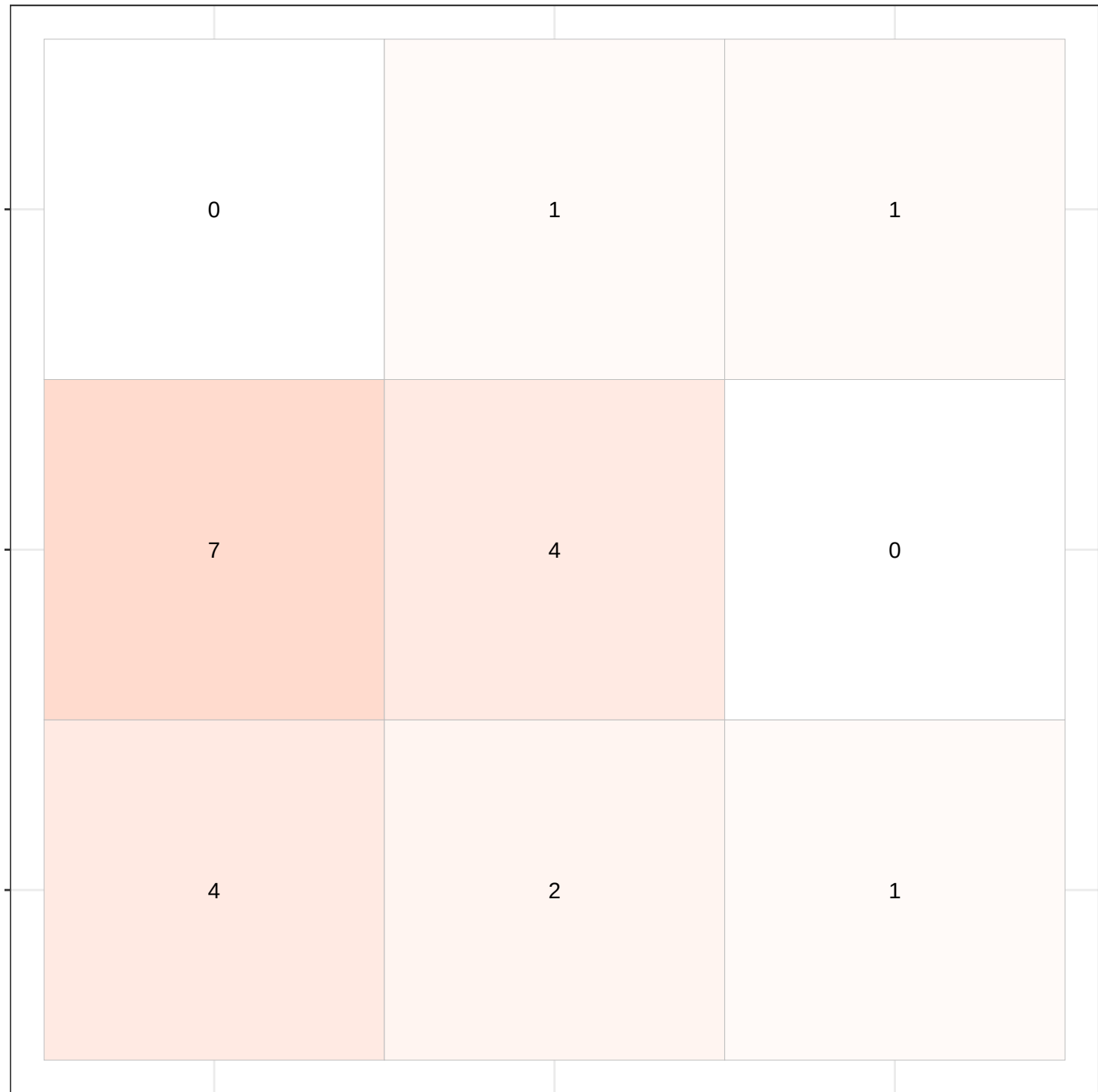
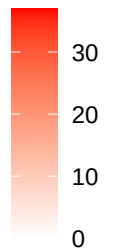
DOFREQUENT

INFREQUENT

RARE

I1.2.Frequency.of.Change

Frequency



I1.2.Type.of.Change_____I1.3.Type.of.Mechanism

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

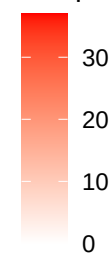
DOFUNCTIONAL

PARAMETRIC

STRUCTURAL

I1.3.Type.of.Mechanism

Frequency



3

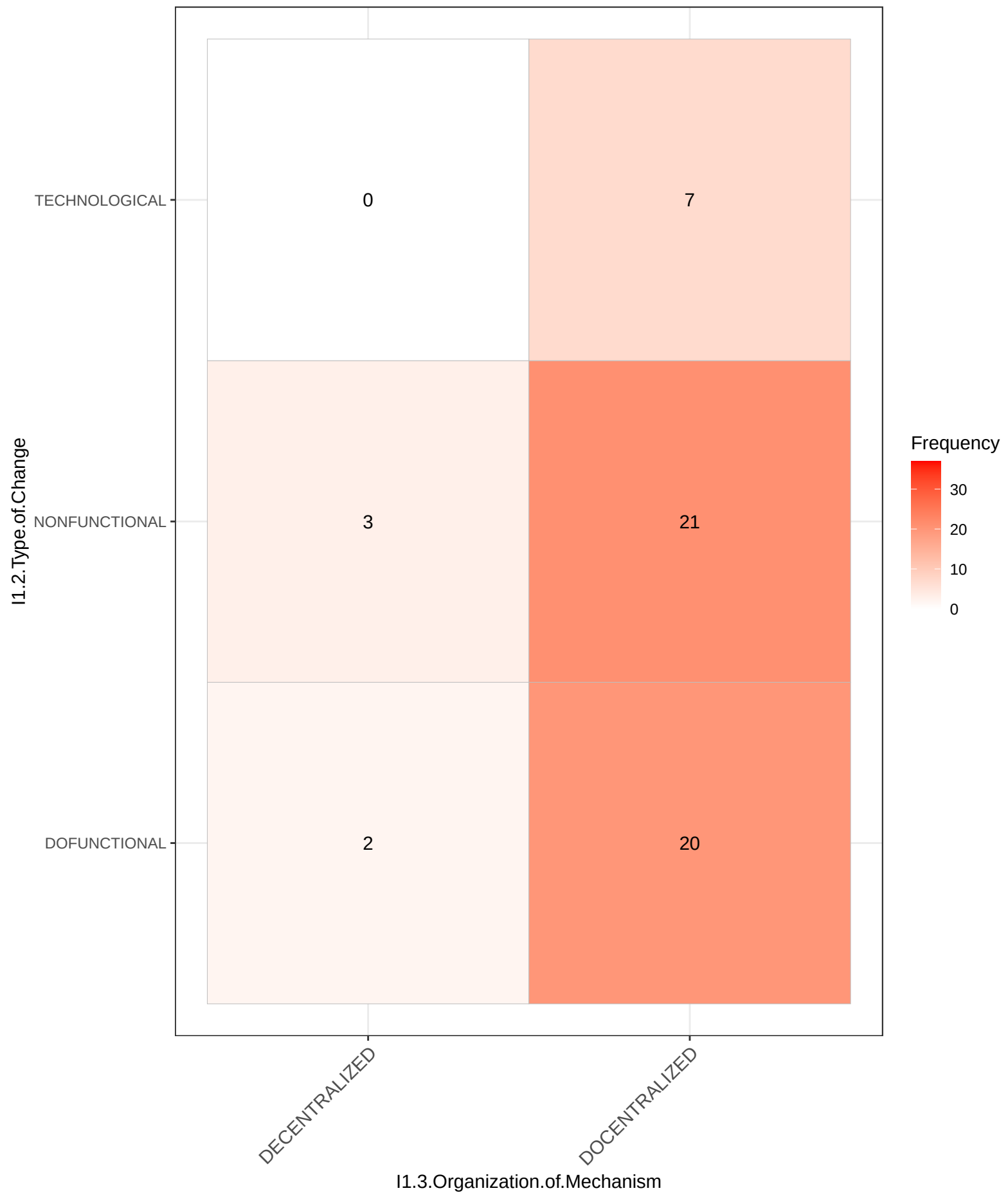
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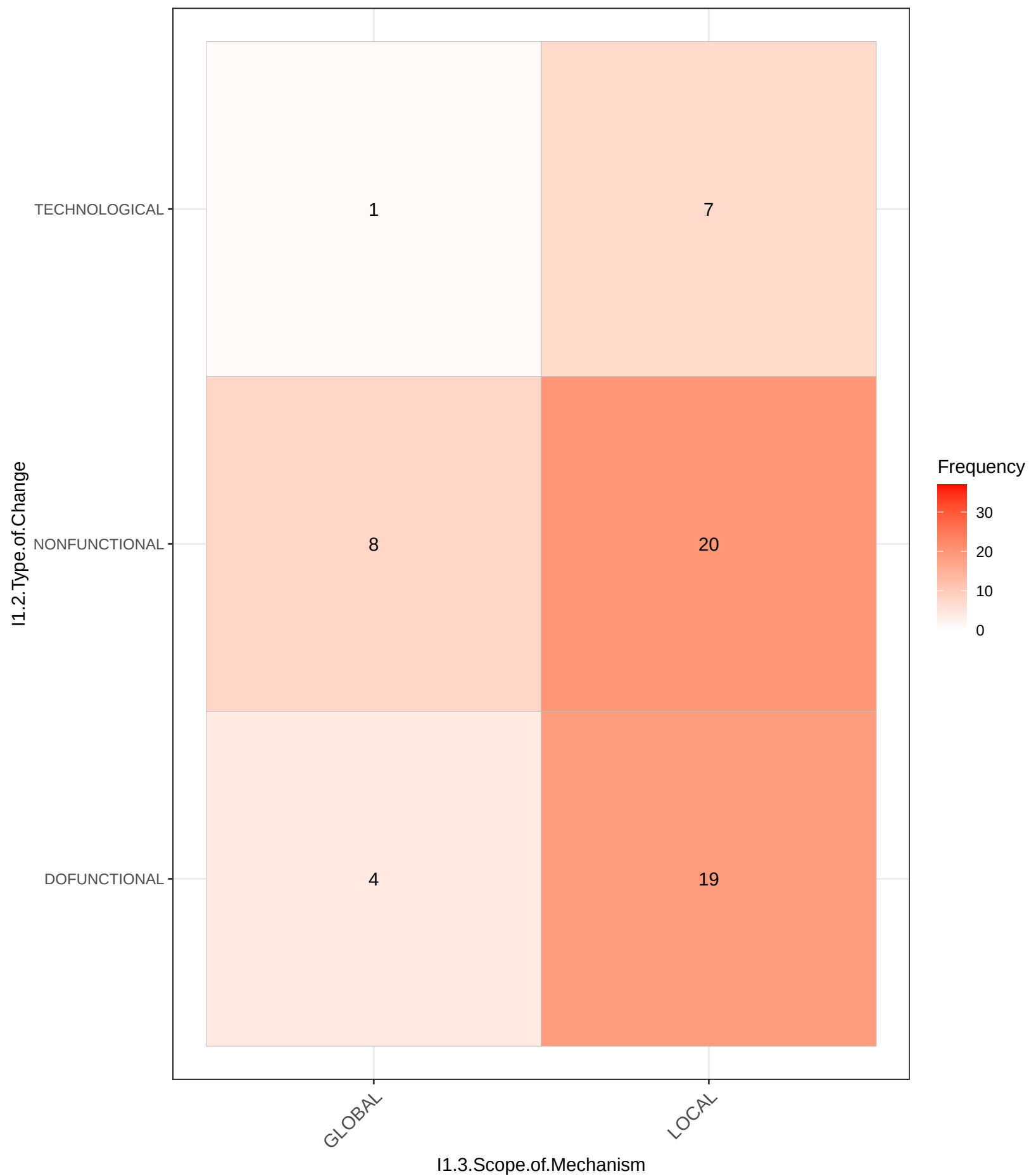
15

21

11

20





I1.2.Type.of.Change_____I1.3.Duration.of.Mechanism

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

DOSHORT

MEDIUM

VERYSHORT

I1.3.Duration.of.Mechanism

Frequency



7

0

0

18

0

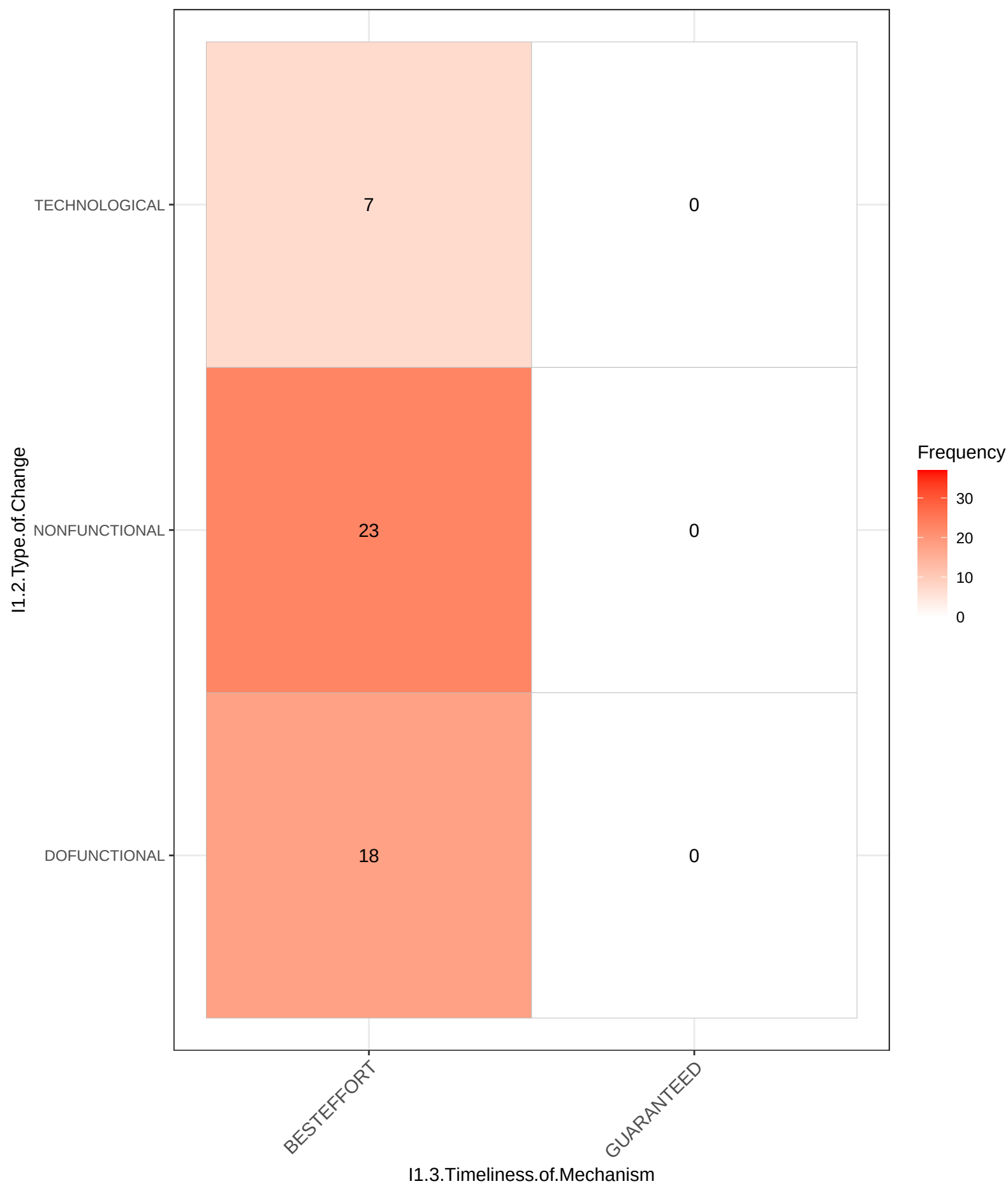
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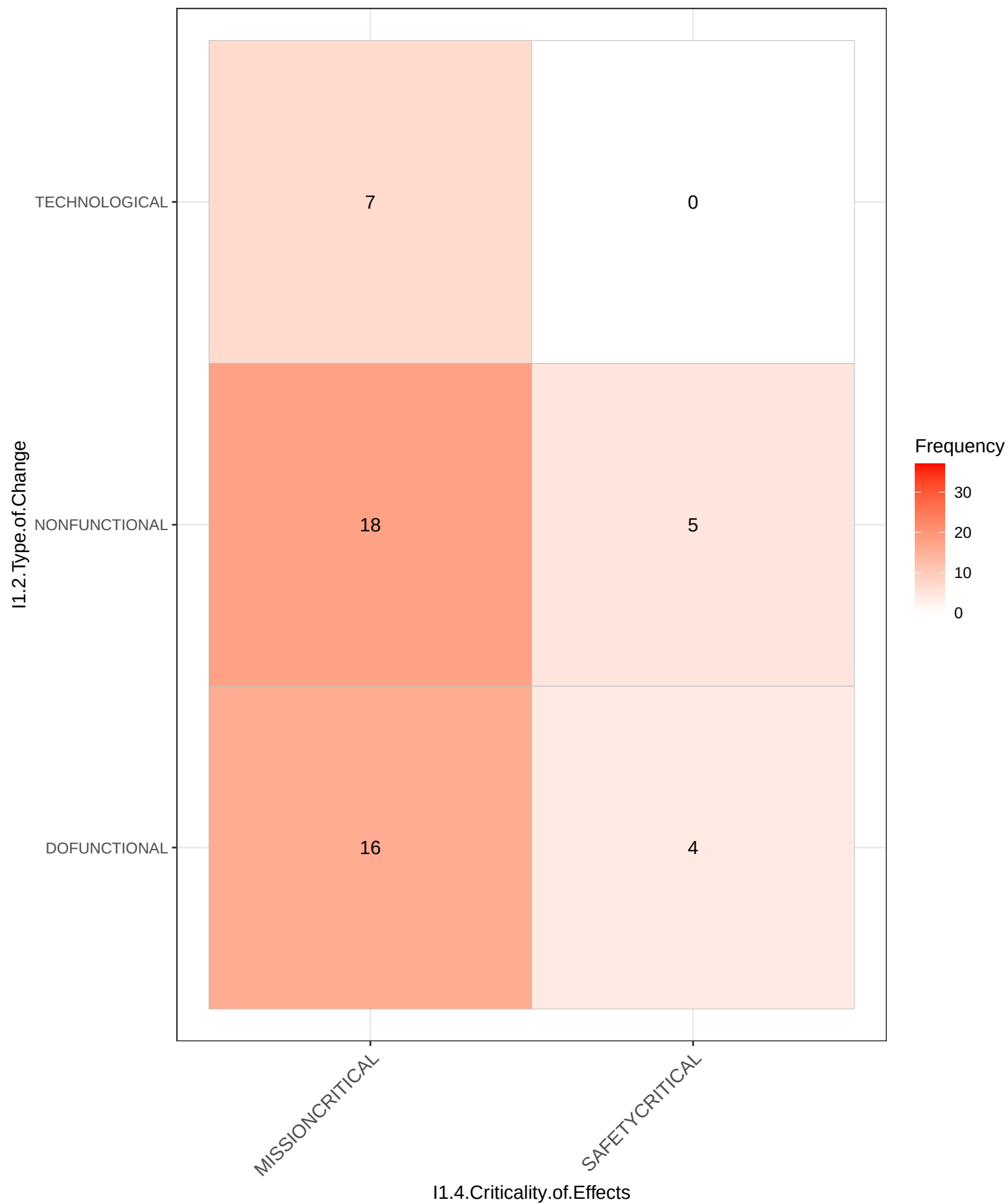
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1

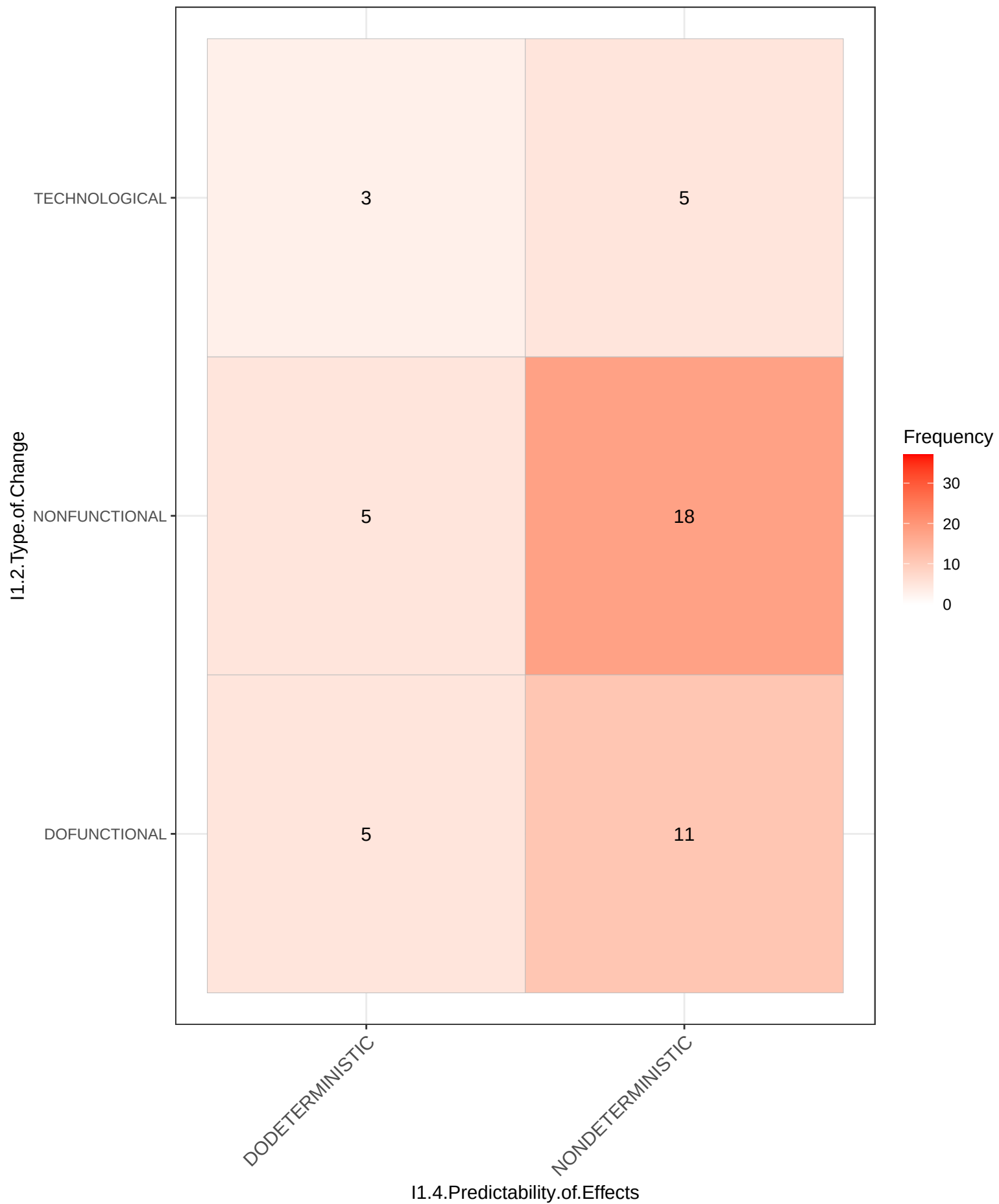
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I1.2.Type.of.Change_____I1.3.Timeliness.of.Mechanism





I1.2.Type.of.Change_____I1.4.Predictability.of.Effects



I1.2.Type.of.Change_____I1.4.Overhead.of.Effects

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

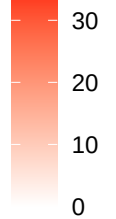
DEPENDENT

DOSIGNIFICANT

INSIGNIFICANT

I1.4.Overhead.of.Effects

Frequency



6

0

0

11

4

4

10

2

5

I1.2.Type.of.Change_____I1.4.Resilience.of.Effects

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

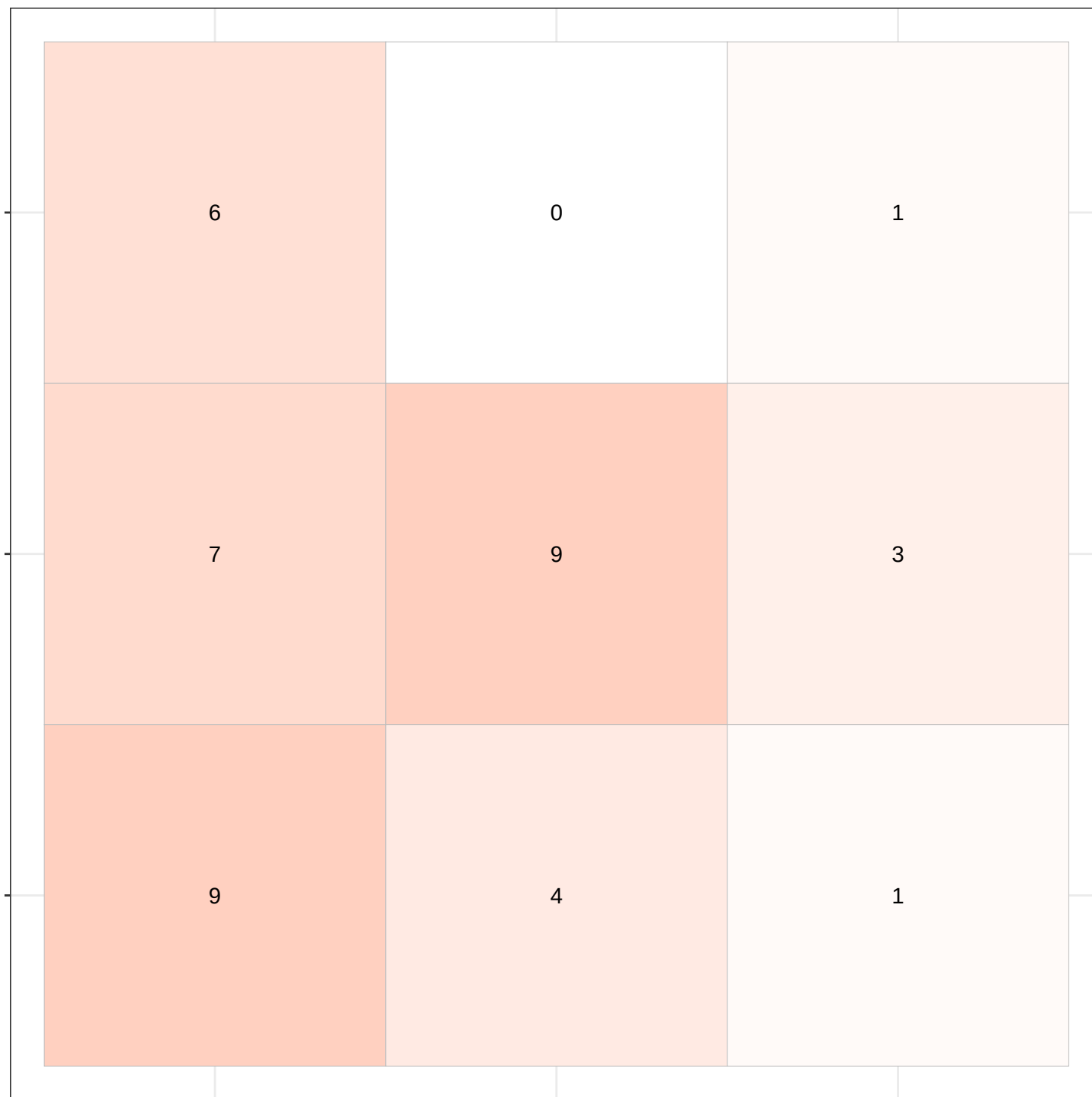
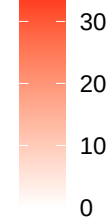
DEPENDENT

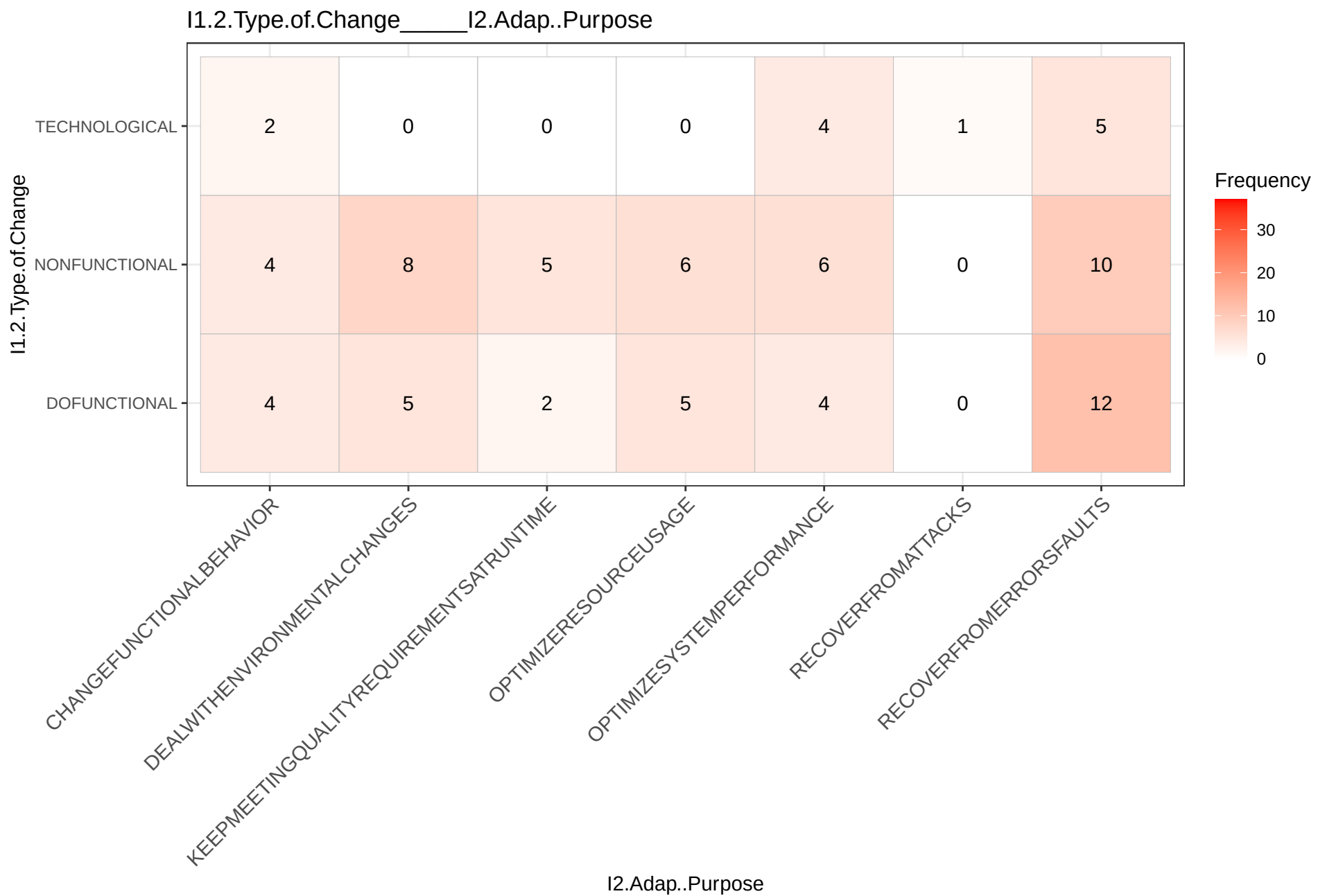
DORESILIENT

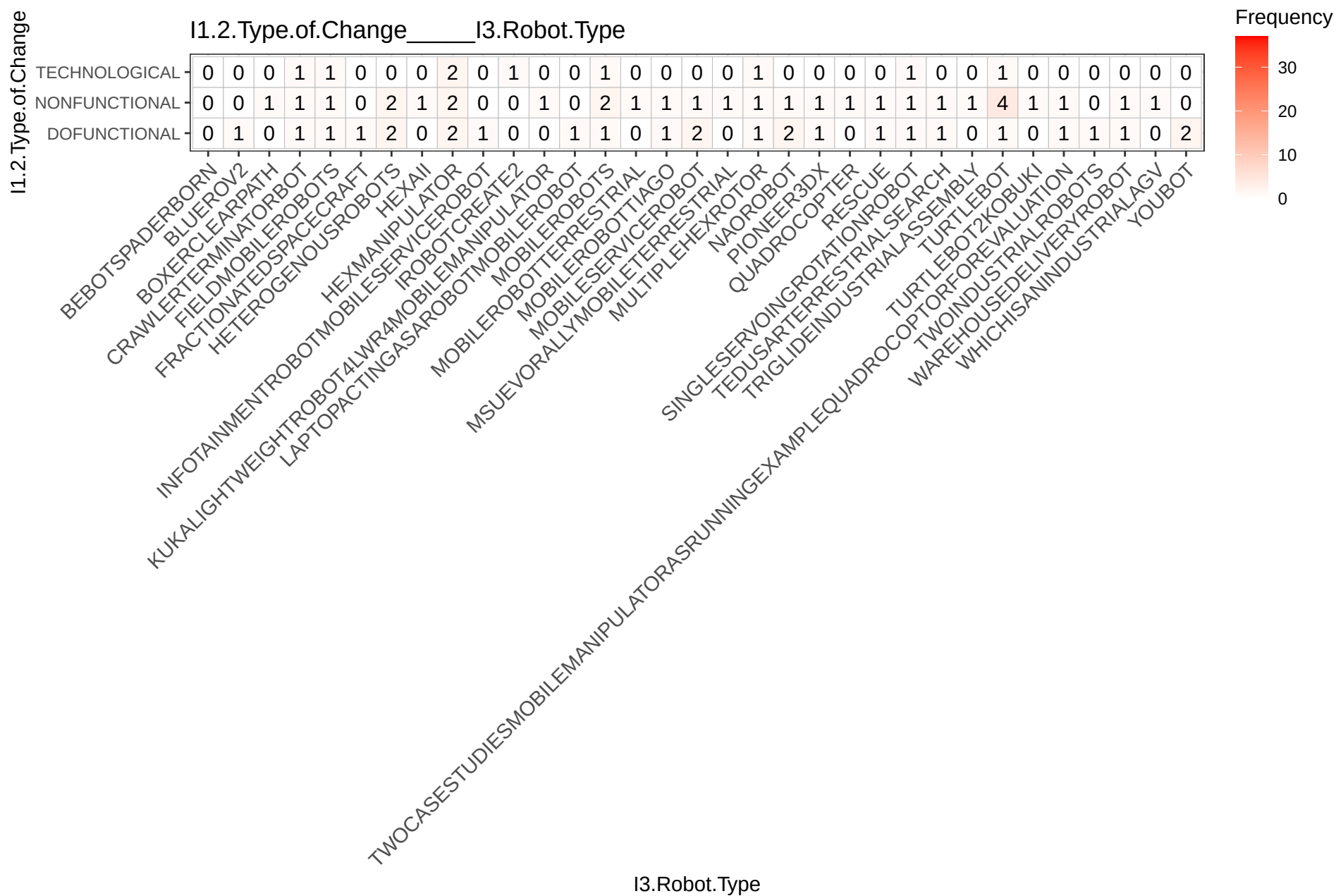
IRRRESILIENT

I1.4.Resilience.of.Effects

Frequency







I1.2.Type.of.Change_____I4.Robo.SW

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

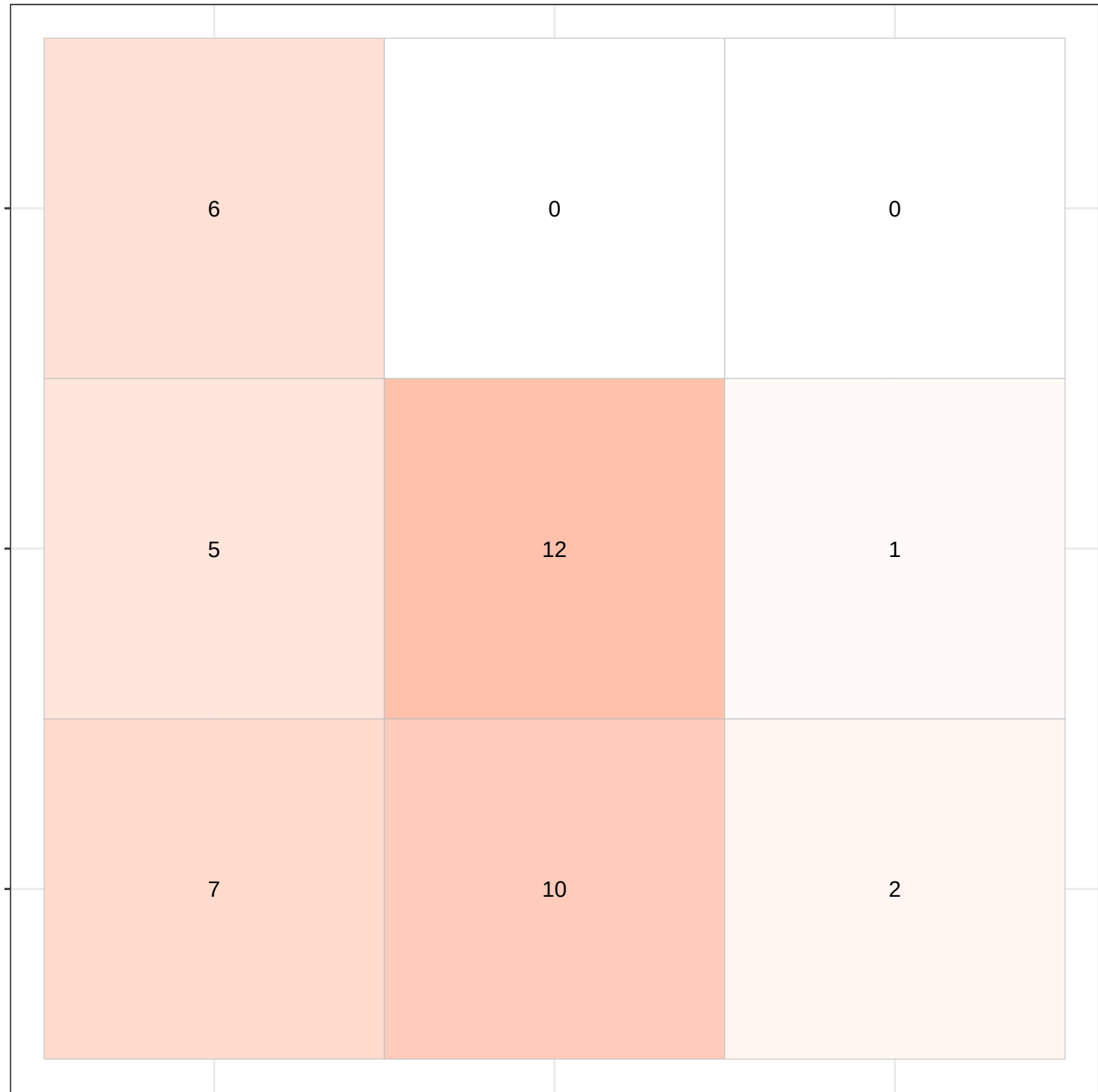
OTHER

ROS1

ROS2

I4.Robo.SW

Frequency



I1.2.Type.of.Change_____I5.QA

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

Frequency



FUNCTIONALSUITABILITY

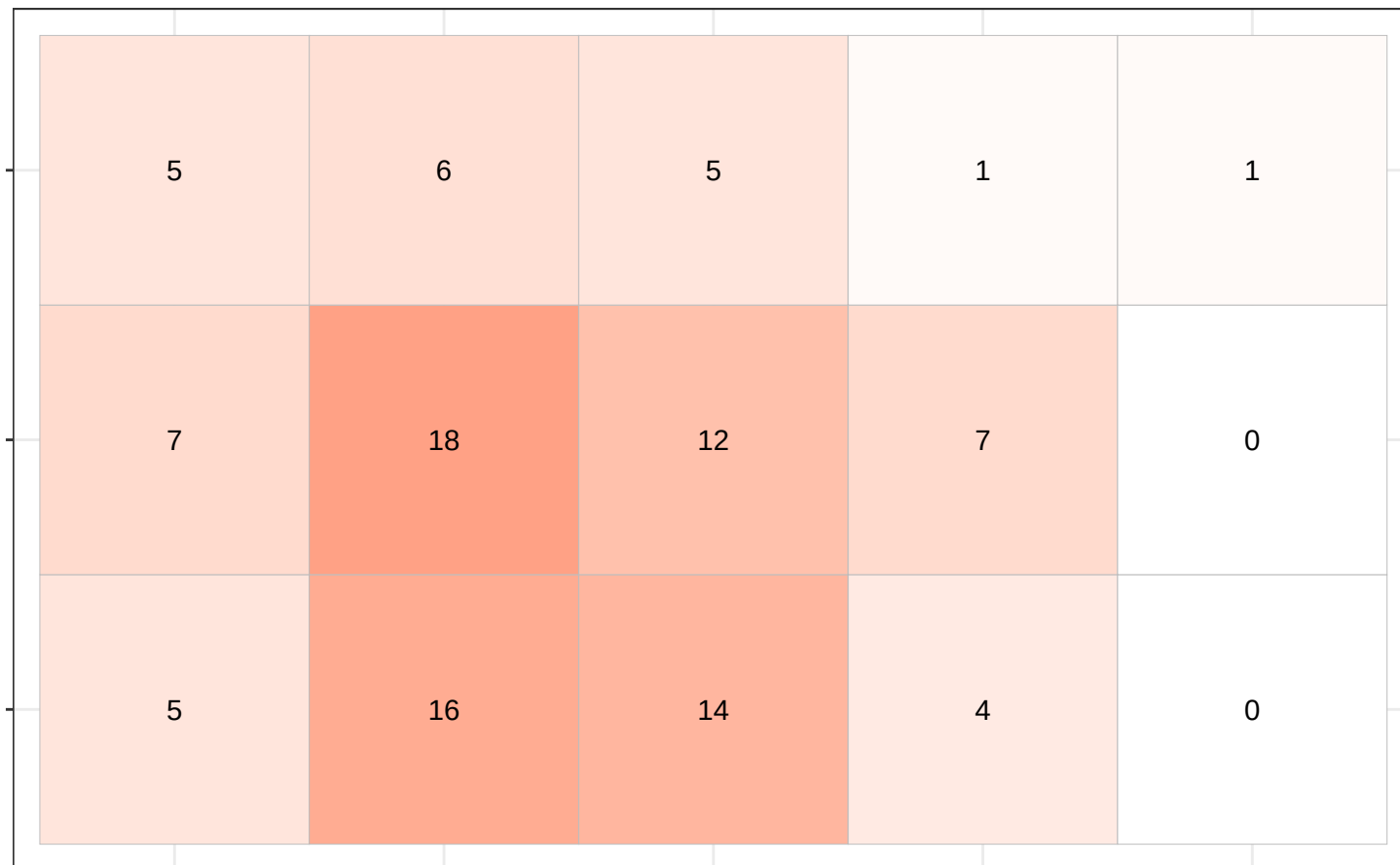
PERFORMANCEEFFICIENCY

RELIABILITY

SAFETY

SECURITY

I5.QA



I1.2.Type.of.Change_____I6.Independence

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

Frequency



1

1

5

7

2

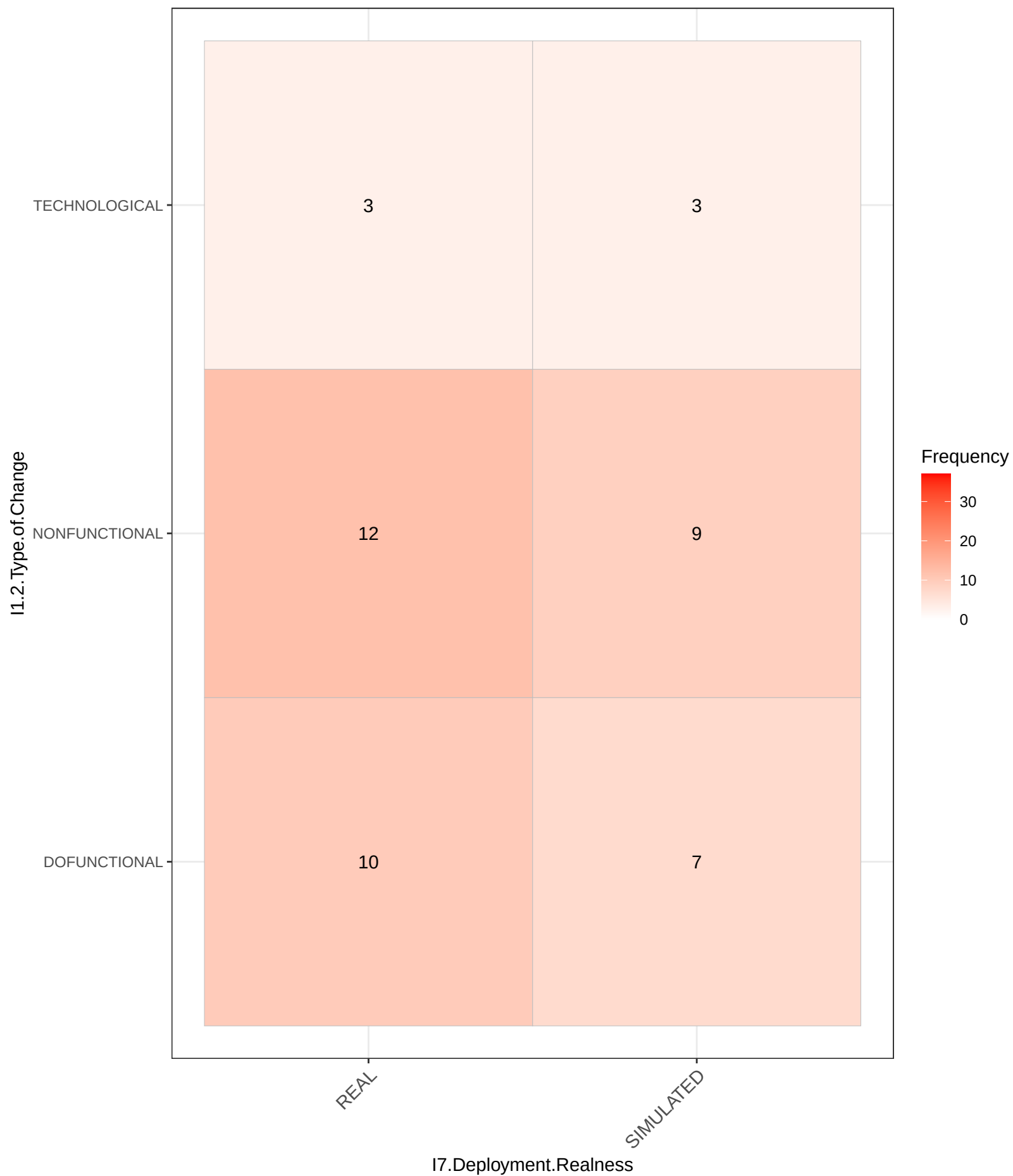
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6

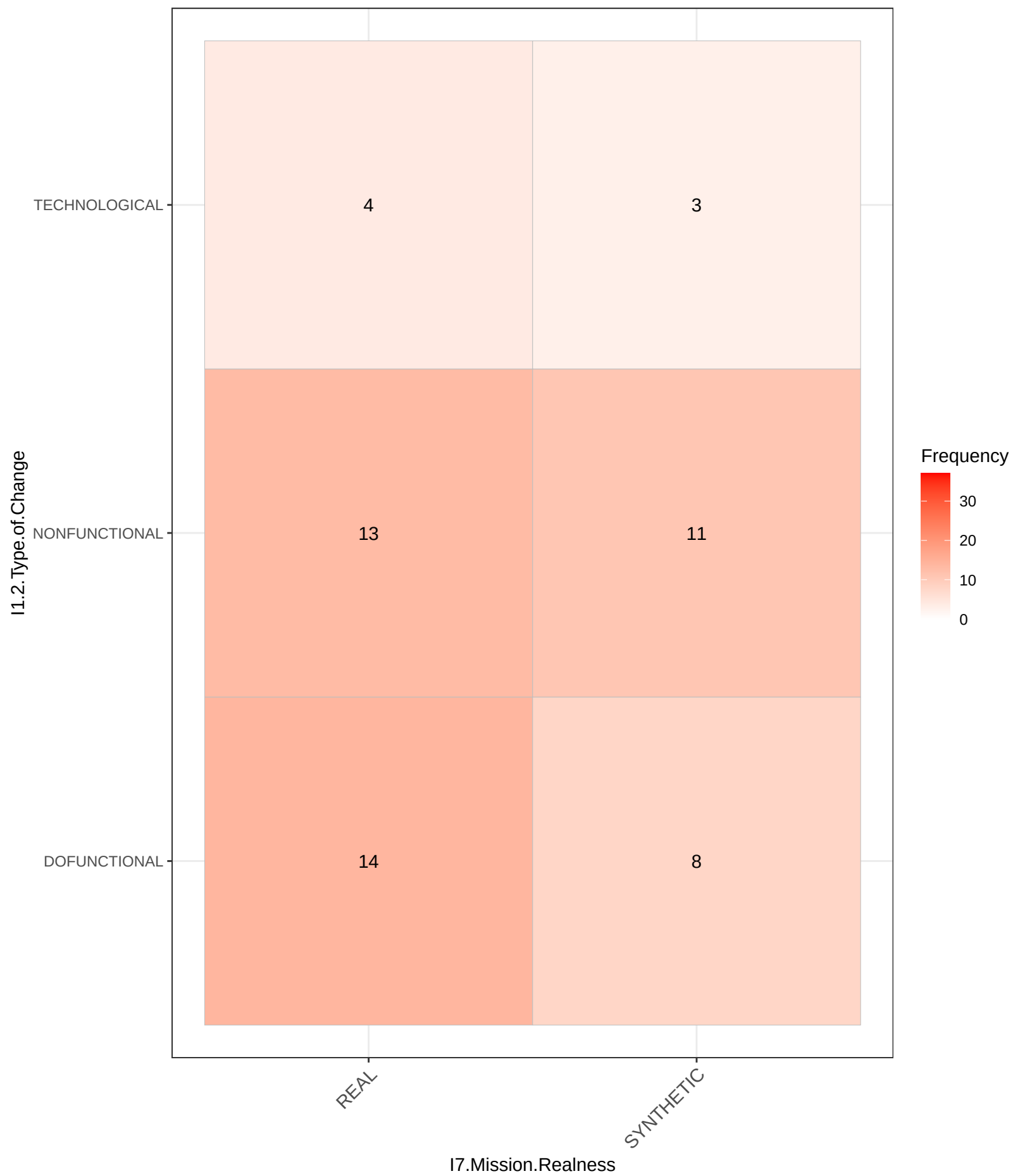
0

16

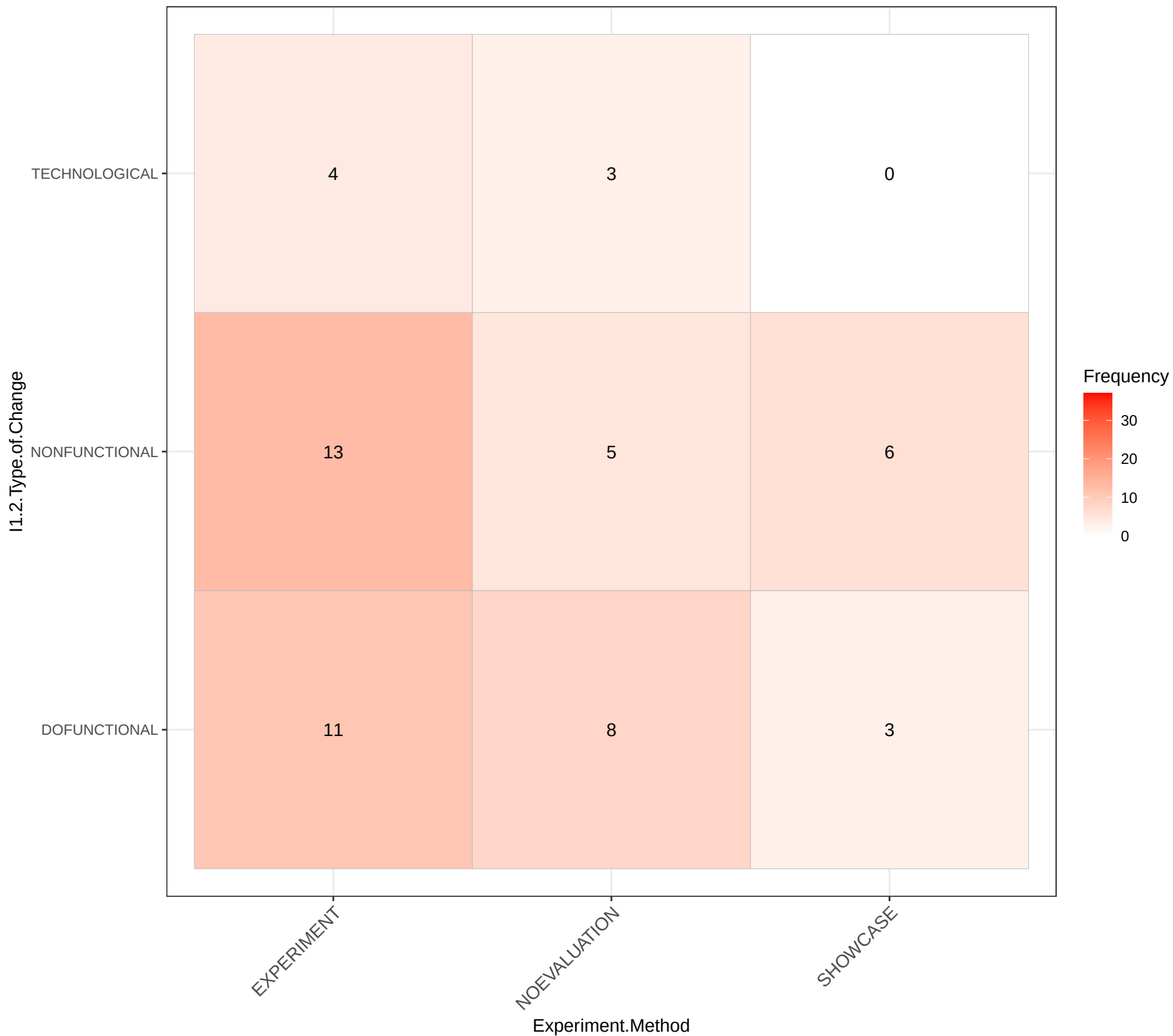
I1.2.Type.of.Change_____I7.Deployment.Realness



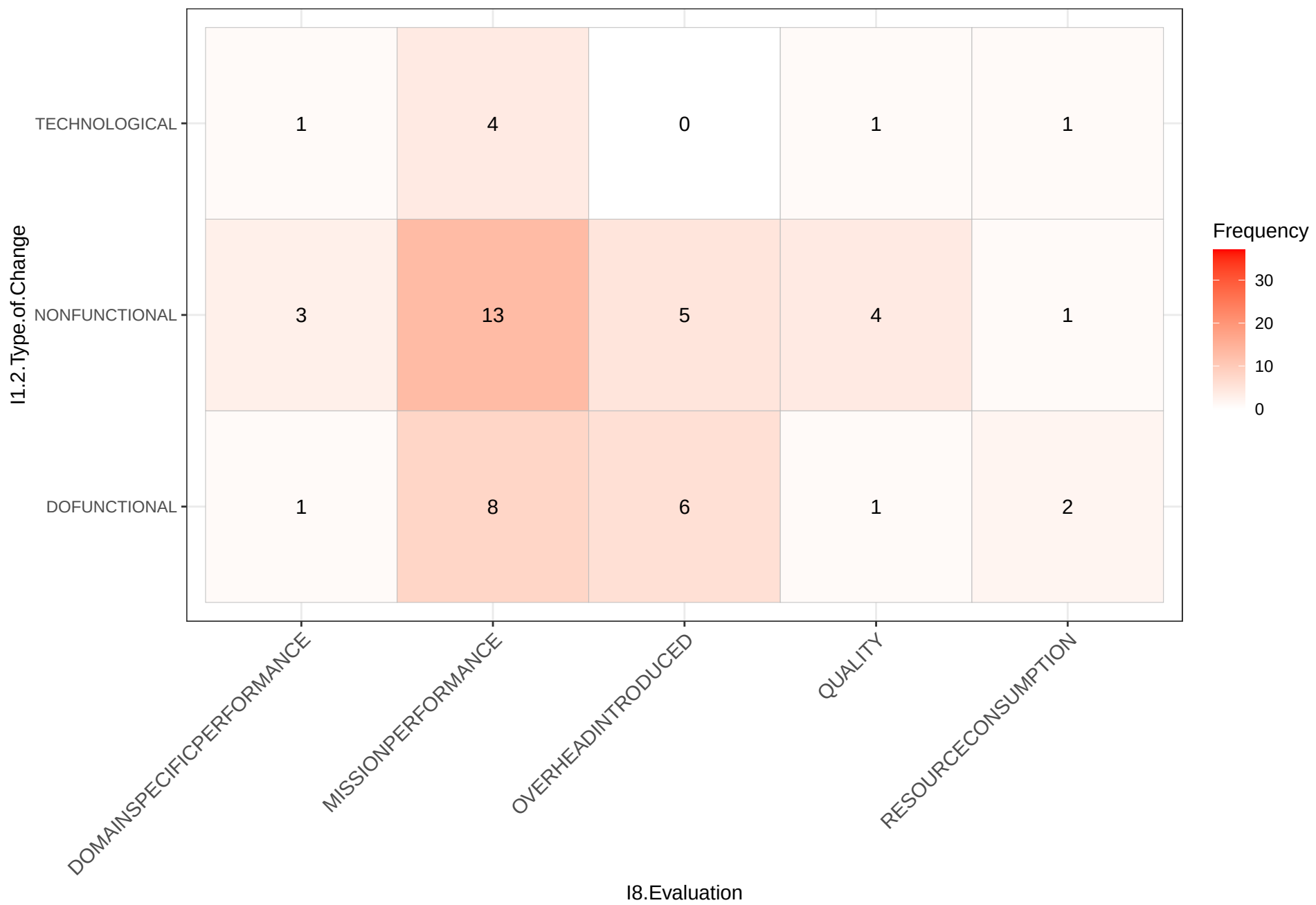
I1.2.Type.of.Change_____I7.Mission.Realness



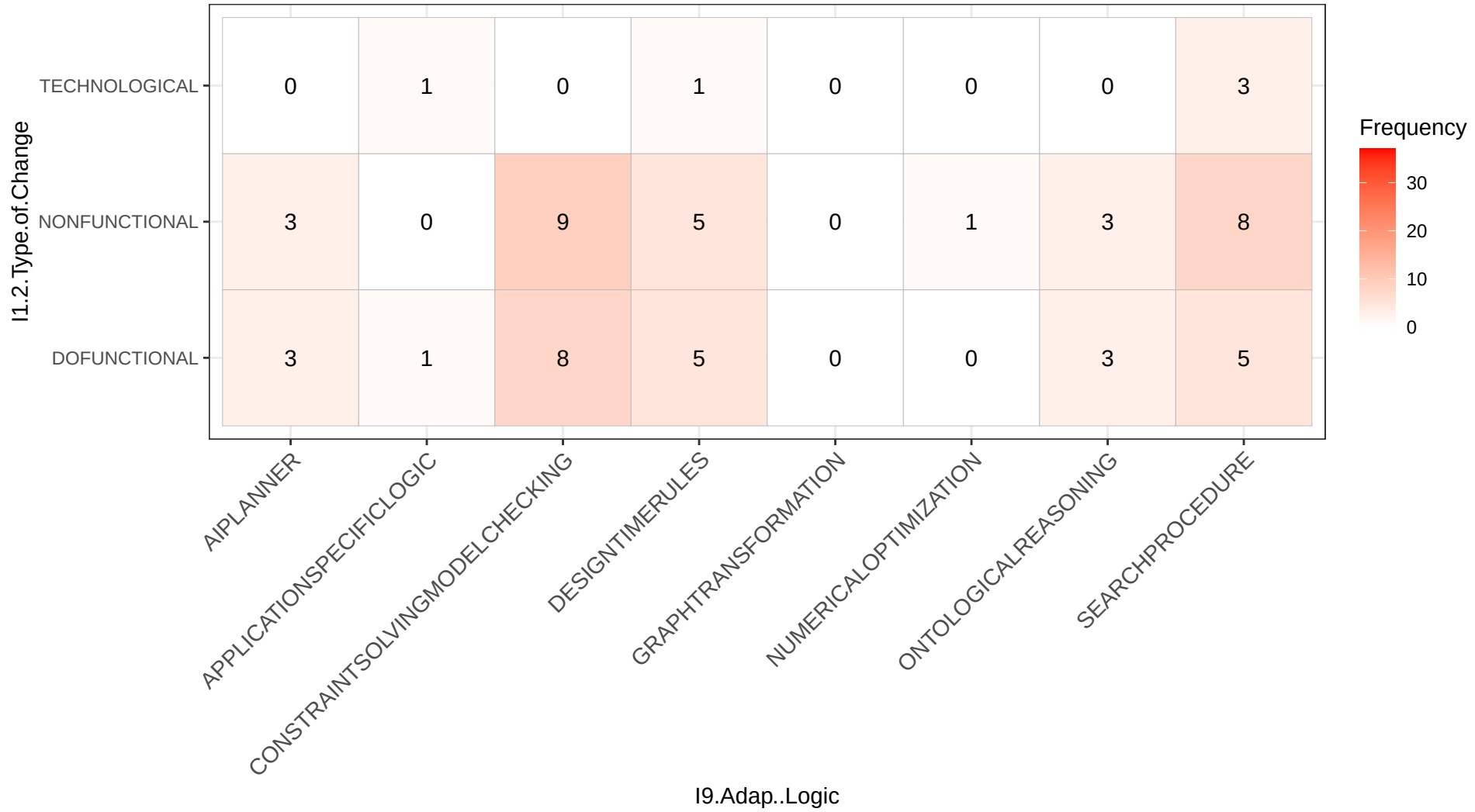
I1.2.Type.of.Change_____Experiment.Method



I1.2.Type.of.Change_____I8.Evaluation

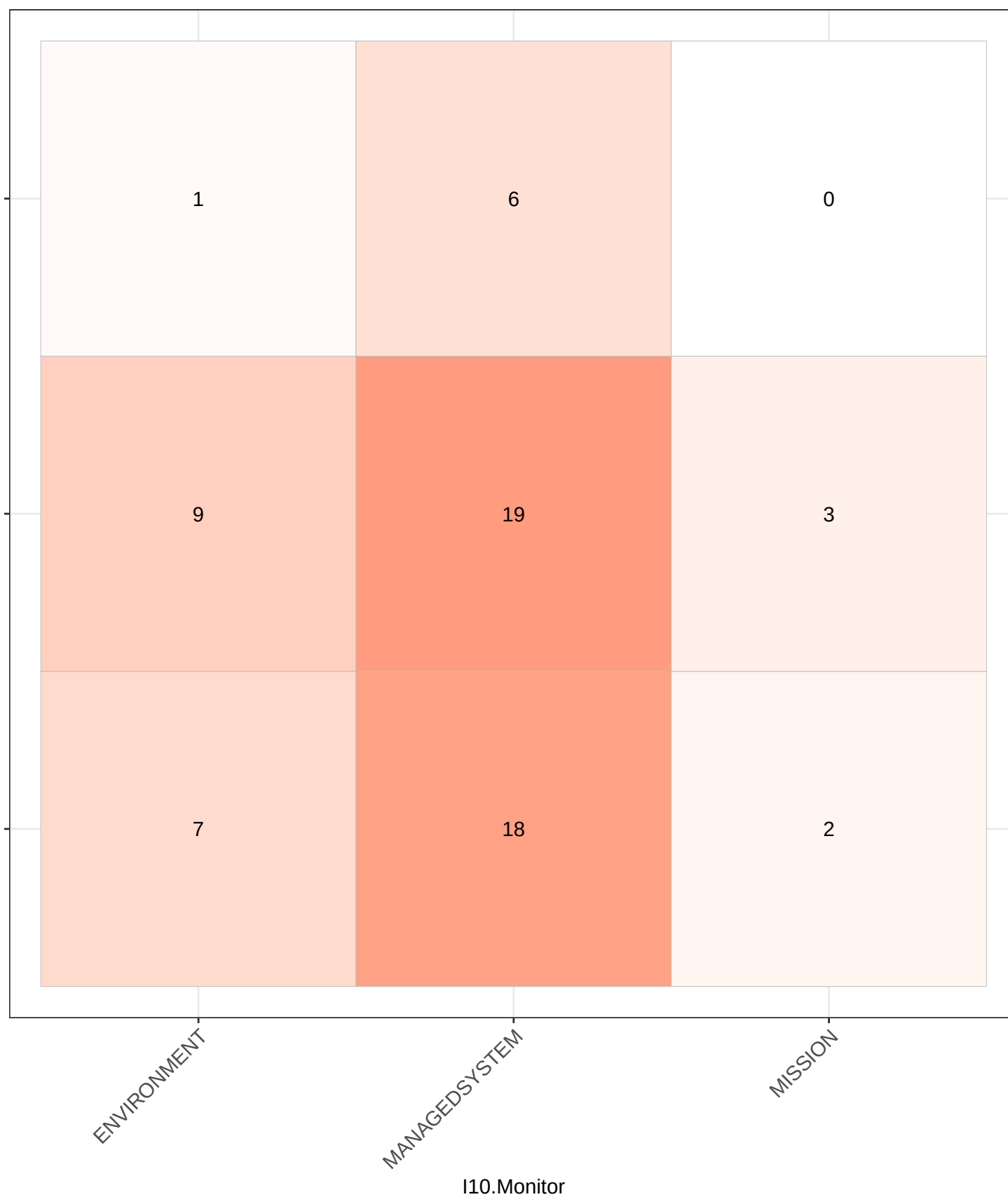


I1.2.Type.of.Change_____I9.Adap..Logic



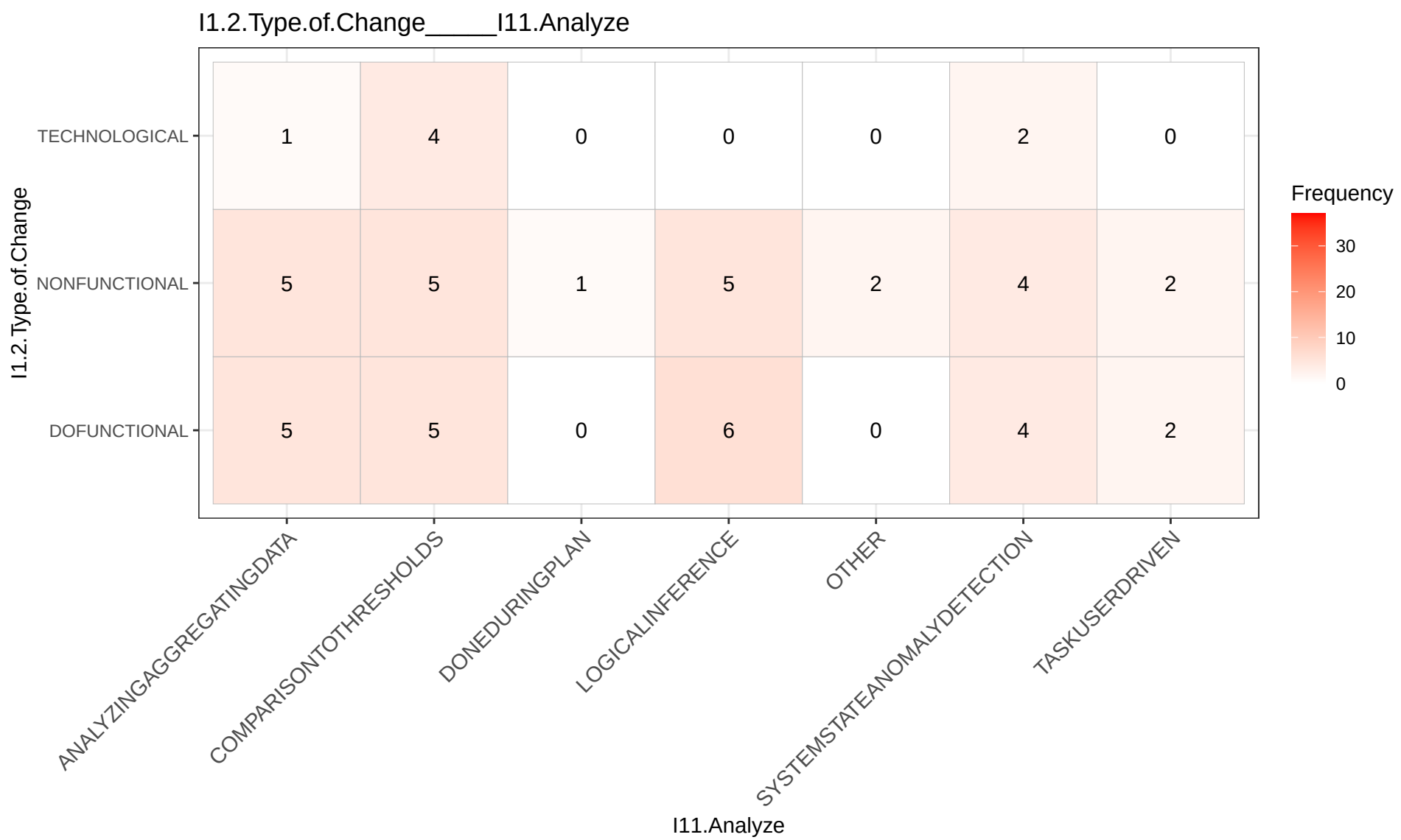
I1.2.Type.of.Change_____I10.Monitor

I1.2.Type.of.Change

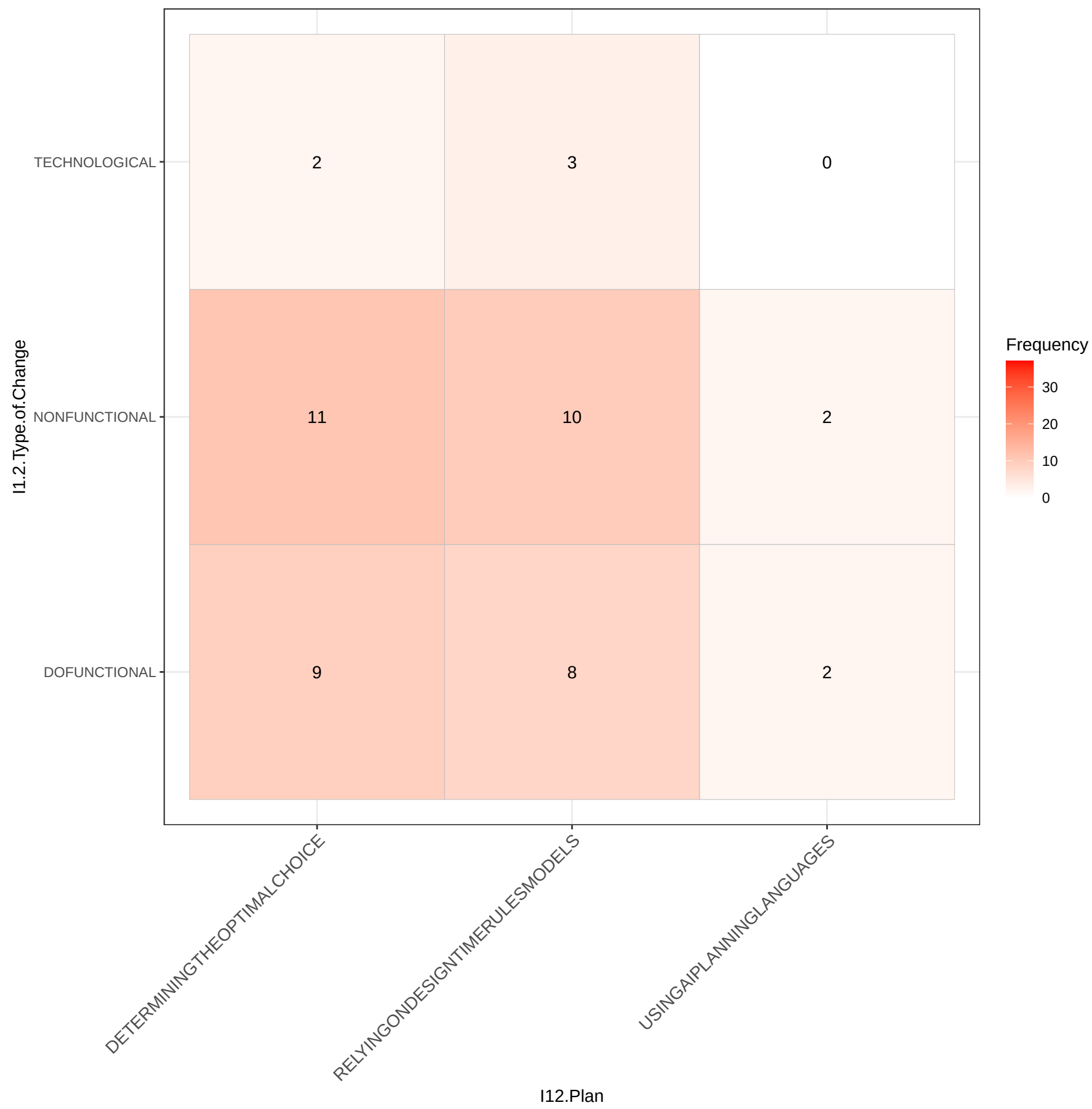


Frequency





I1.2.Type.of.Change_____I12.Plan



I1.2.Type.of.Change_____I13.Execute

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

Frequency

30

20

10

0

ADDITIONORREMOVALOFCOMPONENTS

CHANGEINRELATIONSHIPS BETWEENCOMPONENTS

COMPONENTREDEPLOYMENT

REPARAMETERIZATIONOFCOMPONENTS

I13.Execute

4

1

0

3

12

4

1

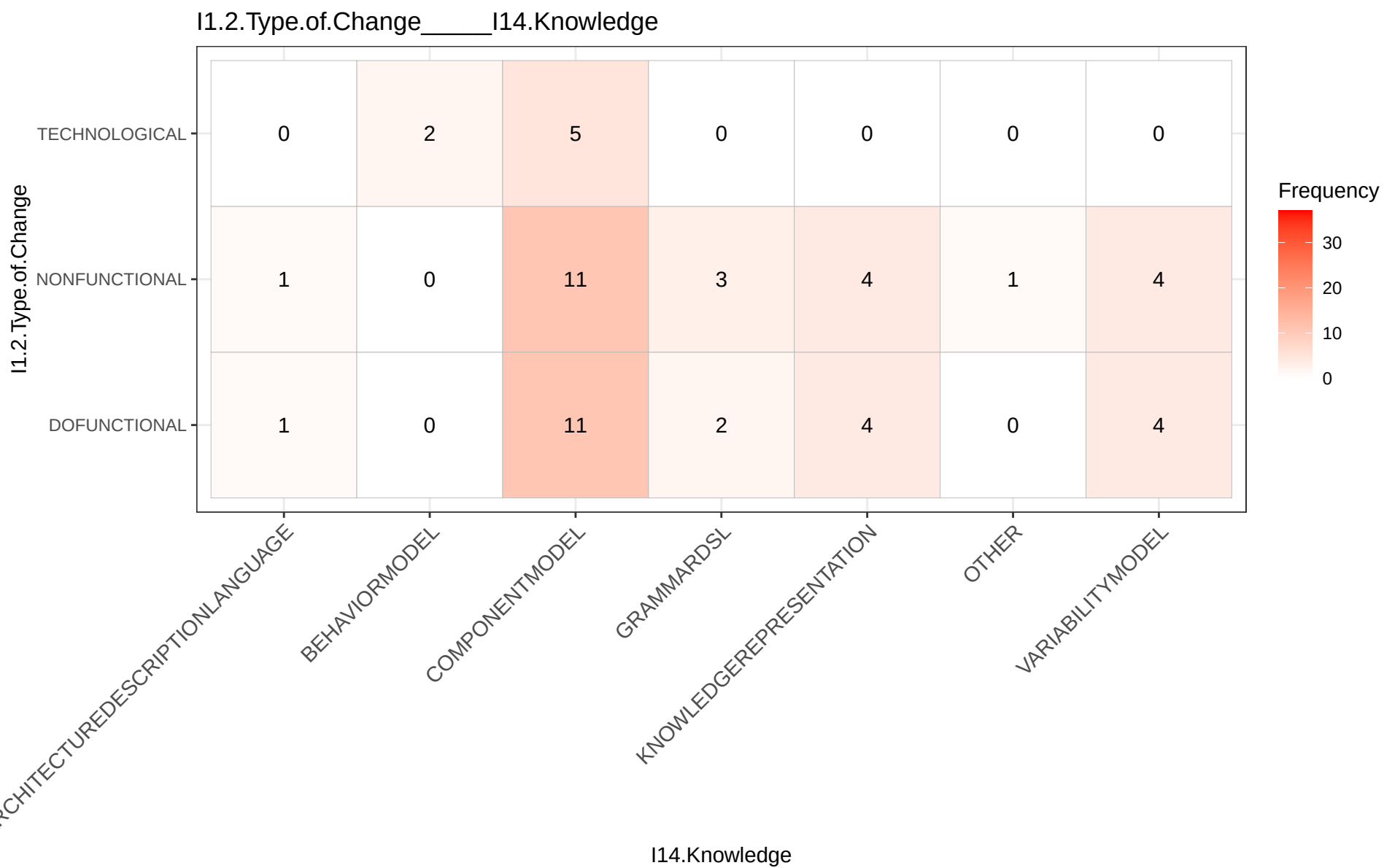
12

12

7

3

10



I1.2.Anticipation.of.Change_____I1.2.Frequency.of.Change

I1.2.Anticipation.of.Change

UNFORESEEN

0

0

0

FORESEEABLE

7

5

1

DOFORESEEN

1

1

1

DOFREQUENT

INFREQUENT

RARE

I1.2.Frequency.of.Change

Frequency



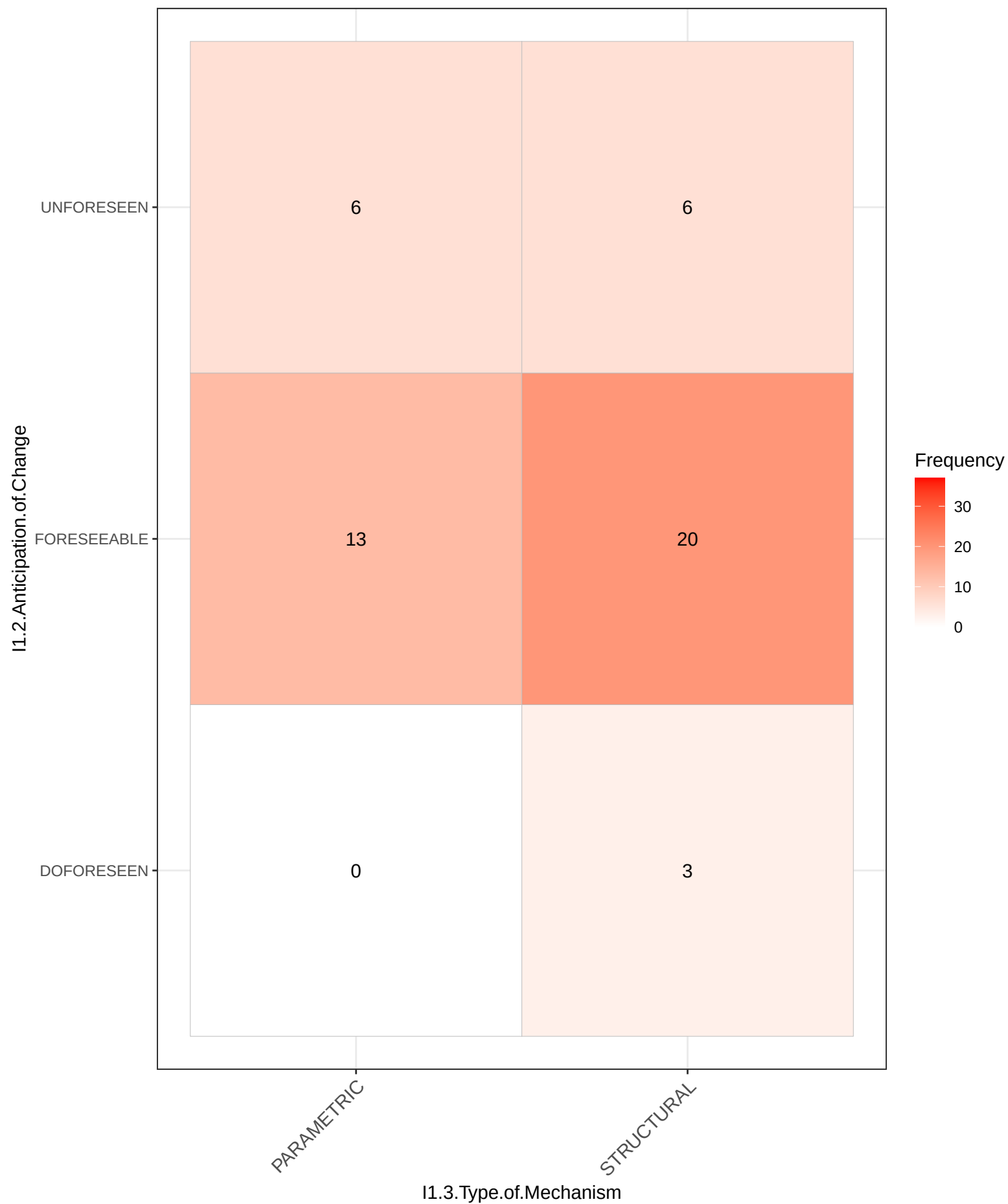
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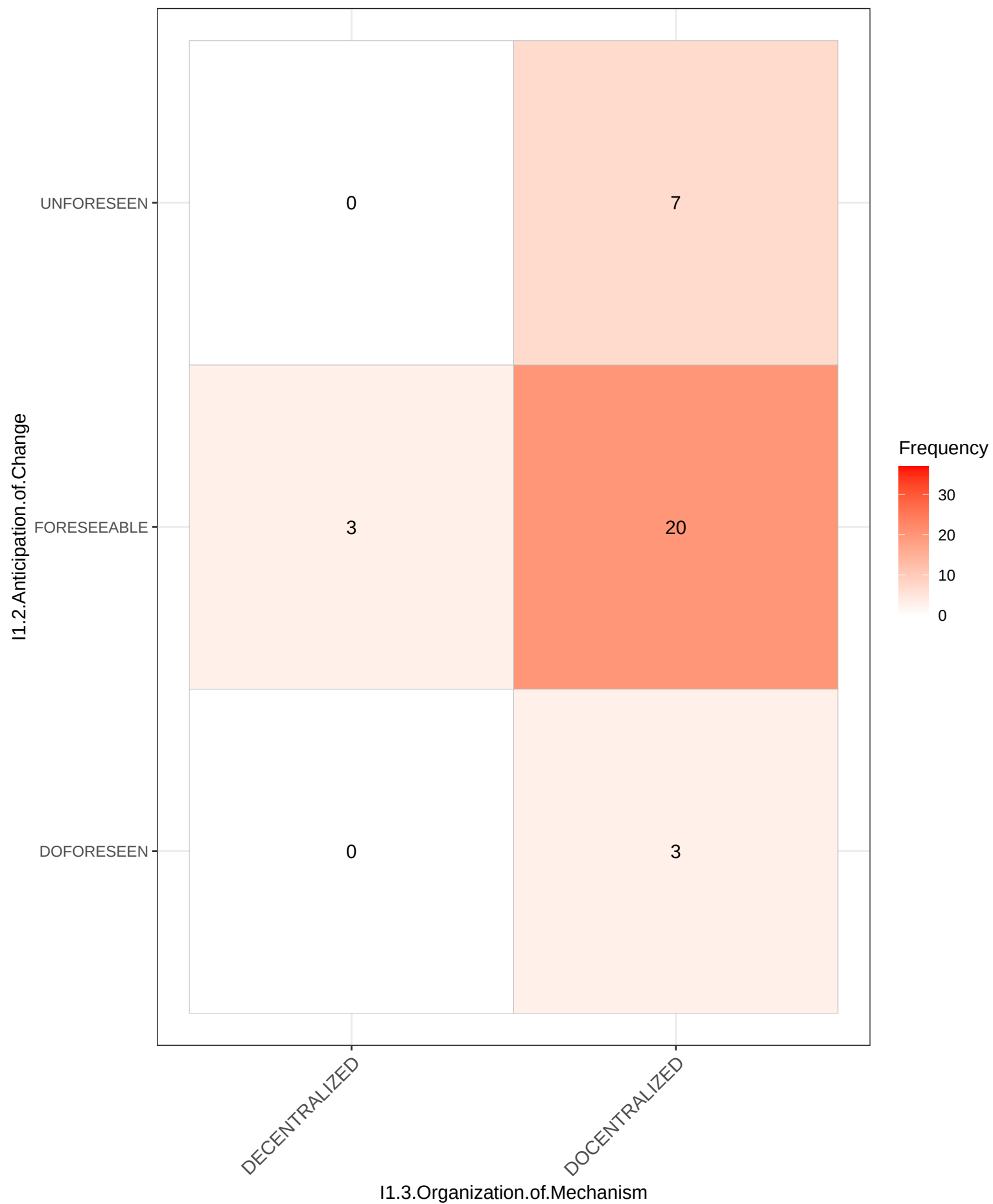
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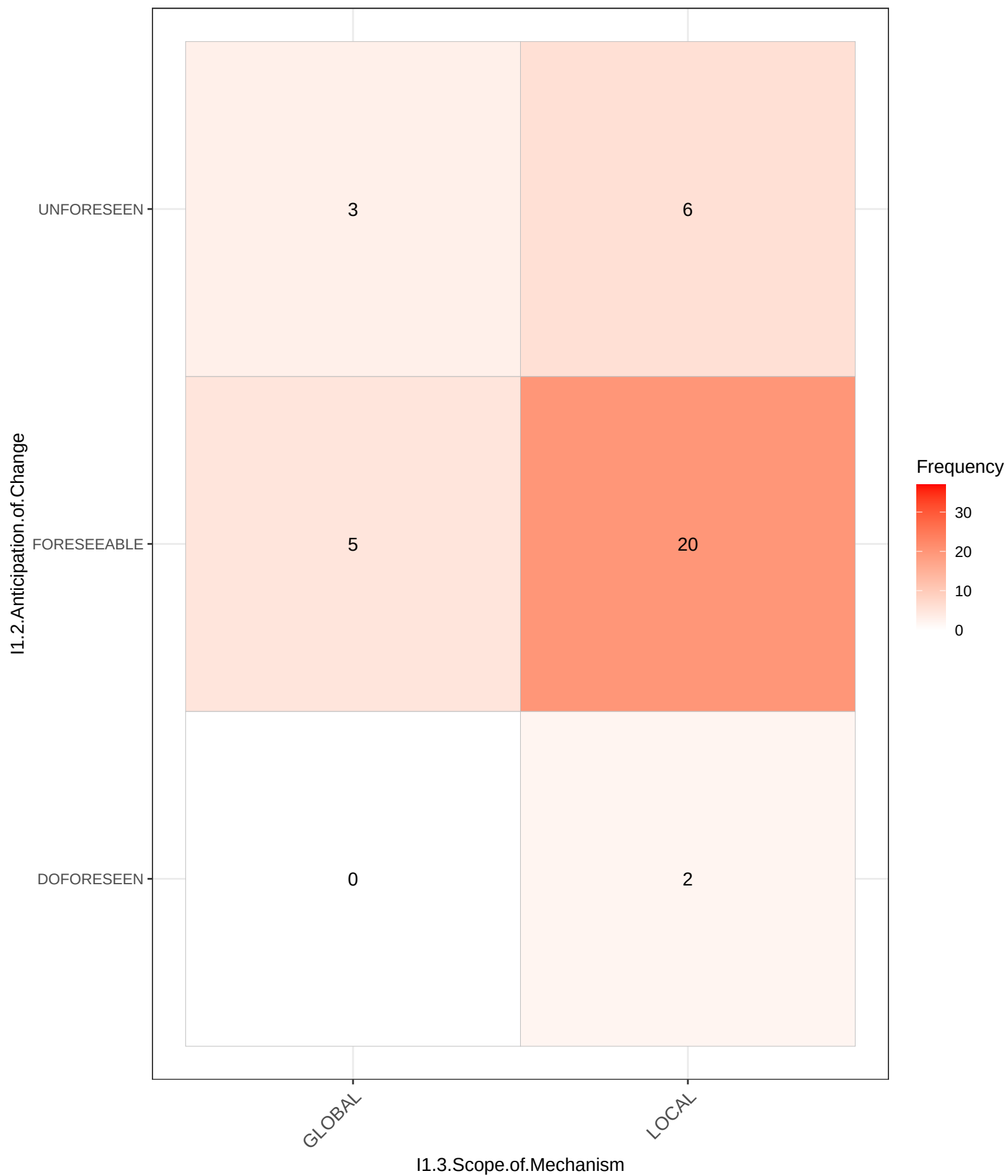
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I1.2.Anticipation.of.Change_____I1.3.Type.of.Mechanism

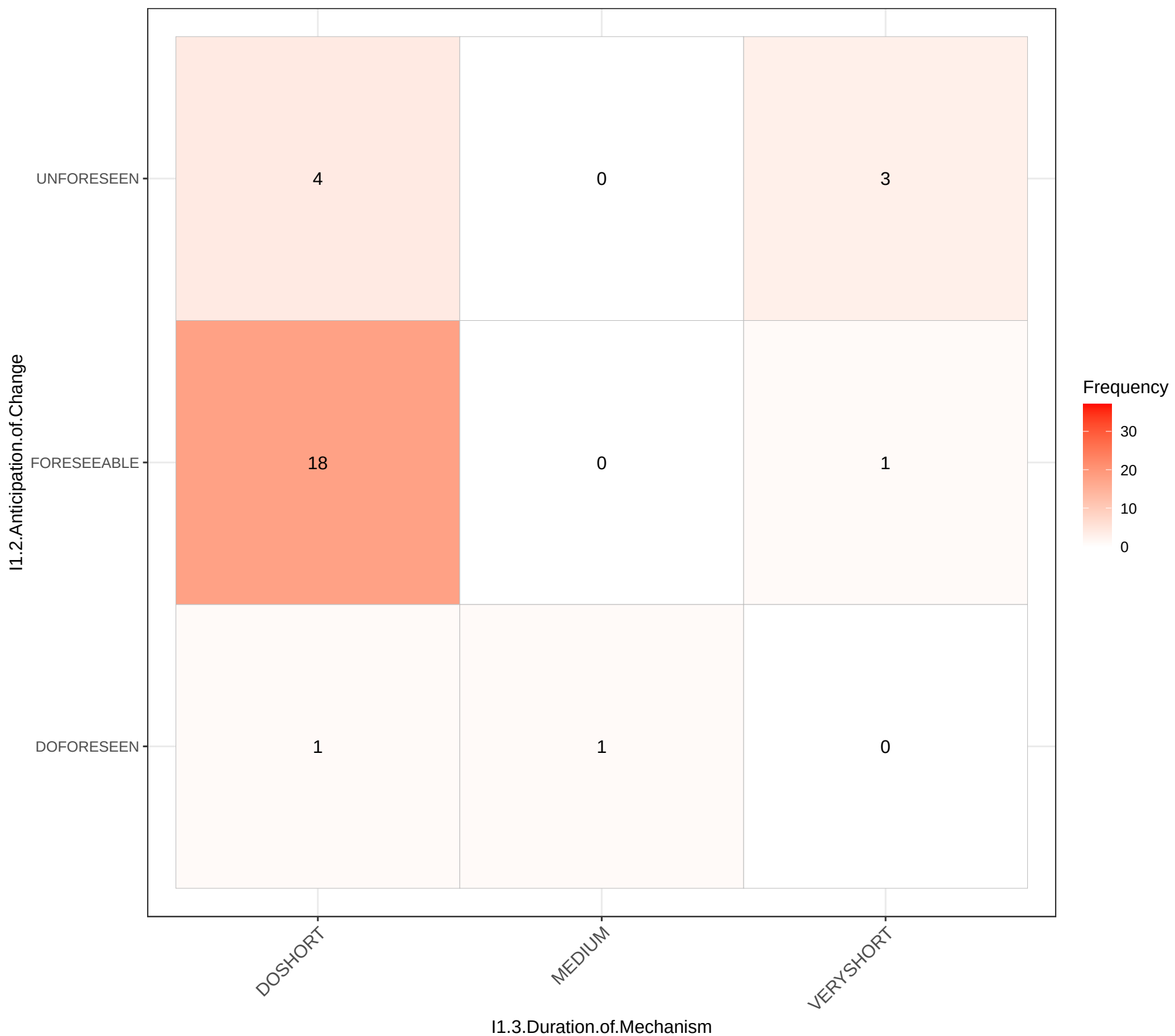


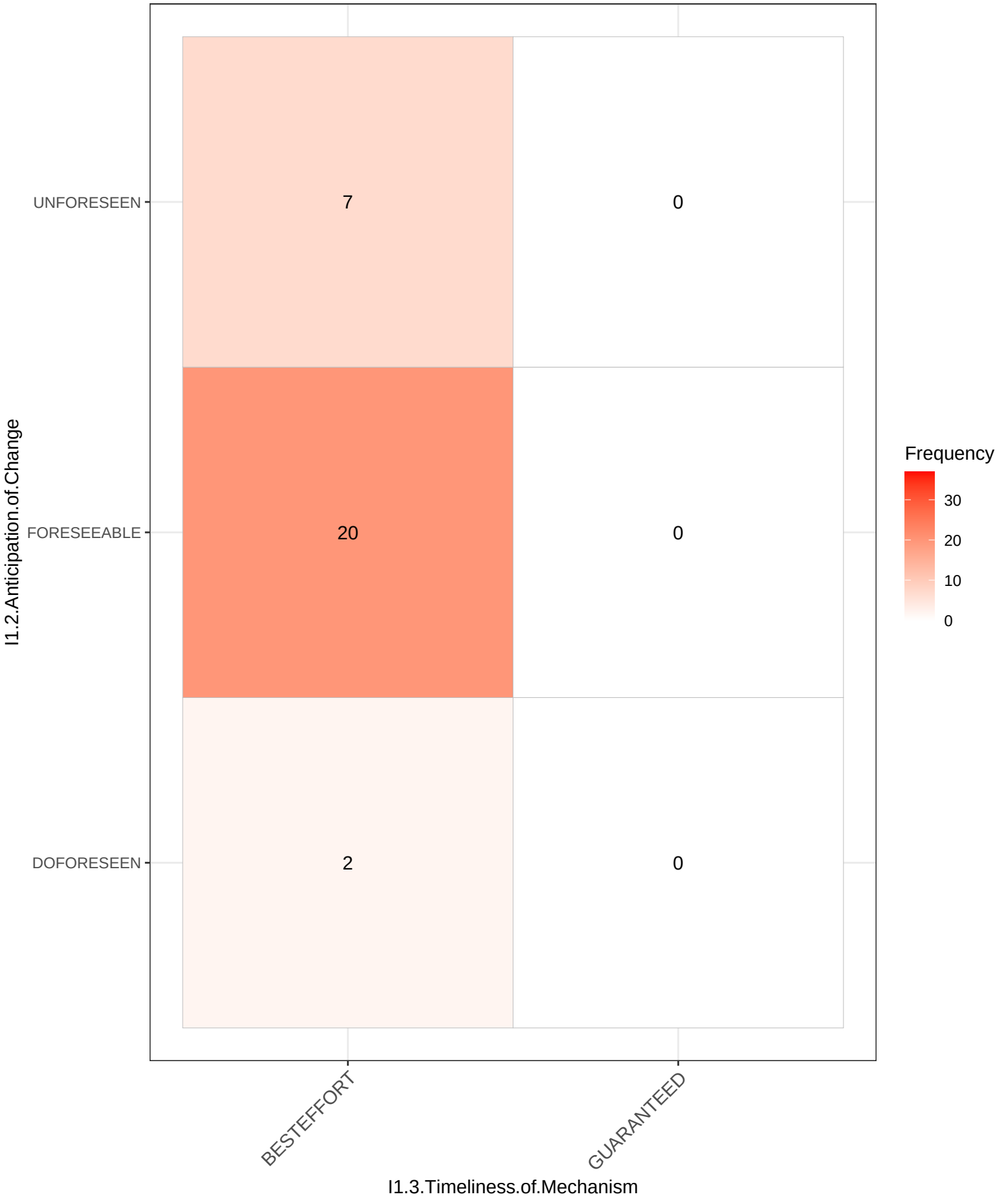


I1.2.Anticipation.of.Change_____I1.3.Scope.of.Mechanism



I1.2.Anticipation.of.Change_____I1.3.Duration.of.Mechanism





I1.2.Anticipation.of.Change_____I1.3.Trigger.of.Mechanism

I1.2.Anticipation.of.Change

UNFORESEEN

7

FORESEEABLE

23

DOFORESEEN

3

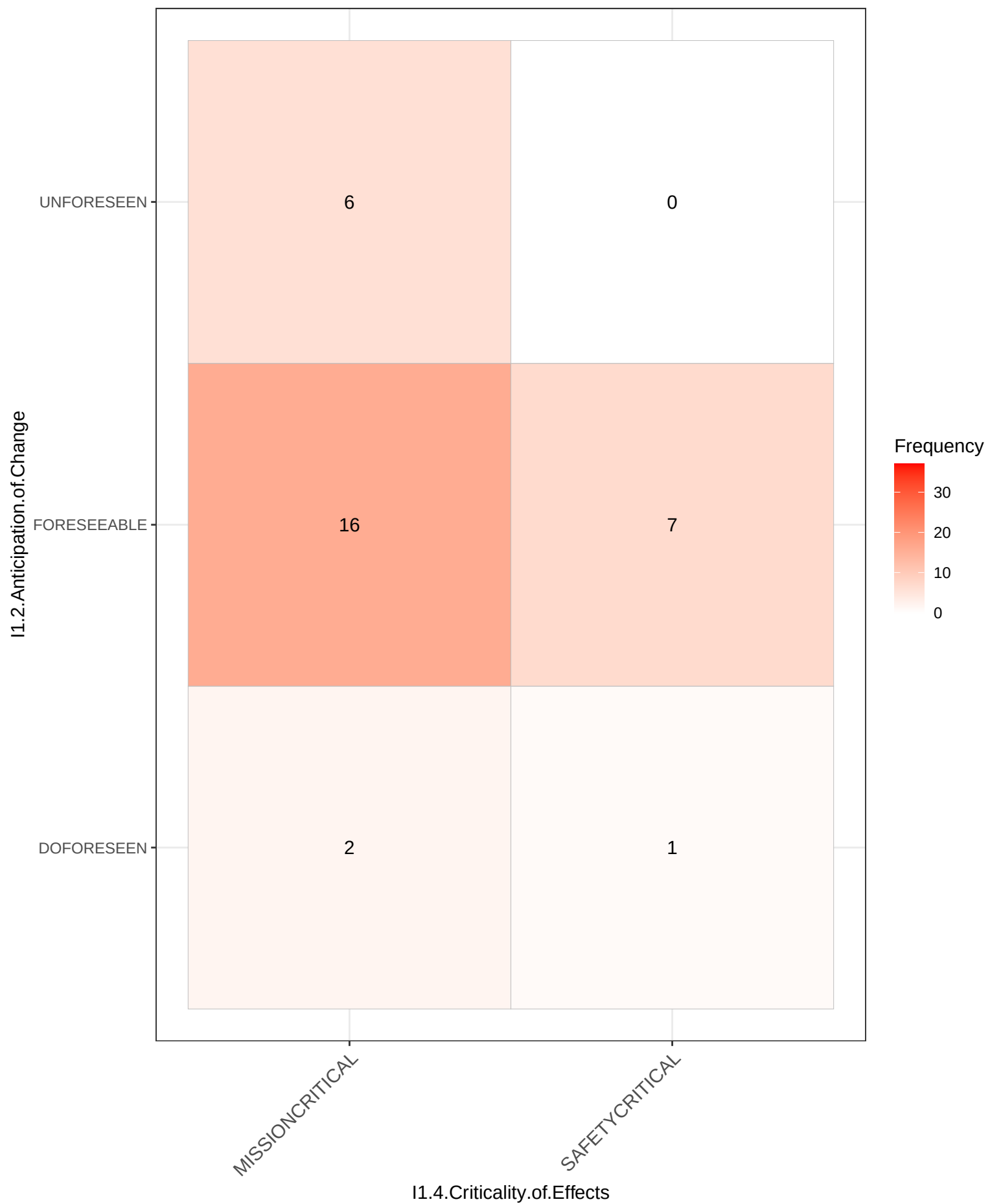
EVENTTRIGGER

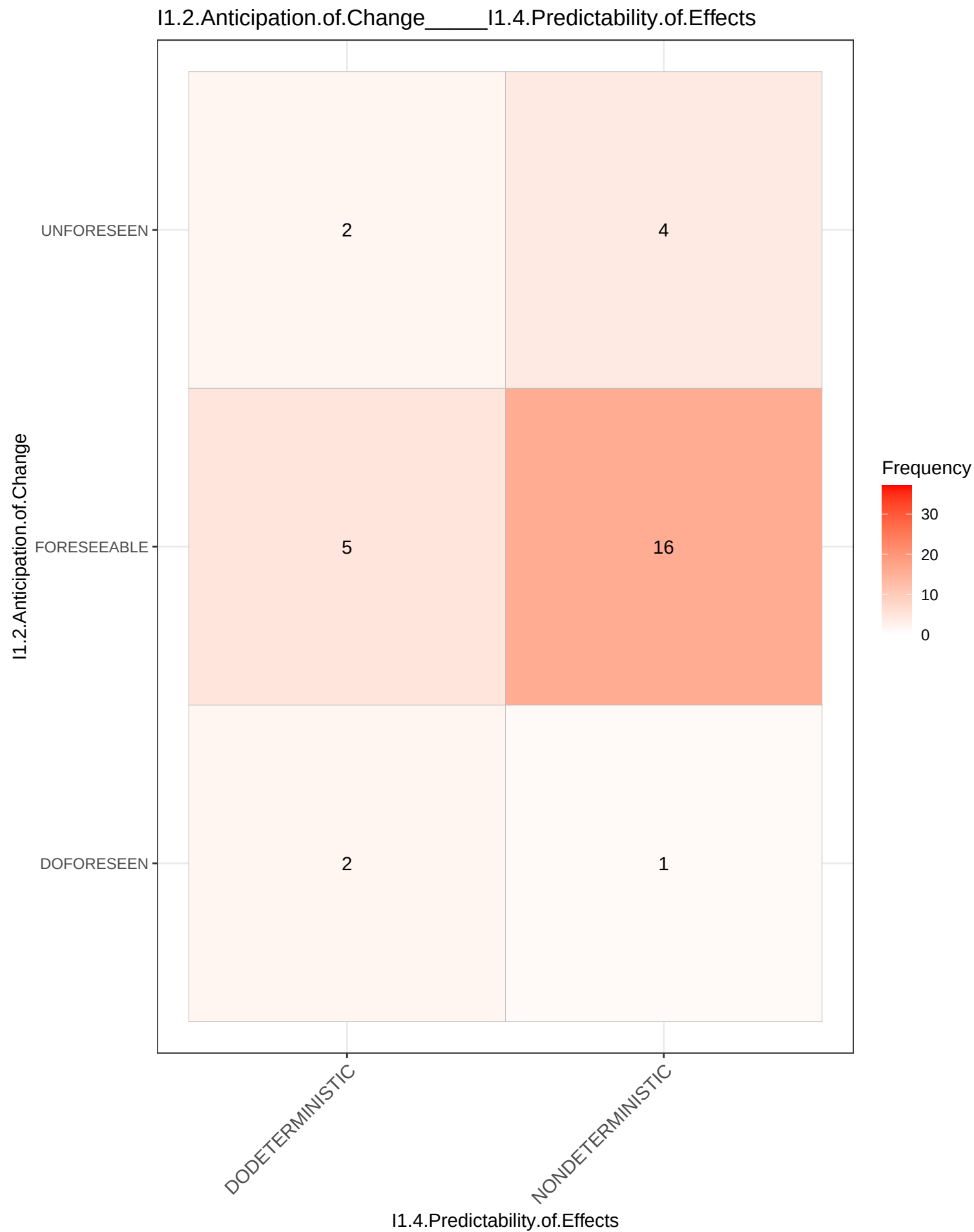
I1.3.Trigger.of.Mechanism

Frequency

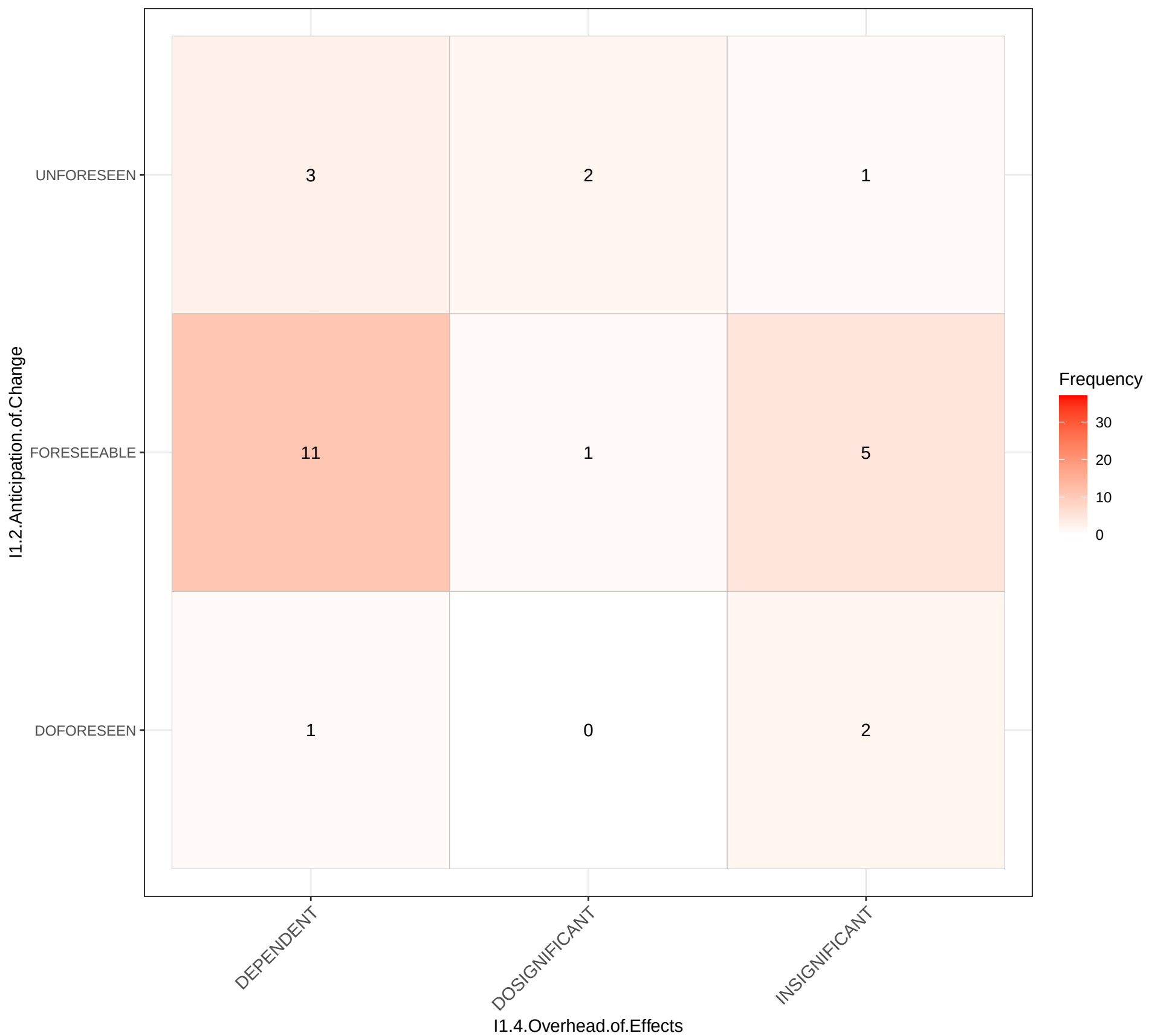
30
20
10
0

I1.2.Anticipation.of.Change_____I1.4.Criticality.of.Effects





I1.2.Anticipation.of.Change_____I1.4.Overhead.of.Effects



I1.2.Anticipation.of.Change_____I1.4.Resilience.of.Effects

I1.2.Anticipation.of.Change

UNFORESEEN

FORESEEABLE

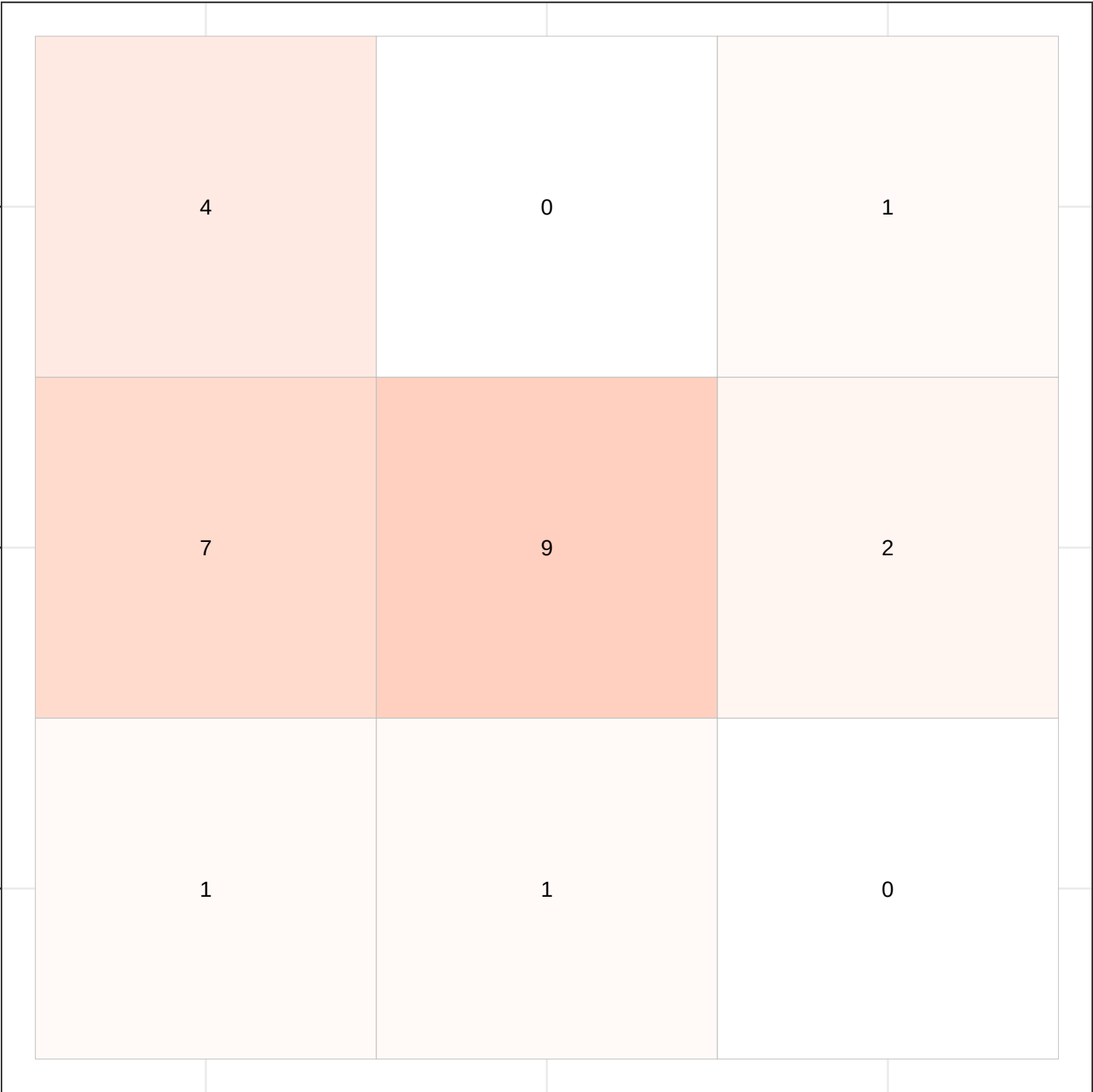
DOFORESEEN

DEPENDENT

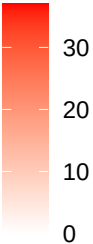
DORESILIENT

IRRESILIENT

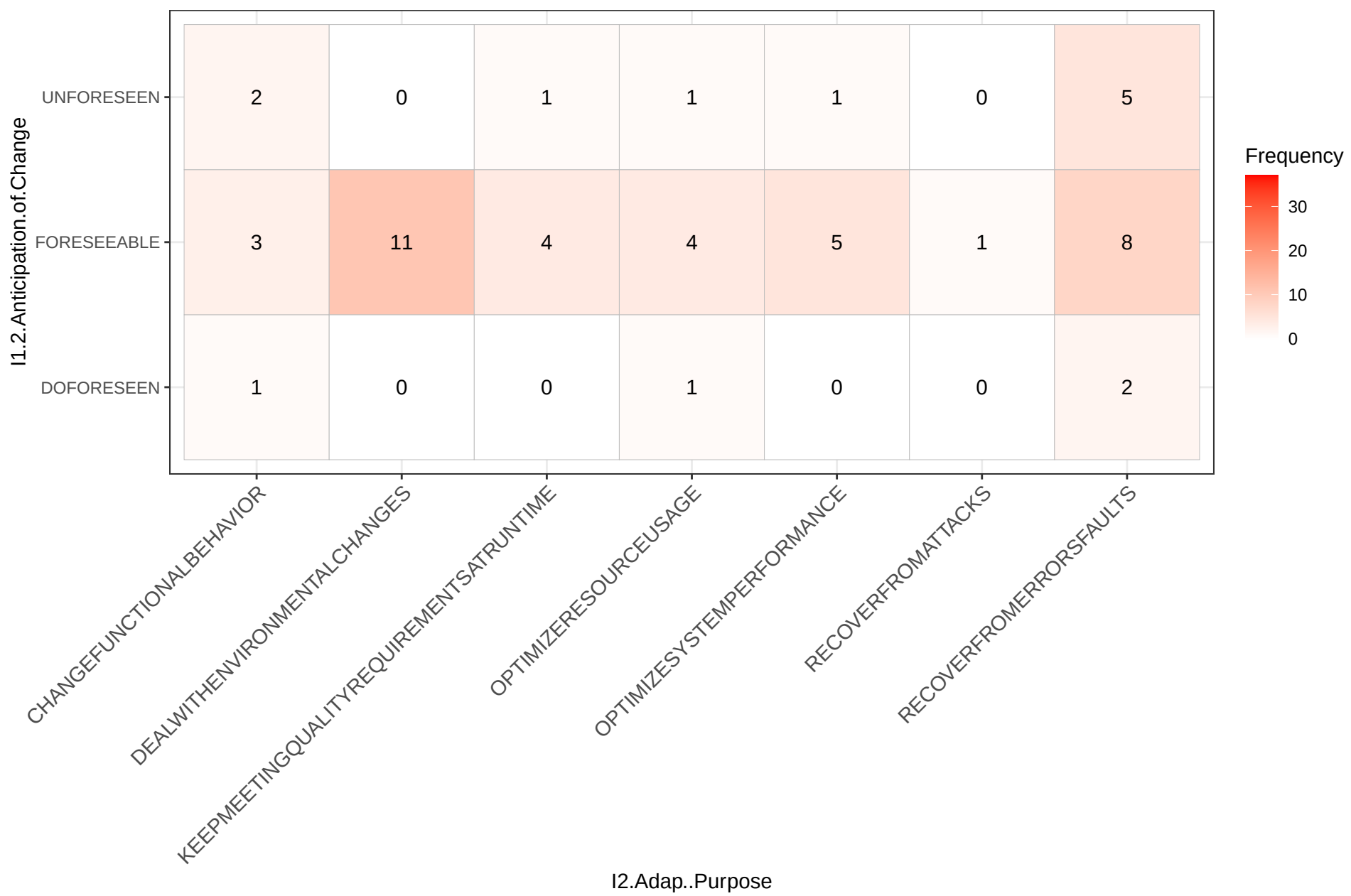
I1.4.Resilience.of.Effects



Frequency



I1.2.Anticipation.of.Change_____I2.Adap..Purpose

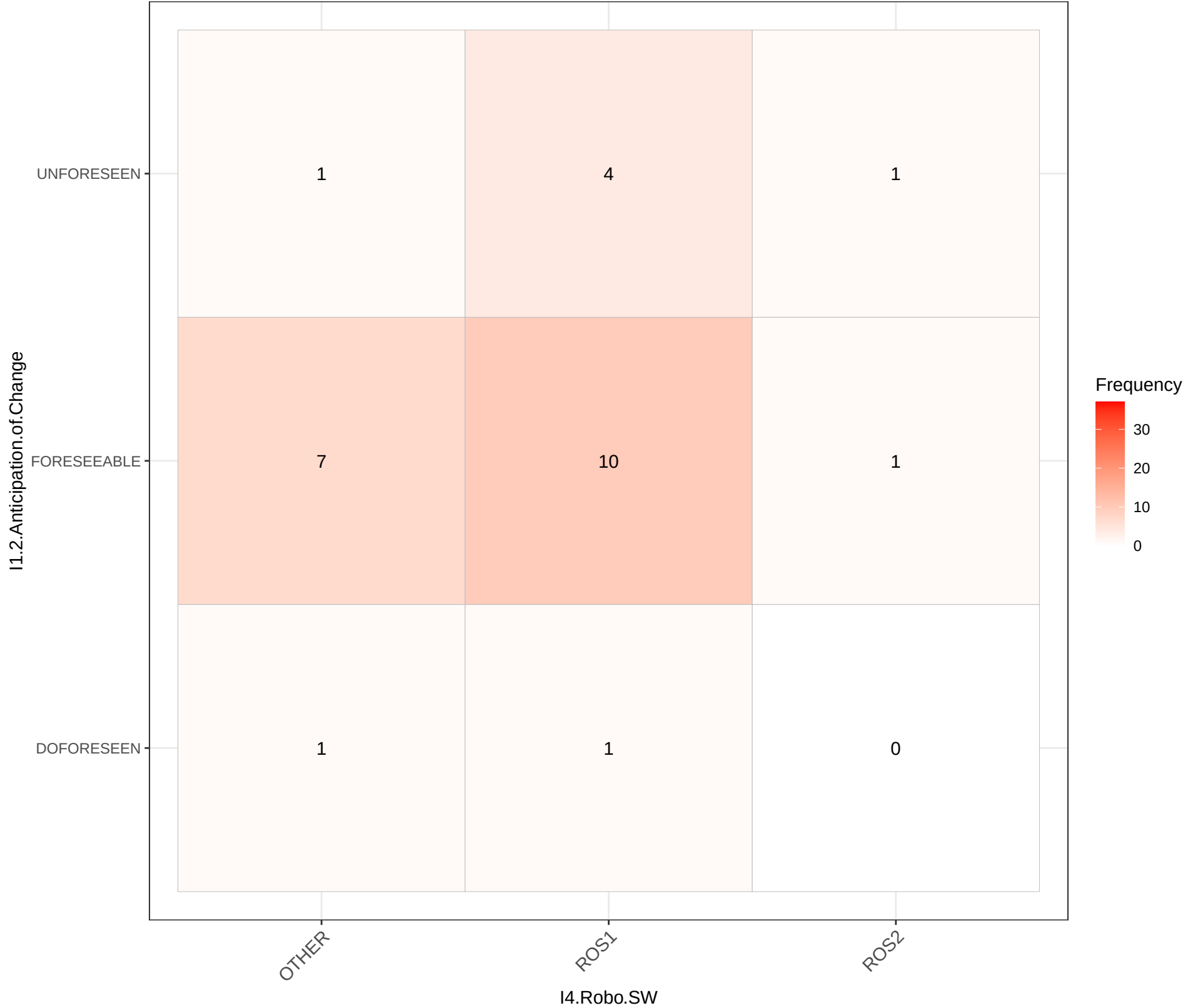


I1.2.Anticipation.of.Change	I3.Robot.Type
-----------------------------	---------------

Frequency

I3.Robot.Type

I1.2.Anticipation.of.Change_____I4.Robo.SW



I1.2.Anticipation.of.Change_____I5.QA

I1.2.Anticipation.of.Change

UNFORESEEN

FORESEEABLE

DOFORESEEN

FUNCTIONALSUITABILITY

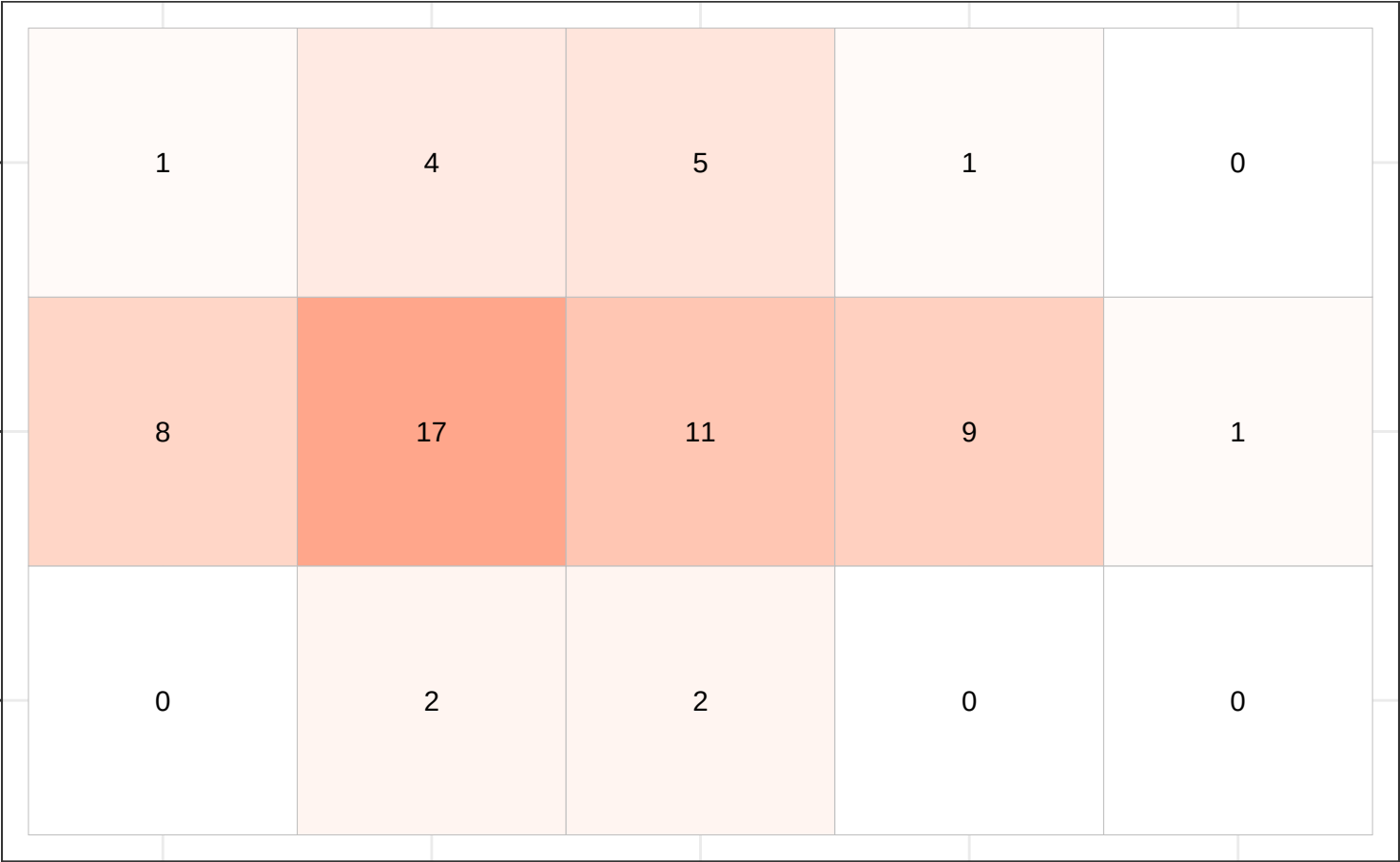
PERFORMANCEEFFICIENCY

RELIABILITY

SAFETY

SECURITY

I5.QA



Frequency



30

20

10

0

I1.2.Anticipation.of.Change_____I6.Independence

I1.2.Anticipation.of.Change

UNFORESEEN

FORESEEABLE

DOFORESEEN

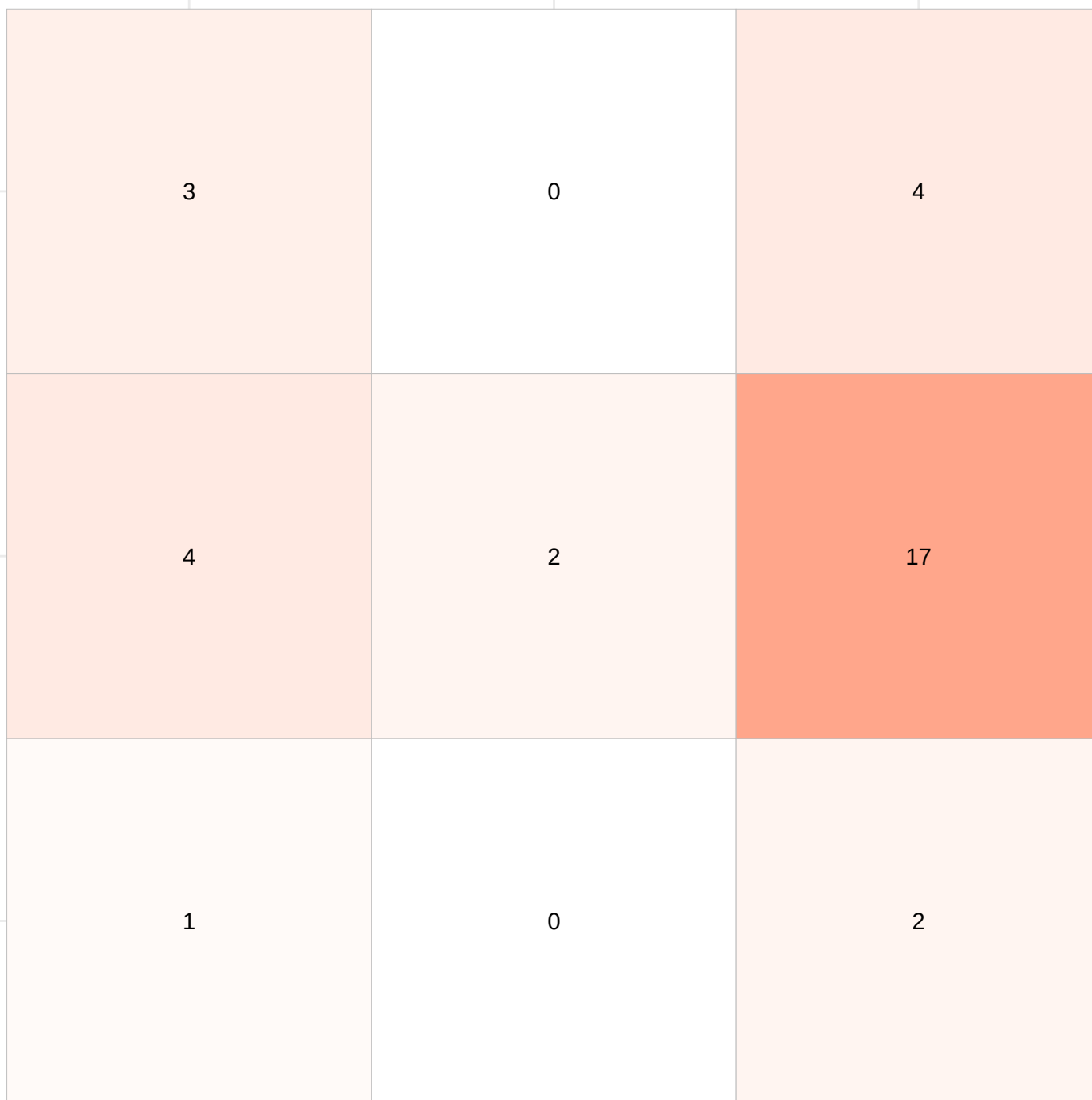
DETACHABLE

INSEPARABLE

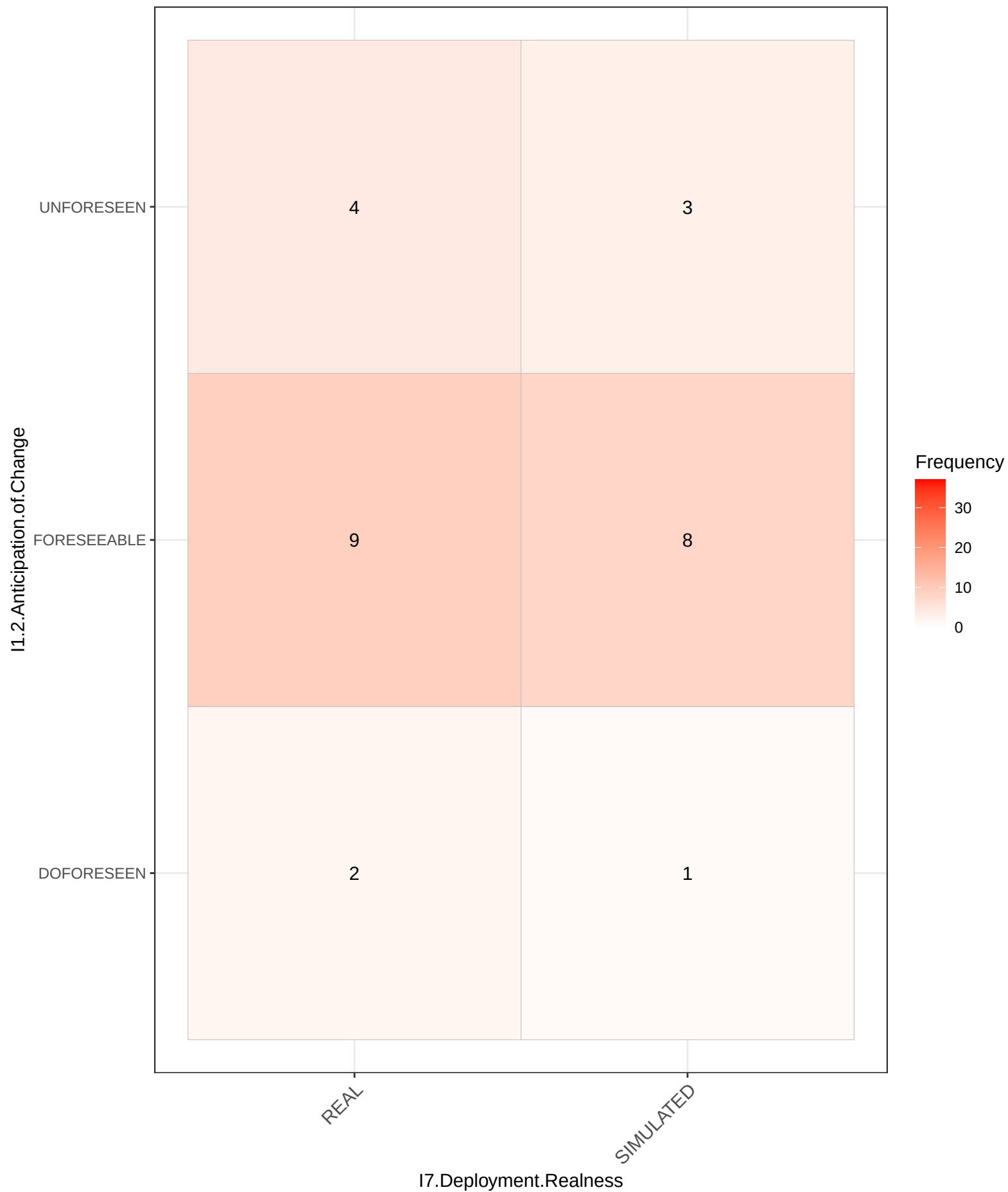
REQUIRESREPRESENTATION

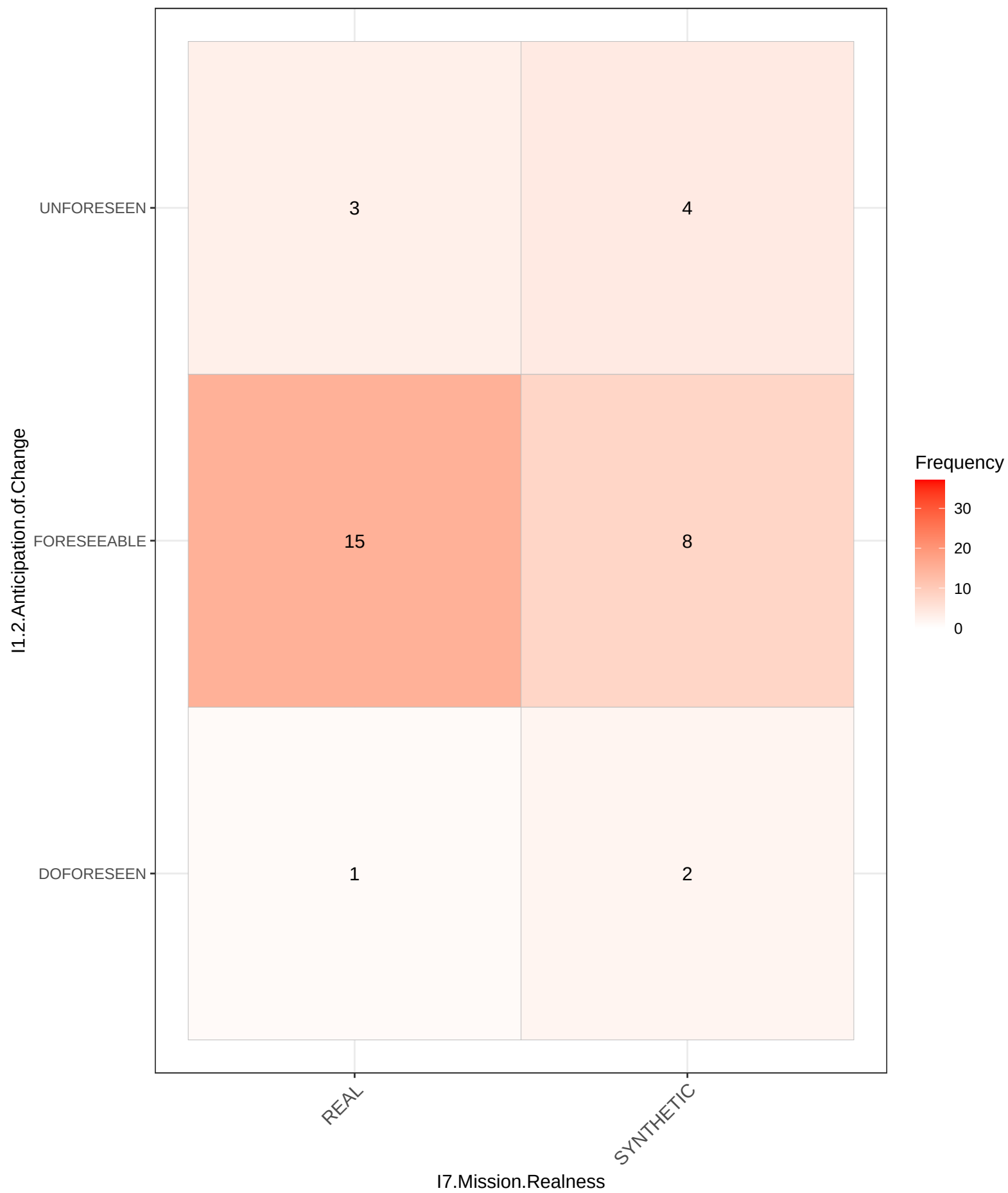
I6.Independence

Frequency

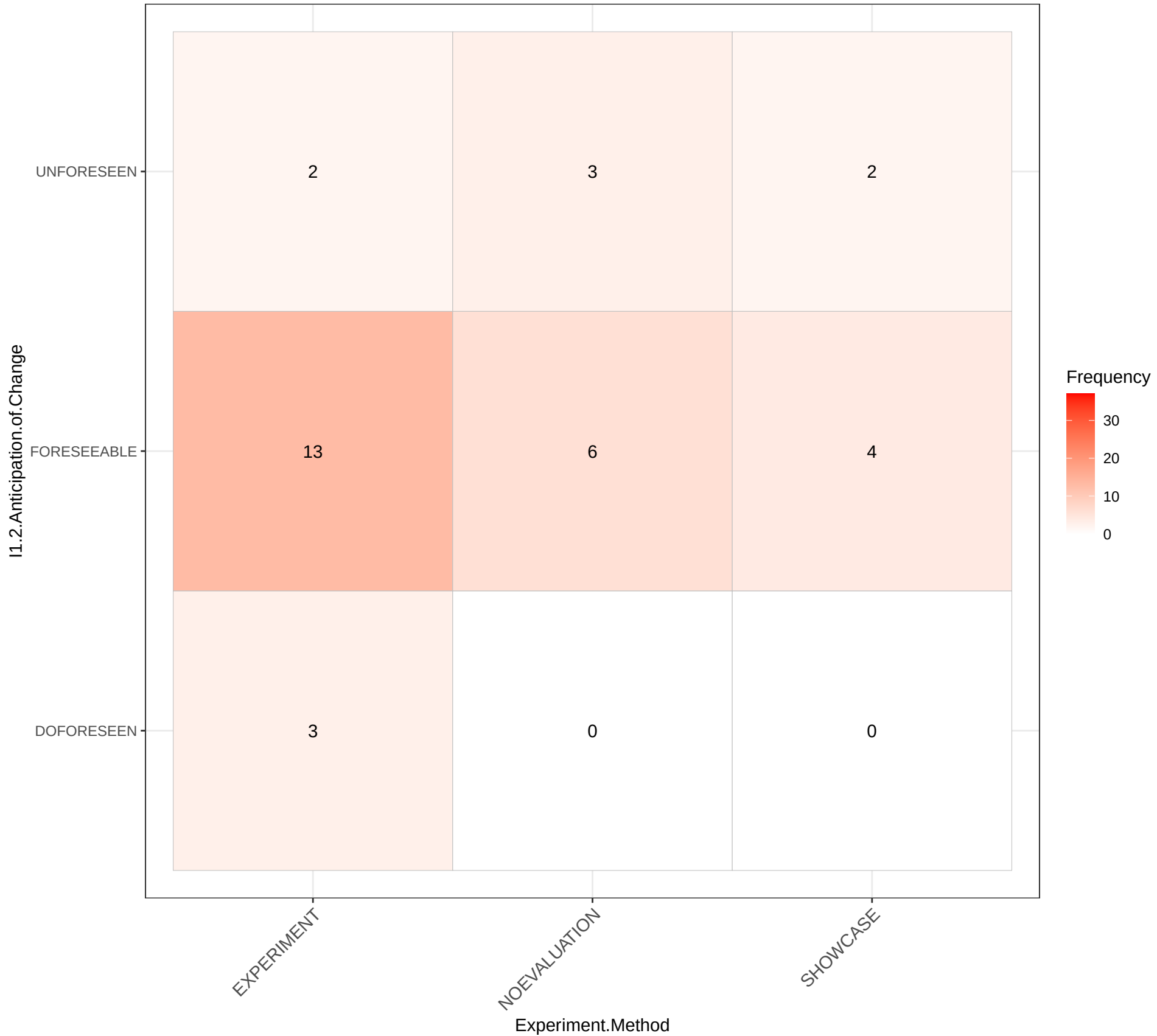


I1.2.Anticipation.of.Change_____I7.Deployment.Realness

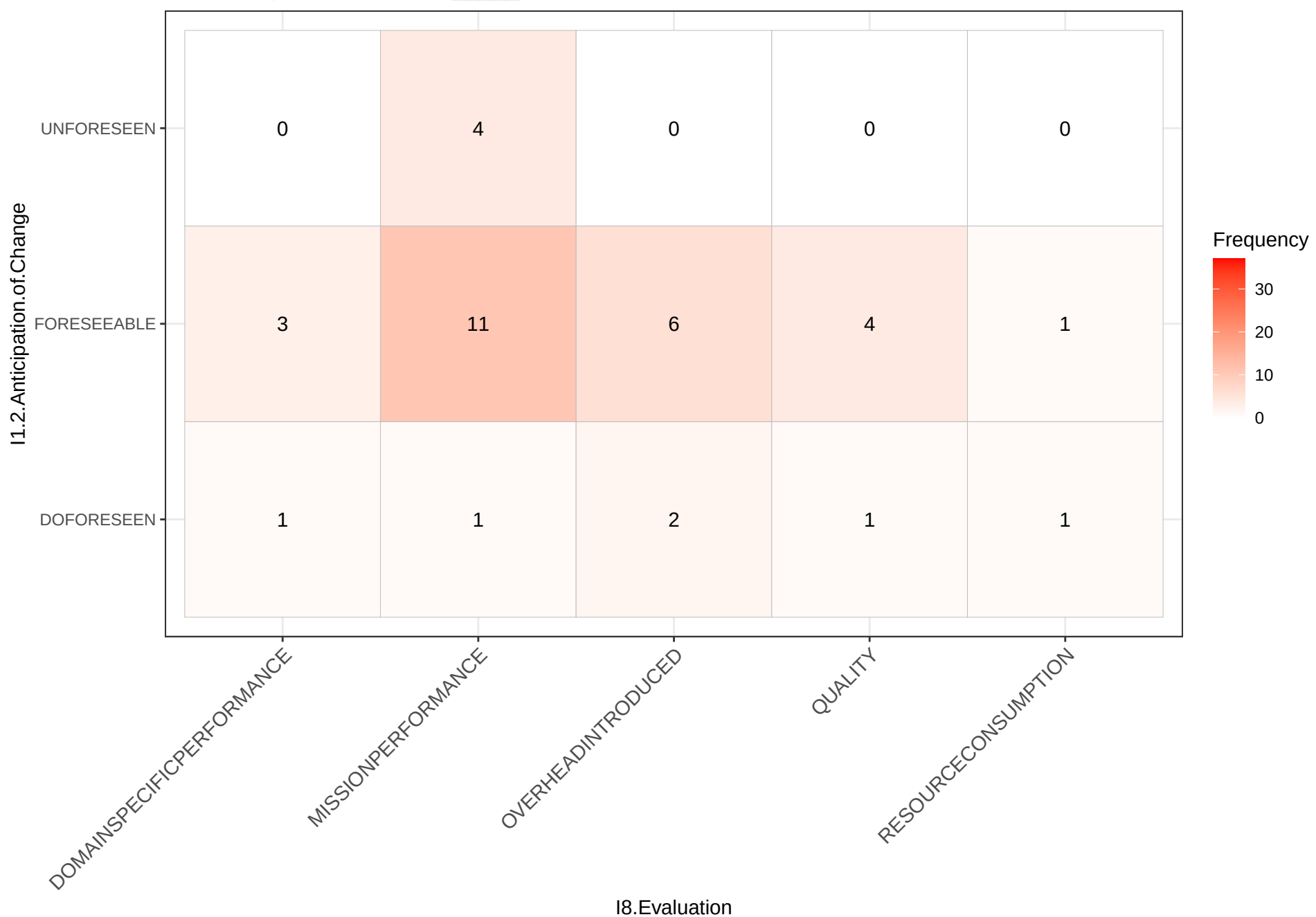


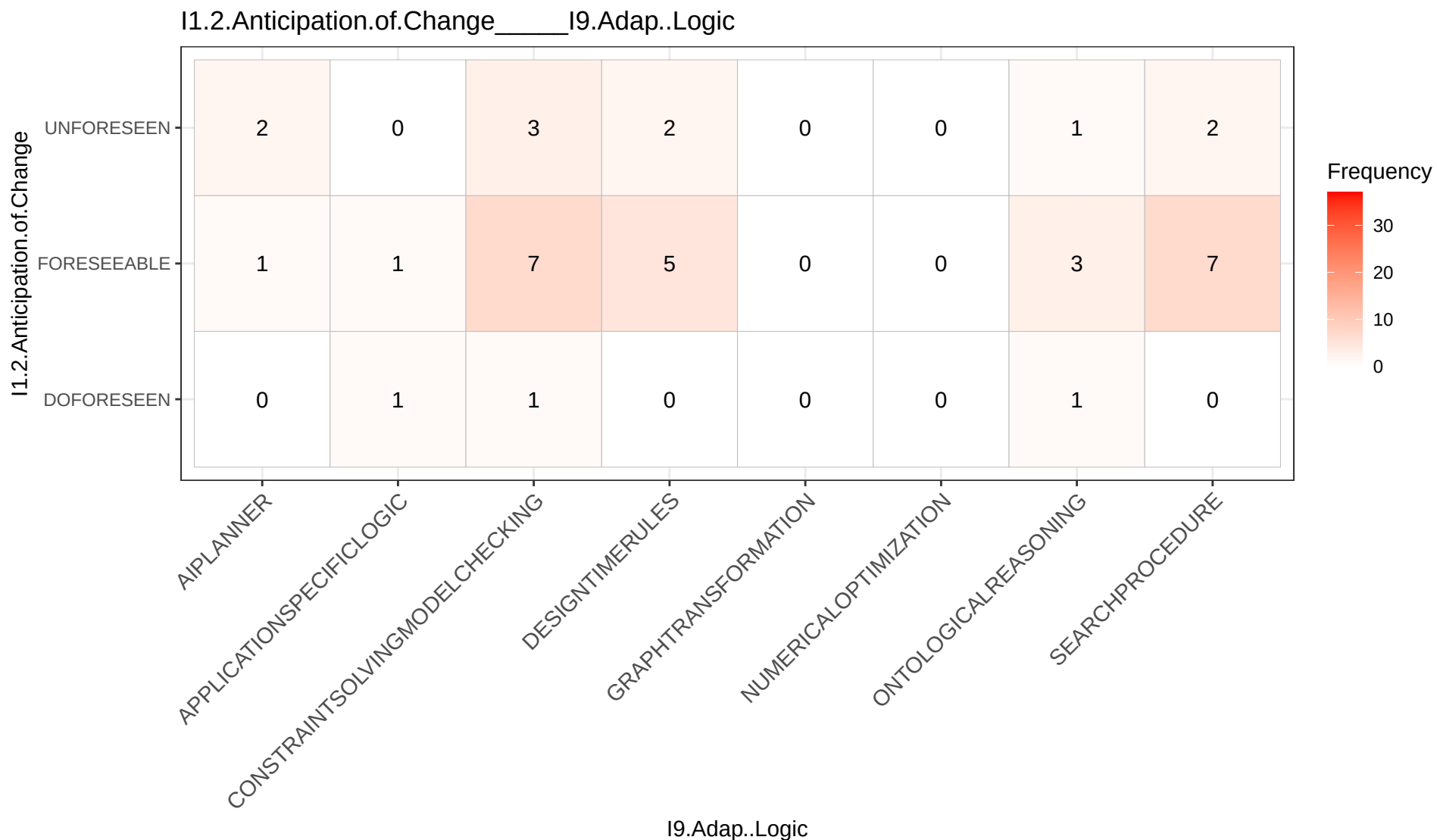


I1.2.Anticipation.of.Change_____Experiment.Method

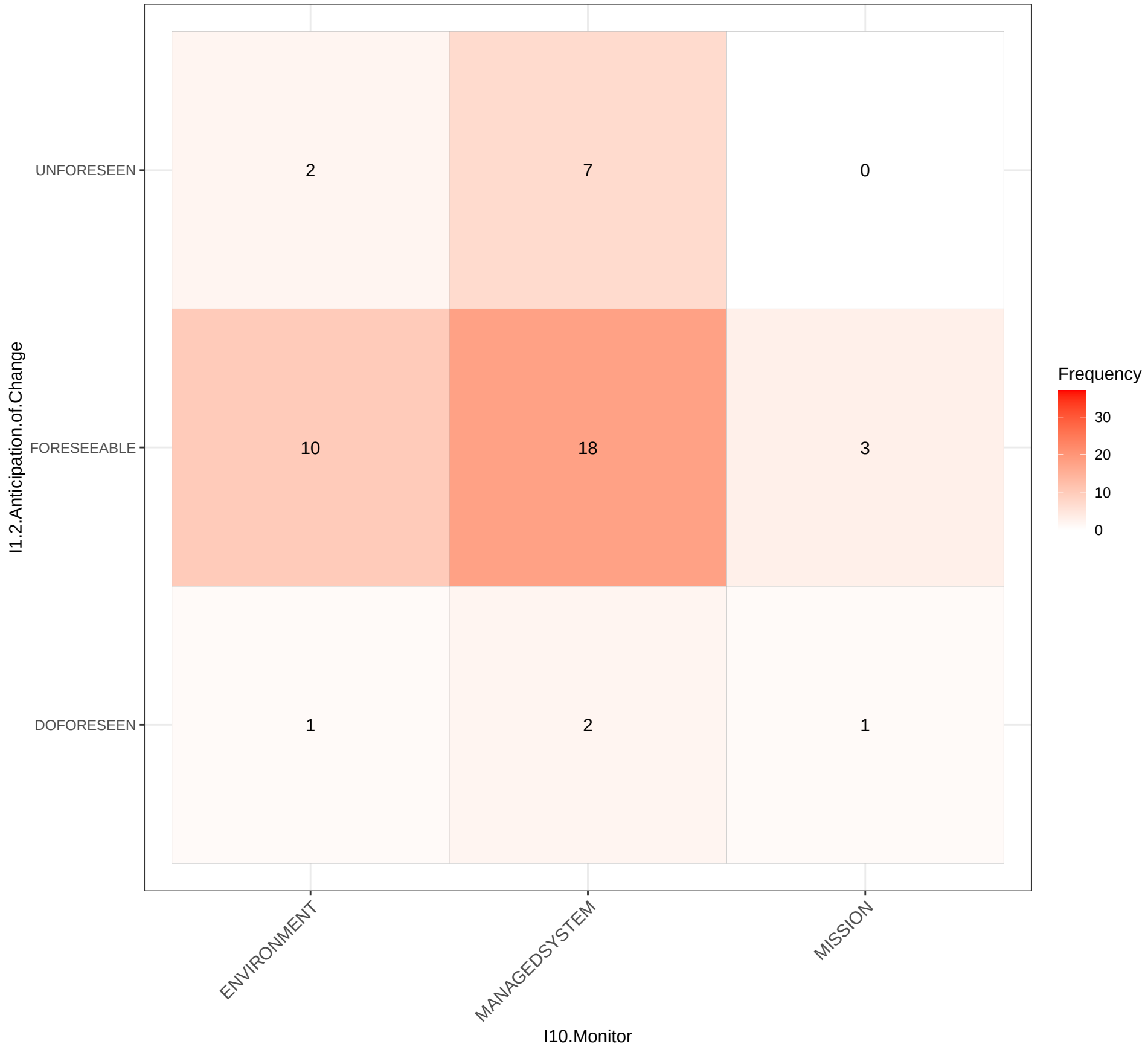


I1.2.Anticipation.of.Change_____I8.Evaluation

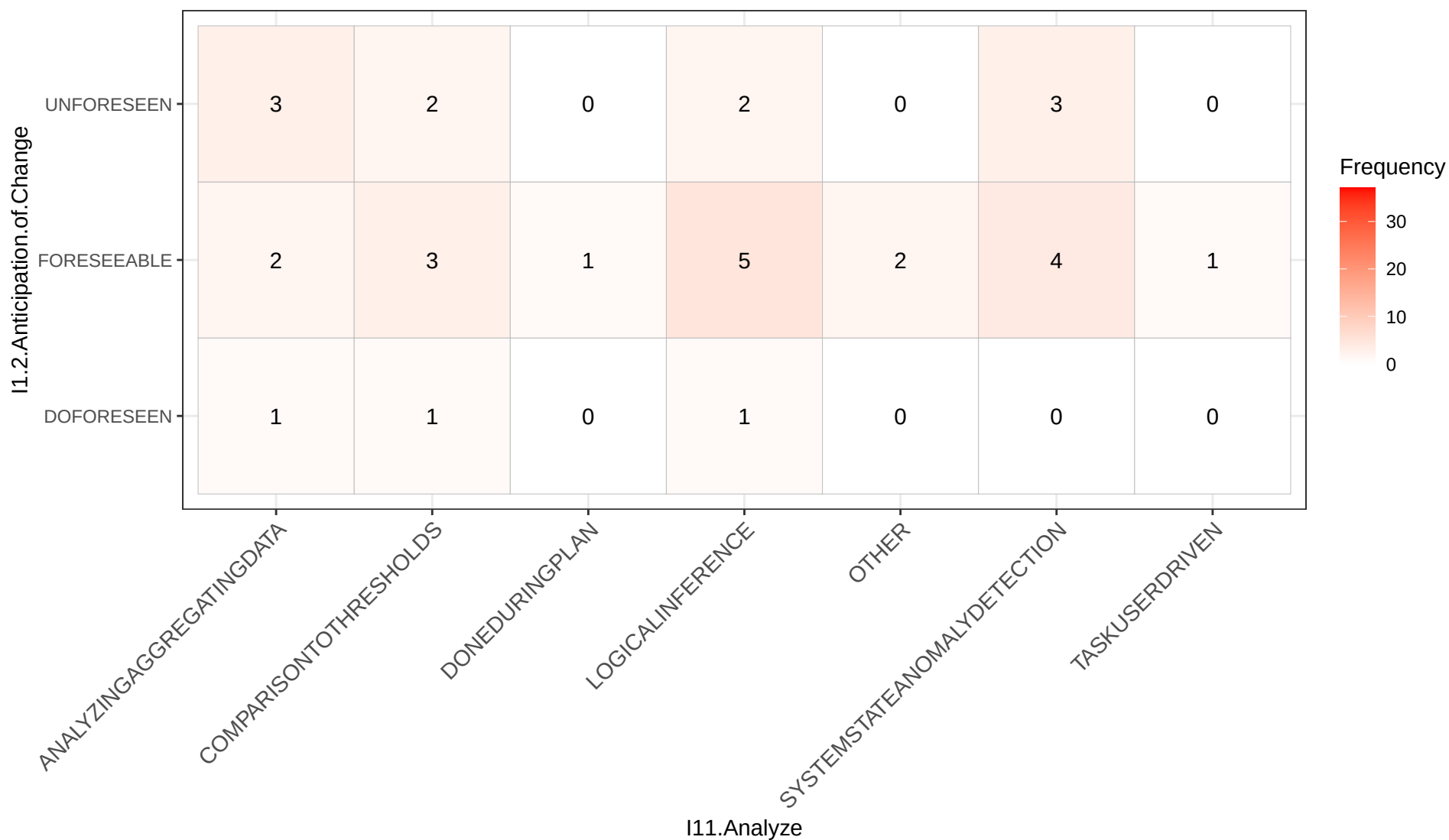




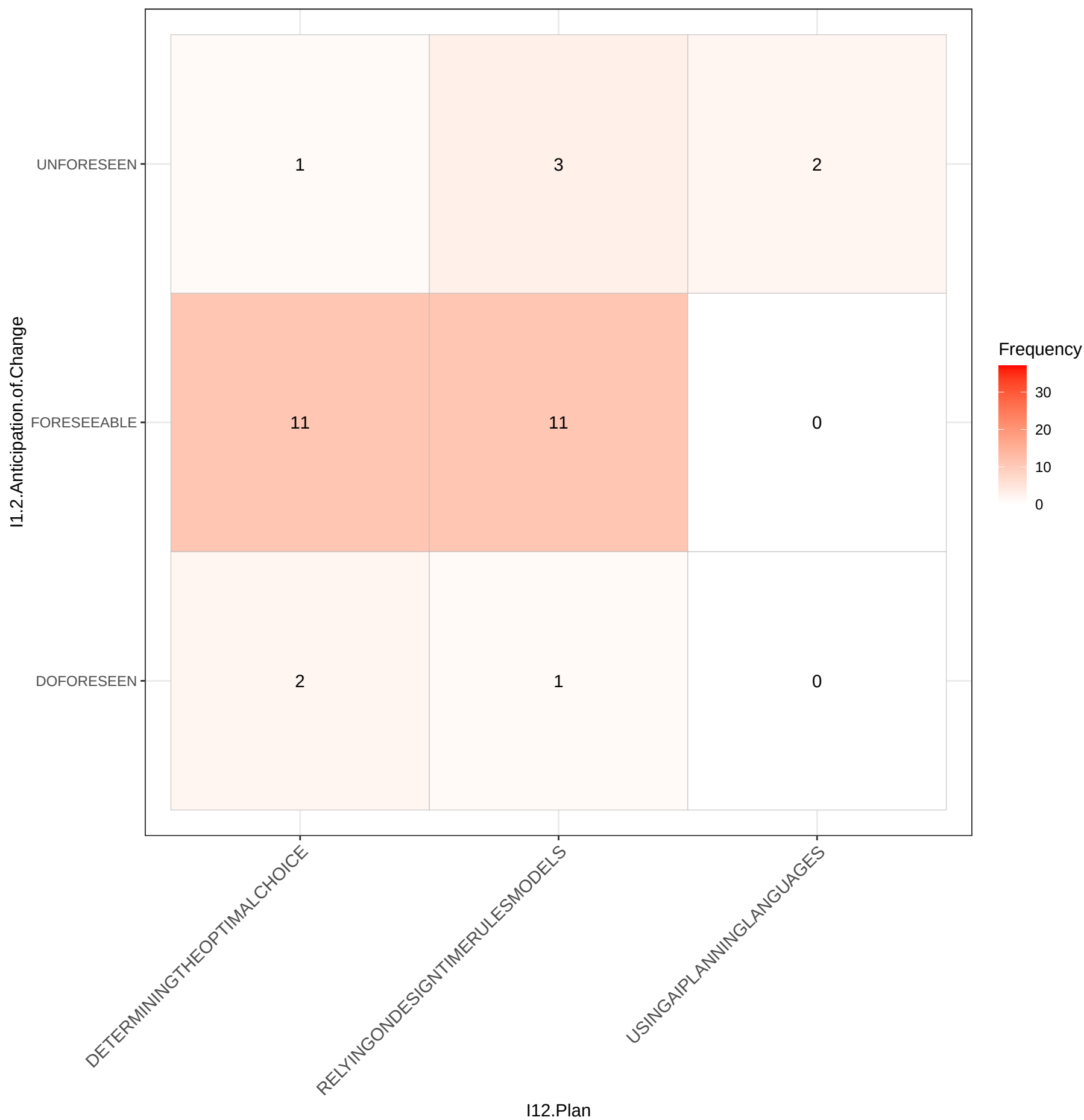
I1.2.Anticipation.of.Change_____I10.Monitor



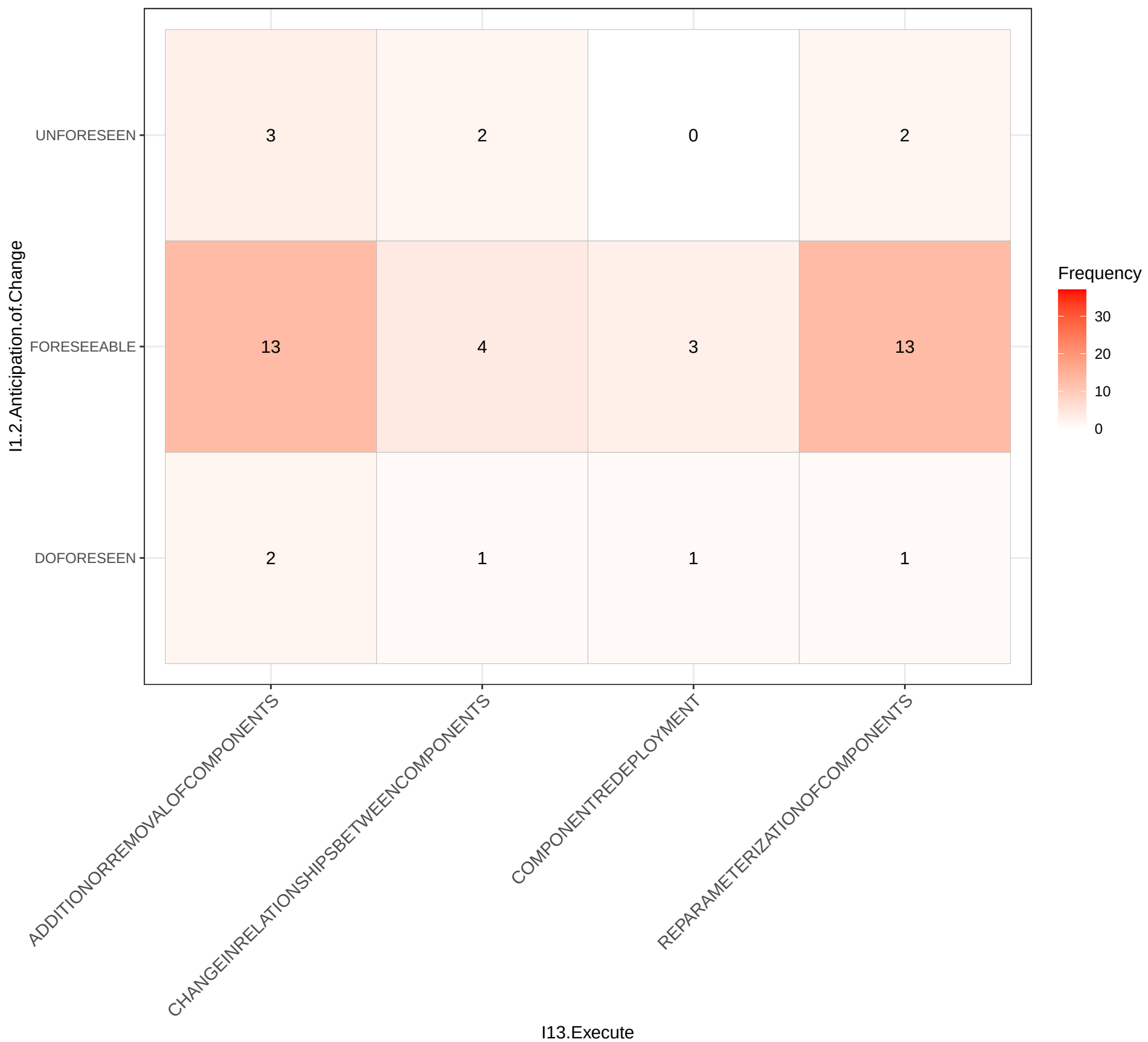
I1.2.Anticipation.of.Change_____I11.Analyze

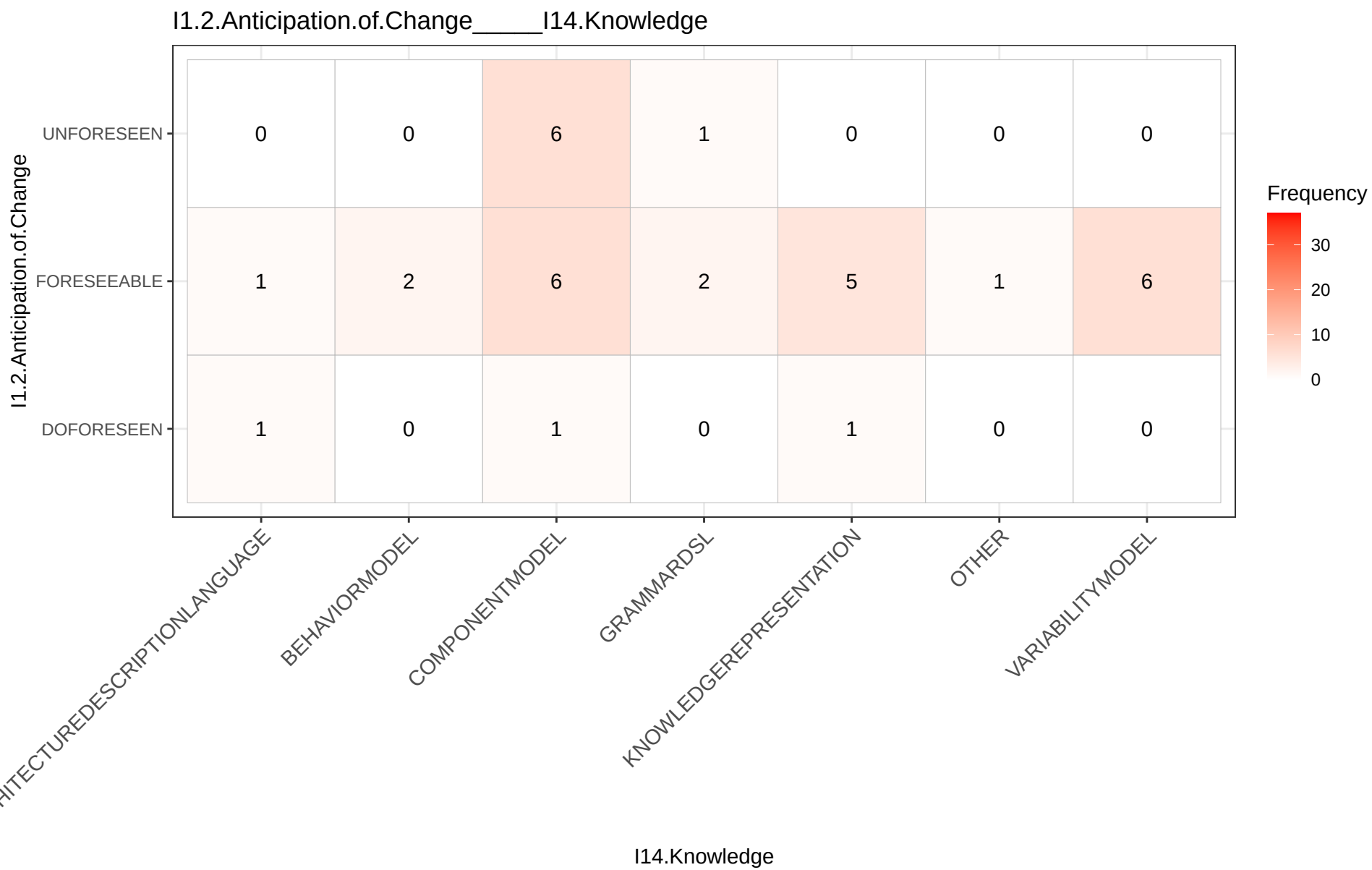


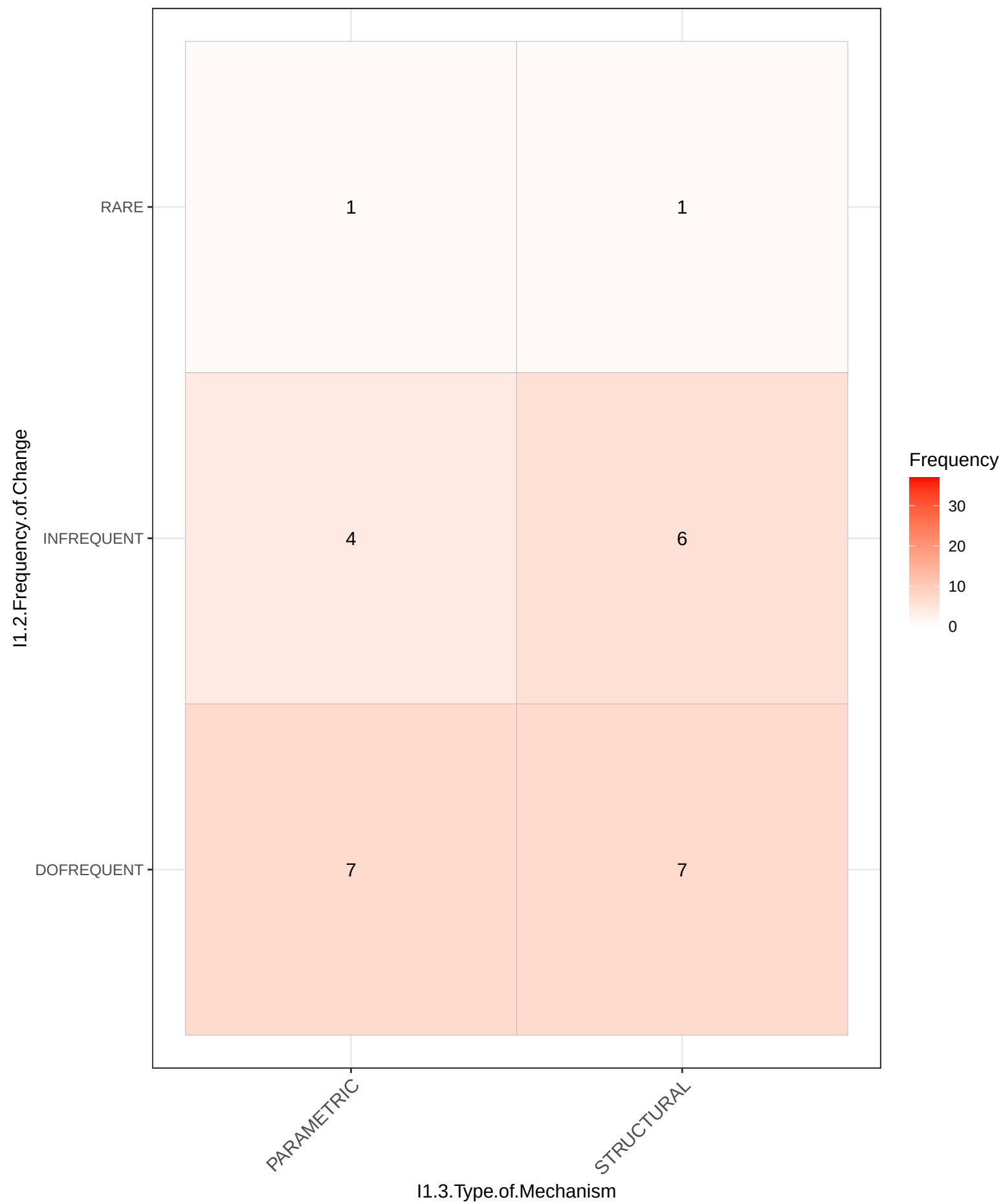
I1.2.Anticipation.of.Change_____I12.Plan



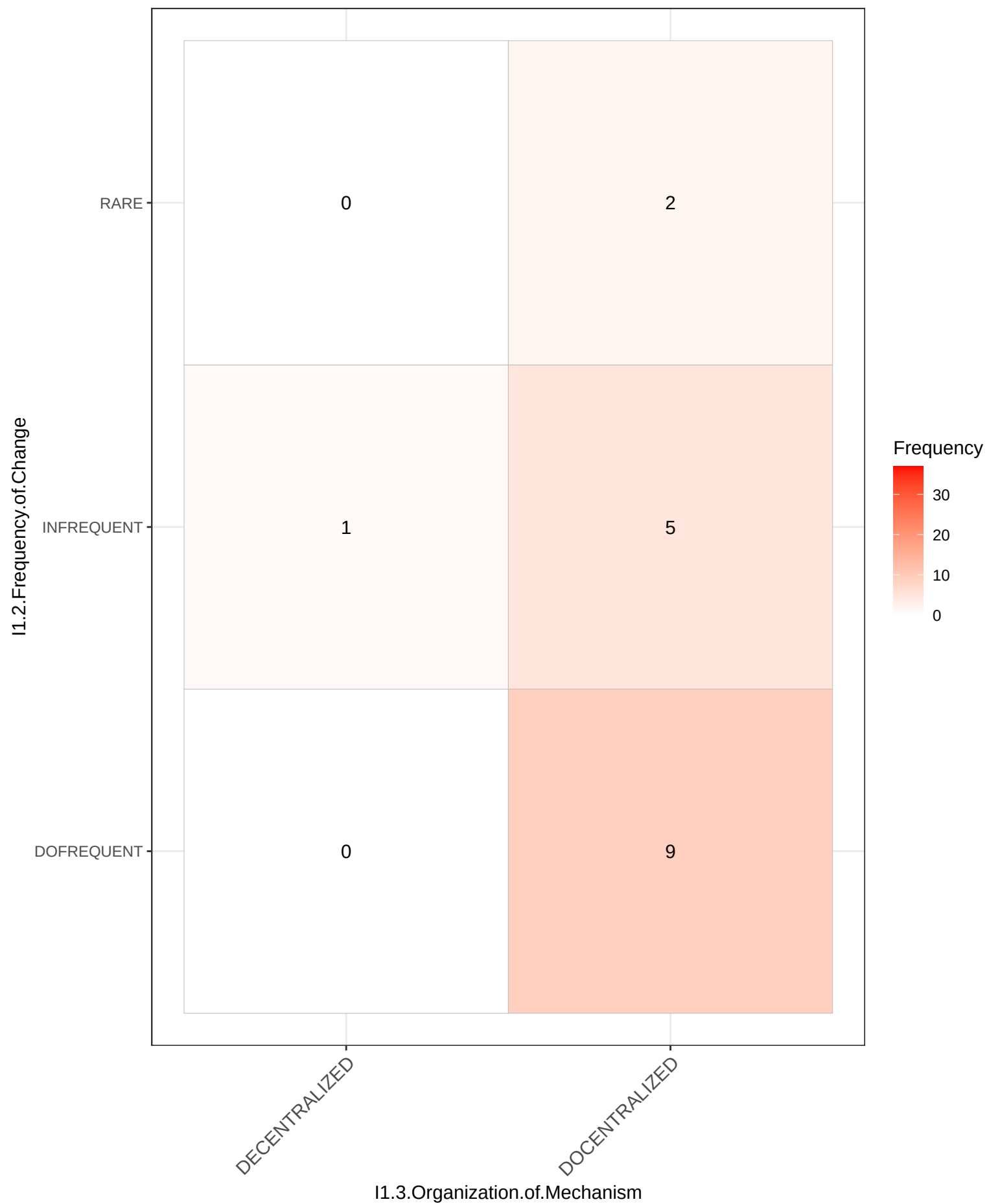
I1.2.Anticipation.of.Change_____I13.Execute







I1.2.Frequency.of.Change_____I1.3.Organization.of.Mechanism



I1.2.Frequency.of.Change_____I1.3.Scope.of.Mechanism

I1.2.Frequency.of.Change

RARE

INFREQUENT

DOFREQUENT

GLOBAL

LOCAL

I1.3.Scope.of.Mechanism

Frequency



0

2

0

5

5

7

I1.2.Frequency.of.Change_____I1.3.Duration.of.Mechanism

I1.2.Frequency.of.Change

RARE

1

1

0

INFREQUENT

4

0

0

DOFREQUENT

7

0

1

DOSHORT

MEDIUM

VERYSHORT

I1.3.Duration.of.Mechanism

Frequency



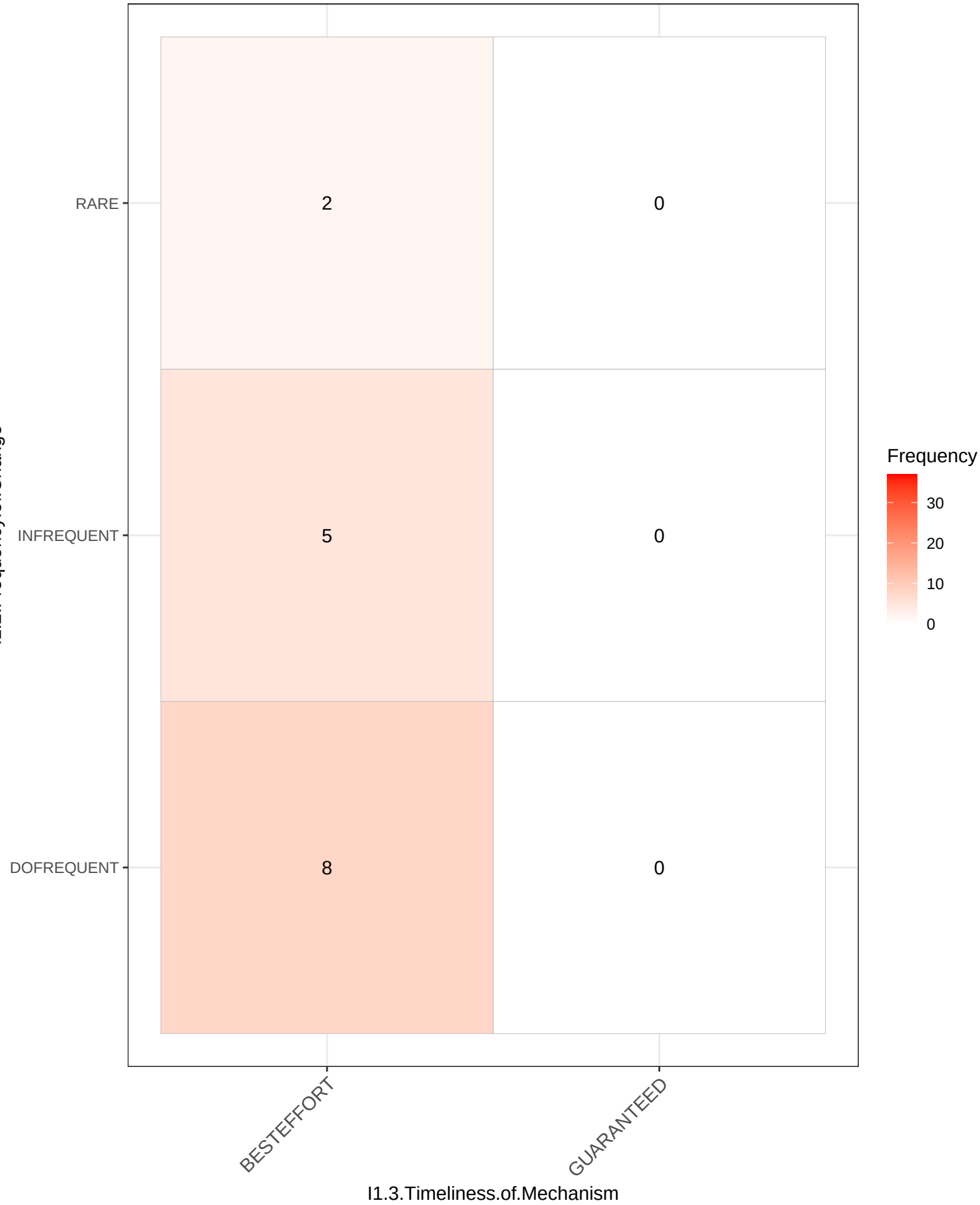
30

20

10

0

I1.2.Frequency.of.Change



I1.2.Frequency.of.Change_____I1.3.Trigger.of.Mechanism

I1.2.Frequency.of.Change

RARE

2

INFREQUENT

6

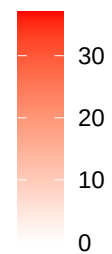
DOFREQUENT

9

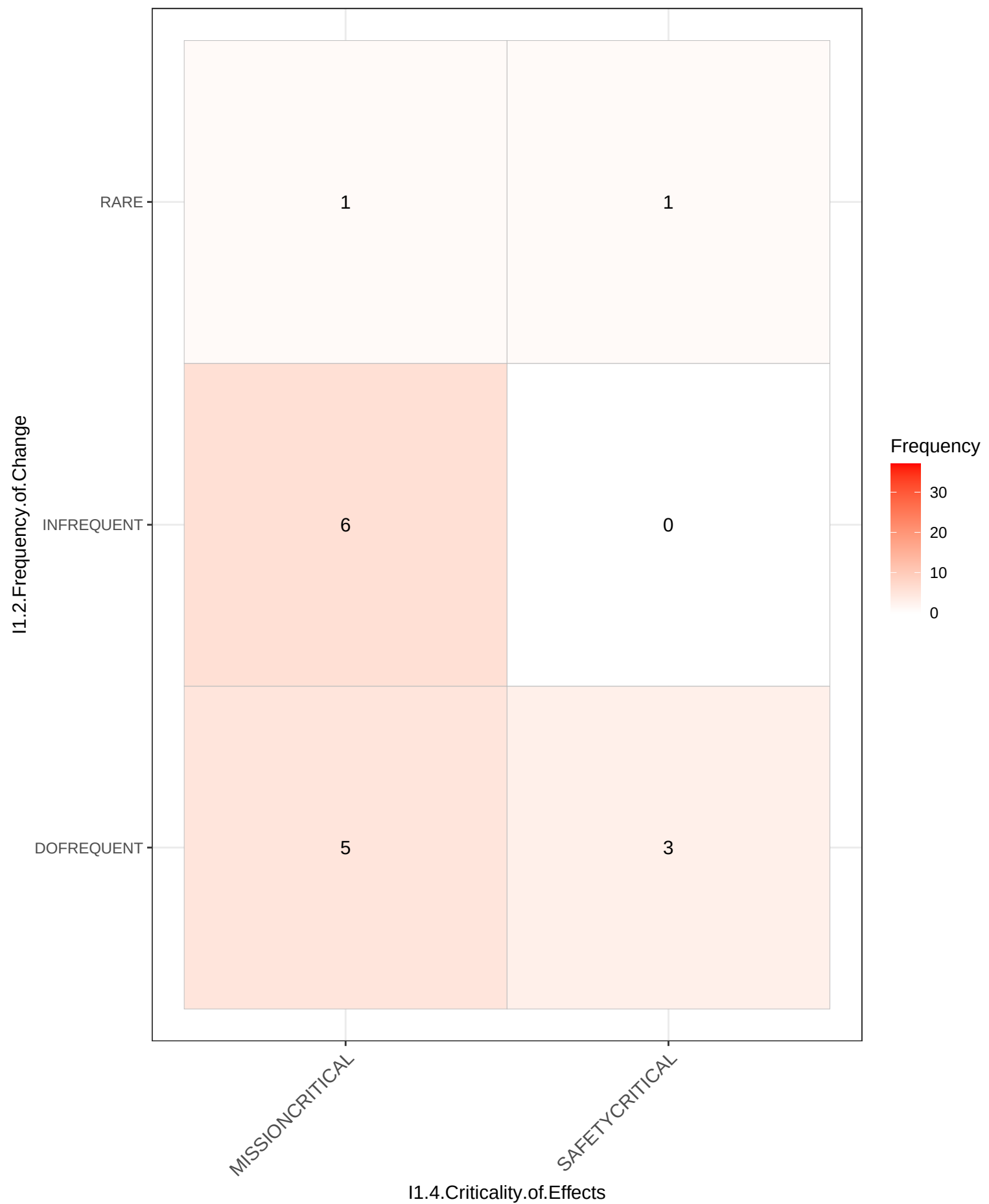
EVENTTRIGGER

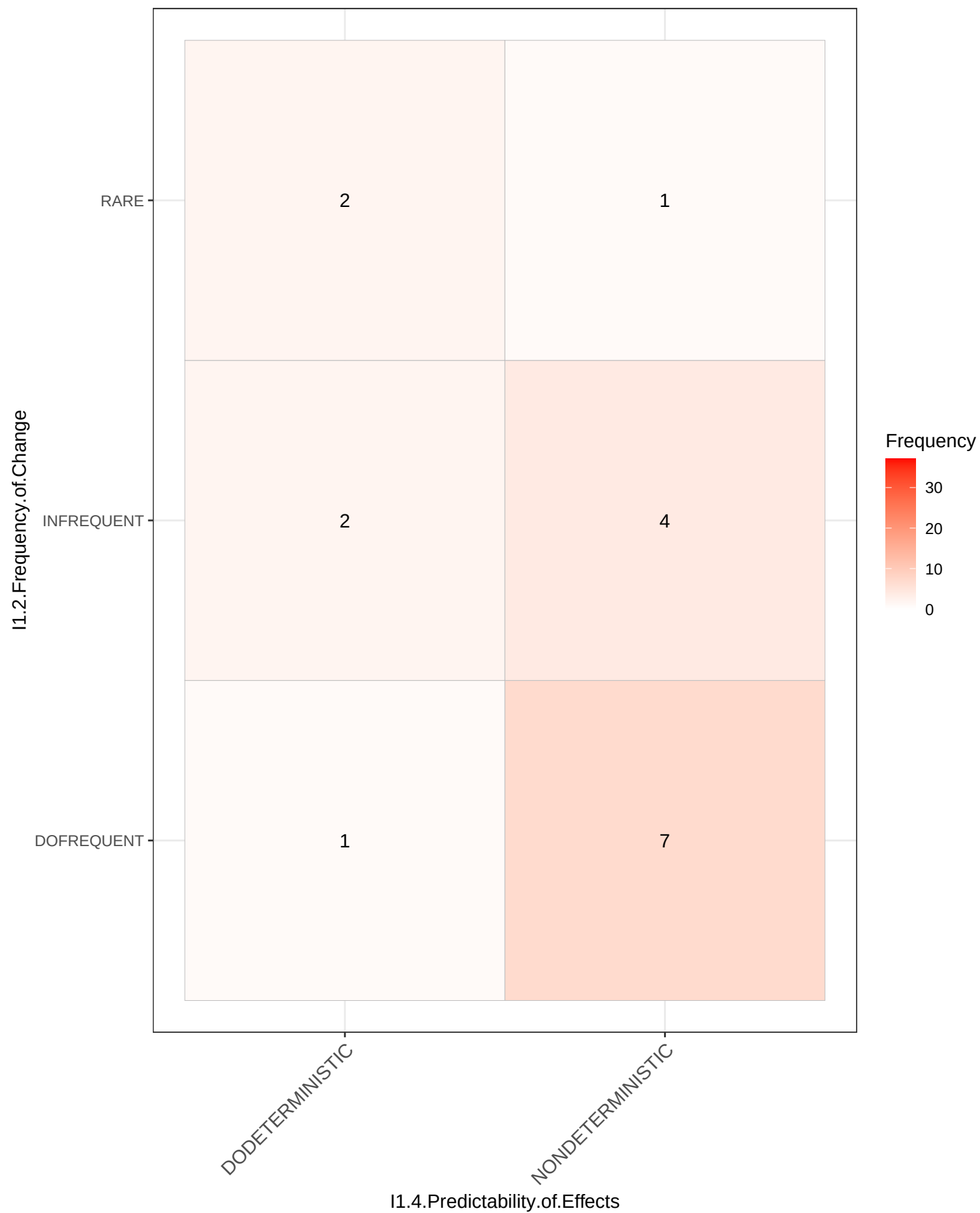
I1.3.Trigger.of.Mechanism

Frequency



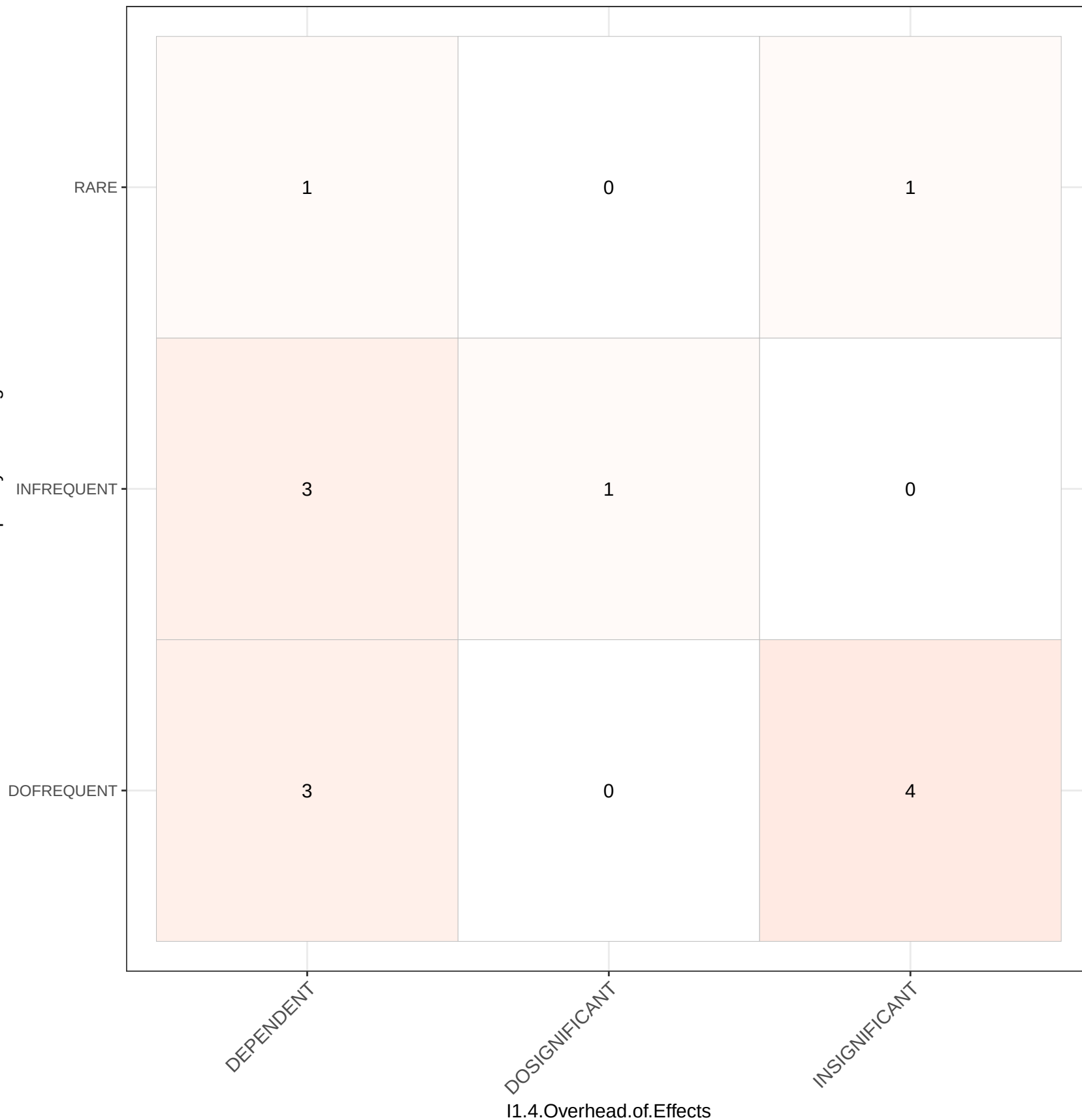
I1.2.Frequency.of.Change_____I1.4.Criticality.of.Effects





I1.2.Frequency.of.Change_____I1.4.Overhead.of.Effects

I1.2.Frequency.of.Change

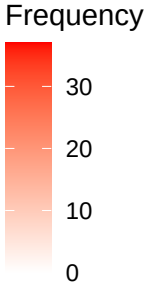


Frequency

30
20
10
0

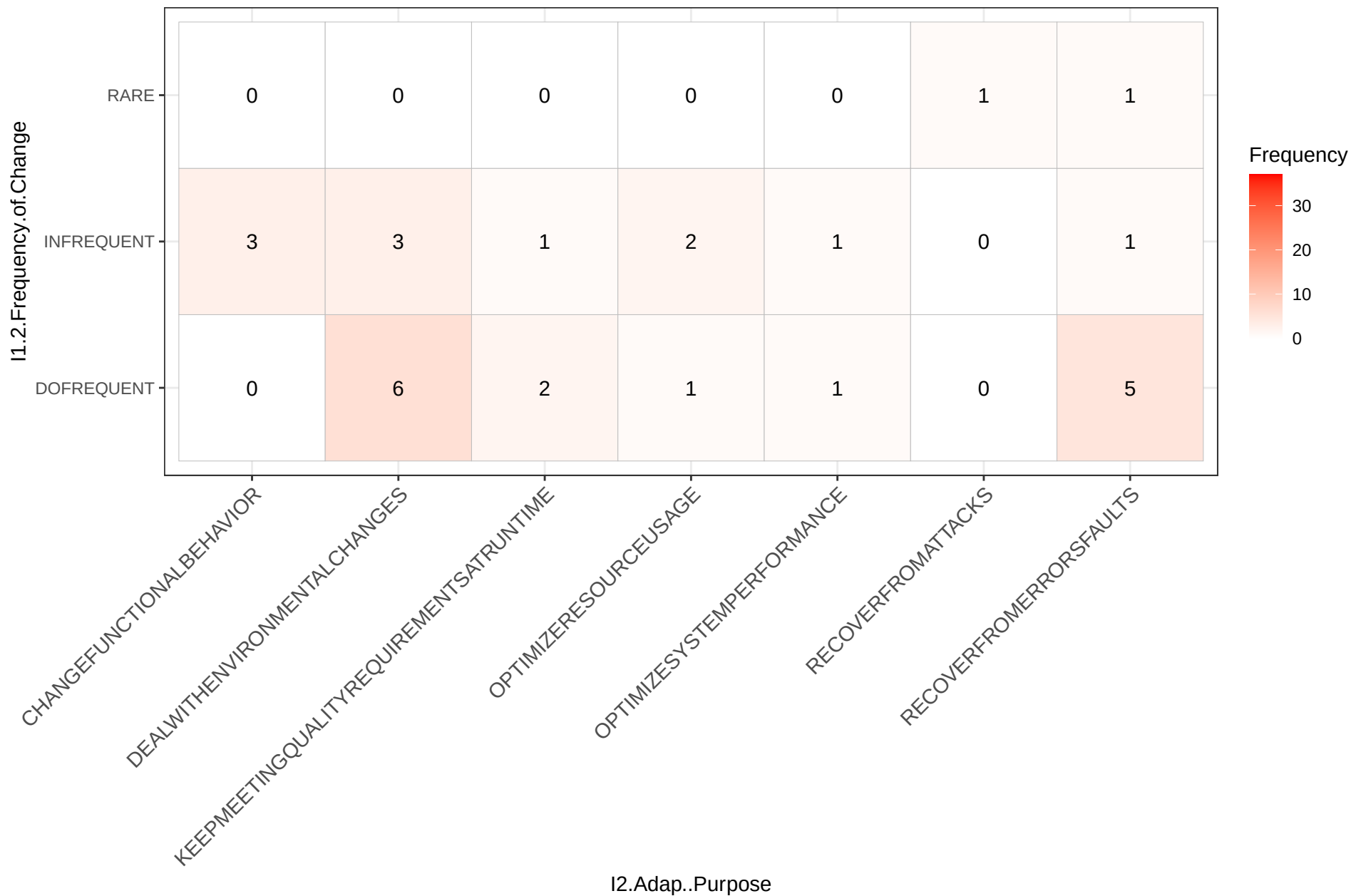
I1.2.Frequency.of.Change_____I1.4.Resilience.of.Effects

I1.2.Frequency.of.Change



I1.4.Resilience.of.Effects

I1.2.Frequency.of.Change_____I2.Adap..Purpose



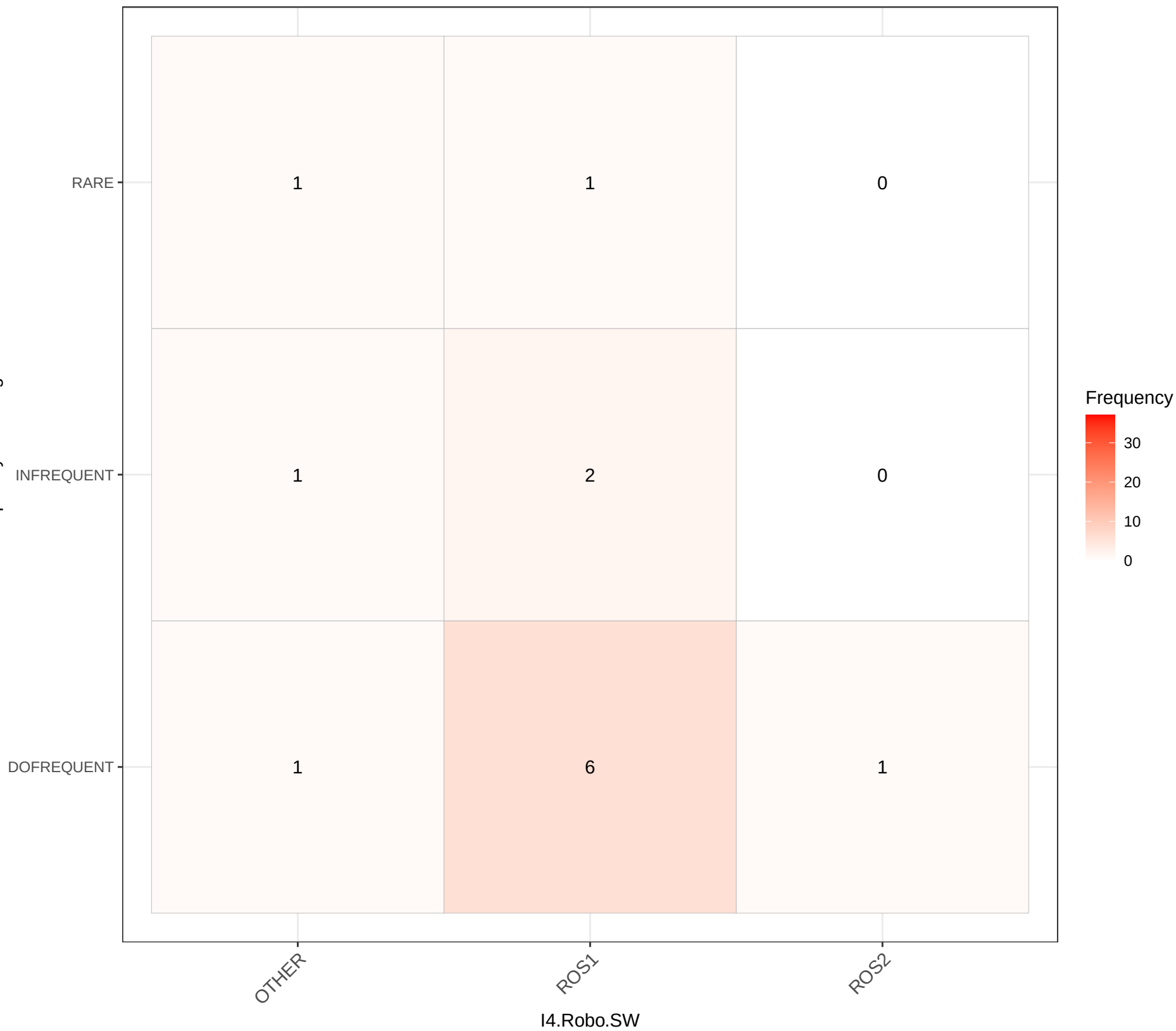
I1.2.Frequency.of.Change	I3.Robot.Type
--------------------------	---------------

Frequency

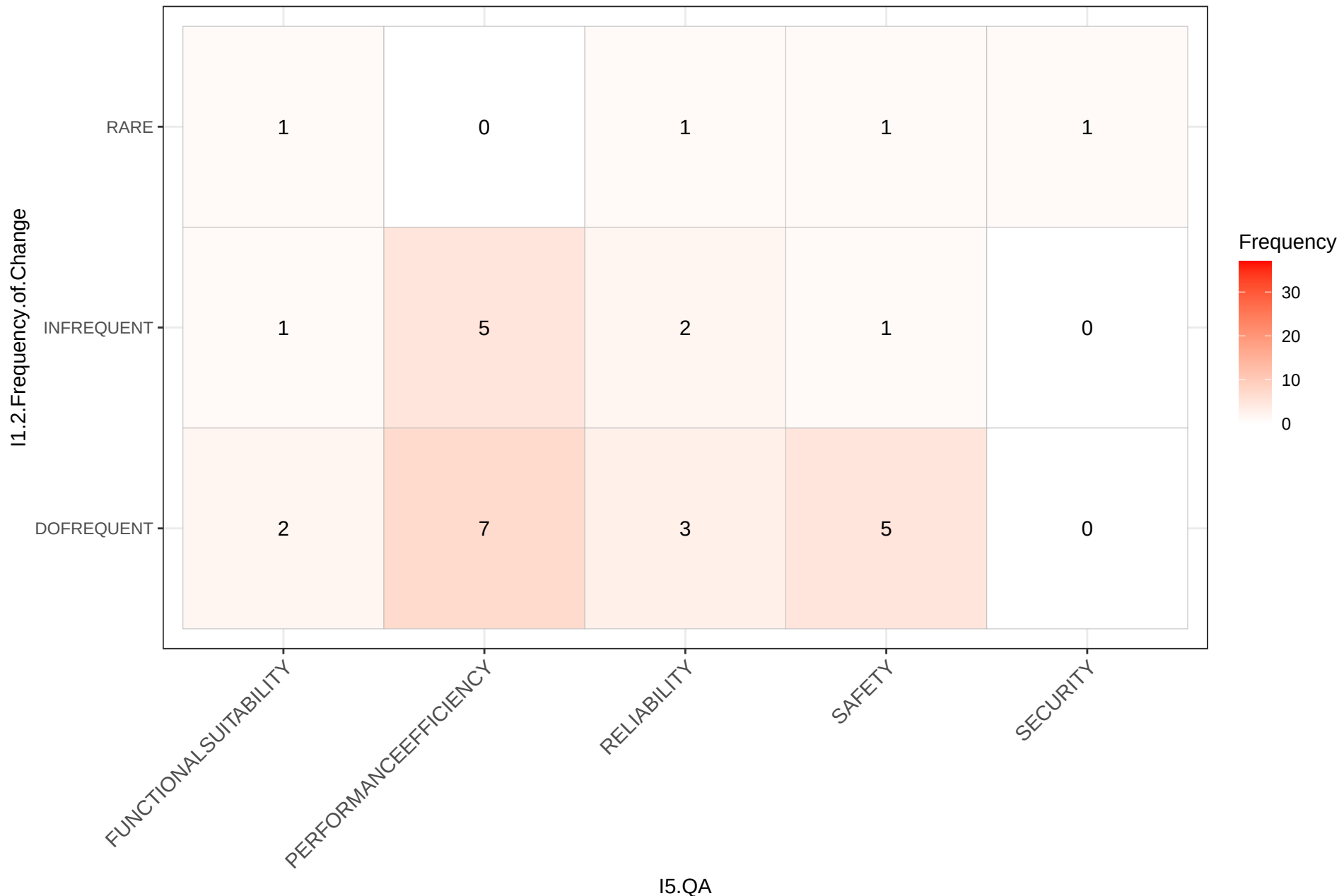
	I3.Robot.Type
BEBOTSPADERBORN	
BLUEROV2	
BOXERCLEARPATH	
CRAWLERTERMINATORBOT	
FIELDMOBILEROBOTS	
FRACTIONATEDSPACECRAFT	
HETEROGENOUSROBOTS	
HEXMANIPULATOR	
HEXASII	
INFOTAINMENTROBOT	
MOBILE-SERVICEROBOT	
IROBOTCREATE2	
MOBILEROBOT	
MOBILEROBOTMOBILEROBOT	
MOBILEROBOTS	
MOBILE-ROBOTS	
MOBILE-SERVICEROBOT	
MULTIPLE-ROBOTS	
NAO-ROTOR	
PIONEER3-DX	
QUADROCOPTER	
RESCUE	
SINGLE-SERVOING-ROBOT	
TEDUSAR-TERRESTRIAL	
TRIGLIDE-INDUSTRIAL	
ASSEMBLY	
TURTLEBOT2	
KOBUKI	
TWOINDUSTRIALROBOTS	
WAREHOUSEDELIVERYROBOT	
WHICHISANINDUSTRIALAGV	
YUBOT	
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLE	
QUADROCOPTOR	
TURTLEBOT	
FOREVALUATION	
INDUSTRIALROBOTS	
DELIVERYROBOT	
INDUSTRIALAGV	
YUBOT	

I1.2.Frequency.of.Change_____I4.Robo.SW

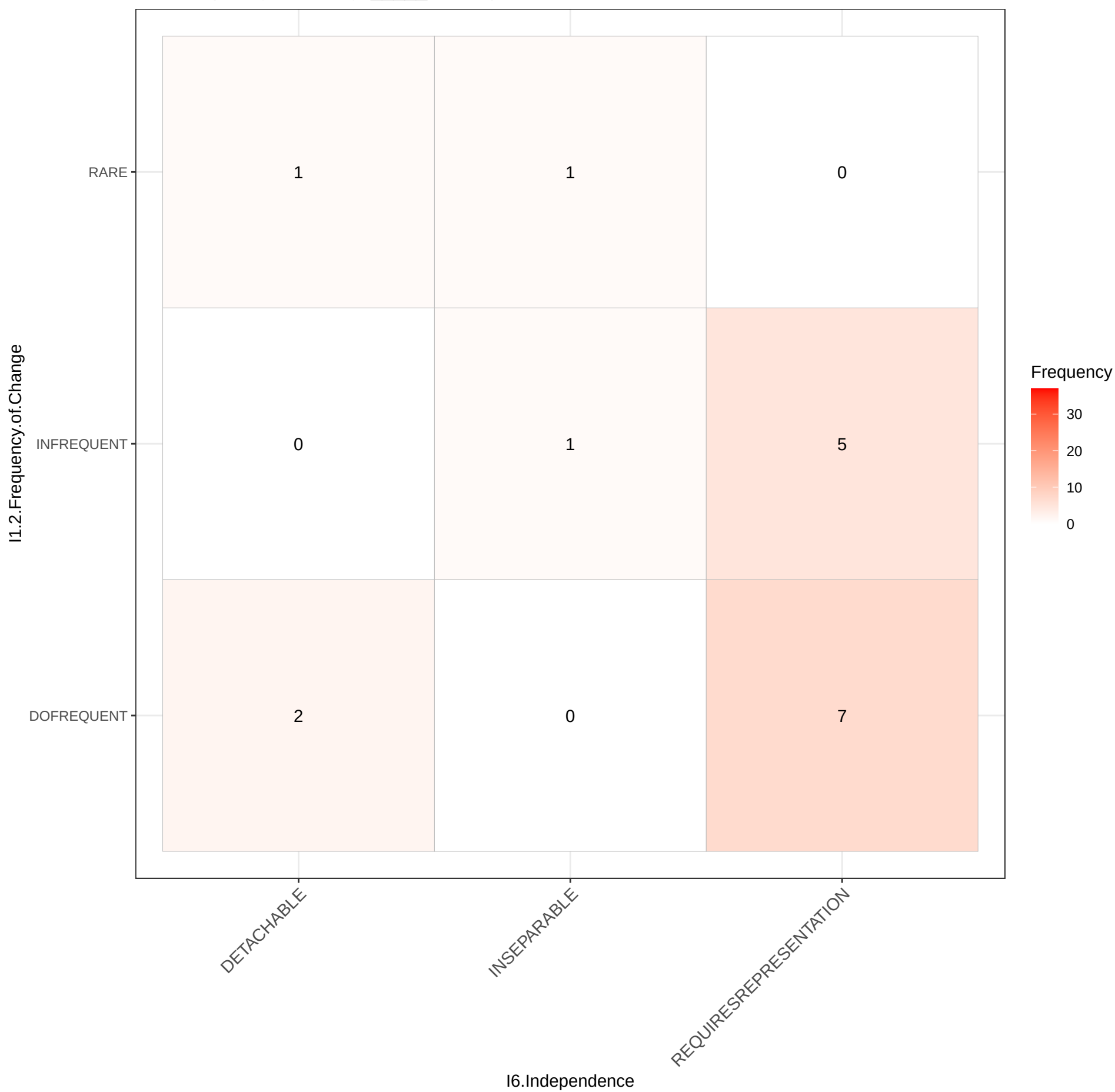
I1.2.Frequency.of.Change



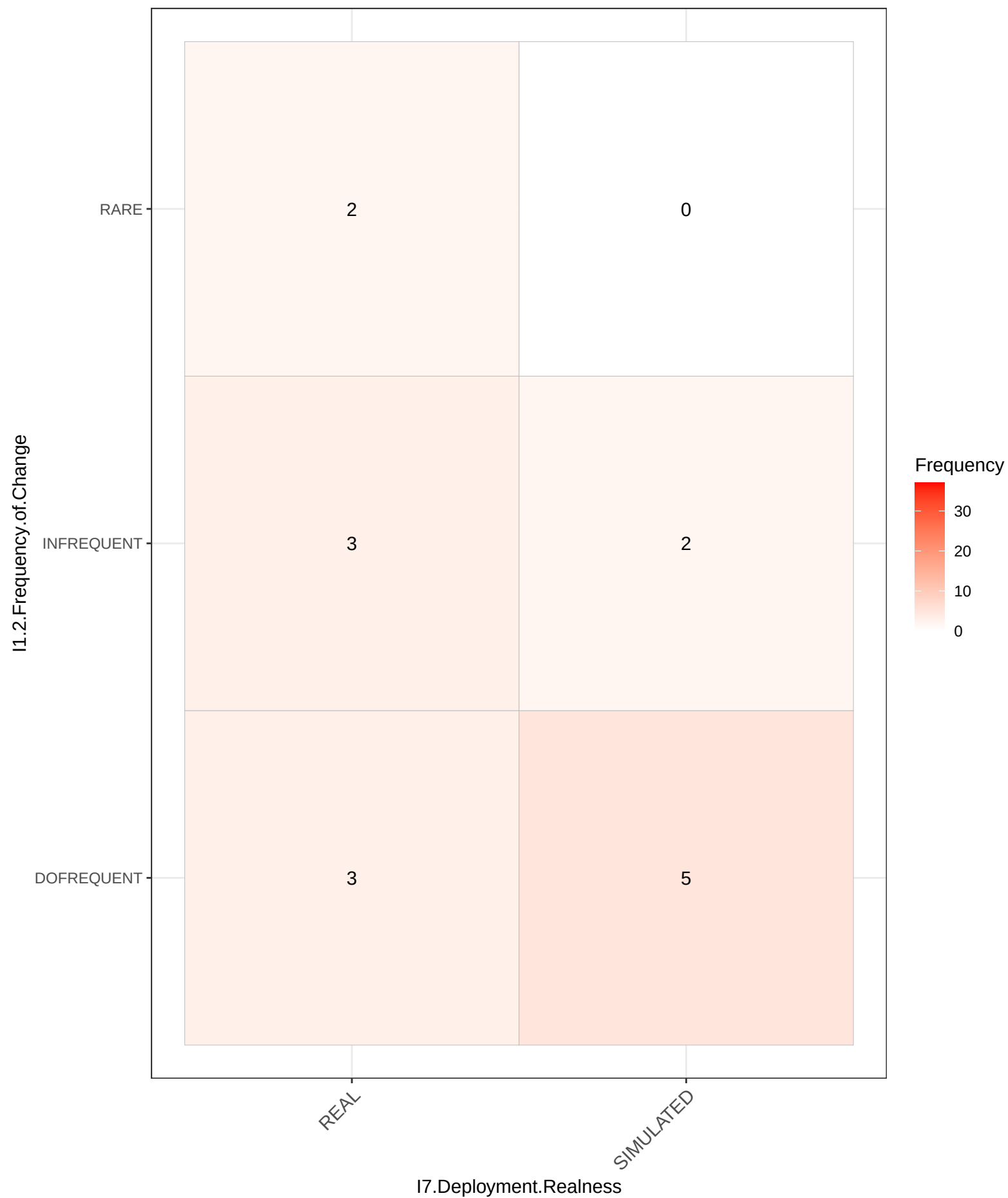
I1.2.Frequency.of.Change_____I5.QA



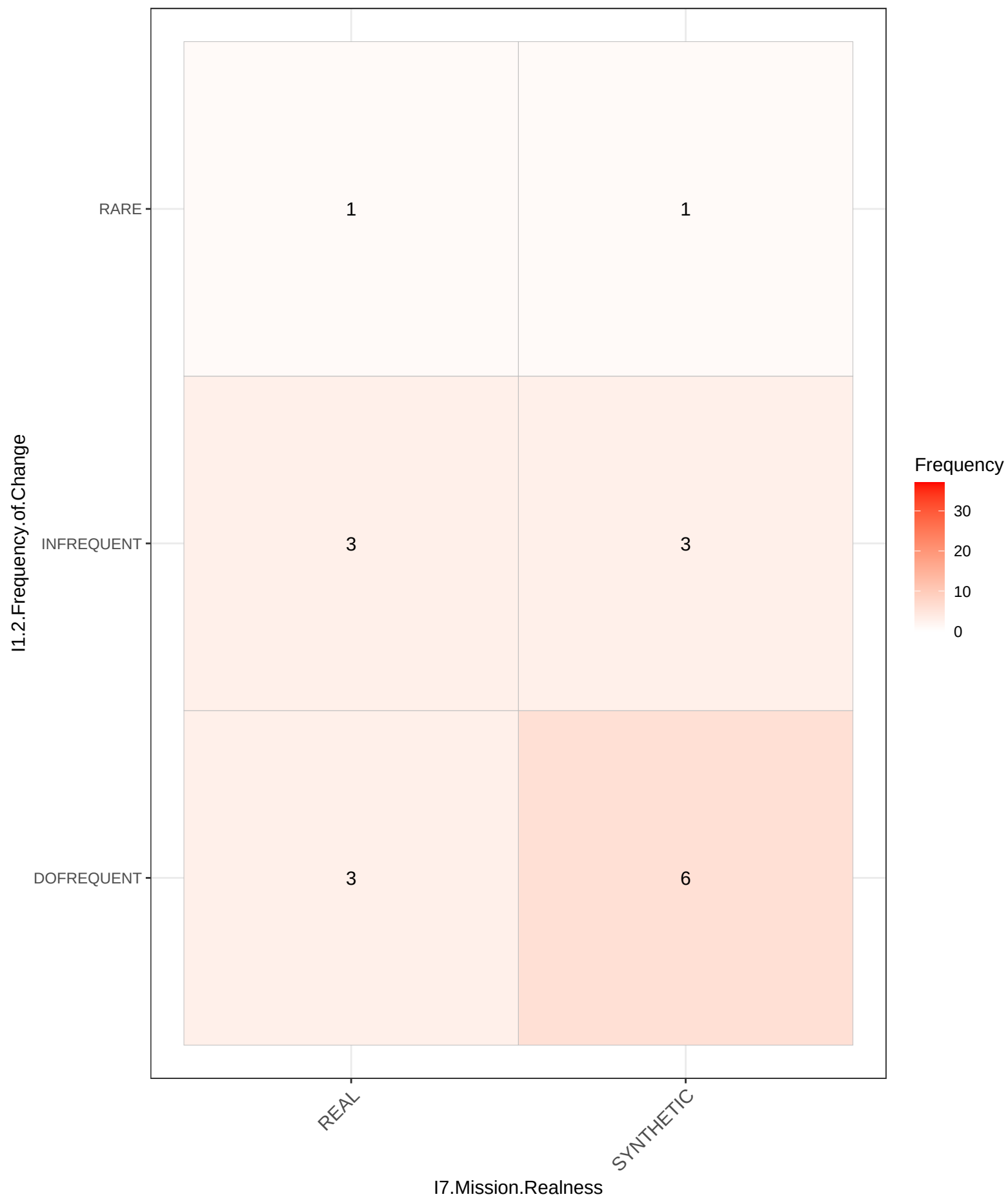
I1.2.Frequency.of.Change_____I6.Independence



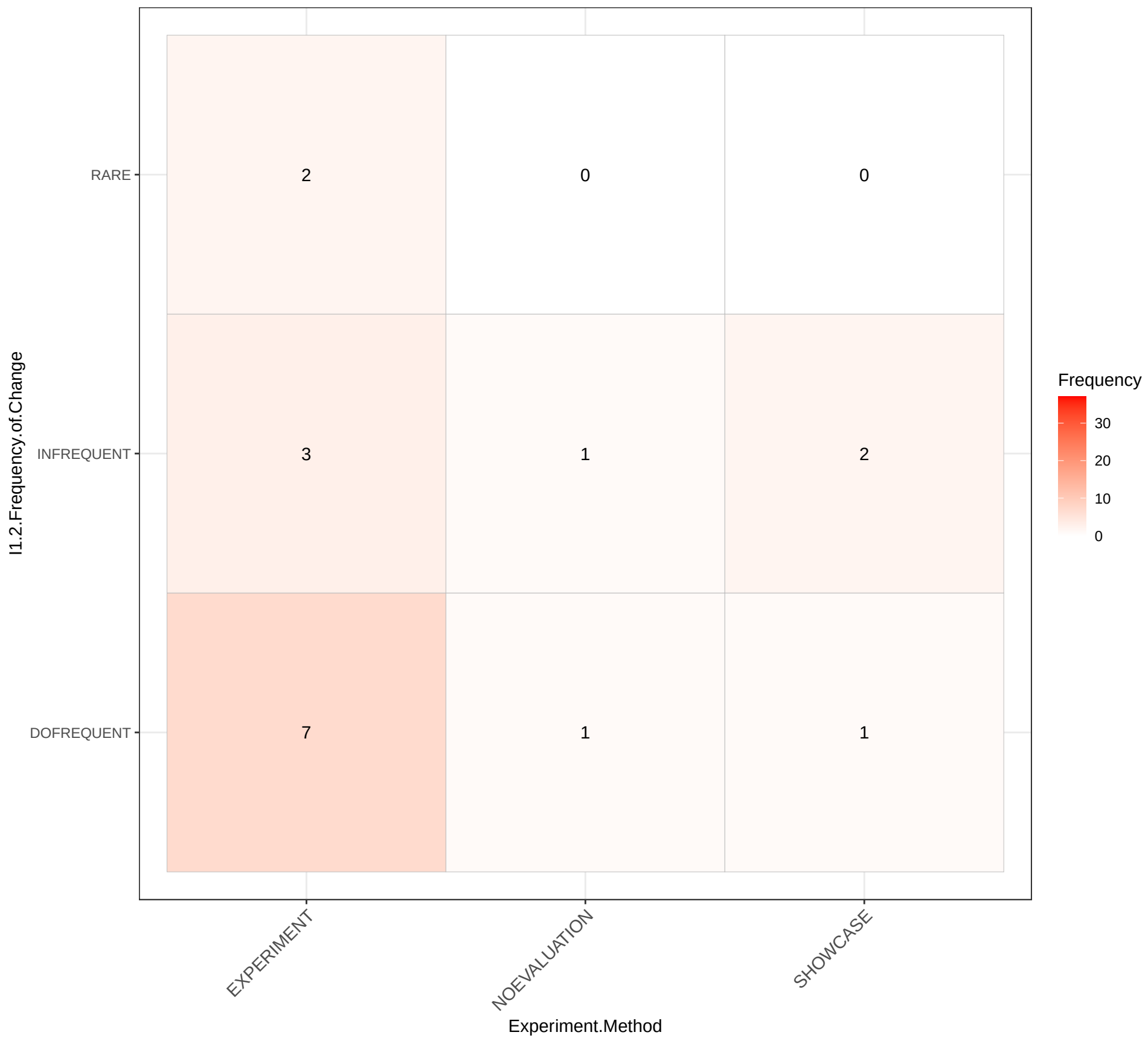
I1.2.Frequency.of.Change_____I7.Deployment.Realness



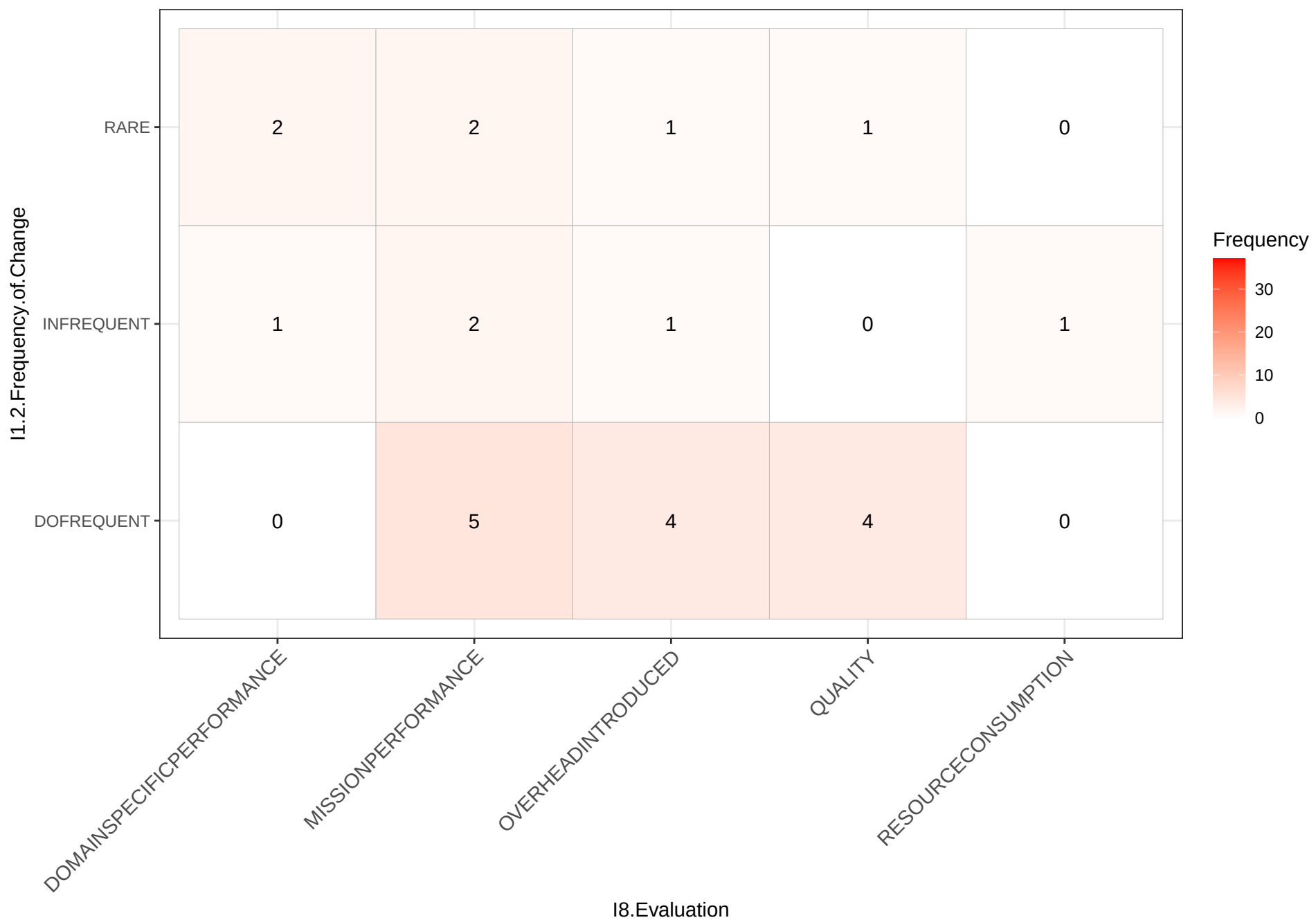
I1.2.Frequency.of.Change_____I7.Mission.Realness



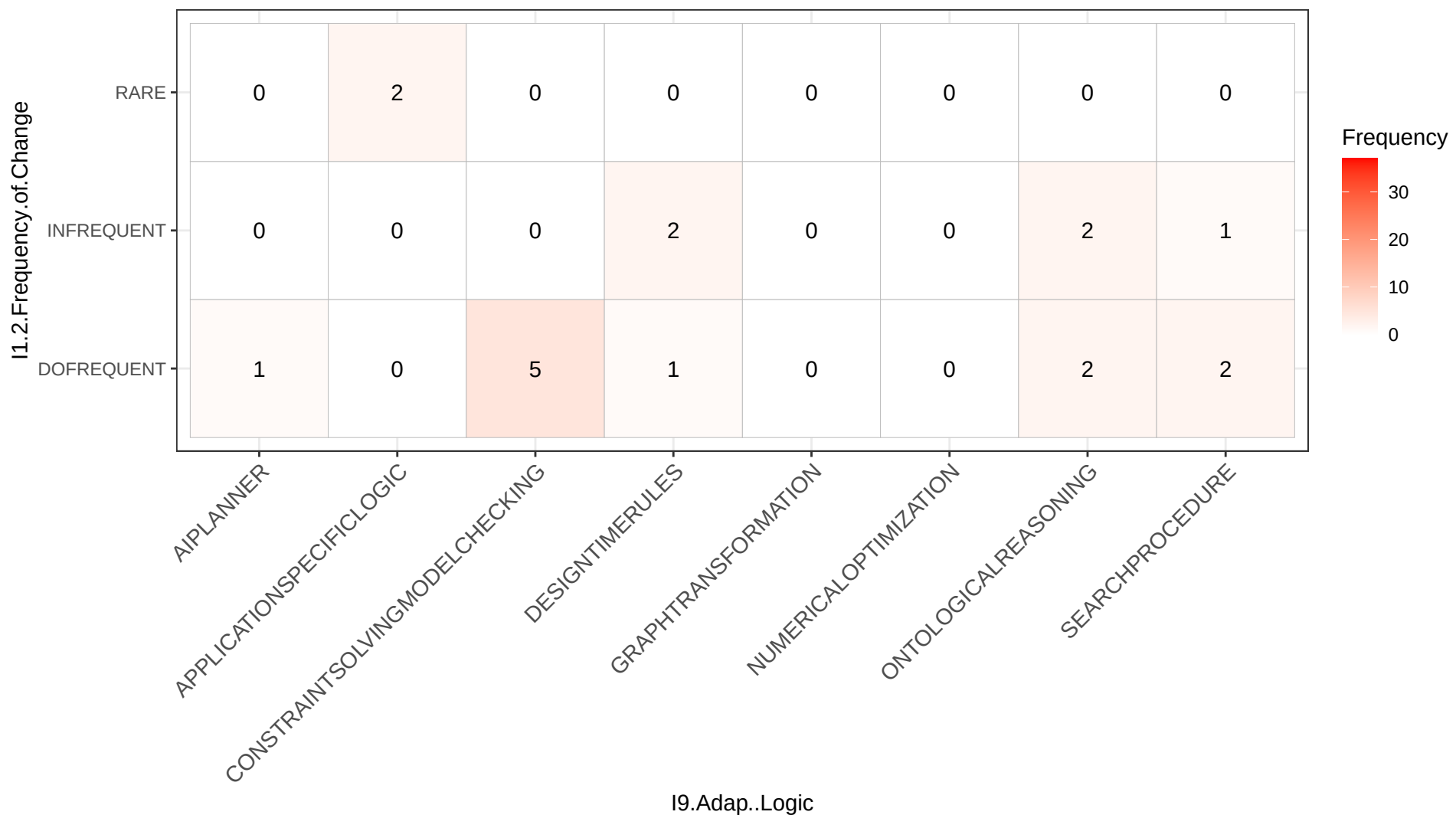
I1.2.Frequency.of.Change_____Experiment.Method



I1.2.Frequency.of.Change_____I8.Evaluation



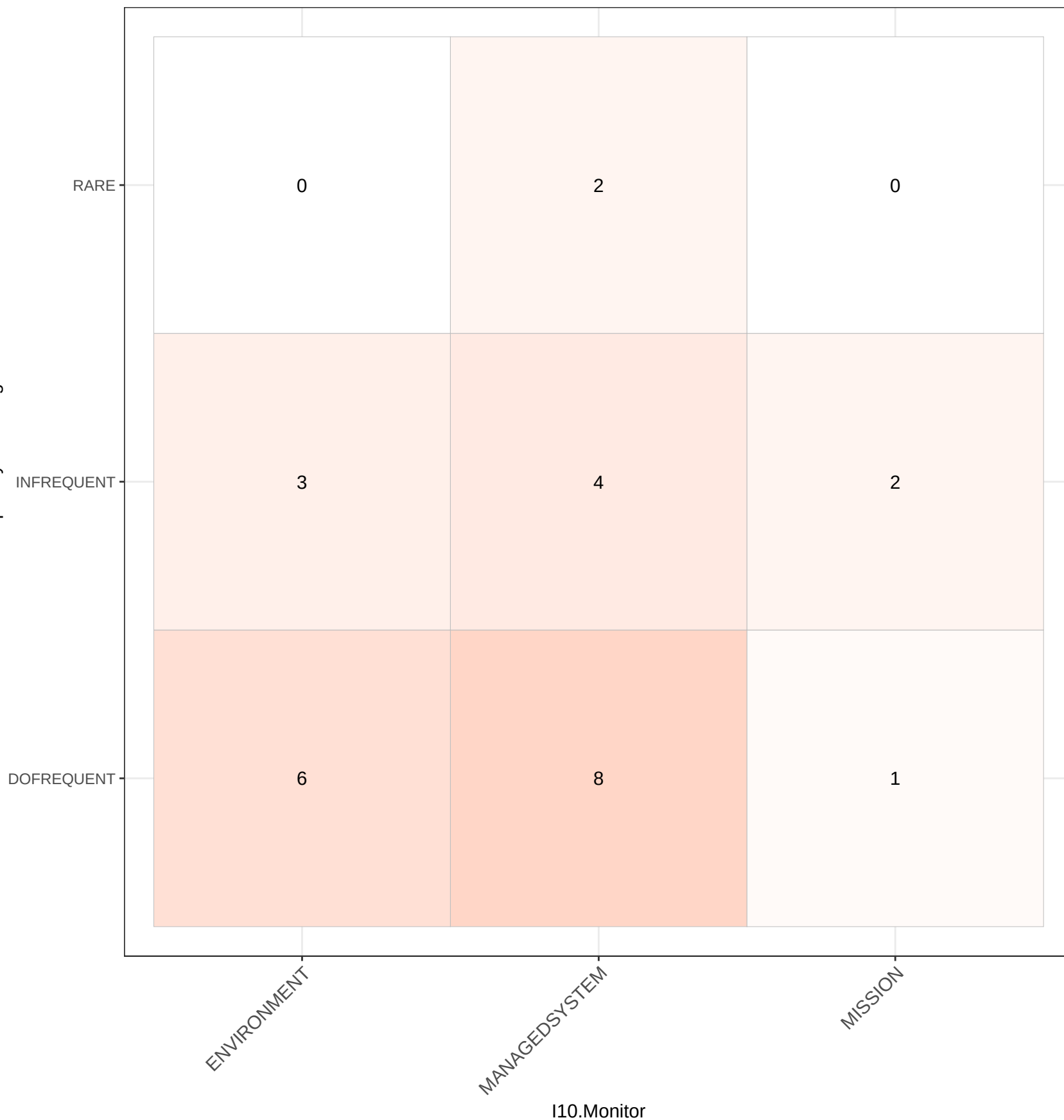
I1.2.Frequency.of.Change_____I9.Adap..Logic



I9.Adap..Logic

I1.2.Frequency.of.Change_____I10.Monitor

I1.2.Frequency.of.Change



Frequency



30

20

10

0

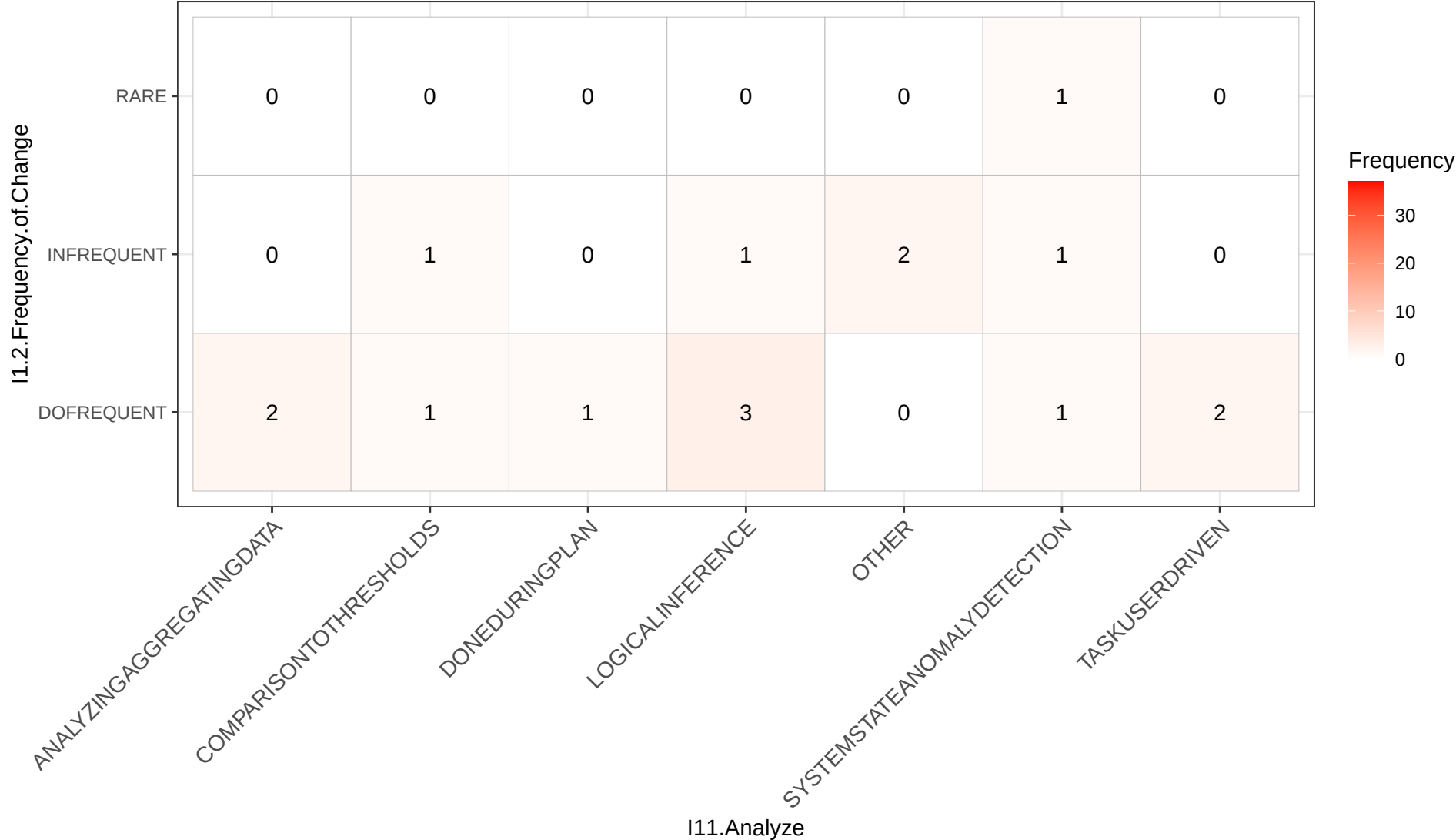
ENVIRONMENT

MANAGEDSYSTEM

MISSION

I10.Monitor

I1.2.Frequency.of.Change_____I11.Analyze



I1.2.Frequency.of.Change

RARE

INFREQUENT

DOFREQUENT

0

2

0

1

5

0

7

1

0

Frequency



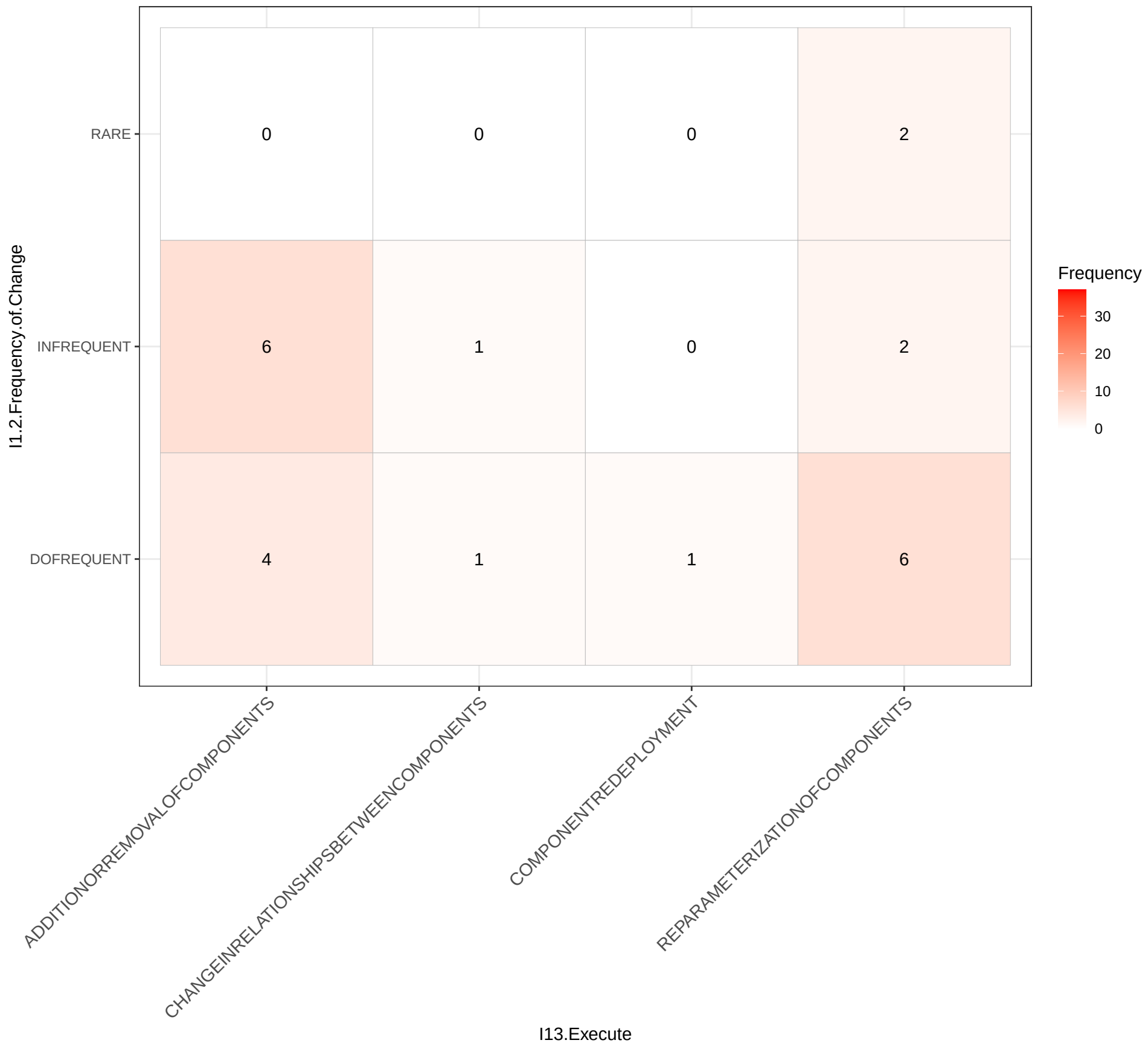
DETERMININGTHEOPTIMALCHOICE

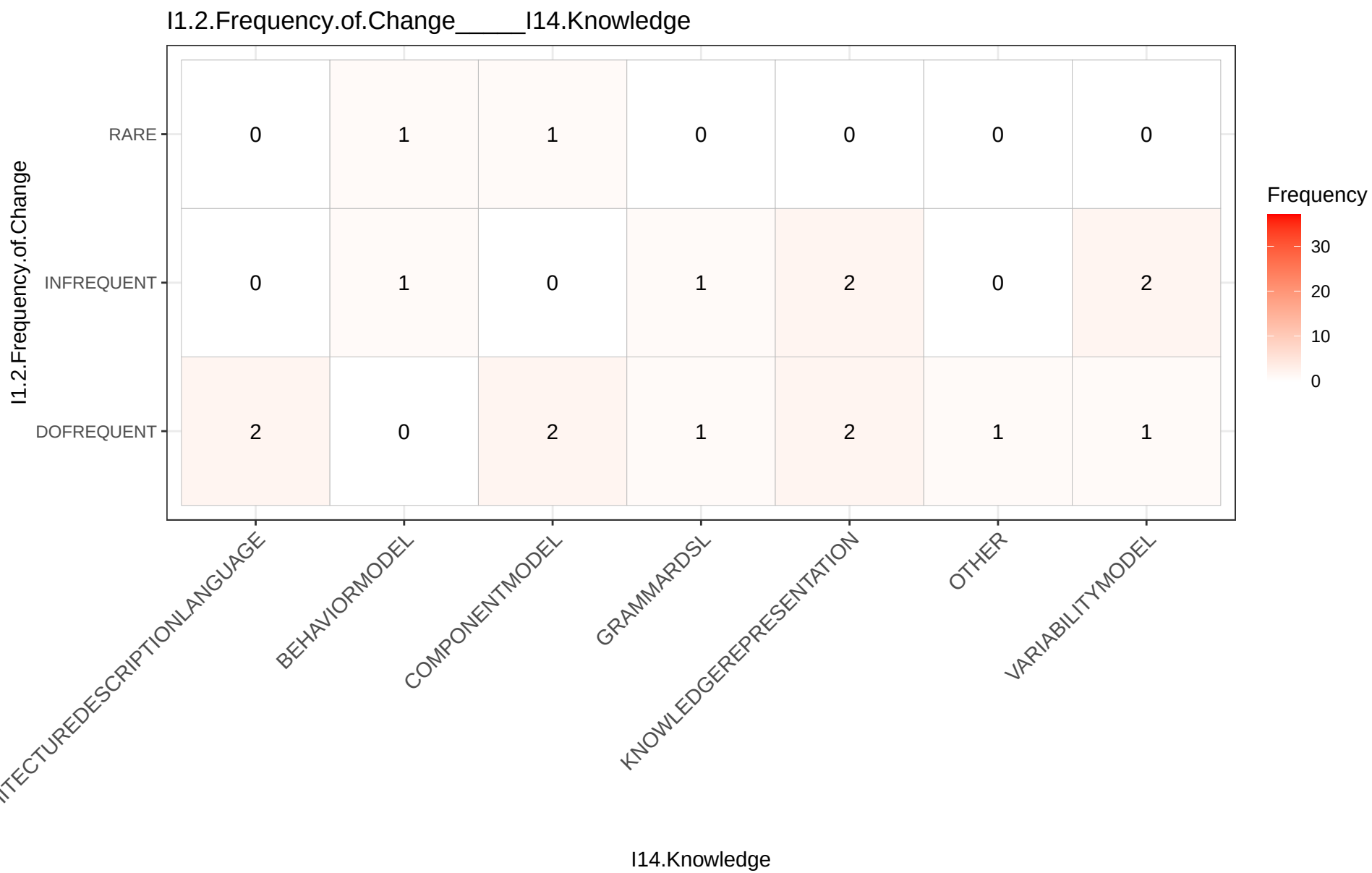
RELYINGONDESIGNTIMERULESMODELS

USINGAIPLANNINGLANGUAGES

I12.Plan

I1.2.Frequency.of.Change_____I13.Execute





I1.3.Type.of.Mechanism_____I1.3.Organization.of.Mechanism

I1.3.Type.of.Mechanism

STRUCTURAL

3

29

PARAMETRIC

1

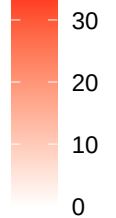
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DECENTRALIZED

DOCENTRALIZED

I1.3.Organization.of.Mechanism

Frequency



I1.3.Type.of.Mechanism_____I1.3.Scope.of.Mechanism

I1.3.Type.of.Mechanism

STRUCTURAL

9

27

PARAMETRIC

8

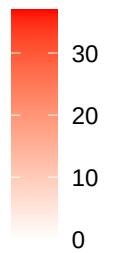
19

GLOBAL

LOCAL

I1.3.Scope.of.Mechanism

Frequency



I1.3.Type.of.Mechanism_____I1.3.Duration.of.Mechanism

I1.3.Type.of.Mechanism

STRUCTURAL

22

1

3

PARAMETRIC

13

0

4

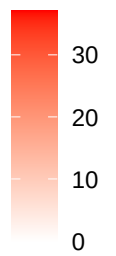
DOSHORT

MEDIUM

VERYSHORT

I1.3.Duration.of.Mechanism

Frequency



I1.3.Type.of.Mechanism_____I1.3.Timeliness.of.Mechanism

I1.3.Type.of.Mechanism

STRUCTURAL

26

1

PARAMETRIC

18

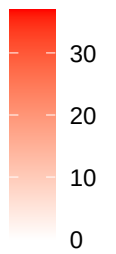
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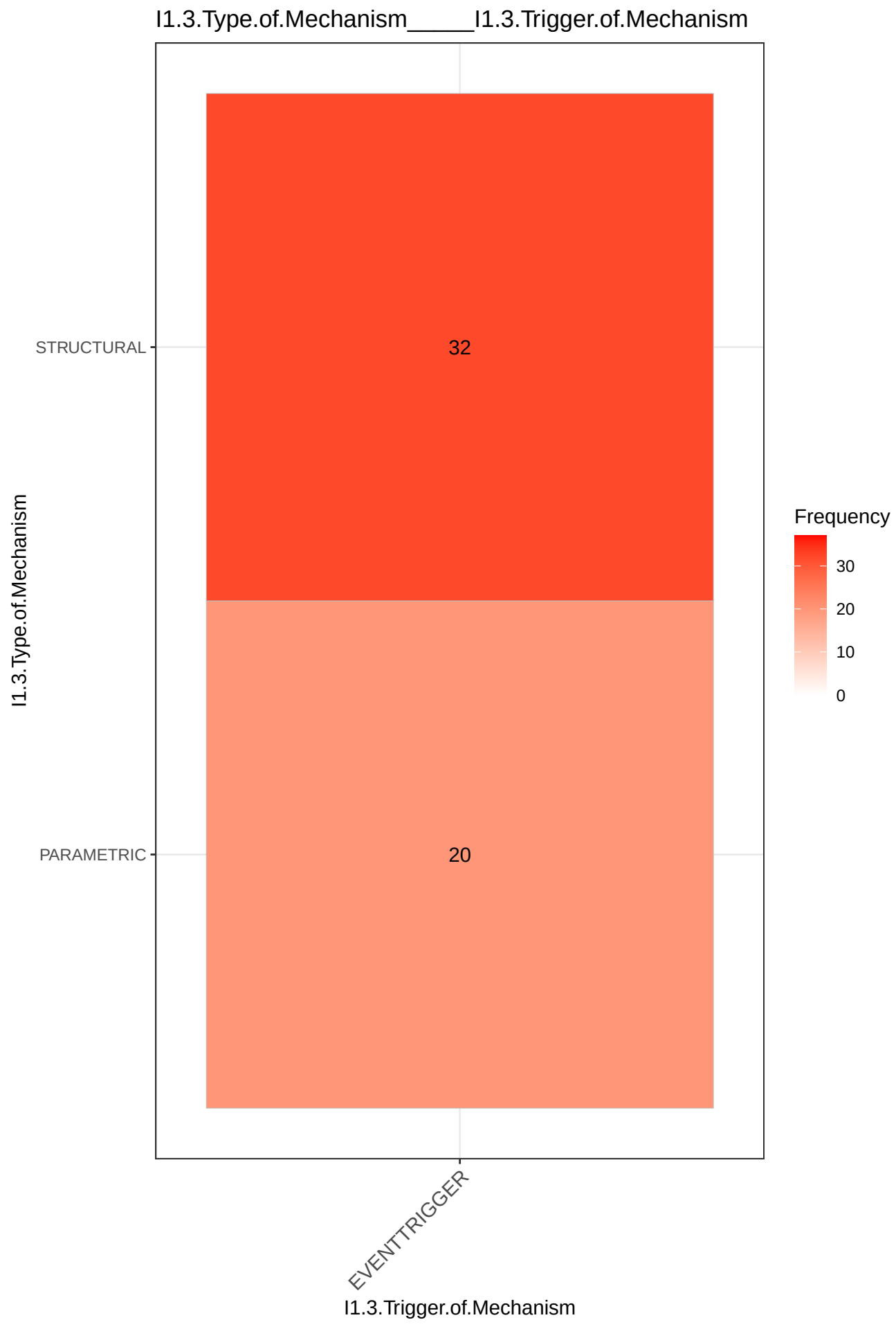
BESTEFFECT

GUARANTEED

I1.3.Timeliness.of.Mechanism

Frequency





I1.3.Type.of.Mechanism_____I1.4.Criticality.of.Effects

I1.3.Type.of.Mechanism

STRUCTURAL

20

9

PARAMETRIC

13

6

MISSIONCRITICAL

SAFETYCRITICAL

I1.4.Criticality.of.Effects

Frequency



I1.3.Type.of.Mechanism_____I1.4.Predictability.of.Effects

I1.3.Type.of.Mechanism

STRUCTURAL

9

17

PARAMETRIC

5

14

DODETERMINISTIC

NONDETERMINISTIC

I1.4.Predictability.of.Effects

Frequency



I1.3.Type.of.Mechanism_____I1.4.Overhead.of.Effects

I1.3.Type.of.Mechanism

STRUCTURAL

16

2

7

PARAMETRIC

9

3

3

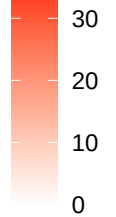
DEPENDENT

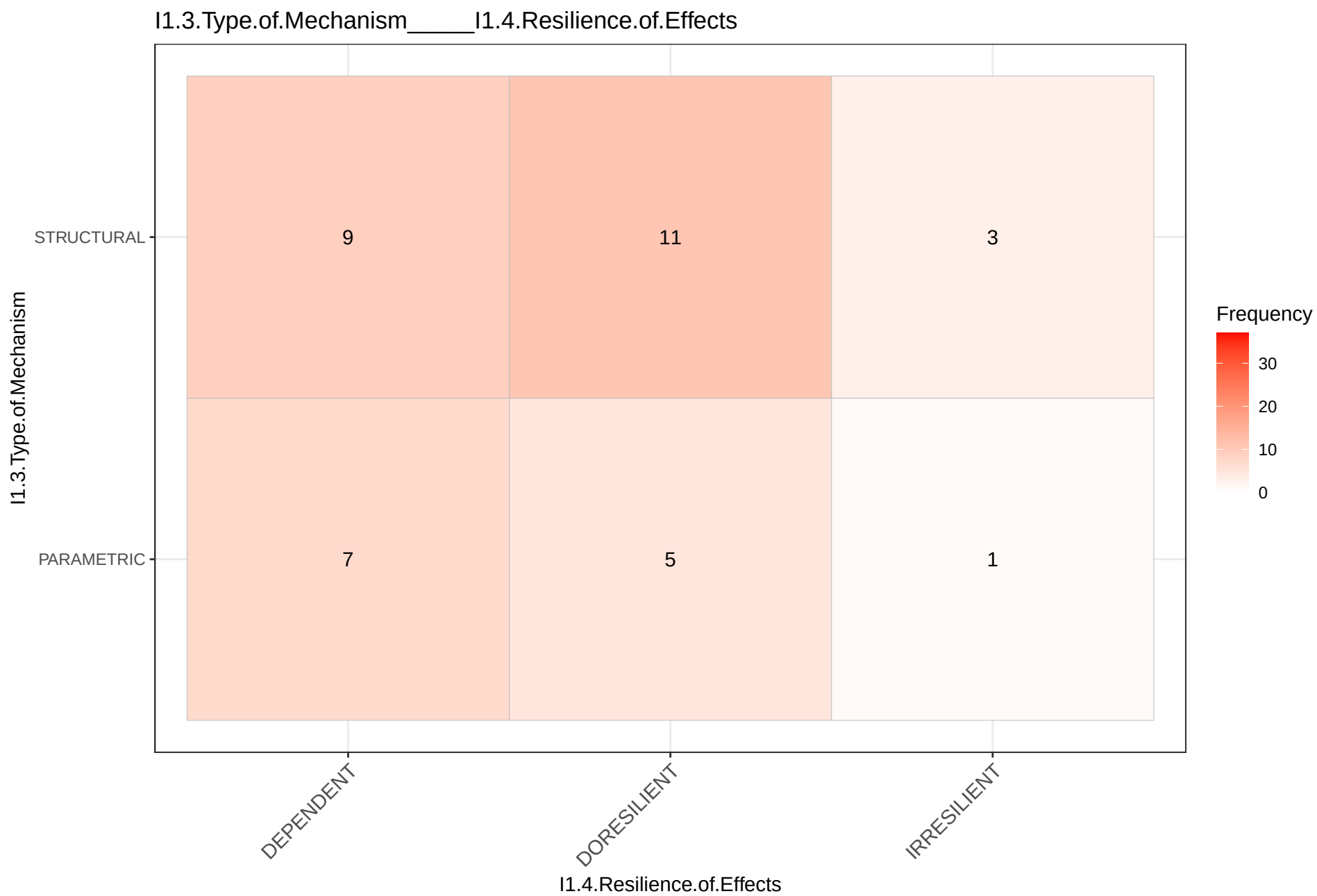
DOSIGNIFICANT

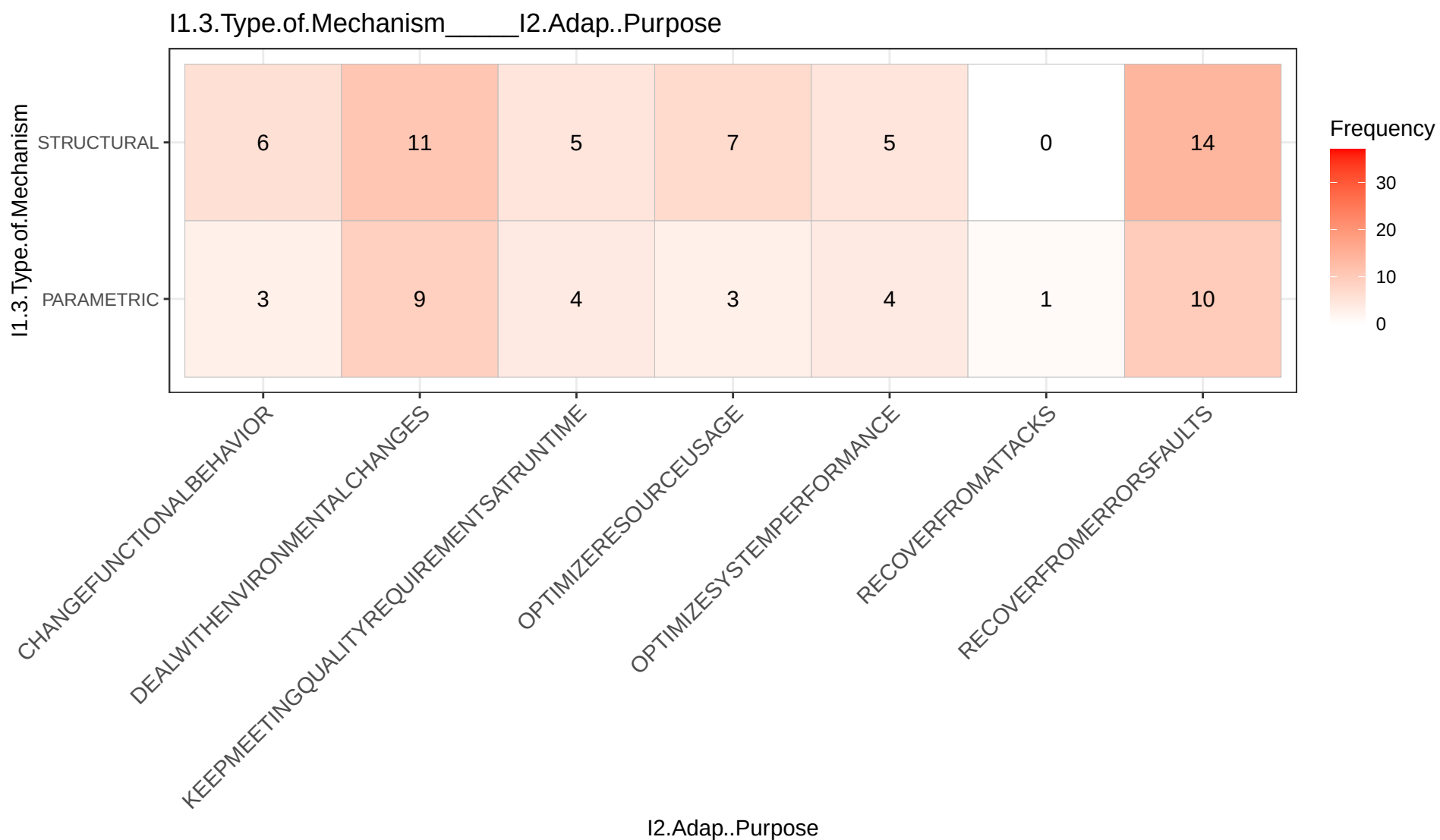
INSIGNIFICANT

I1.4.Overhead.of.Effects

Frequency







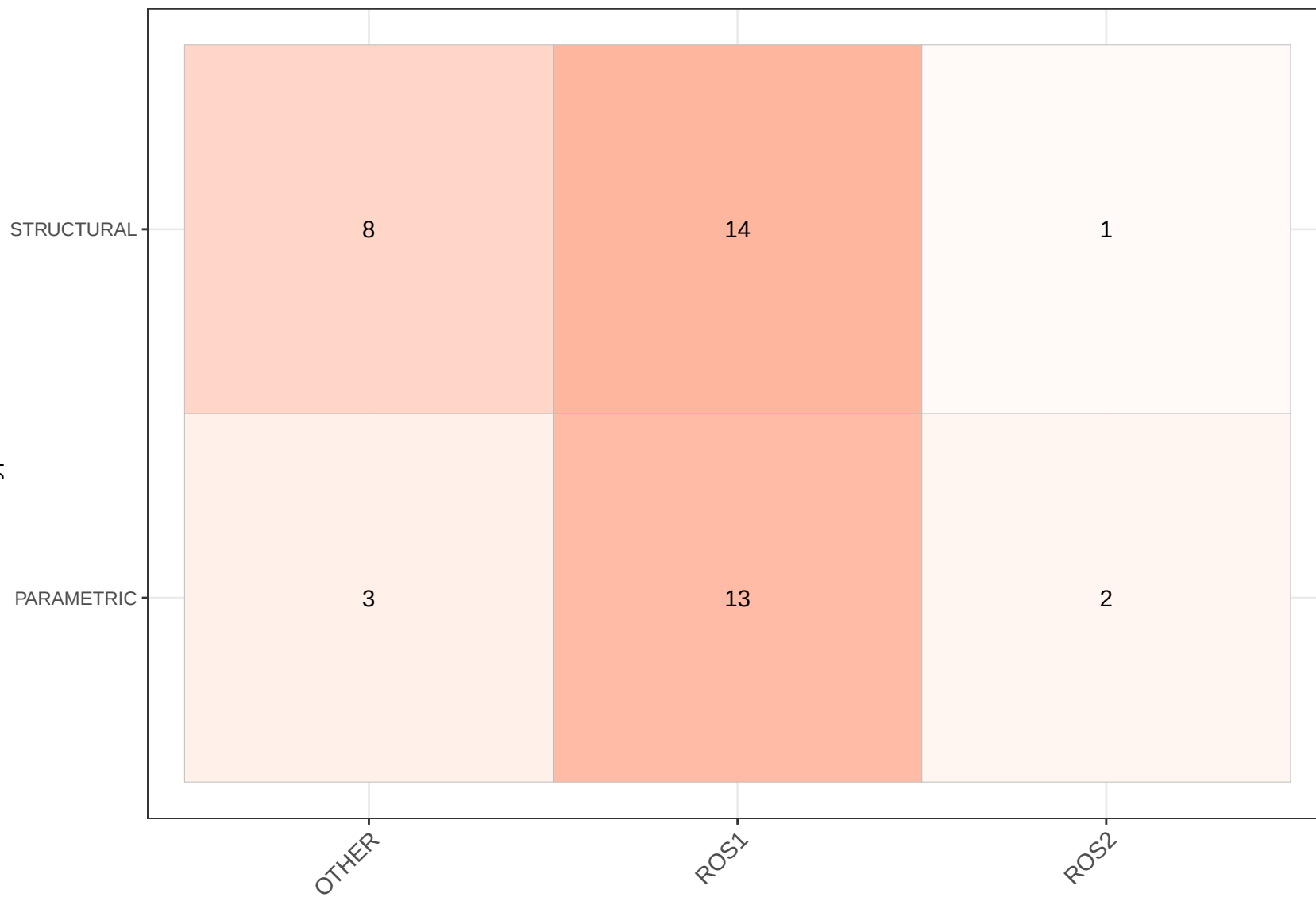
I1.3.Type.of.Mechanism_____I3.Robot.Type

Frequency

13.Robot.Type

I1.3.Type.of.Mechanism____I4.Robo.SW

I1.3.Type.of.Mechanism



PARAMETRIC

OTHER

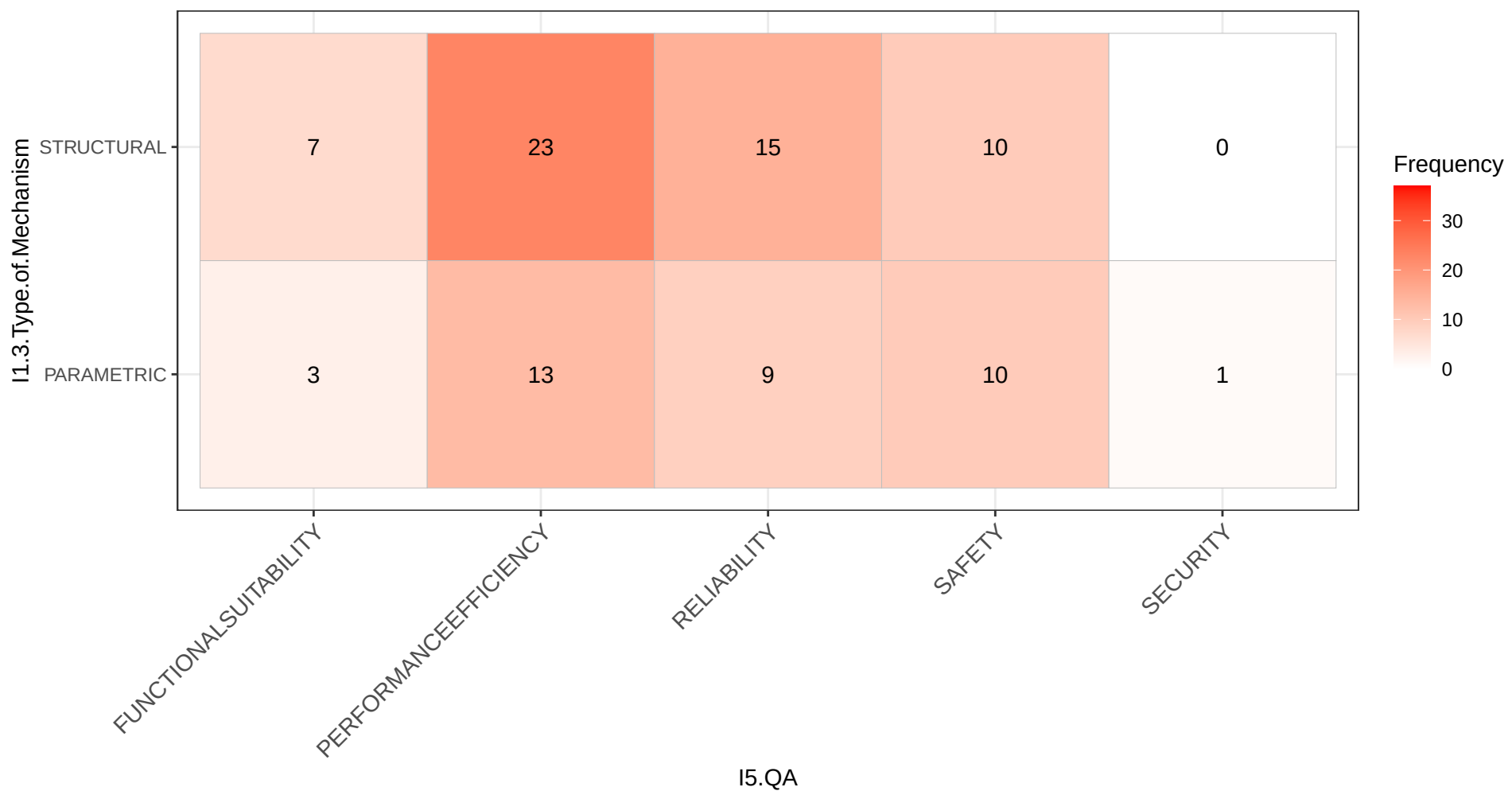
ROS1
I4.Robo.SW

ROS2

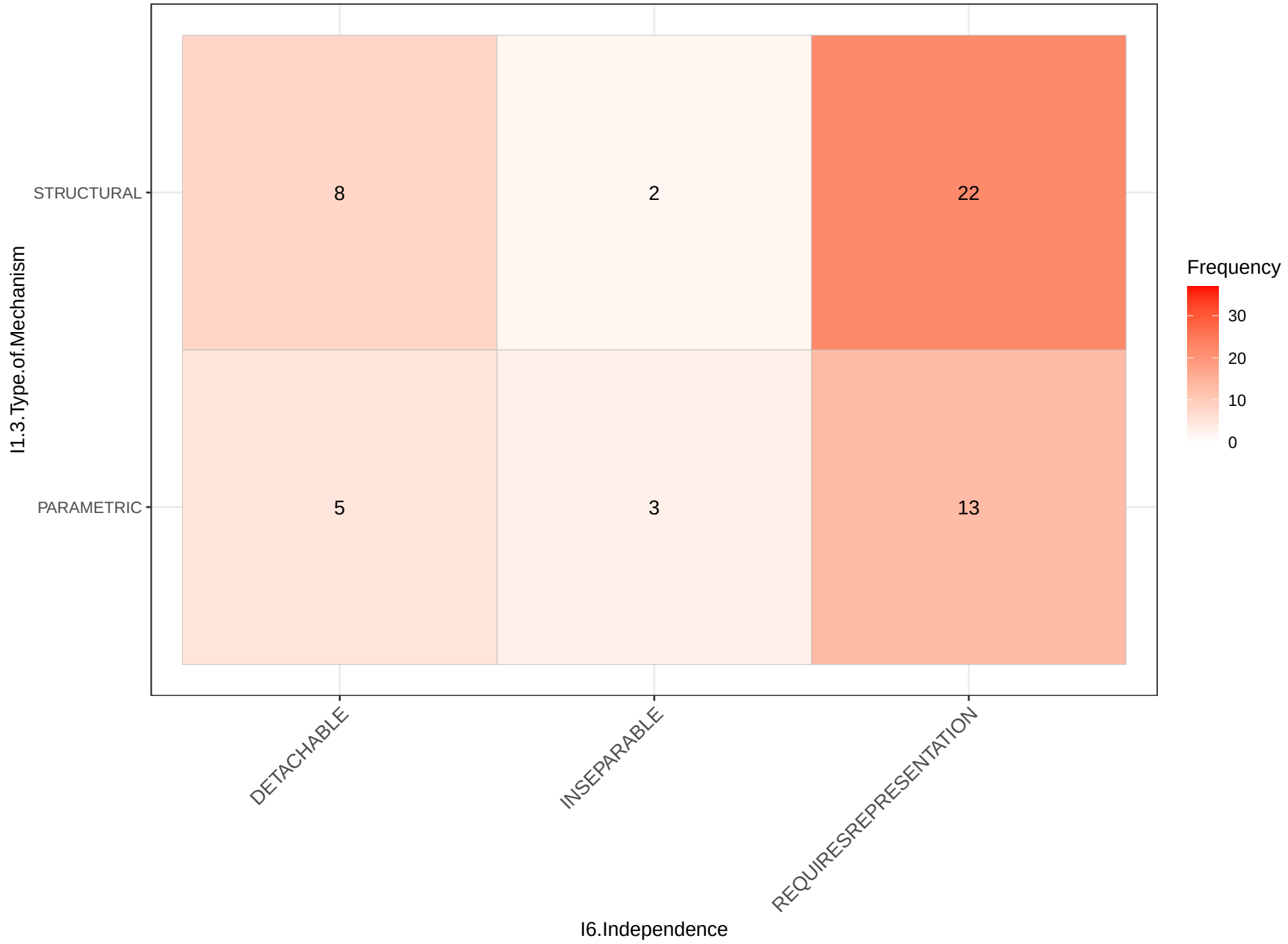
Frequency



I1.3.Type.of.Mechanism_____I5.QA



I1.3.Type.of.Mechanism_____I6.Independence



I1.3.Type.of.Mechanism_____I7.Deployment.Realness

I1.3.Type.of.Mechanism

STRUCTURAL

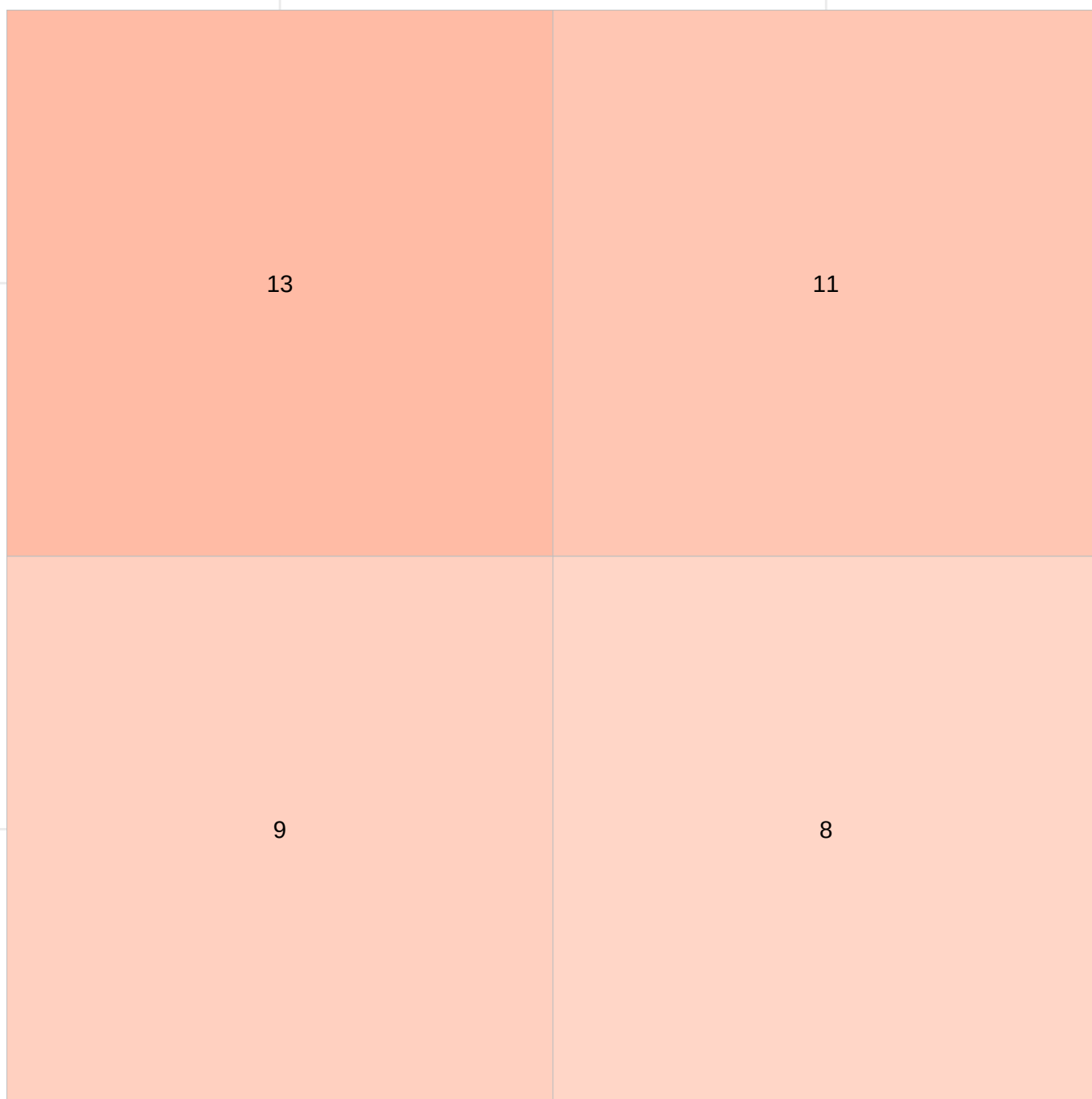
PARAMETRIC

REAL

SIMULATED

I7.Deployment.Realness

Frequency



I1.3.Type.of.Mechanism____I7.Mission.Realness

I1.3.Type.of.Mechanism

STRUCTURAL

PARAMETRIC

REAL

SYNTHETIC

I7.Mission.Realness

Frequency



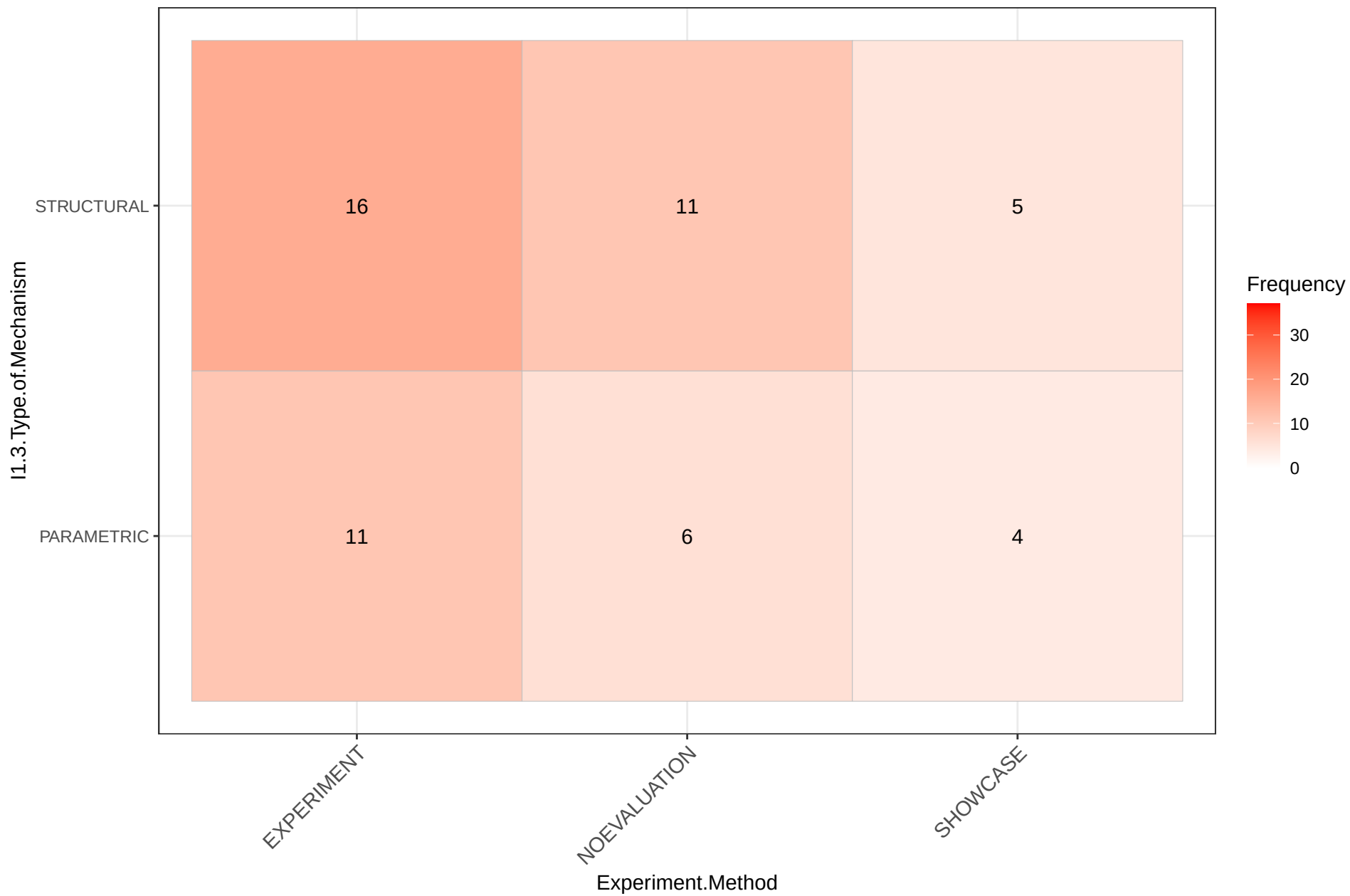
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14

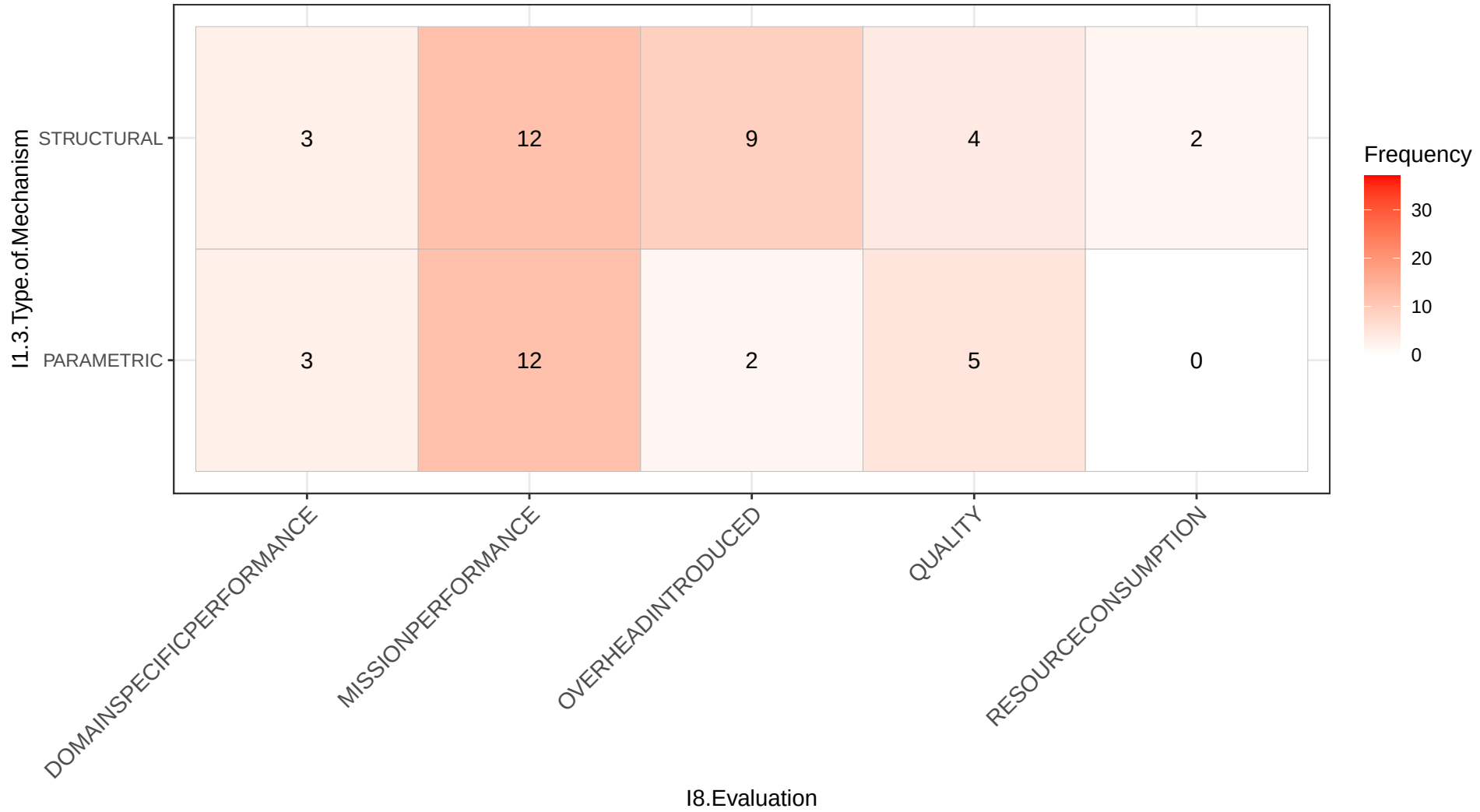
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11

I1.3.Type.of.Mechanism_____Experiment.Method



I1.3.Type.of.Mechanism_____I8.Evaluation

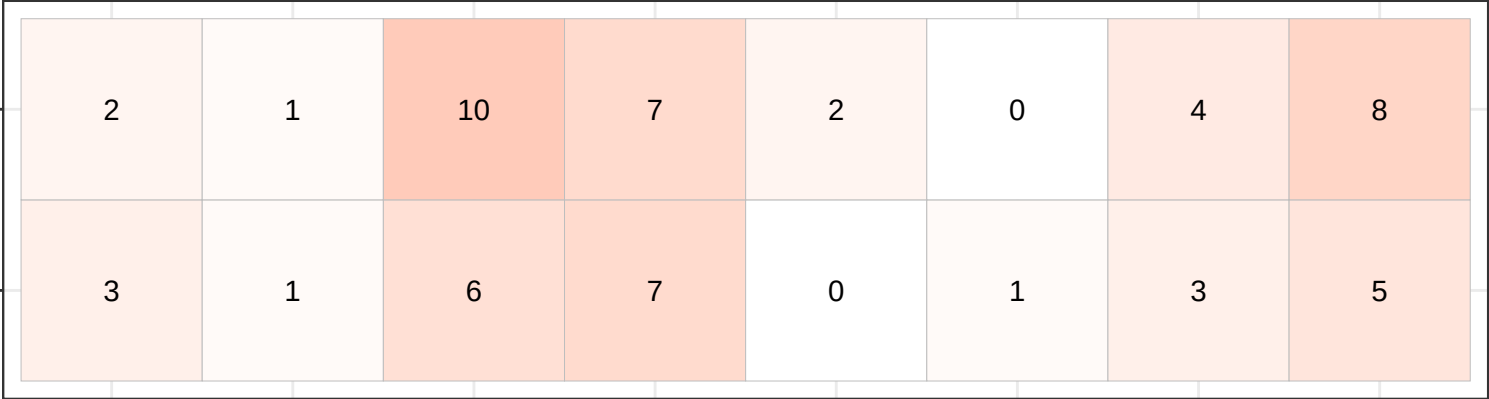


I1.3.Type.of.Mechanism_____I9.Adap..Logic

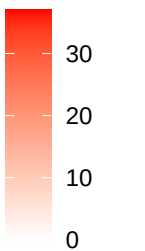
I1.3.Type.of.Mechanism

STRUCTURAL

PARAMETRIC



Frequency



AIPLANNER
APPLICATIONSPECIFICLOGIC
CONSTRAINTSOLVINGMODELCHECKING
DESIGNTIMERULES
GRAPHTRANSFORMATION
NUMERICALOPTIMIZATION
ONTOLOGICALREASONING
SEARCHPROCEDURE

I9.Adap..Logic

I1.3.Type.of.Mechanism_____I10.Monitor

I1.3.Type.of.Mechanism

STRUCTURAL

12

24

4

PARAMETRIC

10

19

3

ENVIRONMENT

MANAGEDSYSTEM

MISSION

I10.Monitor

Frequency

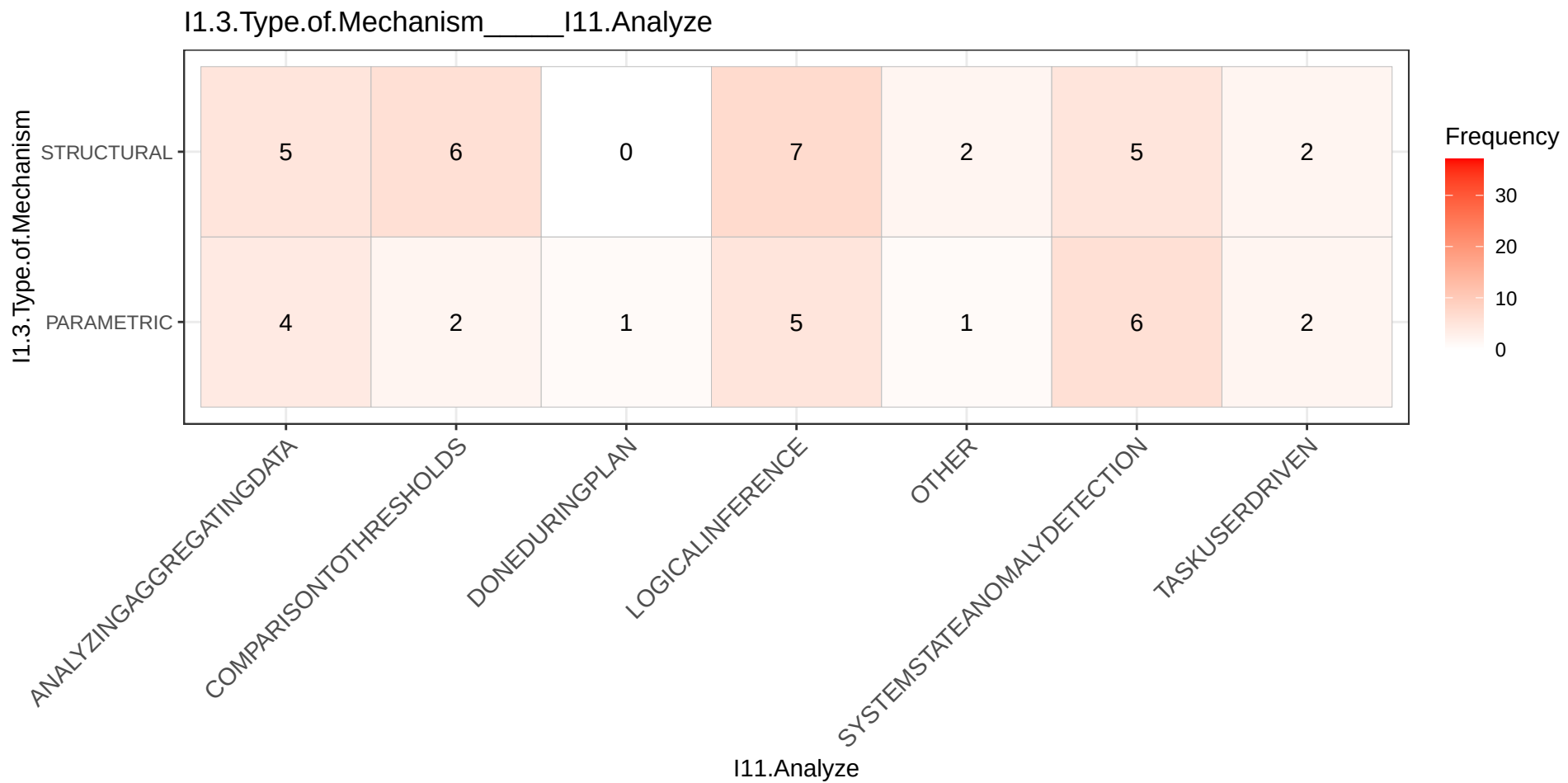


30

20

10

0



I1.3.Type.of.Mechanism_____I12.Plan

I1.3.Type.of.Mechanism

STRUCTURAL

12

15

1

PARAMETRIC

8

9

2

DETERMININGTHEOPTIMALCHOICE

RELYINGONDESIGNTIMERULESMODELS

USINGAIPLANNINGLANGUAGES

I12.Plan

Frequency



30

20

10

0

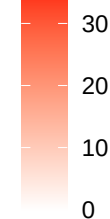
I1.3.Type.of.Mechanism____I13.Execute

I1.3.Type.of.Mechanism

STRUCTURAL

PARAMETRIC

Frequency



19

10

4

13

10

5

0

14

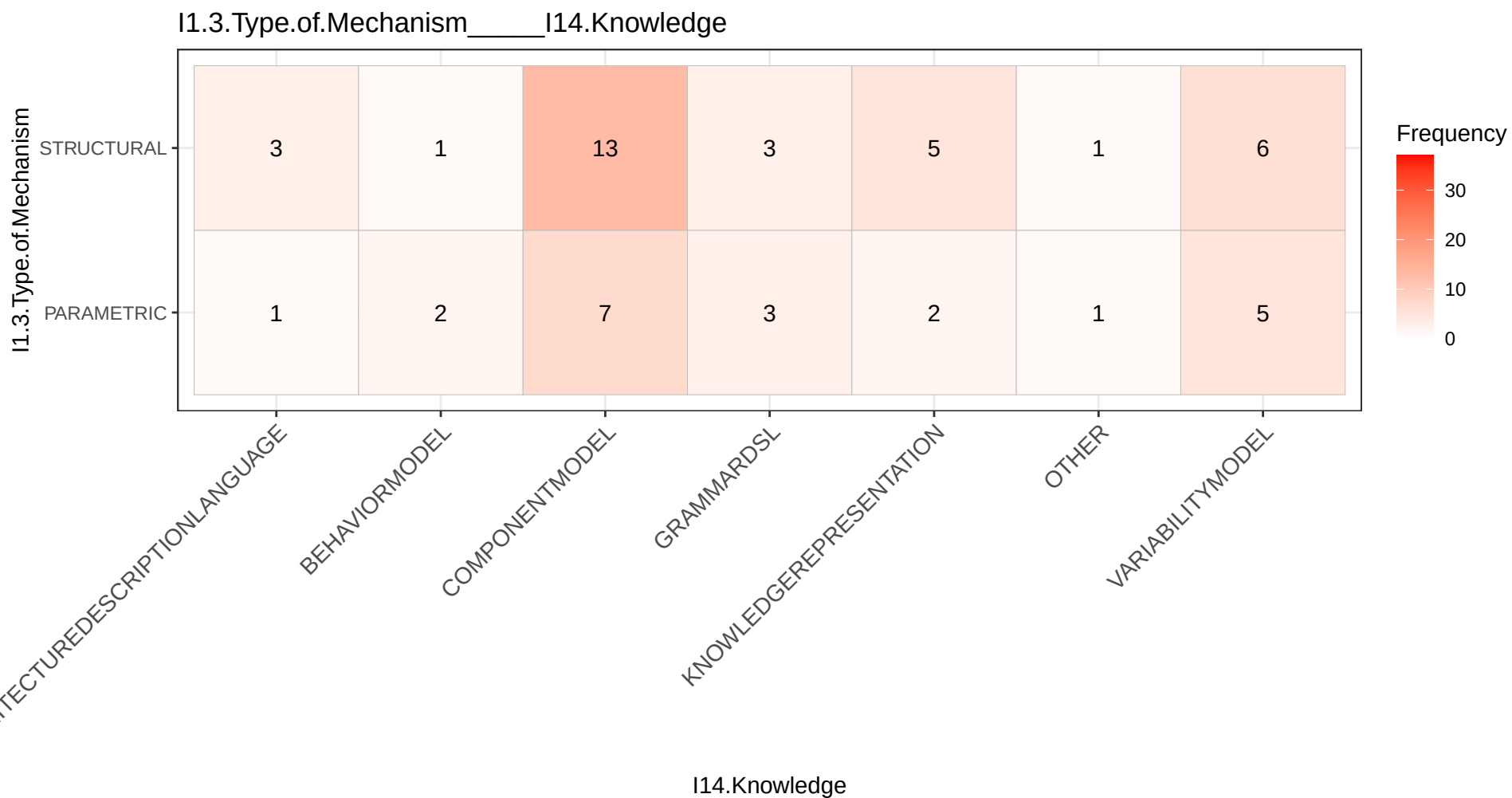
ADDITIONORREMOVALOFCOMPONENTS

CHANGEINRELATIONSHIPSBEWEENCOMPONENTS

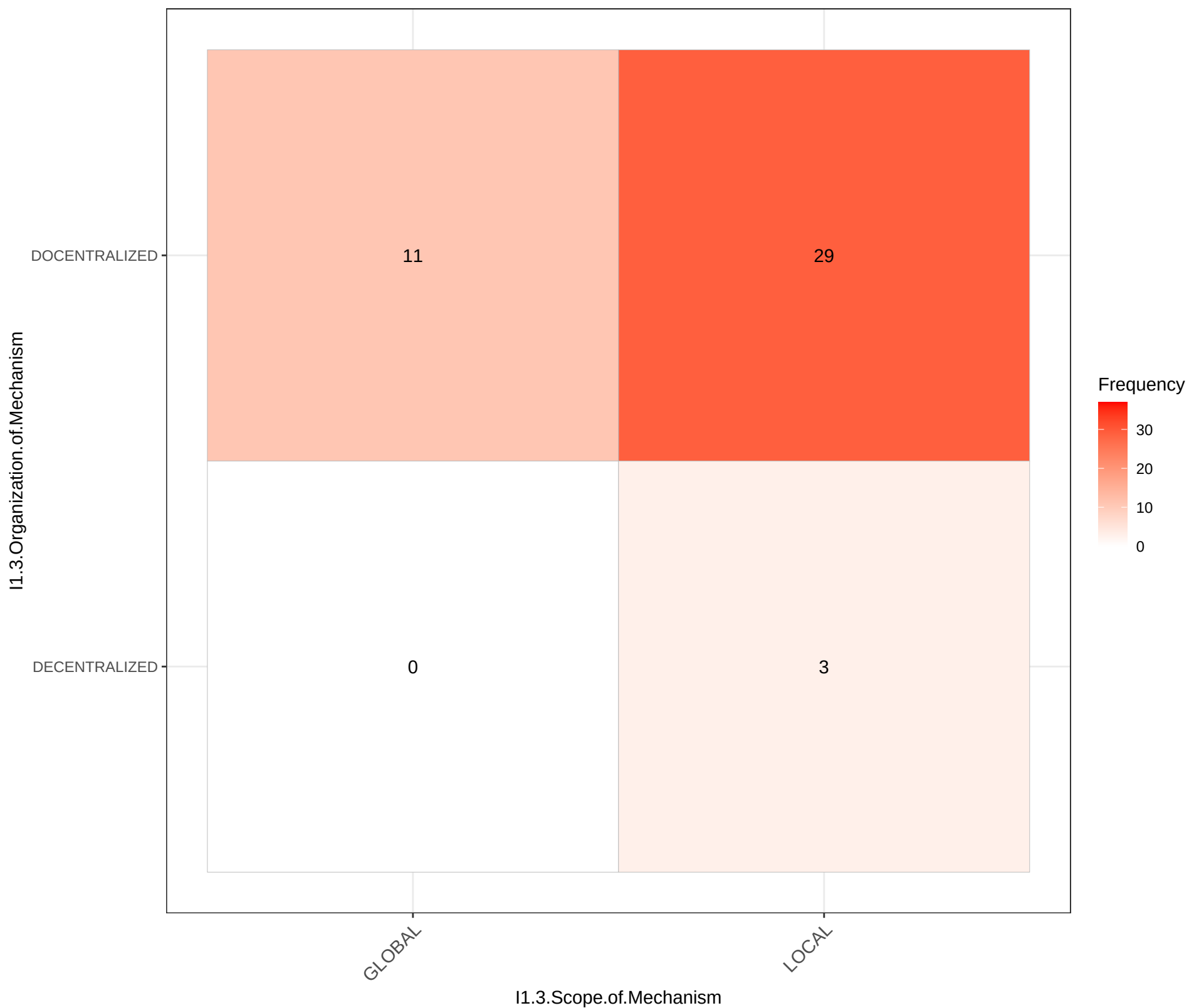
COMPONENTREDEPLOYMENT

REPARAMETERIZATIONOFCOMPONENTS

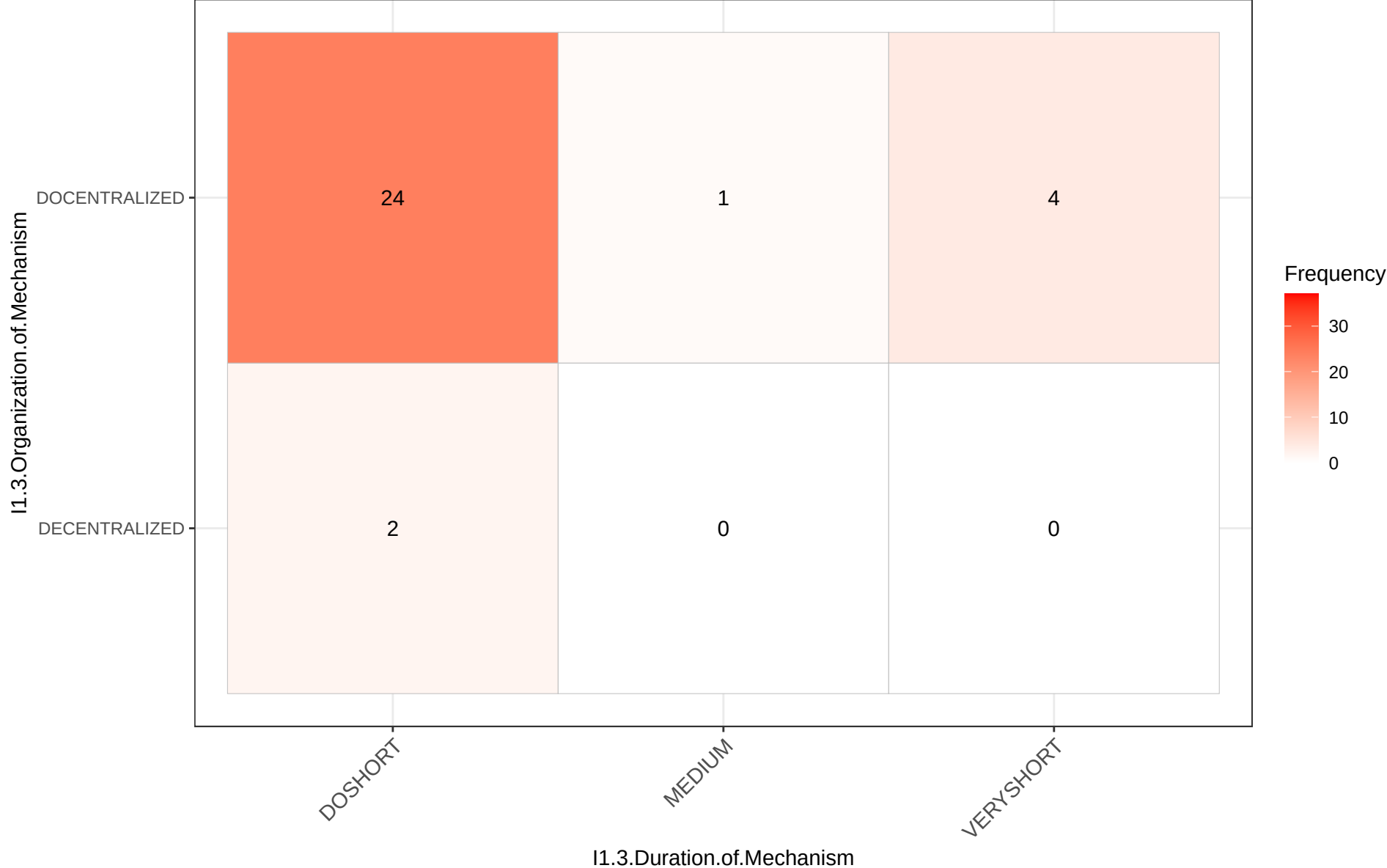
I13.Execute



I1.3.Organization.of.Mechanism_____I1.3.Scope.of.Mechanism



I1.3.Organization.of.Mechanism_____I1.3.Duration.of.Mechanism



I1.3.Organization.of.Mechanism_____I1.3.Timeliness.of.Mechanism

I1.3.Organization.of.Mechanism

DOCENTRALIZED

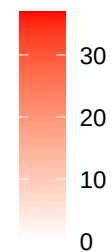
DECENTRALIZED

BESTEFFECT

GUARANTEED

I1.3.Timeliness.of.Mechanism

Frequency



28

1

3

0

I1.3.Organization.of.Mechanism_____I1.3.Trigger.of.Mechanism

I1.3.Organization.of.Mechanism

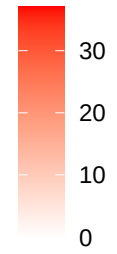
DOCENTRALIZED

DECENTRALIZED

EVENTTRIGGER

I1.3.Trigger.of.Mechanism

Frequency



30

20

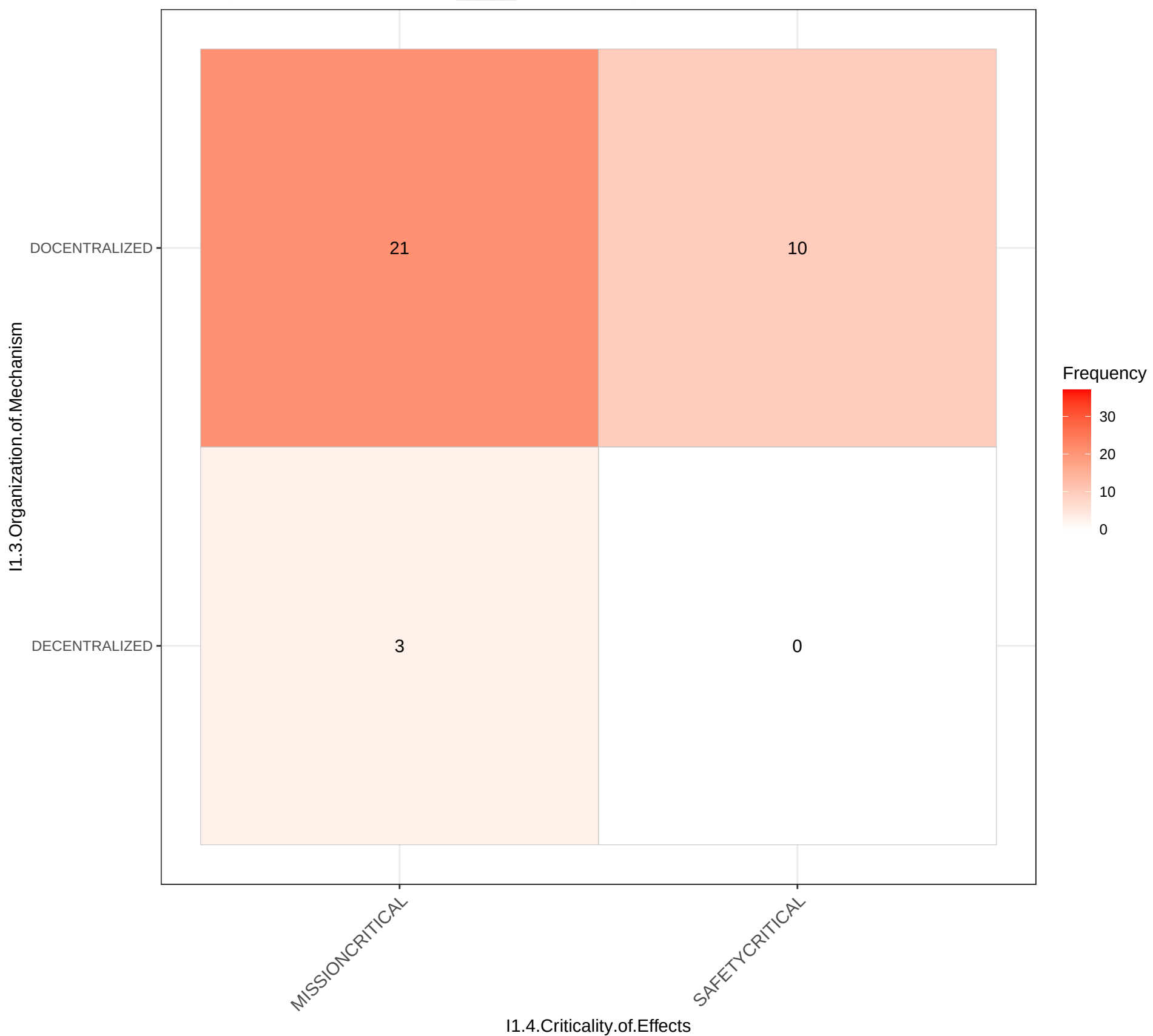
10

0

33

3

I1.3.Organization.of.Mechanism_____I1.4.Criticality.of.Effects



I1.3.Organization.of.Mechanism_____I1.4.Predictability.of.Effects

I1.3.Organization.of.Mechanism

DOCENTRALIZED

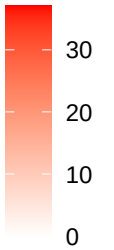
DECENTRALIZED

DODETERMINISTIC

NONDETERMINISTIC

I1.4.Predictability.of.Effects

Frequency

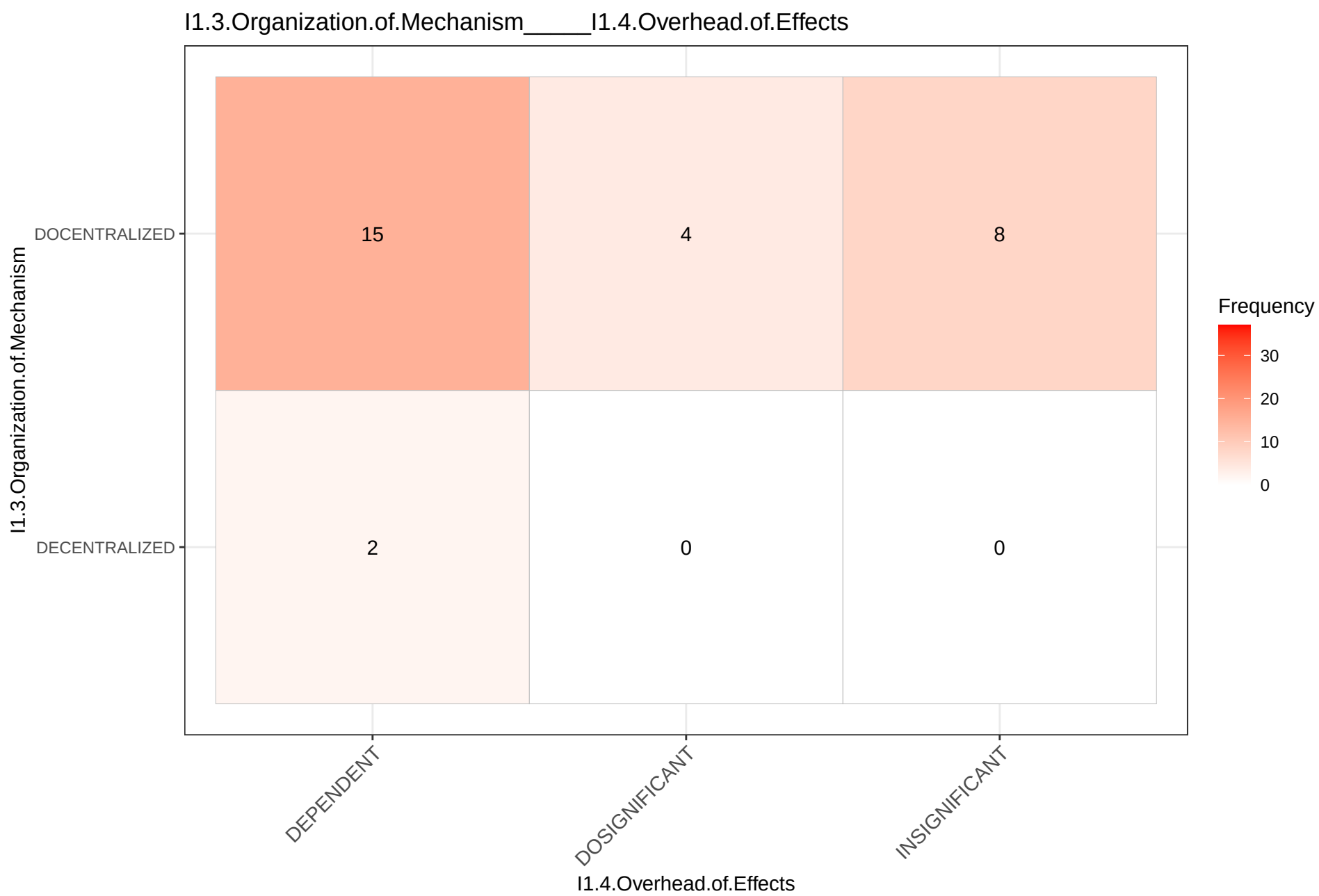


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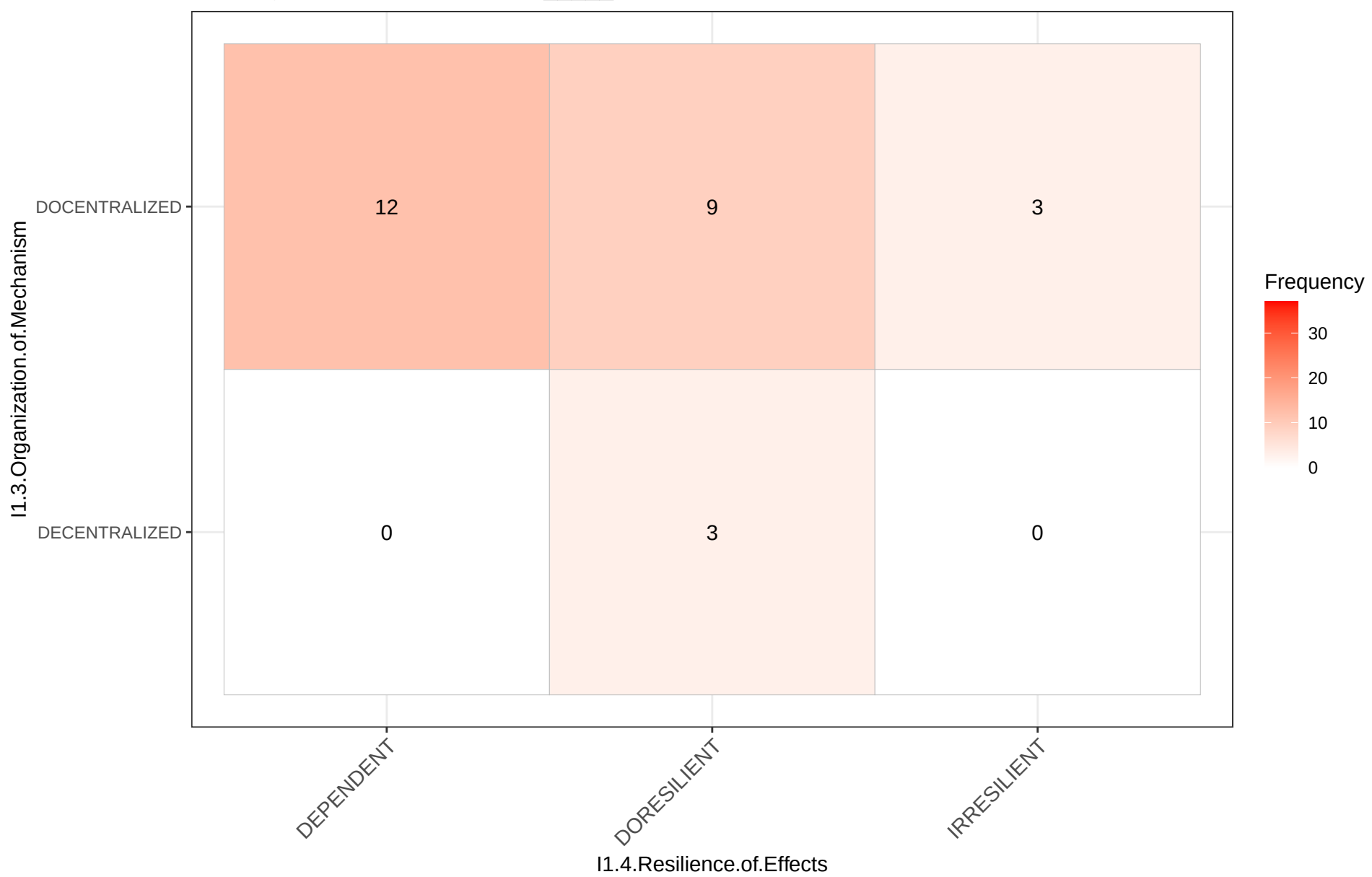
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1

2



I1.3.Organization.of.Mechanism_____I1.4.Resilience.of.Effects



I1.3.Organization.of.Mechanism_____I2.Adap..Purpose

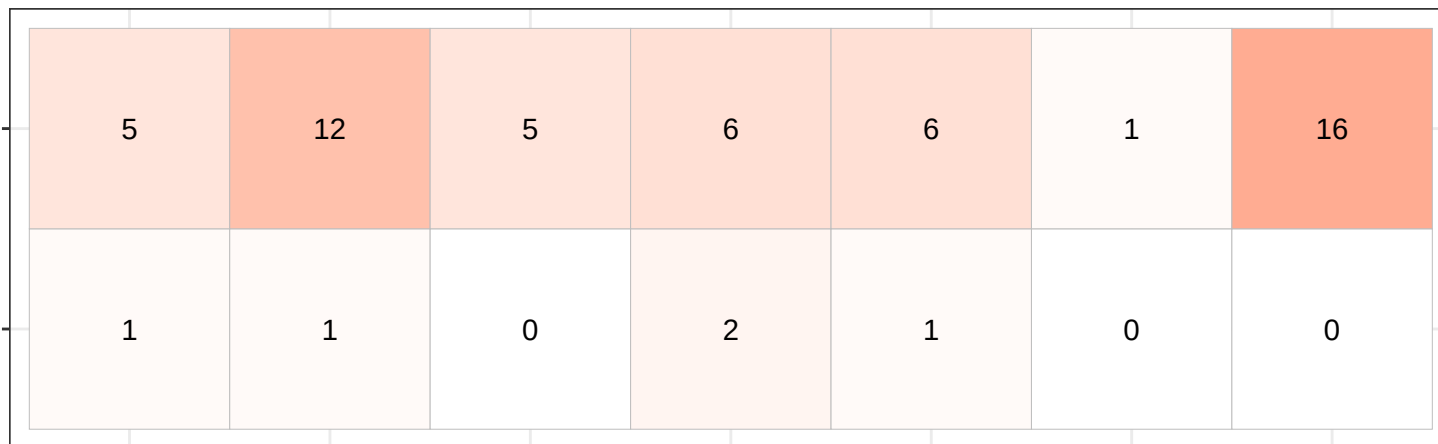
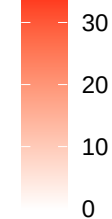
DOCENTRALIZED

DECENTRALIZED

CHANGEFUNCTIONALBEHAVIOR
DEALWITHENVIRONMENTALCHANGES
KEEPMEETINGQUALITYREQUIREMENTSATRUNTIME
OPTIMIZERESOURCEUSAGE
OPTIMIZESYSTEMPERFORMANCE
RECOVERFROMATTACKS
RECOVERFROMERRORSFAULTS

I2.Adap..Purpose

Frequency



I1.3.Organization.of.Mechanism	I3.Robot.Type
--------------------------------	---------------

Frequency

I3.Robot.Type

I1.3.Organization.of.Mechanism_____I4.Robo.SW

I1.3.Organization.of.Mechanism

DOCENTRALIZED

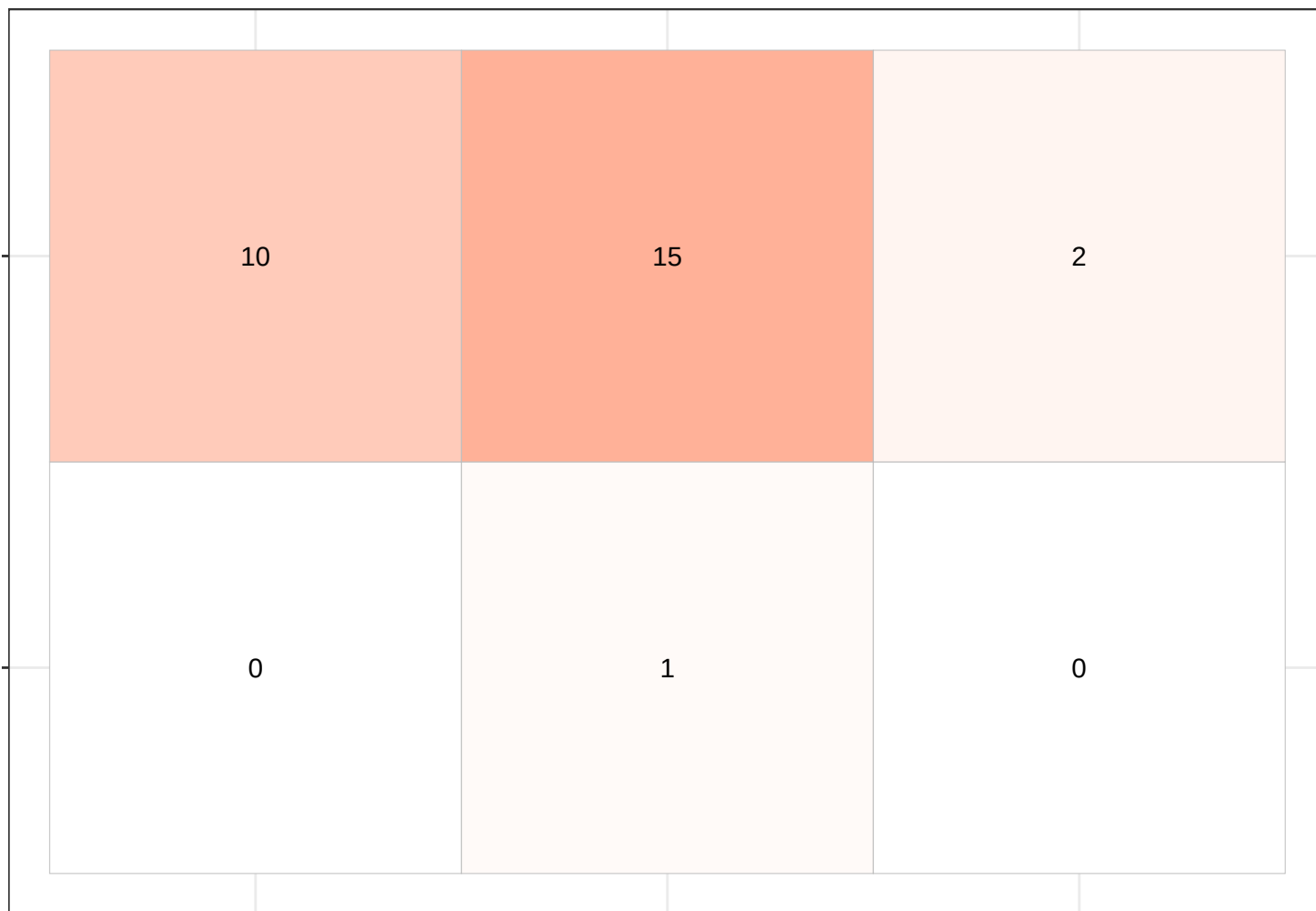
DECENTRALIZED

OTHER

ROS1

ROS2

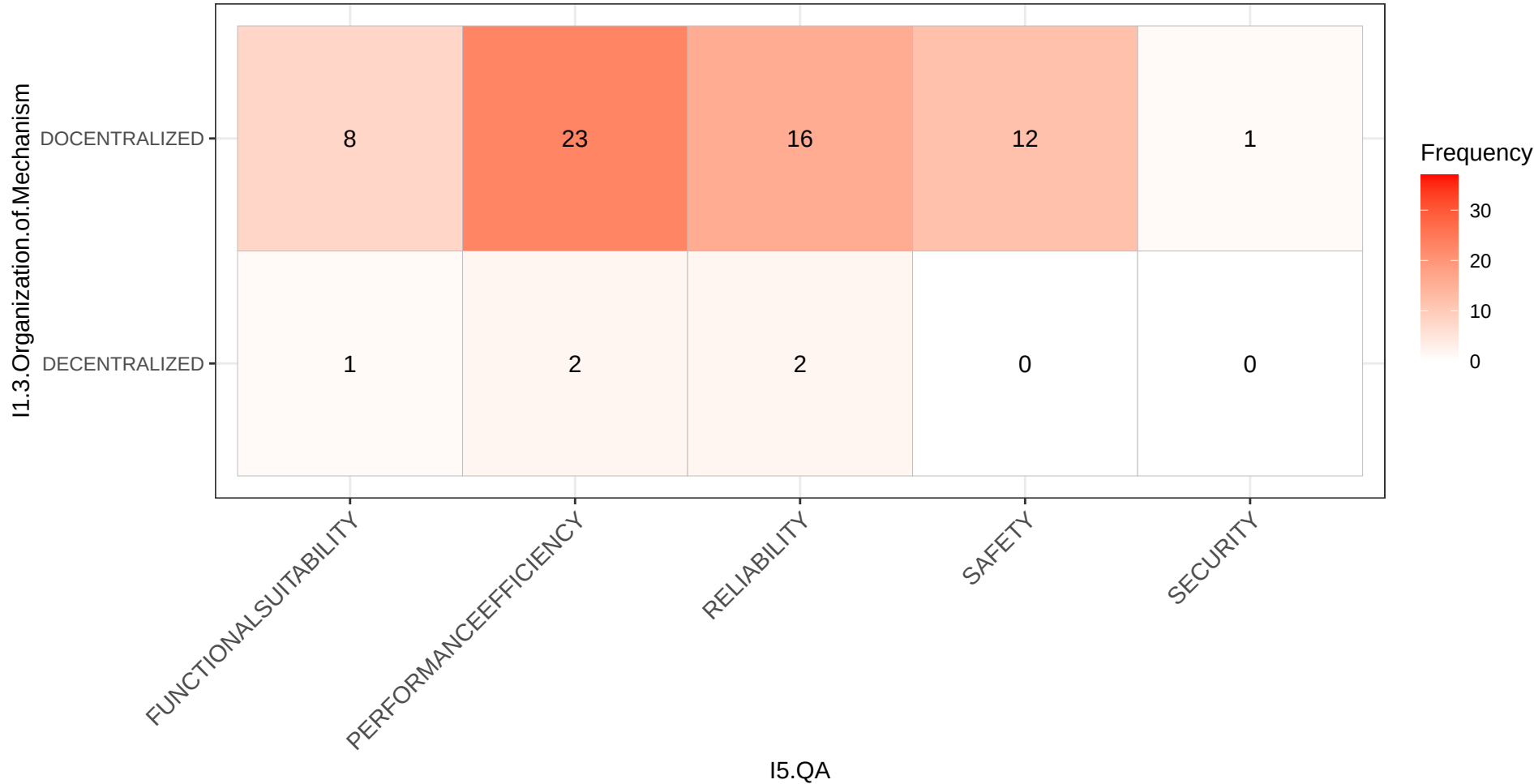
I4.Robo.SW



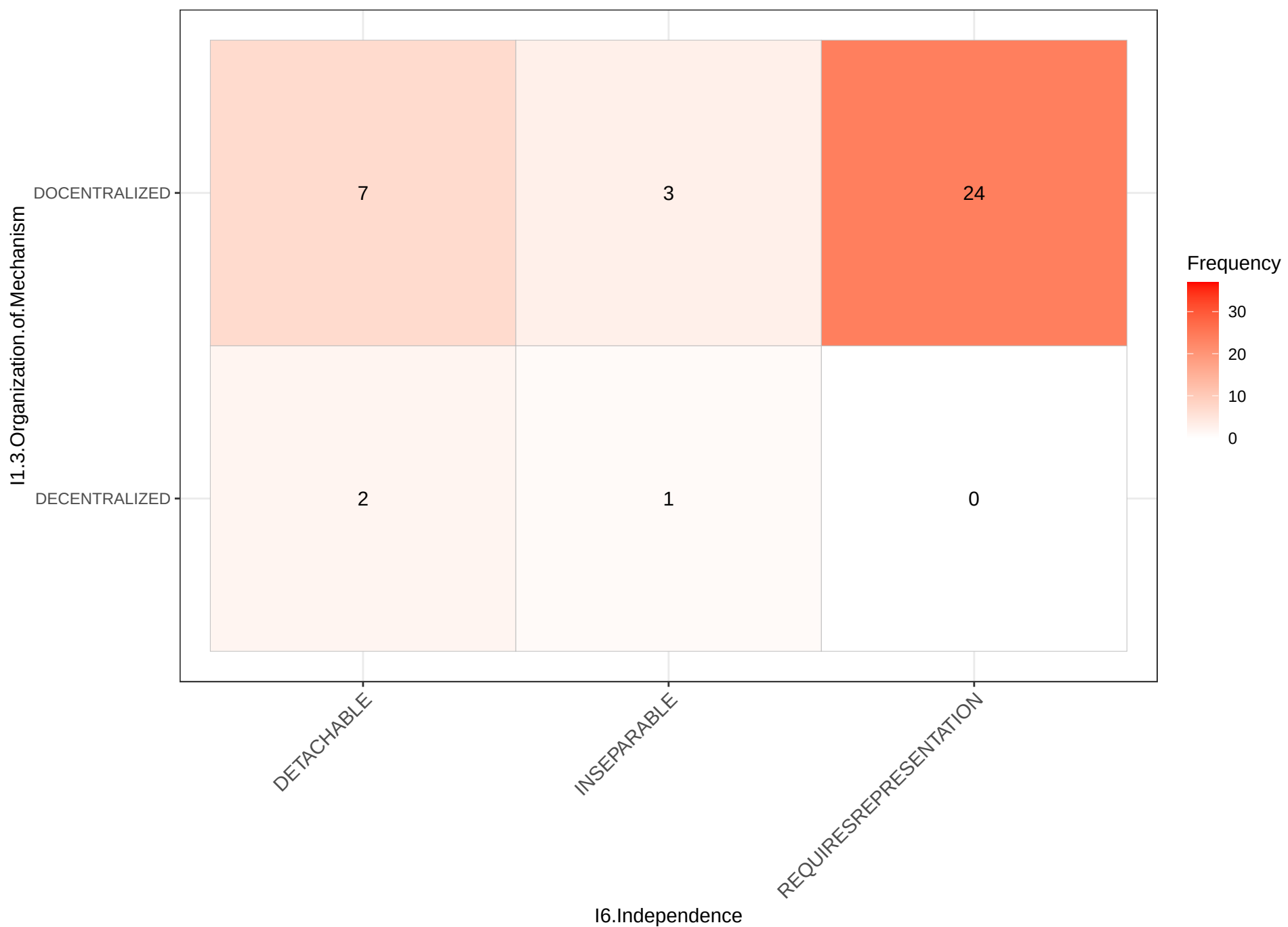
Frequency



I1.3.Organization.of.Mechanism_____I5.QA

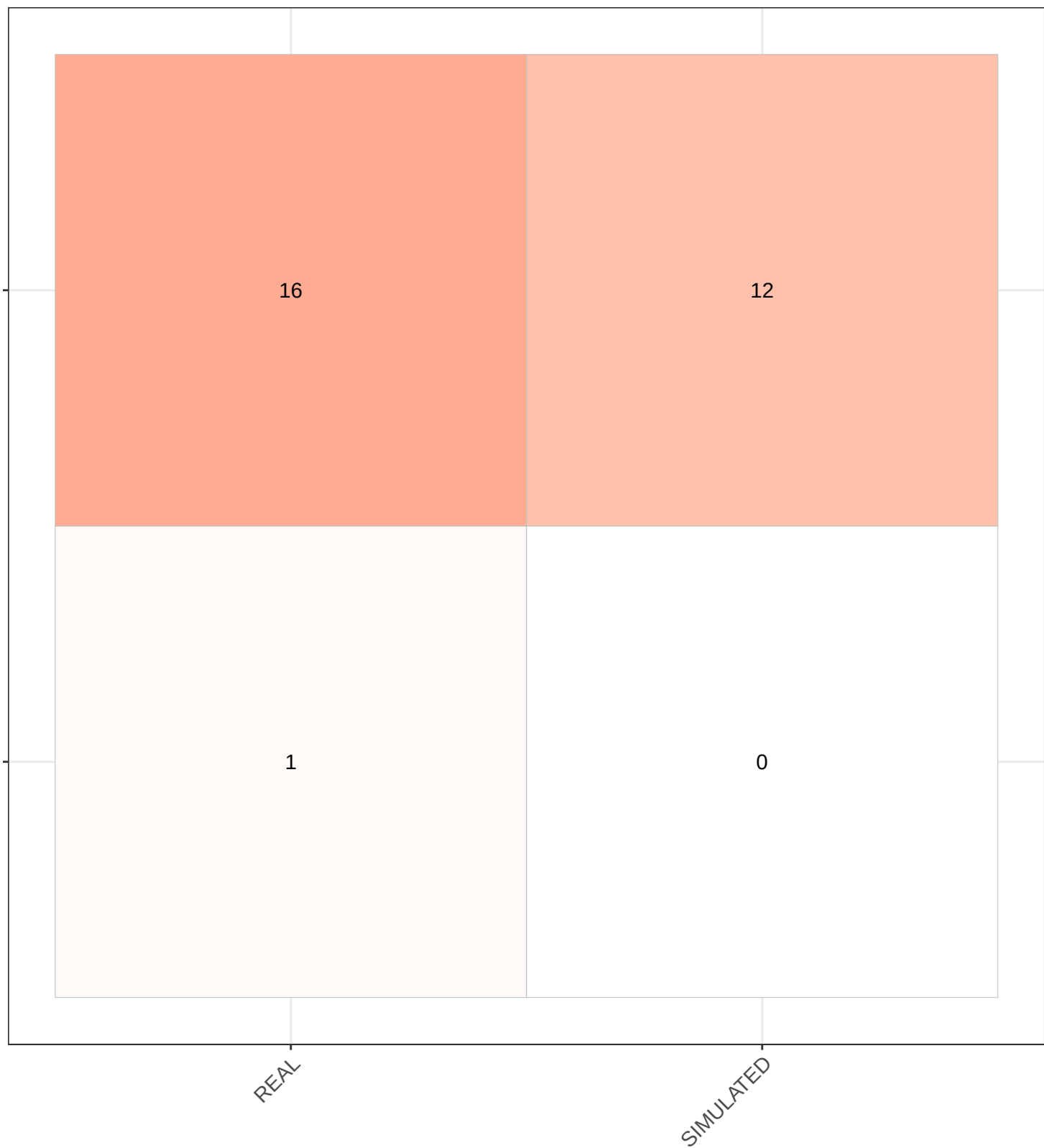


I1.3.Organization.of.Mechanism_____I6.Independence



I1.3.Organization.of.Mechanism_____I7.Deployment.Realness

I1.3.Organization.of.Mechanism



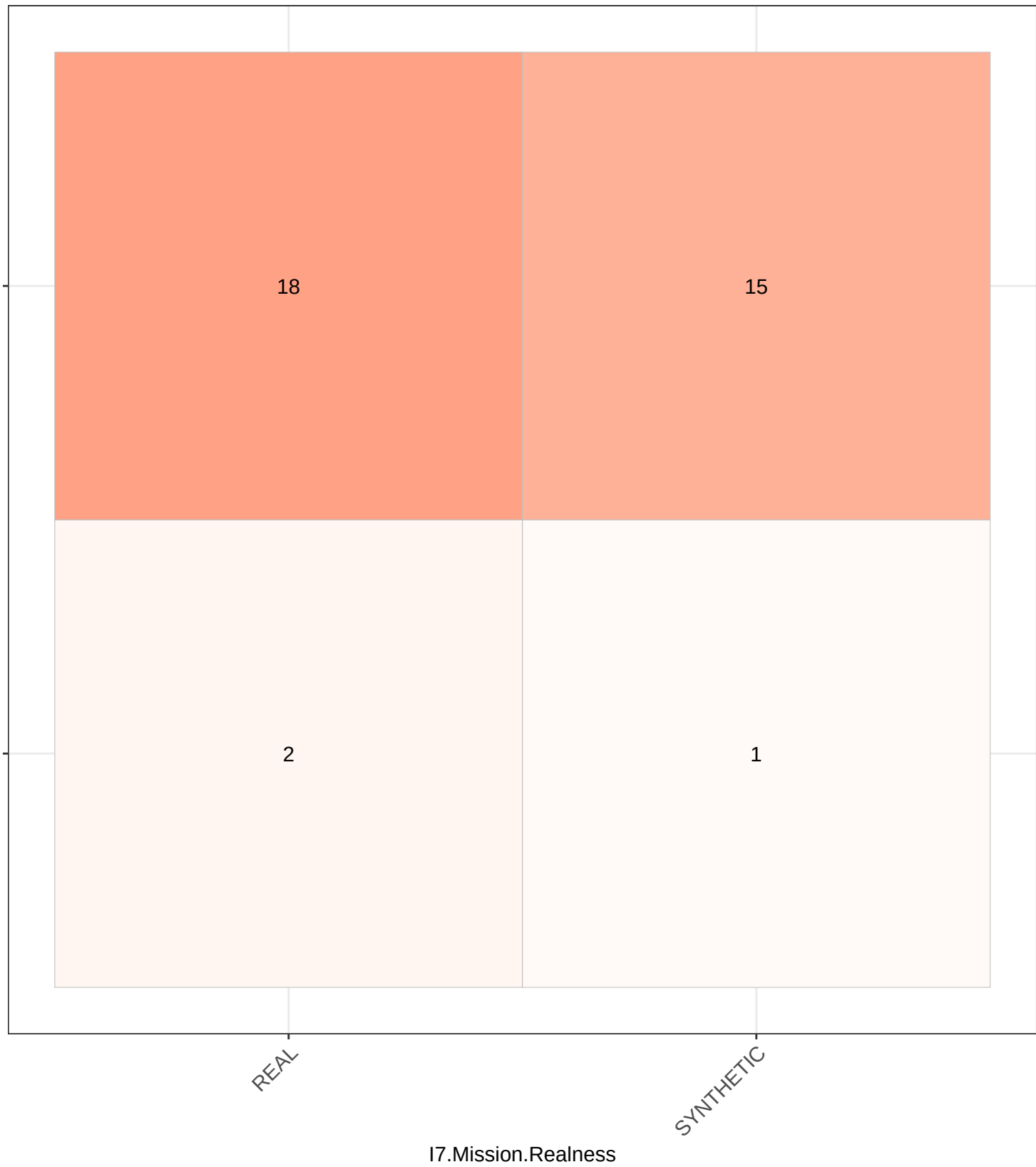
Frequency



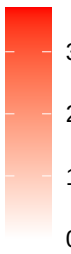
I7.Deployment.Realness

I1.3.Organization.of.Mechanism_____I7.Mission.Realness

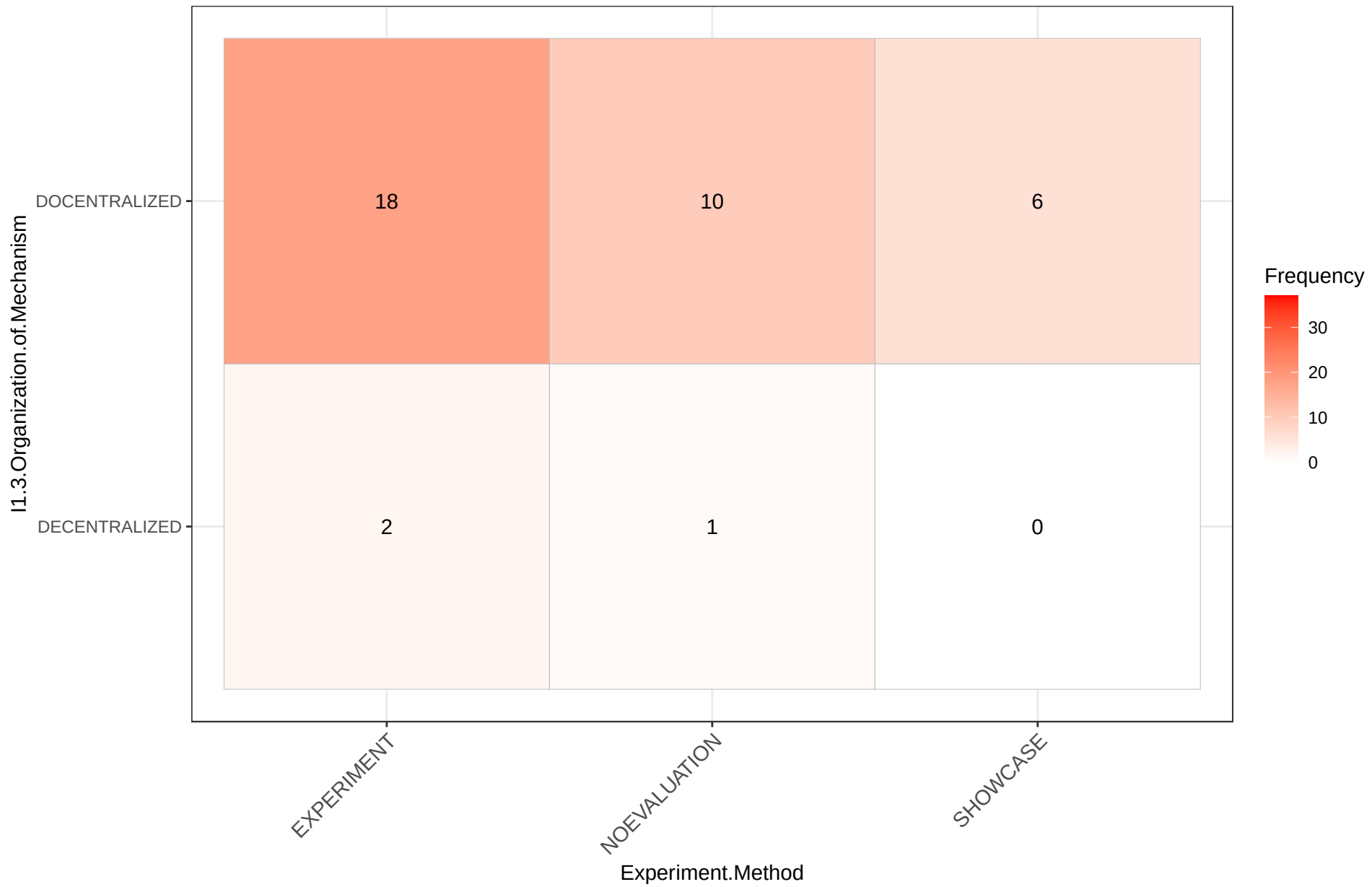
I1.3.Organization.of.Mechanism



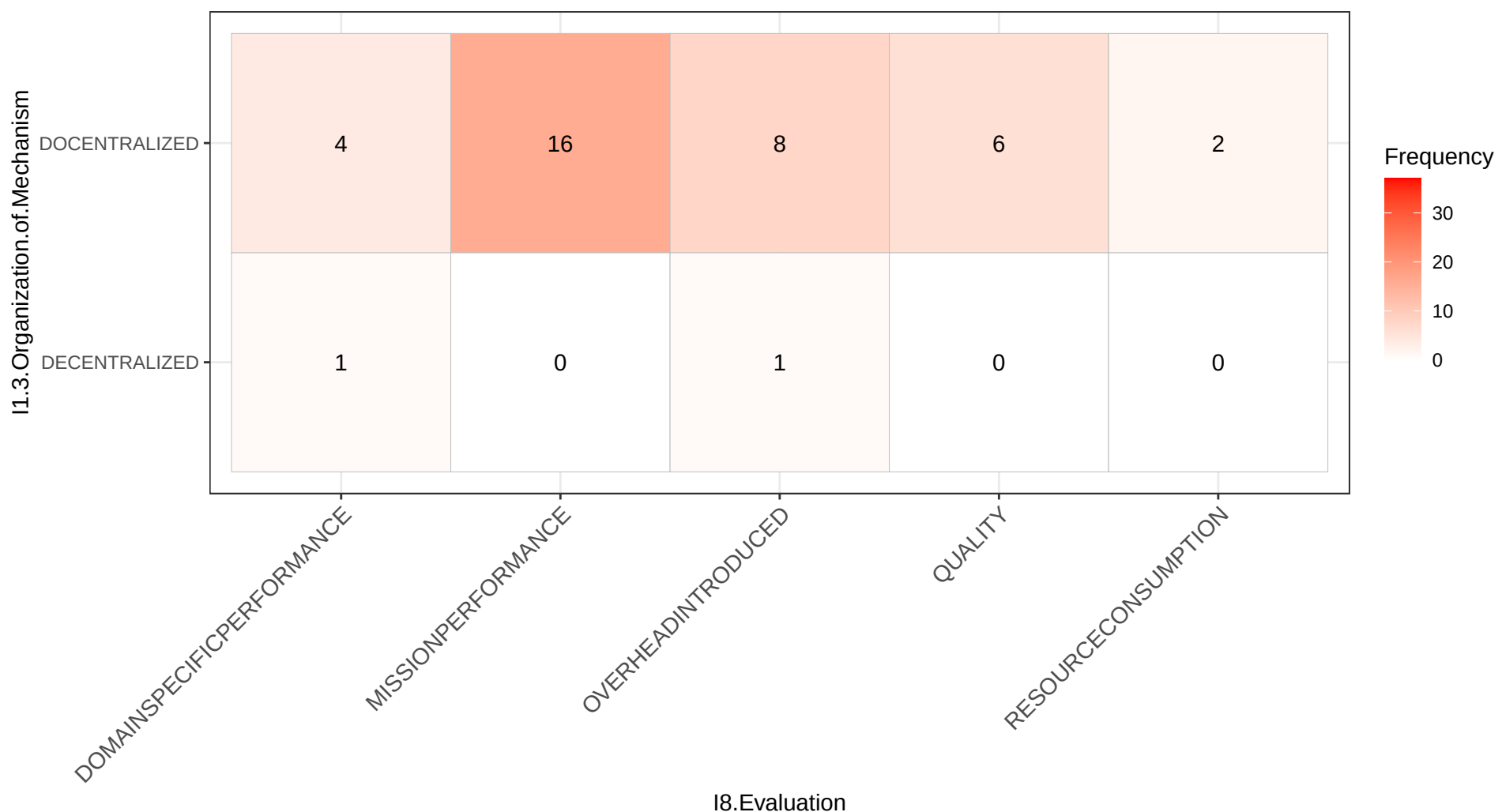
Frequency



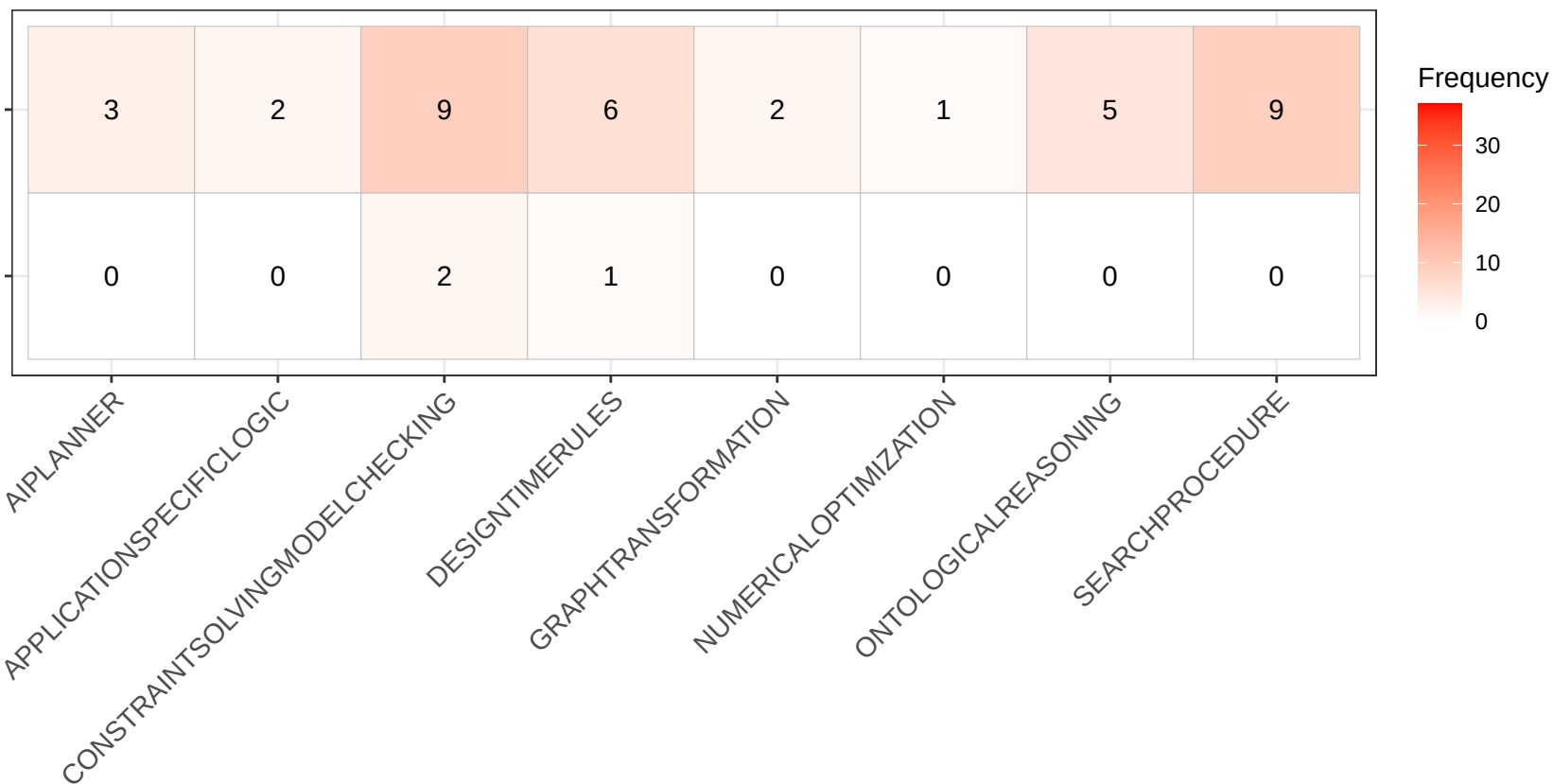
I1.3.Organization.of.Mechanism____Experiment.Method



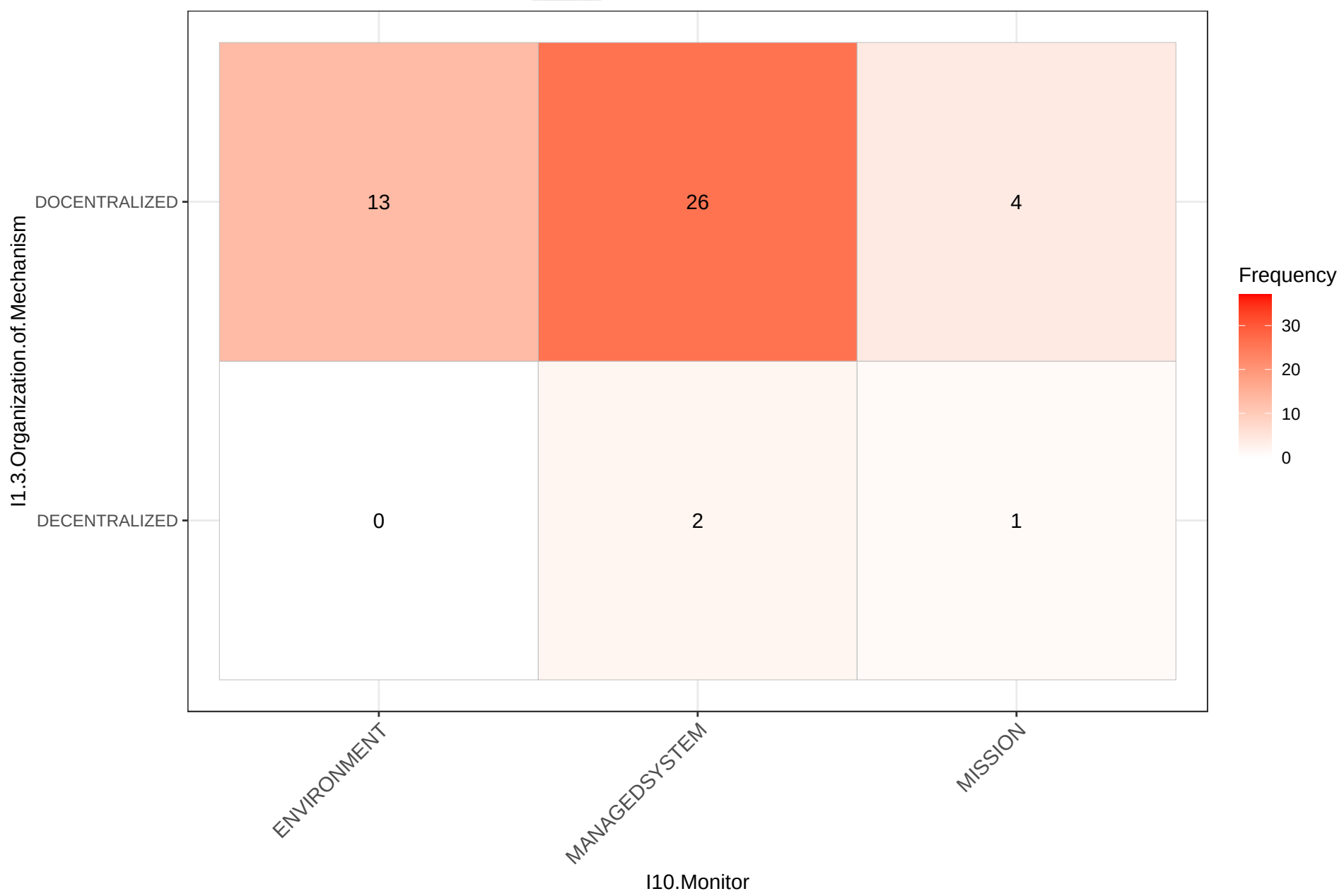
I1.3.Organization.of.Mechanism_____I8.Evaluation



I1.3.Organization.of.Mechanism_____I9.Adap..Logic

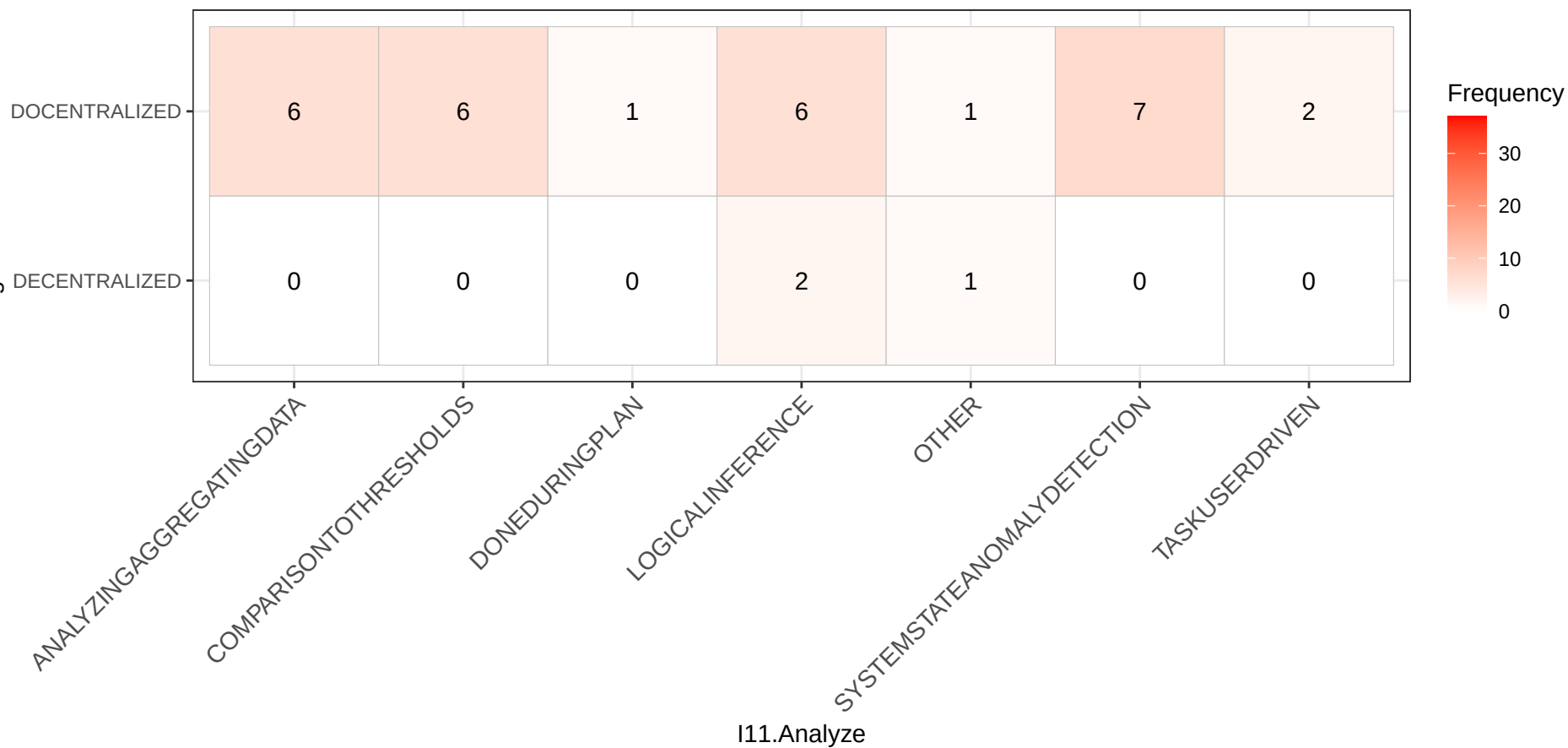


I1.3.Organization.of.Mechanism_____I10.Monitor

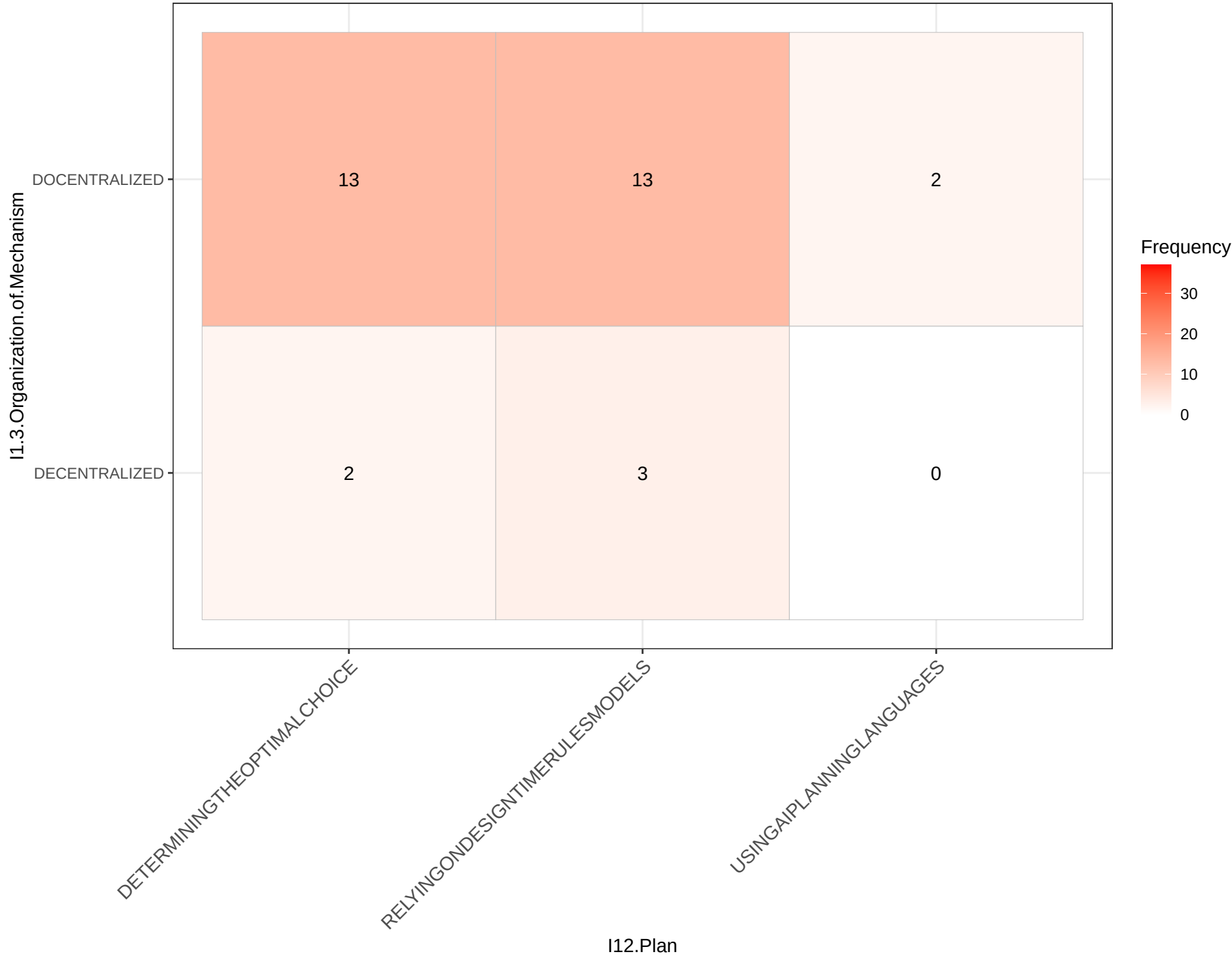


I1.3.Organization.of.Mechanism

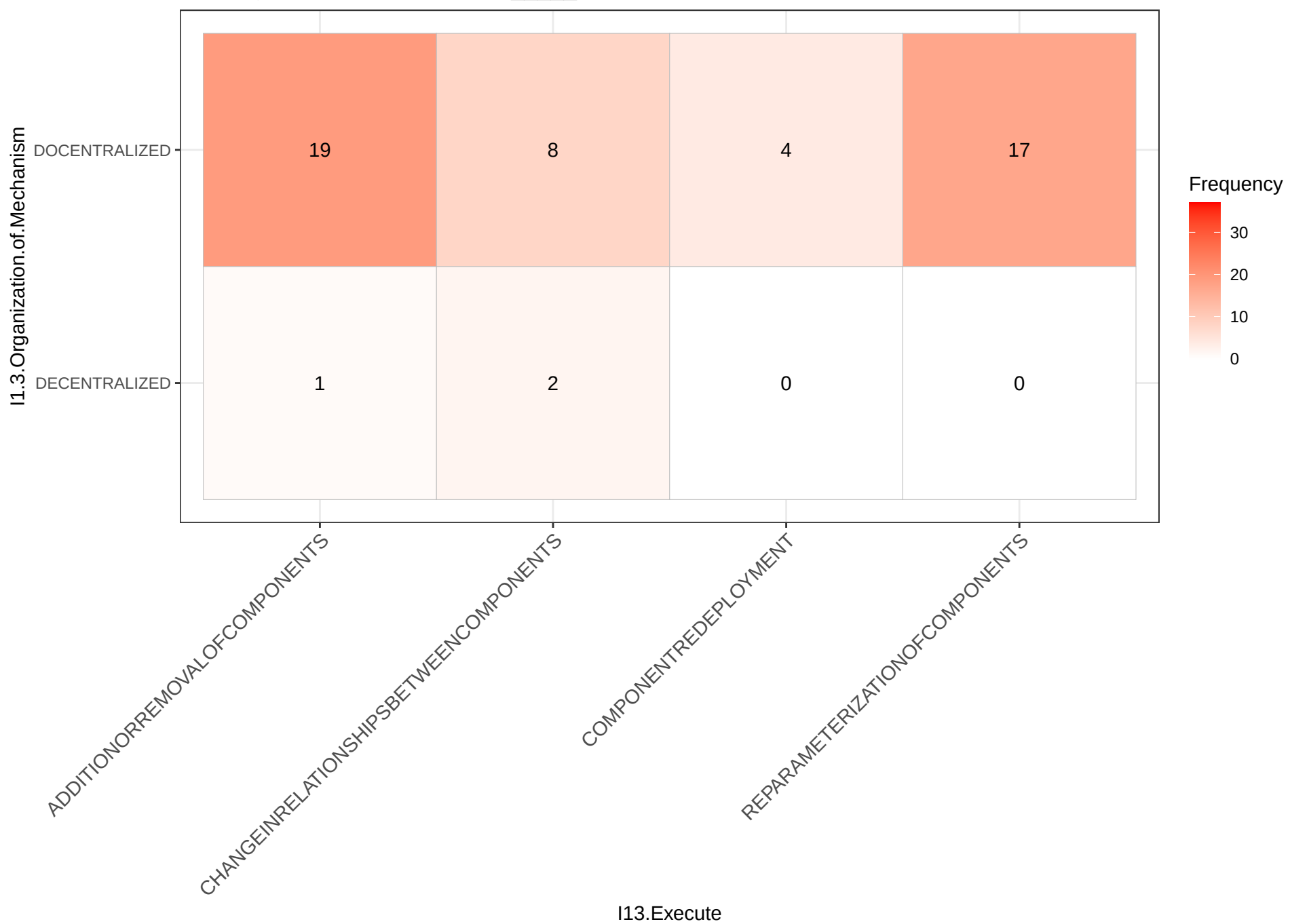
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I1.3.Organization.of.Mechanism_____I12.Plan

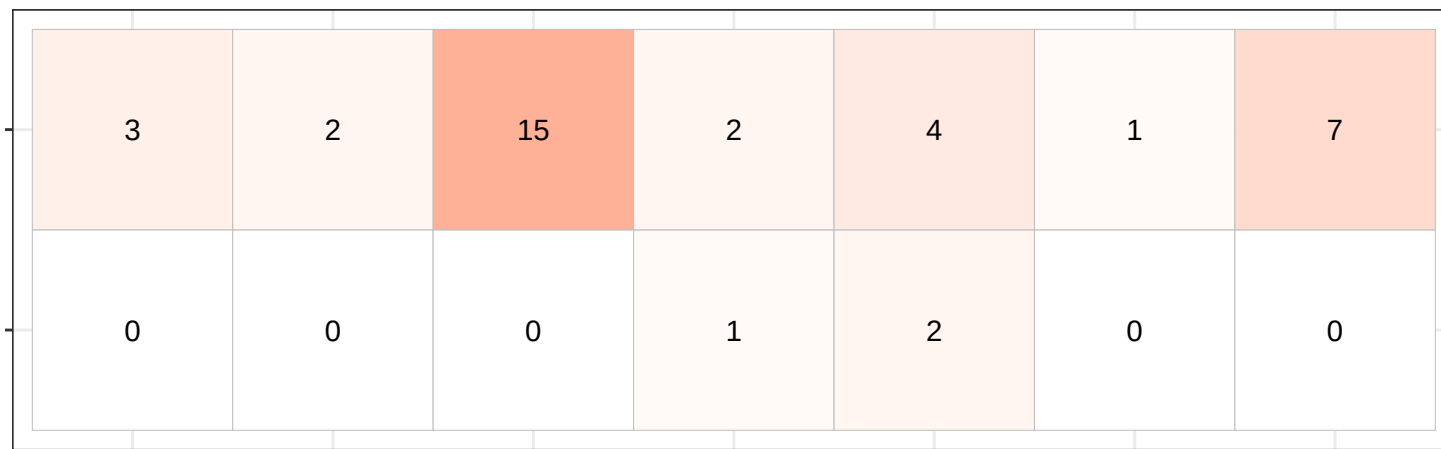


I1.3.Organization.of.Mechanism_____I13.Execute



I1.3.Organization.of.Mechanism

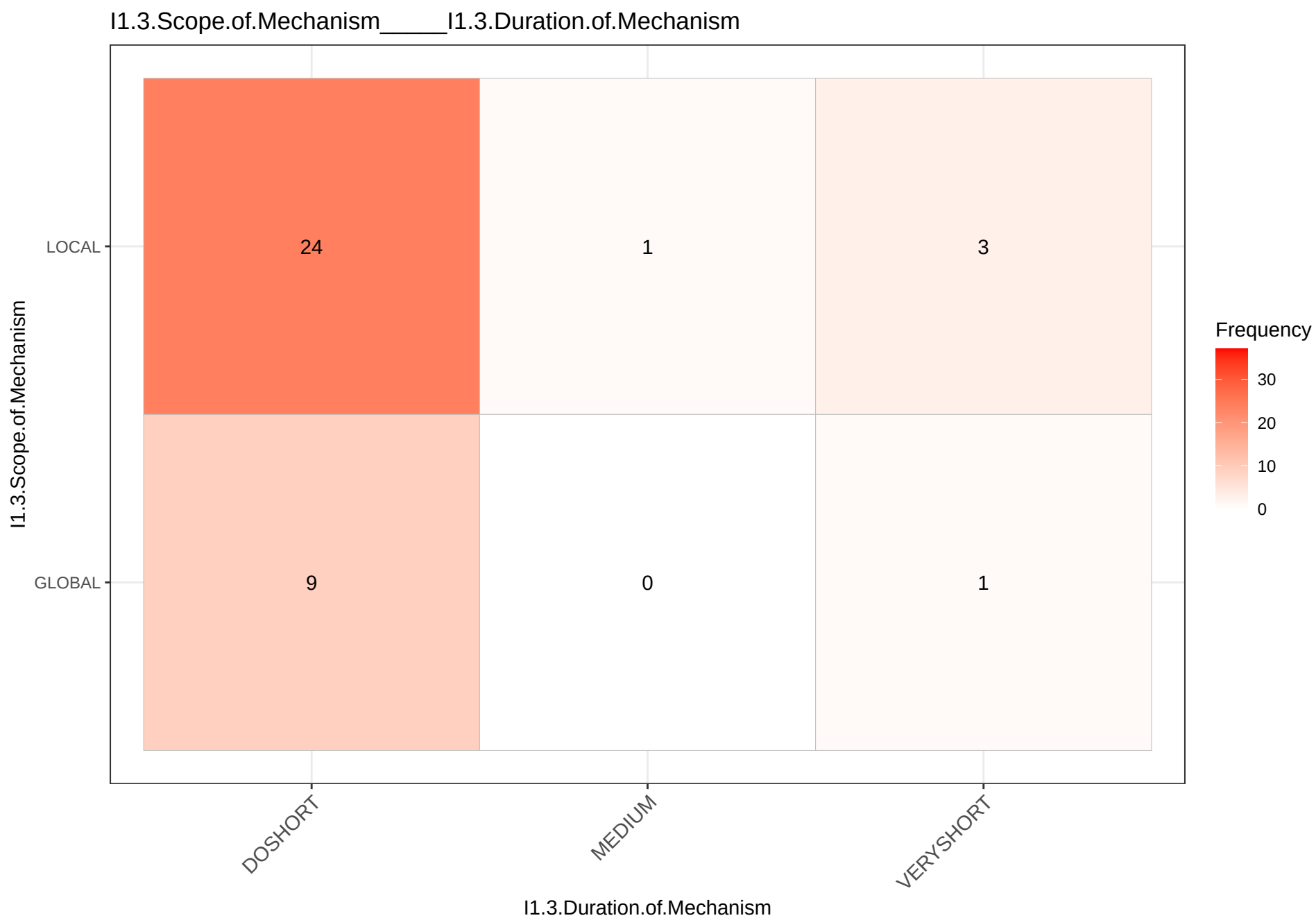
I1.3.Organization.of.Mechanism_____I14.Knowledge



Frequency



I14.Knowledge



I1.3.Scope.of.Mechanism

LOCAL

28

1

GLOBAL

10

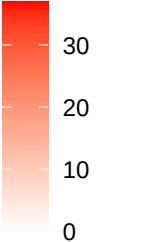
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BESTEFFORT

GUARANTEED

I1.3.Timeliness.of.Mechanism

Frequency



I1.3.Scope.of.Mechanism_____I1.4.Criticality.of.Effects

I1.3.Scope.of.Mechanism

LOCAL

20

9

GLOBAL

4

5

MISSIONCRITICAL

SAFETYCRITICAL

I1.4.Criticality.of.Effects

Frequency



30

20

10

0

I1.3.Scope.of.Mechanism_____I1.4.Predictability.of.Effects

I1.3.Scope.of.Mechanism

LOCAL

6

21

GLOBAL

3

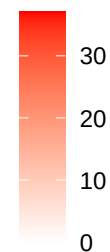
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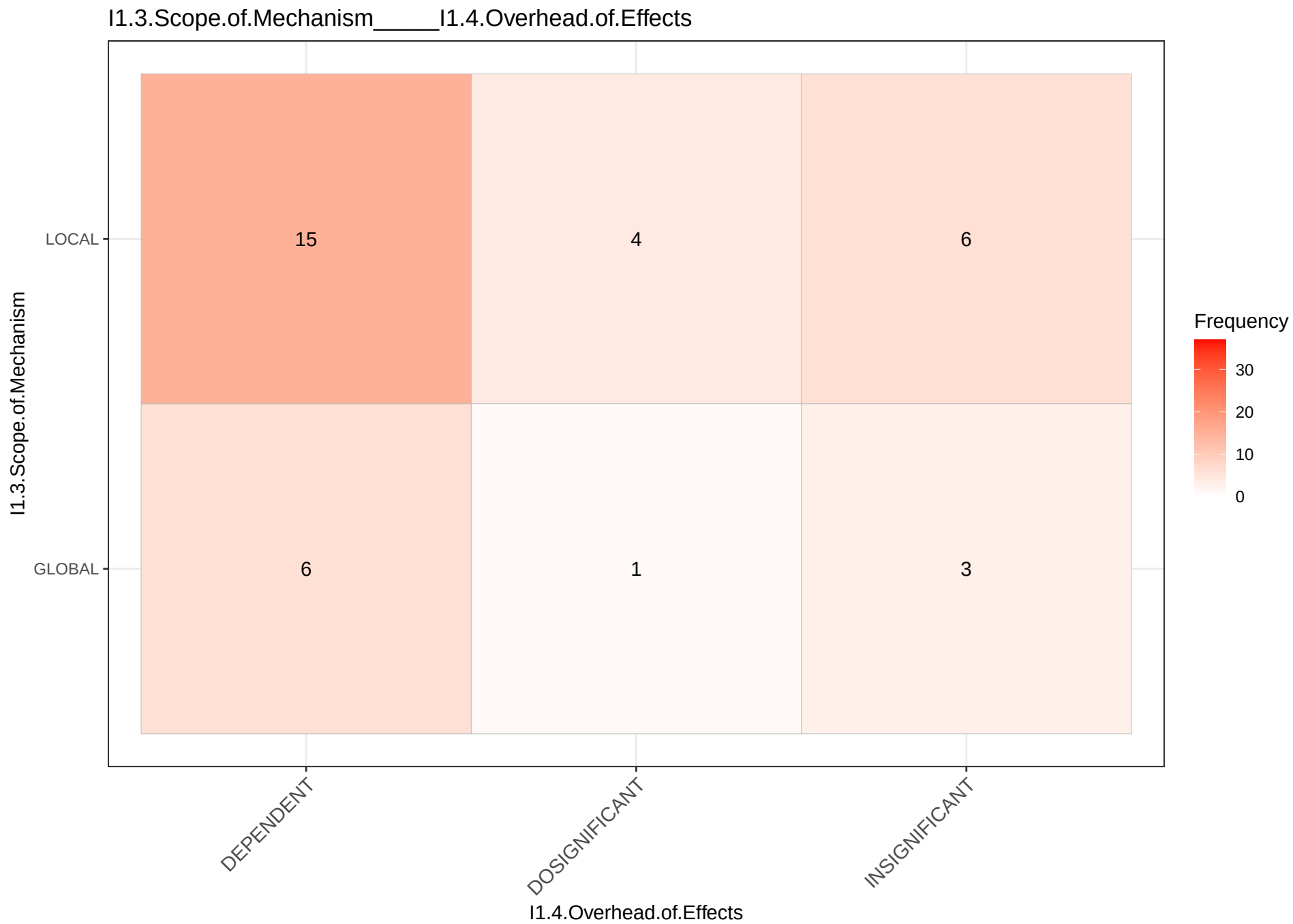
DODETERMINISTIC

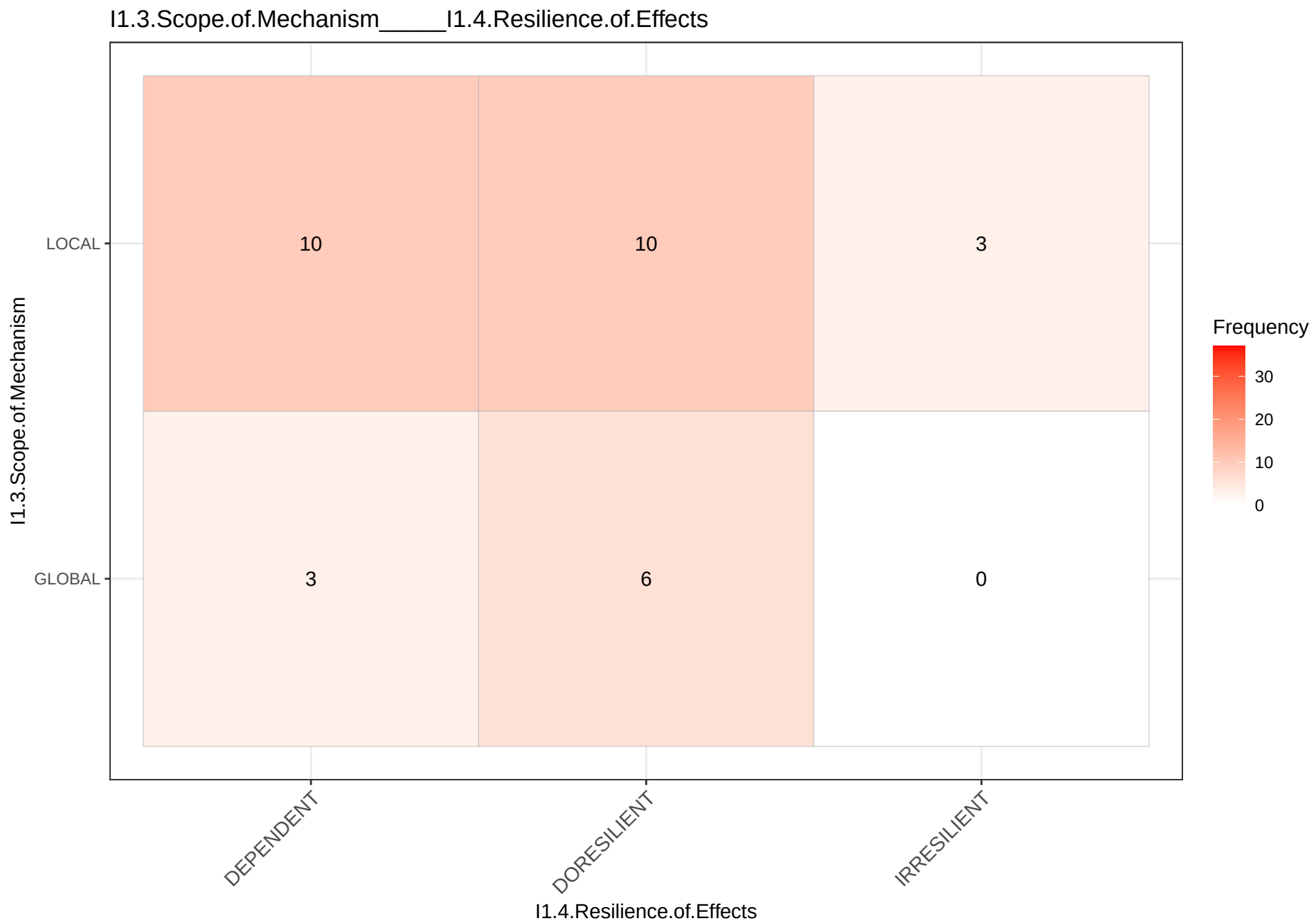
NONDETERMINISTIC

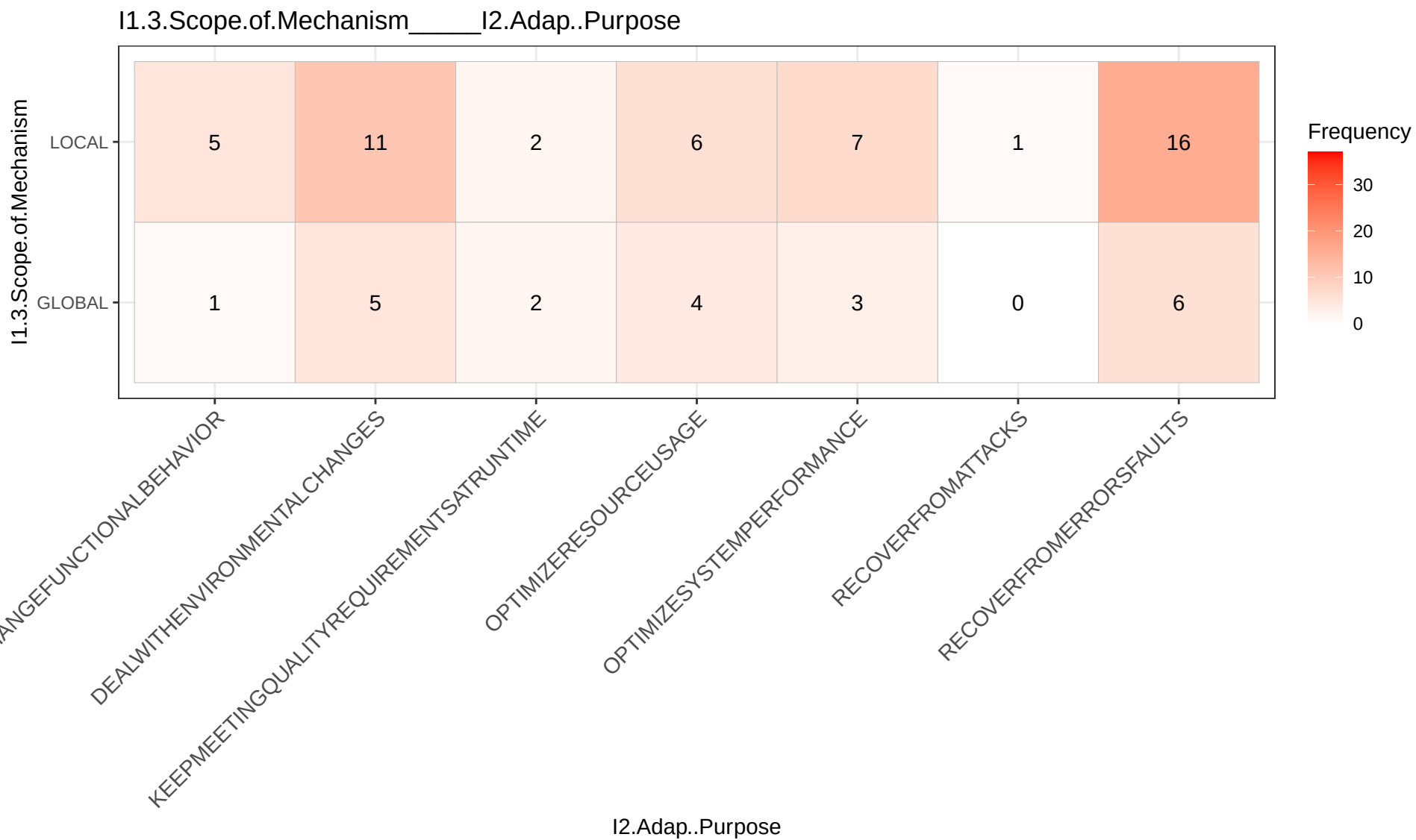
I1.4.Predictability.of.Effects

Frequency









	I1.3.Scope.of.Mechanism										I3.Robot.Type																						
LOCAL	1	1	1	1	1	1	2	0	2	0	1	1	1	2	1	0	0	1	1	2	1	1	1	1	1	0	4	0	1	1	1	1	2
GLOBAL	0	1	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	0	0	2	0	0	0	0	0	1	4	1	0	0	0	0	0
	TSPADERBORN	BLUEROV2	BOXERCLEARPATH	CRAWLERTERMINATORBOT	FIELDMOBILEROBOTS	FRACTIONATEDSPACECRAFT	HETEROGENOUSROBOTS	INFOTAINMENTROBOT	MOBILESERVICEROBOT	HEXMANIPULATOR	HEXAII	IROBOTCREATE2	MOBILEROBOTTERMINATOR	MOBILEROBOT	MOBILEROBOTS	MOBILEROBOTTRIAL	MOBILESERVICEROBOT	MULTIPLEHETEROBOTS	NAOROBOT	PIONEER3DX	QUADROCOPTER	SINGLESERVOINGROTATIONROBOT	TEDUSAR	TERRESTRIALSEARCH	TRIGLIDEINDUSTRIALASSEMBLY	TURTLEBOT2	KOBUKI	WAREHOUSEDELIVERYROBOTS	WHICHISANINDUSTRIALAGV	YOUTBOT			

I3.Robot.Type

I1.3.Scope.of.Mechanism_____I4.Robo.SW

I1.3.Scope.of.Mechanism

LOCAL

GLOBAL

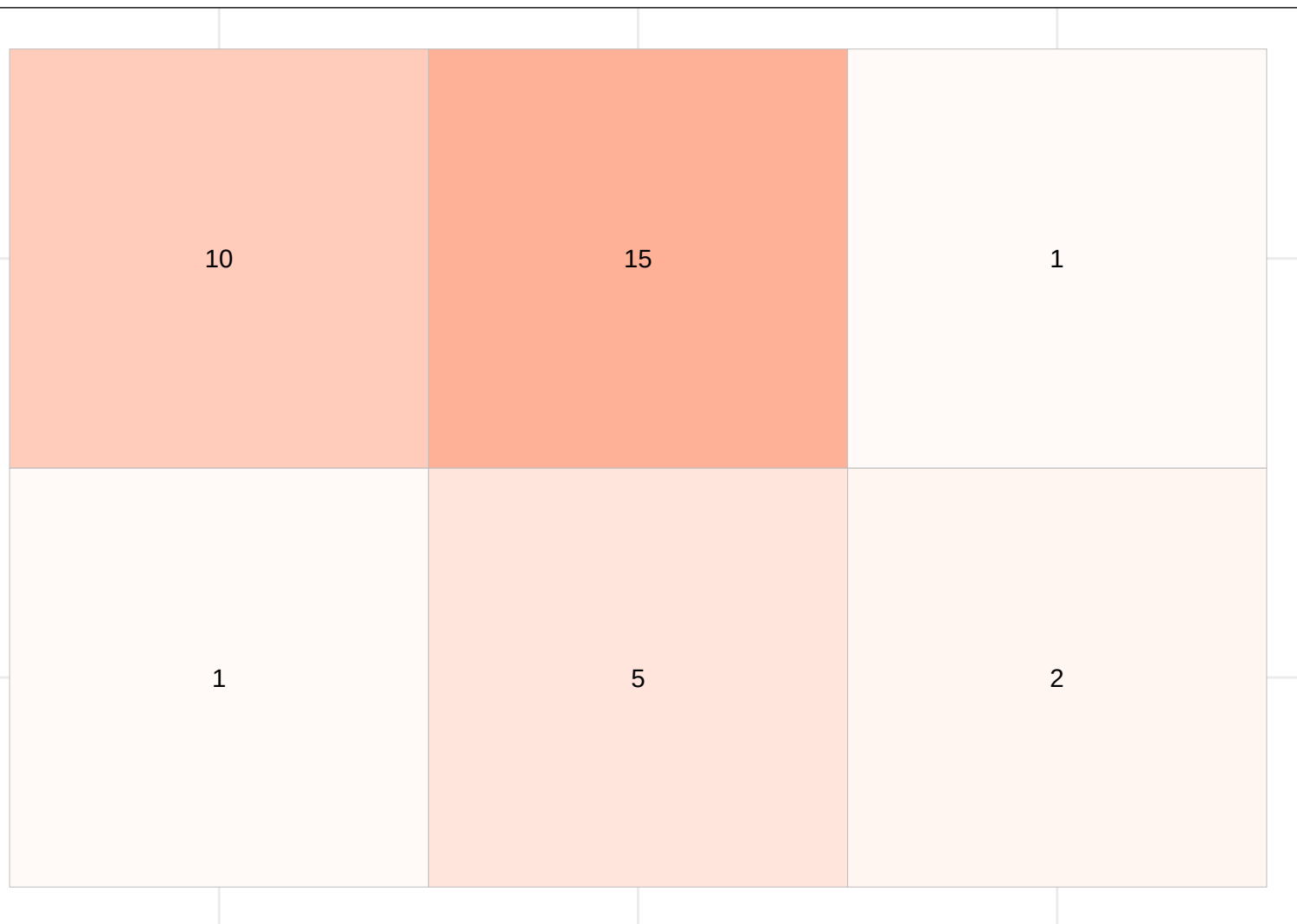
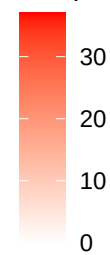
OTHER

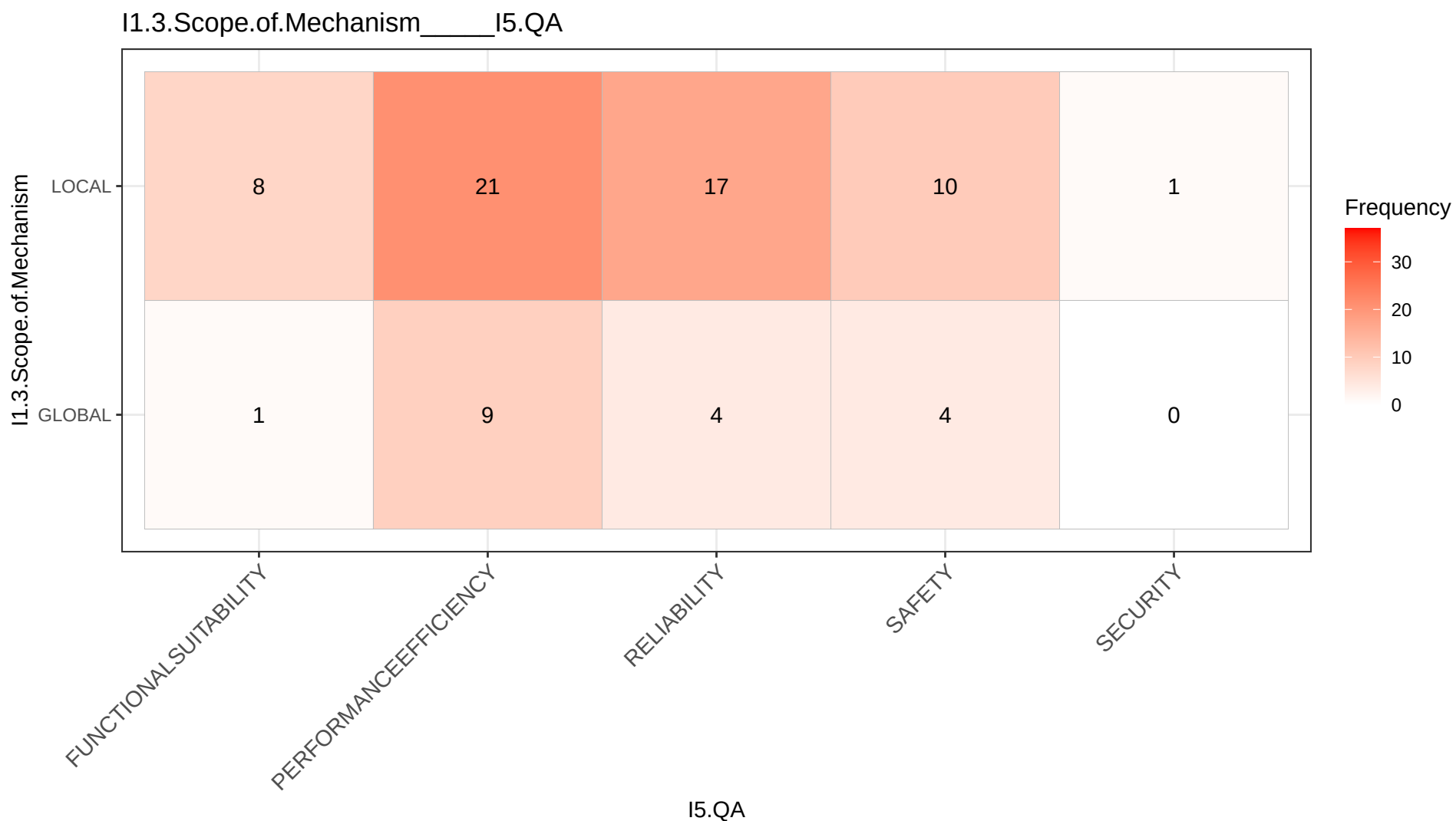
ROS1

ROS2

I4.Robo.SW

Frequency





I1.3.Scope.of.Mechanism_____I6.Independence

I1.3.Scope.of.Mechanism

LOCAL

GLOBAL

DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

Frequency



7

4

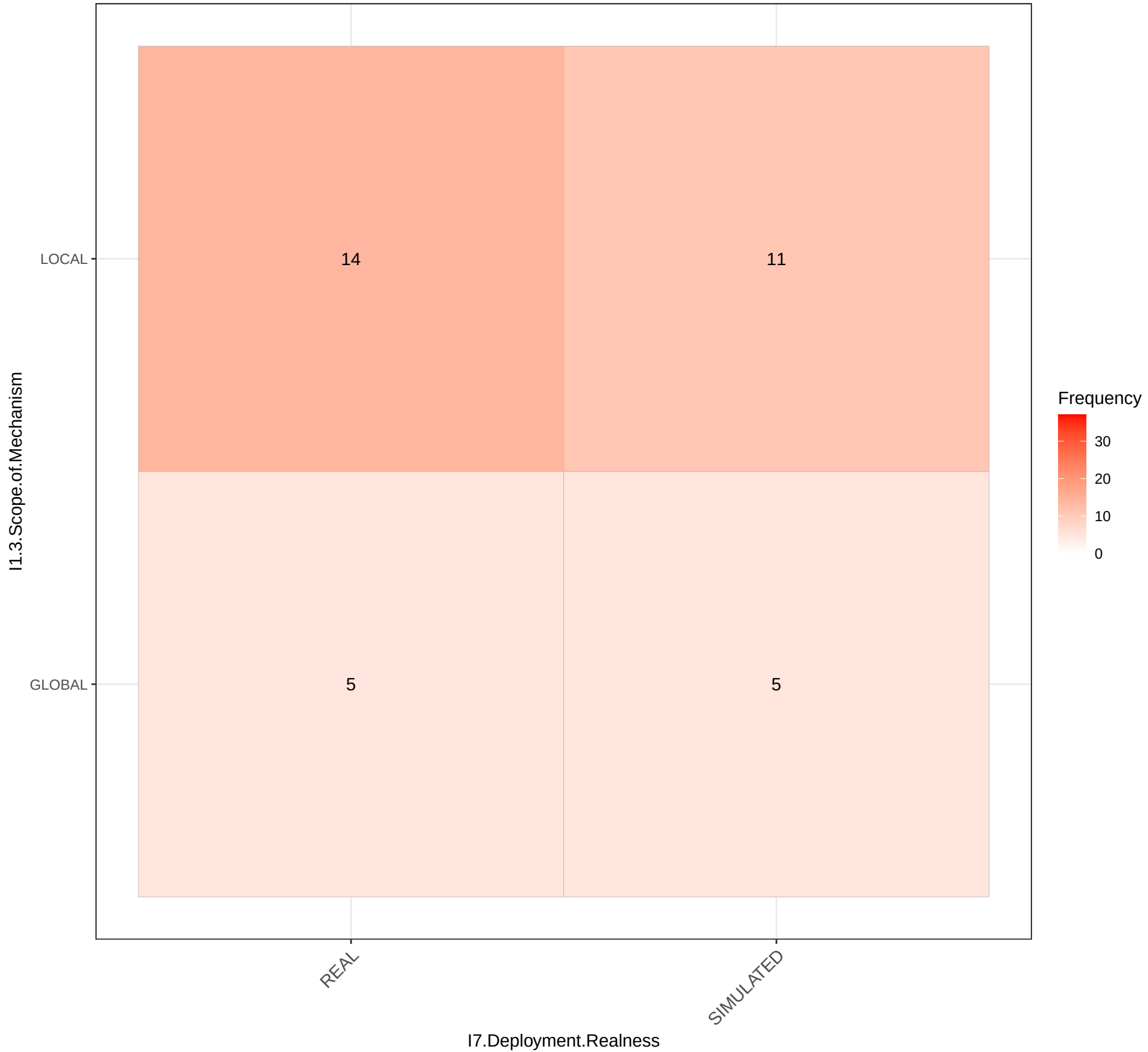
21

5

1

5

I1.3.Scope.of.Mechanism_____I7.Deployment.Realness



I1.3.Scope.of.Mechanism_____I7.Mission.Realness

I1.3.Scope.of.Mechanism

LOCAL

16

15

GLOBAL

4

6

REAL

SYNTHETIC

I7.Mission.Realness

Frequency



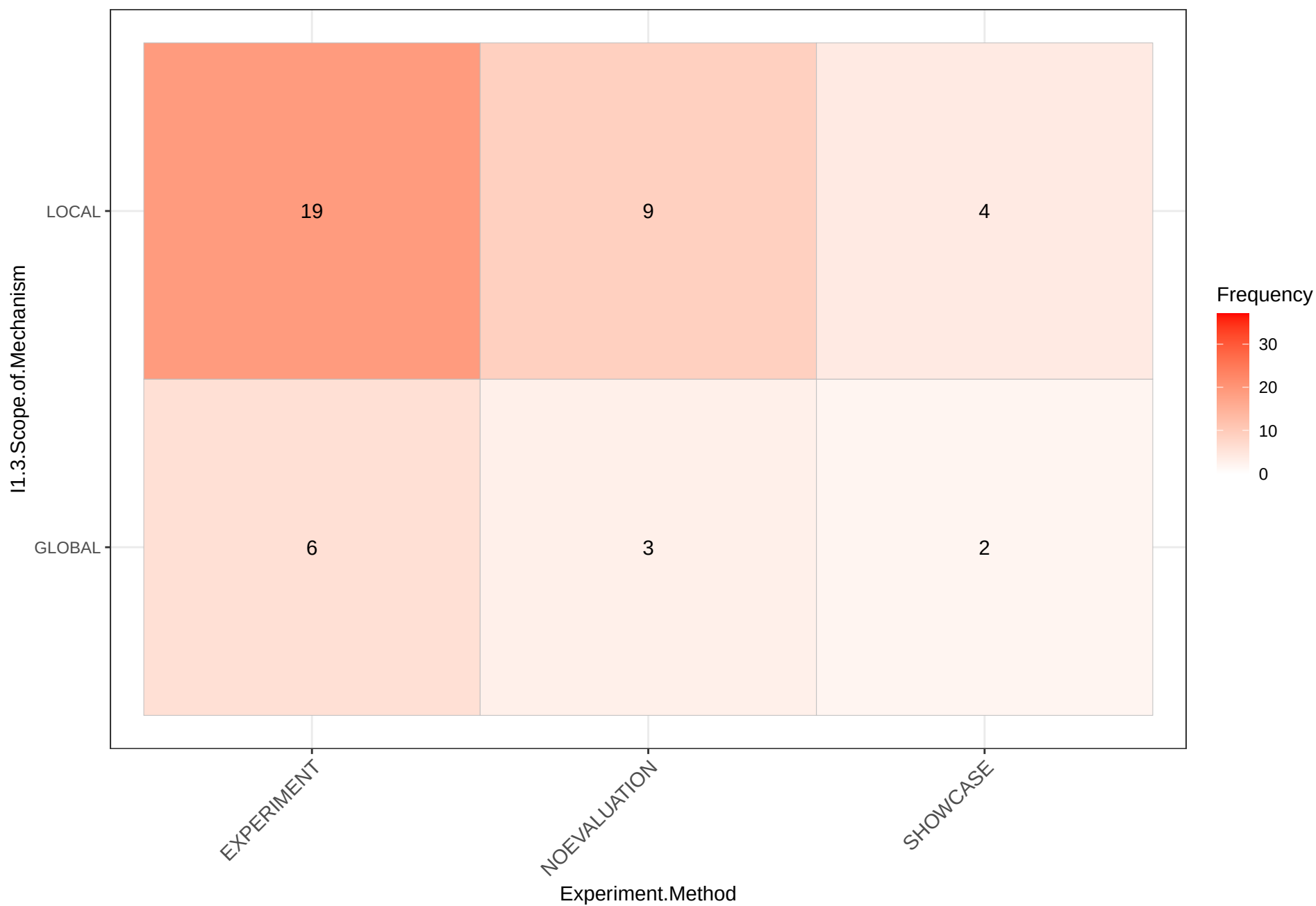
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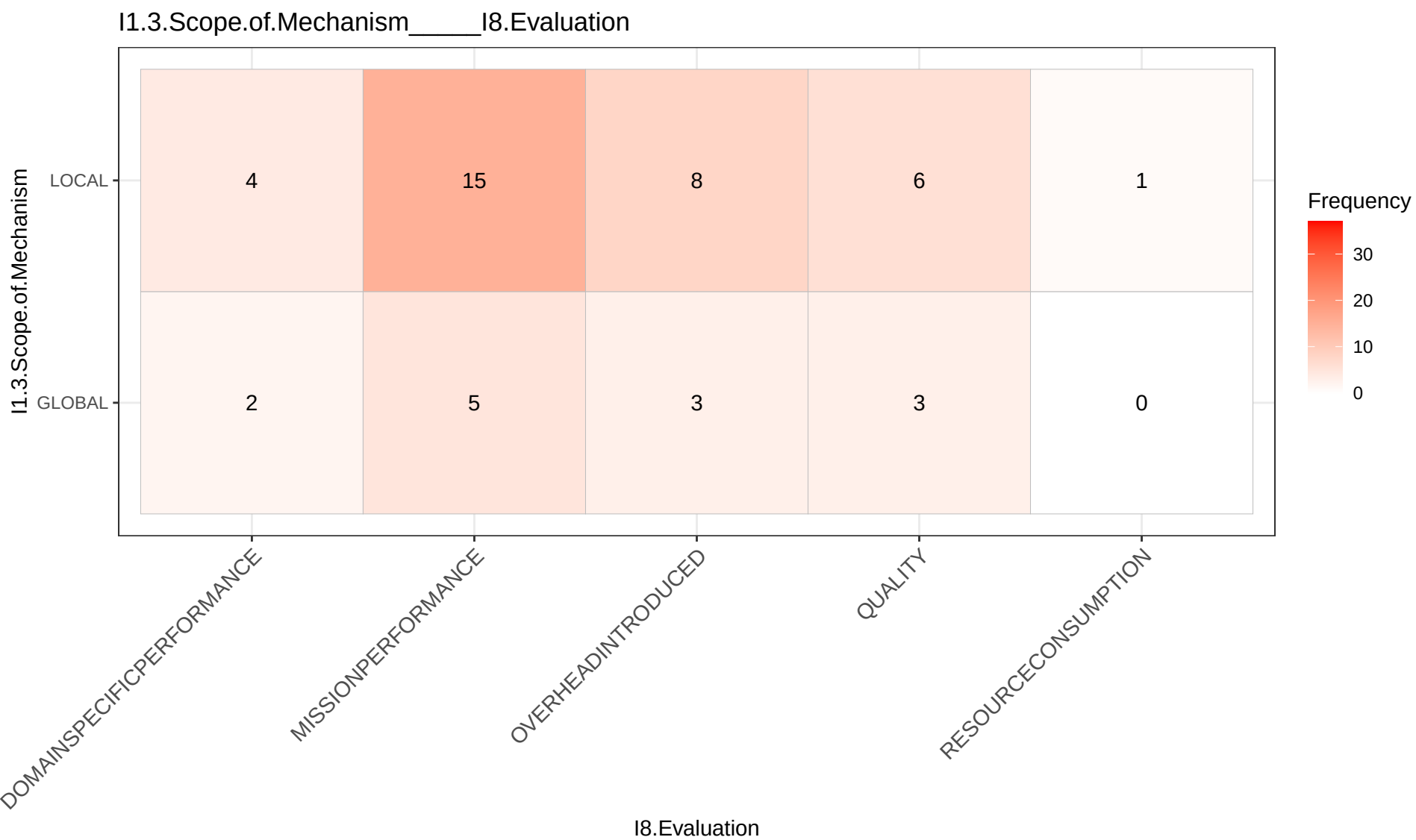
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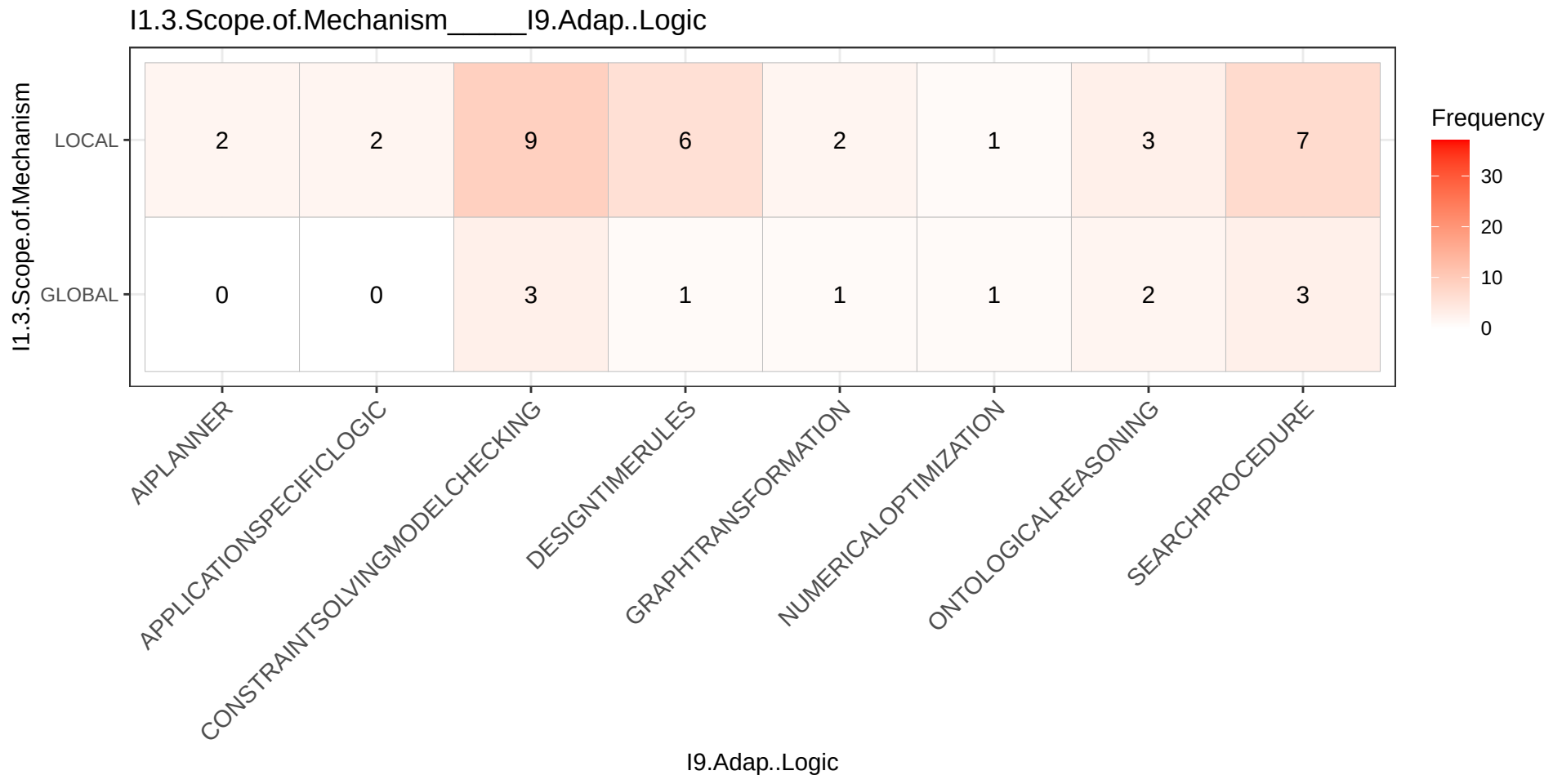
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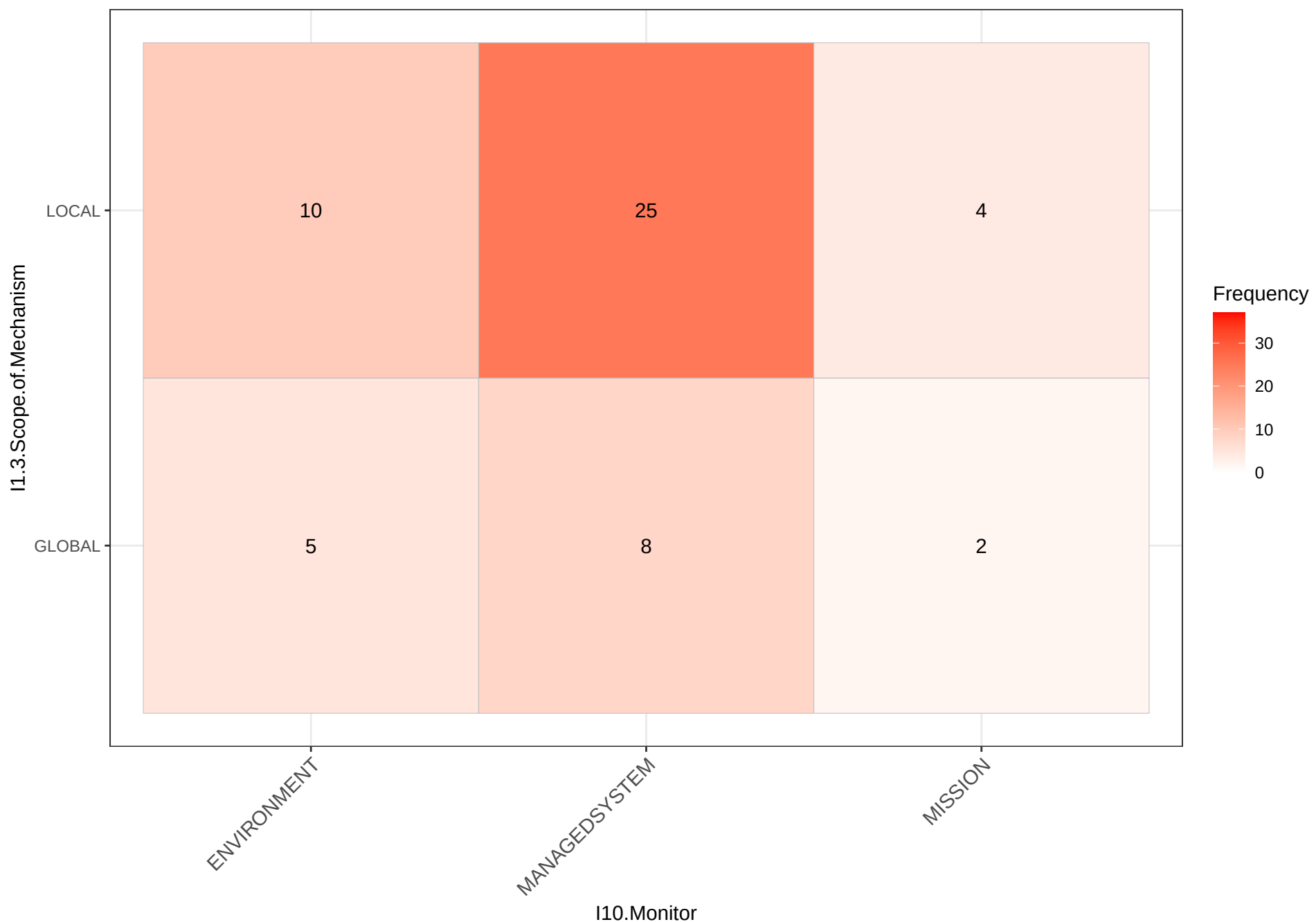
I1.3.Scope.of.Mechanism_____Experiment.Method

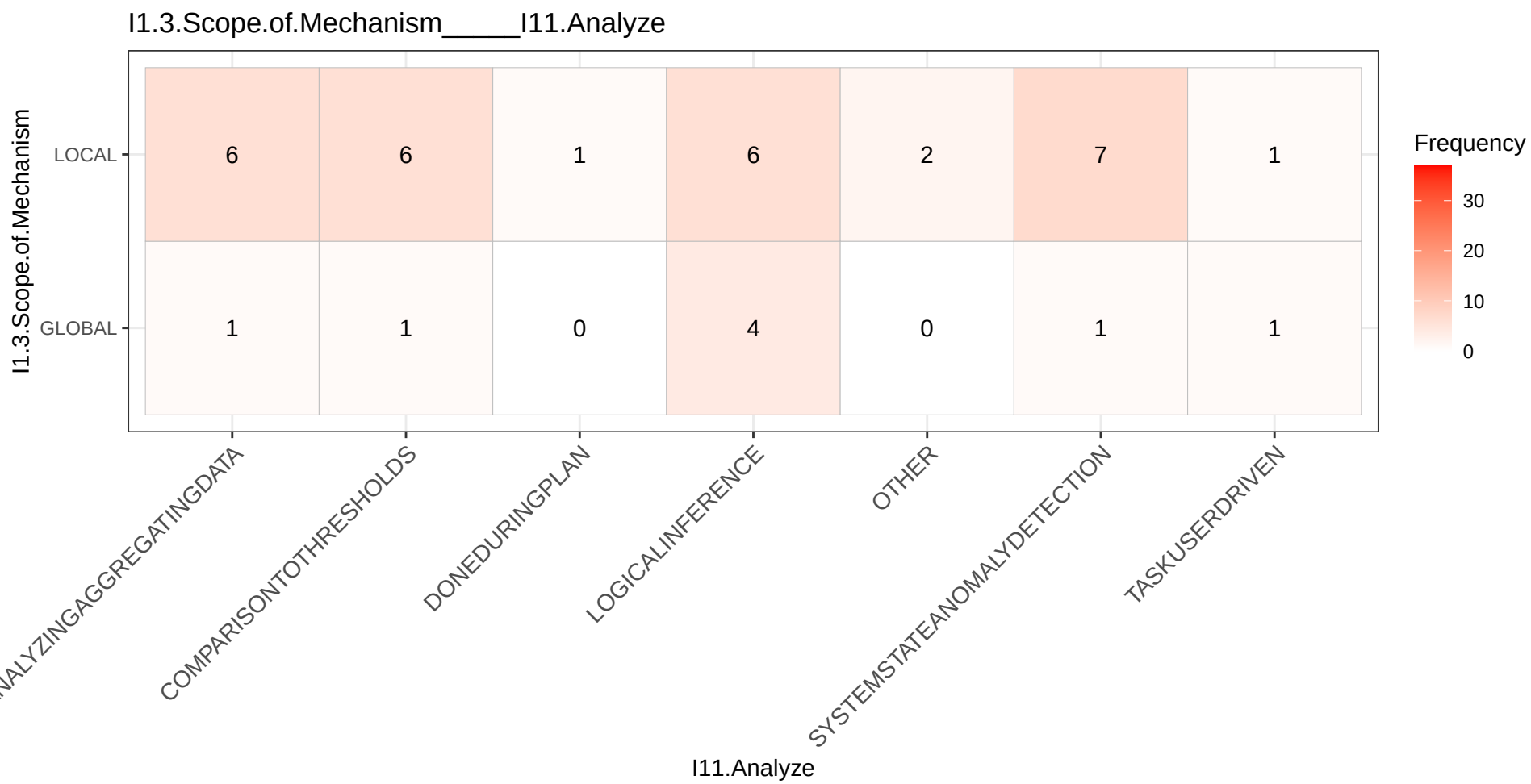




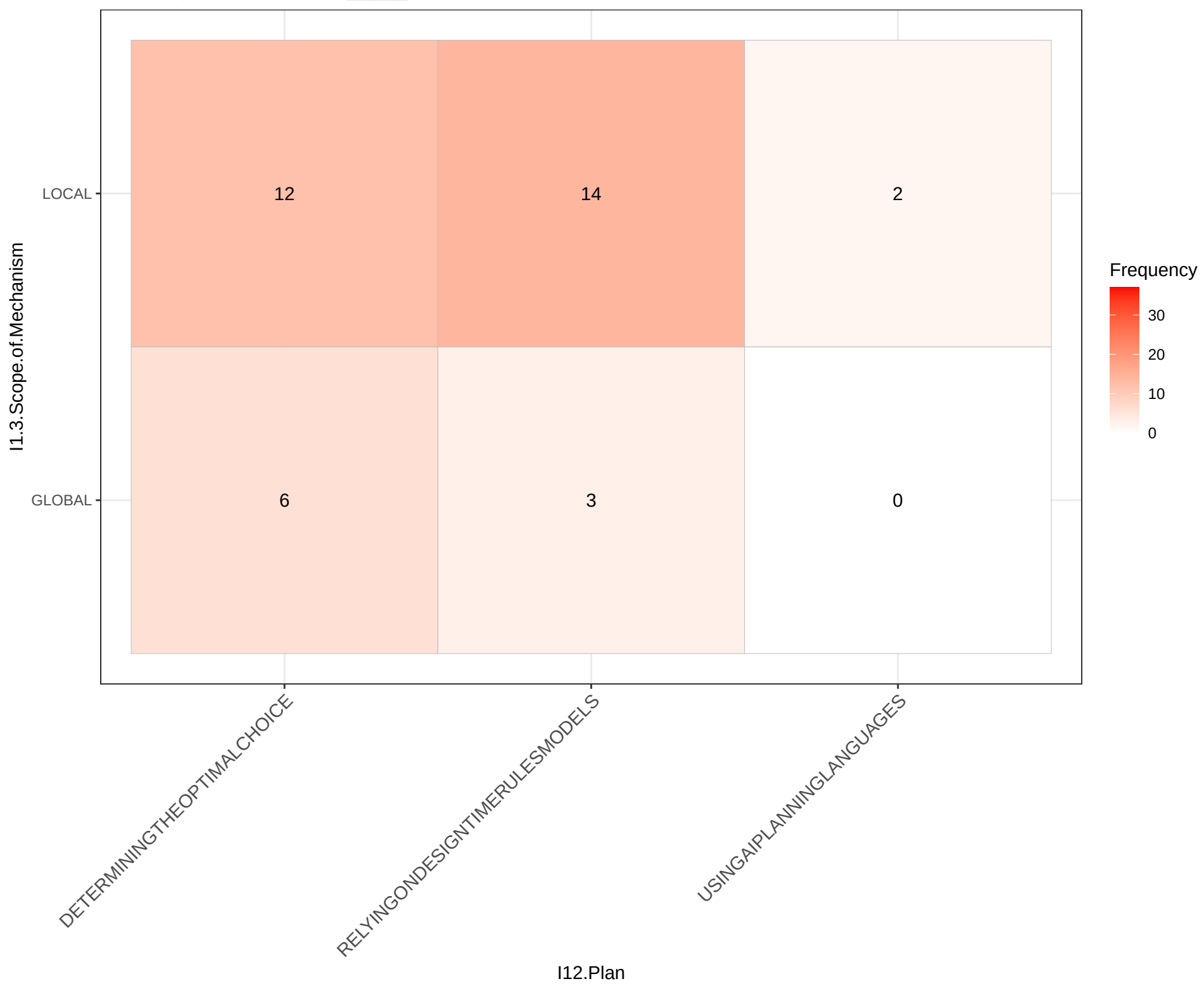


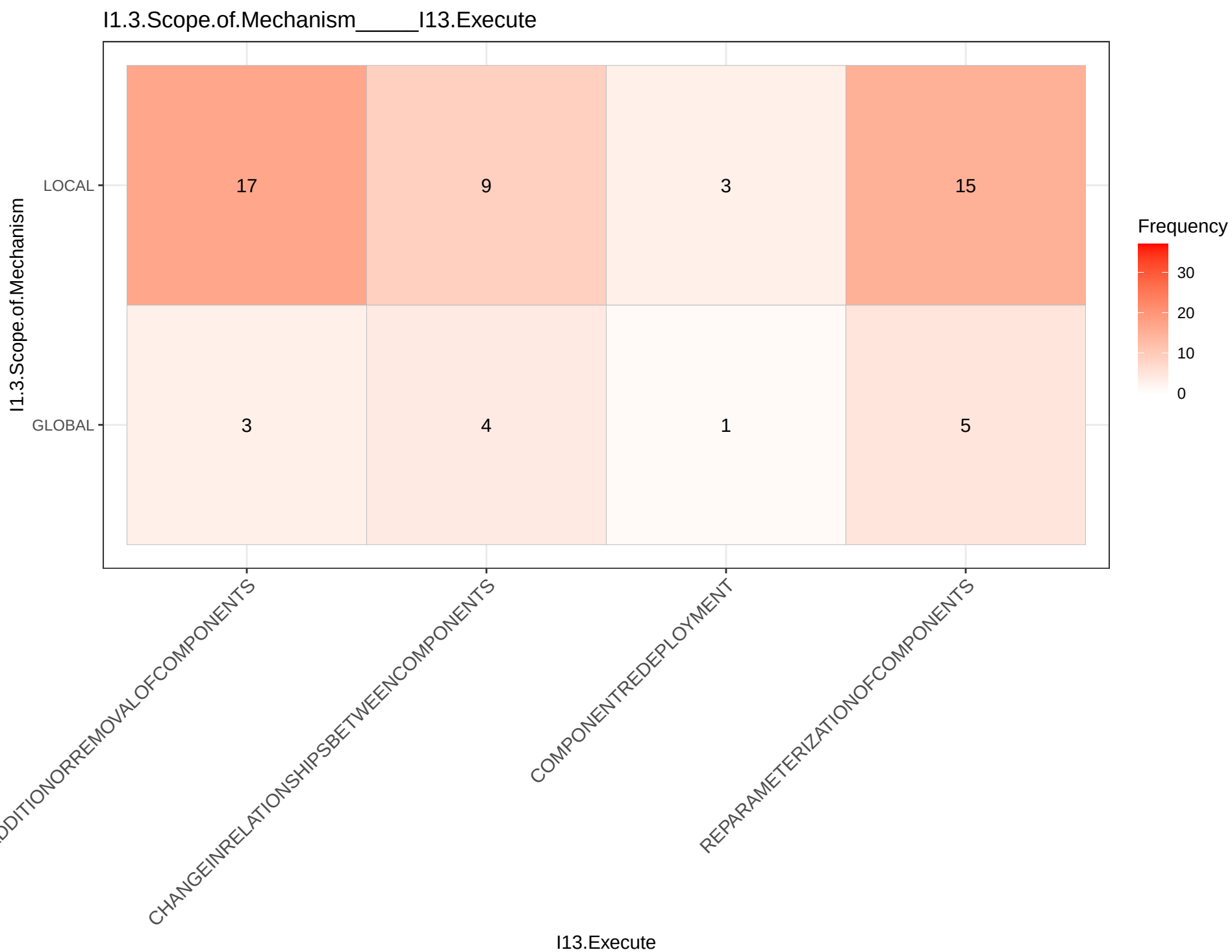
I1.3.Scope.of.Mechanism_____I10.Monitor

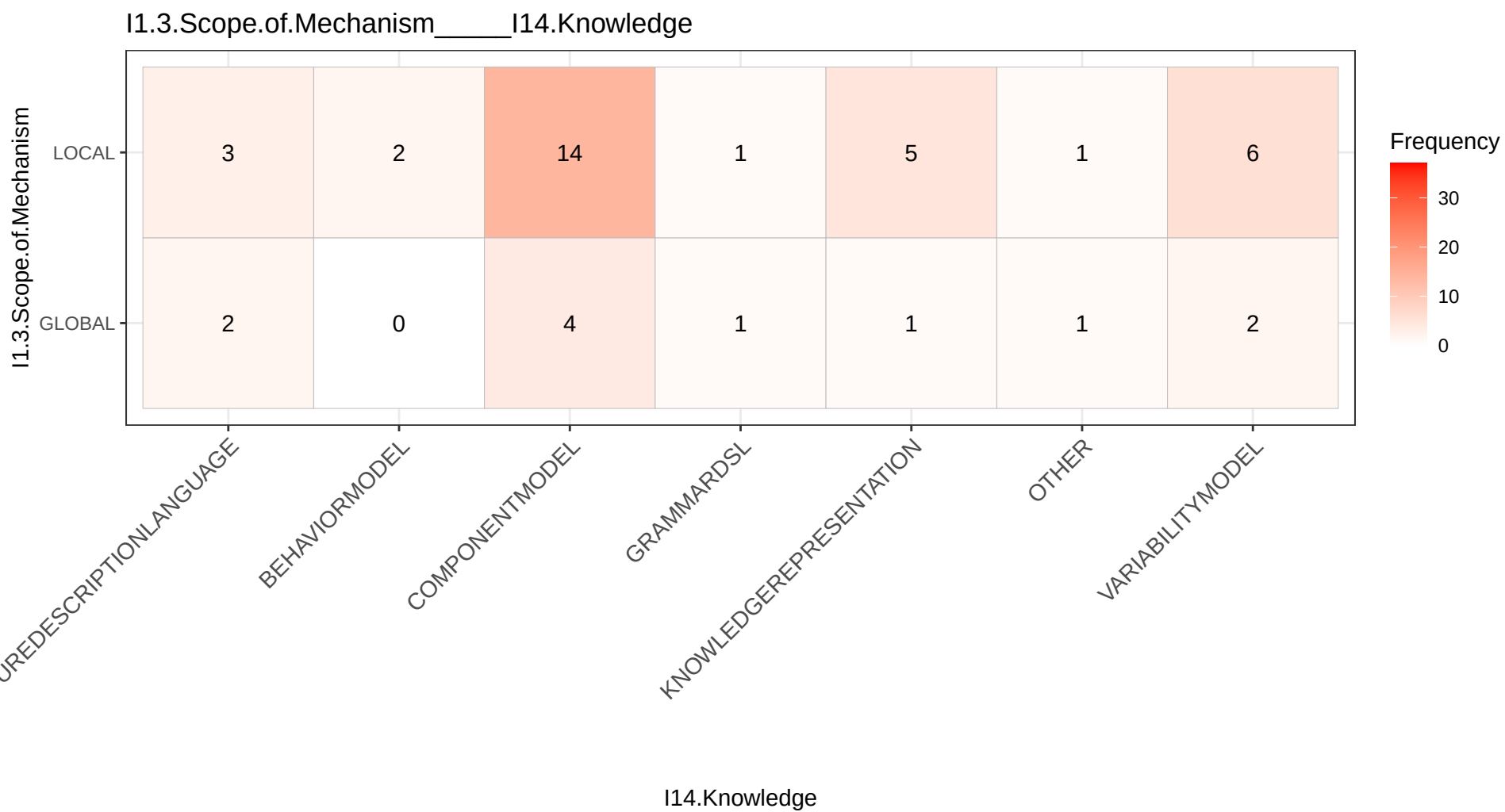




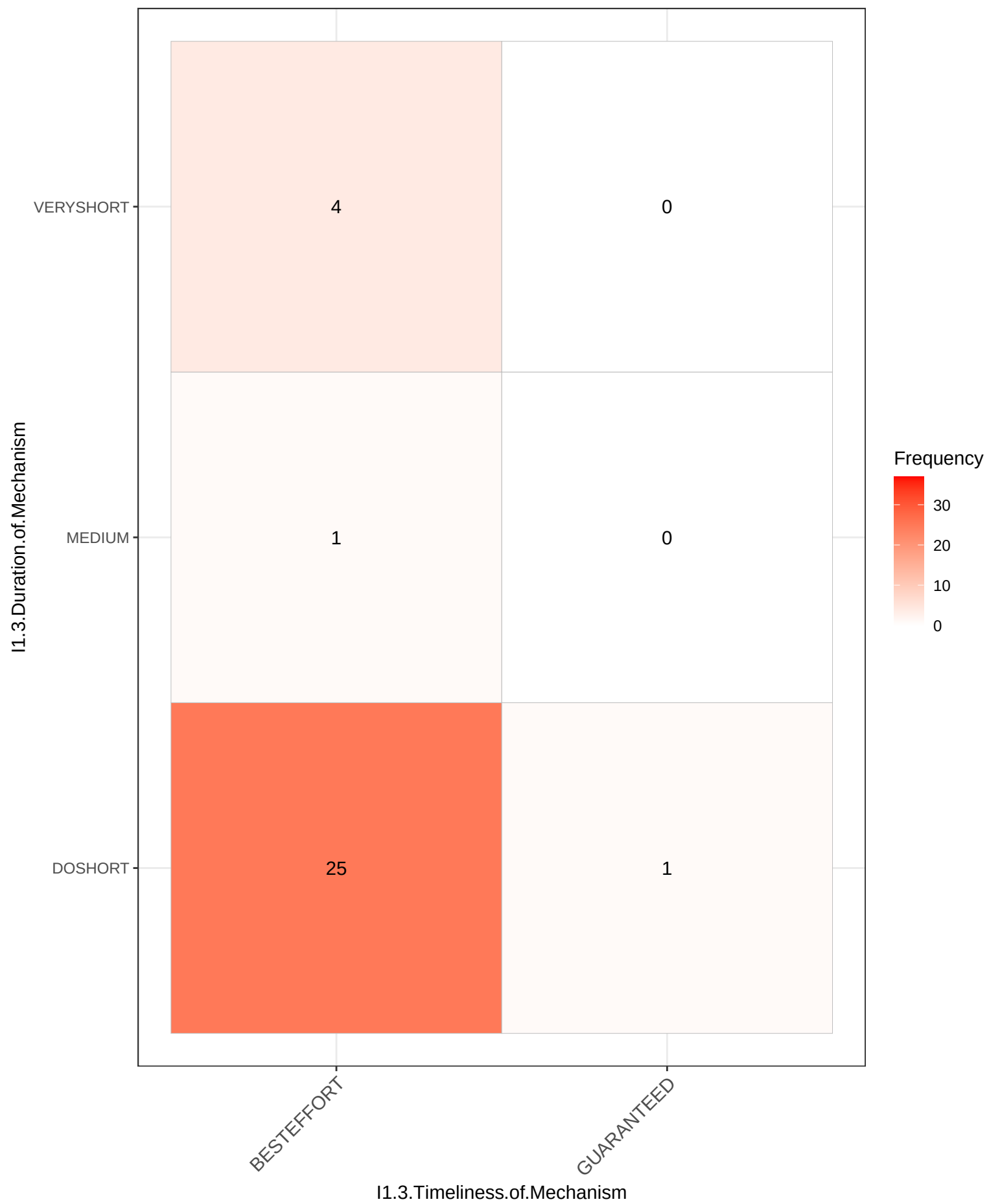
I1.3.Scope.of.Mechanism_____I12.Plan







I1.3.Duration.of.Mechanism_____I1.3.Timeliness.of.Mechanism



I1.3.Duration.of.Mechanism_____I1.3.Trigger.of.Mechanism

I1.3.Duration.of.Mechanism

VERYSHORT

4

MEDIUM

1

DOSHORT

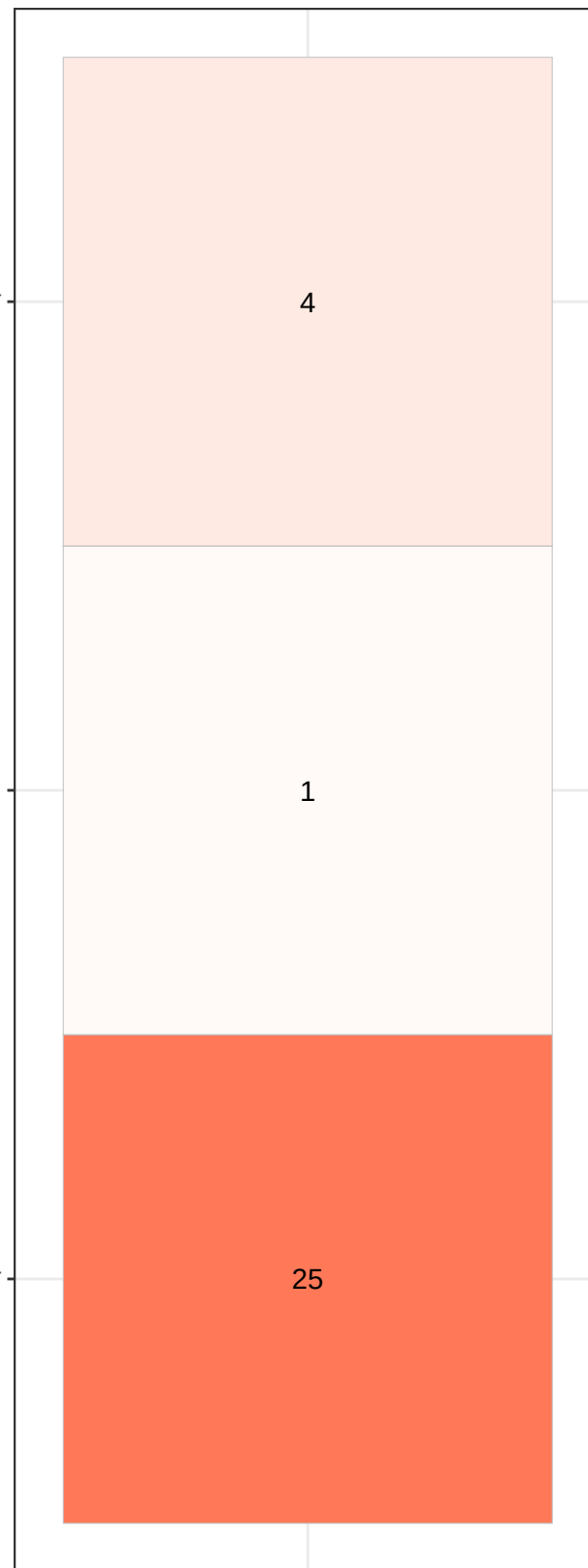
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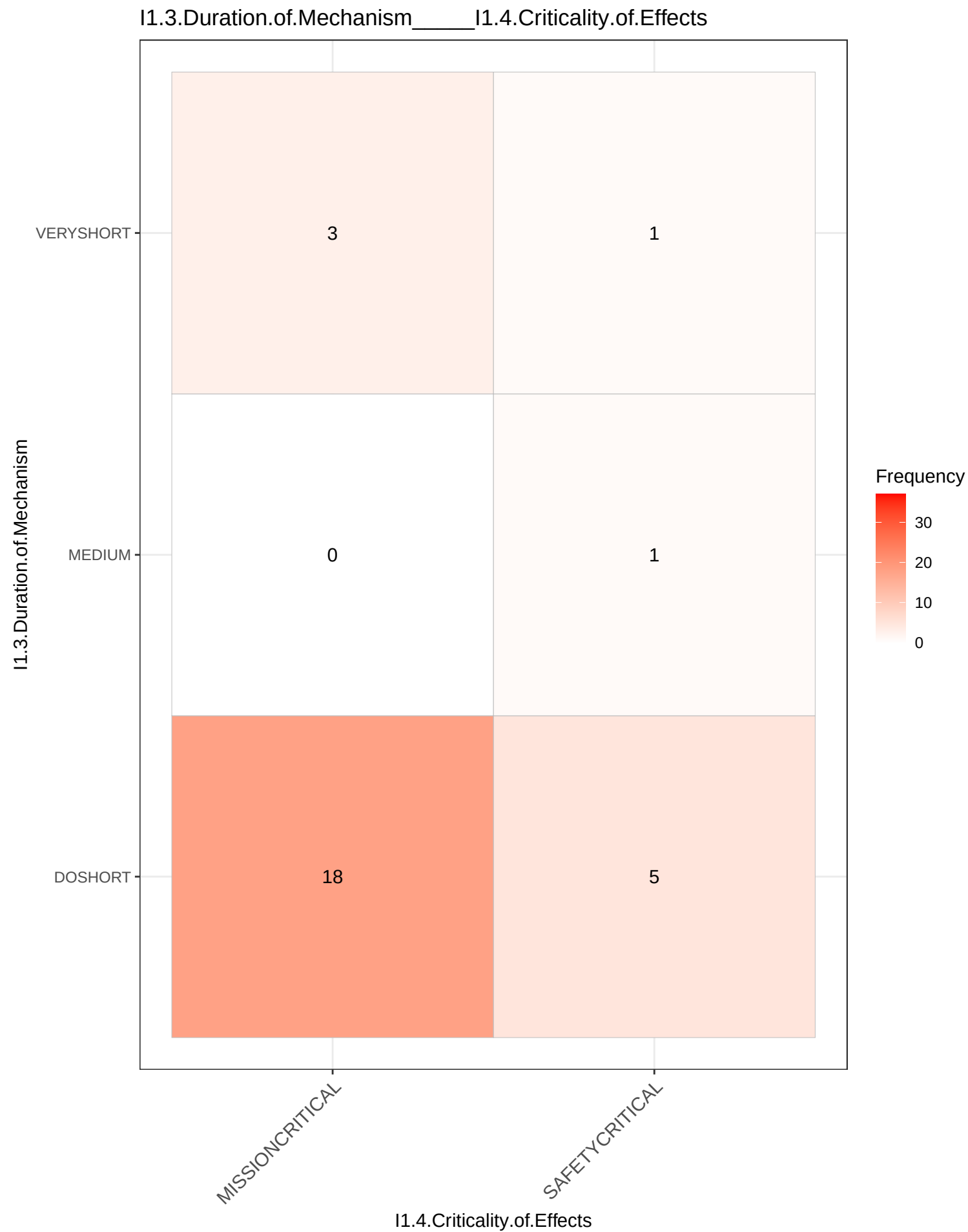
Frequency

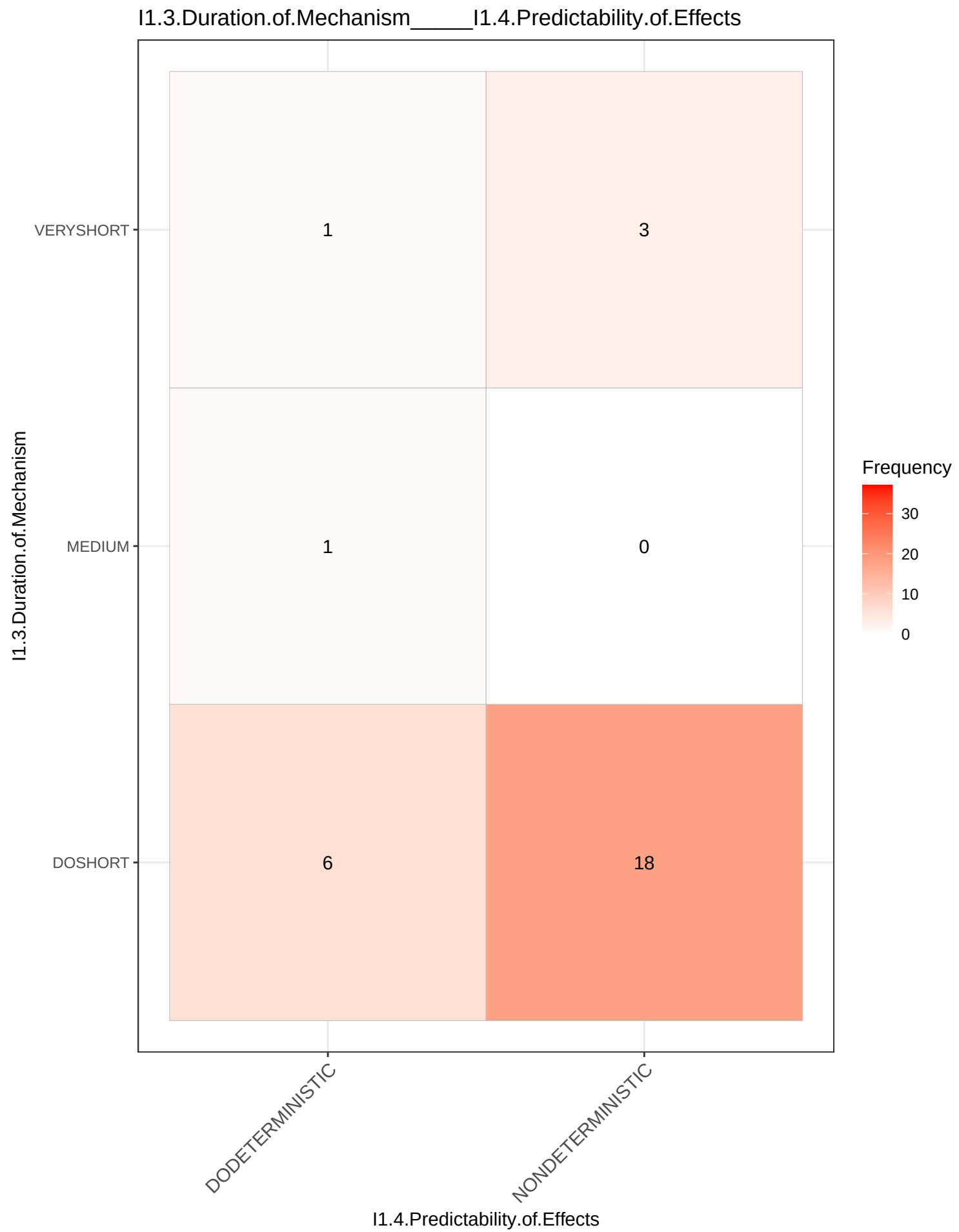
30
20
10
0

EVENTTRIGGER

I1.3.Trigger.of.Mechanism







I1.3.Duration.of.Mechanism_____I1.4.Overhead.of.Effects

I1.3.Duration.of.Mechanism

VERYSHORT

1

1

2

MEDIUM

0

0

1

DOSHORT

15

3

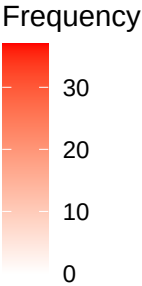
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DEPENDENT

DOSIGNIFICANT

INSIGNIFICANT

I1.4.Overhead.of.Effects



I1.3.Duration.of.Mechanism_____I1.4.Resilience.of.Effects

I1.3.Duration.of.Mechanism

VERYSHORT

2

1

0

MEDIUM

0

1

0

DOSHORT

9

8

3

DEPENDENT

DORESILIENT

IRRRESILIENT

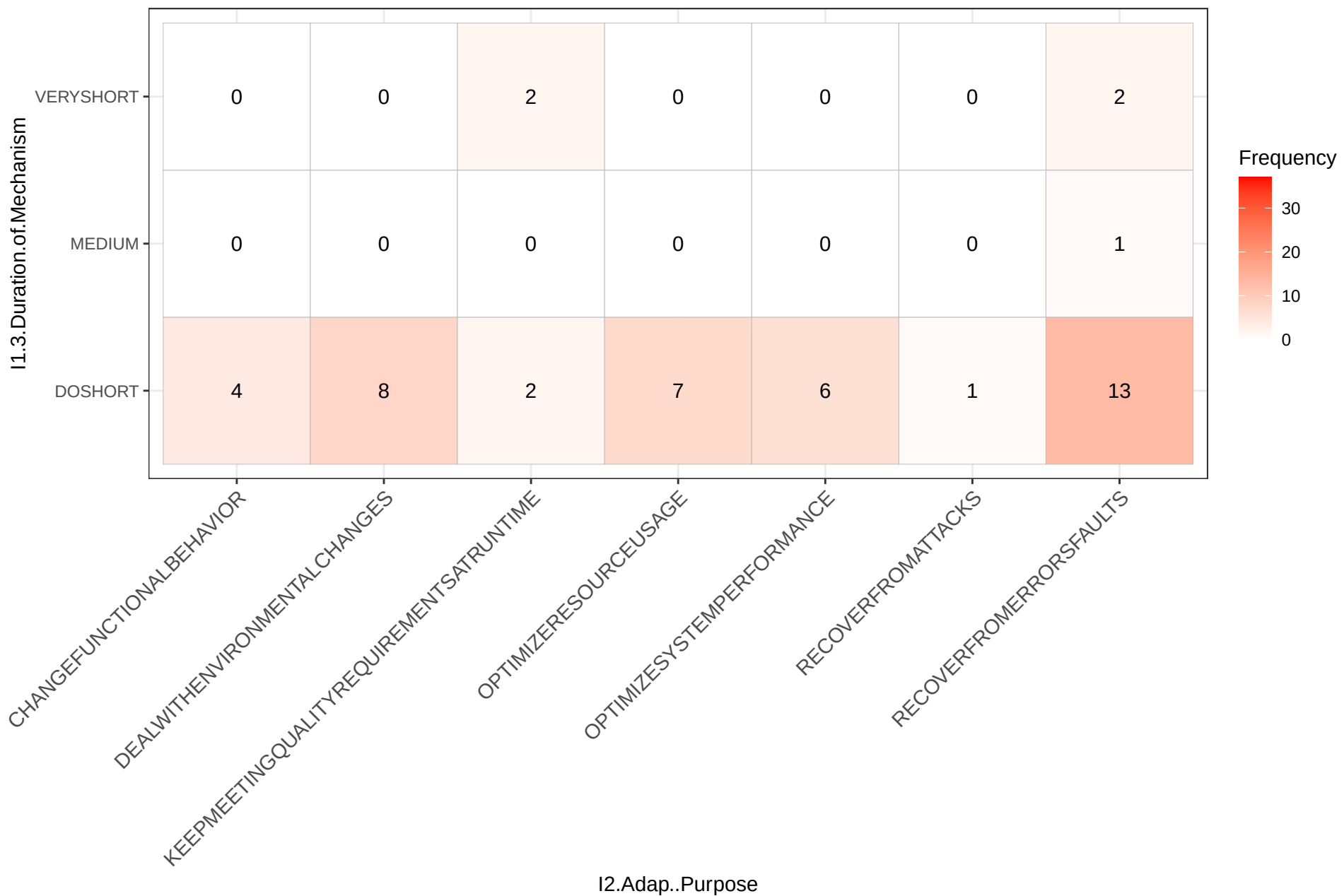
I1.4.Resilience.of.Effects

Frequency



30
20
10
0

I1.3.Duration.of.Mechanism_____I2.Adap..Purpose

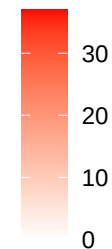


I1.3.Duration.of.Mechanism

I1.3.Duration.of.Mechanism_____I3.Robot.Type

VERYSHORT	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	1	1	0	
MEDIUM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	
DOSHORT	1	1	0	1	1	1	2	1	2	0	1	0	1	2	1	0	0	1	1	2	0	1	1	1	1	1	4	1	1	1	0	0

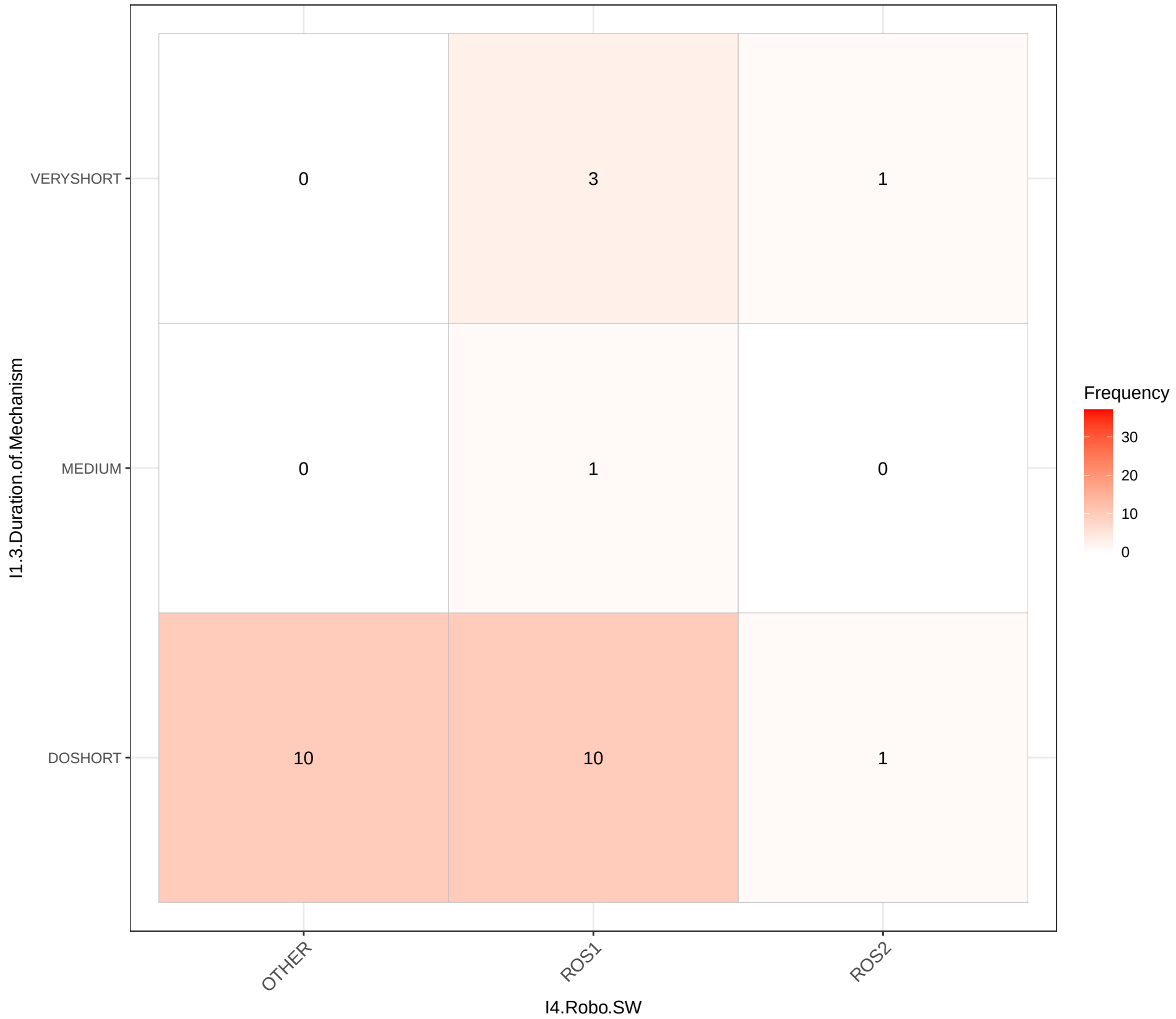
Frequency



BEBOTSPADERBORN
BLUEROV2
BOXERCLEARPATH
CRAWLERTERMINATORBOT
FIELDMOBILEROBOT
FRACTIONATEDSPACECRAFT
HETEROGENOUSROBOTS
HEXMANIPULATOR
HEXAII
INFOTAINMENTROBOTMOBILESERVICEIROBOTCREATE2
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR
LAPTOPACTINGASAROBOTMOBILEROBOTMOBILEROBOTMOBILEROBOT
MOBILEROBOTTERRESTRIAL
MOBILEROBOTTIAGO
MSUEVORALLYMOBILETERRESTRIAL
MULTIPLEHEXROTOR
NAOROBOT
PIONEER3DX
QUADROCOPTER
SINGLESERVINGROTATIONROBOT
TEDUSARTERRESTRIALSEARCH
TRIGLIDEINDUSTRIALASSEMBLY
TURTLEBOT2KOBUKI
TWOINDUSTRIALROBOTS
WAREHOUSEDELIVERYROBOT
WHICHISANINDUSTRIALAGV
YOUBOT

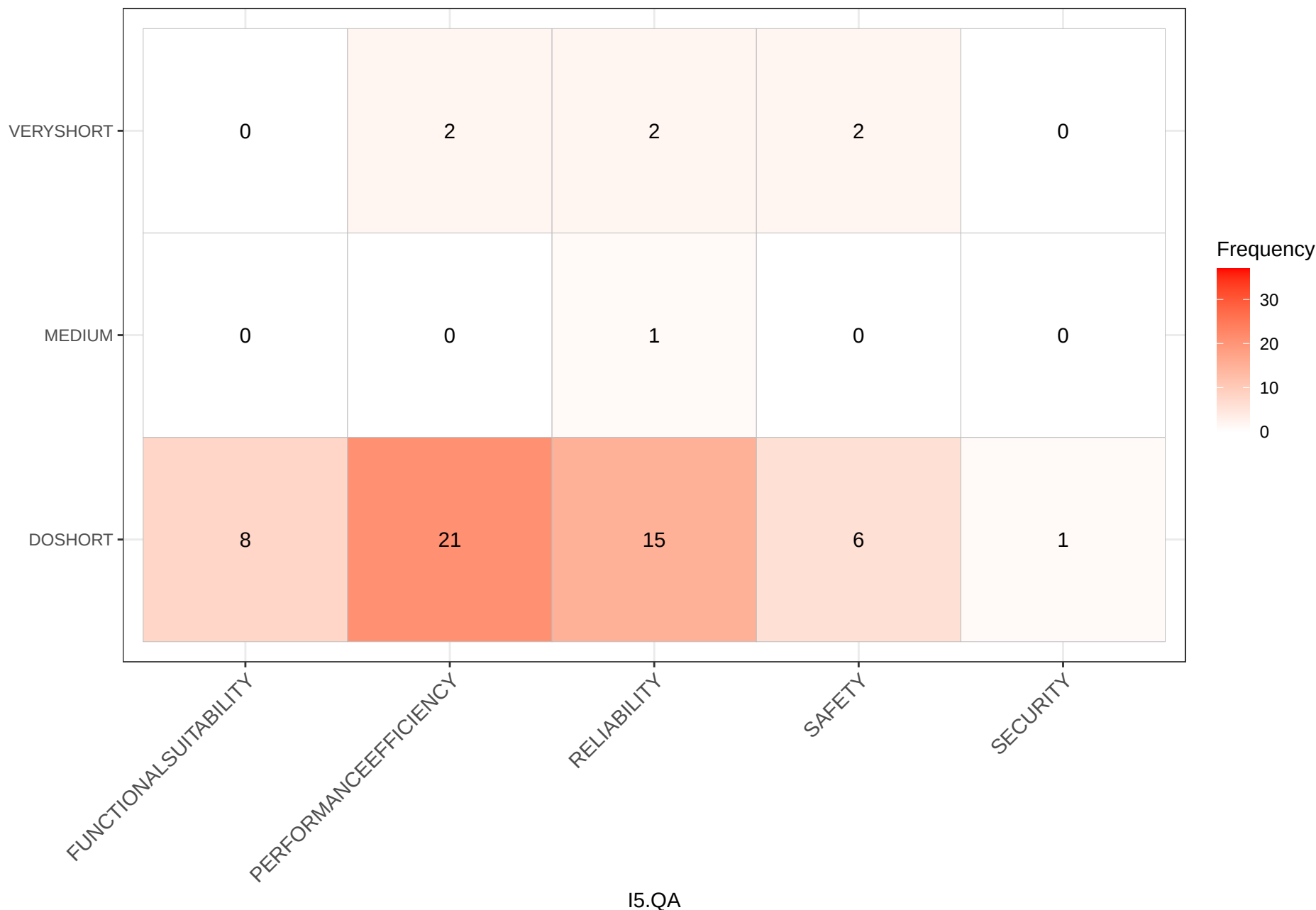
I3.Robot.Type

I1.3.Duration.of.Mechanism_____I4.Robo.SW

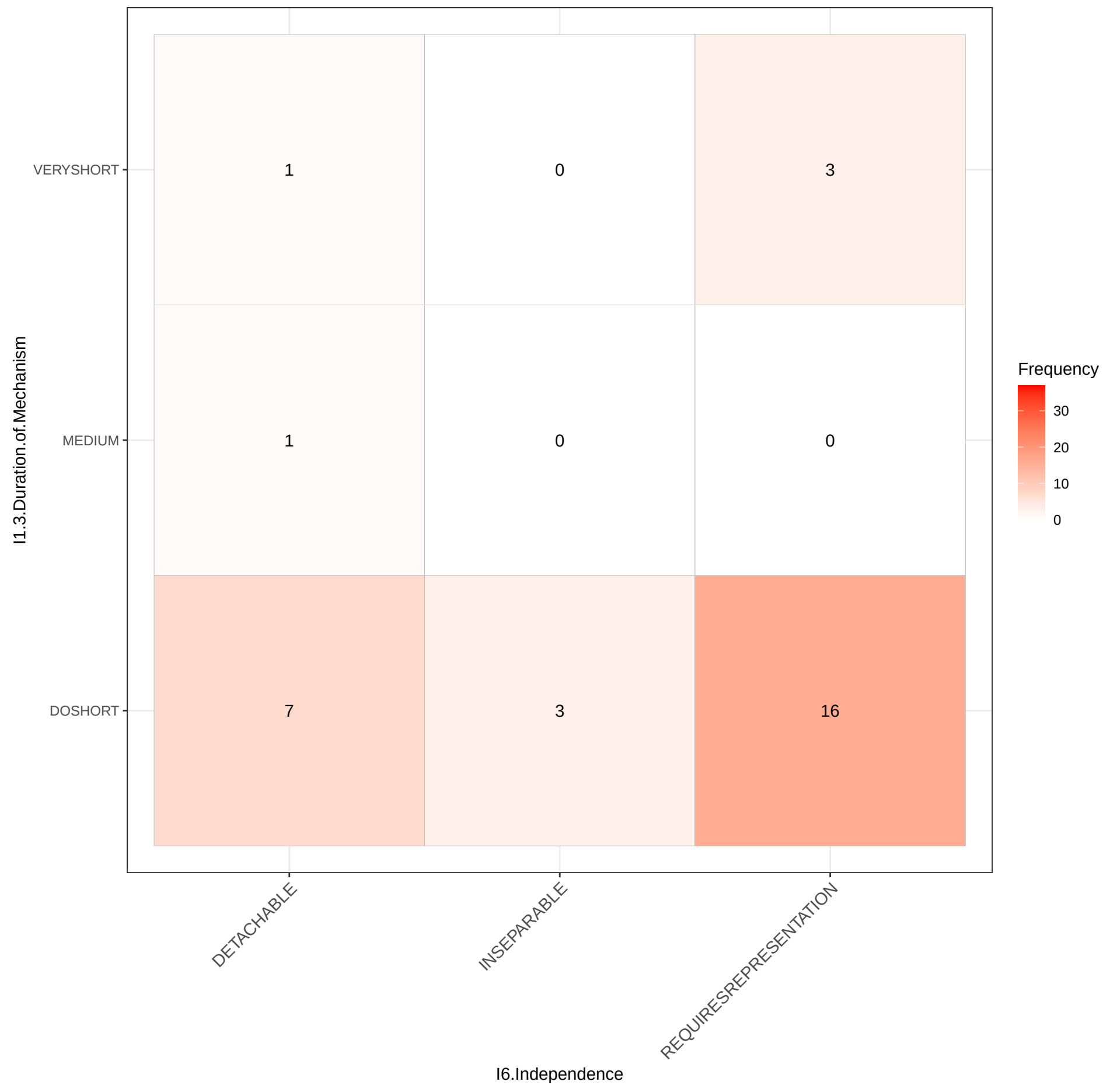


I1.3.Duration.of.Mechanism_____I5.QA

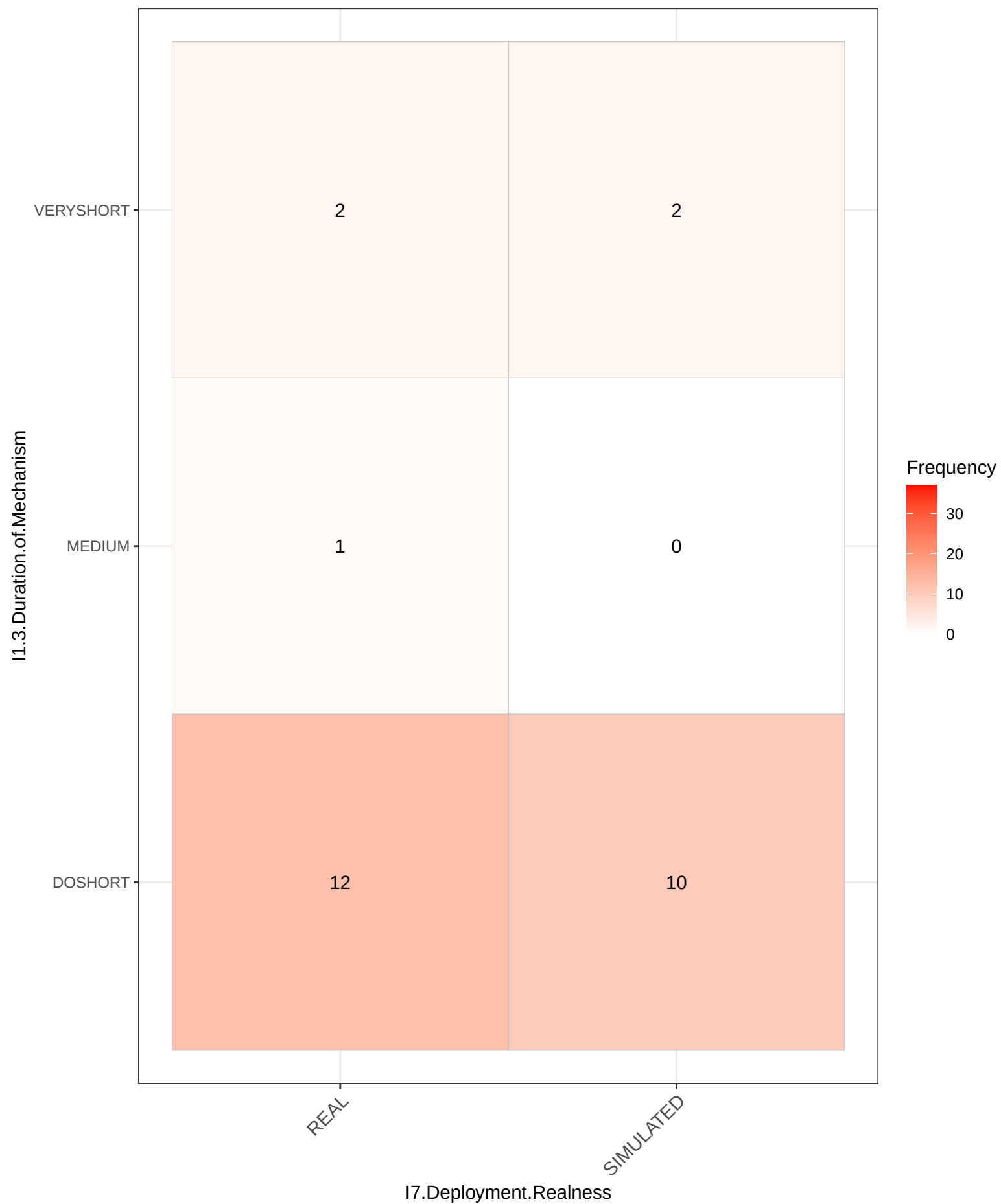
I1.3.Duration.of.Mechanism



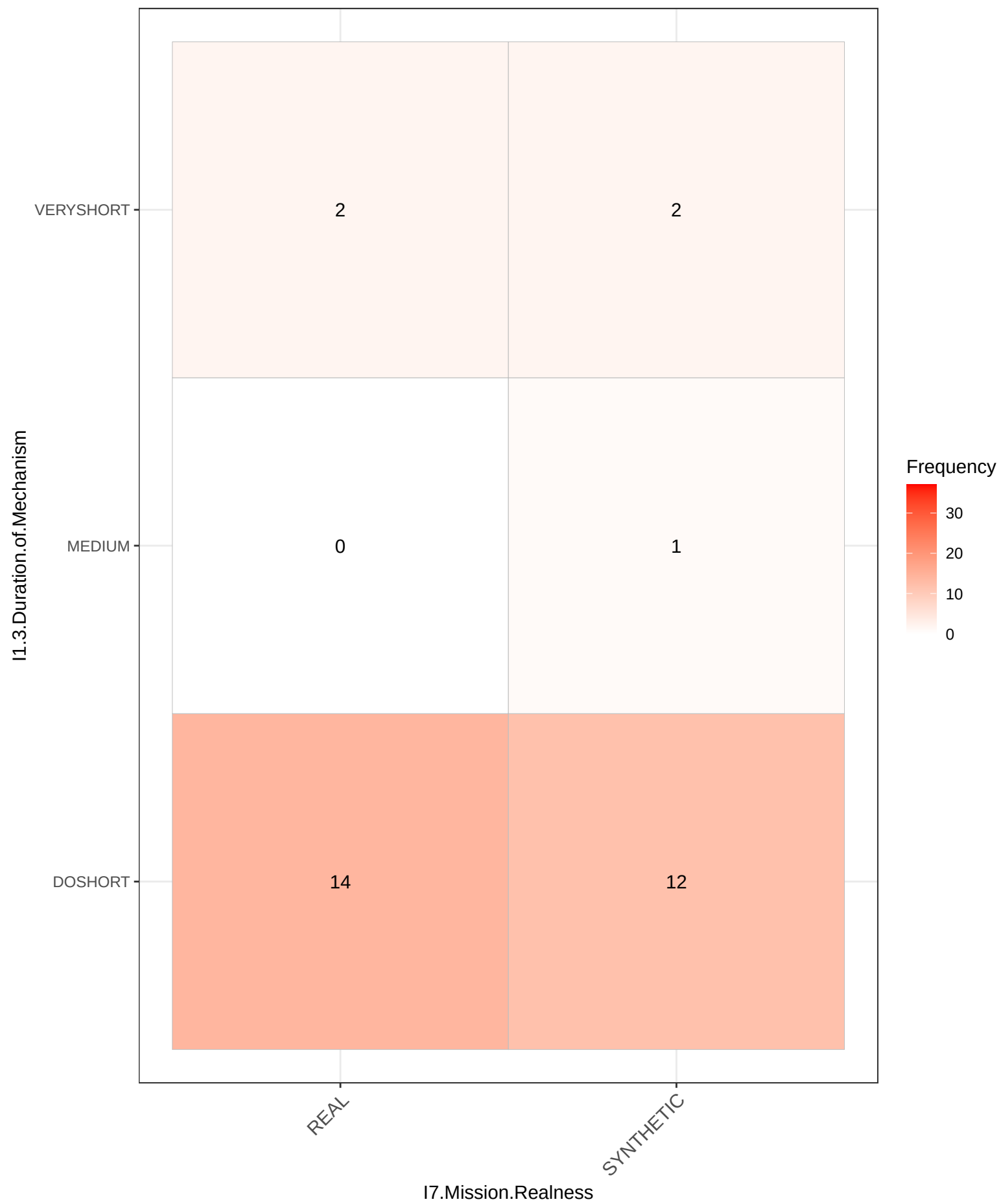
I1.3.Duration.of.Mechanism_____I6.Independence



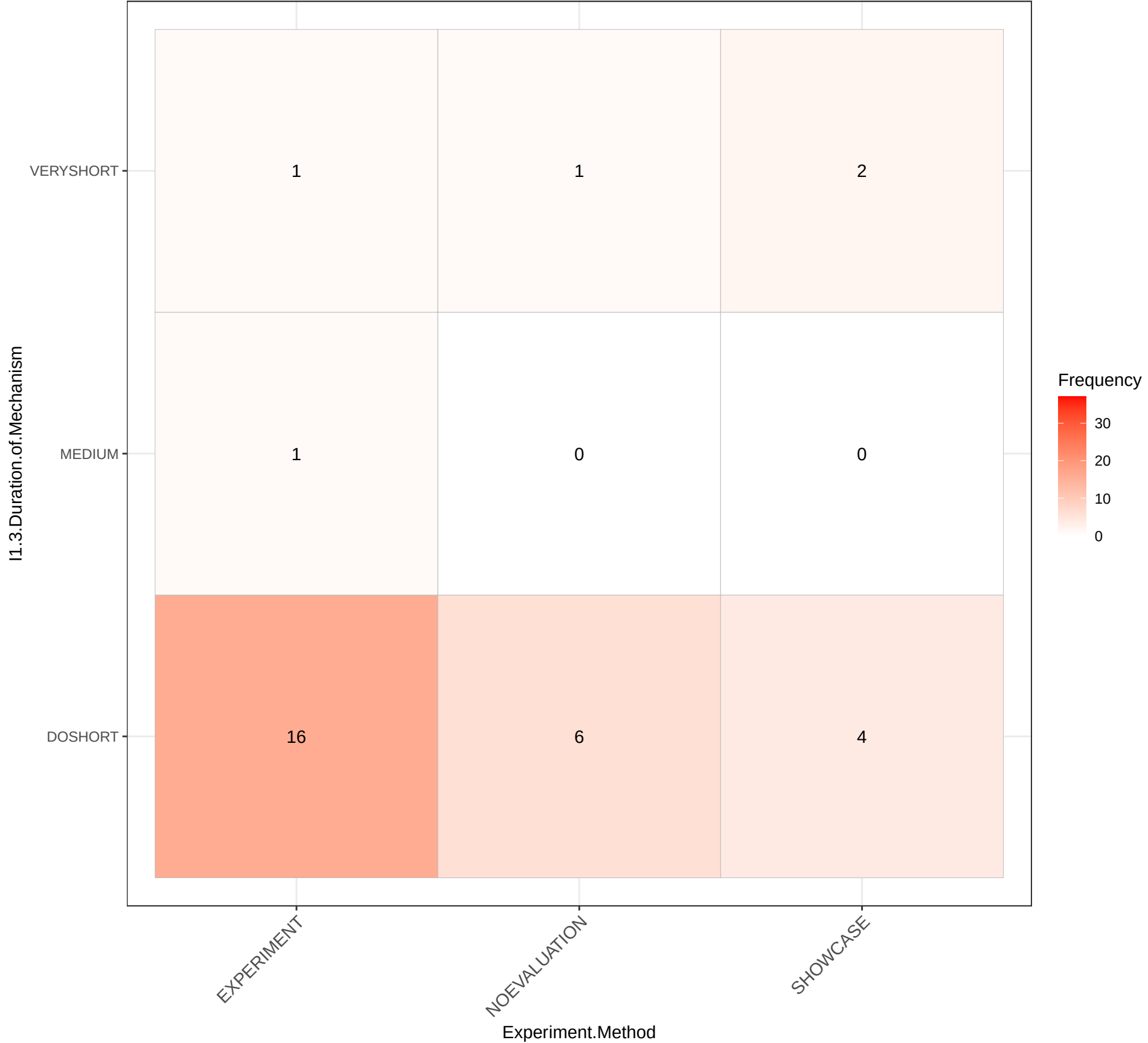
I1.3.Duration.of.Mechanism_____I7.Deployment.Realness



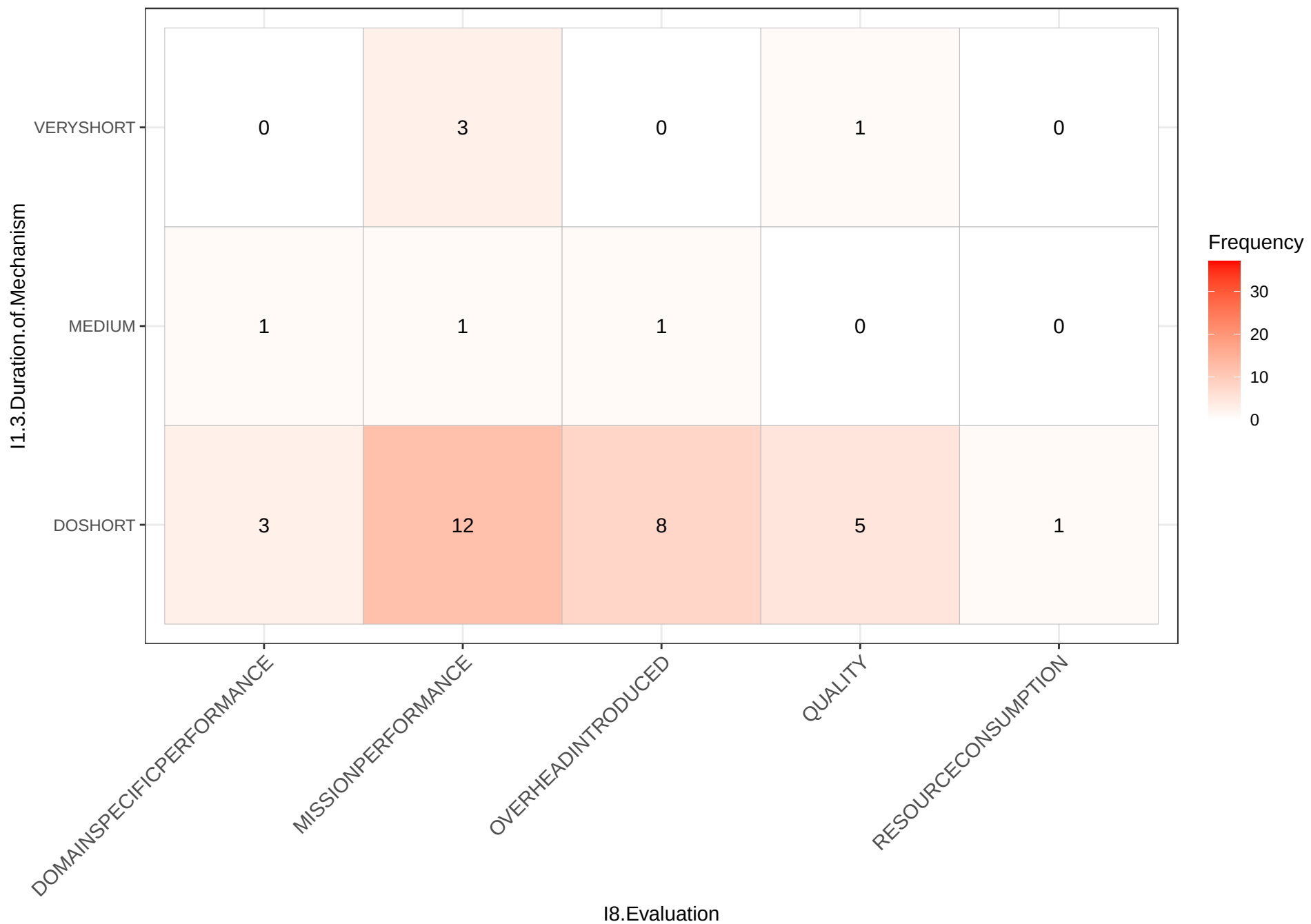
I1.3.Duration.of.Mechanism_____I7.Mission.Realness



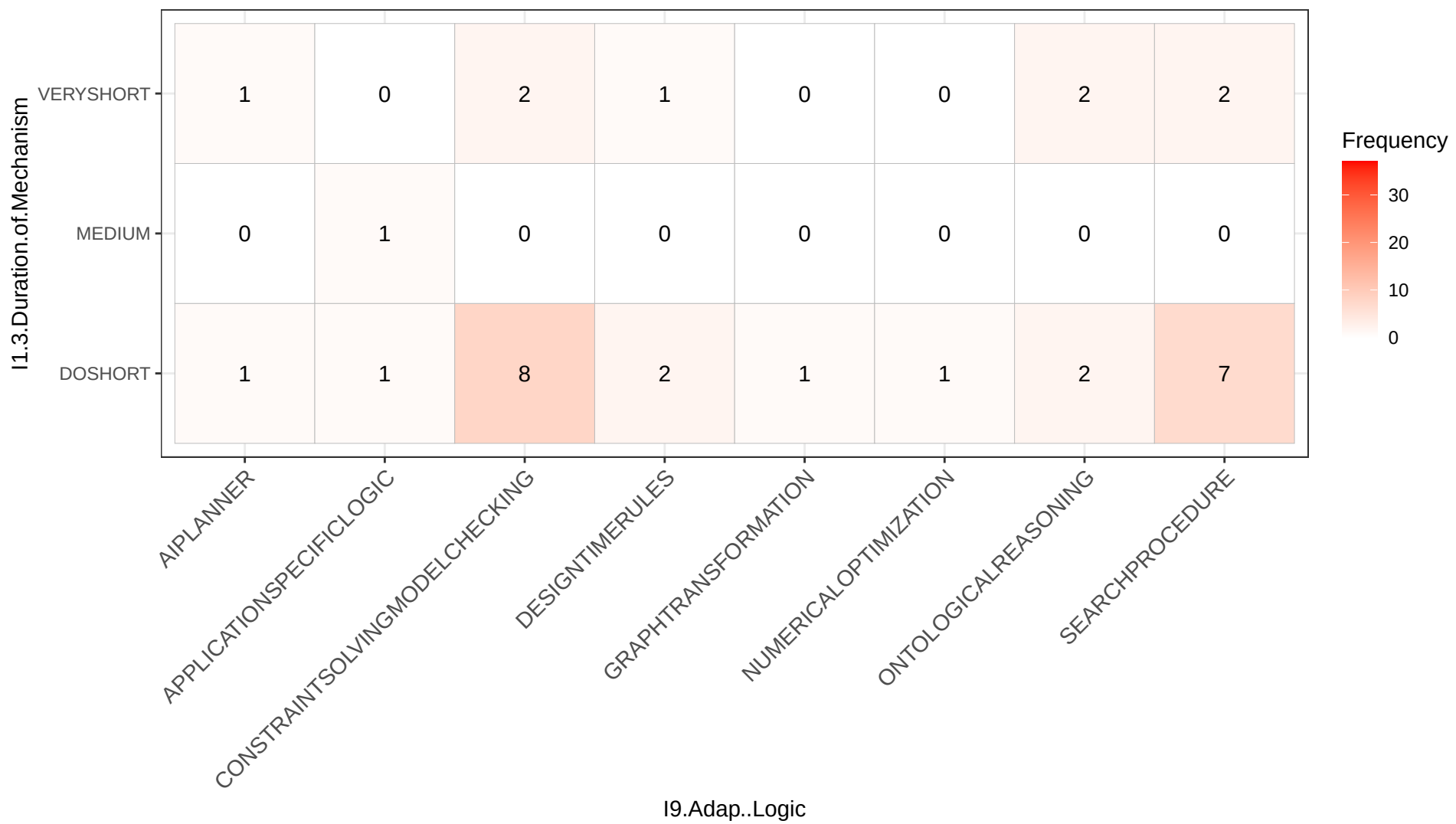
I1.3.Duration.of.Mechanism_____Experiment.Method



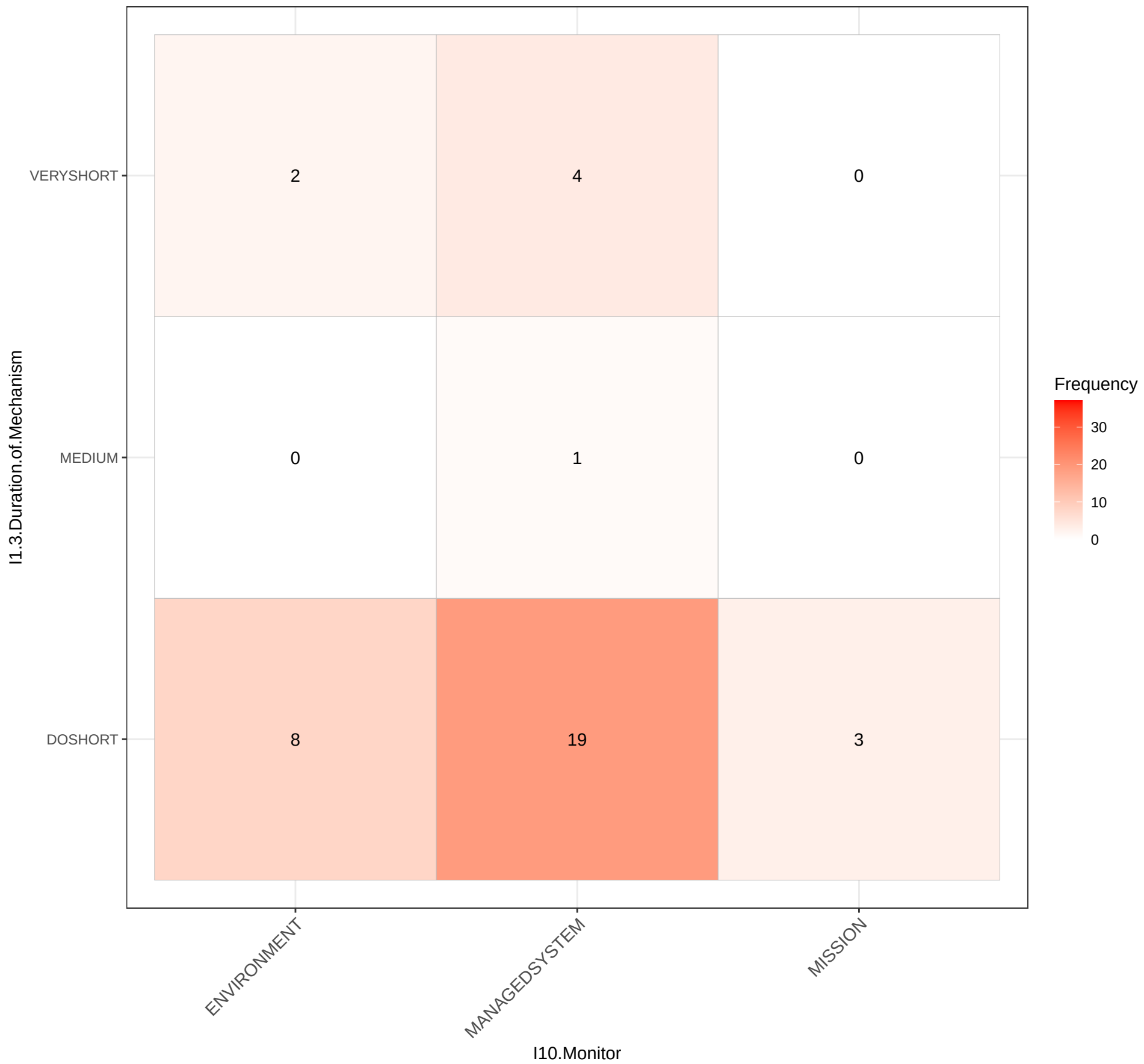
I1.3.Duration.of.Mechanism_____I8.Evaluation



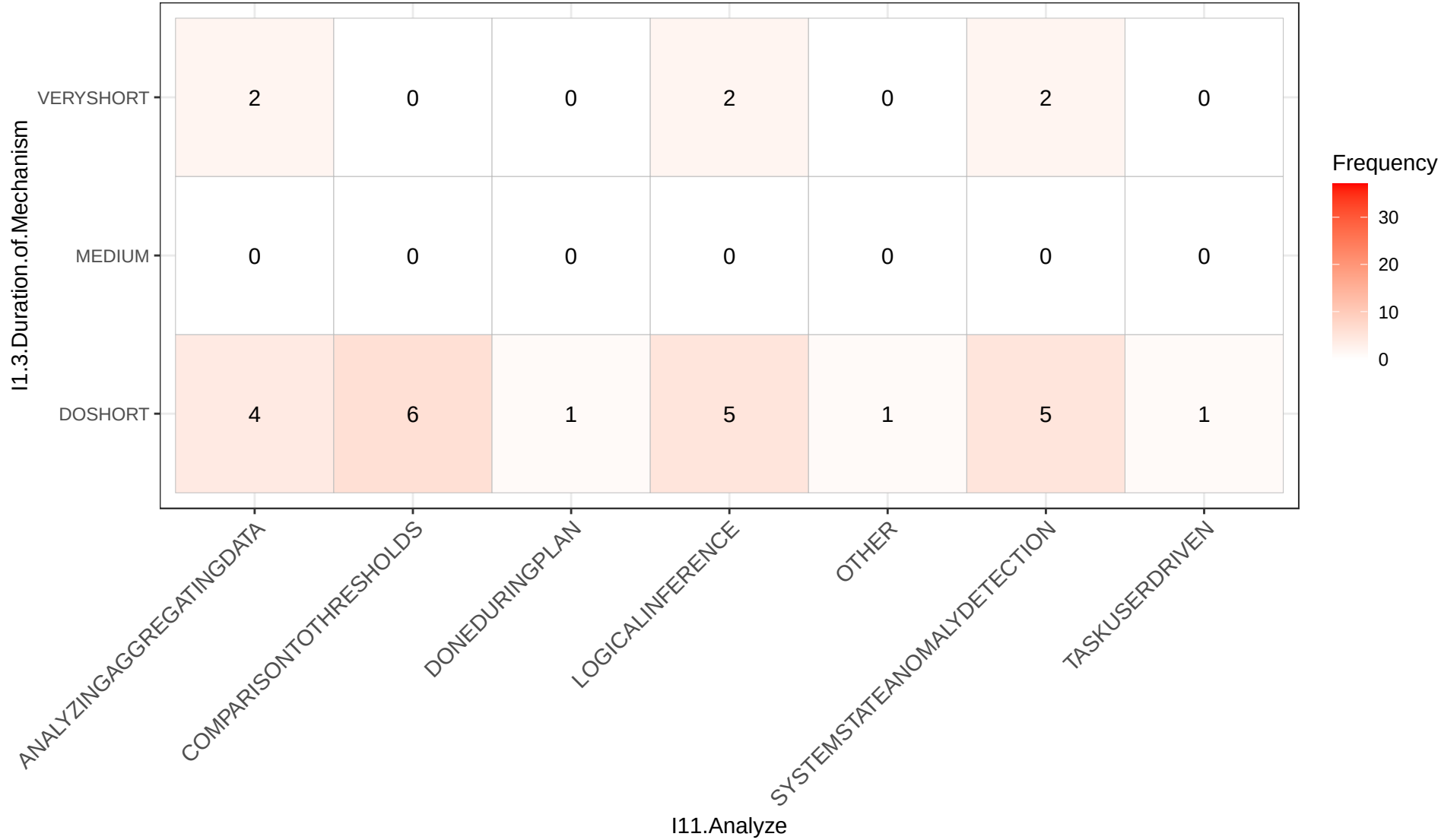
I1.3.Duration.of.Mechanism_____I9.Adap..Logic



I1.3.Duration.of.Mechanism_____I10.Monitor



I1.3.Duration.of.Mechanism_____I11.Analyze



I1.3.Duration.of.Mechanism

VERYSHORT

MEDIUM

DOSHORT

Frequency

30

20

10

0

DETERMININGTHEOPTIMALCHOICE

RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

I12.Plan

1

2

1

0

1

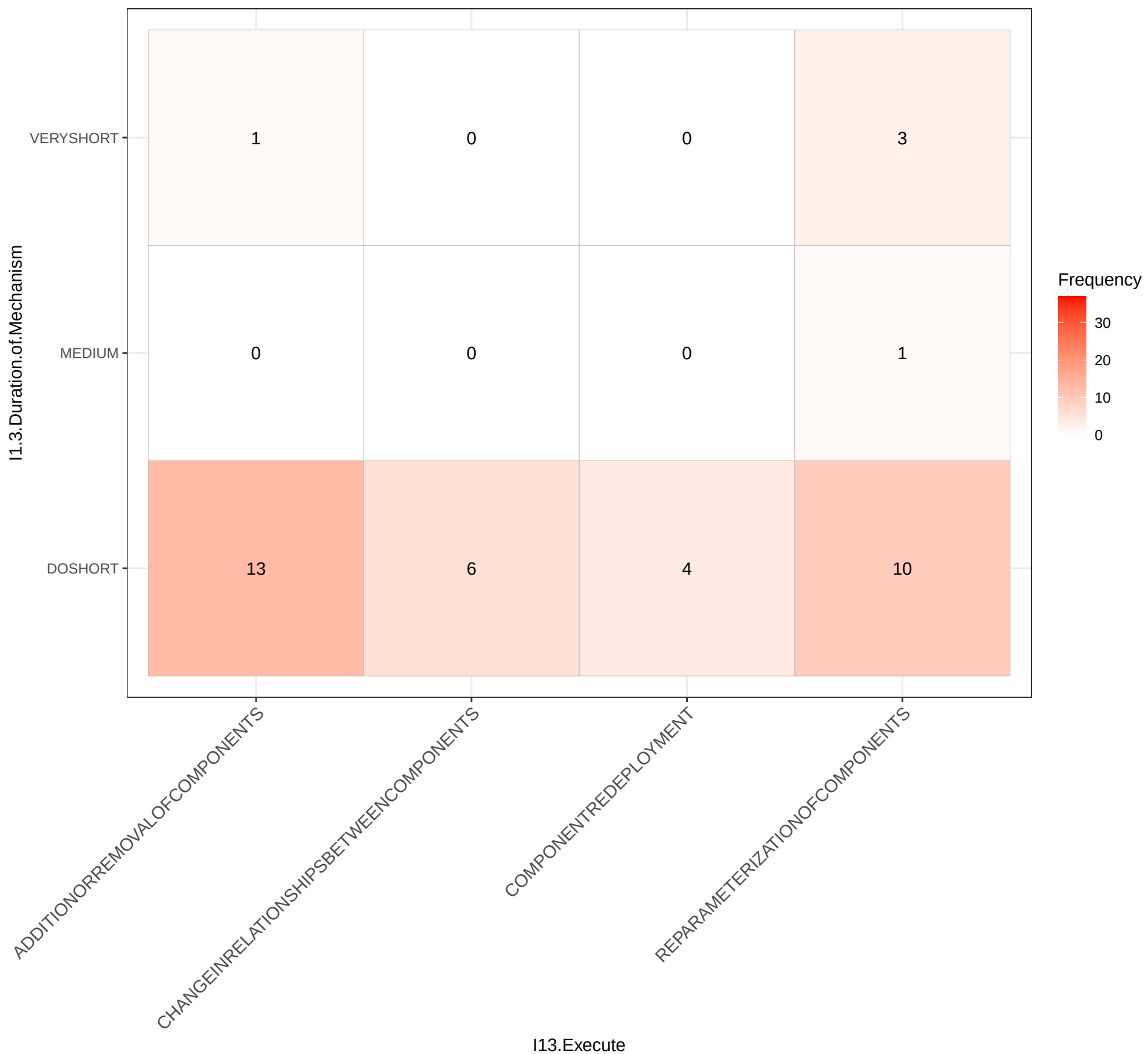
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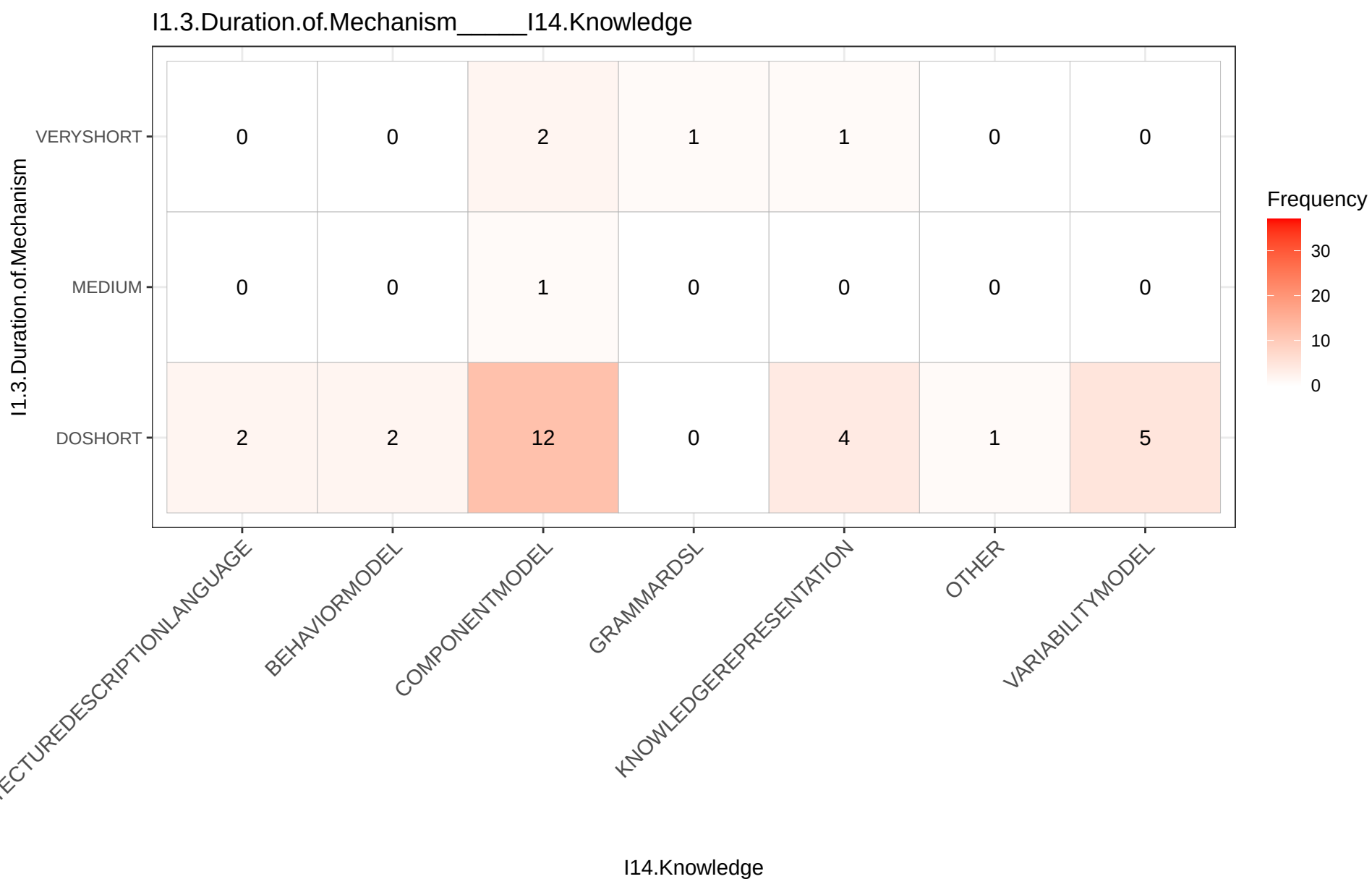
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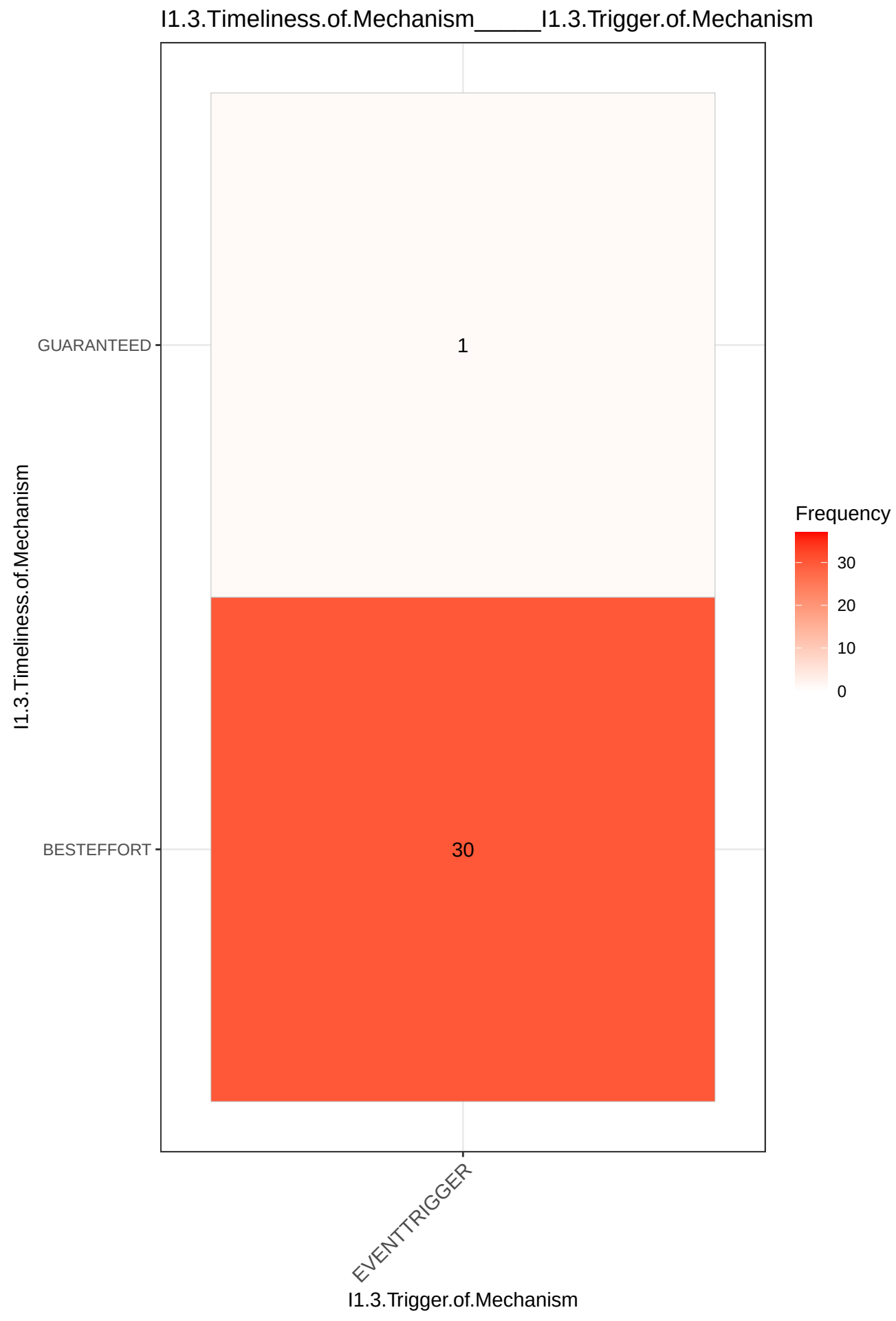
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1

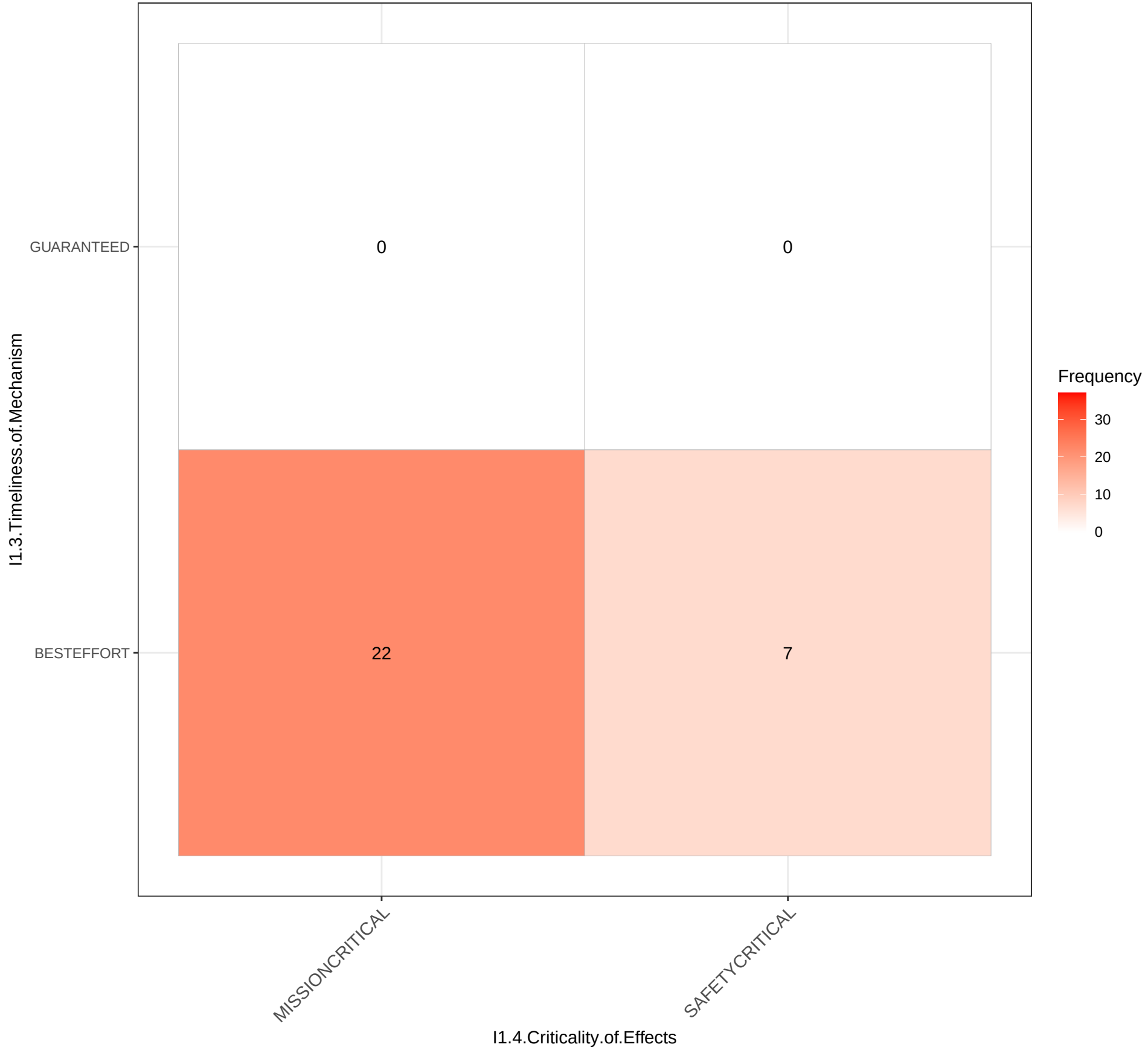
I1.3.Duration.of.Mechanism____I13.Execute







I1.3.Timeliness.of.Mechanism_____I1.4.Criticality.of.Effects



I1.3.Timeliness.of.Mechanism_____I1.4.Predictability.of.Effects

I1.3.Timeliness.of.Mechanism

GUARANTEED

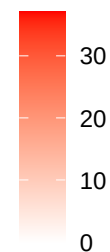
BESTEFFORT

DODETERMINISTIC

NONDETERMINISTIC

I1.4.Predictability.of.Effects

Frequency



1

0

7

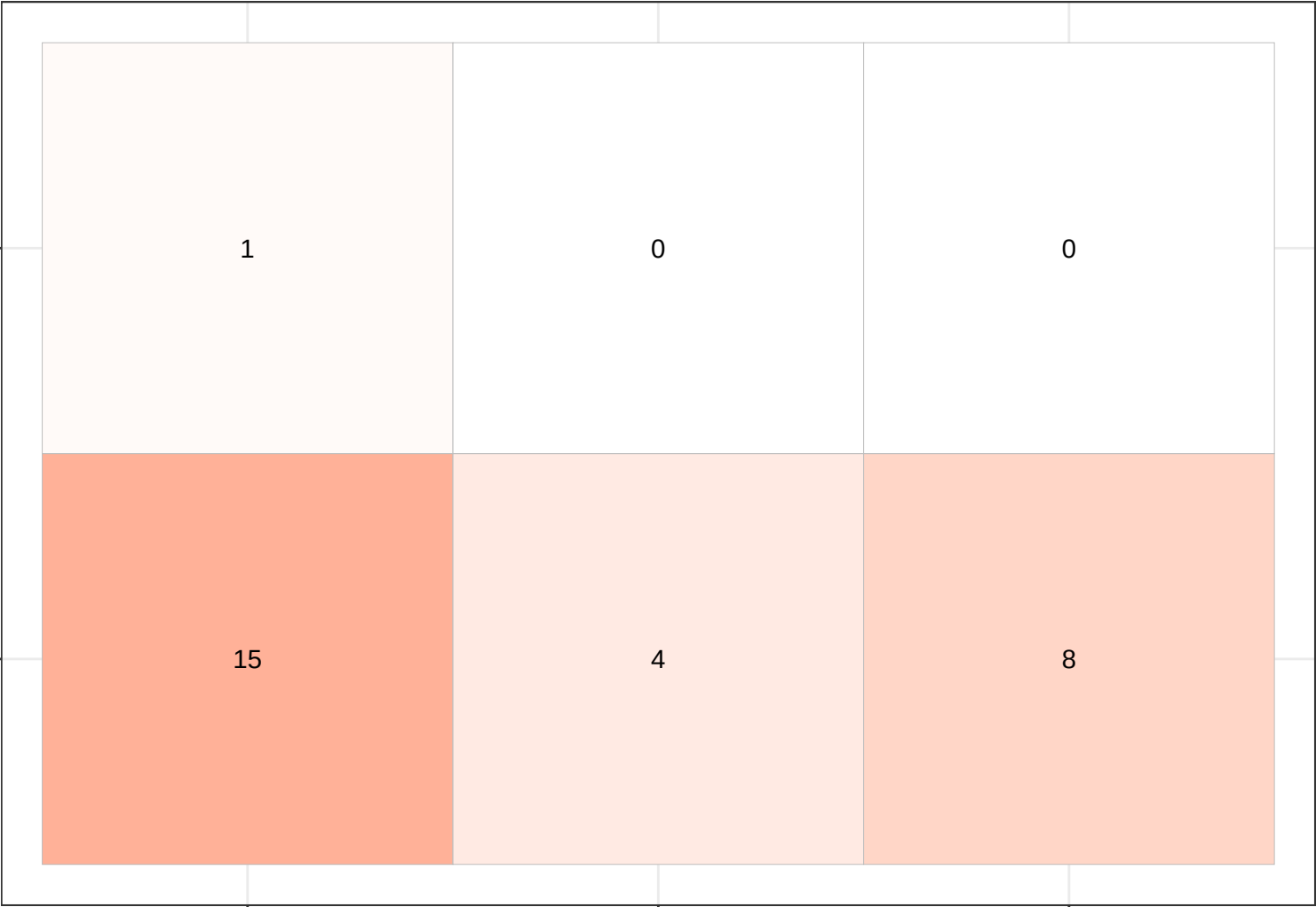
22

I1.3.Timeliness.of.Mechanism_____I1.4.Overhead.of.Effects

I1.3.Timeliness.of.Mechanism

GUARANTEED

BESTEFFECT



DEPENDENT

DOSIGNIFICANT

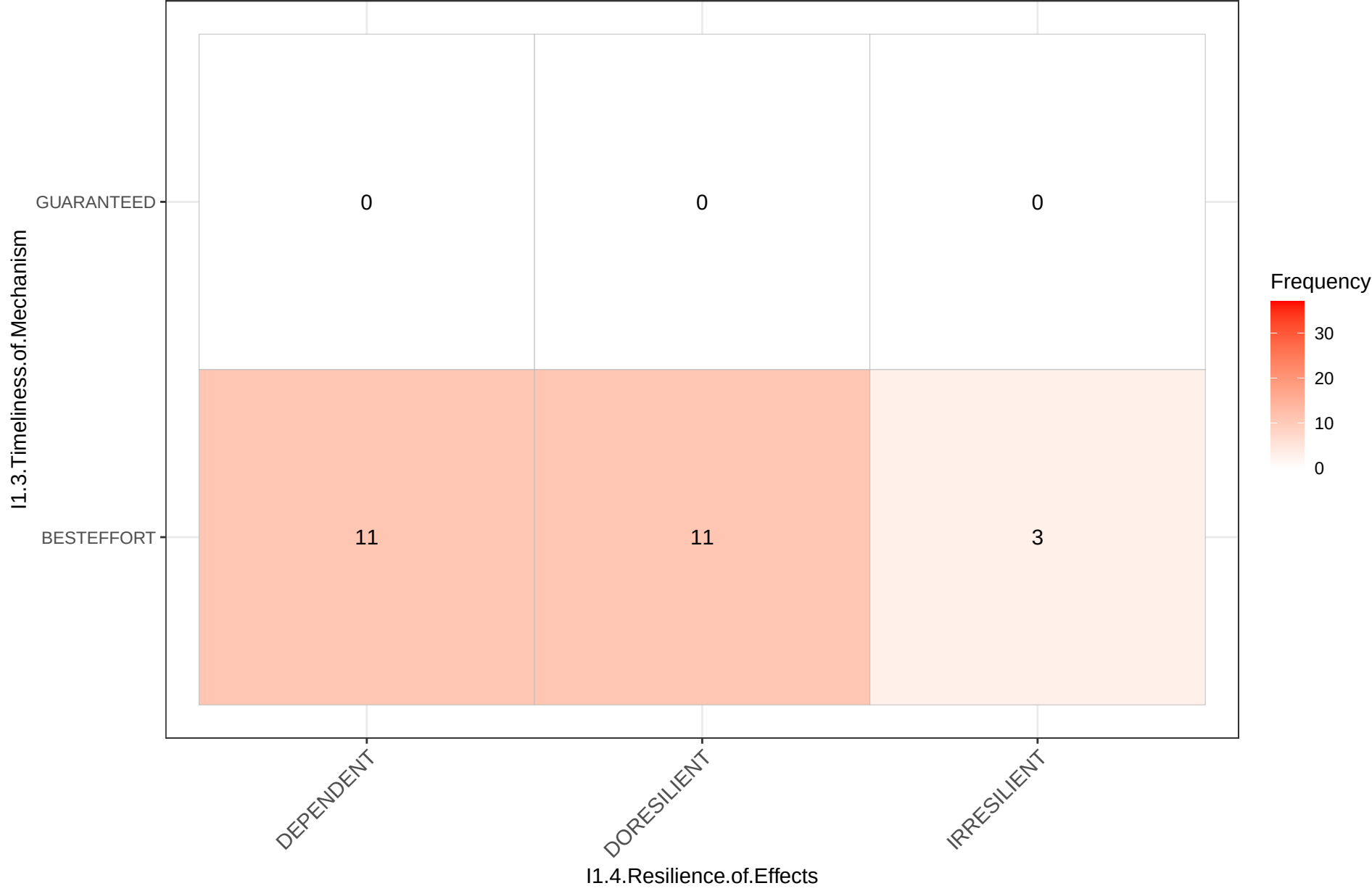
INSIGNIFICANT

I1.4.Overhead.of.Effects

Frequency

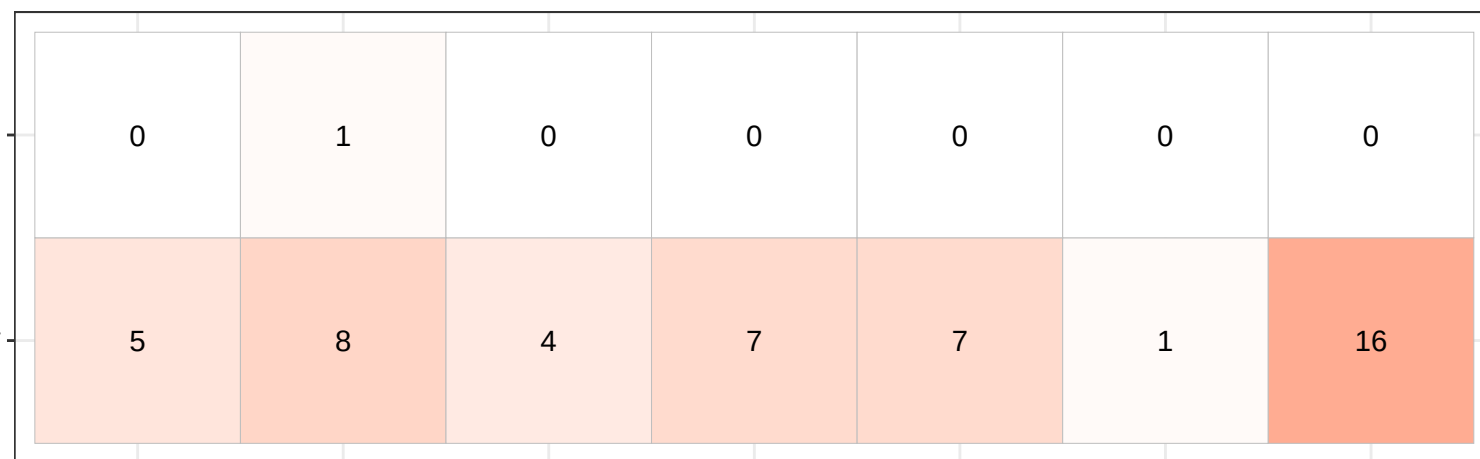


I1.3.Timeliness.of.Mechanism_____I1.4.Resilience.of.Effects

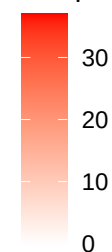


I1.3.Timeliness.of.Mechanism

I1.3.Timeliness.of.Mechanism_____I2.Adap..Purpose



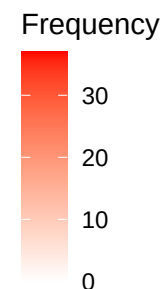
Frequency



I2.Adap..Purpose

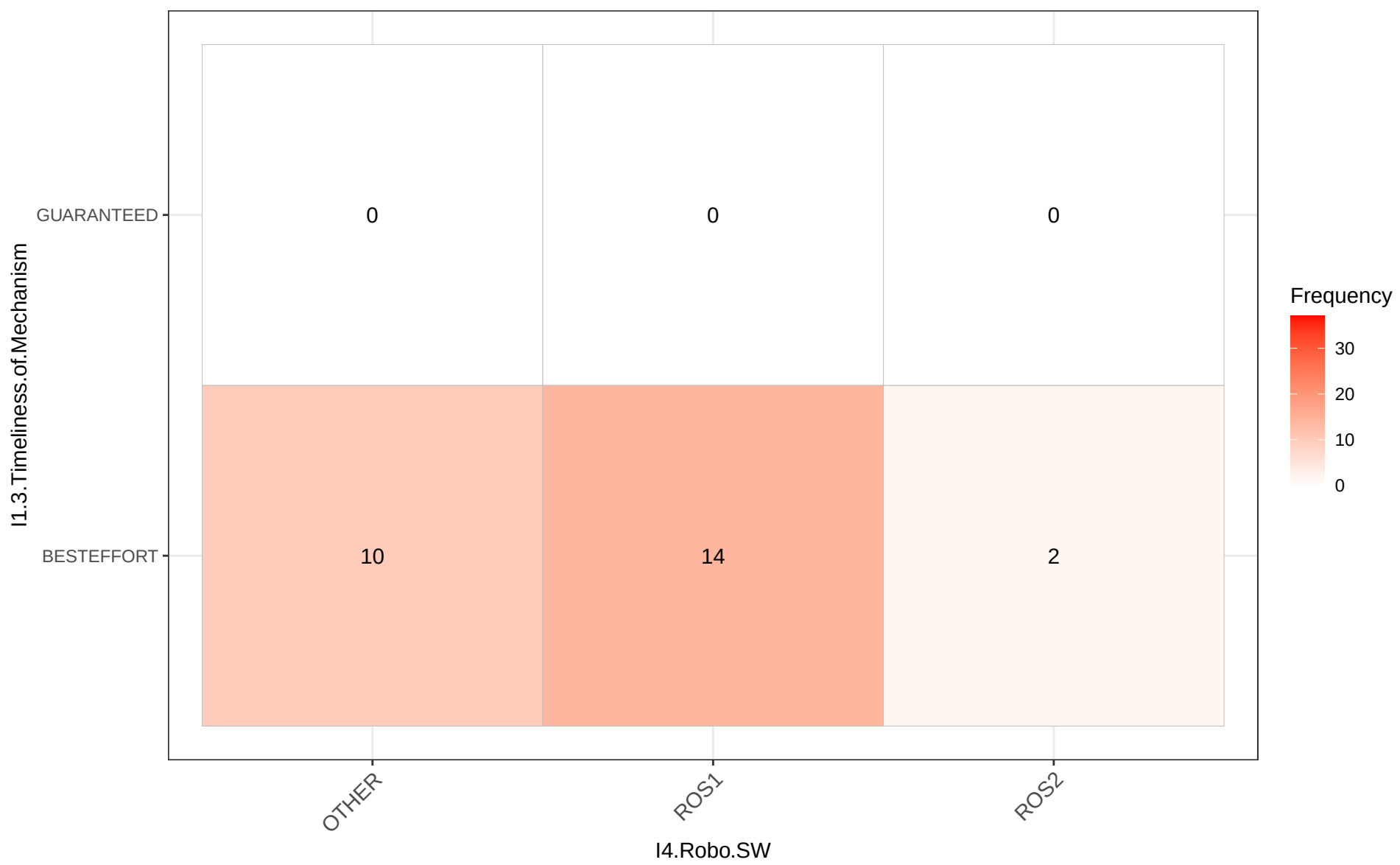
I1.3.Timeliness.of.Mechanism	I3.Robot.Type
------------------------------	---------------

GUARANTEED	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
BESTEFFORT	0	1	1	1	1	1	2	1	2	0	1	1	1	2	1	1	0	1	1	2	1	1	1	1	1	1	5	1	1	1	1	0

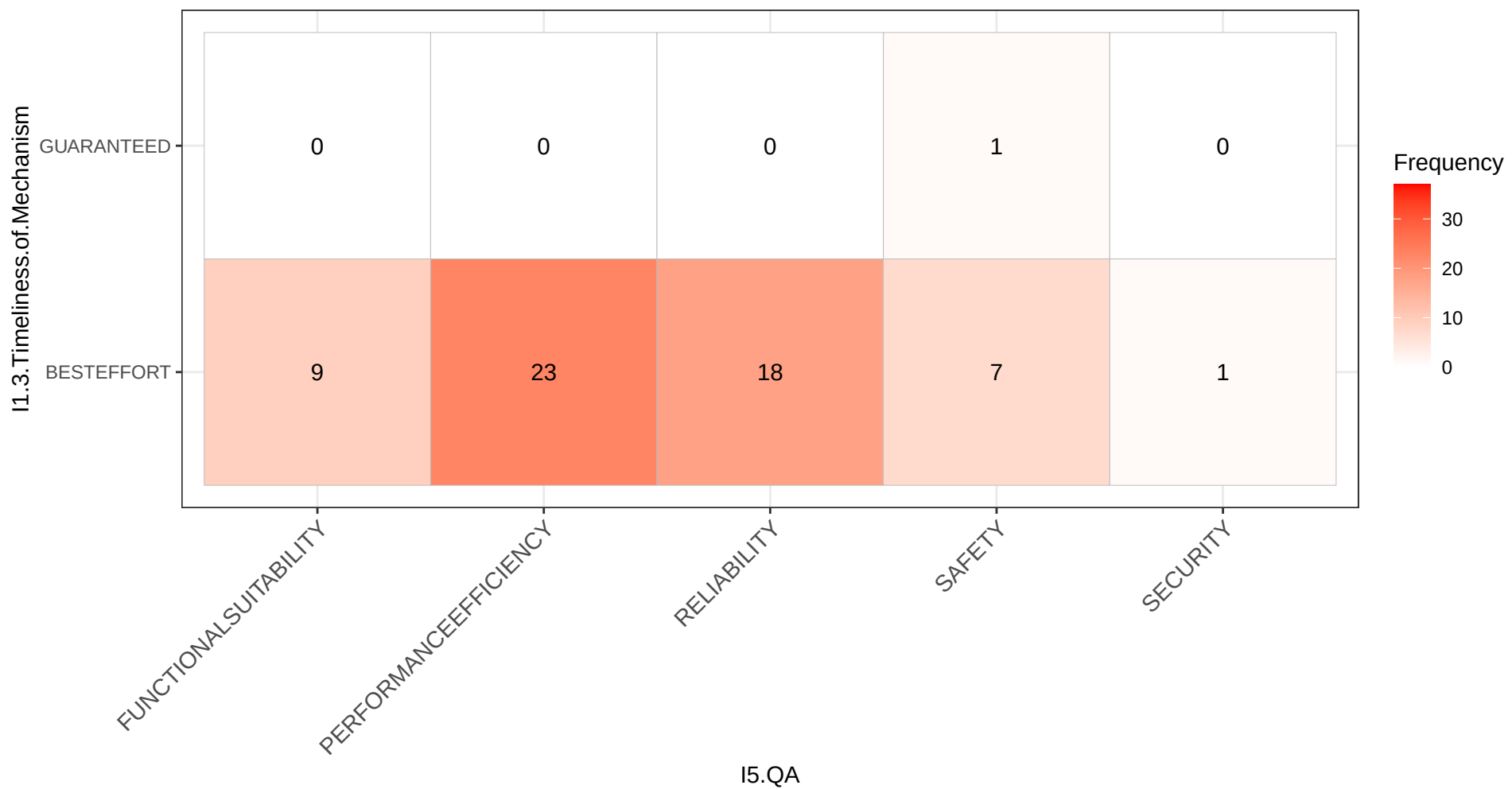


13.Robot.Type

I1.3.Timeliness.of.Mechanism_____I4.Robo.SW



I1.3.Timeliness.of.Mechanism_____I5.QA



I5.QA

I1.3.Timeliness.of.Mechanism_____I6.Independence

I1.3.Timeliness.of.Mechanism

GUARANTEED

BESTEFFORT

DETACHABLE

INSEPARABLE

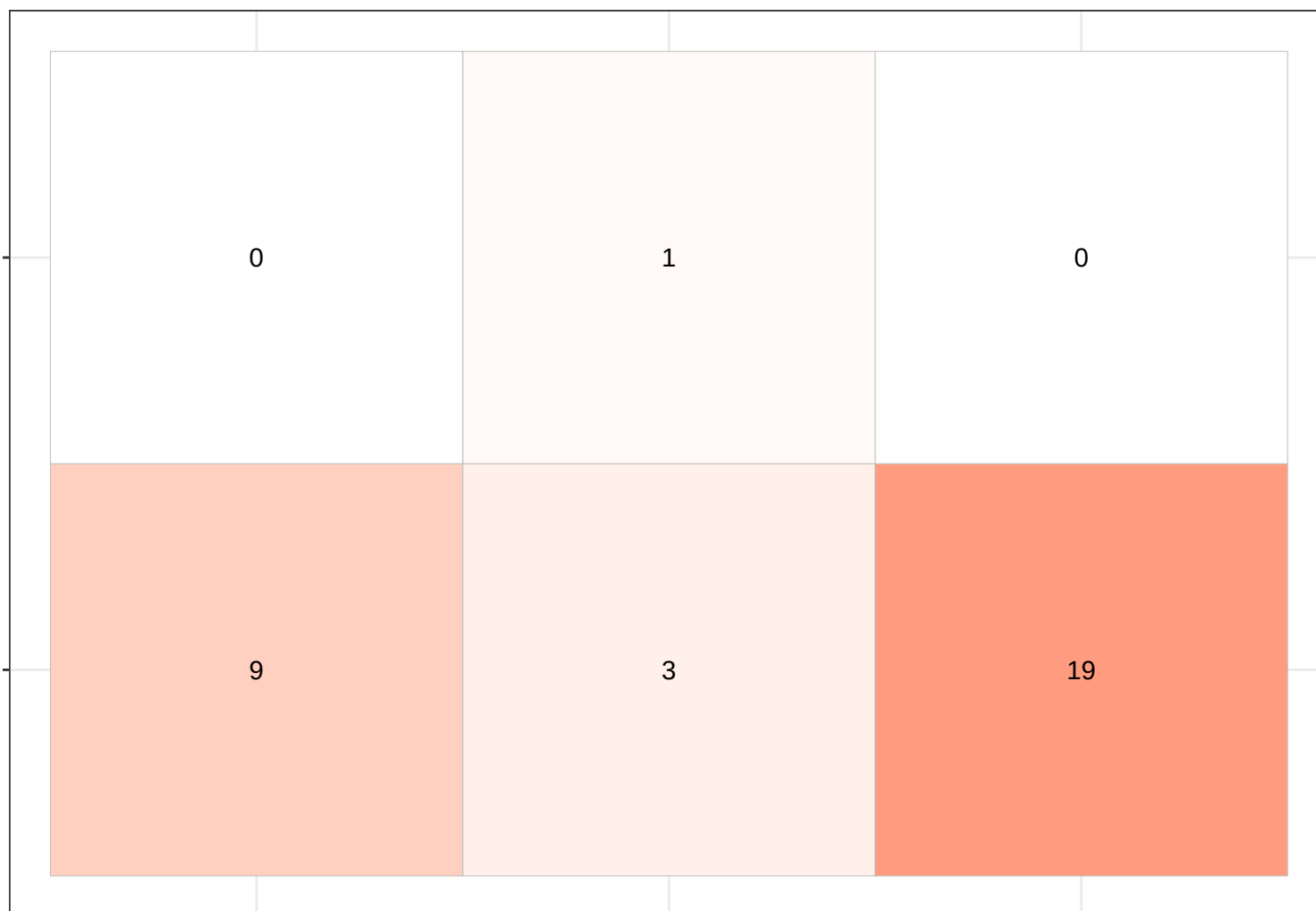
REQUIRESREPRESENTATION

I6.Independence

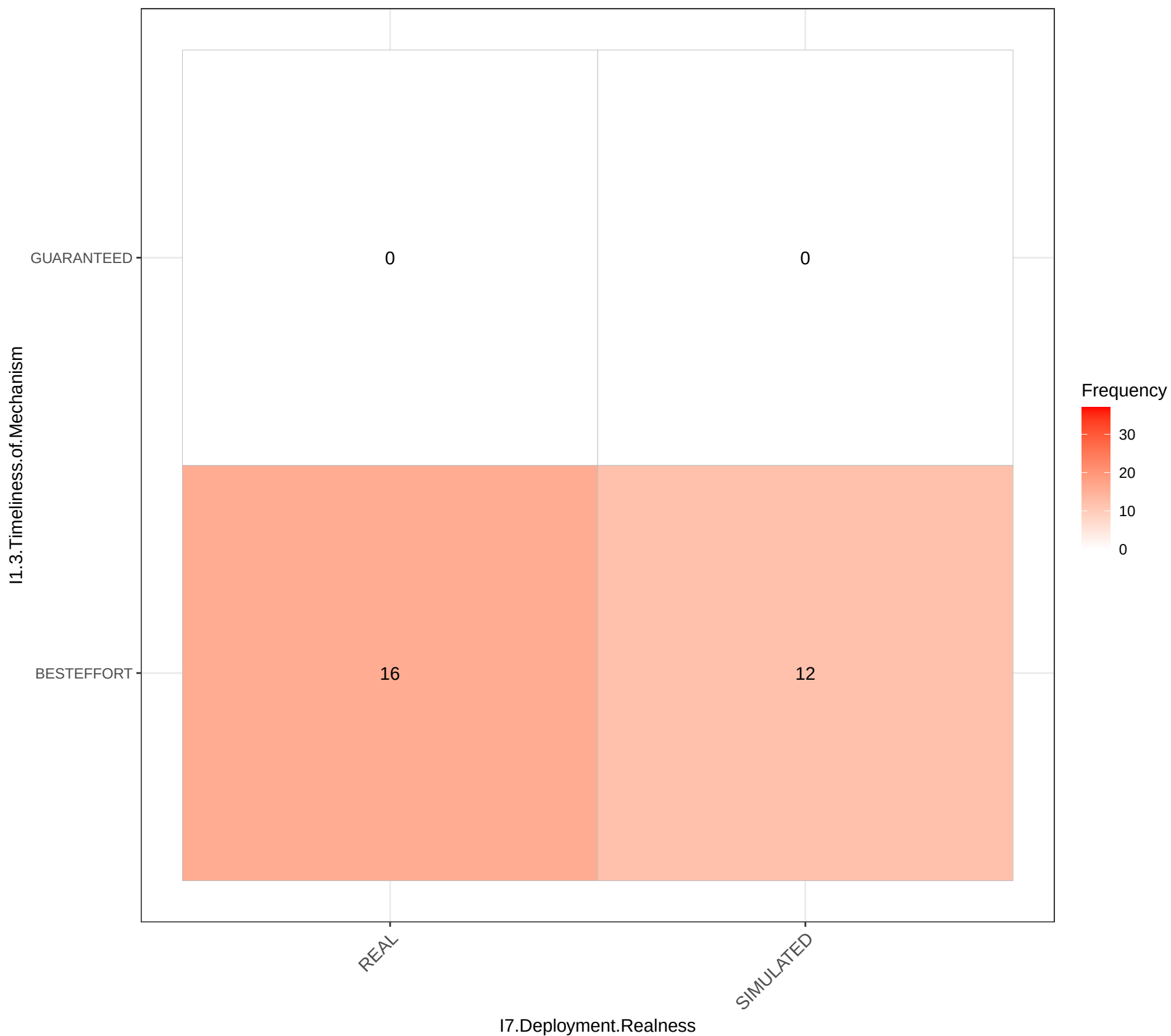
Frequency



30
20
10
0



I1.3.Timeliness.of.Mechanism____I7.Deployment.Realness



I1.3.Timeliness.of.Mechanism_____I7.Mission.Realness

I1.3.Timeliness.of.Mechanism

GUARANTEED

0

1

BESTEFFORT

16

15

REAL

SYNTHETIC

I7.Mission.Realness

Frequency



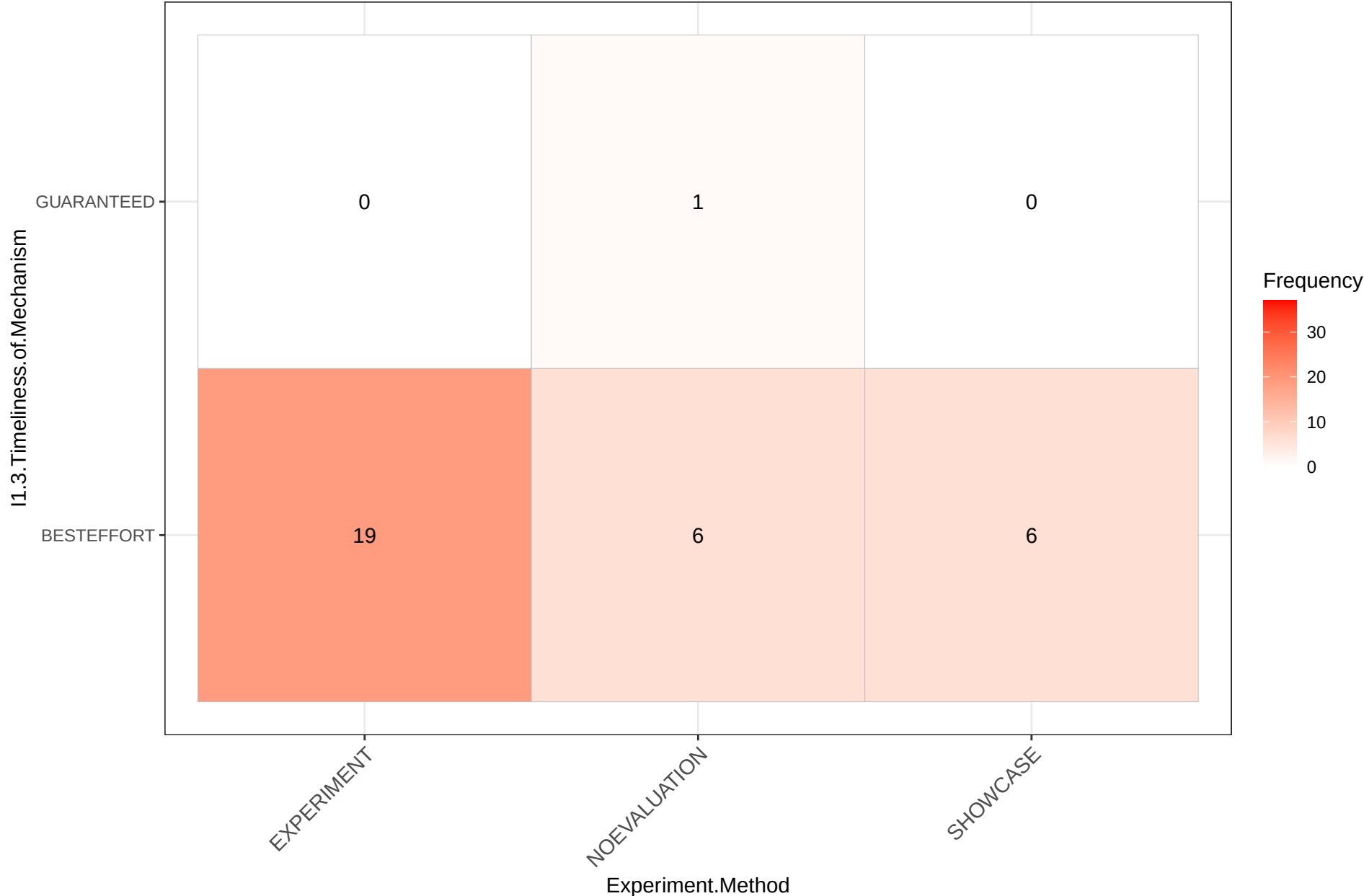
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20

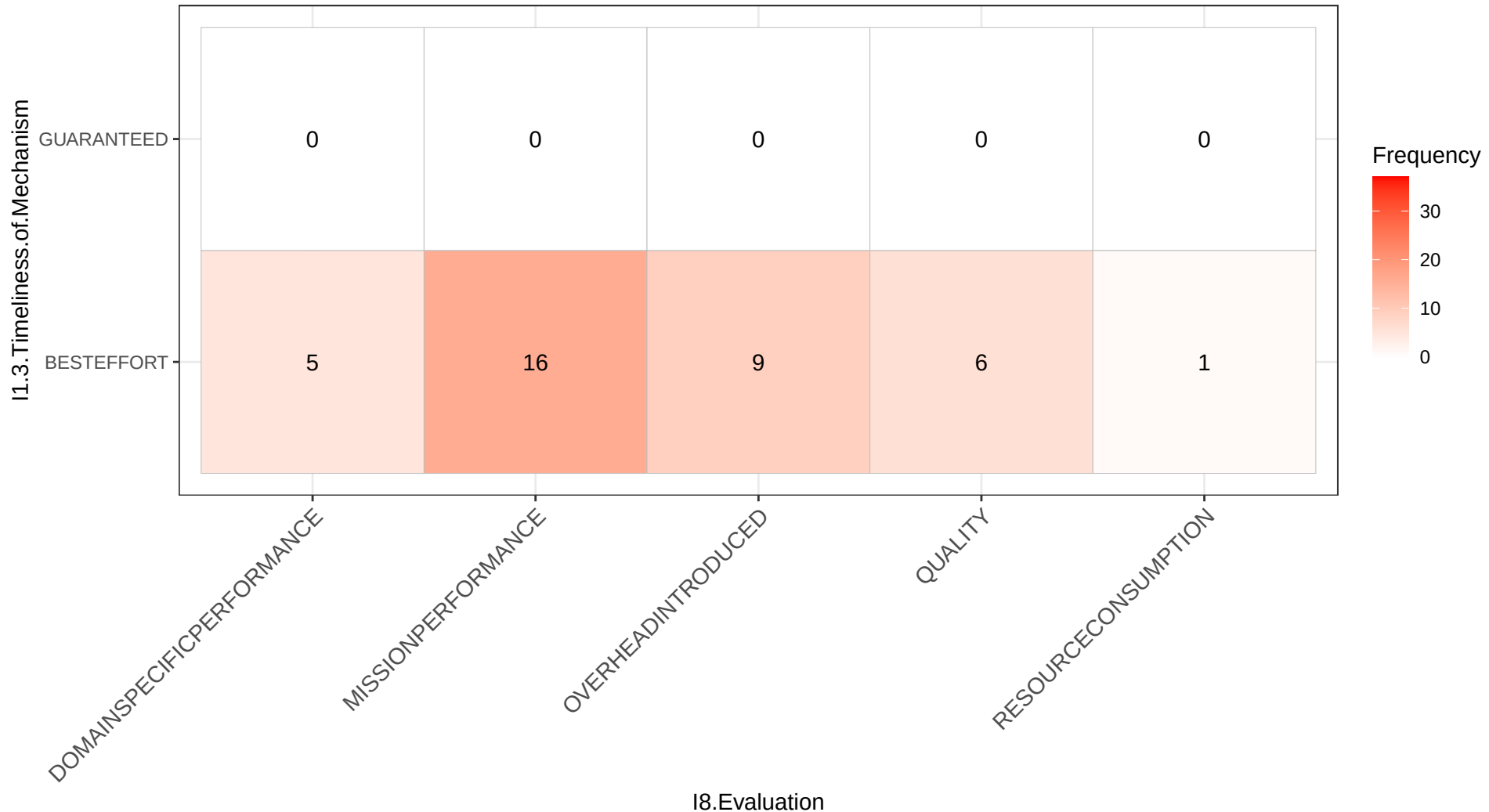
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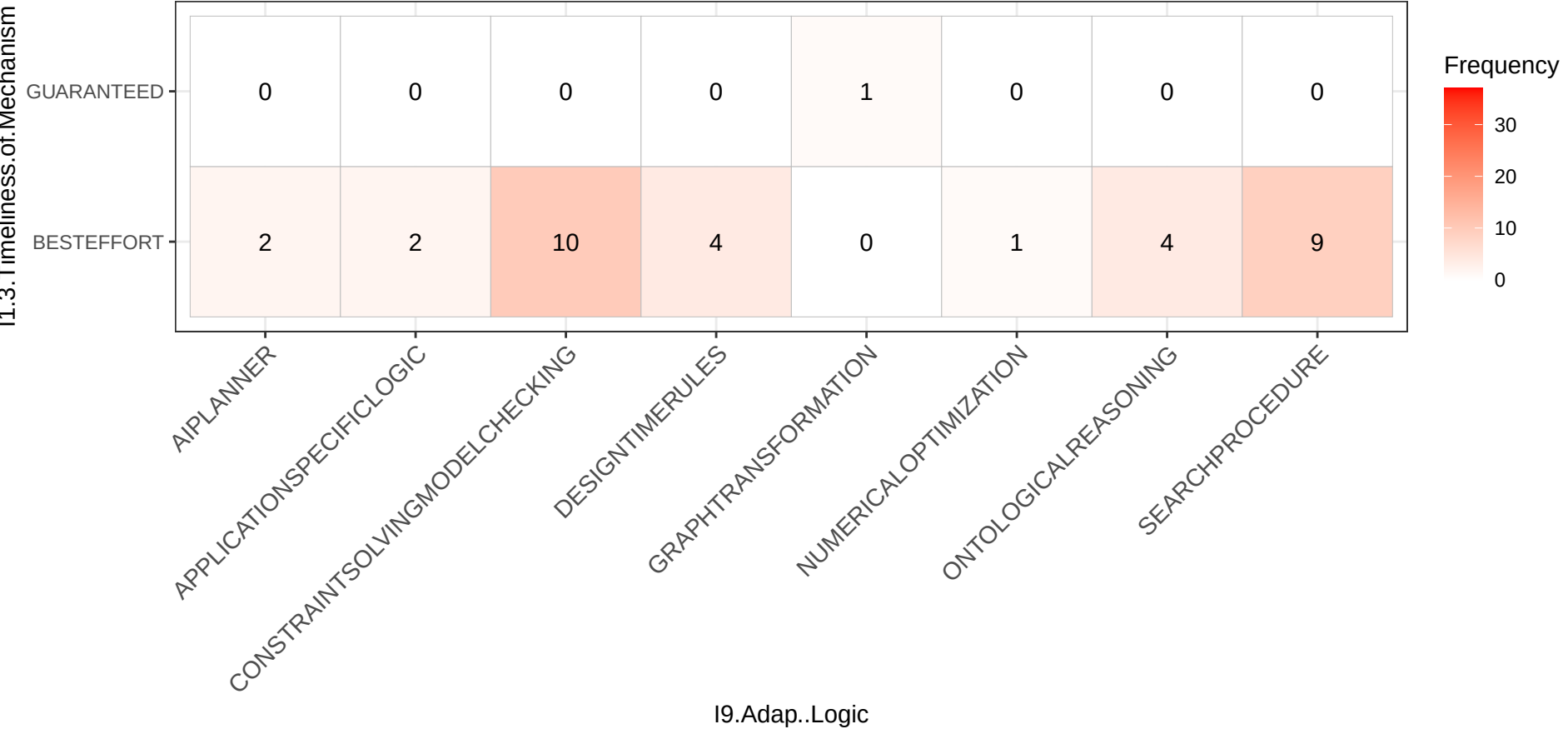
I1.3.Timeliness.of.Mechanism_____Experiment.Method



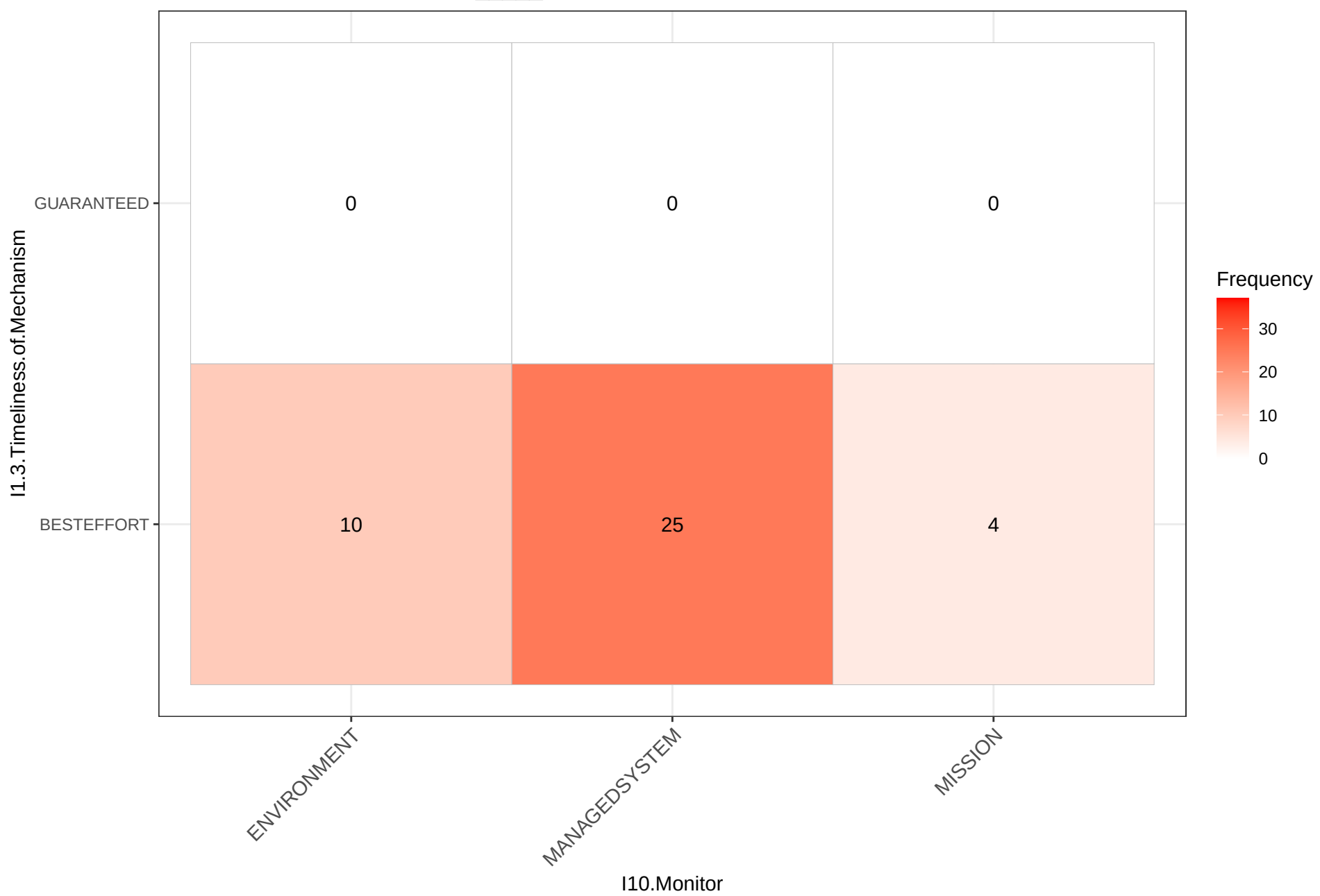
I1.3.Timeliness.of.Mechanism_____I8.Evaluation



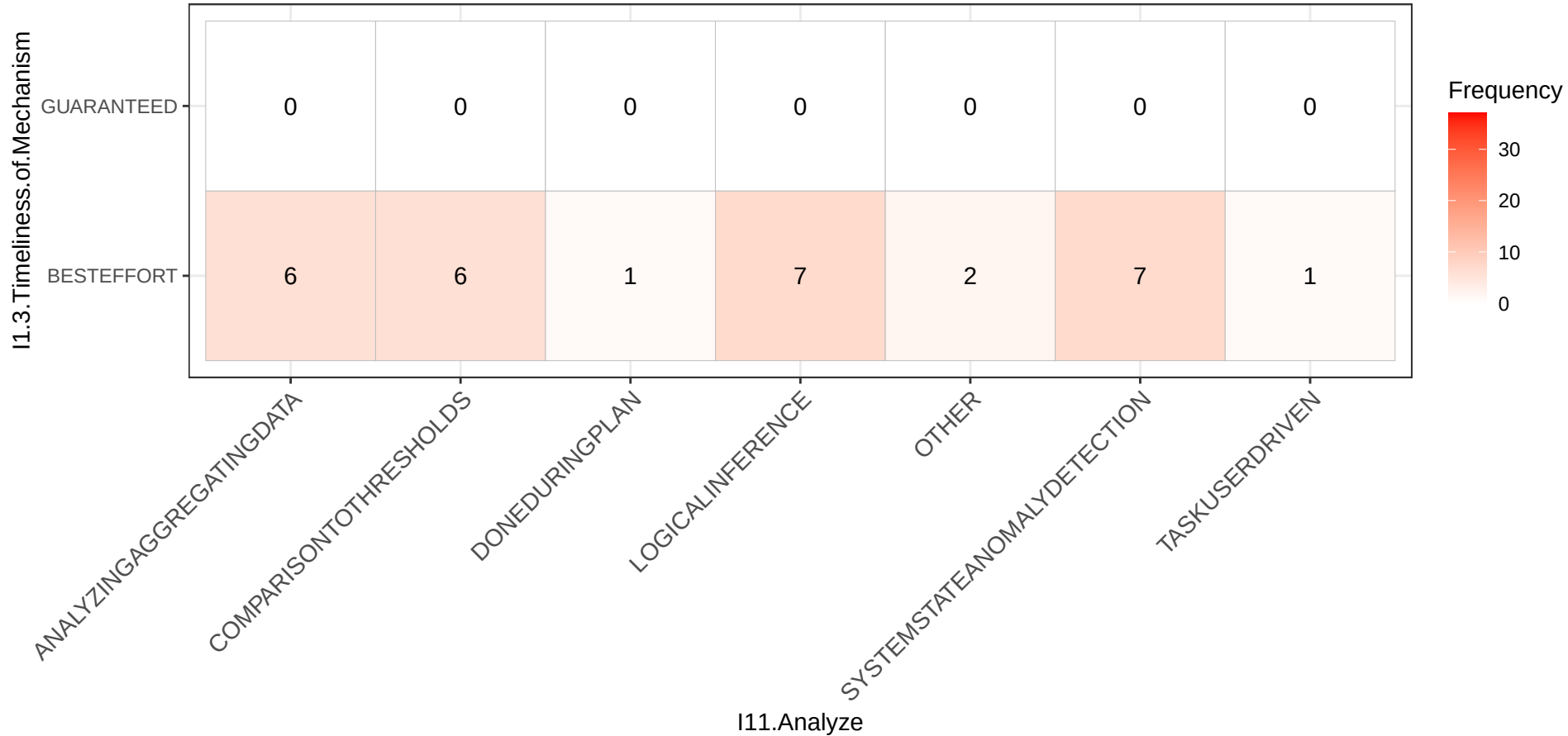
I1.3.Timeliness.of.Mechanism_____I9.Adap..Logic



I1.3.Timeliness.of.Mechanism_____I10.Monitor



I1.3.Timeliness.of.Mechanism_____I11.Analyze



I1.3.Timeliness.of.Mechanism_____I12.Plan

I1.3.Timeliness.of.Mechanism

GUARANTEED

0

1

0

BESTEFFORT

13

14

2

Frequency



30

20

10

0

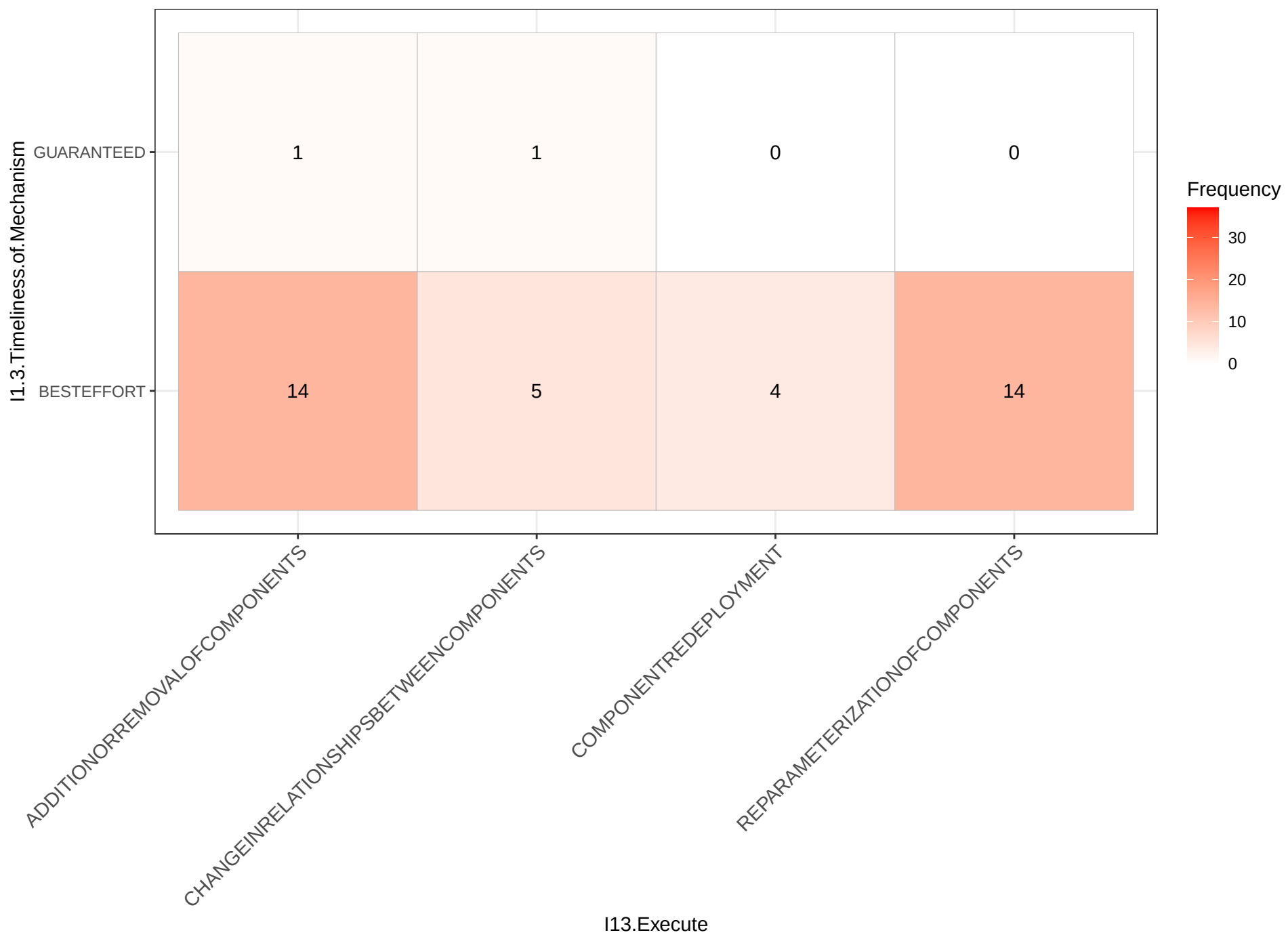
DETERMININGTHEOPTIMALCHOICE

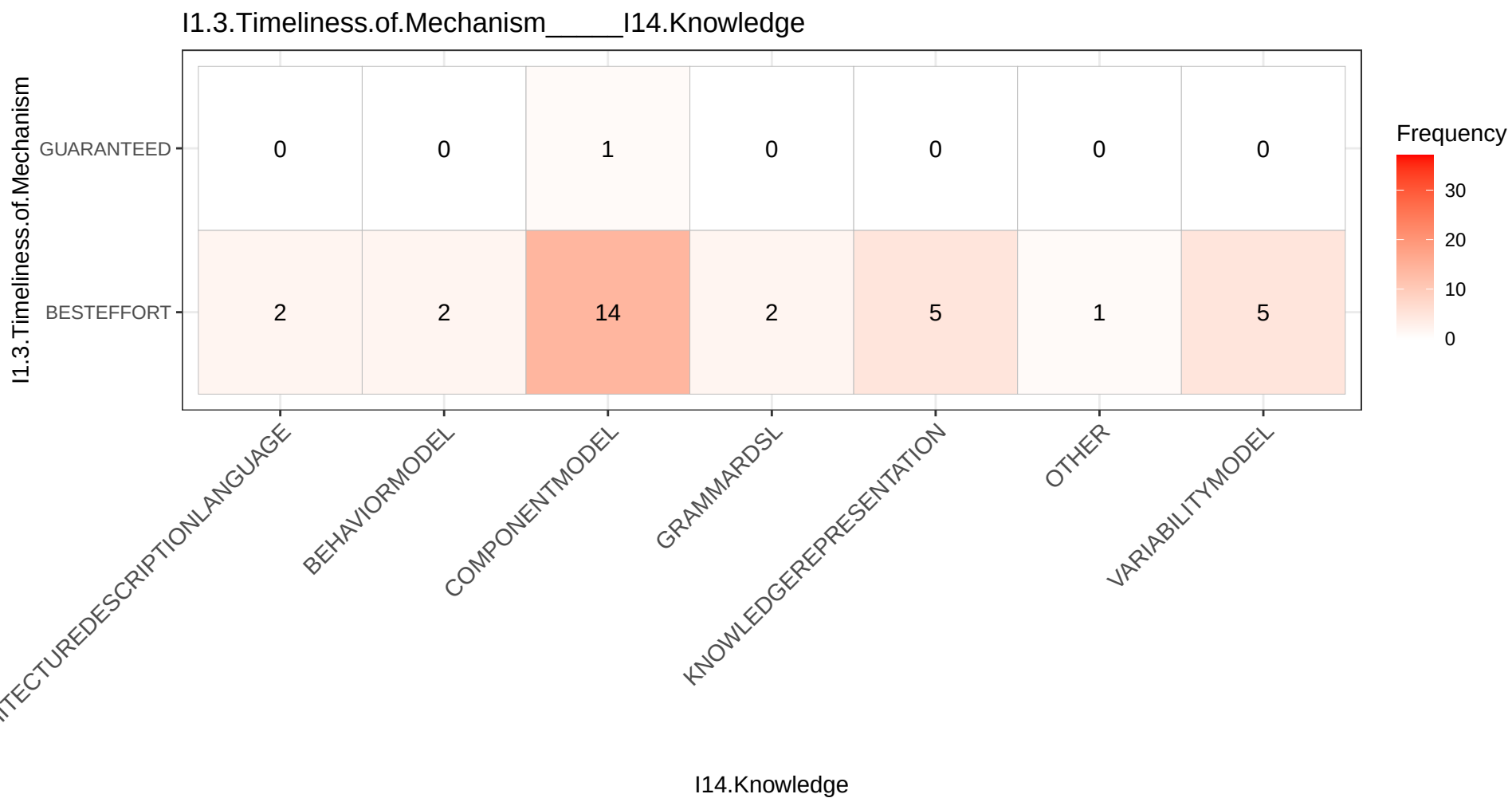
RELYINGONDESIGNTIMERULESMODELS

USINGAIPLANNINGLANGUAGES

I12.Plan

I1.3.Timeliness.of.Mechanism_____I13.Execute





I1.3. Trigger.of.Mechanism

I1.3. Trigger.of.Mechanism_____I1.4. Criticality.of.Effects

EVENTTRIGGER

24

9

MISSIONCRITICAL

SAFETYCRITICAL

I1.4. Criticality.of.Effects

Frequency



30

20

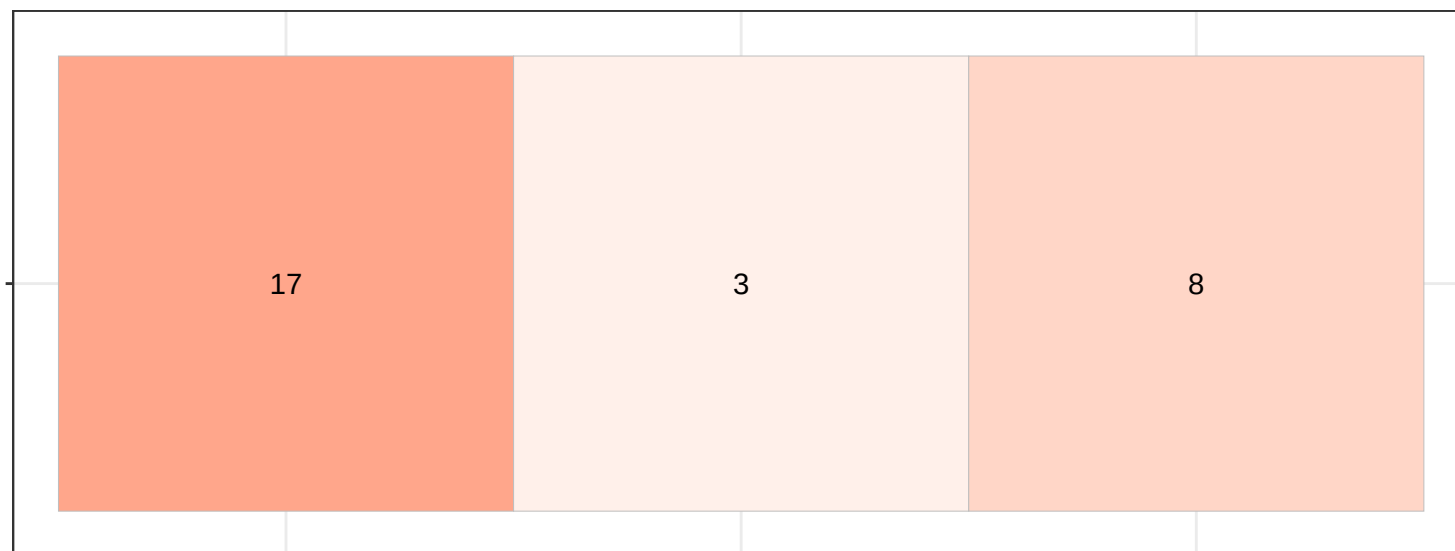
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0

I1.3.Trigger.of.Mechanism

I1.3.Trigger.of.Mechanism_____I1.4.Overhead.of.Effects

EVENTTRIGGER



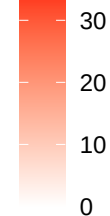
DEPENDENT

DOSIGNIFICANT

INSIGNIFICANT

I1.4.Overhead.of.Effects

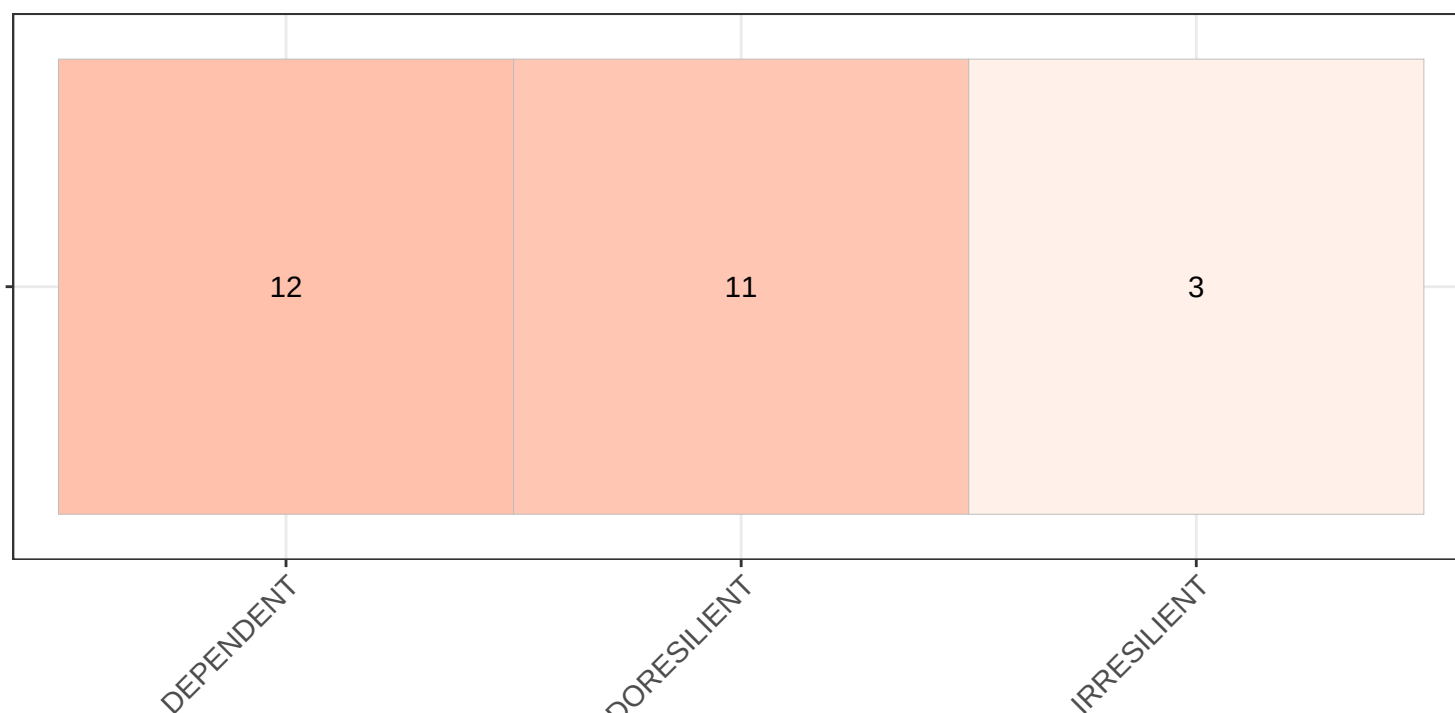
Frequency



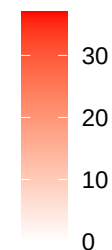
I1.3.Trigger.of.Mechanism

I1.3.Trigger.of.Mechanism_____I1.4.Resilience.of.Effects

EVENTTRIGGER



Frequency



I1.3.Trigger.of.Mechanism_____I3.Robot.Type

1	1	1	1	1	1	2	1	2	1	1	1	1	1	1	2	1	1	2	1	1	1	1	1	1	5	1	1	1	1	1	2
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



20

10

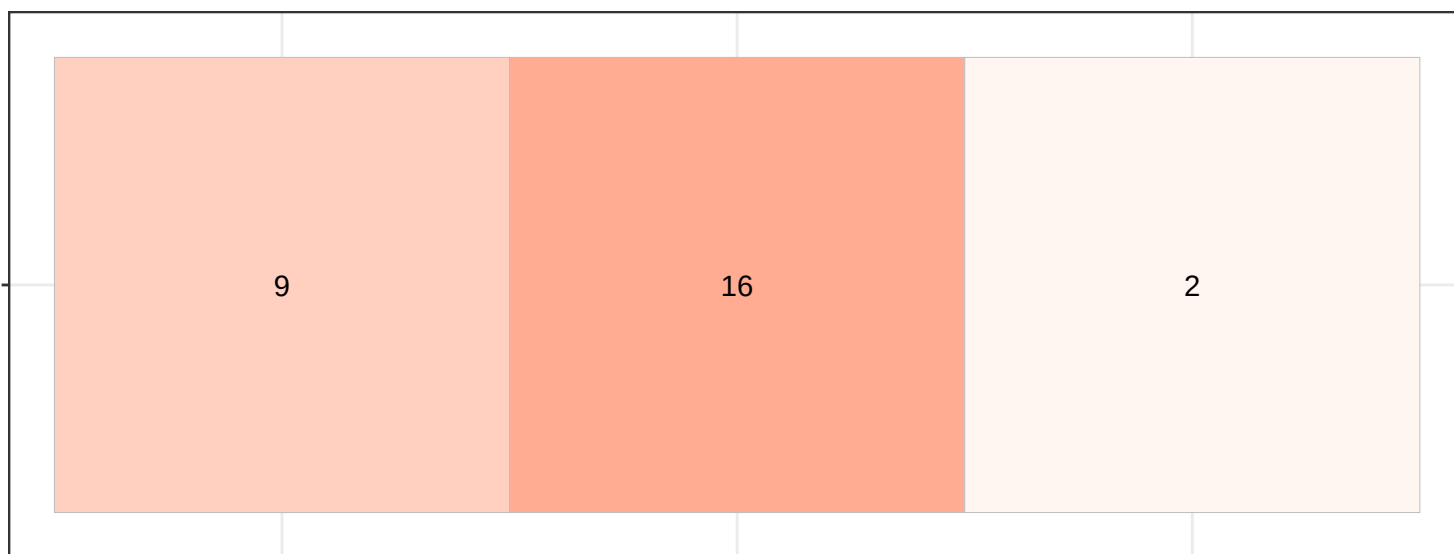
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13.Robot.Type

I1.3. Trigger.of.Mechanism

I1.3.Trigger.of.Mechanism____I4.Robo.SW

EVENTTRIGGER



OTHER

ROS1

ROS2

I4.Robo.SW

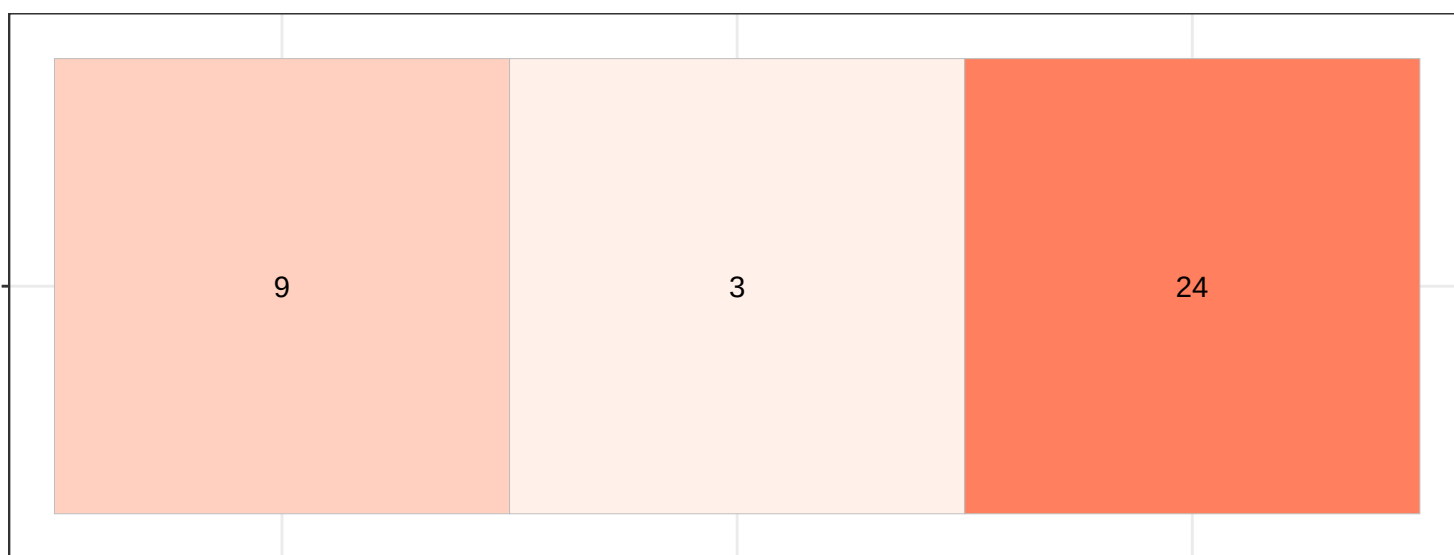
Frequency



I1.3.Trigger.of.Mechanism

I1.3.Trigger.of.Mechanism_____I6.Independence

EVENTTRIGGER



Frequency

30

20

10

0

DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

I1.3.Trigger.of.Mechanism

I1.3.Trigger.of.Mechanism_____I7.Deployment.Realness

EVENTTRIGGER

16

12

REAL

SIMULATED

I7.Deployment.Realness

Frequency



30

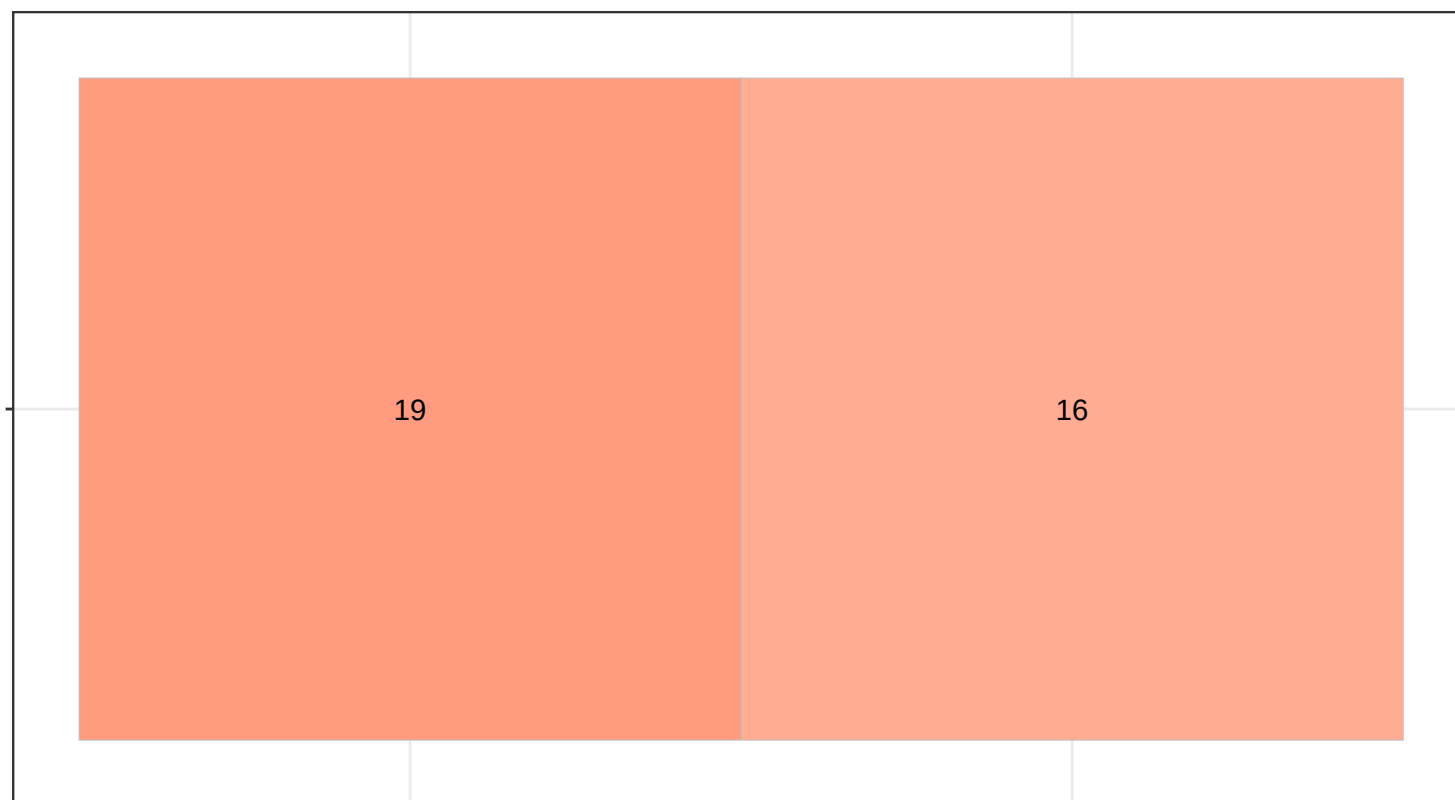
20

10

0

I1.3.Trigger.of.Mechanism

I1.3.Trigger.of.Mechanism_____I7.Mission.Realness



EVENTTRIGGER

REAL

SYNTHETIC

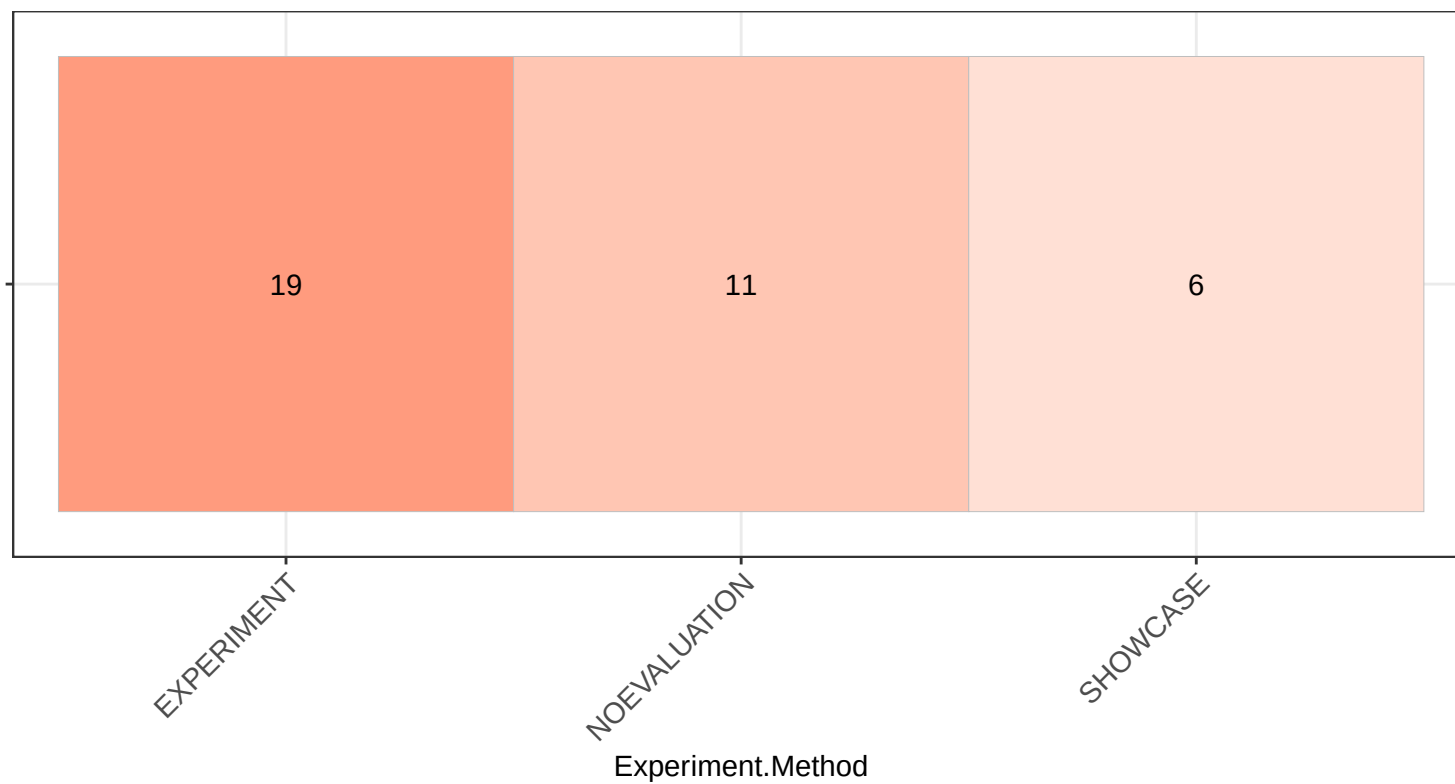
I7.Mission.Realness

Frequency



I1.3.Trigger.of.Mechanism

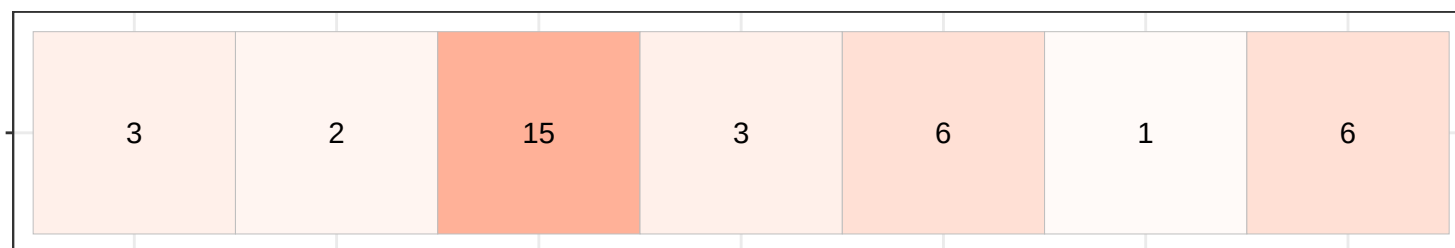
I1.3.Trigger.of.Mechanism_____Experiment.Method



I1.3. Trigger.of.Mechanism

I1.3.Trigger.of.Mechanism____I14.Knowledge

EVENTTRIGGER



Frequency



I14.Knowledge

I1.4.Criticality.of.Effects_____I1.4.Predictability.of.Effects

I1.4.Criticality.of.Effects

SAFETYCRITICAL

2

4

MISSIONCRITICAL

7

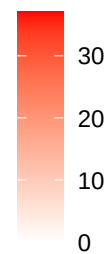
18

DODETERMINISTIC

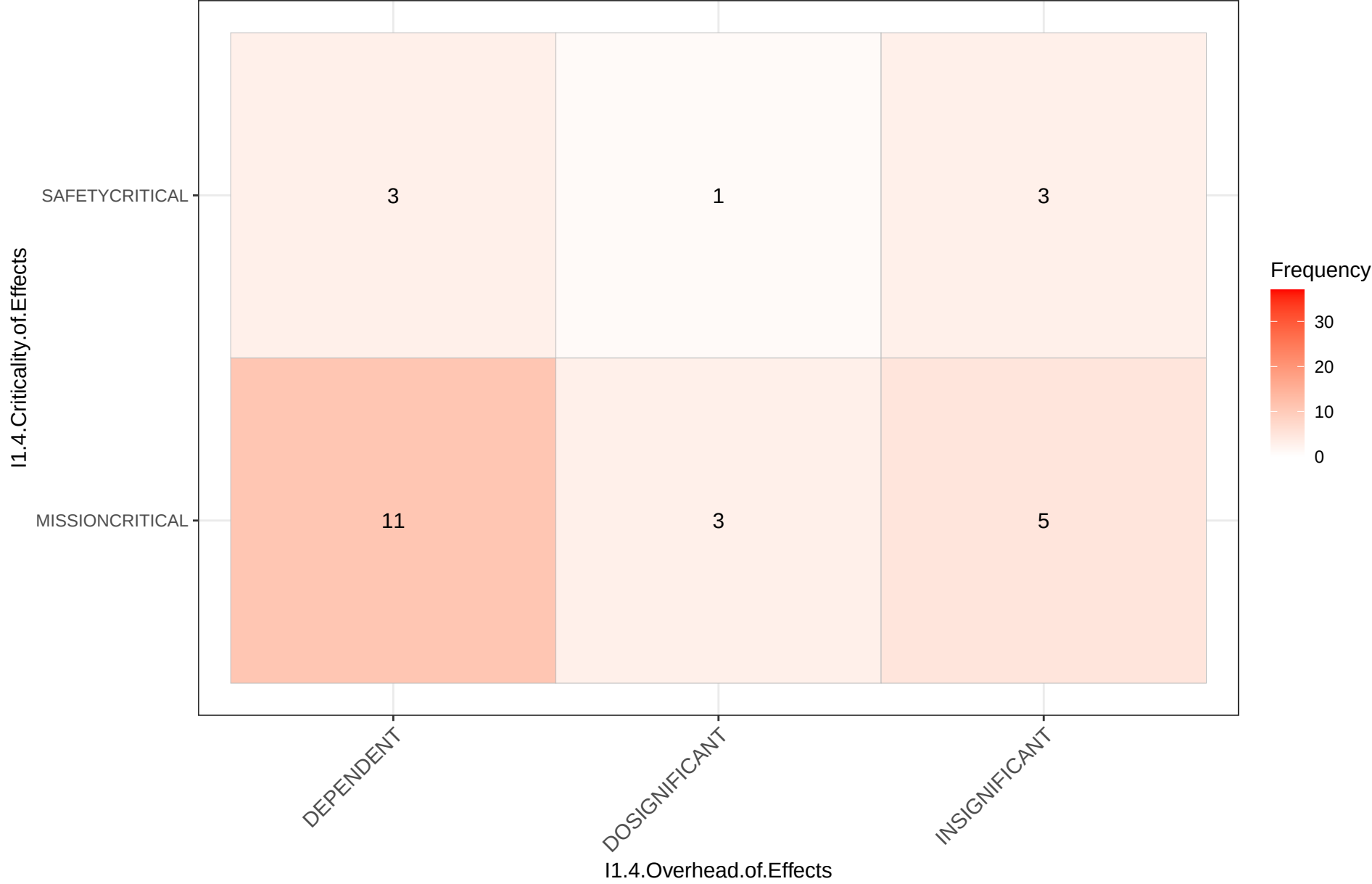
NONDETERMINISTIC

I1.4.Predictability.of.Effects

Frequency



I1.4.Criticality.of.Effects_____I1.4.Overhead.of.Effects



I1.4.Criticality.of.Effects_____I1.4.Resilience.of.Effects

I1.4.Criticality.of.Effects

SAFETYCRITICAL

MISSIONCRITICAL

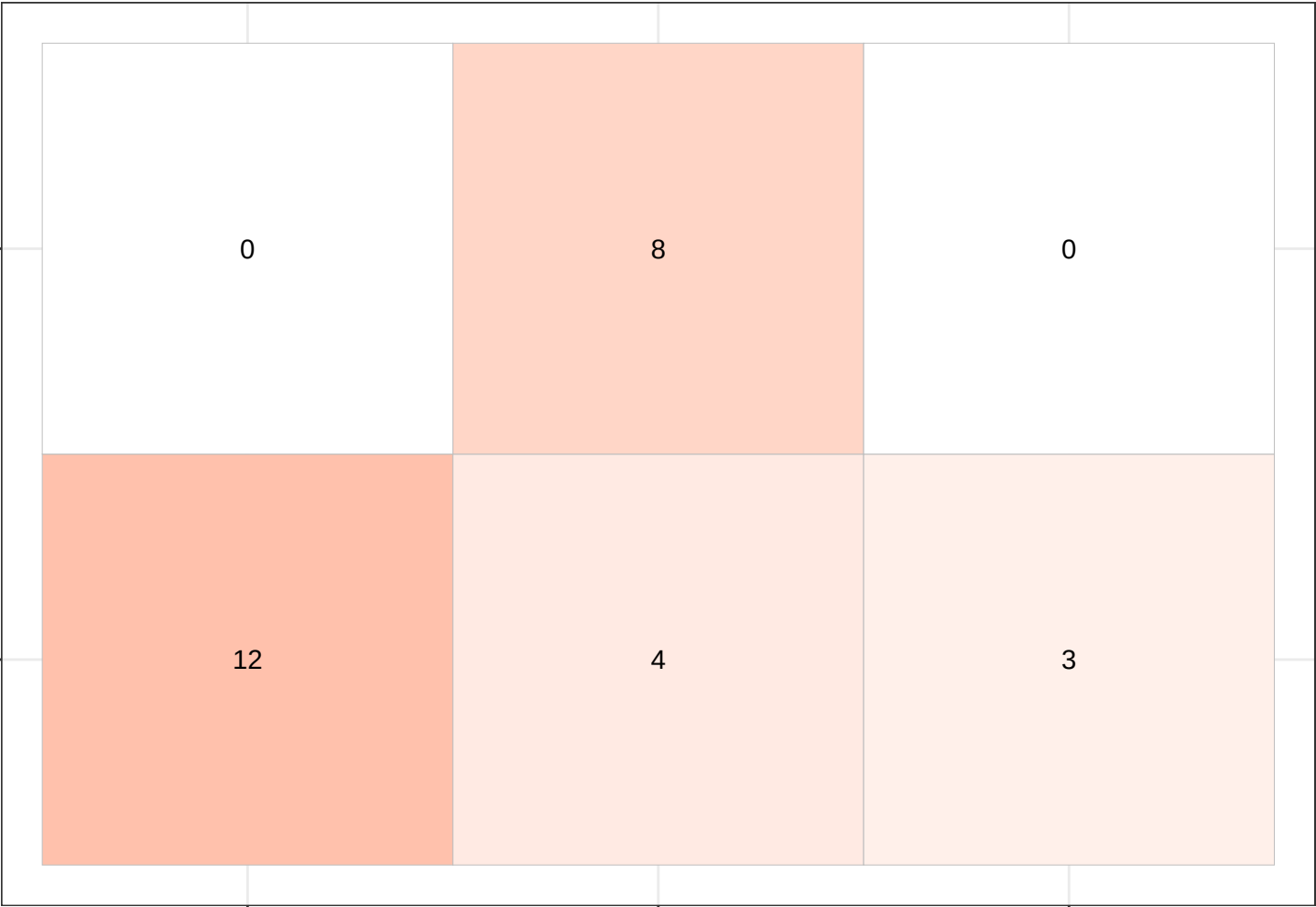
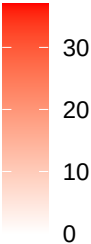
DEPENDENT

DORESILIENT

IRRESILIENT

I1.4.Resilience.of.Effects

Frequency



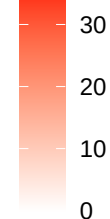
I1.4.Criticality.of.Effects

I1.4.Criticality.of.Effects_____I2.Adap..Purpose

SAFETYCRITICAL

MISSIONCRITICAL

Frequency



CHANGEFUNCTIONALBEHAVIOR	DEALWITHENVIRONMENTALCHANGES	KEEPMEETINGQUALITYREQUIREMENTSATRUNTIME	OPTIMIZERESOURCEUSAGE	OPTIMIZESYSTEMPERFORMANCE	RECOVERFROMATTACKS	RECOVERFROMERRORSFAULTS
0	5	2	2	1	0	4
6	7	3	4	6	1	10

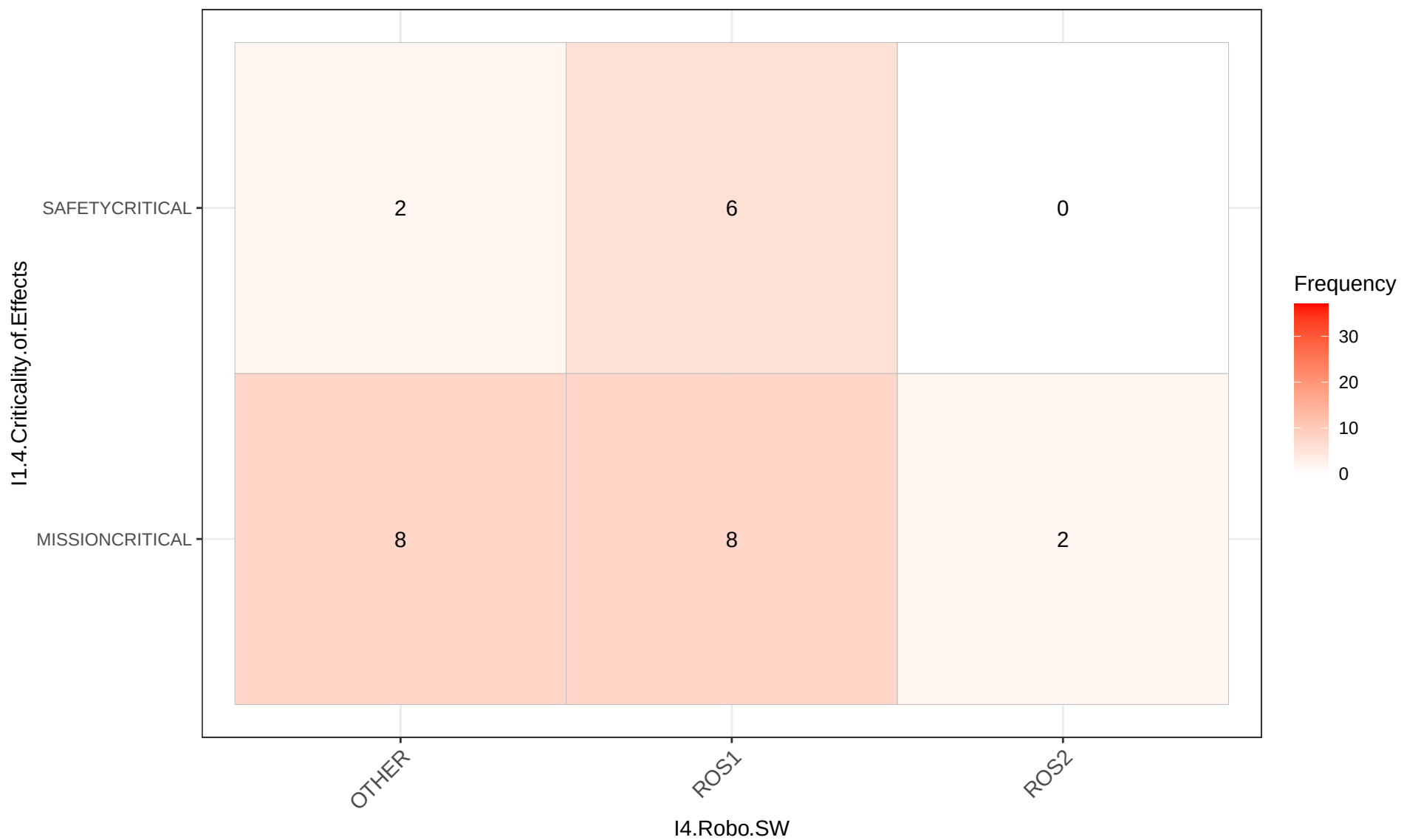
I2.Adap..Purpose

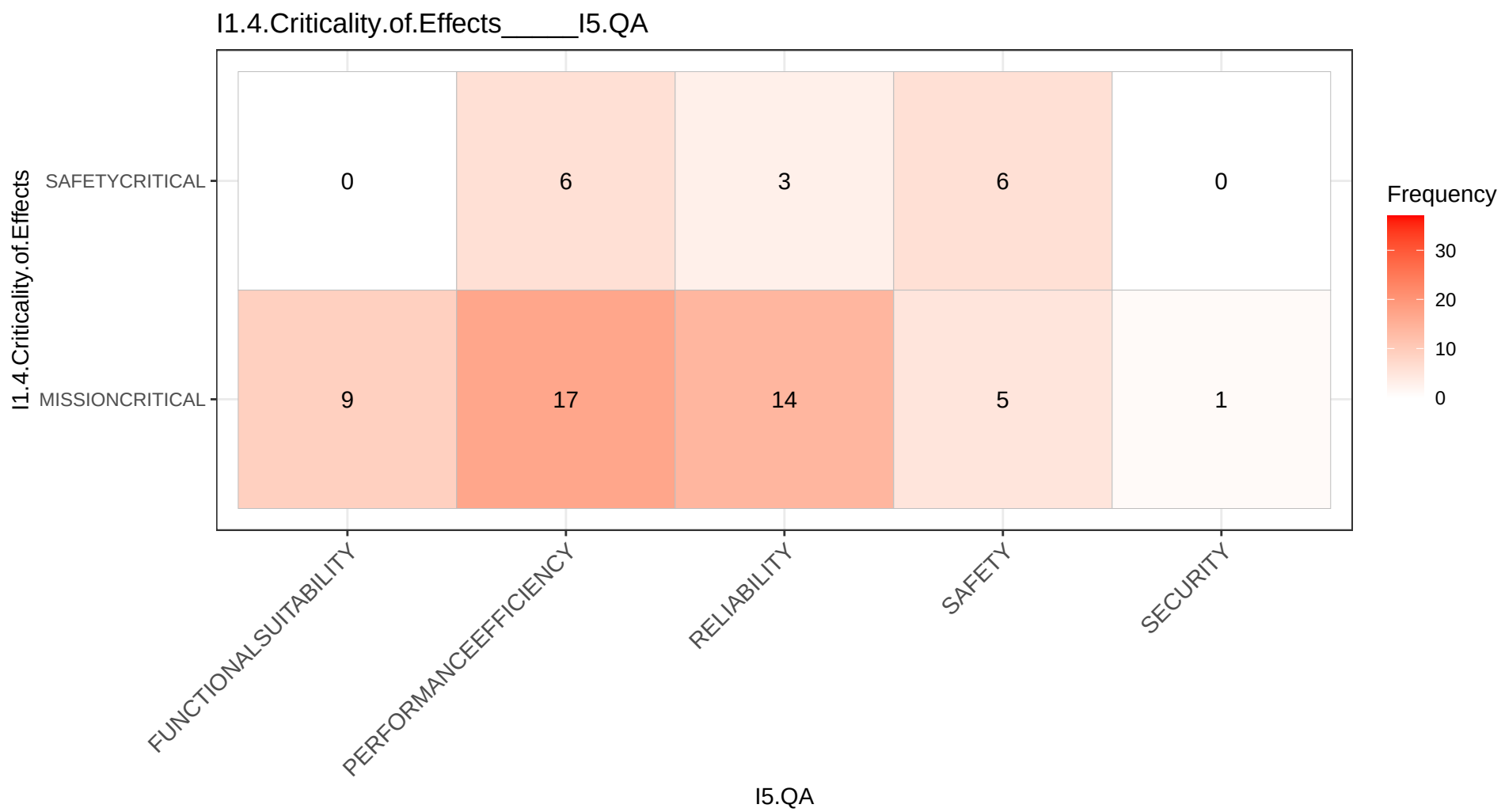
I1.4.Criticality.of.Effects_____I3.Robot.Type

Frequency

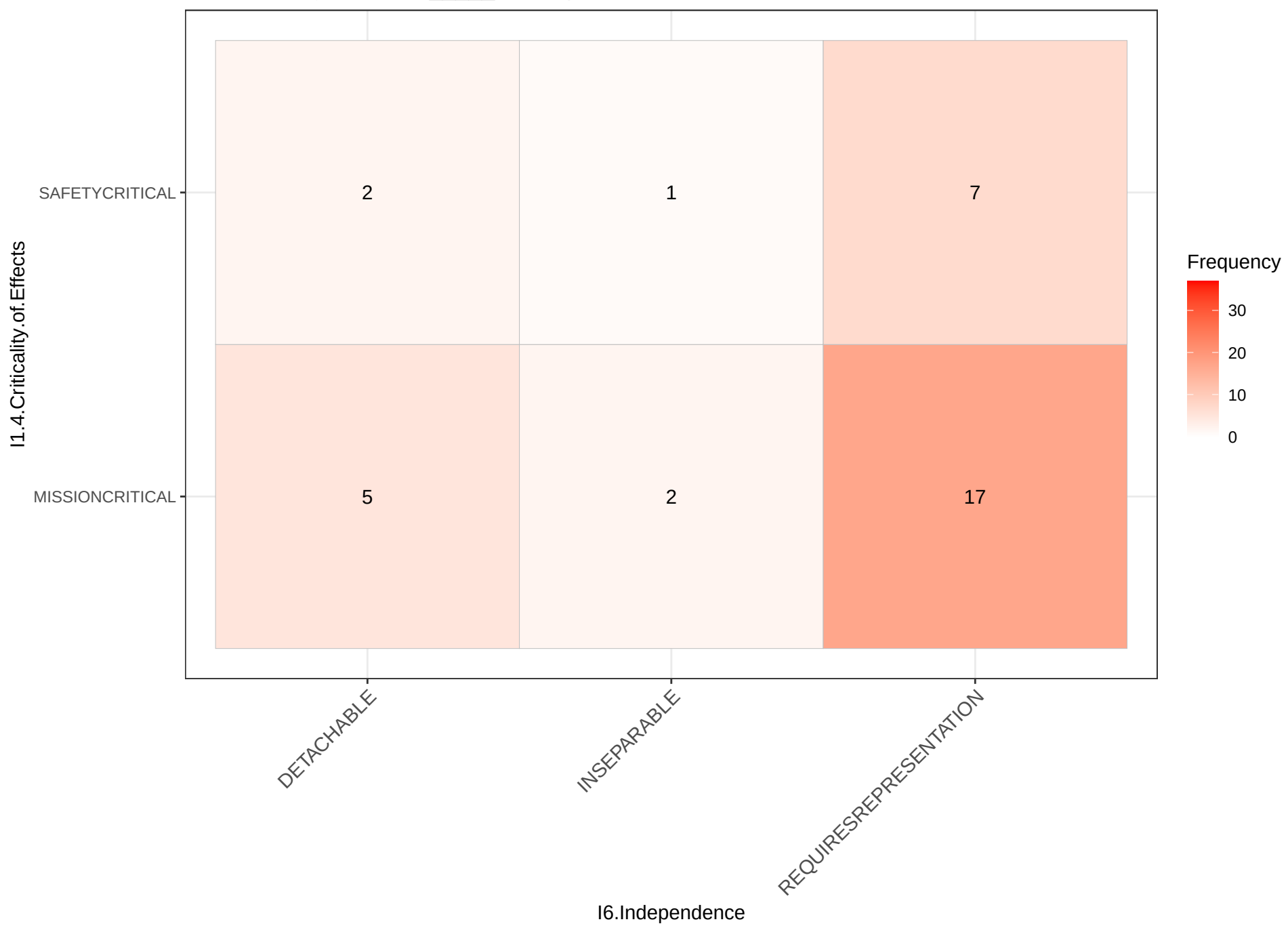
13.Robot.Type

I1.4.Criticality.of.Effects_____I4.Robo.SW





I1.4.Criticality.of.Effects_____I6.Independence



I1.4.Criticality.of.Effects_____I7.Deployment.Realness

I1.4.Criticality.of.Effects

SAFETYCRITICAL

MISSIONCRITICAL

REAL

SIMULATED

I7.Deployment.Realness

Frequency



3

4

12

8

I1.4.Criticality.of.Effects_____I7.Mission.Realness

I1.4.Criticality.of.Effects

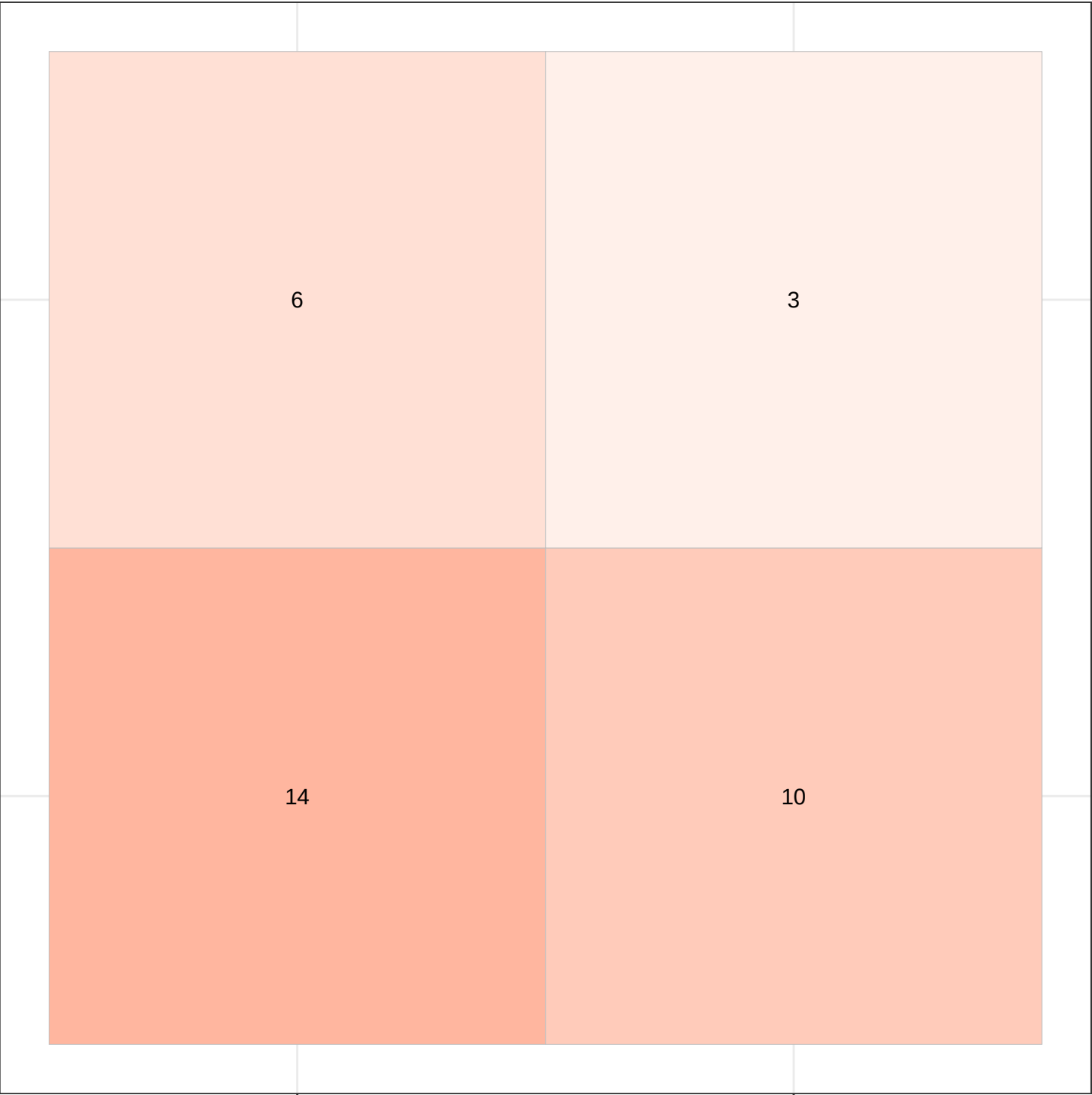
SAFETYCRITICAL

MISSIONCRITICAL

REAL

SYNTHETIC

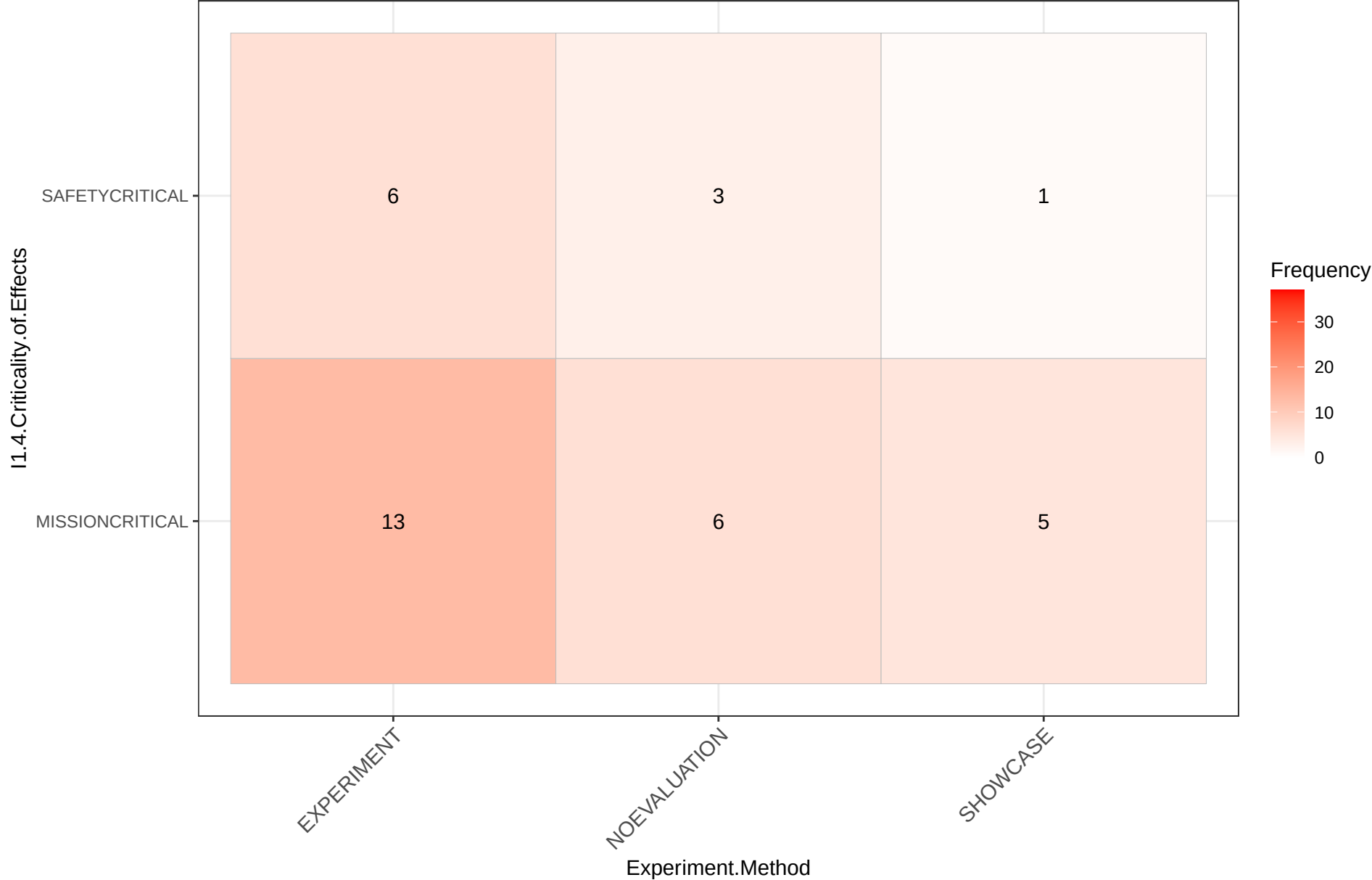
I7.Mission.Realness



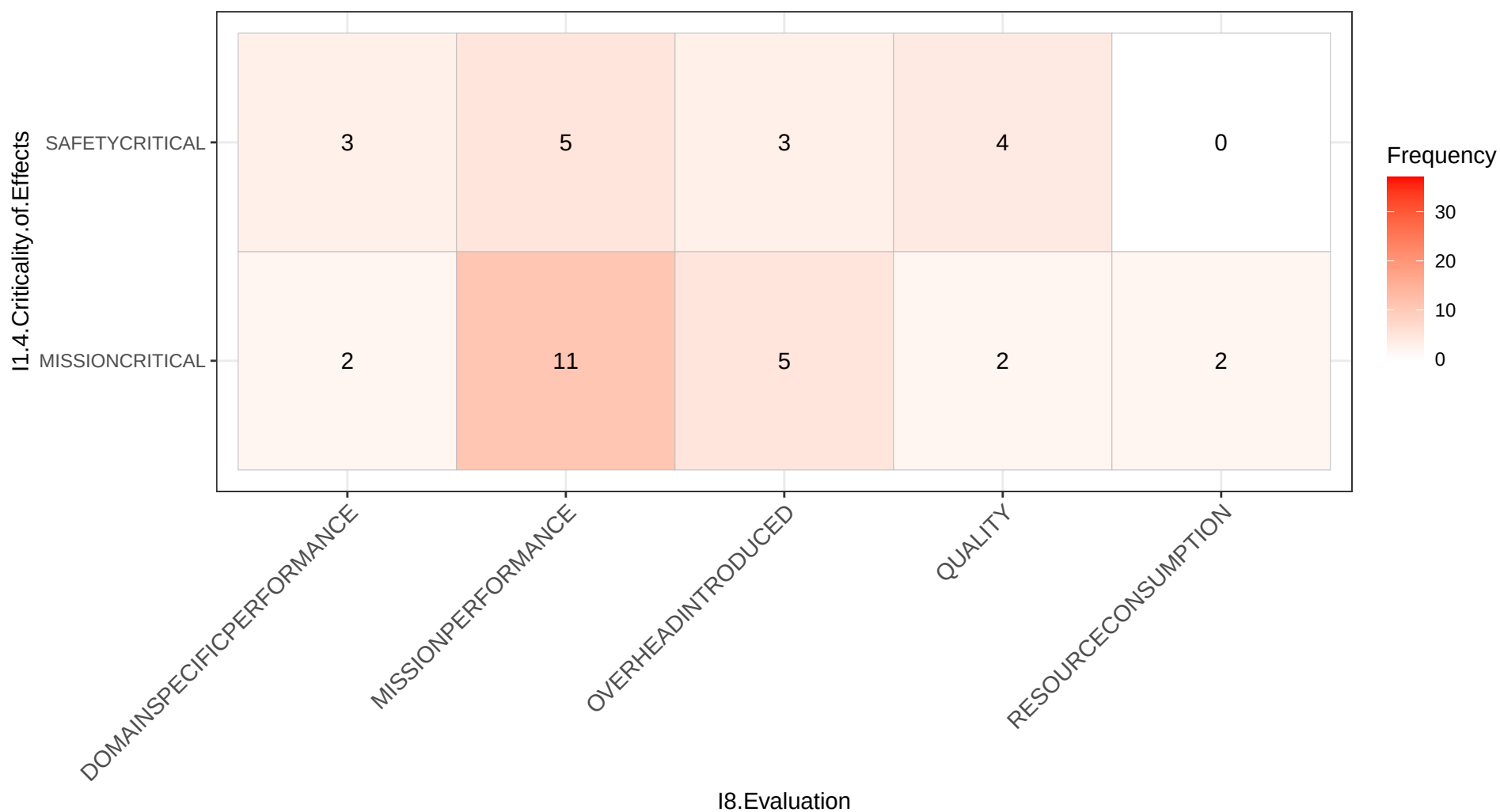
Frequency

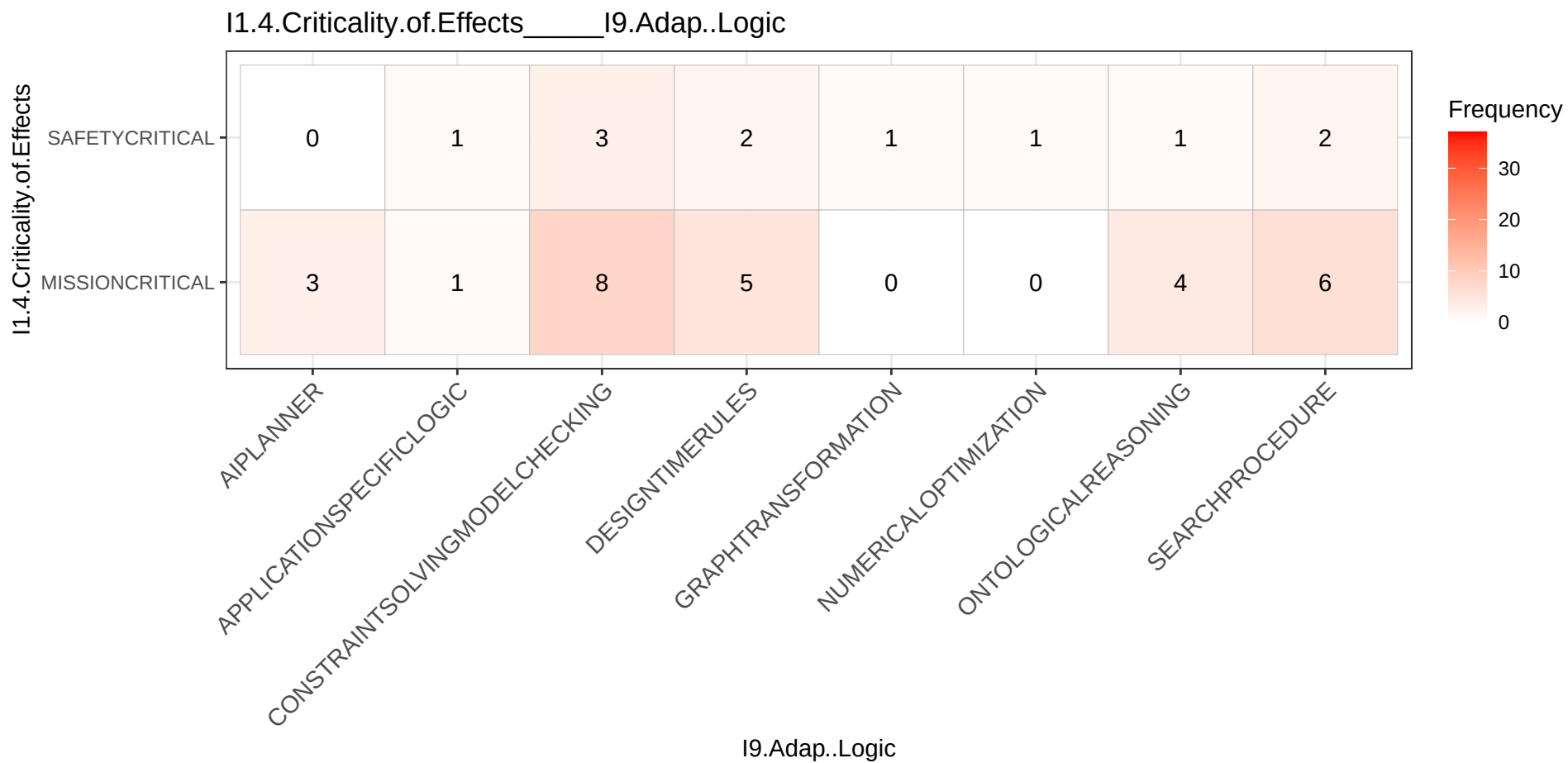


I1.4.Criticality.of.Effects_____Experiment.Method



I1.4.Criticality.of.Effects_____I8.Evaluation



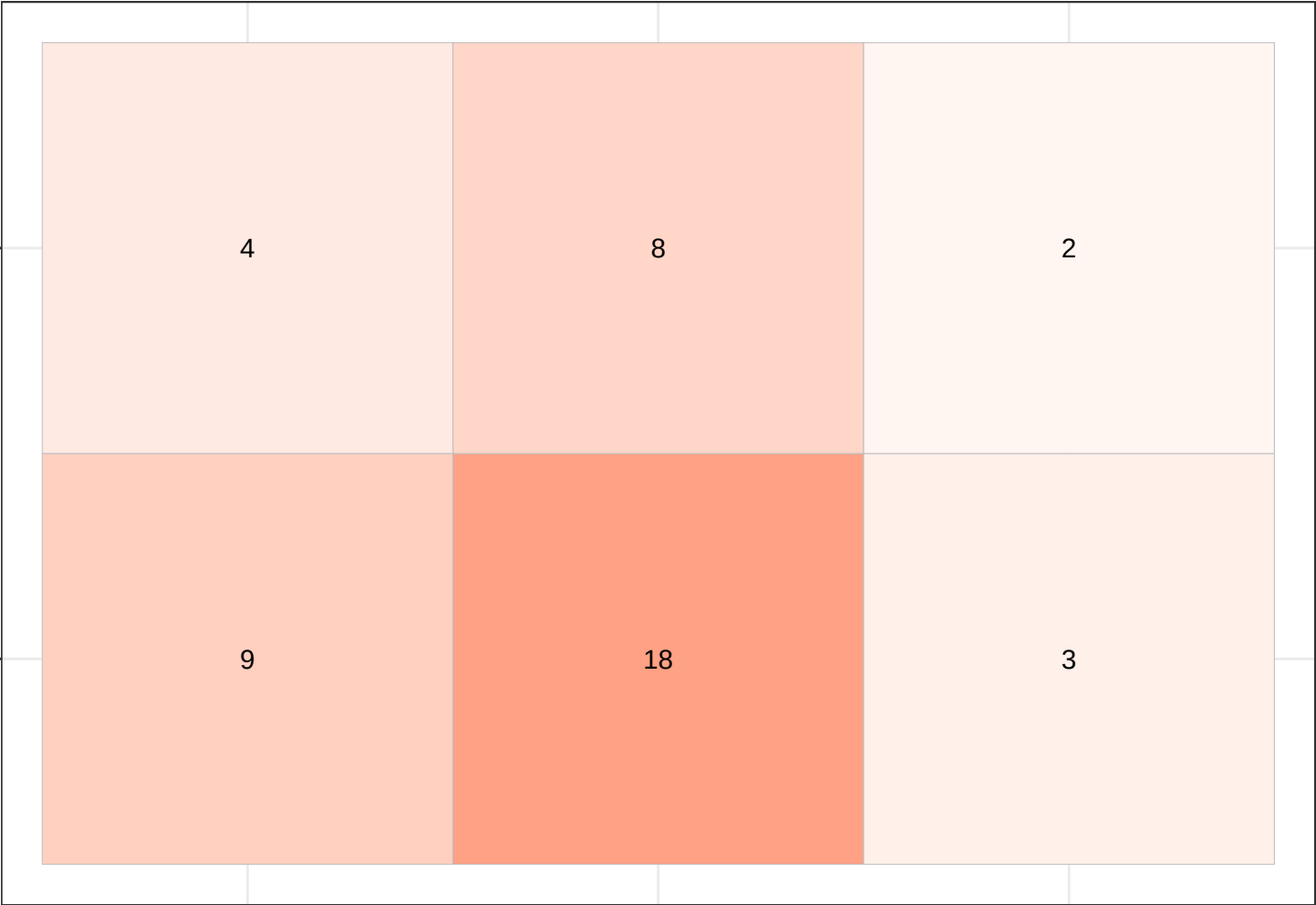


I1.4.Criticality.of.Effects_____I10.Monitor

I1.4.Criticality.of.Effects

SAFETYCRITICAL

MISSIONCRITICAL

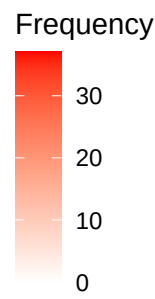


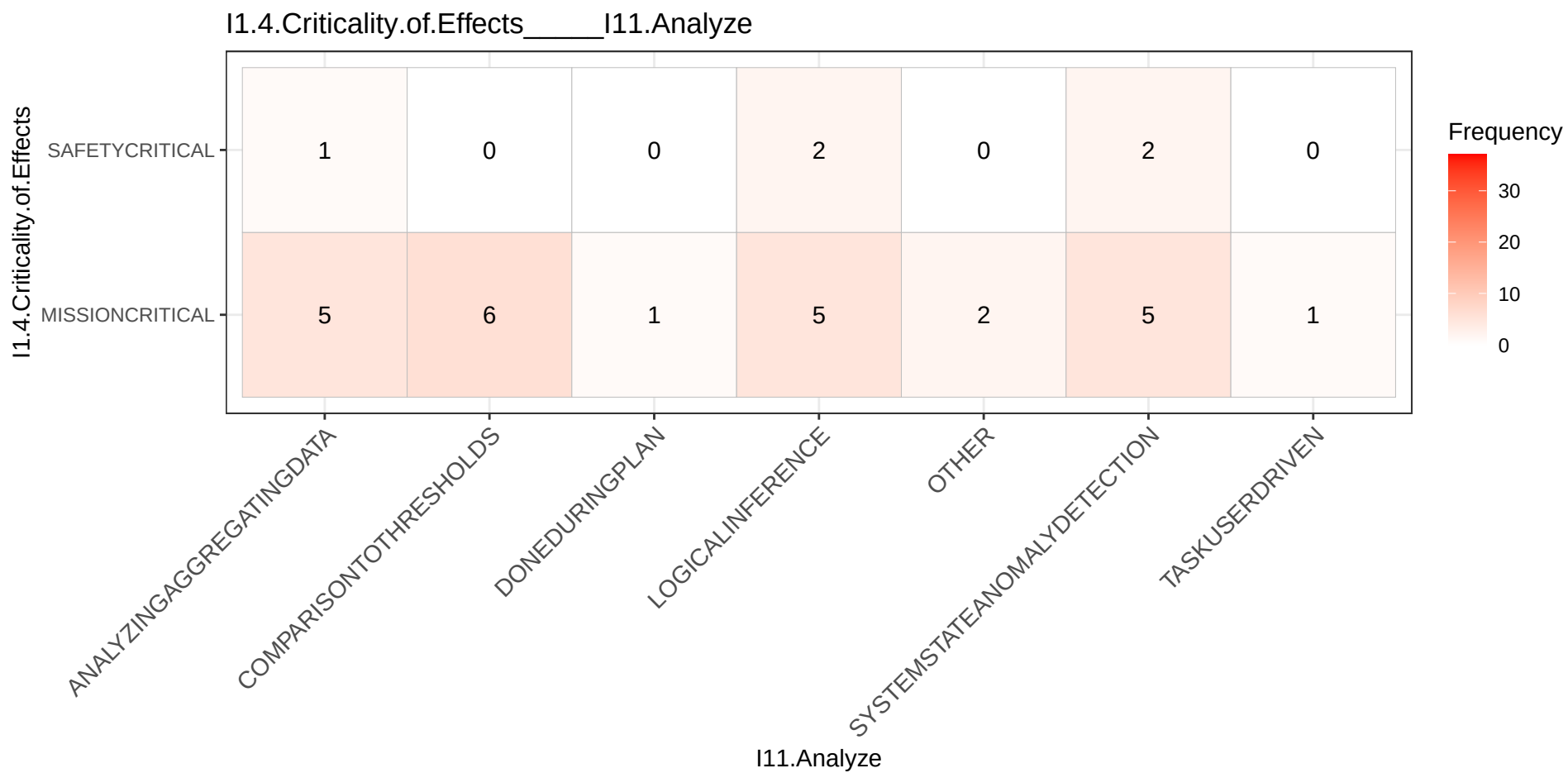
ENVIRONMENT

MANAGEDSYSTEM

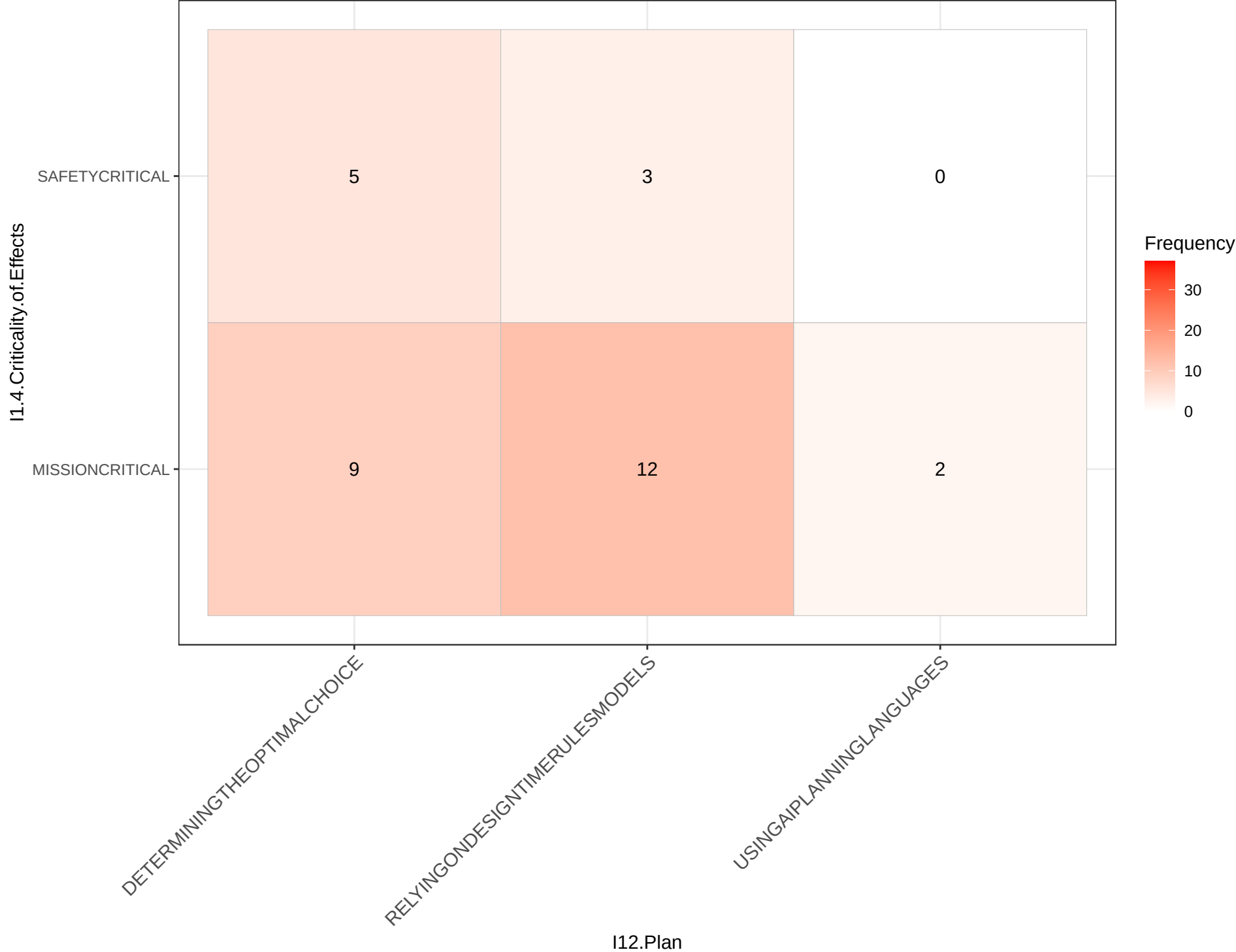
MISSION

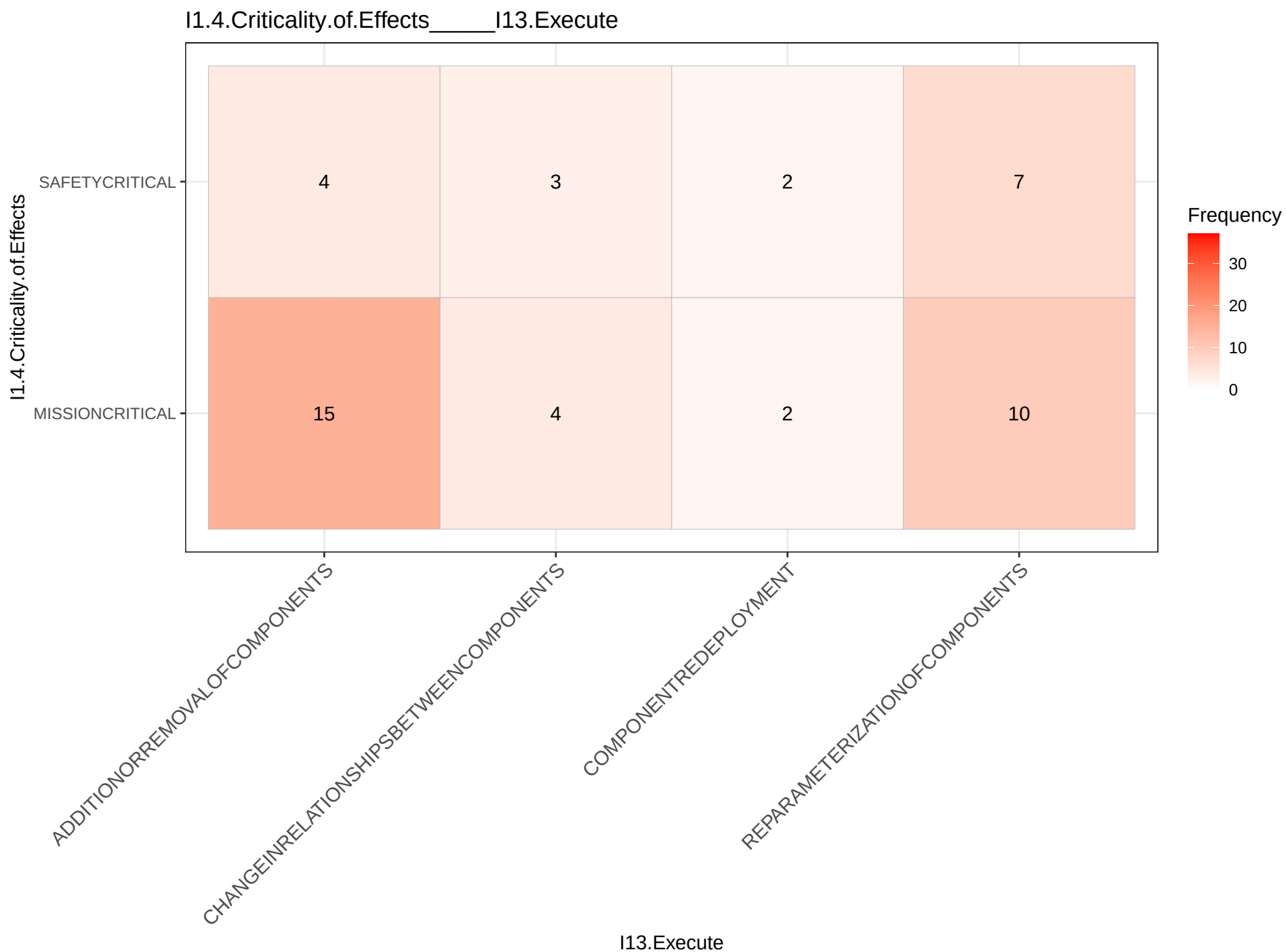
I10.Monitor

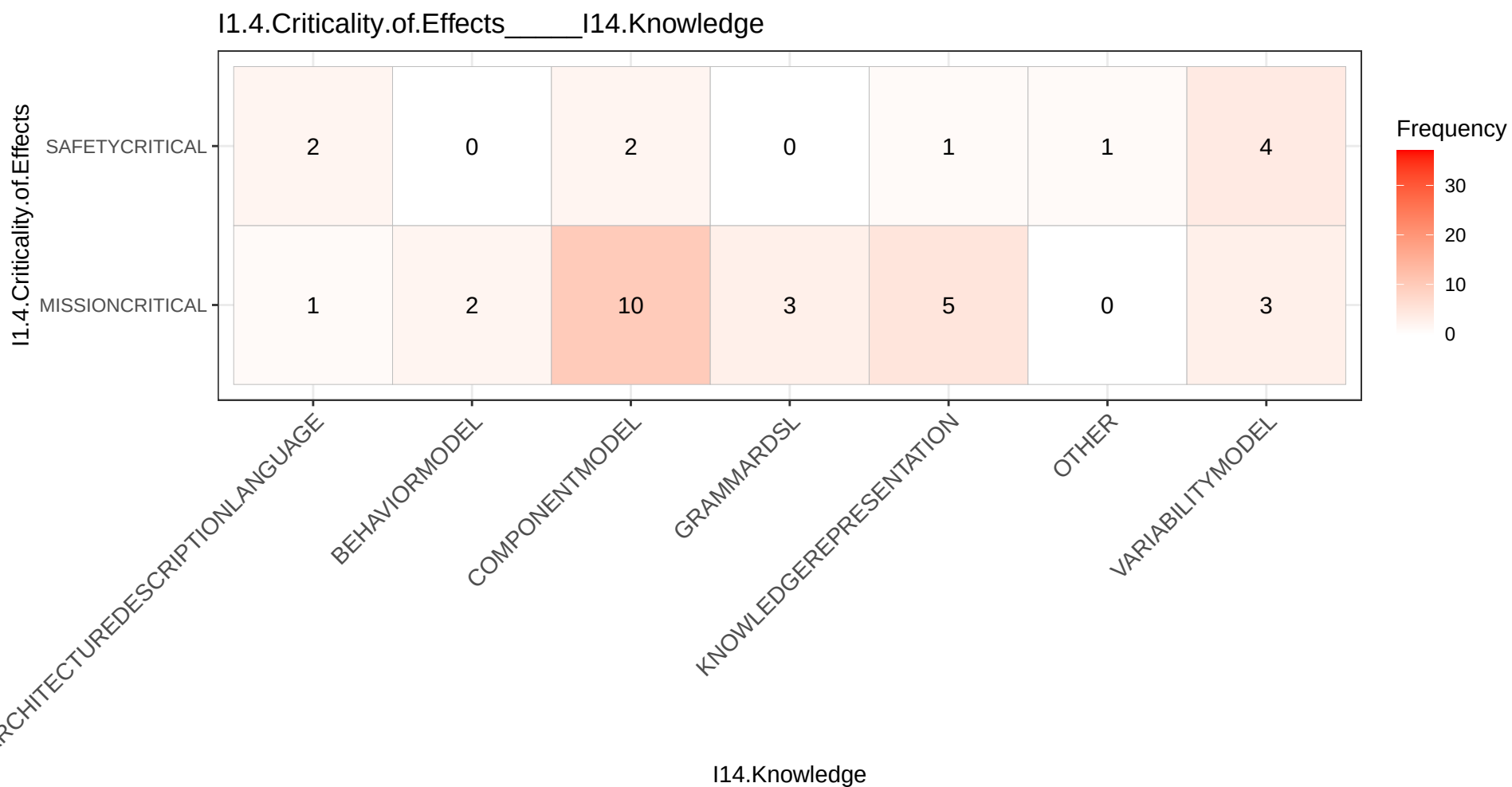




I1.4.Criticality.of.Effects_____I12.Plan







I1.4.Predictability.of.Effects_____I1.4.Overhead.of.Effects

I1.4.Predictability.of.Effects

NONDETERMINISTIC

9

4

6

DODETERMINISTIC

7

0

2

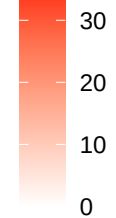
DEPENDENT

DOSIGNIFICANT

INSIGNIFICANT

I1.4.Overhead.of.Effects

Frequency



I1.4.Predictability.of.Effects_____I1.4.Resilience.of.Effects

I1.4.Predictability.of.Effects

NONDETERMINISTIC

9

7

2

DODETERMINISTIC

5

3

0

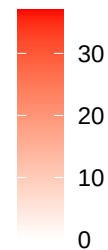
DEPENDENT

DORESILIENT

IRRESILIENT

I1.4.Resilience.of.Effects

Frequency



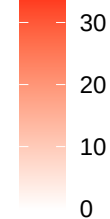
I1.4.Predictability.of.Effects

I1.4.Predictability.of.Effects_____I2.Adap..Purpose

NONDETERMINISTIC

DODETERMINISTIC

Frequency



CHANGEFUNCTIONALBEHAVIOR

DEALWITHENVIRONMENTALCHANGES

KEEPMEETINGQUALITYREQUIREMENTSATRUNTIME

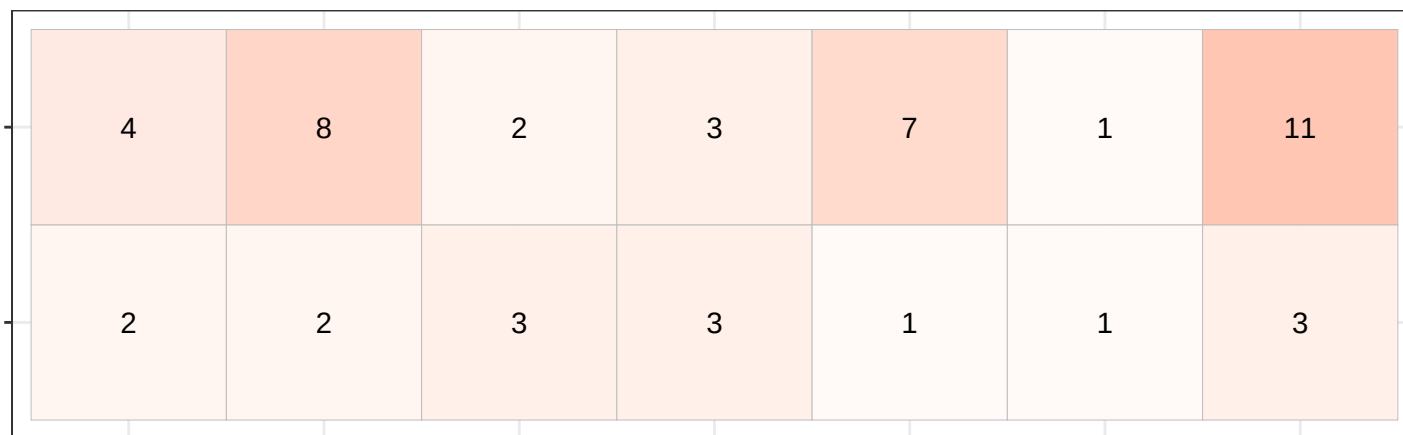
OPTIMIZERESOURCEUSAGE

OPTIMIZESEYSTEMPERFORMANCE

RECOVERFROMATTACKS

RECOVERFROMERRORSFAULTS

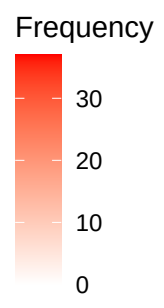
I2.Adap..Purpose



I1.4.Predictability.of.Effects

I1.4.Predictability.of.EffectsI3.Robot.Type

NONDETERMINISTIC	0	1	1	0	1	0	1	0	2	0	1	1	1	2	1	0	0	1	0	0	1	1	1	1	1	0	4	1	1	1	1	1	0
DODETERMINISTIC	1	0	0	0	0	0	1	1	0	1	1	0	0	0	0	1	2	0	0	0	0	0	0	0	0	1	2	0	0	0	0	0	0



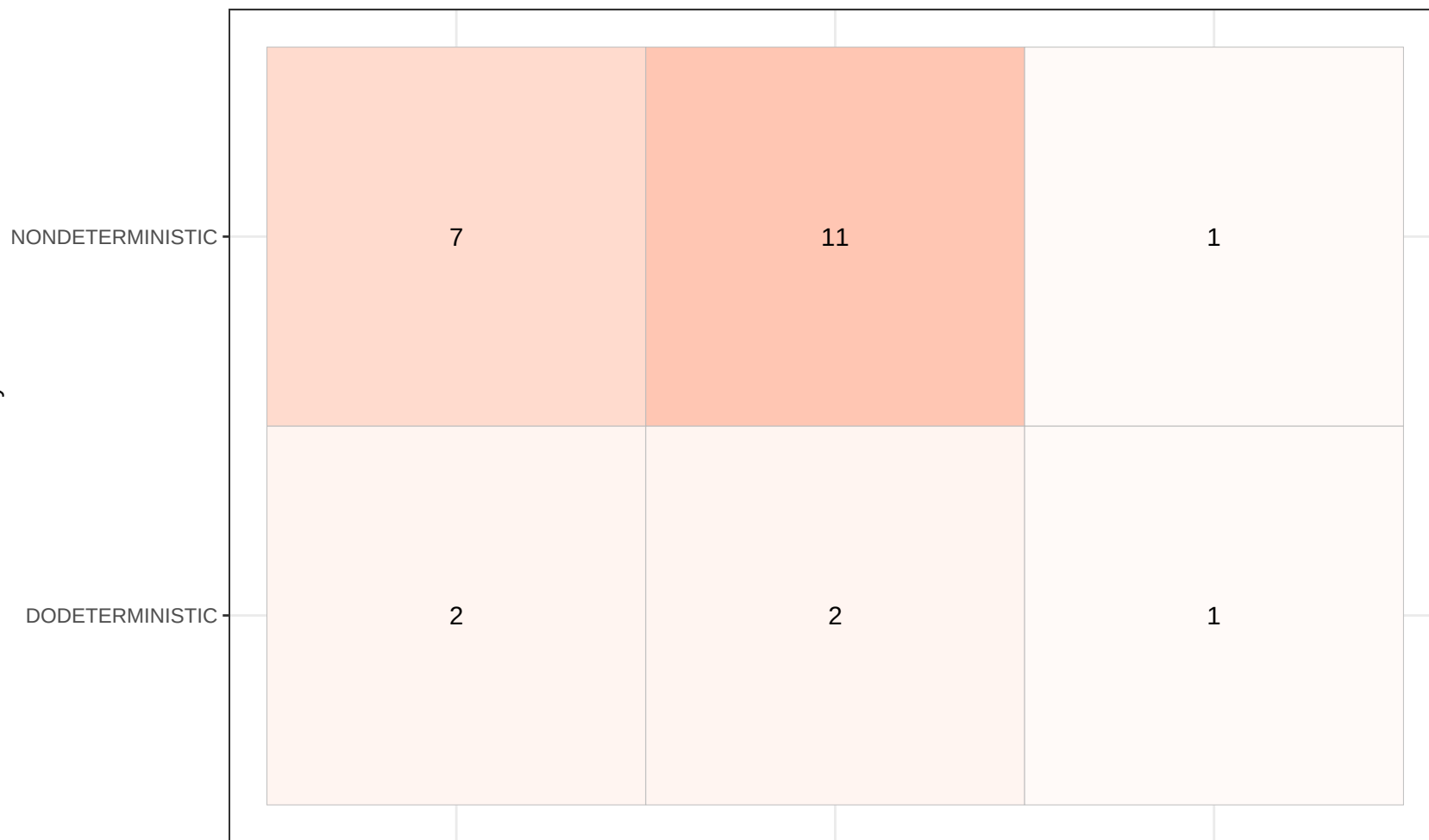
BEBOTSPADERBORN
BLUEROV2
BOXERCLEARPATH
CRAWLERTERMINATORBOT
FIELDMOBILEROBOT
FRACTIONATEDSPACECRAFT
HETEROGENOUSROBOTS
HEXMANIPULATOR
HEXAII
IROBOTSERVICEROBOT
IROBOTCREATE2
MOBILEROBOT
MOBILEROBOT
MOBILEROBOTS
MOBILEROBOTTIAGO
MOBILESERVICEROBOT
MOBILETERRESTRIAL
MULTIPLEHEXROTOR
NAOROBOT
PIONEER3DX
QUADROCOPTER
SINGLESERVINGROTATIONROBOT
TEDUSARTERRESTRIAL
TRIGLIDEINDUSTRIAL
TURTLEBOT
TURTLEBOT2KOBUKI
TWOINDUSTRIALROBOTS
WAREHOUSEDELIVERYROBOT
WHICHISANINDUSTRIALAGV
YUBOT

TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLE

I3.Robot.Type

I1.4.Predictability.of.Effects_____I4.Robo.SW

I1.4.Predictability.of.Effects



DODETERMINISTIC

OTHER

ROS1

ROS2

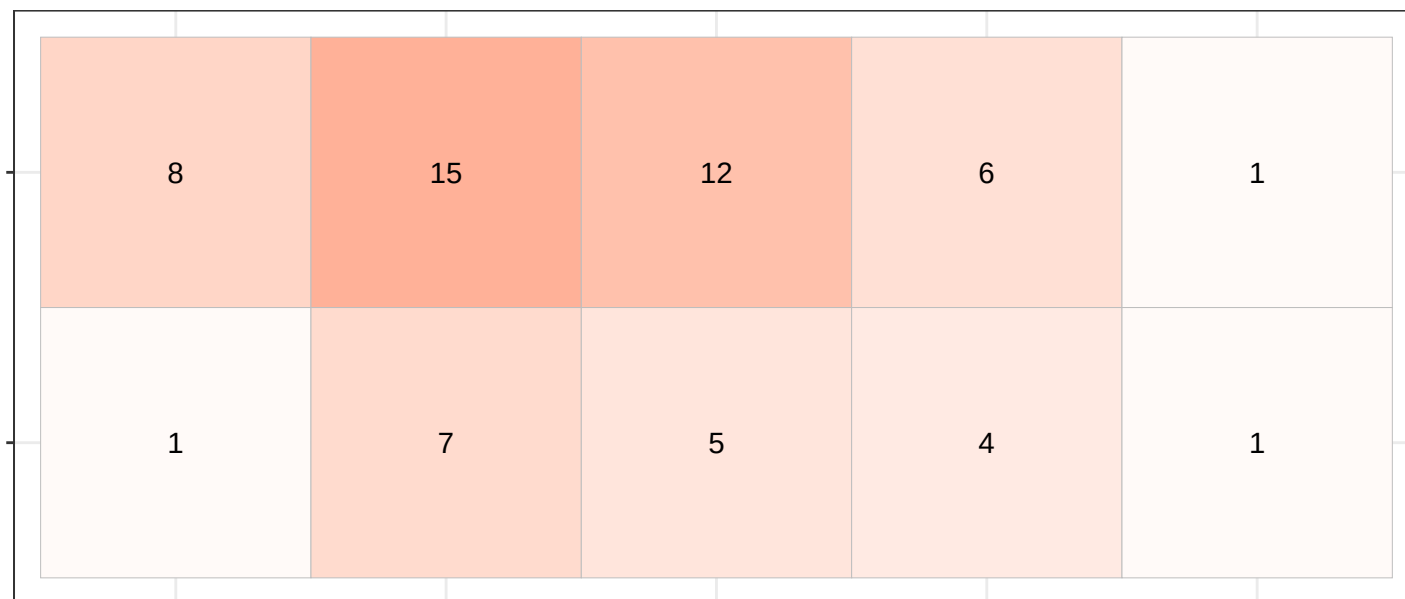
I4.Robo.SW

Frequency



I1.4. Predictability of Effects

I1.4. Predictability of Effects _____ I5. QA



Frequency



FUNCTIONALSUITABILITY

PERFORMANCEEFFICIENCY

RELIABILITY

SAFETY

SECURITY

I5. QA

I1.4.Predictability.of.Effects_____I6.Independence

I1.4.Predictability.of.Effects

NONDETERMINISTIC

DODETERMINISTIC

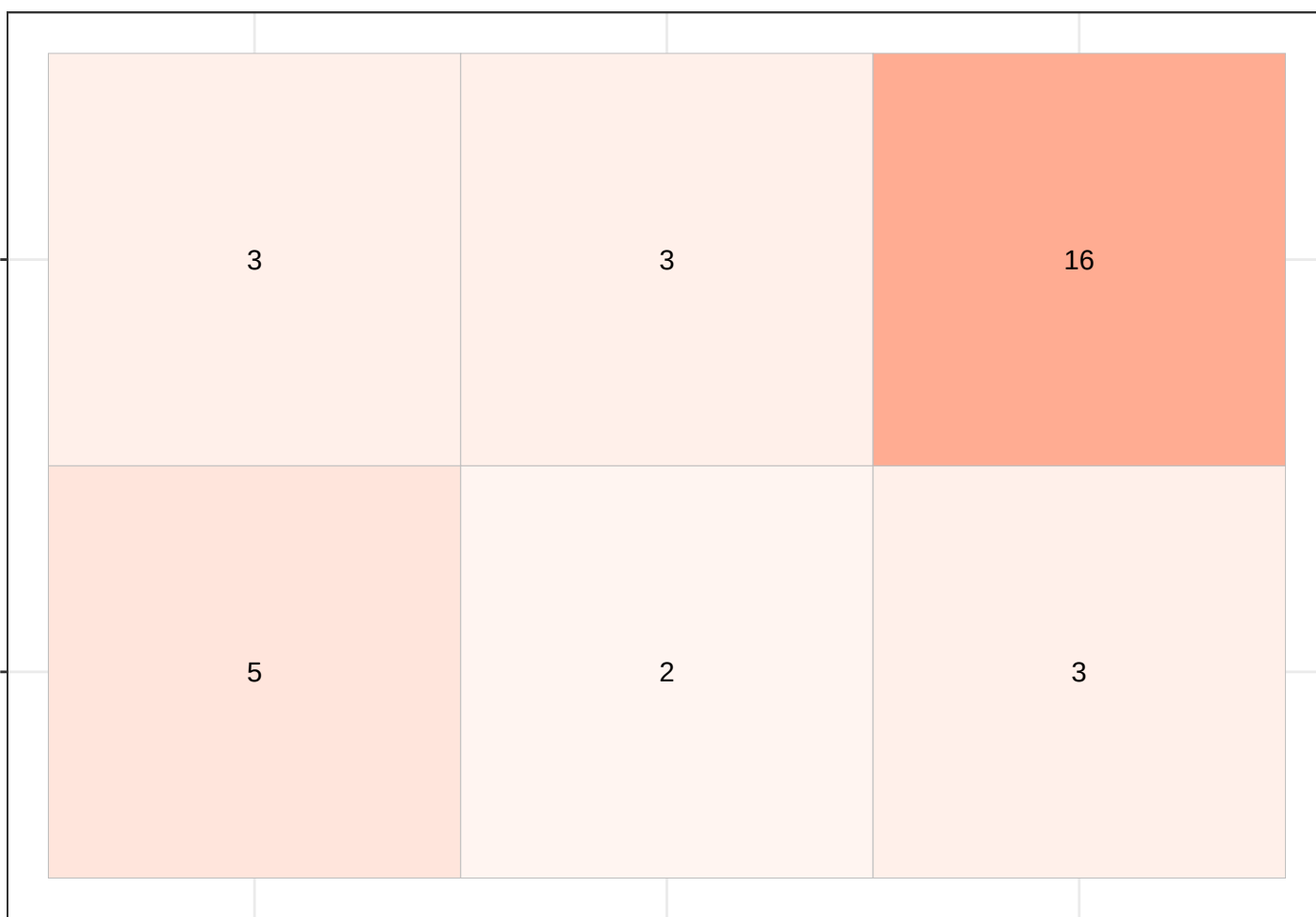
DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

Frequency



I1.4.Predictability.of.Effects_____I7.Deployment.Realness

I1.4.Predictability.of.Effects

NONDETERMINISTIC

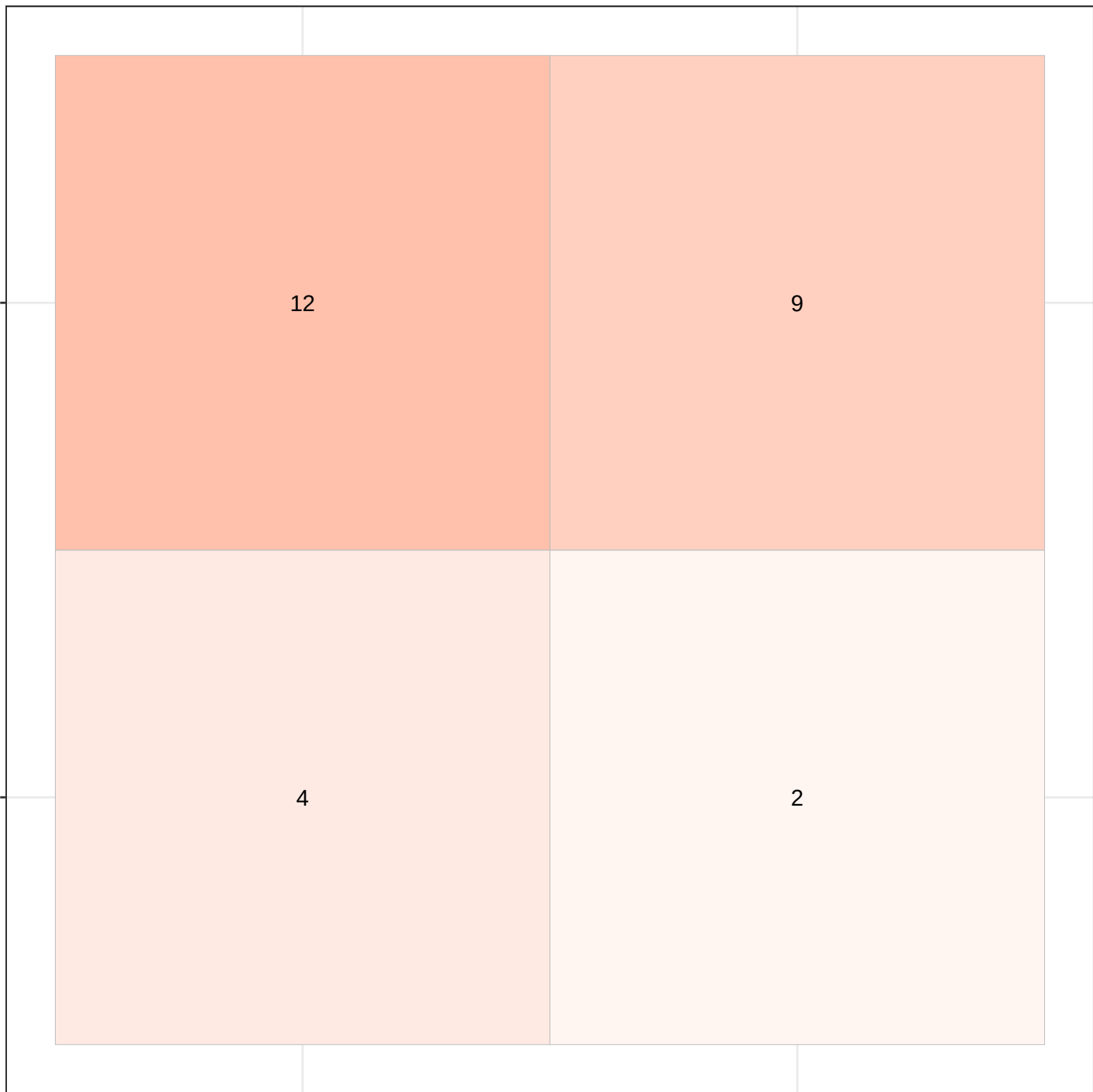
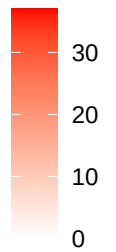
DODETERMINISTIC

REAL

SIMULATED

I7.Deployment.Realness

Frequency



I1.4.Predictability.of.Effects_____I7.Mission.Realness

I1.4.Predictability.of.Effects

NONDETERMINISTIC

11

11

DODETERMINISTIC

6

4

REAL

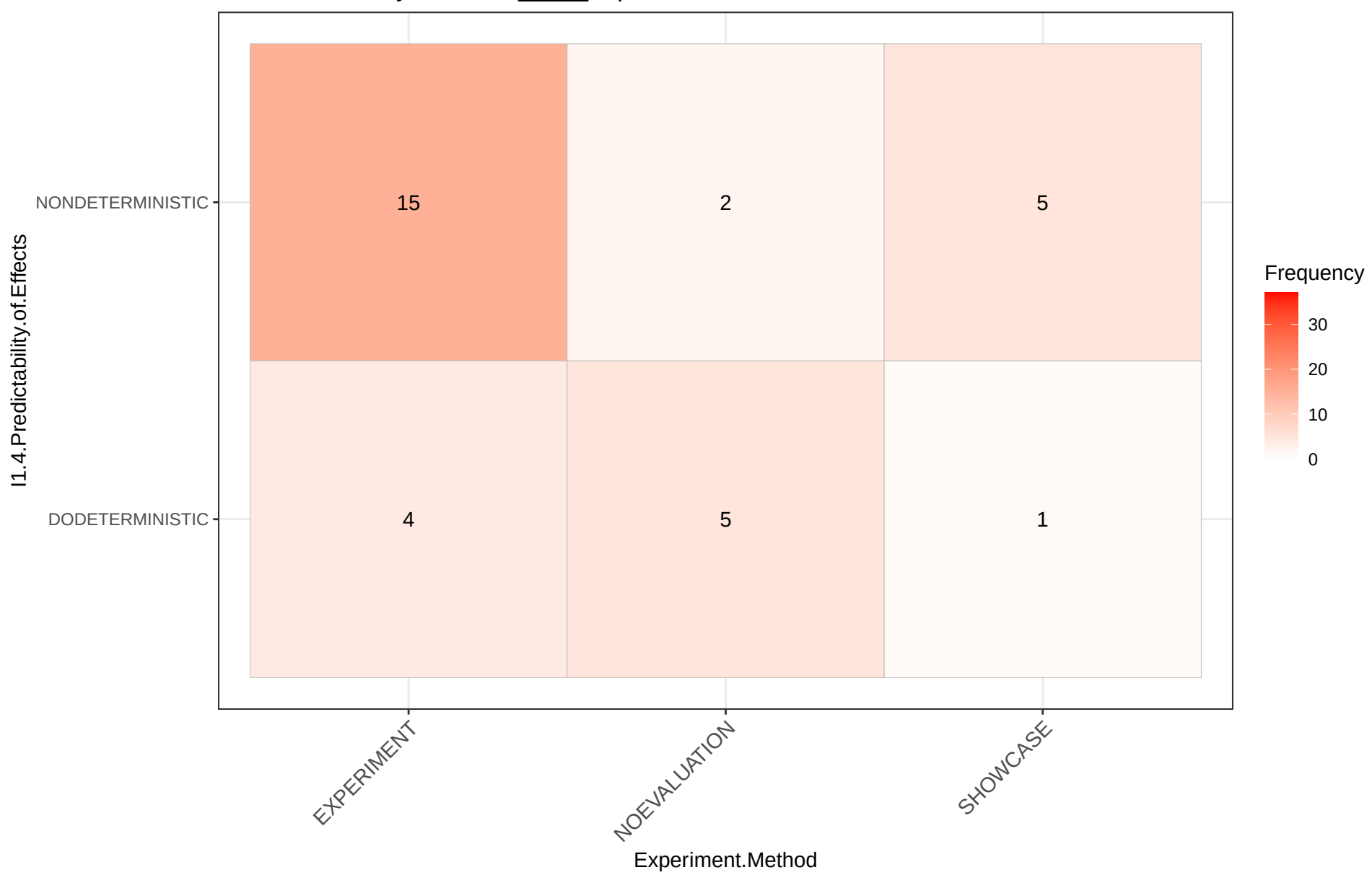
SYNTHETIC

I7.Mission.Realness

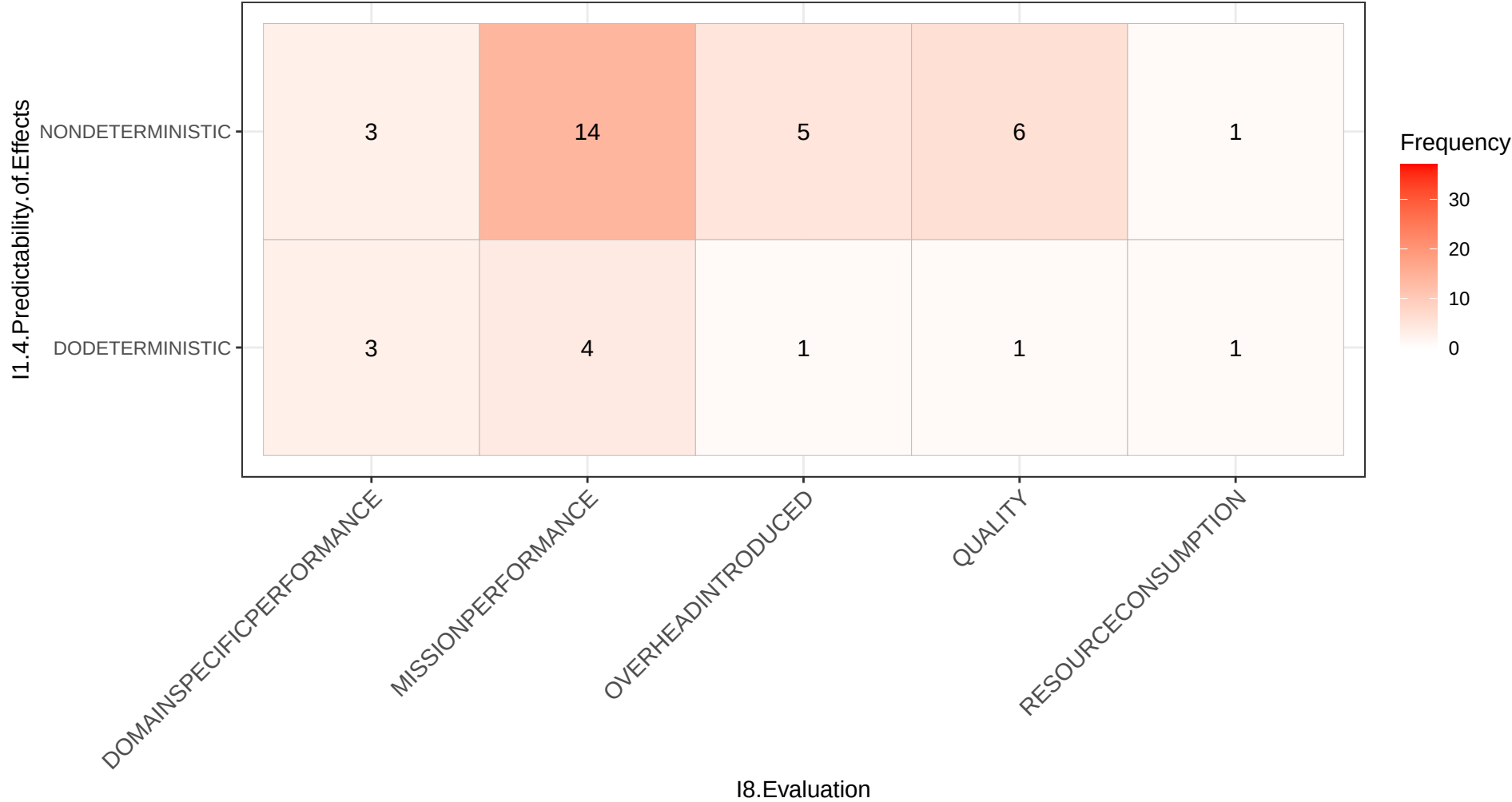
Frequency



I1.4.Predictability.of.Effects_____Experiment.Method



I1.4.Predictability.of.Effects_____I8.Evaluation

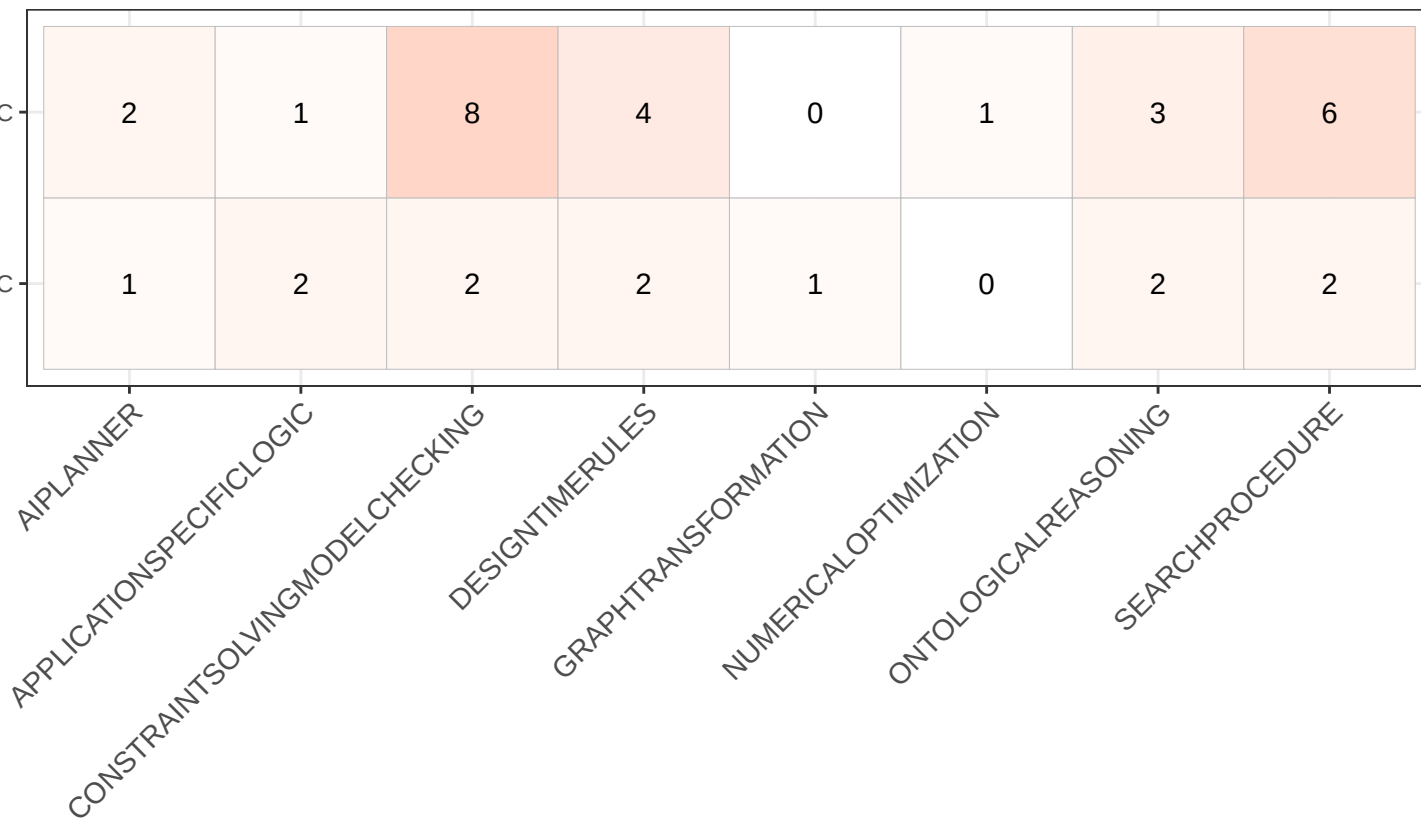


I1.4.Predictability.of.Effects

I1.4.Predictability.of.Effects_____I9.Adap..Logic

NONDETERMINISTIC

DODETERMINISTIC



Frequency



I9.Adap..Logic

I1.4.Predictability.of.Effects_____I10.Monitor

I1.4.Predictability.of.Effects

NONDETERMINISTIC

DODETERMINISTIC

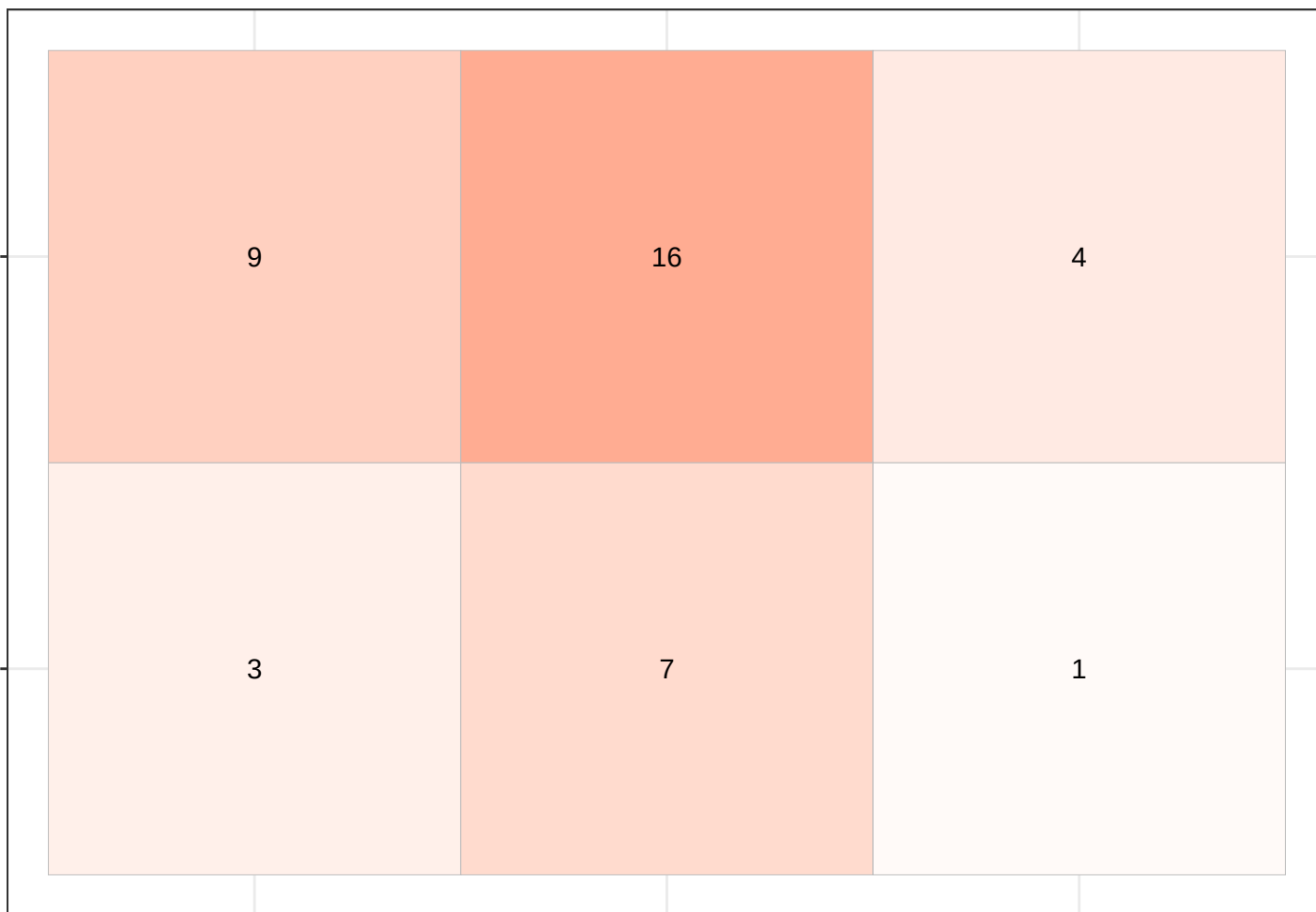
ENVIRONMENT

MANAGEDSYSTEM

MISSION

I10.Monitor

Frequency



I1.4.Predictability.of.Effects

I1.4.Predictability.of.Effects_____I11.Analyze

NONDETERMINISTIC

DODETERMINISTIC

Frequency



ANALYZINGAGGREGATINGDATA

COMPARISONTOHRESHOLDS

DONEDURINGPLAN

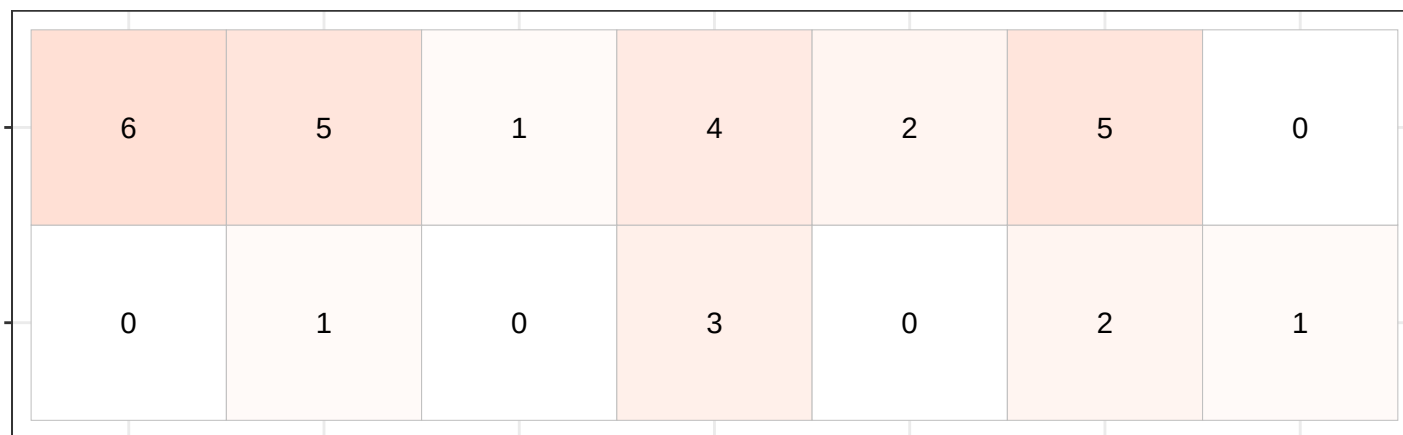
LOGICALINFERENCE

OTHER

SYSTEMSTATEANOMALYDETECTION

TASKUSERDRIVEN

I11.Analyze



I1.4.Predictability.of.Effects_____I12.Plan

I1.4.Predictability.of.Effects

NONDETERMINISTIC

DODETERMINISTIC

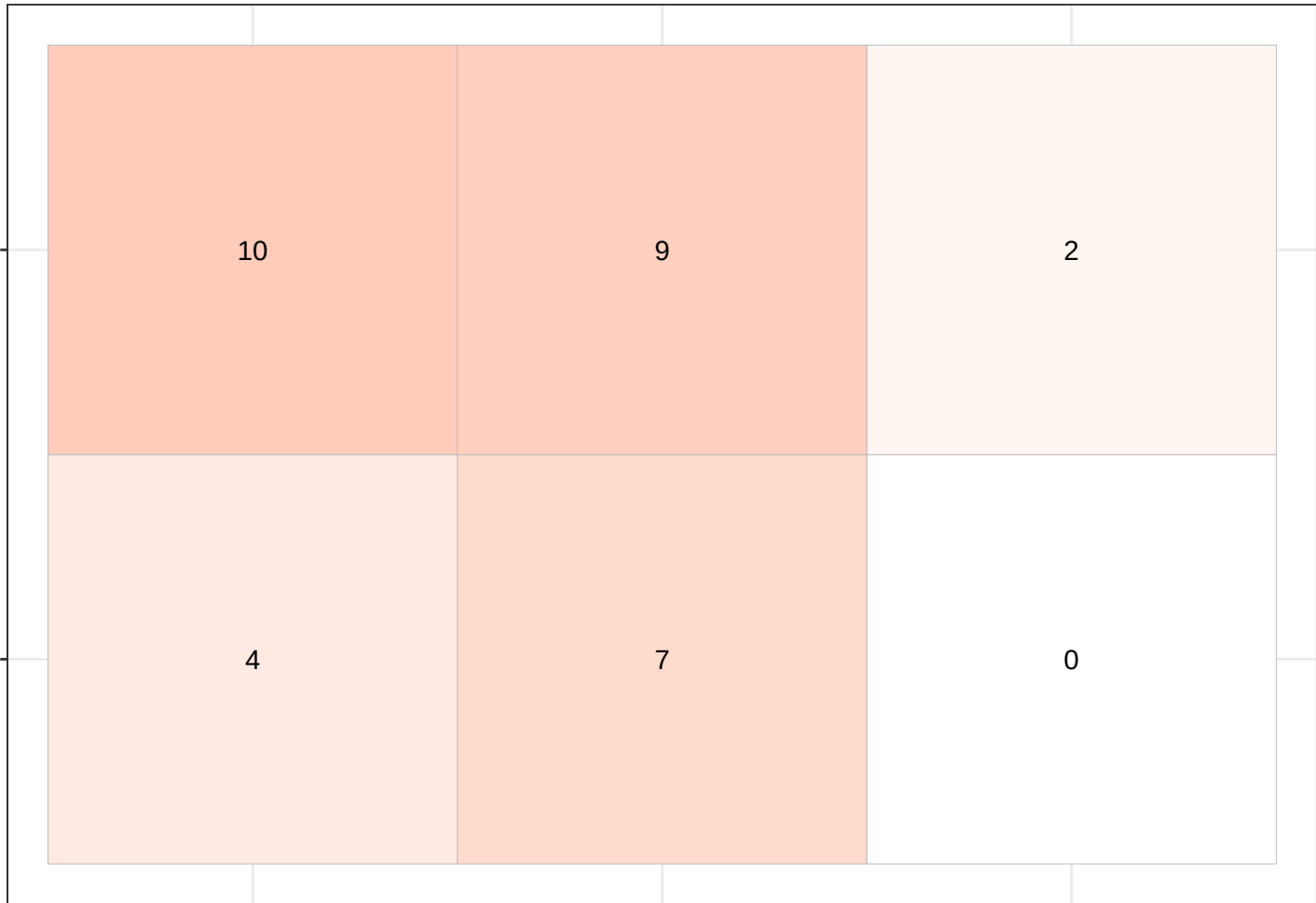
DETERMININGTHEOPTIMALCHOICE

RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

I12.Plan

Frequency



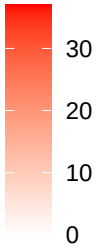
I1.4.Predictability.of.Effects_____I13.Execute

I1.4.Predictability.of.Effects

NONDETERMINISTIC

DODETERMINISTIC

Frequency



12

2

2

12

4

4

1

4

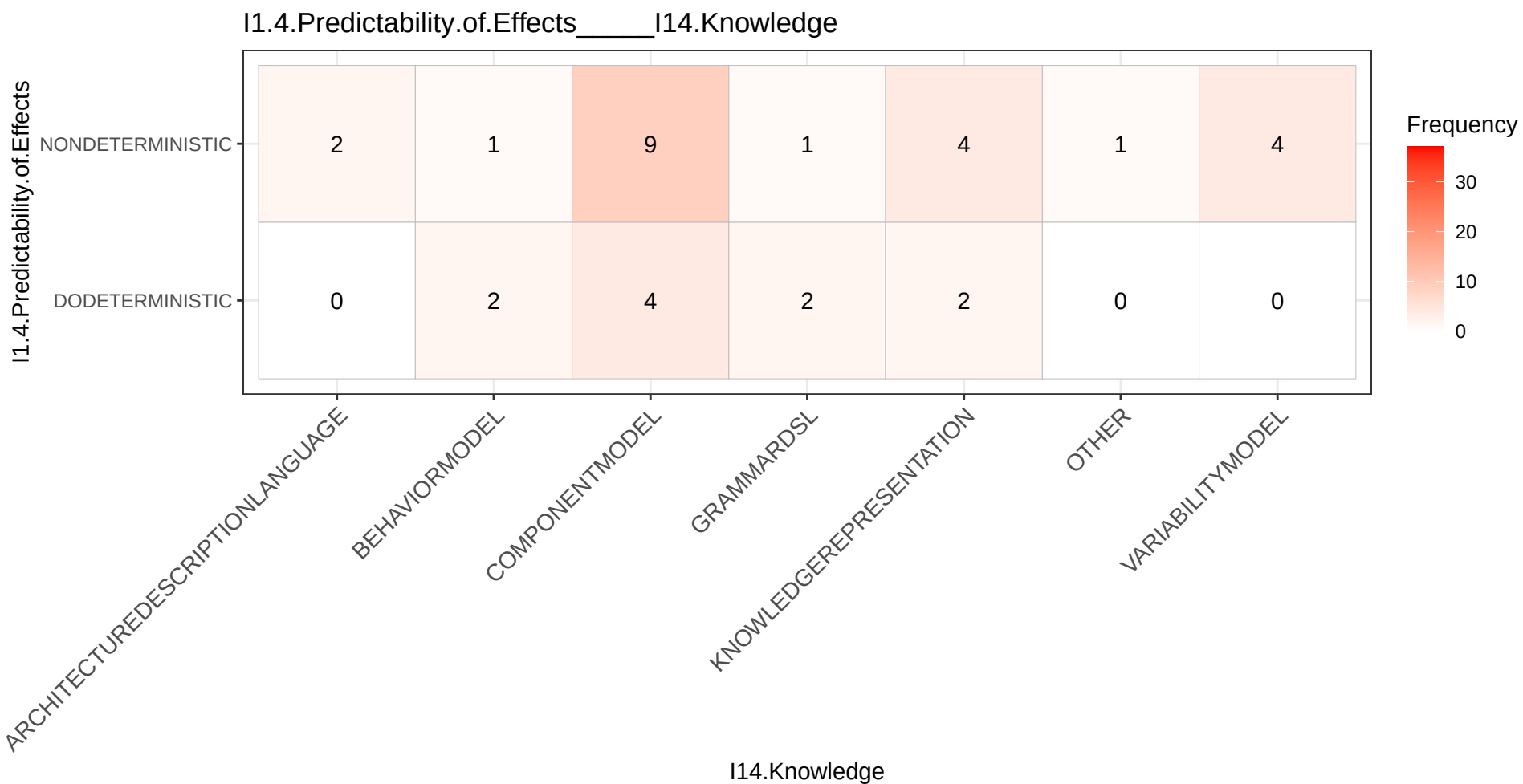
ADDITIONORREMOVALOFCOMPONENTS

CHANGEINRELATIONSHIPSBETWEENCOMPONENTS

COMPONENTREDEPLOYMENT

REPARAMETERIZATIONOFCOMPONENTS

I13.Execute



I1.4.Overhead.of.Effects_____I1.4.Resilience.of.Effects

I1.4.Overhead.of.Effects

INSIGNIFICANT

2

4

0

DOSIGNIFICANT

2

1

1

DEPENDENT

8

5

1

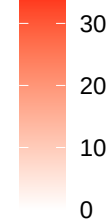
DEPENDENT

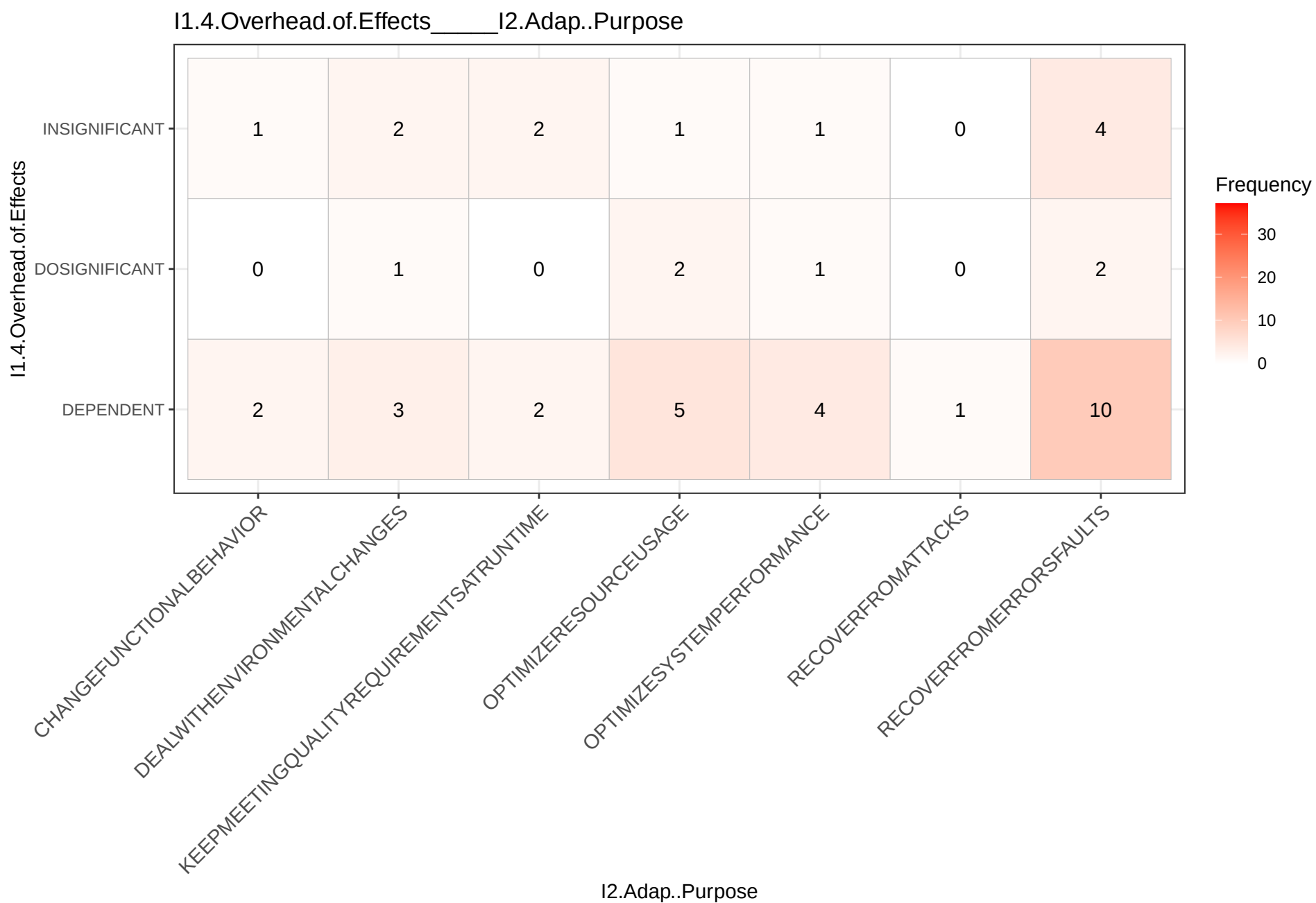
DORESILIENT

IRRESILIENT

I1.4.Resilience.of.Effects

Frequency

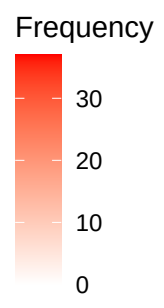




I1.4.Overhead.of.Effects

I1.4.Overhead.of.Effects_____I3.Robot.Type

INSIGNIFICANT	0	1	1	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	2	1	0	1	1	1	0
DOSIGNIFICANT	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0
DEPENDENT	1	0	0	0	1	1	2	0	2	1	1	0	0	1	1	1	1	0	0	2	0	0	0	1	0	0	3	0	0	0	0

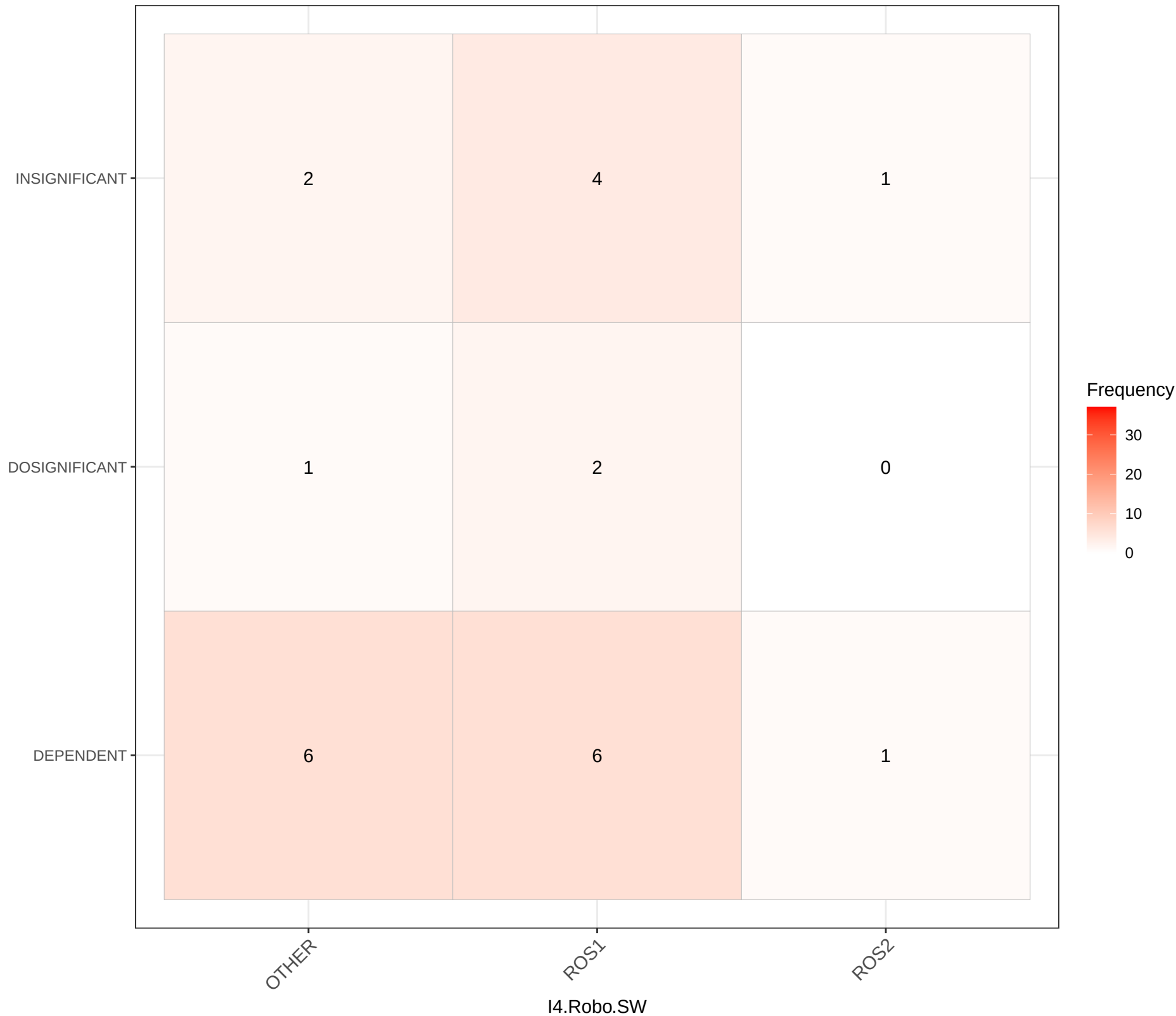


BEBOTSPADERBORN
BLUEROV2
BOXERCLEARPATH
CRAWLERTERMINATORBOT
FIELDMOBILEROBOTS
FRACTIONATEDSPACECRAFT
HETEROGENOUSROBOTS
HEXMANIPULATOR
HEXSERVICEROBOT
IROBOTCREATE2
MOBILEROBOT
MOBILEROBOT
MOBILEROBOTTERRESTRIAL
MOBILEROBOTTERRESTRIAL
MOBILEROBOTTERRESTRIAL
MULTIPLEHEXROBOT
NAOROBOT
PIONEER3DX
QUADROCOPTER
SINGLESERVOINGROTATIONROBOT
TEDUSARTERRESTRIAL
TRIGLIDEINDUSTRIALASSEMBLY
TURTLEBOT2
TURTLEBOT2
TWOINDUSTRIALROBOTS
WAREHOUSEDELIVERYROBOT
WHICHISANINDUSTRIALAGV
YOUBOT

I3.Robot.Type

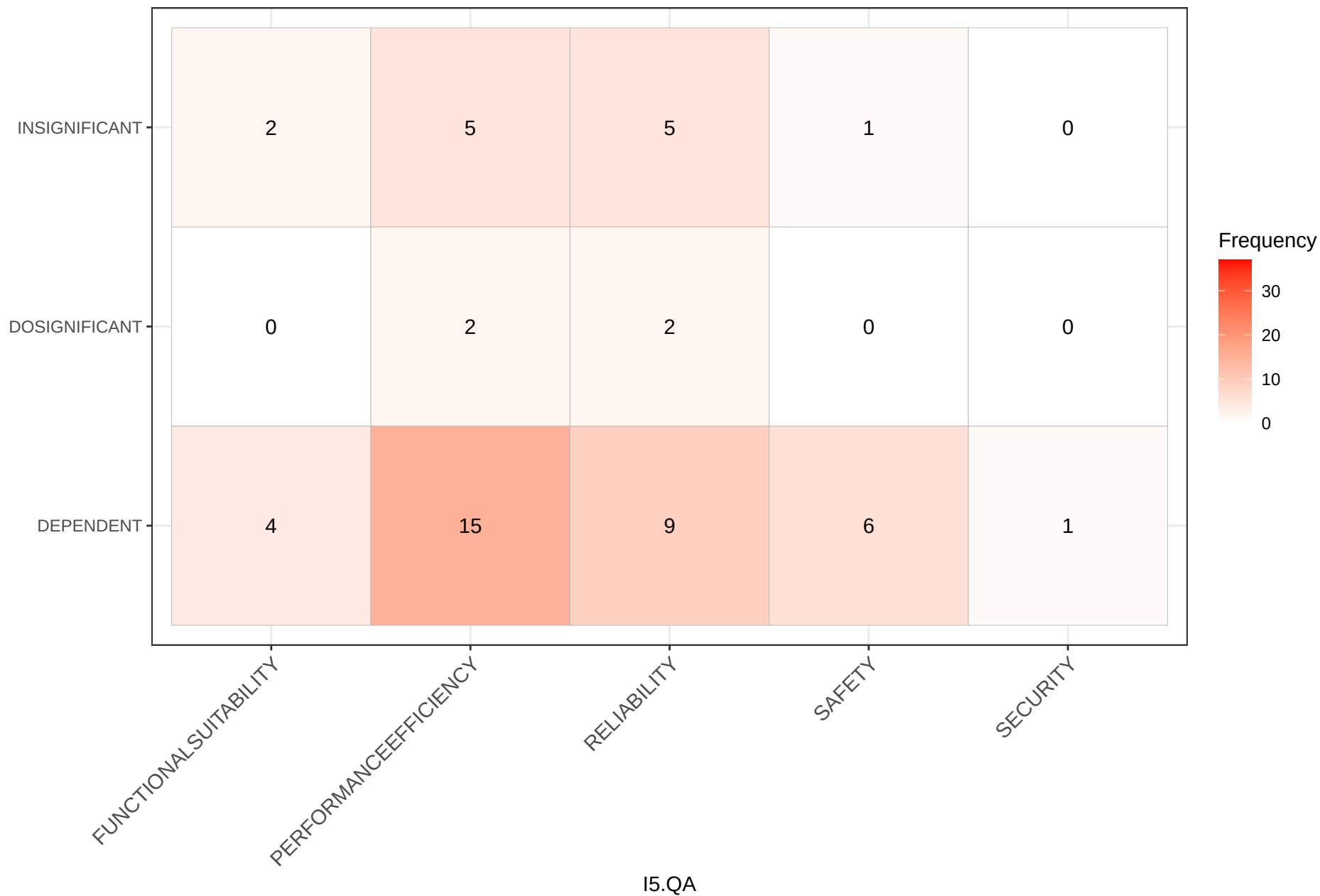
I1.4.Overhead.of.Effects_____I4.Robo.SW

I1.4.Overhead.of.Effects

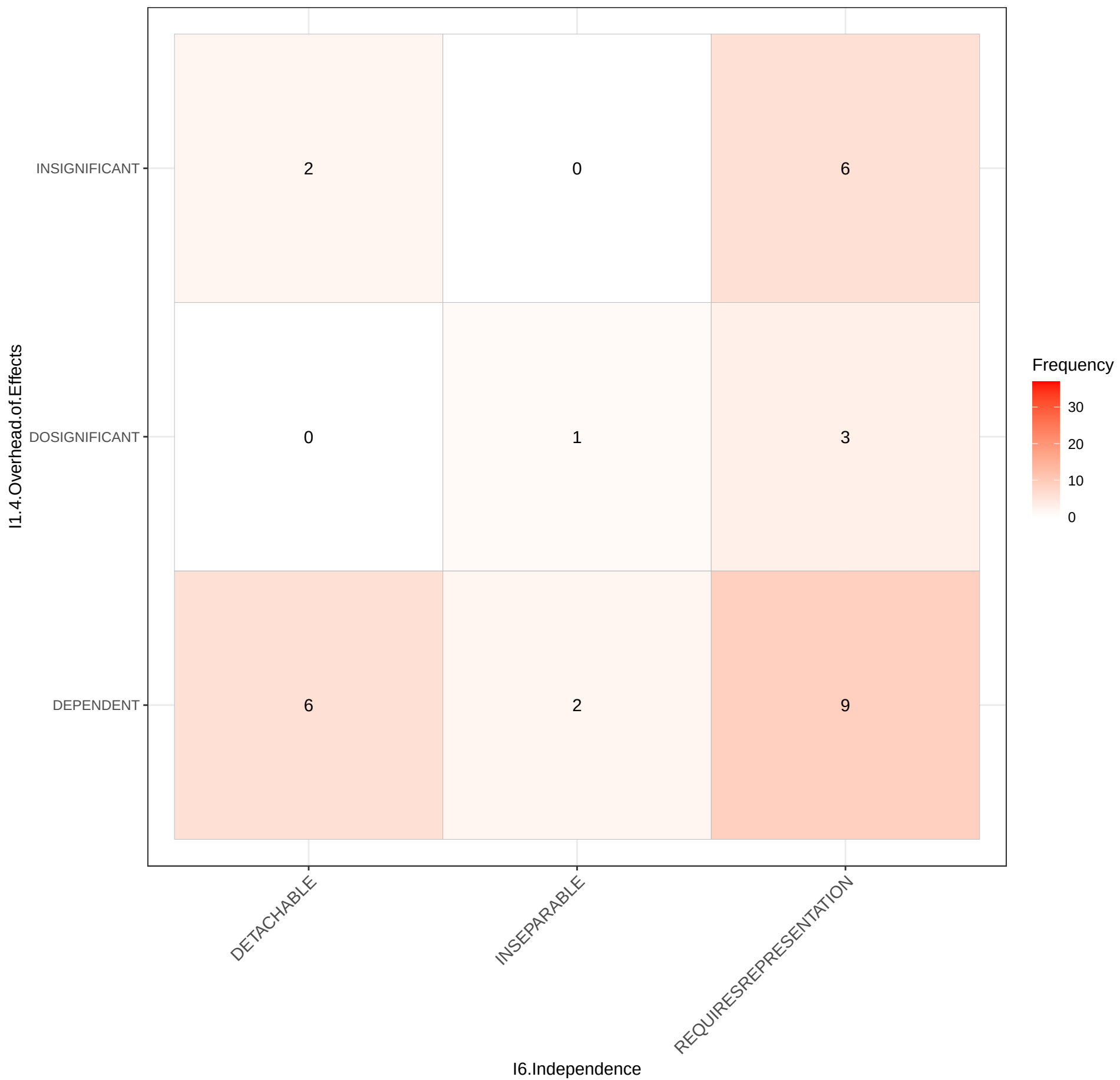


I1.4.Overhead.of.Effects_____I5.QA

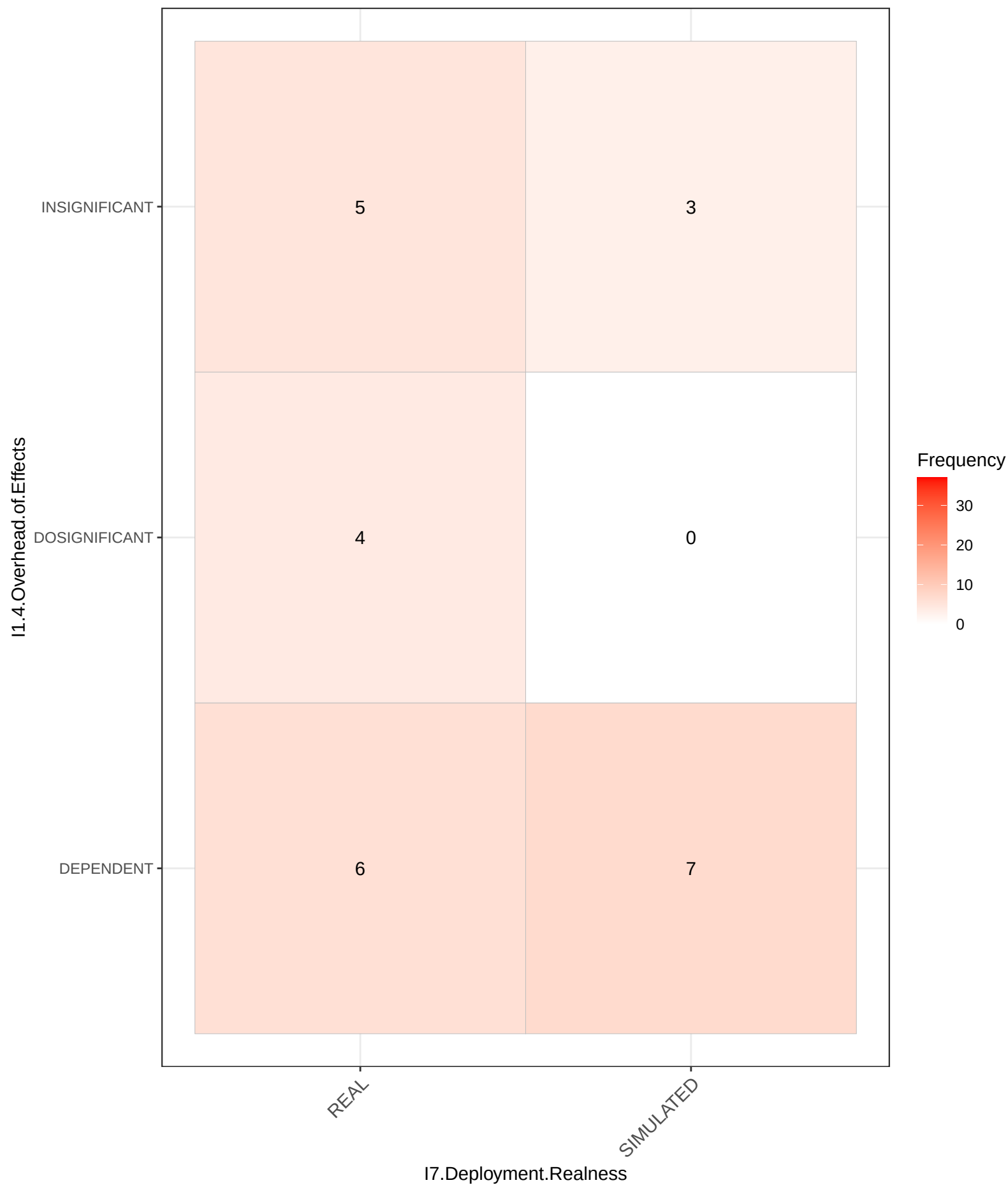
I1.4.Overhead.of.Effects



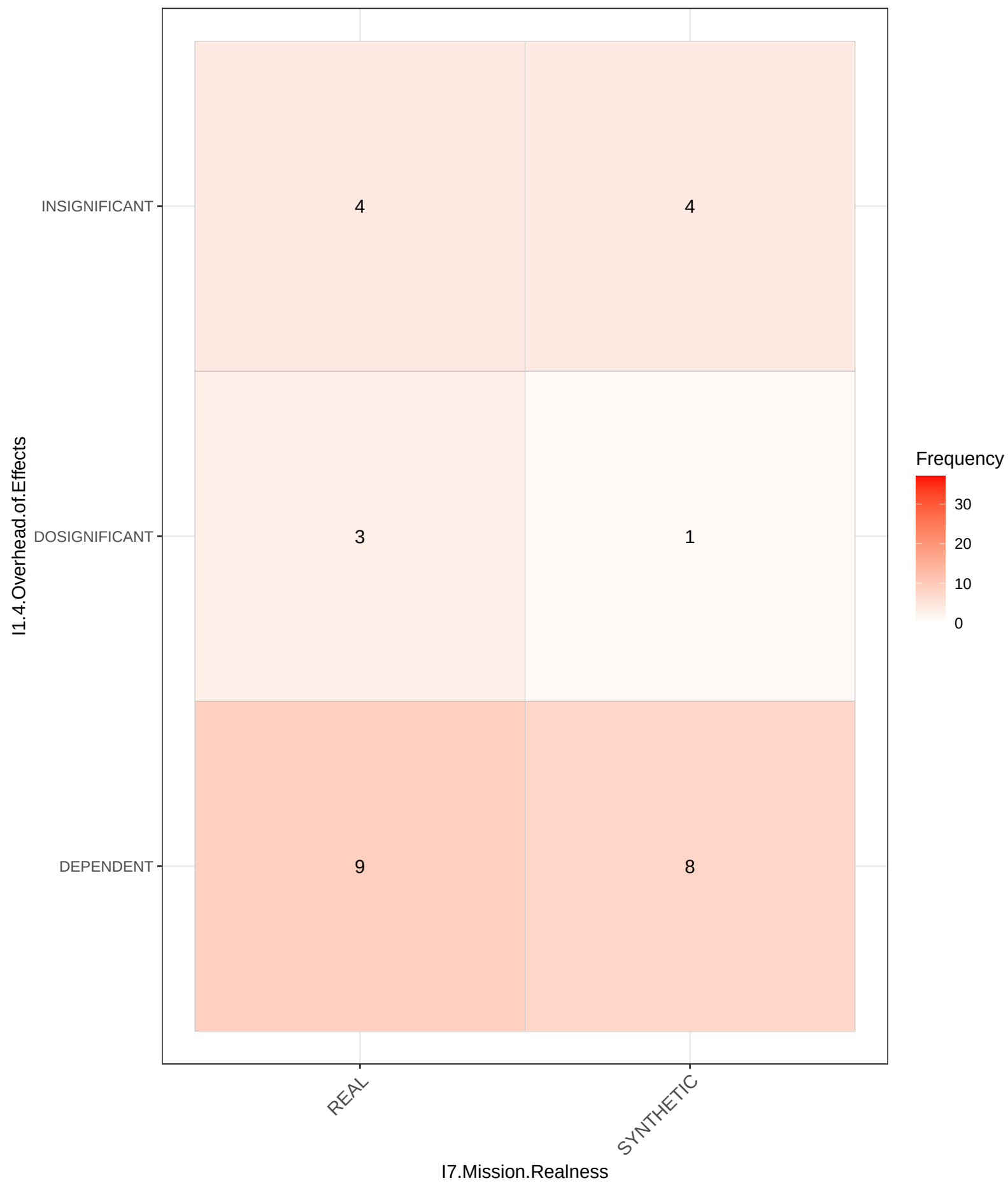
I1.4.Overhead.of.Effects_____I6.Independence



I1.4.Overhead.of.Effects_____I7.Deployment.Realness

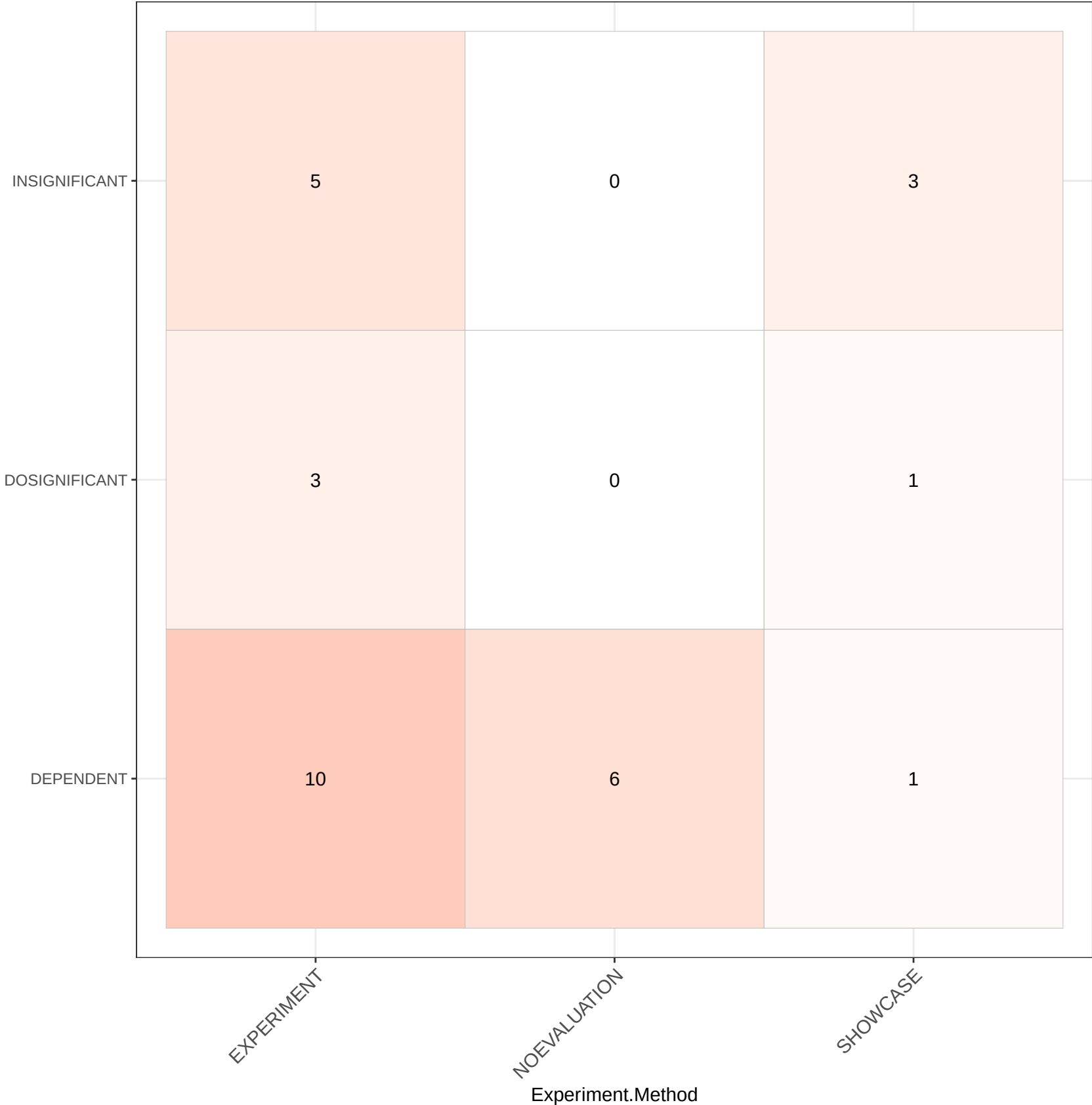


I1.4.Overhead.of.Effects_____I7.Mission.Realness

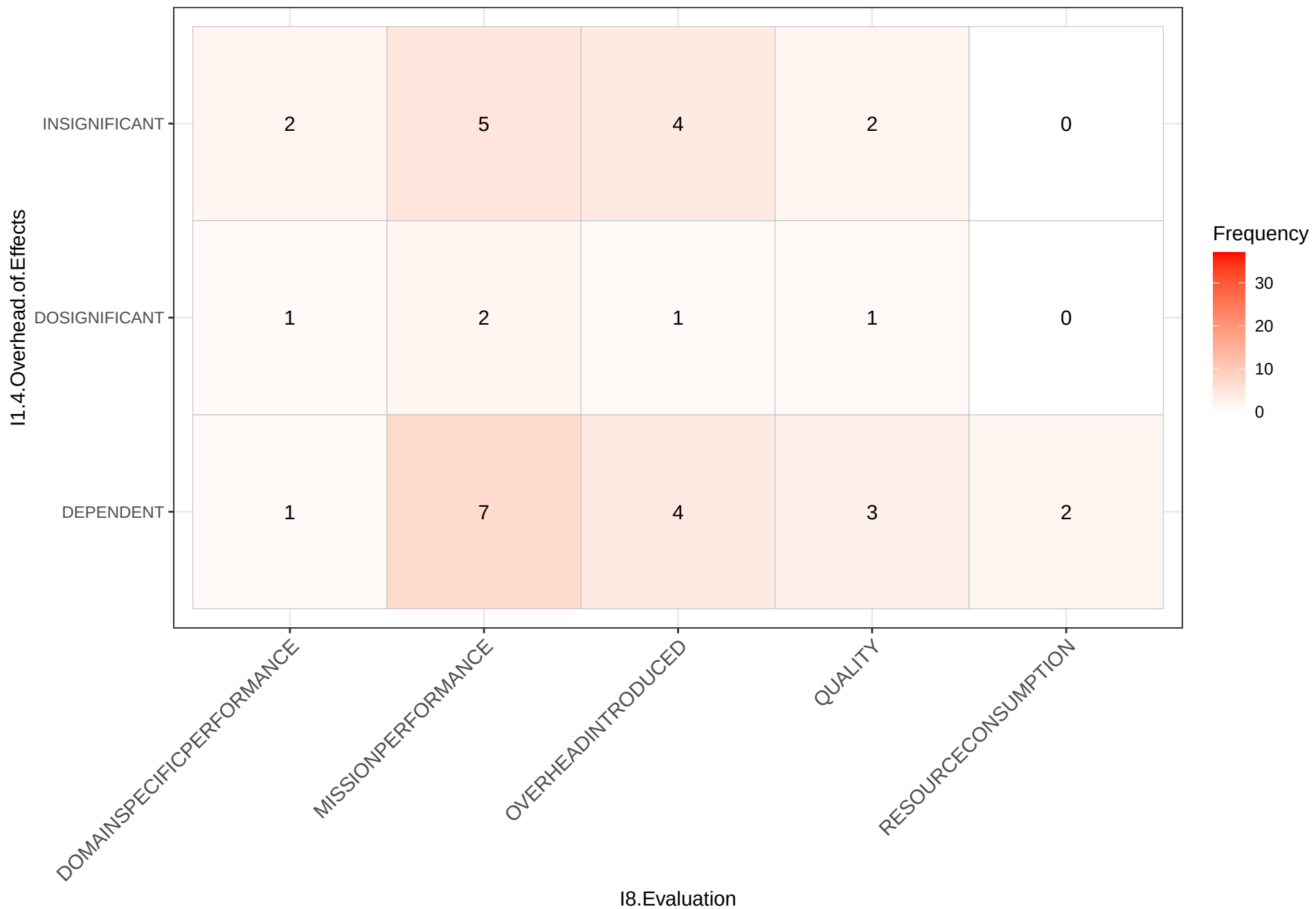


I1.4.Overhead.of.Effects_____Experiment.Method

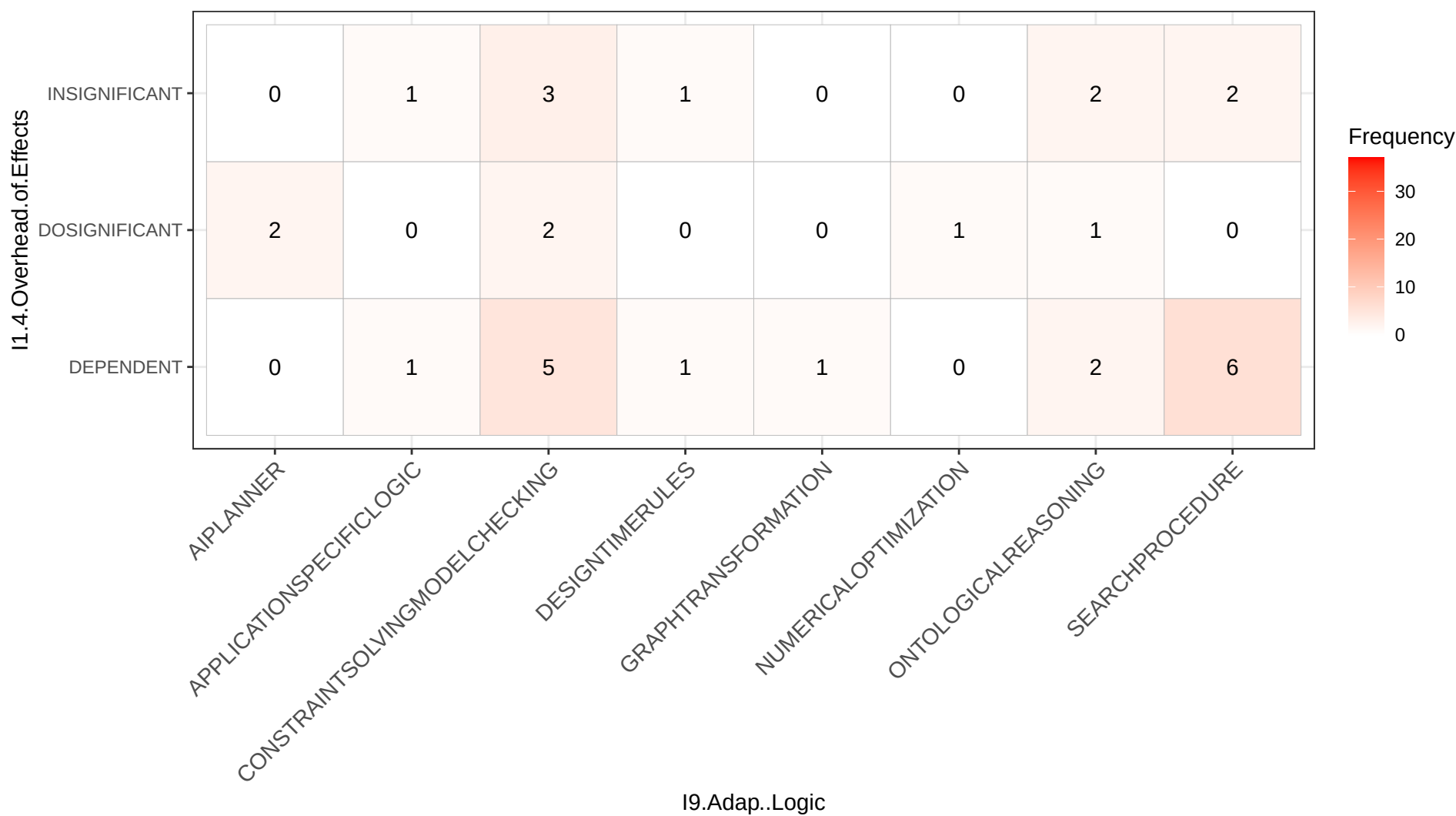
I1.4.Overhead.of.Effects



I1.4.Overhead.of.Effects_____I8.Evaluation

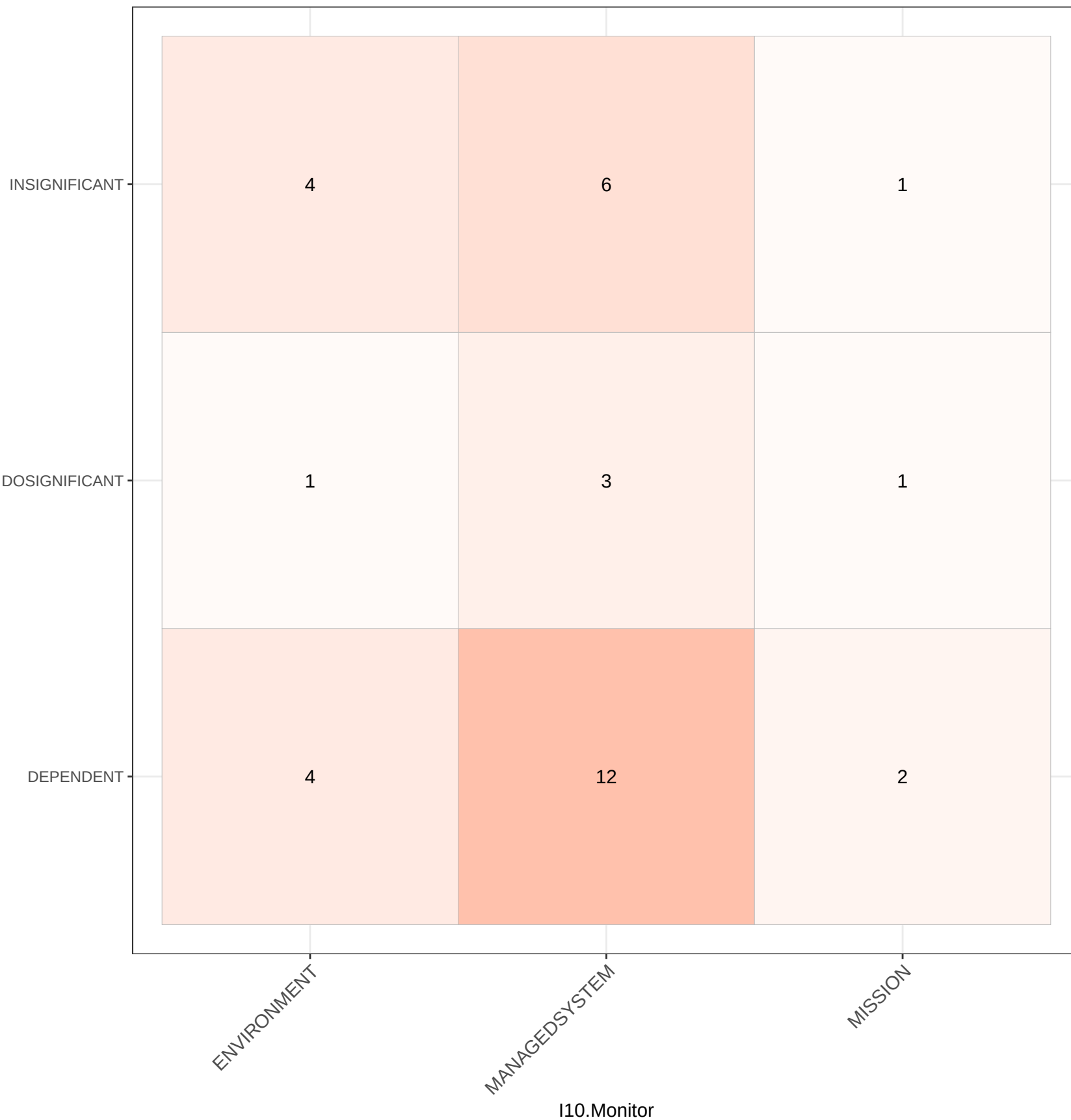


I1.4.Overhead.of.Effects_____I9.Adap..Logic

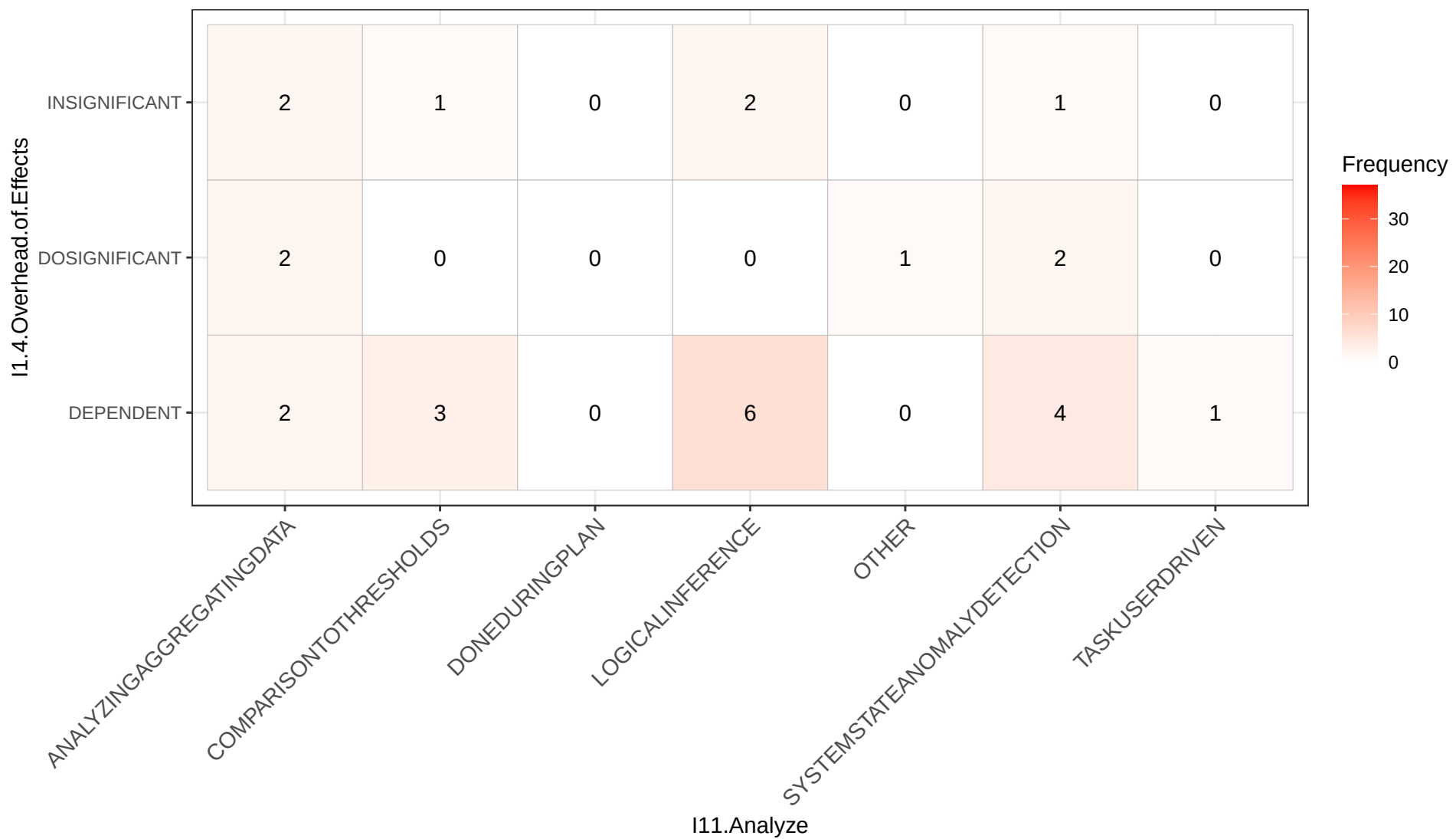


I1.4.Overhead.of.Effects_____I10.Monitor

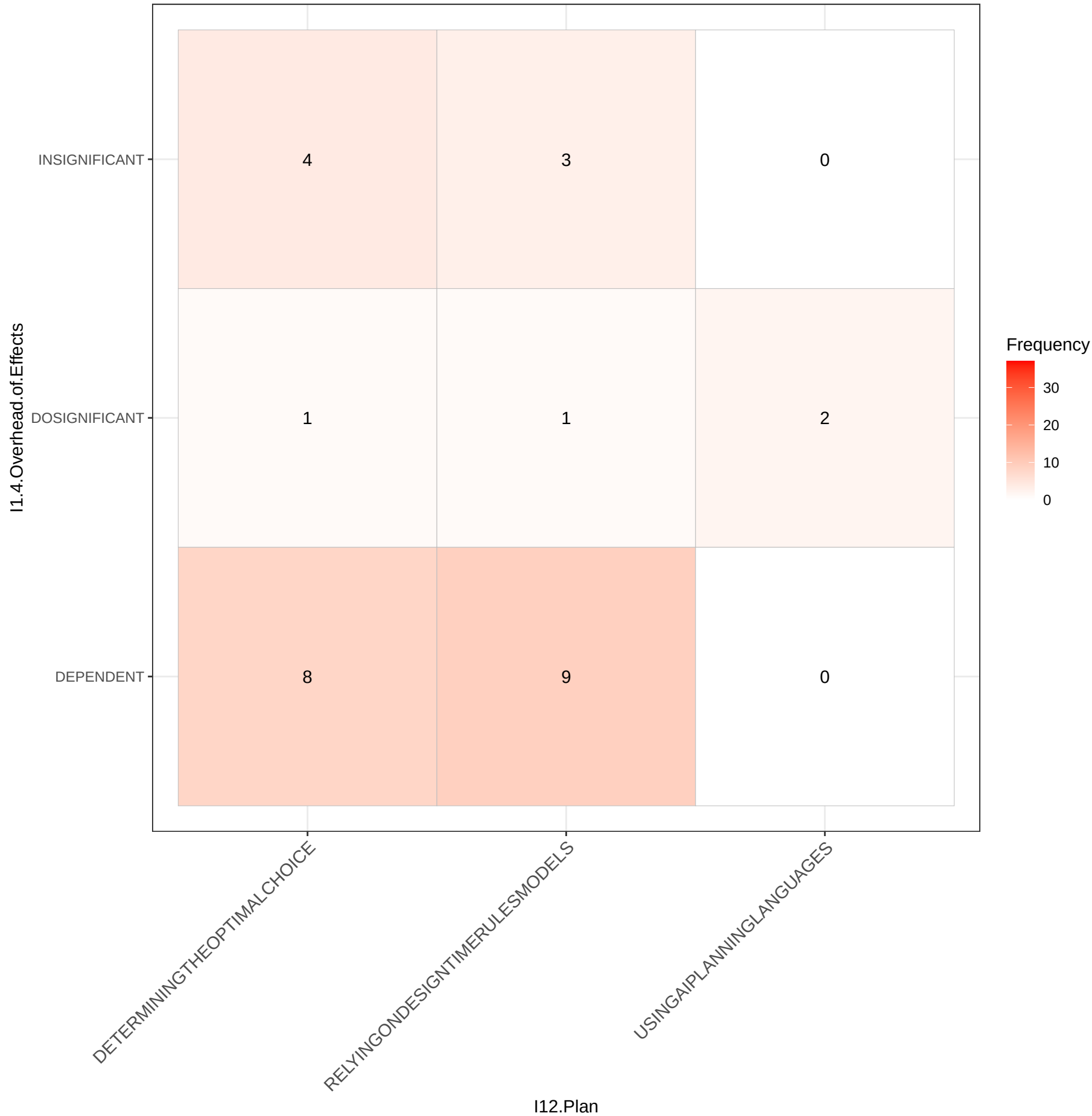
I1.4.Overhead.of.Effects



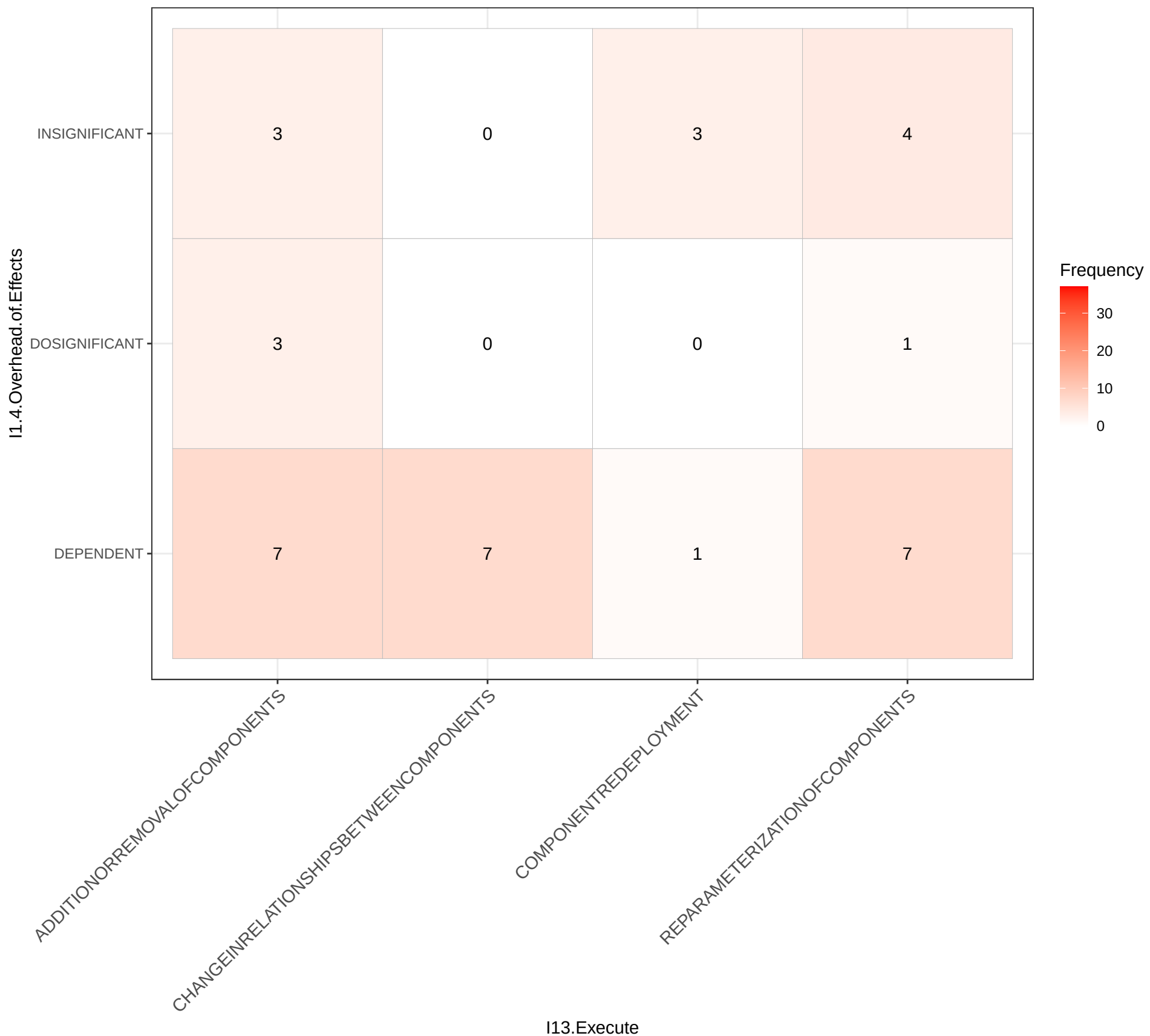
I1.4.Overhead.of.Effects_____I11.Analyze

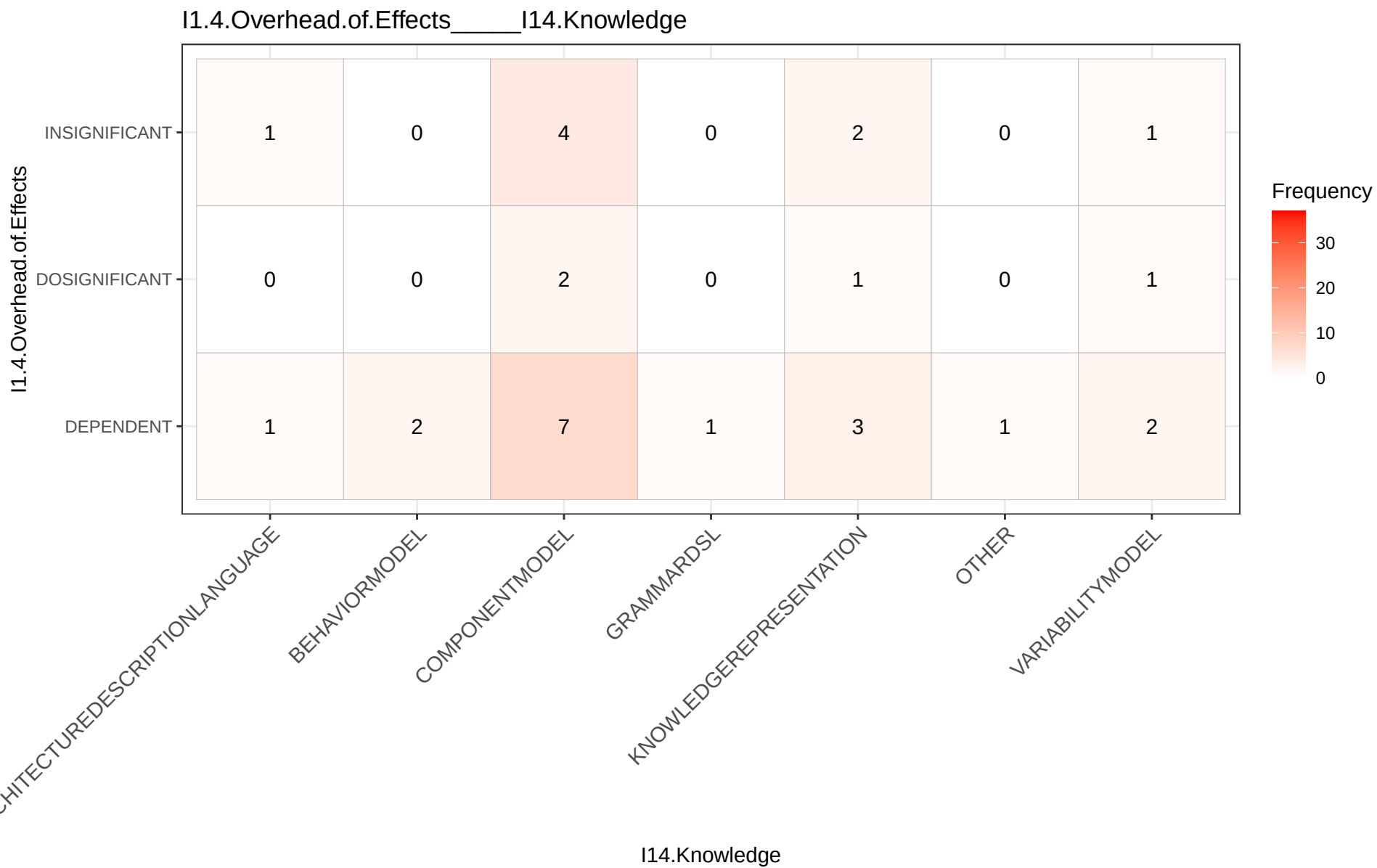


I1.4.Overhead.of.Effects_____I12.Plan

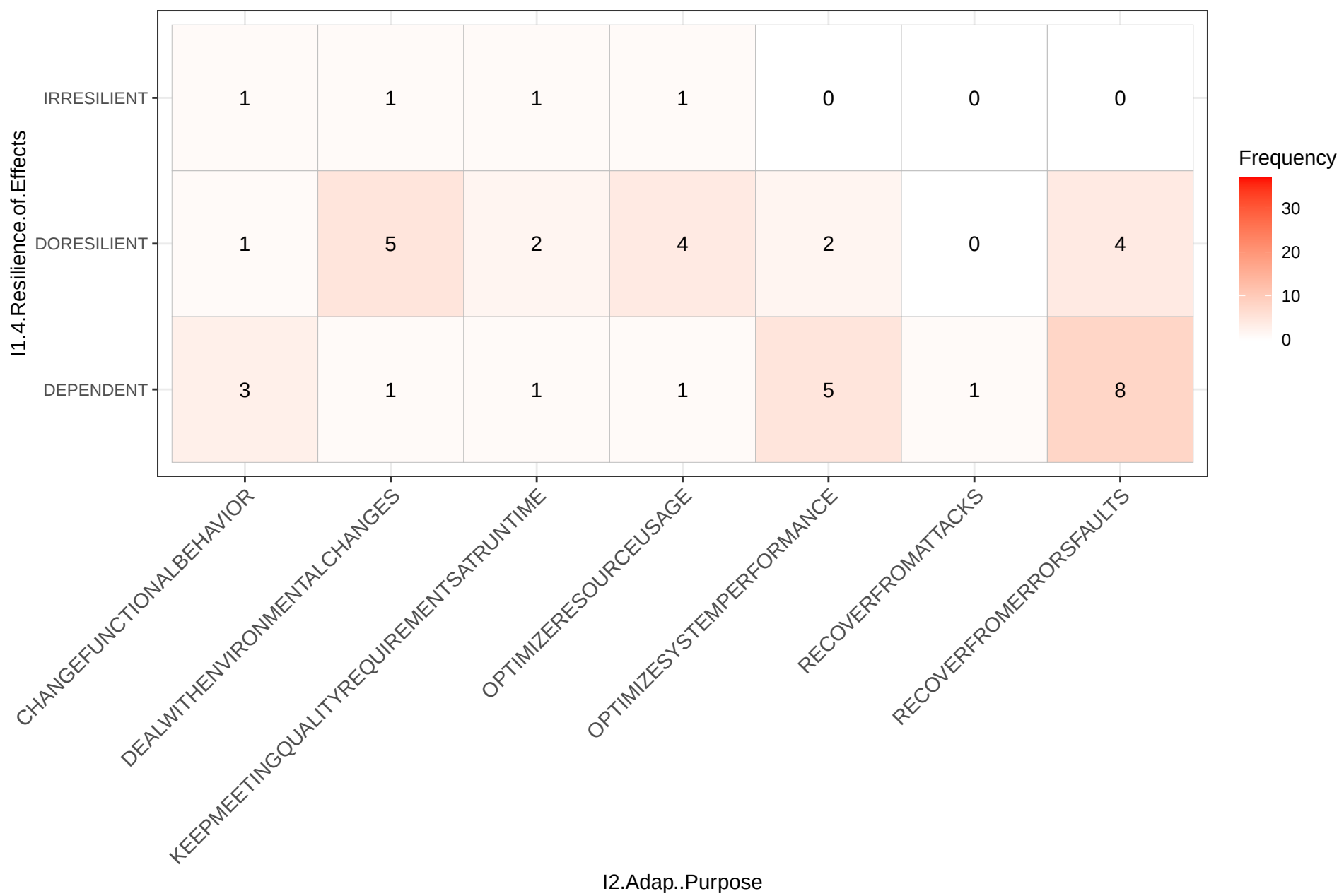


I1.4.Overhead.of.Effects_____I13.Execute





I1.4.Resilience.of.Effects_____I2.Adap..Purpose



I1.4.Resilience.of.Effects

I1.4.Resilience.of.Effects_____I3.Robot.Type

IRRESILIENT
DORESILIENT
DEPENDENT

0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	1	2	1	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	4	1	0	0	0	1	0
0	1	0	0	1	0	0	0	2	1	1	0	0	1	0	1	1	0	0	0	1	0	1	1	1	0	1	0	0	1	0	0

BEBOTSPADERBORN
BLUEROV2
BOXERCLEARPATH
CRAWLERTERMINATORBOT
FIELDMOBILEROBOTS
FRACTIONATEDSPACECRAFT
HETEROGENOUSROBOTS
HEXMANIPULATOR
HEXSERVICEROBOT
IROBOTCREATE2
MOBILEROBOT
MOBILEROBOT
MOBILEROBOTS
MOBILEROBOTTIAGO
MOBILESERVICEROBOT
MULTIPLEHEXROTOR
NAOROBOT
PIONEER3DX
QUADROCOPTER
SINGLESERVINGROTATIONROBOT
TEDUSARTERRESTRIAL
TRIGLIDEINDUSTRIAL
TURTLEBOT
TURTLEBOT2KOBUKI
TWOINDUSTRIALROBOTS
WAREHOUSEDELIVERYROBOT
WHICHISANINDUSTRIALAGV
YUBOT

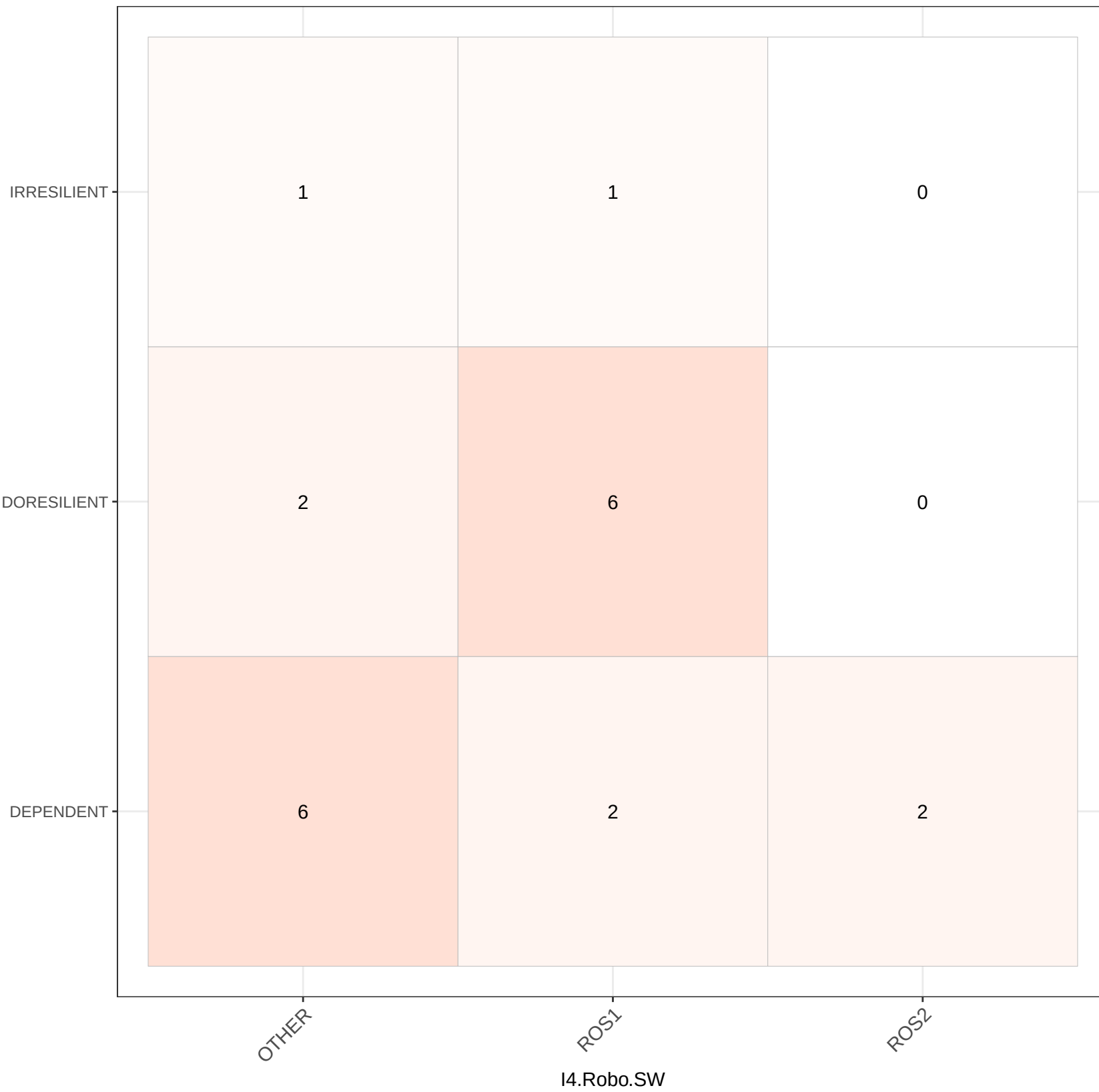
Frequency



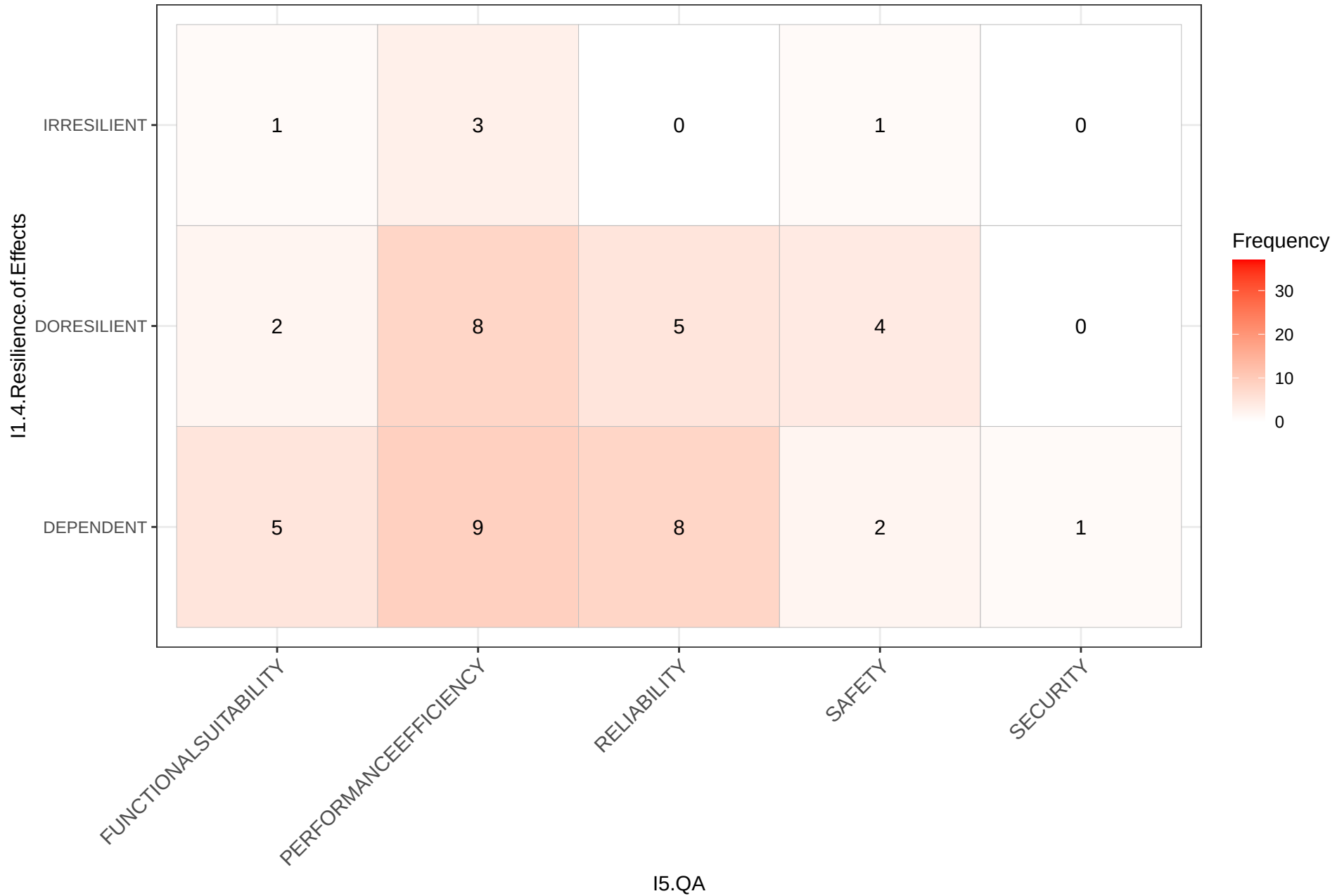
I3.Robot.Type

I1.4.Resilience.of.Effects_____I4.Robo.SW

I1.4.Resilience.of.Effects

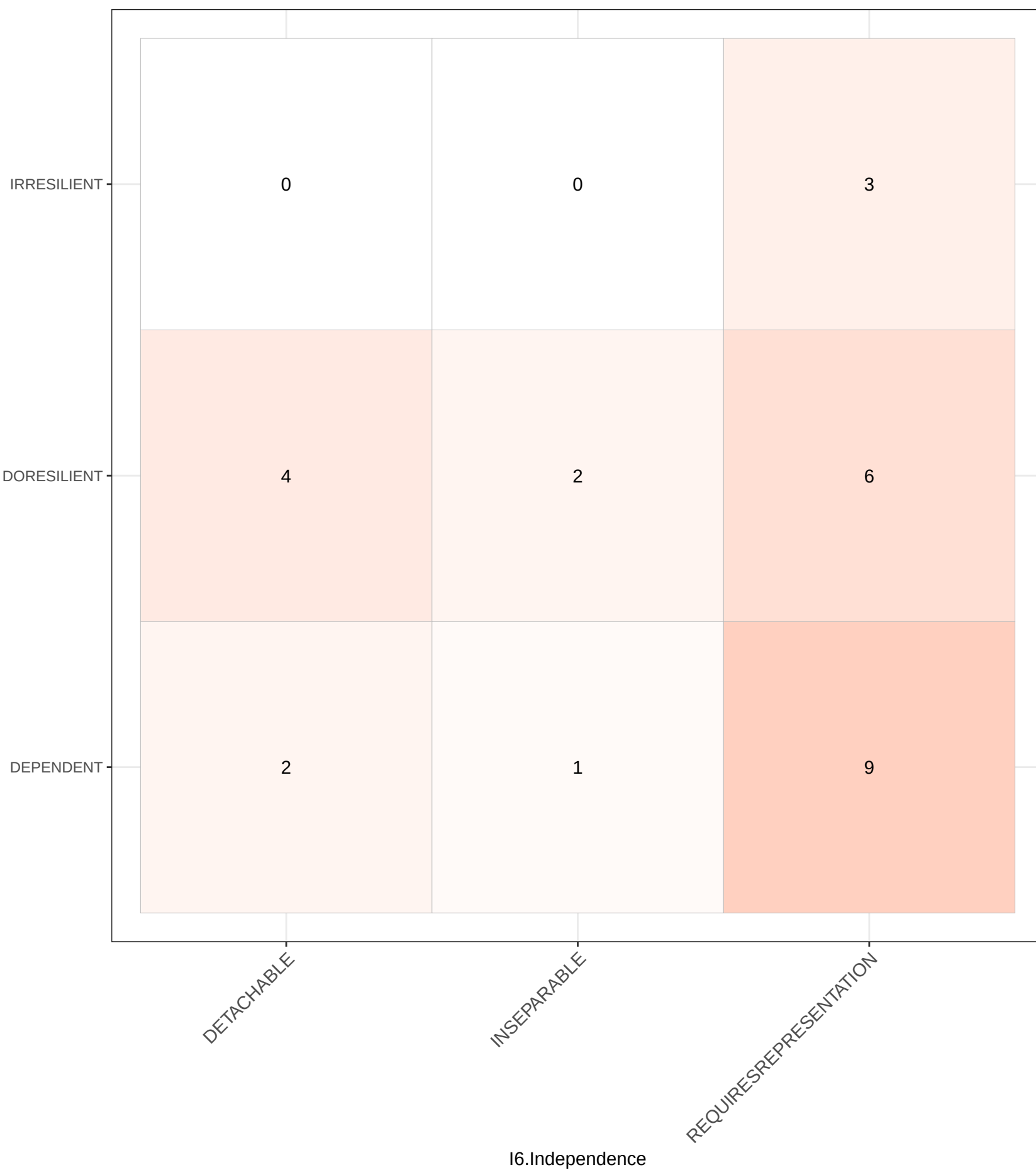


I1.4.Resilience.of.Effects_____I5.QA

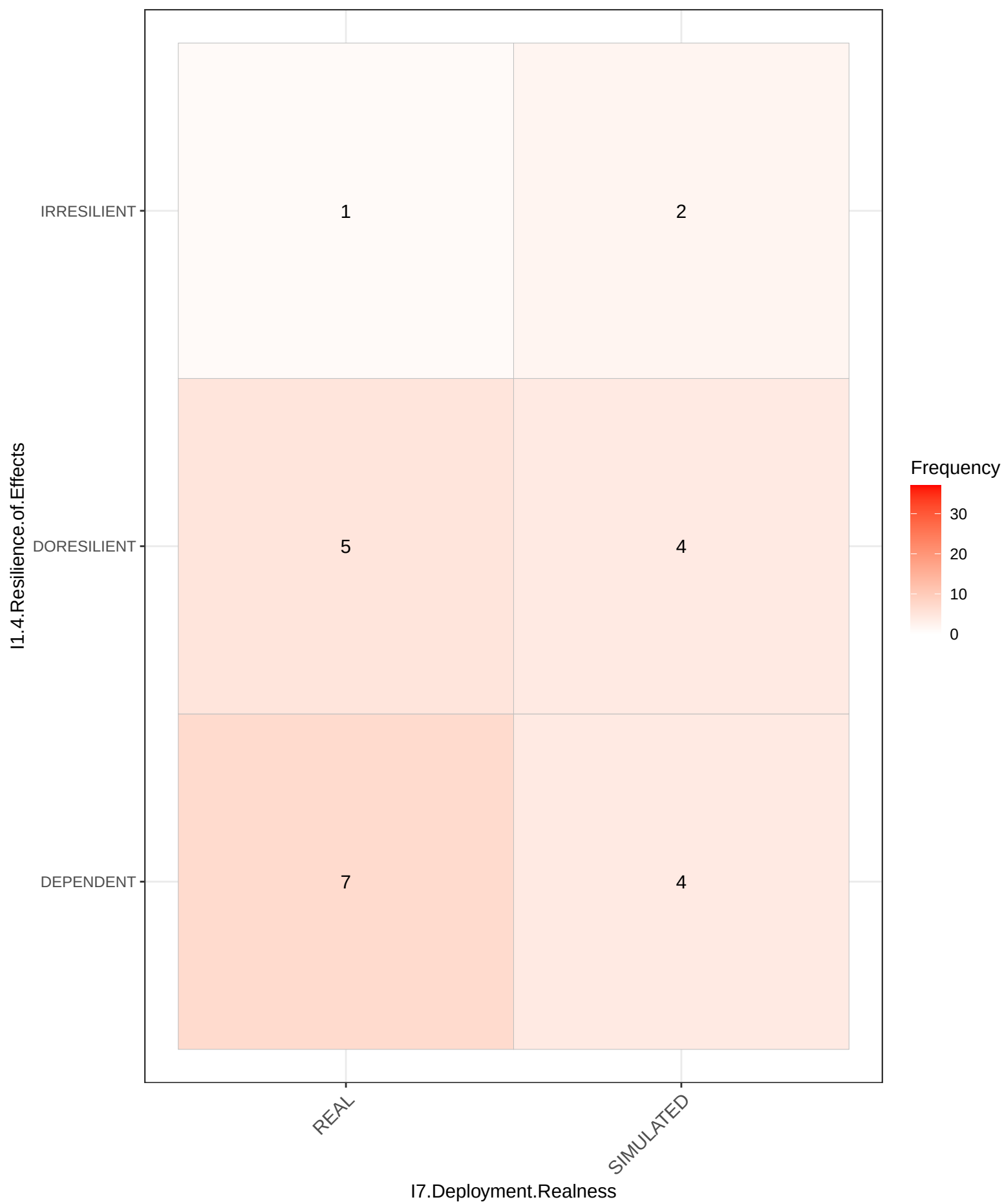


I1.4.Resilience.of.Effects_____I6.Independence

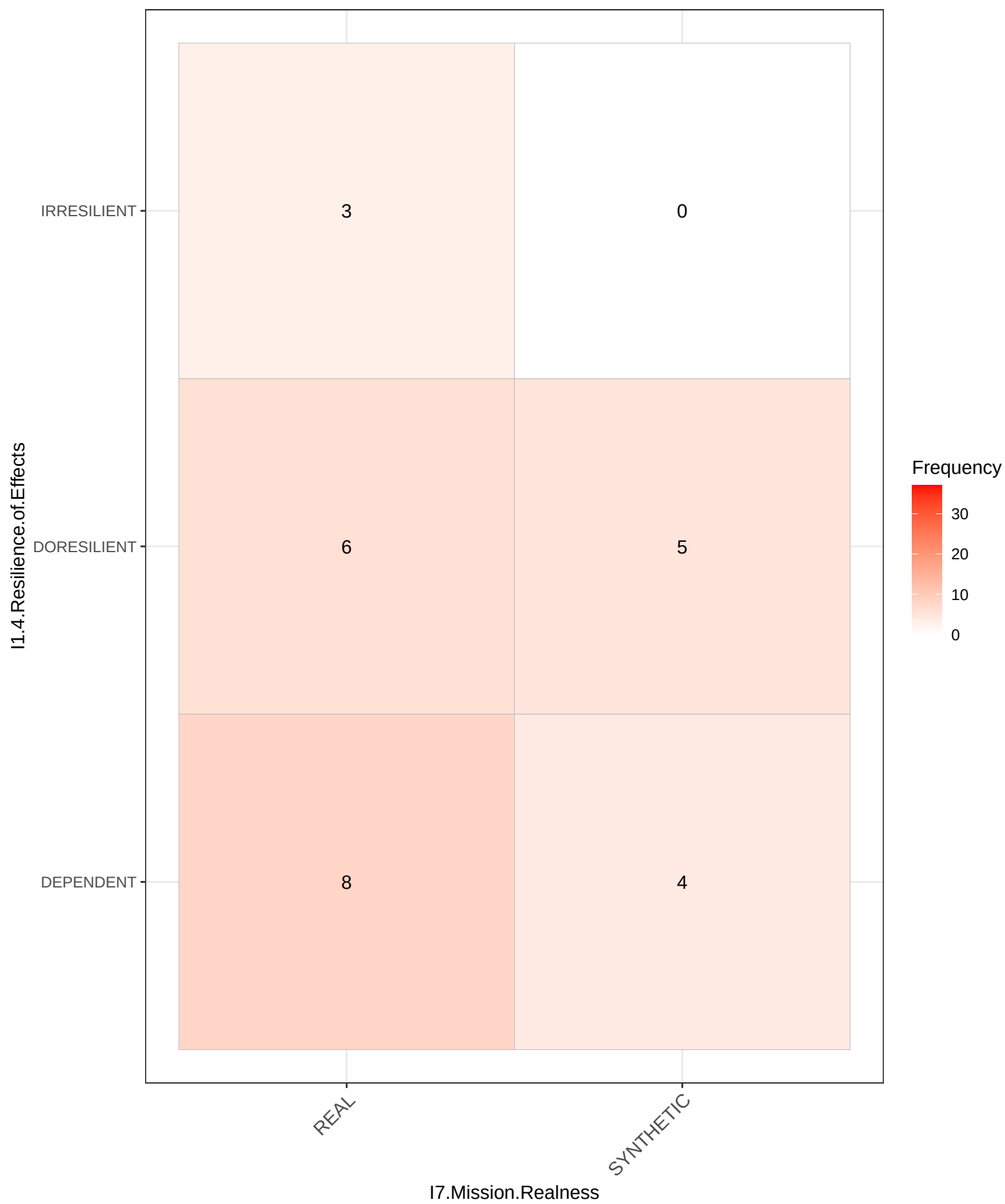
I1.4.Resilience.of.Effects



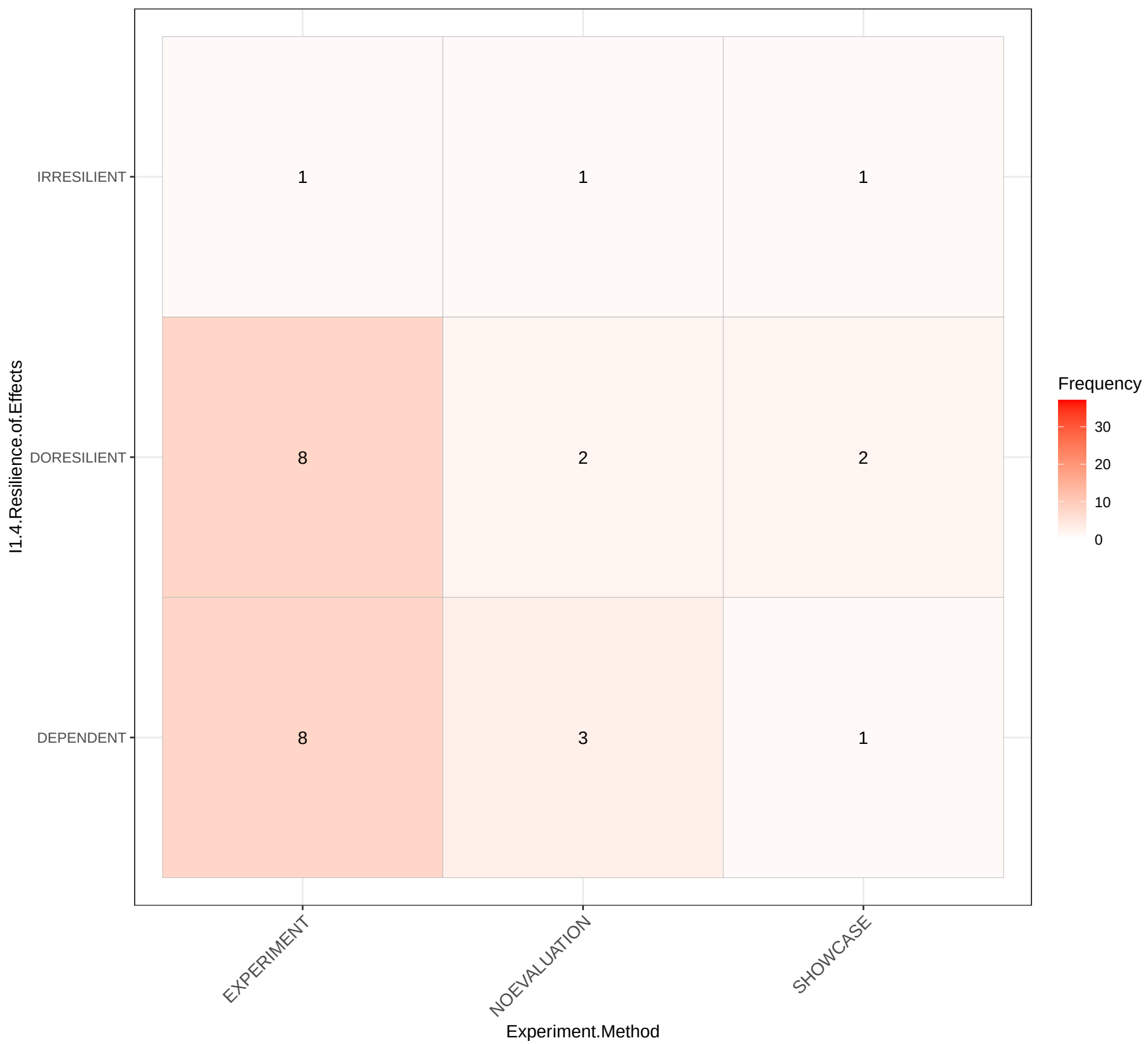
I1.4.Resilience.of.Effects_____I7.Deployment.Realness



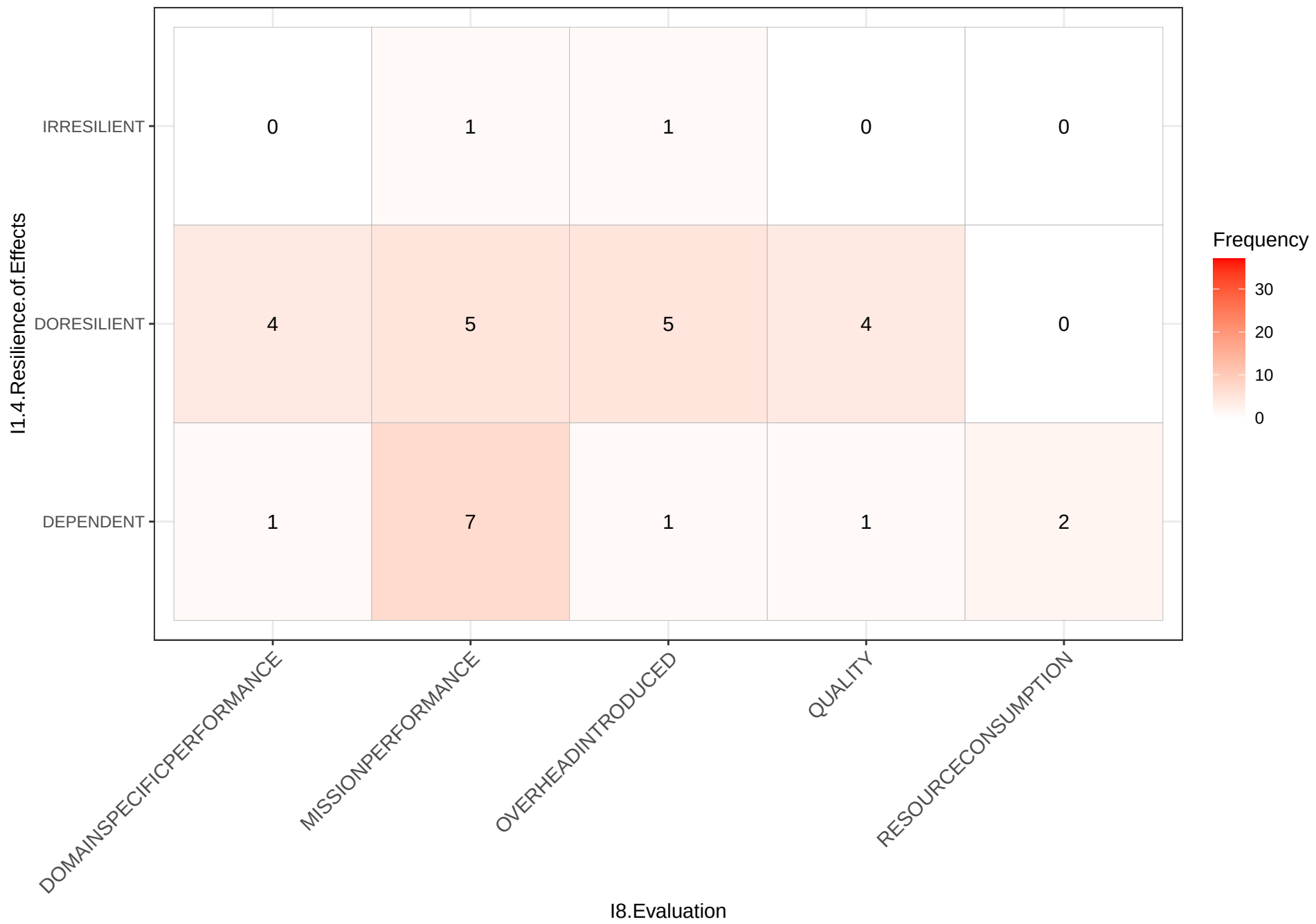
I1.4.Resilience.of.Effects_____I7.Mission.Realness



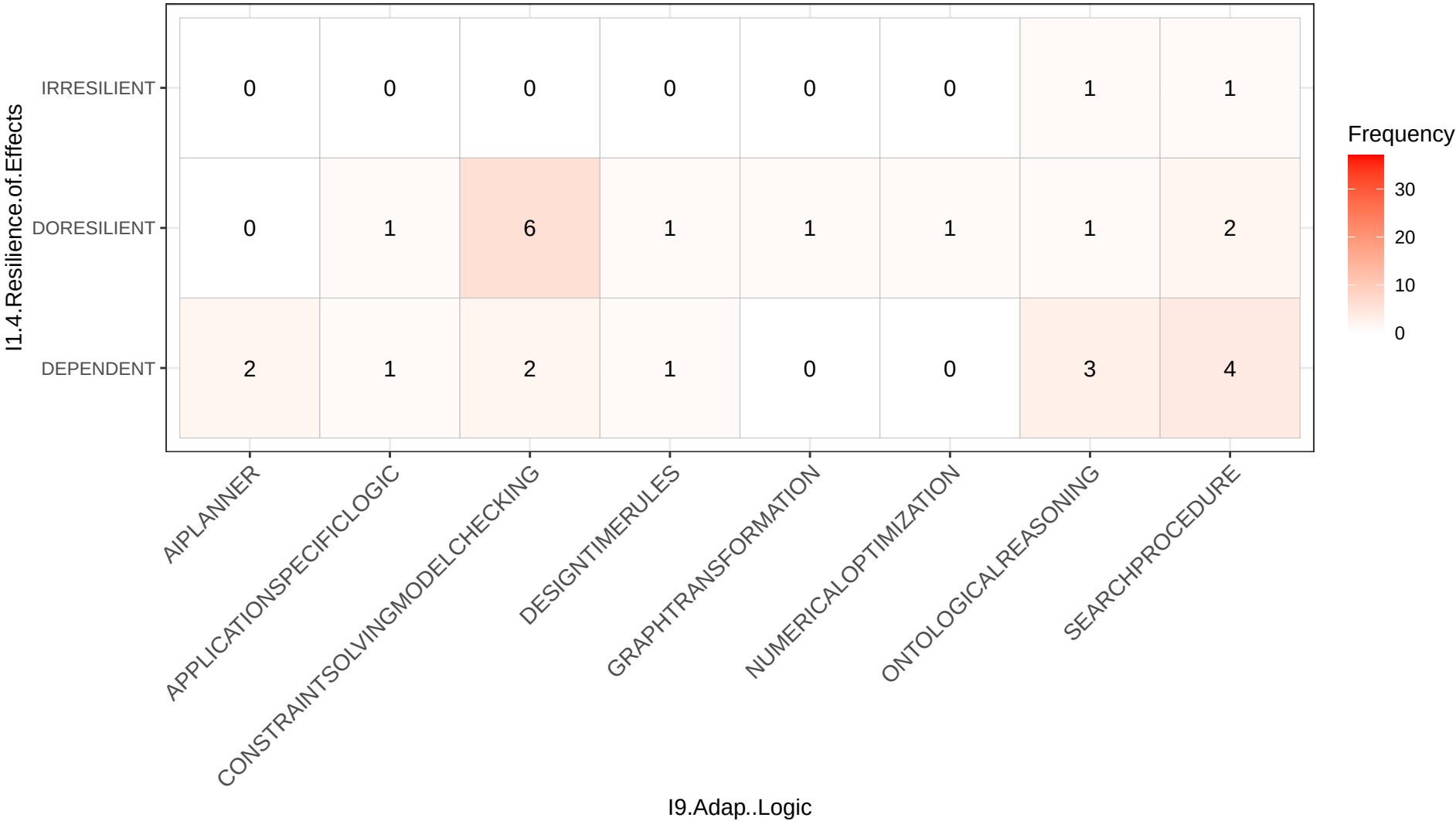
I1.4.Resilience.of.Effects_____Experiment.Method



I1.4.Resilience.of.Effects_____I8.Evaluation

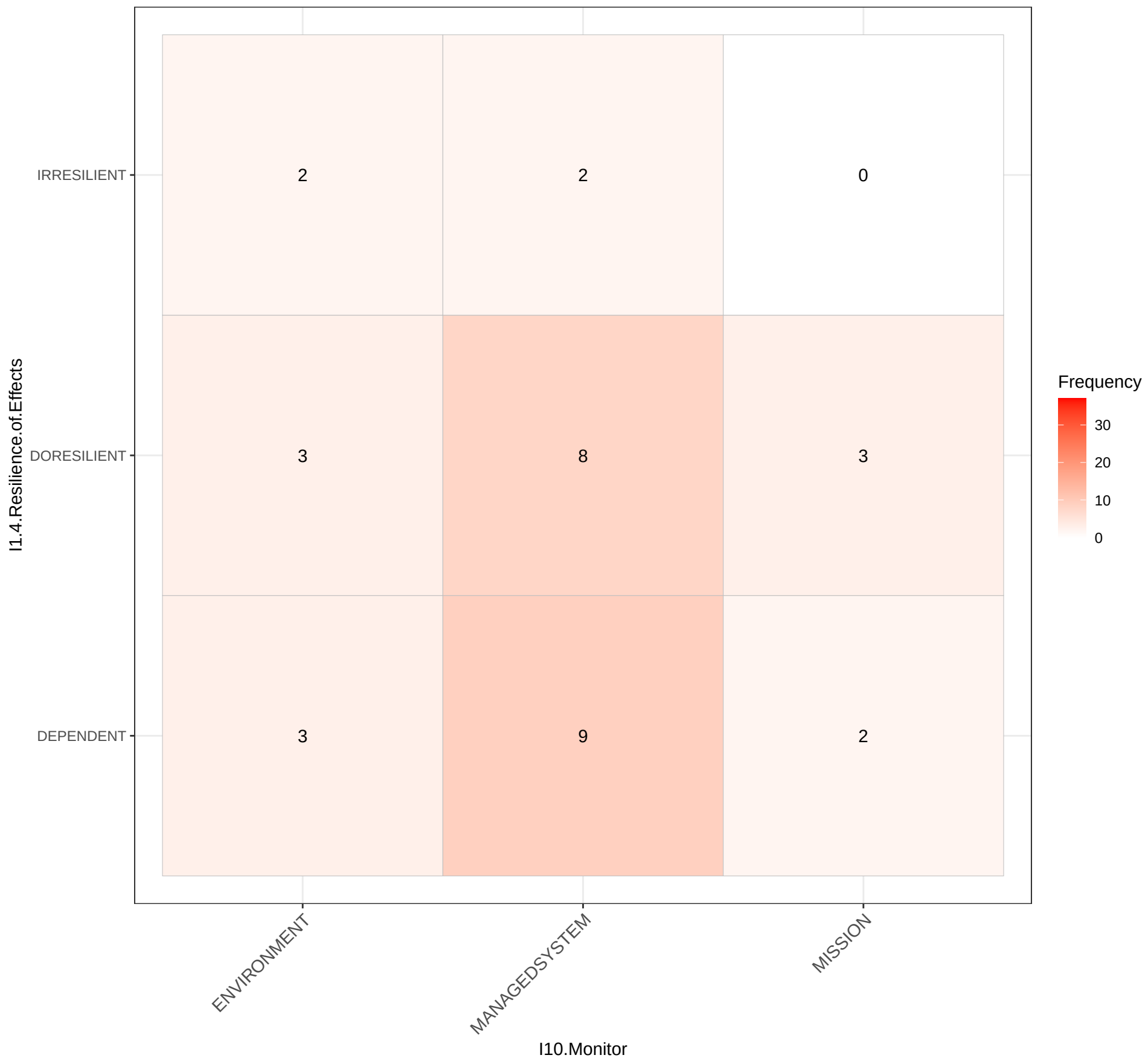


I1.4.Resilience.of.Effects_____I9.Adap..Logic

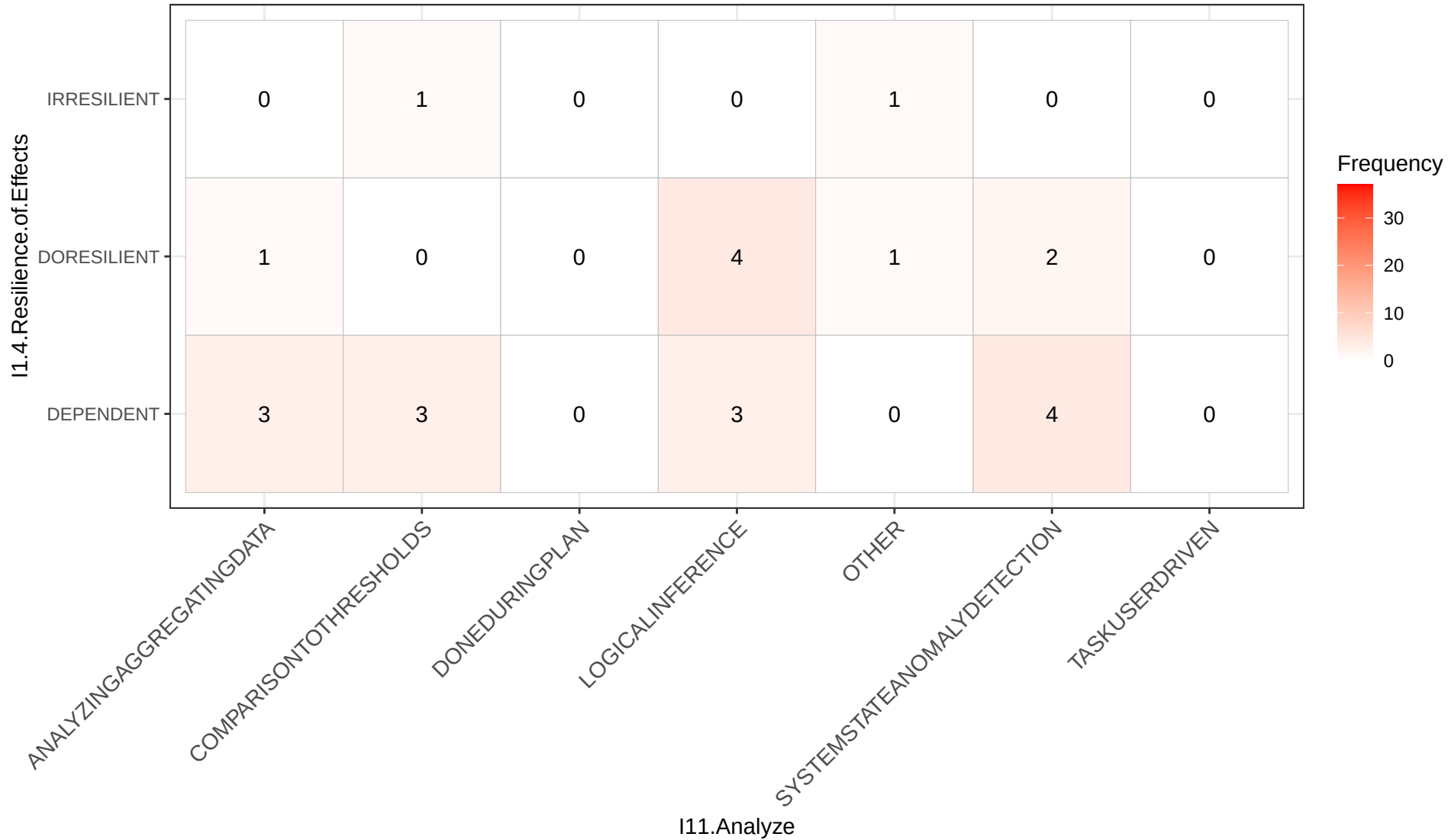


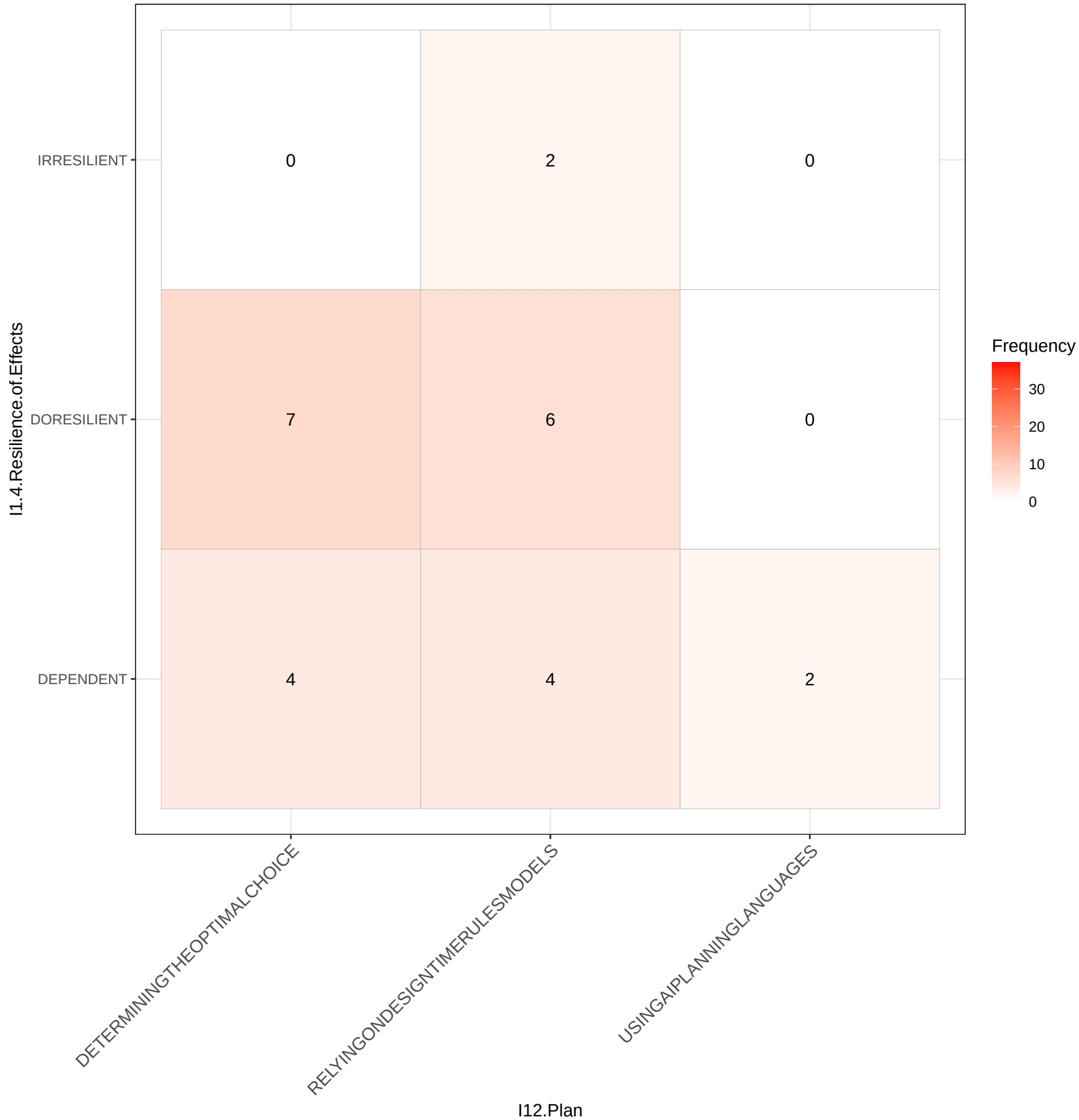
I9.Adap..Logic

I1.4.Resilience.of.Effects_____I10.Monitor



I1.4.Resilience.of.Effects_____I11.Analyze





I1.4.Resilience.of.Effects_____I13.Execute

I1.4.Resilience.of.Effects

IRRESILIENT

DORESILIENT

DEPENDENT

Frequency

30

20

10

0

3

0

0

1

4

3

2

5

7

2

1

5

ADDITIONORREMOVALOFCOMPONENTS

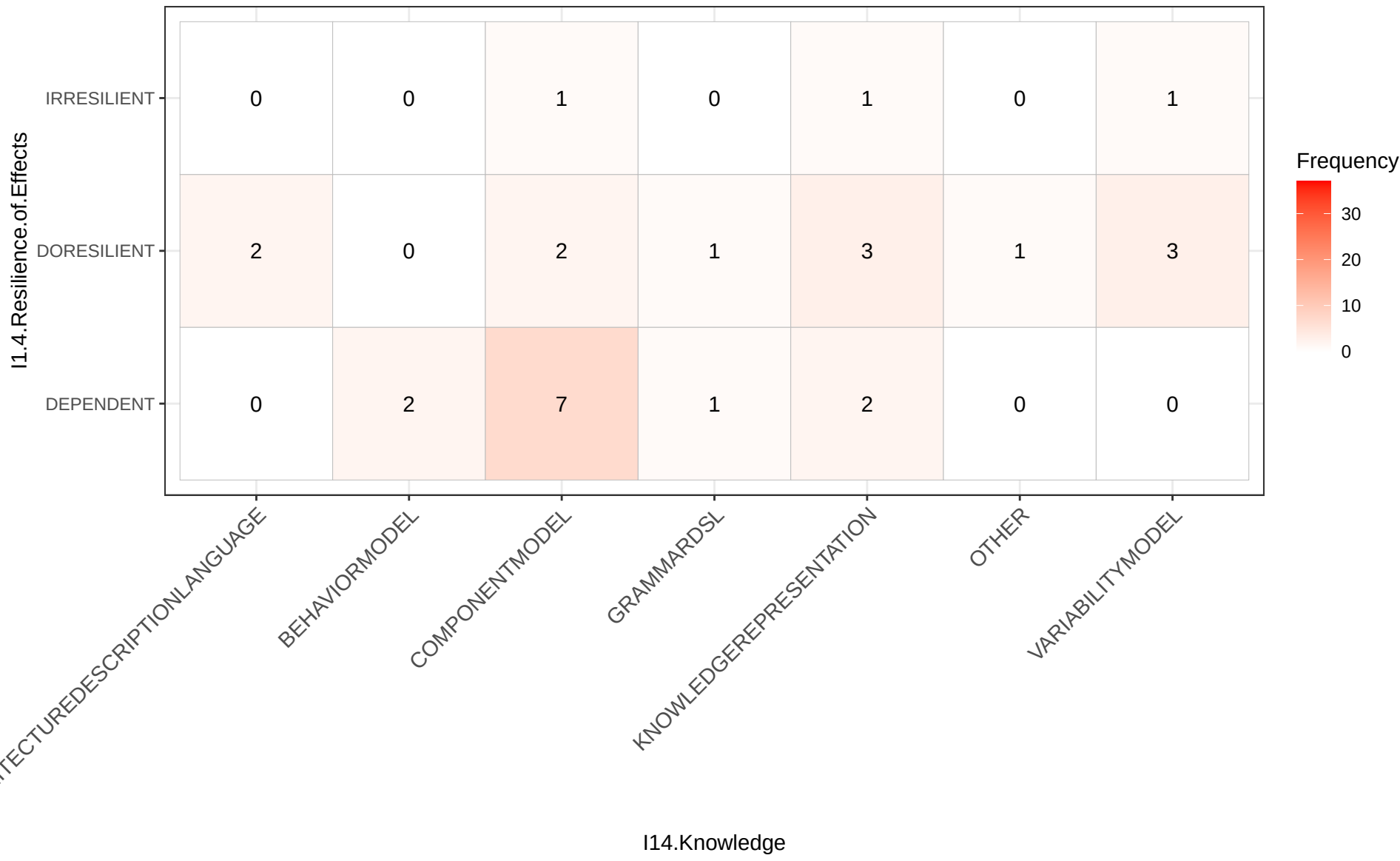
CHANGEINRELATIONSHIPS BETWEENCOMPONENTS

COMPONENTREDEPLOYMENT

REPARAMETERIZATIONOFCOMPONENTS

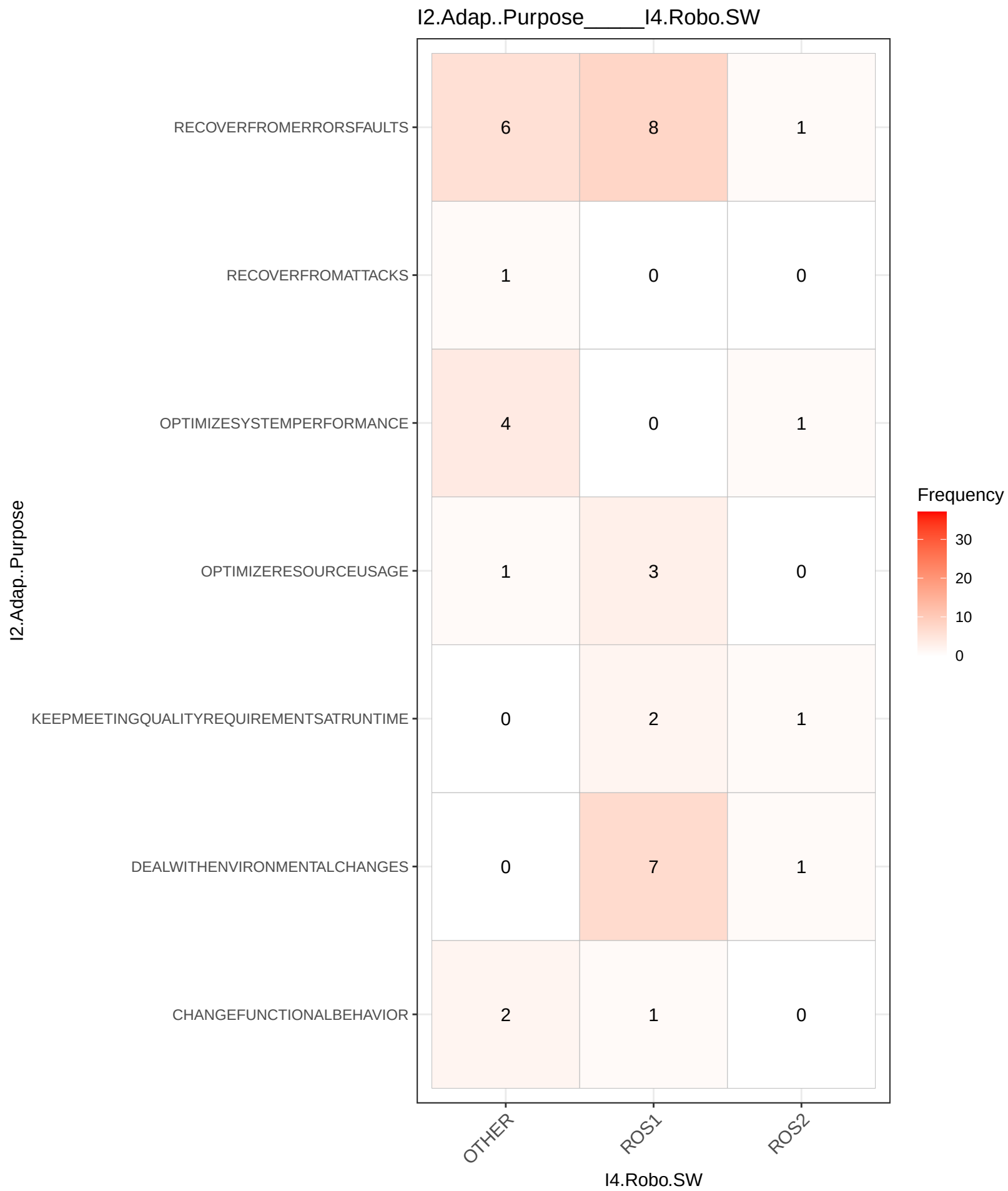
I13.Execute

I1.4.Resilience.of.Effects_____I14.Knowledge



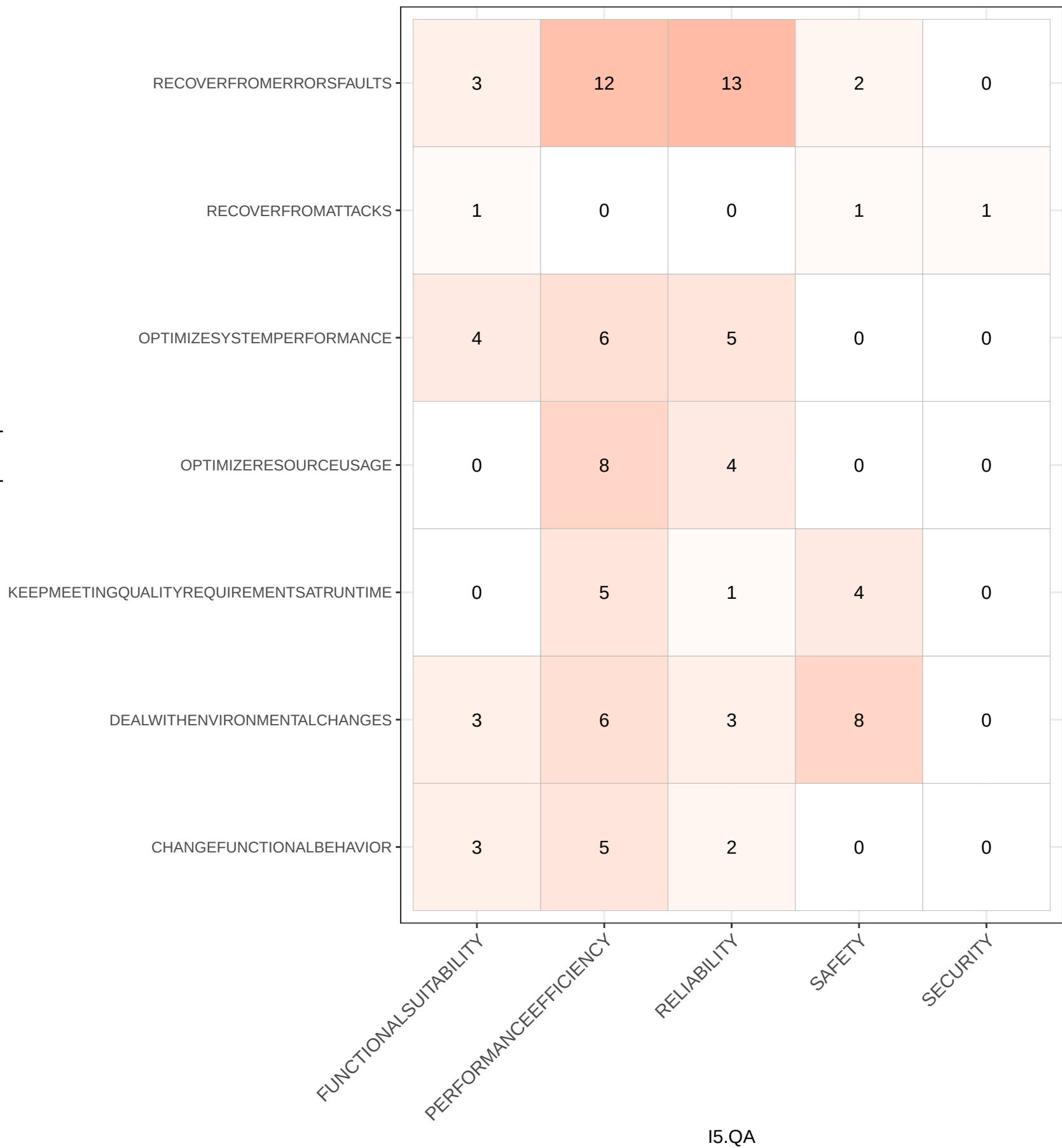
	I2.Adap..Purpose												I3.Robot.Type																			
RECOVERFROMERRORSFAULTS	0	1	0	0	1	1	0	0	2	0	0	0	1	1	0	0	0	0	2	1	0	1	1	1	0	4	0	0	0	1	0	0
RECOVERFROMATTACKS	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
OPTIMIZESYSTEMPERFORMANCE	0	1	0	0	1	0	0	0	2	0	0	1	0	2	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0
OPTIMIZERESOURCEUSAGE	0	0	0	0	0	0	2	1	0	1	0	0	0	1	0	0	1	0	2	0	1	0	0	0	1	0	0	0	0	0	0	0
KEEPMEETINGQUALITYREQUIREMENTSATRUNTIME	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0
DEALWITHENVIRONMENTALCHANGES	1	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	0	0	1	0	0	0	3	1	1	0	0	0	2
CHANGEFUNCTIONALBEHAVIOR	0	0	0	1	0	0	0	0	0	1	0	1	0	0	0	0	1	0	1	0	0	0	0	0	0	1	0	1	1	0	0	0
BEBOTSPADERBORN																																
BLUEPROV2																																
BOXERCLEANPATH																																
CRAWLERTERMINATORBOT																																
FIELDMOBILEROBOT																																
FRACTIONATEDSPACECRAFT																																
HETEROGENOUSROBOT																																
HEXMANIPULATOR																																
IRBOTCRAI																																
MOBILEROBOTMANIPULATOR																																
MOBILEROBOTCREATOR																																
MOBILEROBOTMANIPULATOR2																																
MOBILEROBOT																																
MOBILEROBOT																																
MOBILEROBOT																																
MOBILEROBOT																																
MOBILEROBOT																																
MOBILESERVICERESTRIALS					</																											

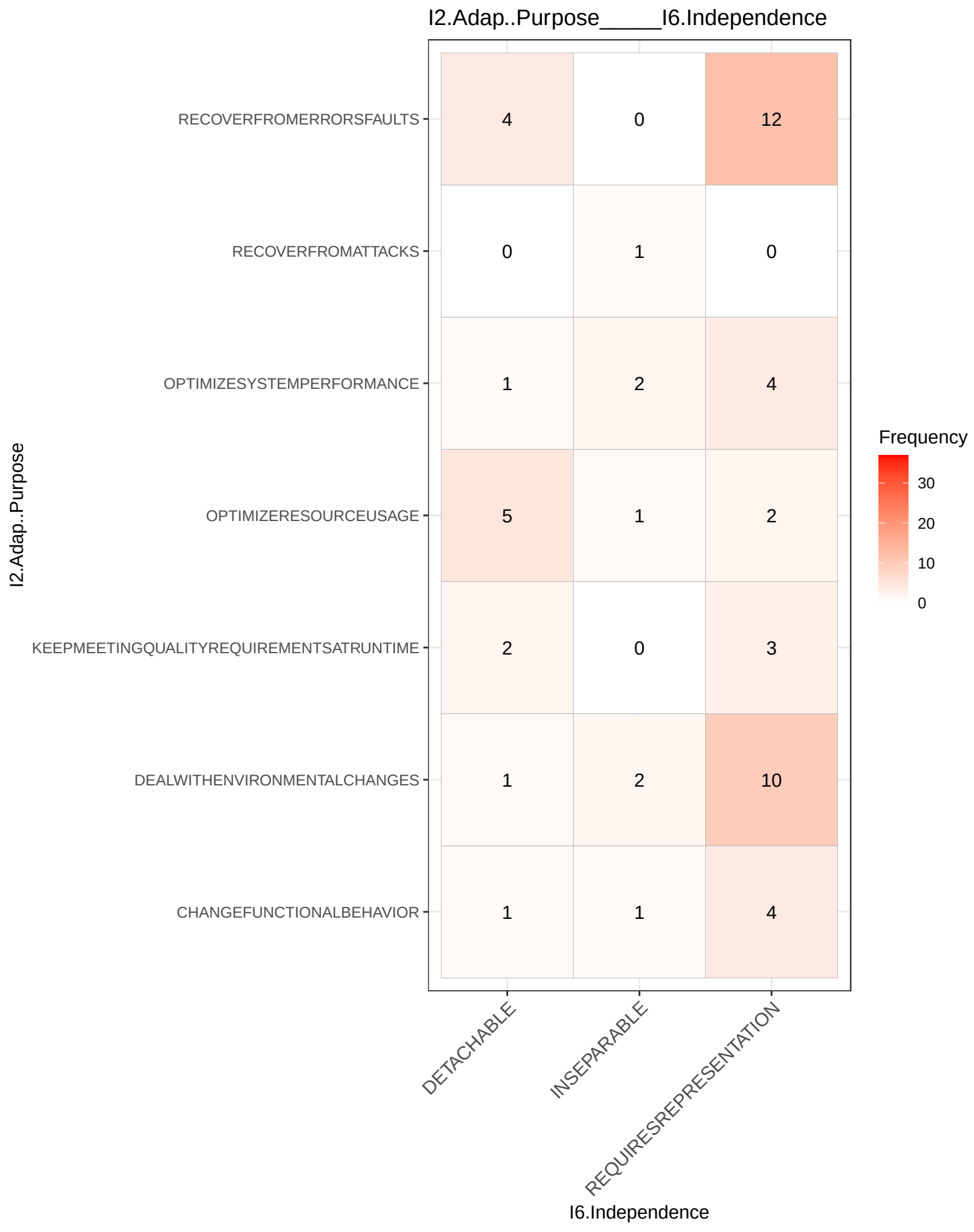
I3.Robot.Type

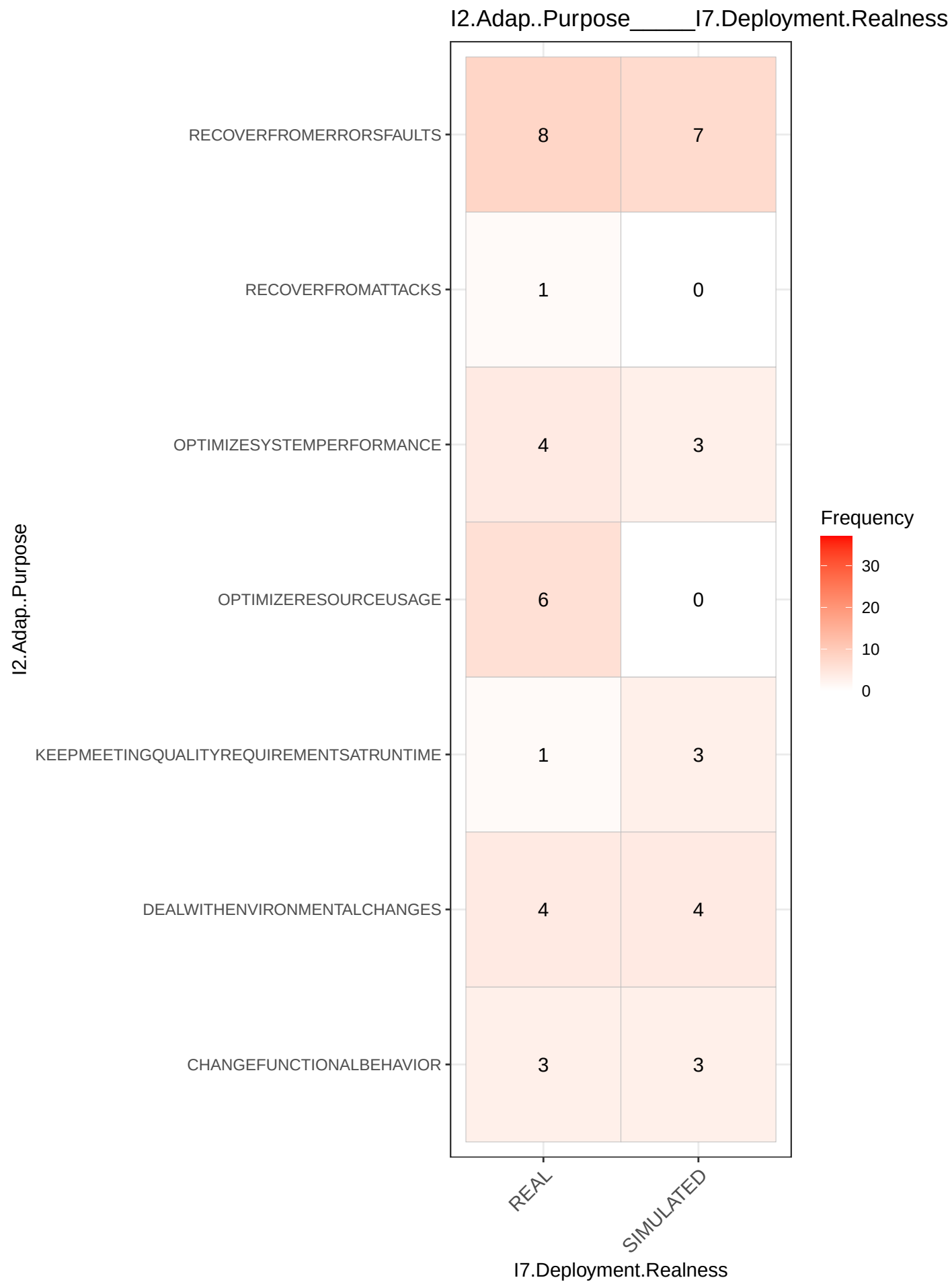


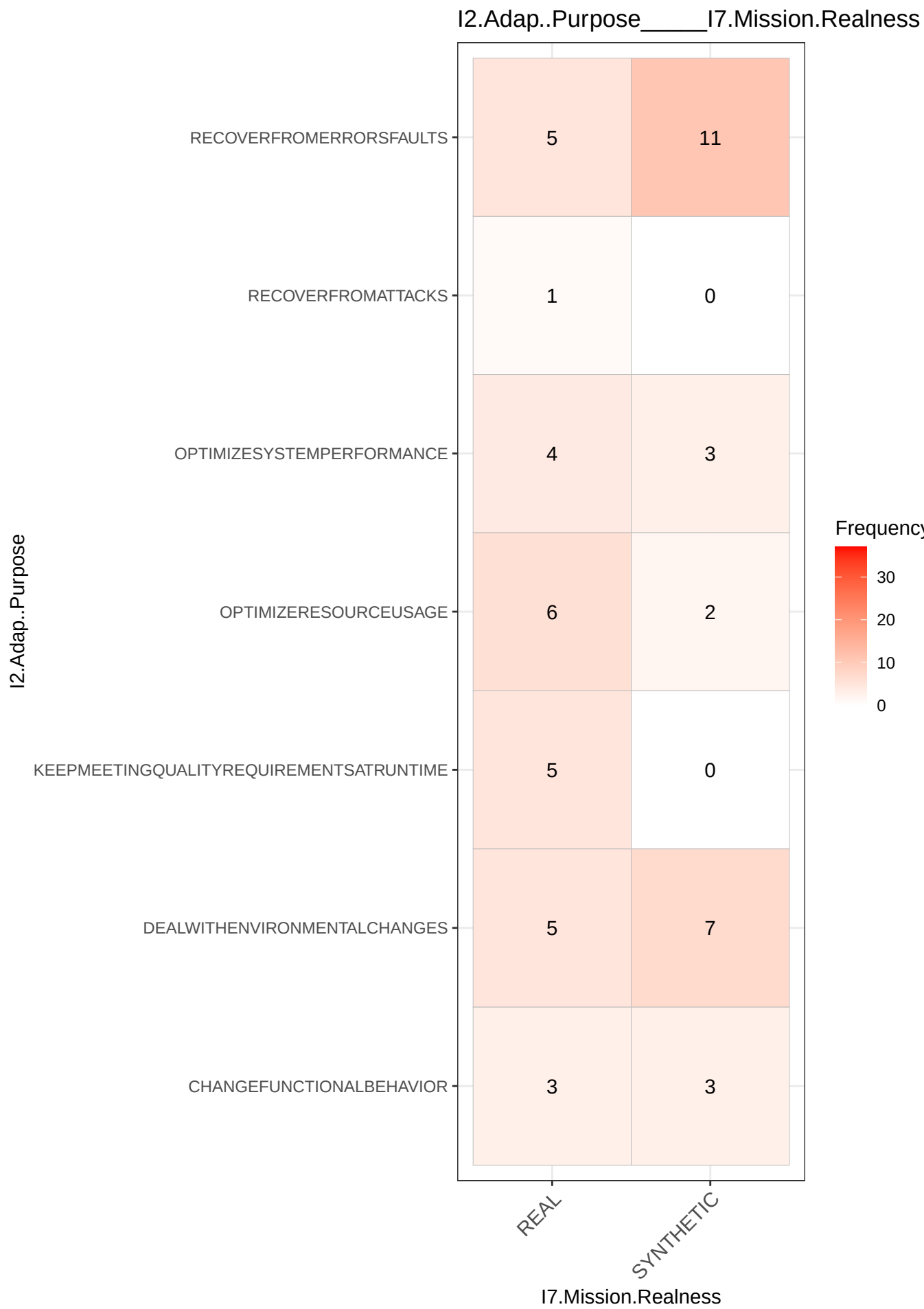
I2.Adap..Purpose_____I5.QA

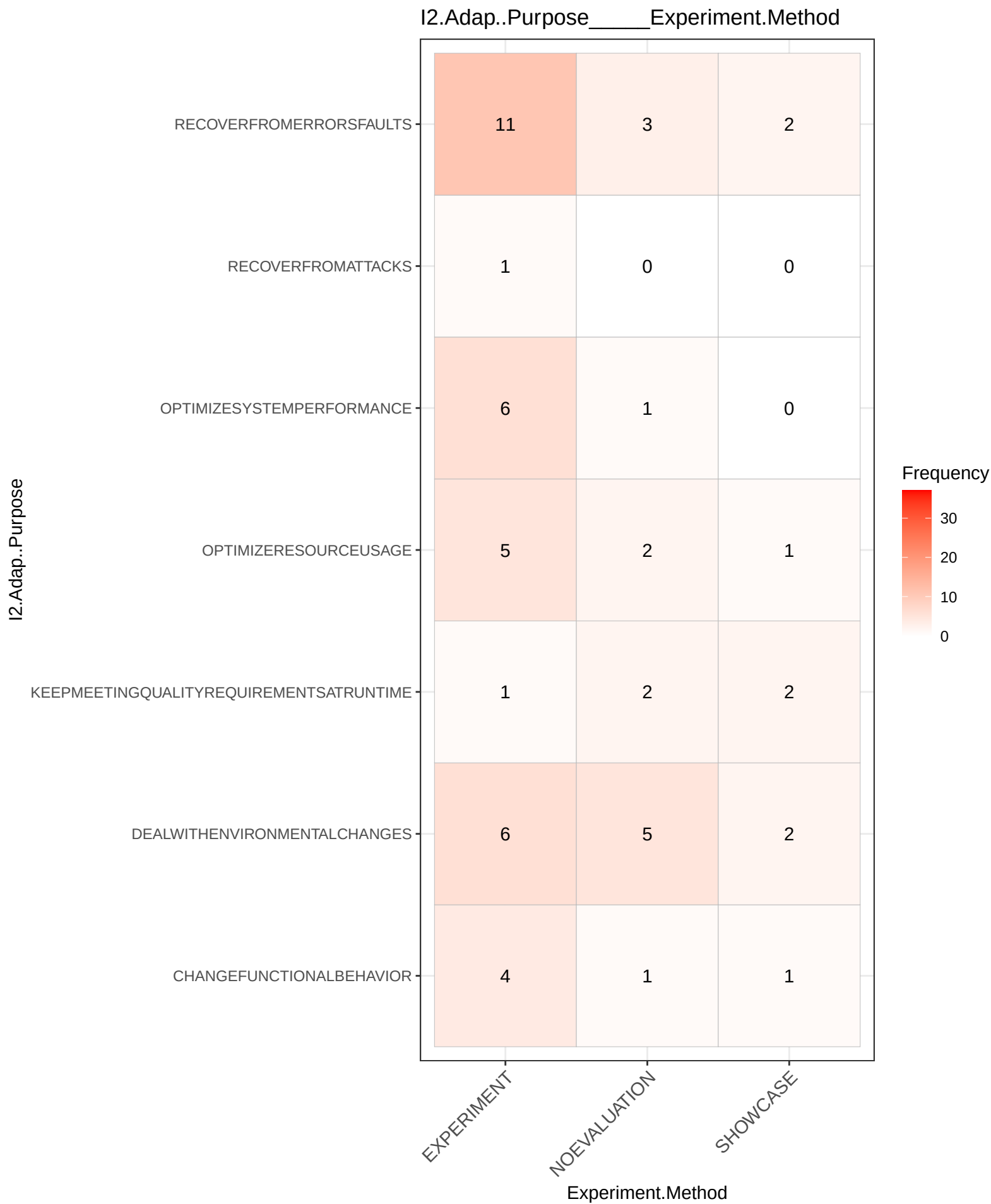
I2.Adap..Purpose

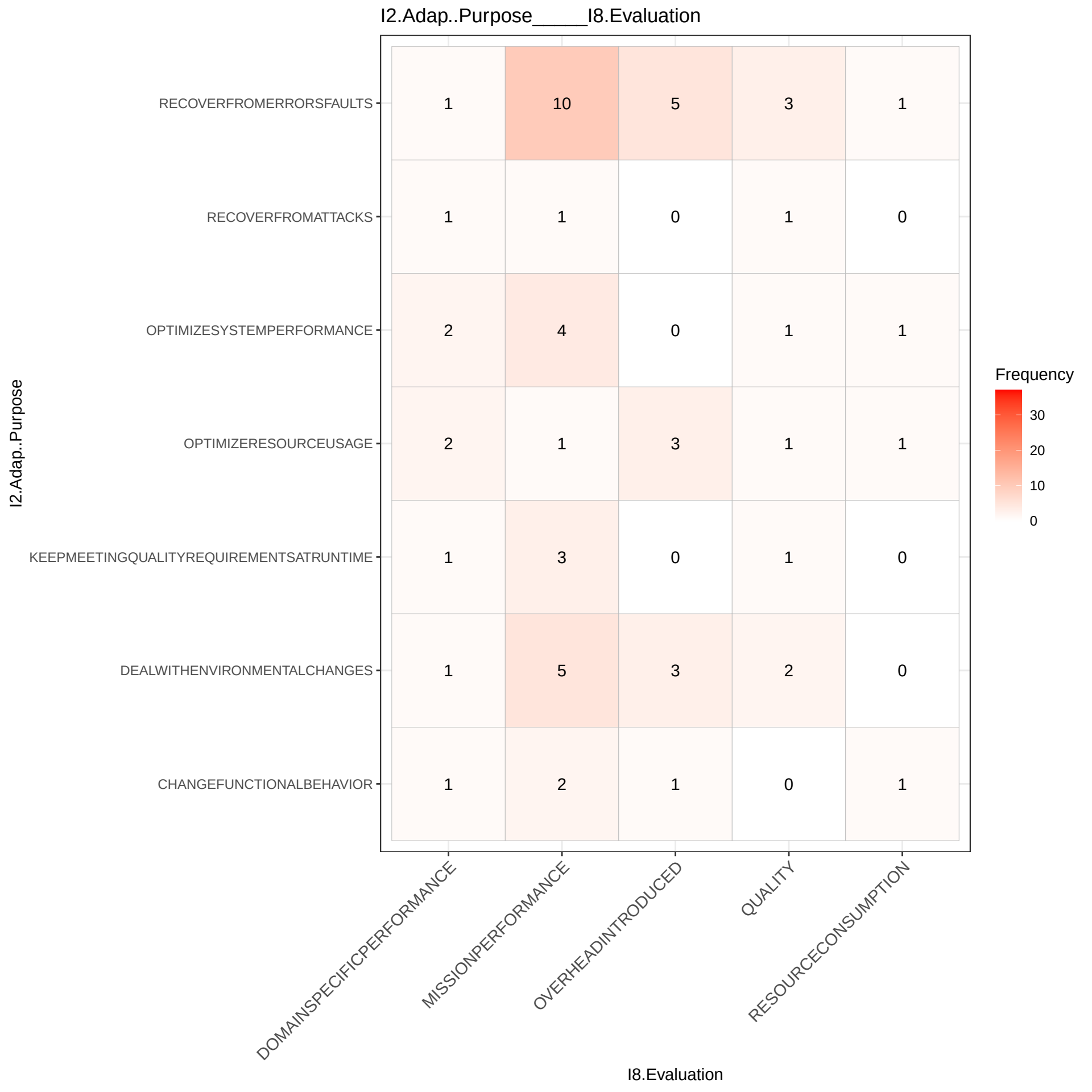


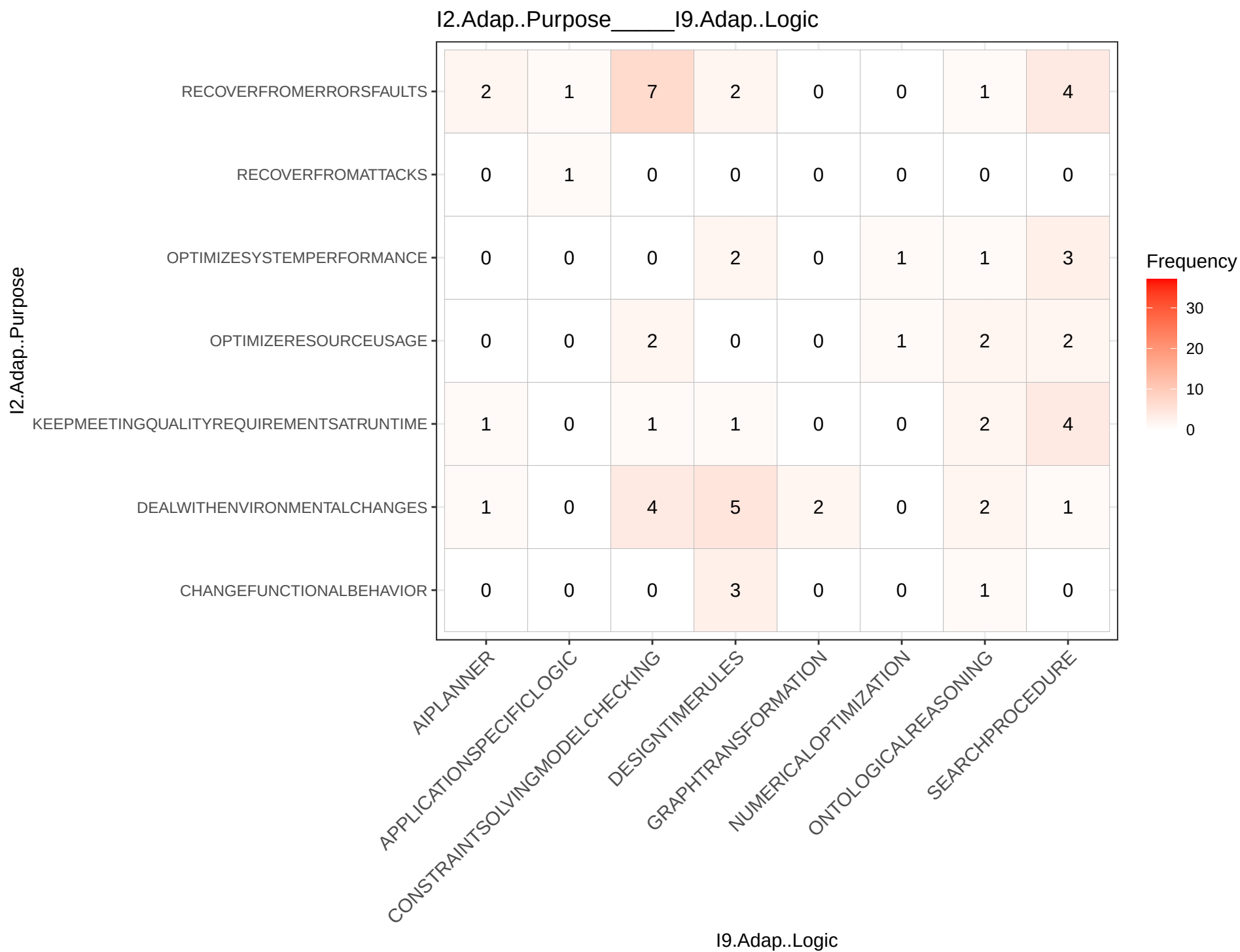




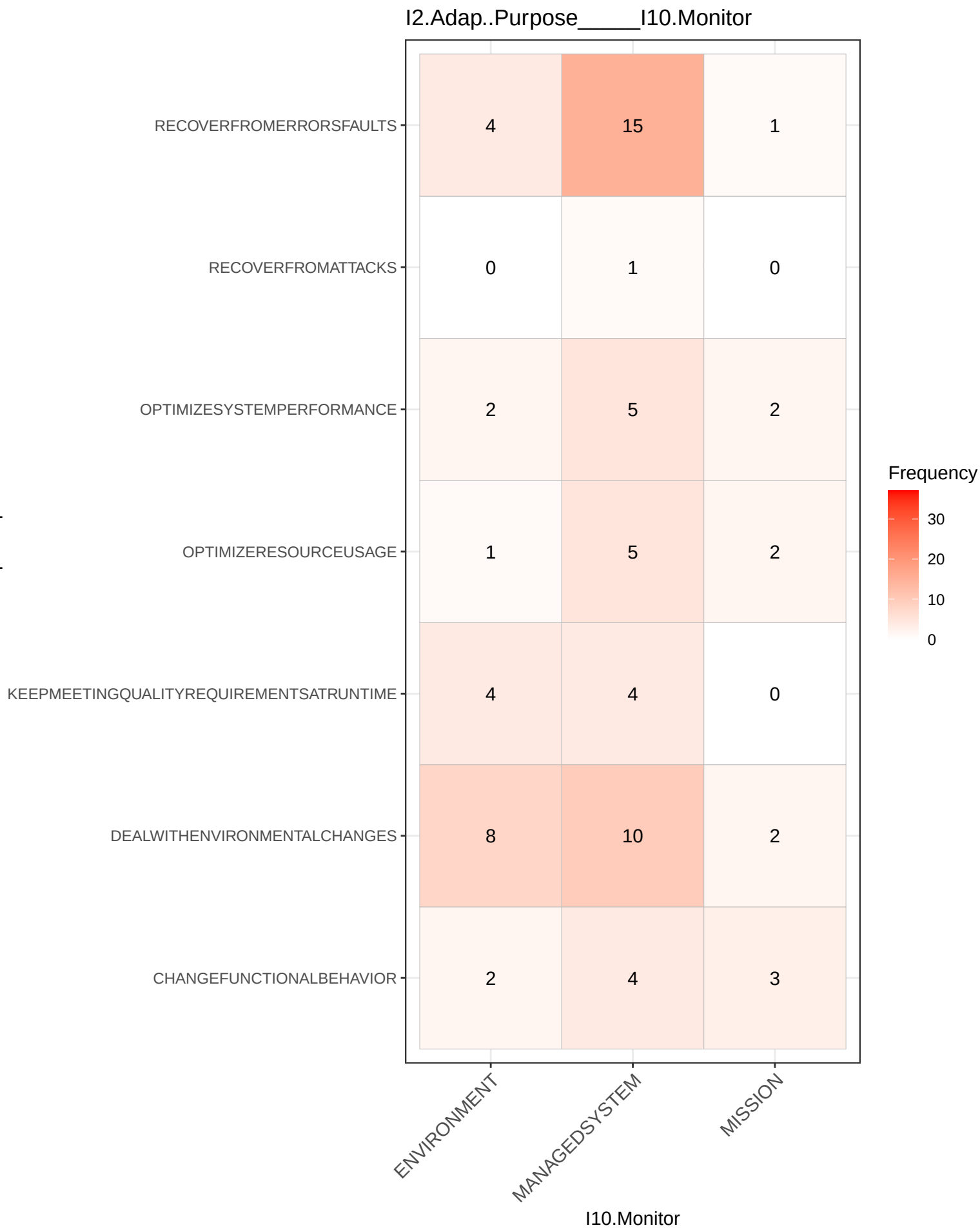


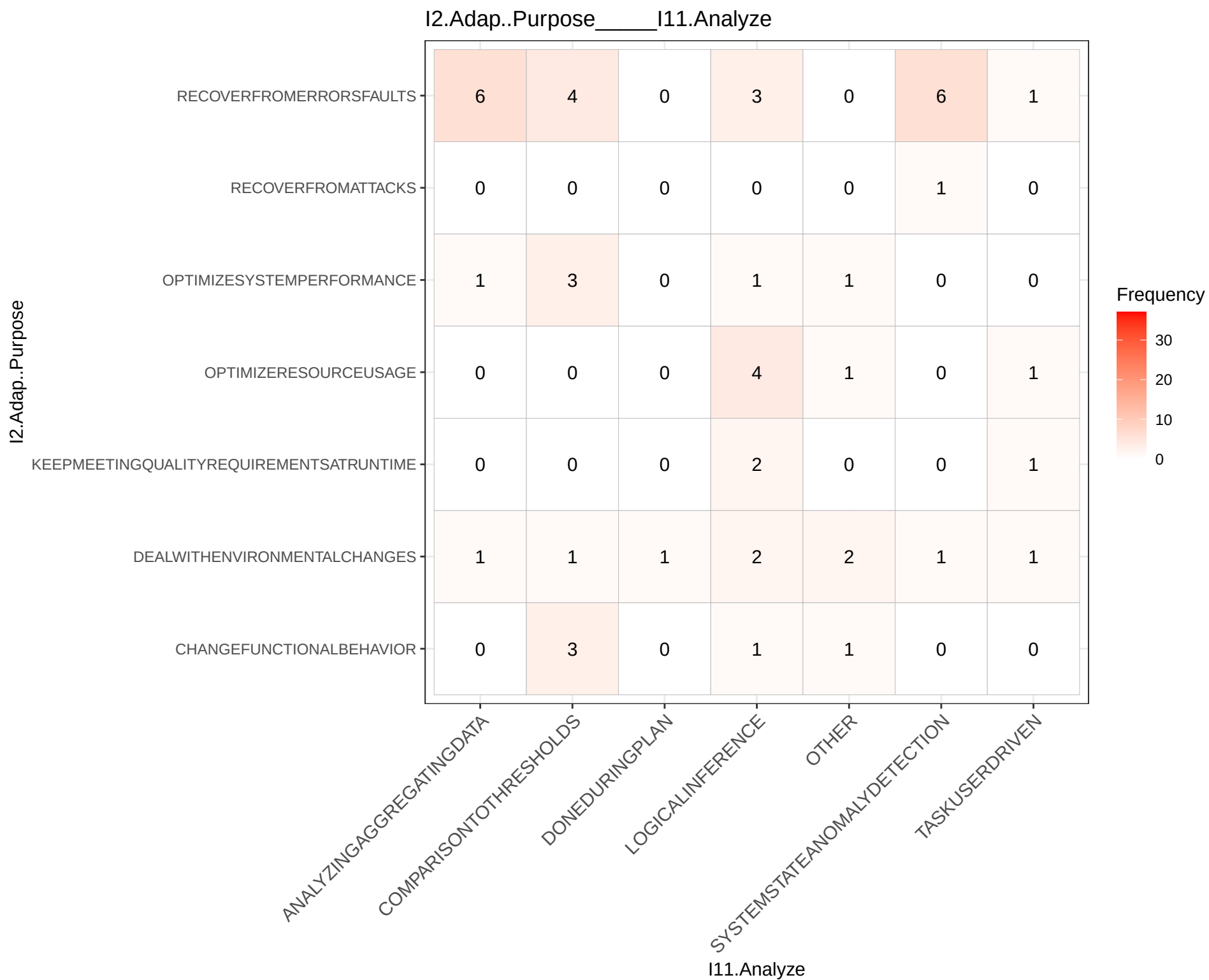


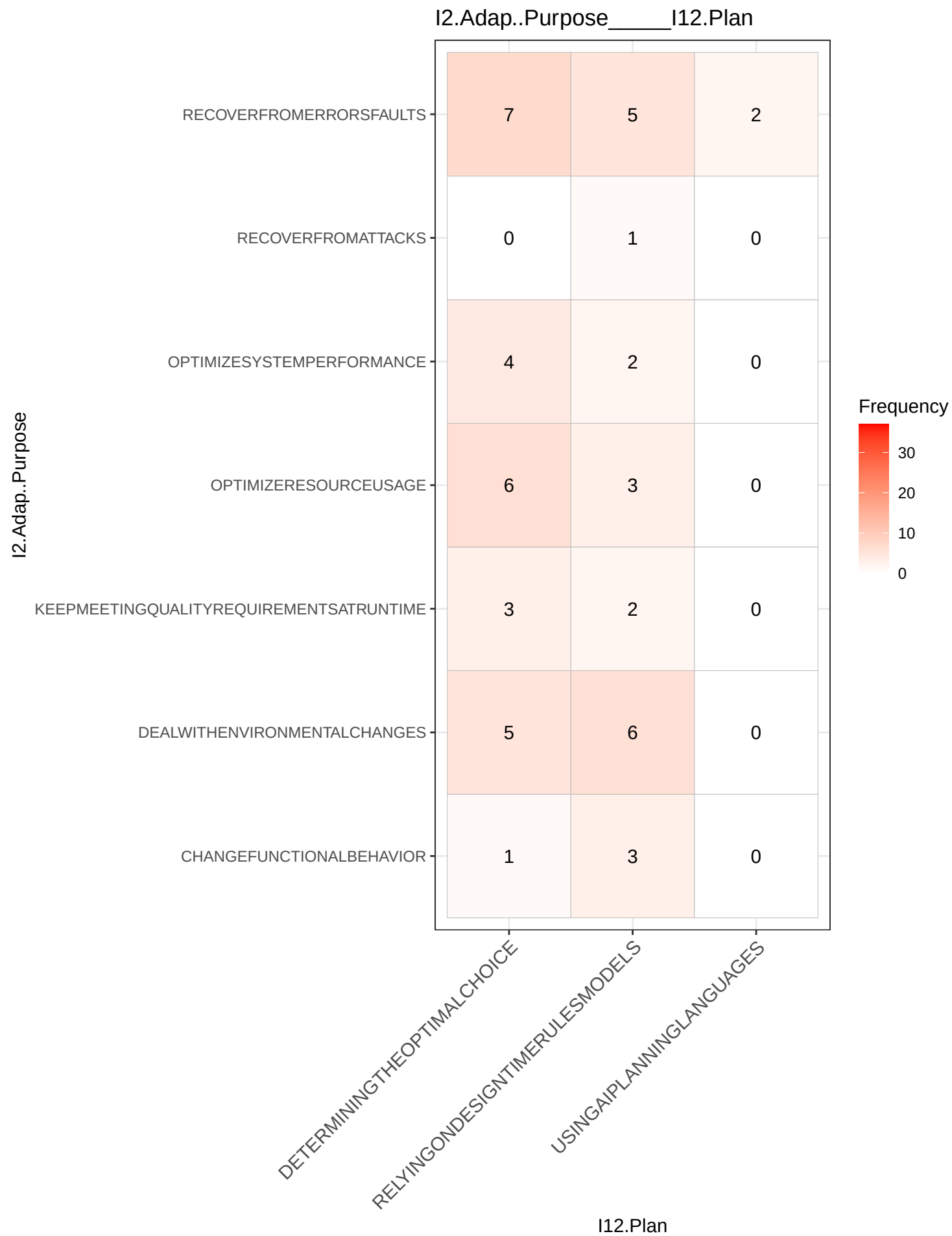


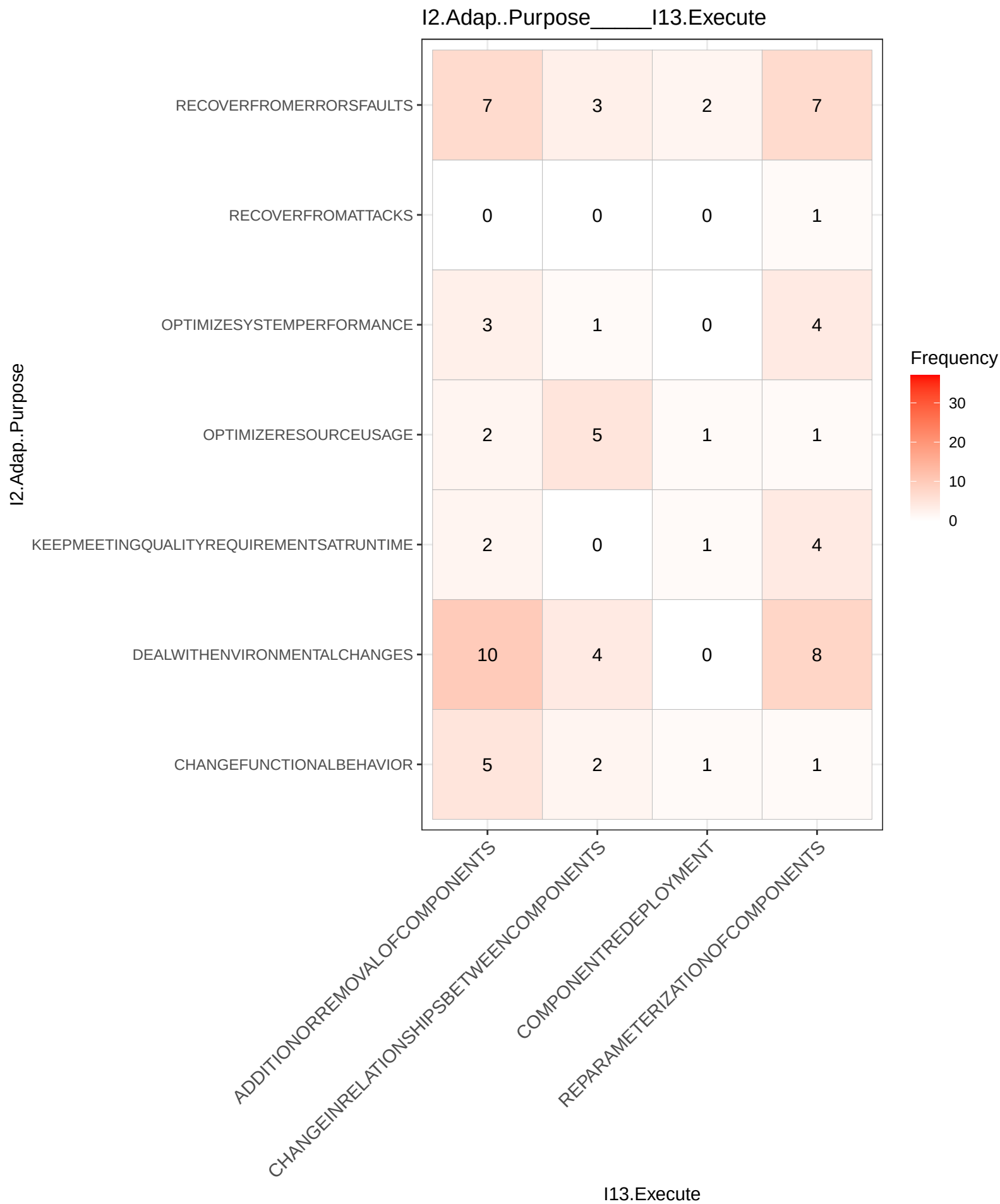


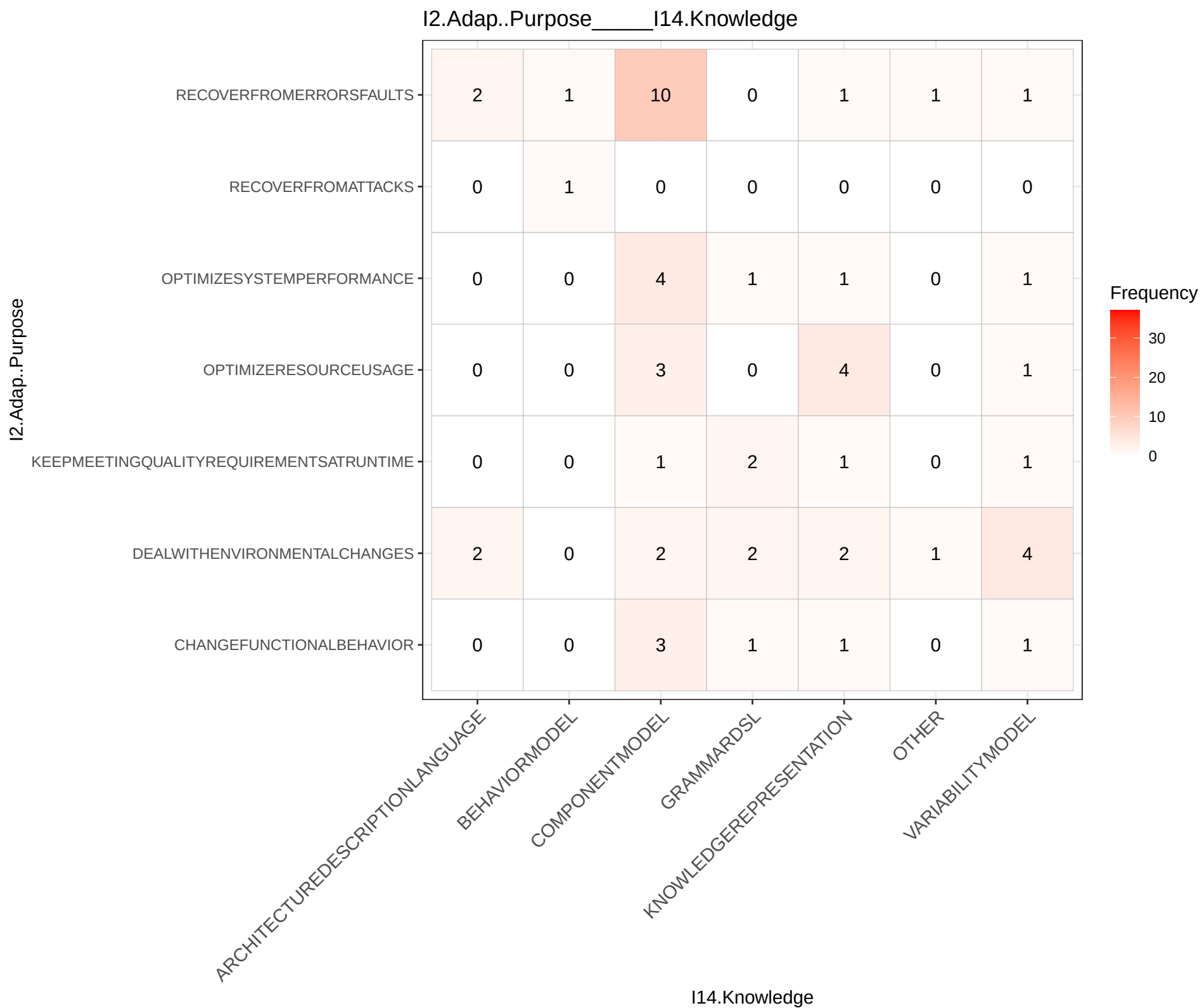
I2.Adap..Purpose









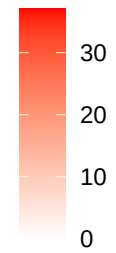


I3.Robot.Type

I3.Robot.Type_____I4.Robo.SW

YOUBOT	0	2	0
WHICHISANINDUSTRIALAGV	0	1	0
WAREHOUSEDELIVERYROBOT	0	1	0
TWOINDUSTRIALROBOTS	1	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	0	1	0
TURTLEBOT2KOBUKI	0	1	0
TURTLEBOT	0	4	0
TRIGLIDEINDUSTRIALASSEMBLY	0	0	0
TEDUSARTERRESTRIALSEARCH	0	1	0
SINGLESERVOINGROTATIONROBOT	1	0	0
RESCUE	0	1	0
QUADROCOPTER	0	0	0
PIONEER3DX	0	1	0
NAOROBOT	0	2	0
MULTIPLEHEXROTOR	1	0	0
MSUEVORALLYMOBILETERRESTRIAL	0	1	0
MOBILESERVICEROBOT	0	0	0
MOBILEROBOTTIAGO	0	0	1
MOBILEROBOTTERRESTRIAL	0	1	0
MOBILEROBOTS	2	0	0
LAPTOPACTINGASAROBOTMOBILEROBOT	1	0	0
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	0	0	0
IROBOTCREATE2	1	0	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	0	0	0
HEXMANIPULATOR	2	0	0
HEXAII	0	0	0
HETEROGENOUSROBOTS	0	1	0
FRACTIONATEDSPACECRAFT	1	0	0
FIELDMOBILEROBOTS	1	0	0
CRAWLERTERMINATORBOT	1	0	0
BOXERCLEARPATH	0	1	0
BLUEROV2	0	0	1
BEBOTSPADERBORN	0	0	0

Frequency



OTHER
ROS1
ROS2

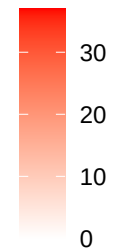
I4.Robo.SW

I3.Robot.Type

I3.Robot.TypeI5.QA

YOUBOT	0	0	0	2	0
WHICHISANINDUSTRIALAGV	0	1	0	1	0
WAREHOUSEDELIVERYROBOT	0	0	1	0	0
TWOINDUSTRIALROBOTS	1	1	0	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	0	1	1	0	0
TURTLEBOT2KOBUKI	1	0	0	0	0
TURTLEBOT	1	3	2	2	0
TRIGLIDEINDUSTRIALASSEMBLY	0	1	1	0	0
TEDUSARTERRESTRIALSEARCH	0	0	1	0	0
SINGLESERVOINGROTATIONROBOT	1	1	1	0	0
RESCUE	0	0	1	0	0
QUADROCOPTER	0	1	0	0	0
PIONEER3DX	0	0	1	0	0
NAOROBOT	0	2	1	0	0
MULTIPLEHEXROTOR	1	1	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	1	0	1	1	0
MOBILESERVICEROBOT	0	2	0	1	0
MOBILEROBOTTIAGO	0	1	0	1	0
MOBILEROBOTTERRESTRIAL	0	1	0	1	0
MOBILEROBOTS	1	2	1	0	0
LAPTOPACTINGASAROBOTMOBILEROBOT	0	1	1	0	0
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	1	0	0	0	0
IROBOTCREATE2	1	0	0	1	1
INFOTAINMENTROBOTMOBILESERVICEROBOT	0	1	0	0	0
HEXMANIPULATOR	2	2	2	0	0
HEXAII	0	1	1	0	0
HETEROGENOUSROBOTS	0	2	2	0	0
FRACTIONATEDSPACECRAFT	0	1	1	0	0
FIELDMOBILEROBOTS	1	1	1	0	0
CRAWLERTERMINATORBOT	1	1	0	0	0
BOXERCLEARPATH	0	1	0	1	0
BLUEROV2	0	1	1	0	0
BEBOTSPADERBORN	0	0	0	1	0

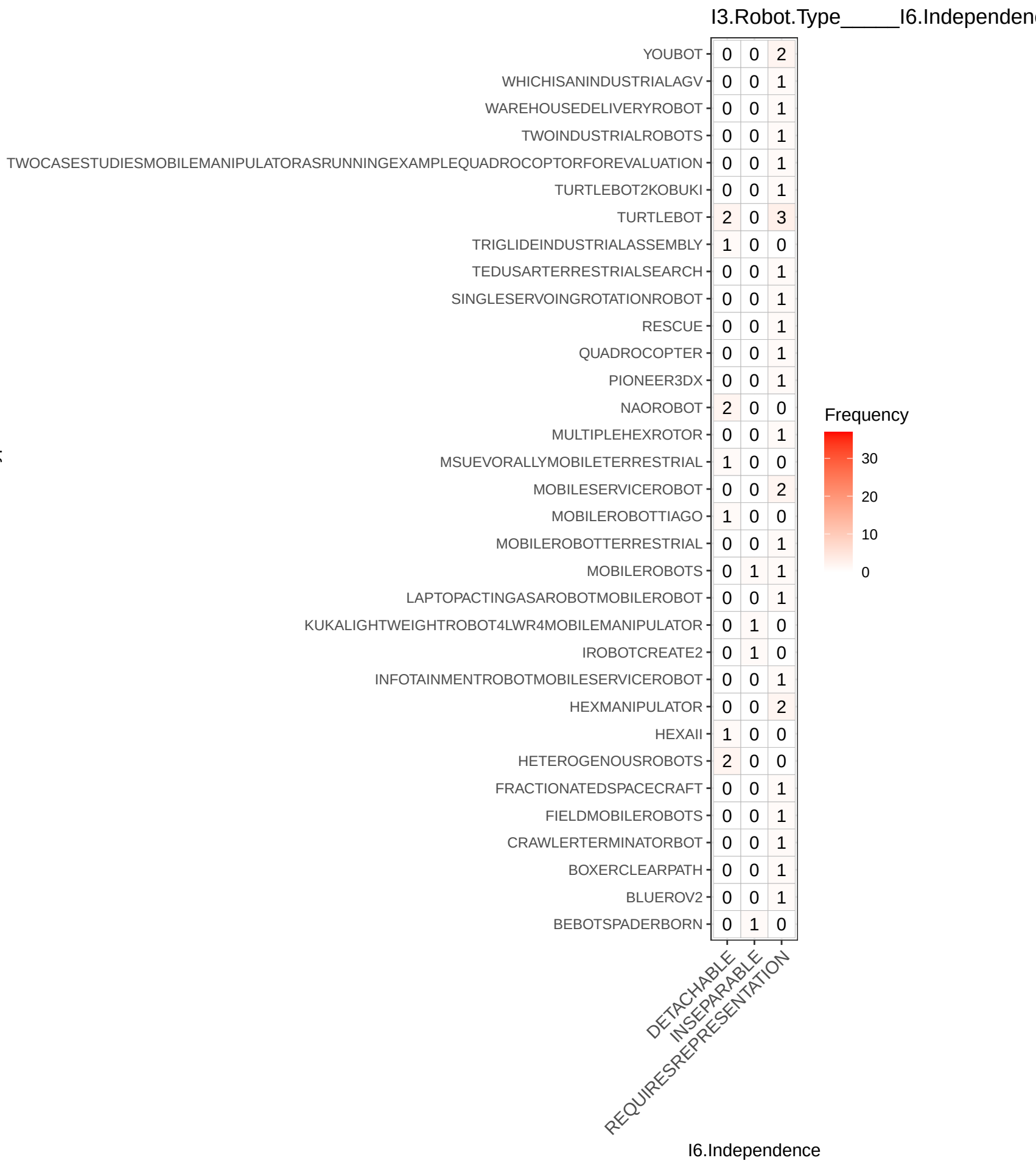
Frequency

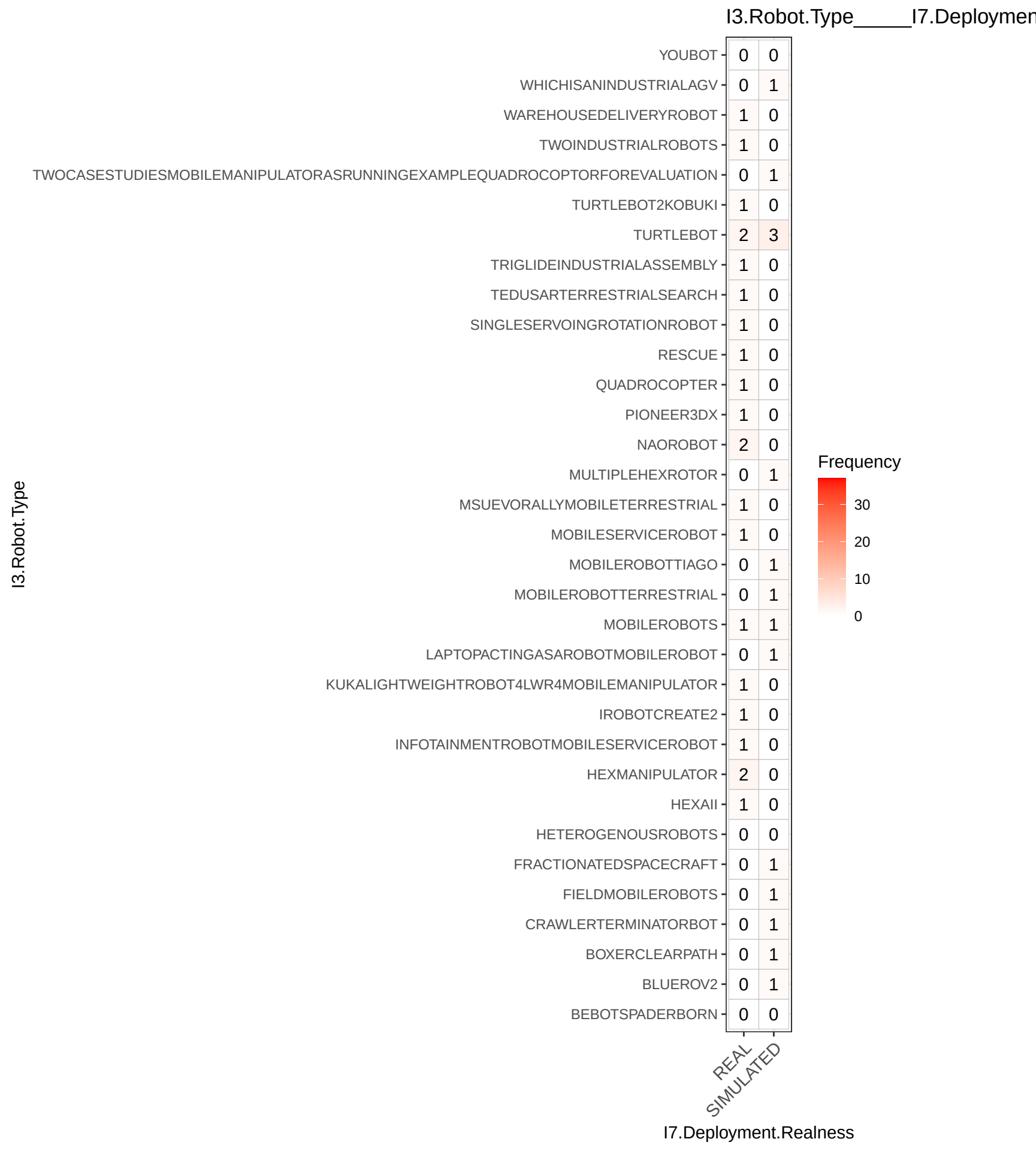


FUNCTIONALSUITABILITY
PERFORMANCEEFFICIENCY
RELIABILITY
SAFETY
SECURITY

I5.QA

I3.Robot.Type





I3.Robot.Type

TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION

KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR

INFOTAINMENTROBOTMOBILESERVICEROBOT

FRACTIONATEDSPACECRAFT

CRAWLERTERMINATORBOT

BOXERCLEARPATH

BLUEROV2

BEBOTSPADERBORN

SINGLESERVOINGROTATIONROBOT

TEDUSARTERRESTRIALSEARCH

TRIGLIDEINDUSTRIALASSEMBLY

TURTLEBOT

TURTLEBOT2KOBUKI

TWOINDUSTRIALROBOTS

WAREHOUSEDELIVERYROBOT

WHICHISANINDUSTRIALAGV

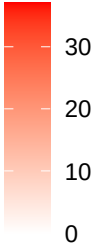
YOUBOT

REAL
SYNTHETIC

I3.Robot.TypeI7.Mission.Re

2	0
1	0
0	1
1	0
0	1
0	1
0	5
1	0
1	0
1	0
1	0
1	0
0	1
0	2
1	0
0	1
2	0
1	0
1	0
2	0
0	1
0	1
1	0
1	0
1	1
1	0
2	0
1	0
1	0
1	0
1	0
1	0
0	1

Frequency



I7.Mission.Realness

I3.Robot.Type

TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION

KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR

INFOTAINMENTROBOTMOBILESERVICEROBOT

FRACTIONATEDSPACECRAFT

CRAWLERTERMINATORBOT

BOXERCLEARPATH

BLUEROV2

BEBOTSPADERBORN

TRIGLIDEINDUSTRIALASSEMBLY

TEDUSARTERRESTRIALSEARCH

SINGLESERVOINGROTATIONROBOT

RESCUE

QUADROCOPTER

PIONEER3DX

NAOROBOT

MULTIPLEHEXROTOR

MSUEVORALLYMOBILETERRESTRIAL

MOBILESERVICEROBOT

MOBILEROBOTTIAGO

MOBILEROBOTTERRESTRIAL

MOBILEROBOTS

LAPTOPACTINGASAROBOTMOBILEROBOT

IROBOTCREATE2

HEXMANIPULATOR

HEXAII

HETEROGENOUSROBOTS

FRACTIONATEDSPACECRAFT

FIELDMOBILEROBOTS

CRAWLERTERMINATORBOT

BOXERCLEARPATH

BLUEROV2

BEBOTSPADERBORN

WAREHOUSEDELIVERYROBOT

WHICHISANINDUSTRIALAGV

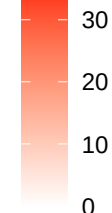
YOUBOT

TURTLEBOT2KOBUKI

TURTLEBOT

I3.Robot.TypeExperiment.Me

Frequency



EXPERIMENT
NOEVALUATION
SHOWCASE

Experiment.Method

0	2	0
1	0	0
0	0	1
1	0	0
0	0	1
0	0	1
4	0	1
0	0	1
1	0	0
1	0	0
1	0	0
1	0	0
0	0	1
1	1	0
0	1	0
1	0	0
1	1	0
0	1	0
0	0	1
2	0	0
1	0	0
1	0	0
1	0	0
1	1	0
0	0	1
1	1	0
1	0	0
1	0	0
1	0	0
0	1	0
1	0	0
1	0	0
0	1	0
1	0	0
0	1	0

I3.Robot.Type

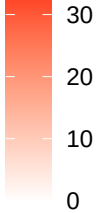
I3.Robot.TypeI8.Evaluation

TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCTORFOREVALUATION

KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR

DOMAINSPECIFICPERFORMANCE
MISSIONPERFORMANCE
OVERHEADINTRODUCED
RESOURCECONSUMPTION

Frequency



I8.Evaluation

YOUBOT	0	0	0	0	0
WHICHISANINDUSTRIALAGV	0	1	0	1	0
WAREHOUSEDELIVERYROBOT	0	1	0	0	0
TWOINDUSTRIALROBOTS	0	0	1	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCTORFOREVALUATION	0	1	0	0	0
TURTLEBOT2KOBUKI	0	0	1	0	0
TURTLEBOT	1	4	3	2	0
TRIGLIDEINDUSTRIALASSEMBLY	1	1	0	0	0
TEDUSARTERRESTRIALSEARCH	0	1	0	0	0
SINGLESERVOINGROTATIONROBOT	0	1	0	0	1
RESCUE	0	1	0	0	0
QUADROCOPTER	0	0	1	0	0
PIONEER3DX	0	1	0	0	0
NAOROBOT	0	0	1	0	0
MULTIPLEHEXROTOR	0	0	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	0	1	0	0	0
MOBILESERVICEROBOT	0	0	0	0	1
MOBILEROBOTTIAGO	0	0	0	0	0
MOBILEROBOTTERRESTRIAL	0	1	0	0	0
MOBILEROBOTS	1	1	0	1	0
LAPTOPACTINGASAROBOTMOBILEROBOT	0	0	1	1	0
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	1	0	0	0	0
IROBOTCREATE2	1	1	0	1	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	0	0	0	0	1
HEXMANIPULATOR	0	1	0	0	1
HEXAII	1	1	0	0	0
HETEROGENOUSROBOTS	0	0	1	0	0
FRACTIONATEDSPACECRAFT	0	0	1	0	0
FIELDMOBILEROBOTS	0	1	0	0	0
CRAWLERTERMINATORBOT	0	0	0	0	0
BOXERCLEARPATH	0	1	0	1	0
BLUEROV2	0	1	0	0	0
BEBOTSPADERBORN	0	0	0	0	0

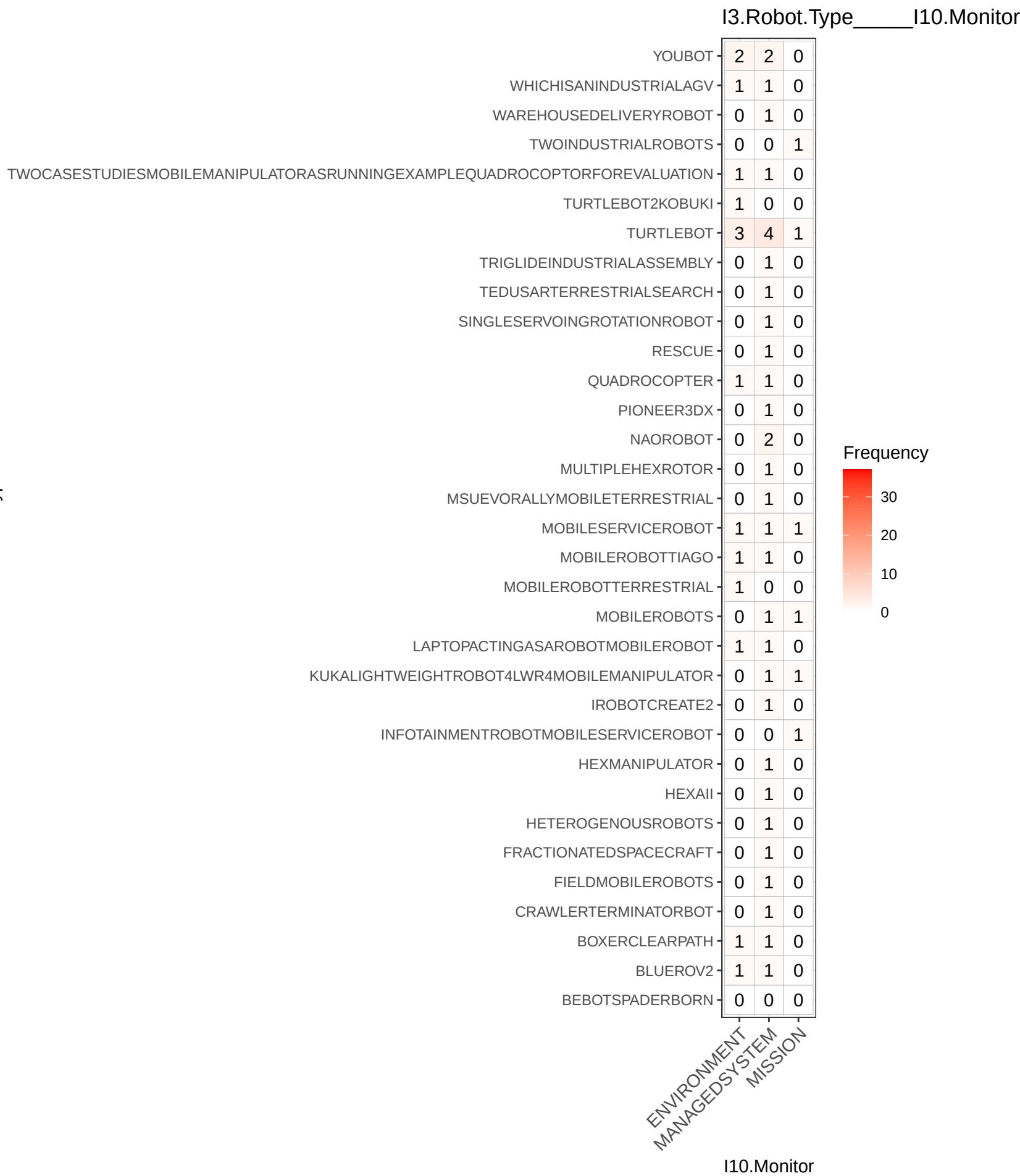
I3.Robot.Type

TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION

KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR



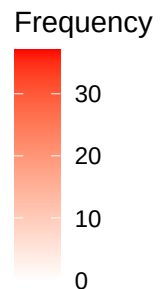
I3.Robot.Type



I3.Robot.Type

I3.Robot.TypeI11.Analyze

YOUBOT	0	0	0	0	0	0	0
WHICHISANINDUSTRIALAGV	0	0	0	1	0	0	0
WAREHOUSEDELIVERYROBOT	1	0	0	0	0	1	0
TWOINDUSTRIALROBOTS	0	0	0	0	0	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	0	1	0	0	0	0	0
TURTLEBOT2KOBUKI	0	0	0	0	0	0	0
TURTLEBOT	1	1	0	1	0	1	0
TRIGLIDEINDUSTRIALASSEMBLY	0	0	0	0	0	0	0
TEDUSARTERRESTRIALSEARCH	1	0	0	0	0	1	0
SINGLESERVOINGROTATIONROBOT	1	1	0	0	0	0	0
RESCUE	1	0	0	0	0	1	0
QUADROCOPTER	0	0	0	0	1	0	0
PIONEER3DX	1	0	0	0	0	1	0
NAOROBOT	0	0	0	1	0	0	1
MULTIPLEHEXROTOR	0	1	0	0	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	0	0	1	0	0	0	0
MOBILESERVICEROBOT	0	0	0	1	0	0	1
MOBILEROBOTTIAGO	0	0	0	1	0	0	0
MOBILEROBOTTERRESTRIAL	0	0	0	0	0	0	0
MOBILEROBOTS	0	1	0	0	0	0	0
LAPTOPACTINGASAROBOTMOBILEROBOT	1	1	0	0	0	0	0
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	0	0	0	0	1	0	0
IROBOTCREATE2	0	0	0	0	0	1	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	0	0	0	1	0	0	0
HEXMANIPULATOR	1	1	0	0	0	0	0
HEXAII	0	0	0	0	0	0	0
HETEROGENOUSROBOTS	0	0	0	2	0	0	0
FRACTIONATEDSPACECRAFT	0	0	0	0	0	1	0
FIELDMOBILEROBOTS	0	1	0	0	0	0	0
CRAWLERTERMINATORBOT	0	1	0	0	0	0	0
BOXERCLEARPATH	0	0	0	1	0	0	0
BLUEROV2	0	0	0	1	0	0	0
BEBOTSPADERBORN	0	0	0	0	0	0	0



ANALYZINGAGGREGATINGDATA
COMPARISONTOHRESHOLDS
DONEDURINGPLAN
LOGICALINFERENCE
SYSTEMSTATEANOMALYDETECTION
TASKUSERDRIVEN

I11.Analyze

I3.Robot.Type

TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION

KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR

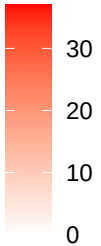
DETERMININGTHEOPTIMALCHOICE
RELYINGONDESIGNTIMERULESMODELS
USINGAIPLANNINGLANGUAGES

I3.Robot.TypeI12.Plan

I12.Plan

YUBOT	0	1	0
WHICHISANINDUSTRIALAGV	1	0	0
WAREHOUSEDELIVERYROBOT	0	1	0
TWOINDUSTRIALROBOTS	0	0	0
TURTLEBOT2KOBUKI	0	1	0
TURTLEBOT	2	3	0
TRIGLIDEINDUSTRIALASSEMBLY	1	0	0
TEDUSARTERRESTRIALSEARCH	0	0	1
SINGLESERVOINGROTATIONROBOT	1	0	0
RESCUE	0	0	1
QUADROCOPTER	0	1	0
PIONEER3DX	0	0	1
NAOROBOT	1	0	0
MULTIPLEHEXROTOR	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	1	0	0
MOBILESERVICEROBOT	2	0	0
MOBILEROBOTTIAGO	0	1	0
MOBILEROBOTTERRESTRIAL	0	1	0
MOBILEROBOTS	2	0	0
LAPTOPACTINGASAROBOTMOBILEROBOT	1	0	0
IROBOTCREATE2	0	1	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	1	0	0
HEXMANIPULATOR	1	0	0
HEXAII	1	0	0
HETEROGENOUSROBOTS	2	2	0
FRACTIONATEDSPACECRAFT	0	1	0
FIELDMOBILEROBOTS	1	0	0
CRAWLERTERMINATORBOT	0	0	0
BOXERCLEARPATH	1	0	0
BLUEROV2	1	0	0
BEBOTSPADERBORN	0	1	0

Frequency

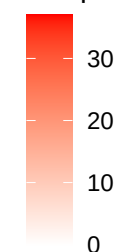


I3.Robot.Type

I3.Robot.Type_____I3.Execute

YOUBOT	2	2	0	2
WHICHISANINDUSTRIALAGV	0	0	0	1
WAREHOUSEDELIVERYROBOT	0	0	0	1
TWOINDUSTRIALROBOTS	1	0	1	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	1	0	0	1
TURTLEBOT2KOBUKI	1	0	0	0
TURTLEBOT	2	1	0	3
TRIGLIDEINDUSTRIALASSEMBLY	0	0	1	0
TEDUSARTERRESTRIALSEARCH	1	0	0	0
SINGLESERVOINGROTATIONROBOT	1	0	0	1
RESCUE	1	0	0	0
QUADROCOPTER	1	0	0	0
PIONEER3DX	1	0	0	0
NAOROBOT	0	2	0	0
MULTIPLEHEXROTOR	1	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	0	0	0	1
MOBILESERVICEROBOT	2	1	0	1
MOBILEROBOTTIAGO	0	0	0	1
MOBILEROBOTTERRESTRIAL	1	0	0	1
MOBILEROBOTS	1	0	0	2
LAPTOPACTINGASAROBOTMOBILEROBOT	1	0	1	0
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	1	0	0	0
IROBOTCREATE2	0	0	0	1
INFOTAINMENTROBOTMOBILESERVICEROBOT	1	1	0	0
HEXMANIPULATOR	1	0	0	1
HEXAII	0	0	1	0
HETEROGENOUSROBOTS	0	2	0	0
FRACTIONATEDSPACECRAFT	0	0	1	0
FIELDMOBILEROBOTS	1	0	0	1
CRAWLERTERMINATORBOT	1	0	0	0
BOXERCLEARPATH	0	0	0	1
BLUEROV2	0	0	0	1
BEBOTSPADERBORN	1	1	0	0

Frequency



CHANGEINRELATIONSHIPSBETWEENCOMPONENTS
ADDITIONORREMOVALOFCOMPONENTS
COMPONENTREDEPLOYMENT
REPARAMETERIZATIONOFCOMPONENTS

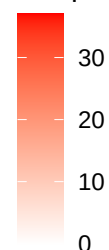
I3.Execute

I3.Robot.Type

I3.Robot.TypeI14.Knowledge

YOUBOT	0	0	0	0	0	0	2
WHICHISANINDUSTRIALAGV	0	0	0	0	1	0	0
WAREHOUSEDELIVERYROBOT	0	0	1	0	0	0	0
TWOINDUSTRIALROBOTS	0	0	1	0	0	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	0	0	0	0	0	0	1
TURTLEBOT2KOBUKI	0	0	0	0	0	0	1
TURTLEBOT	1	0	2	0	0	1	1
TRIGLIDEINDUSTRIALASSEMBLY	0	0	1	0	0	0	0
TEDUSARTERRESTRIALSEARCH	0	0	1	0	0	0	0
SINGLESERVOINGROTATIONROBOT	0	0	1	0	0	0	0
RESCUE	0	0	1	0	0	0	0
QUADROCOPTER	0	0	0	0	1	0	0
PIONEER3DX	0	0	1	0	0	0	0
NAOROBOT	0	0	2	0	0	0	0
MULTIPLEHEXROTOR	0	0	1	0	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	0	0	1	0	0	0	0
MOBILESERVICEROBOT	0	0	0	1	1	0	0
MOBILEROBOTTIAGO	0	0	0	1	0	0	0
MOBILEROBOTTERRESTRIAL	0	0	0	0	0	0	1
MOBILEROBOTS	0	0	1	0	0	0	1
LAPTOPACTINGASAROBOTMOBILEROBOT	1	0	0	0	0	0	0
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	0	0	0	1	0	0	0
IROBOTCREATE2	0	1	0	0	0	0	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	0	0	0	0	1	0	0
HEXMANIPULATOR	0	0	2	0	0	0	0
HEXAII	0	0	1	0	0	0	0
HETEROGENOUSROBOTS	0	0	0	0	2	0	0
FRACTIONATEDSPACECRAFT	0	0	0	0	0	0	1
FIELDMOBILEROBOTS	0	0	1	0	0	0	0
CRAWLERTERMINATORBOT	0	0	1	0	0	0	0
BOXERCLEARPATH	0	0	0	0	1	0	0
BLUEROV2	0	0	0	0	1	0	0
BEBOTSPADERBORN	0	0	1	0	0	0	0

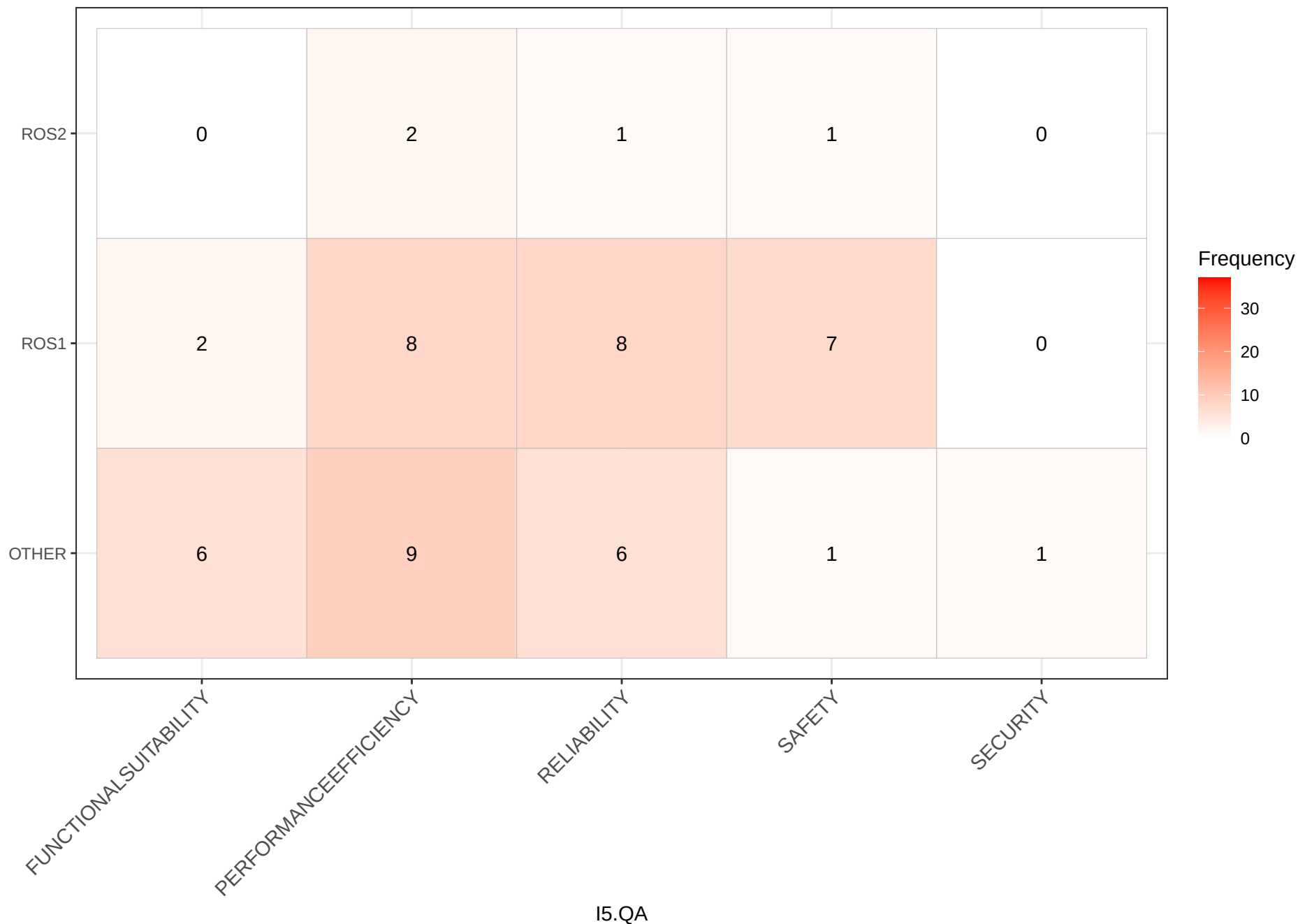
Frequency



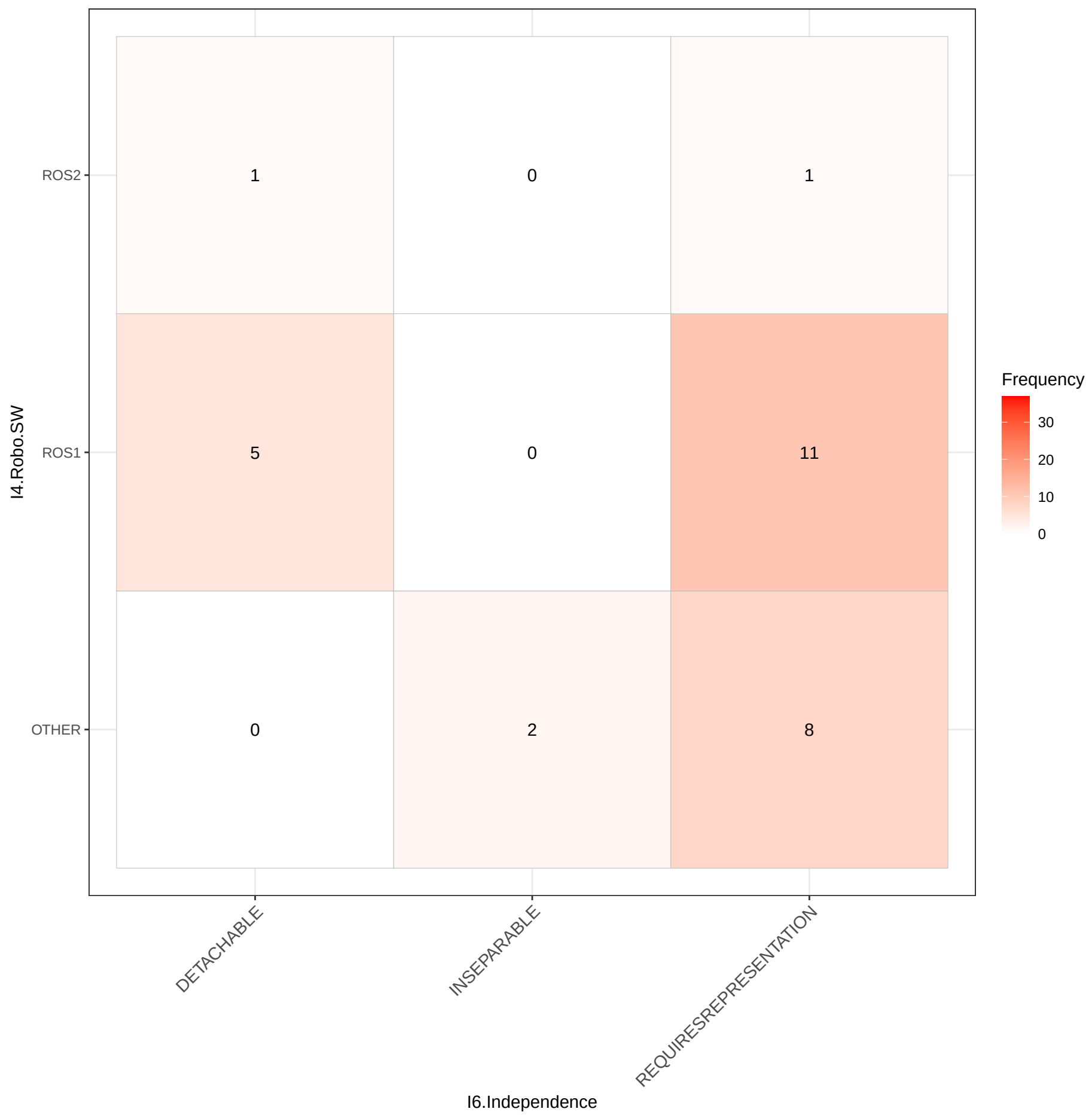
ARCHITECTUREDESCRIPTIONLANGUAGE
BEHAVIORMODEL
COMPONENTMODEL
KNOWLEDGEREPRESENTATION
GRAMMARDSL
OTHER
VARIABILITYMODEL

I14.Knowledge

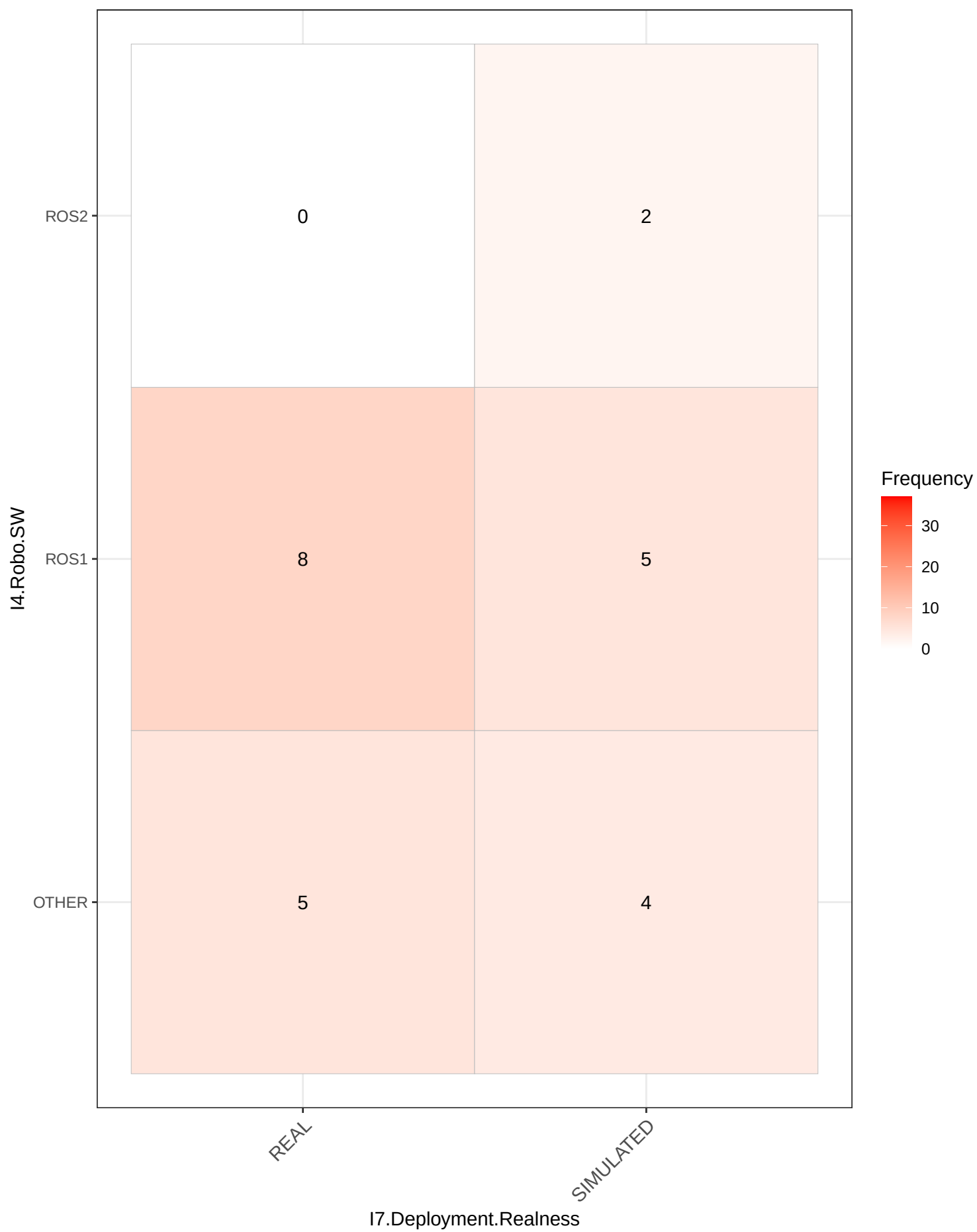
I4.Robo.SW _____ I5.QA



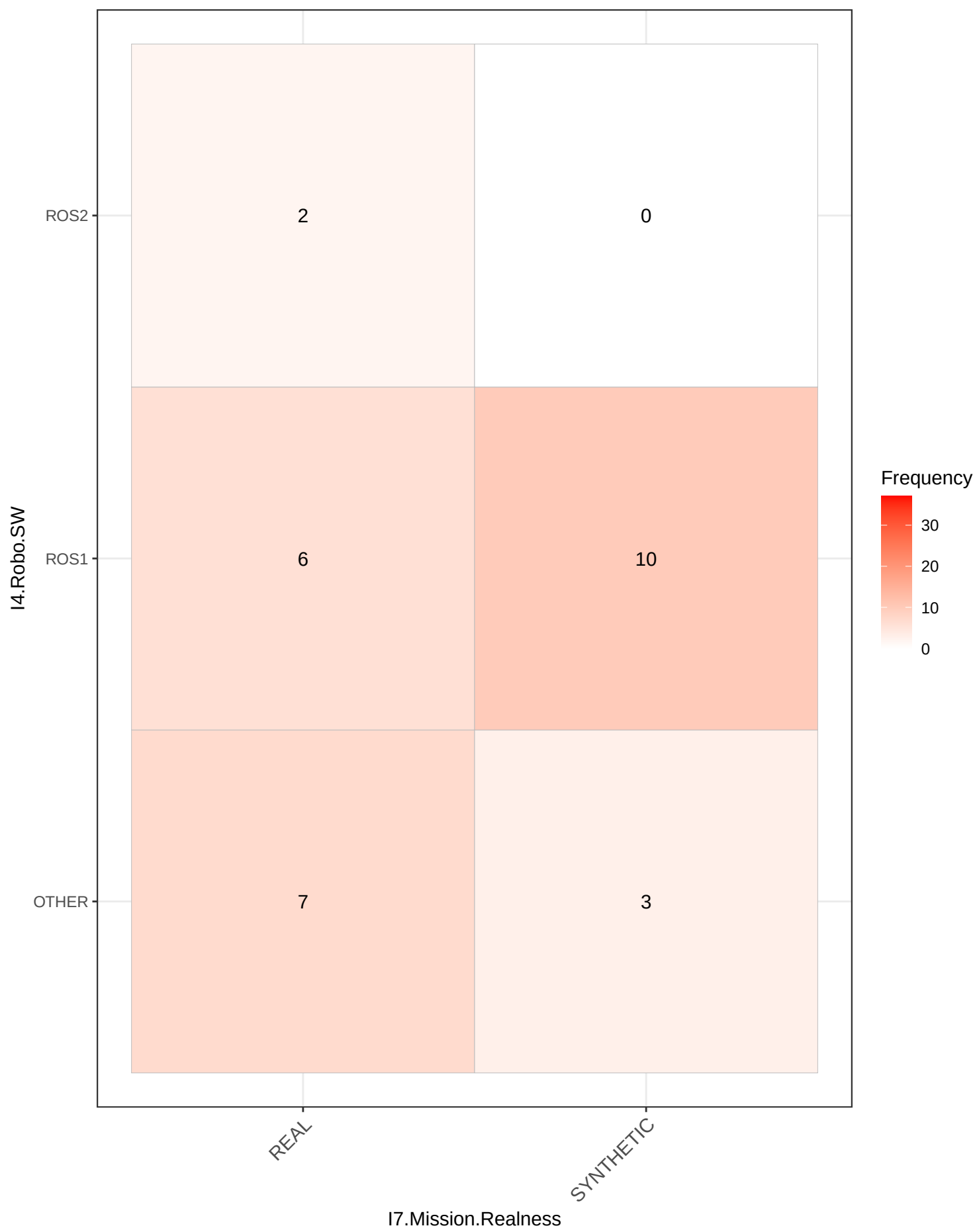
I4.Robo.SW_____I6.Independence



I4.Robo.SW_____I7.Deployment.Realness



I4.Robo.SW _____ I7.Mission.Realness



I4.Robo.SW_____Experiment.Method

I4.Robo.SW

ROS2

ROS1

OTHER

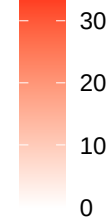
EXPERIMENT

NOEVALUATION

SHOWCASE

Experiment.Method

Frequency



1

1

0

7

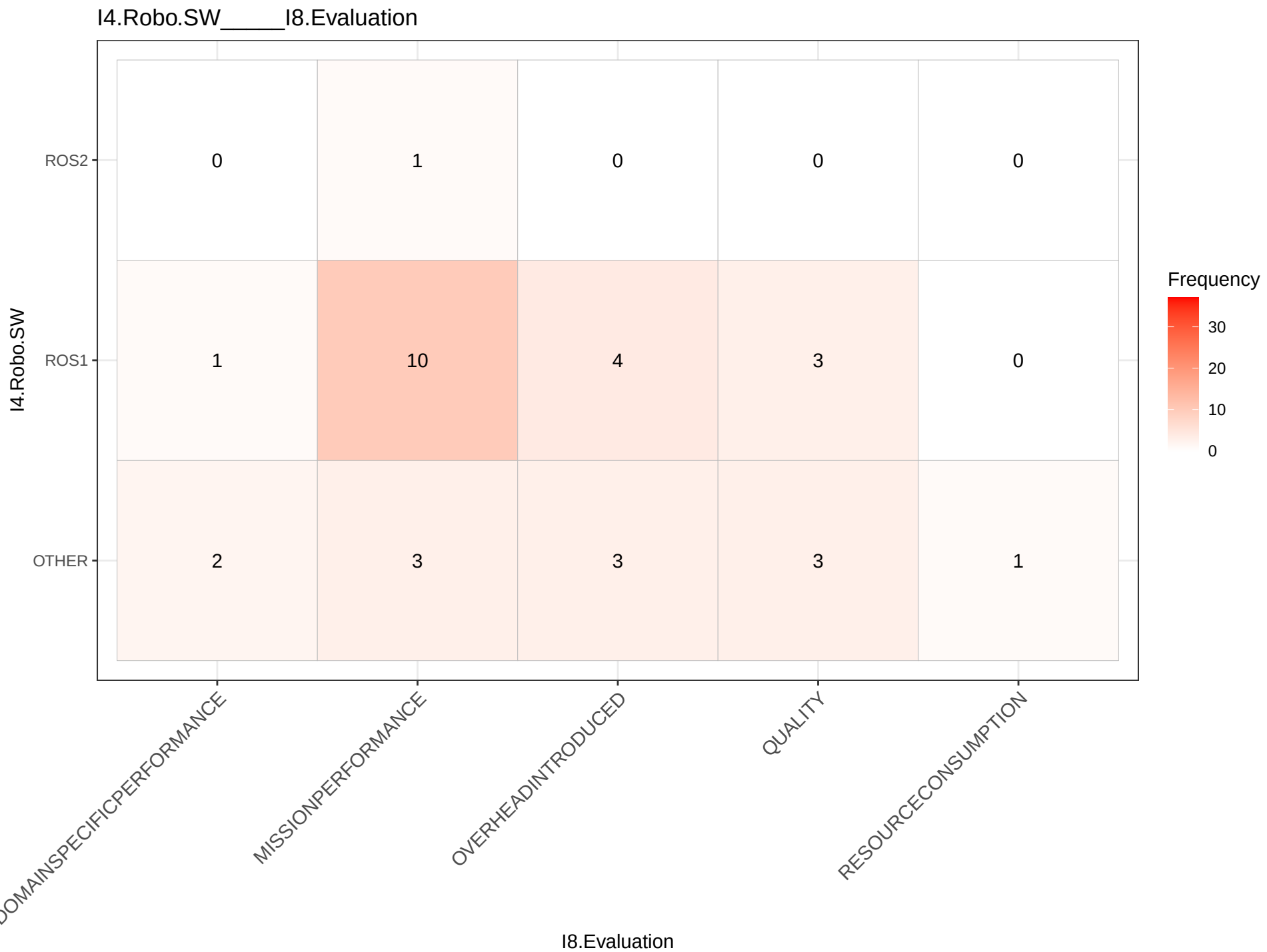
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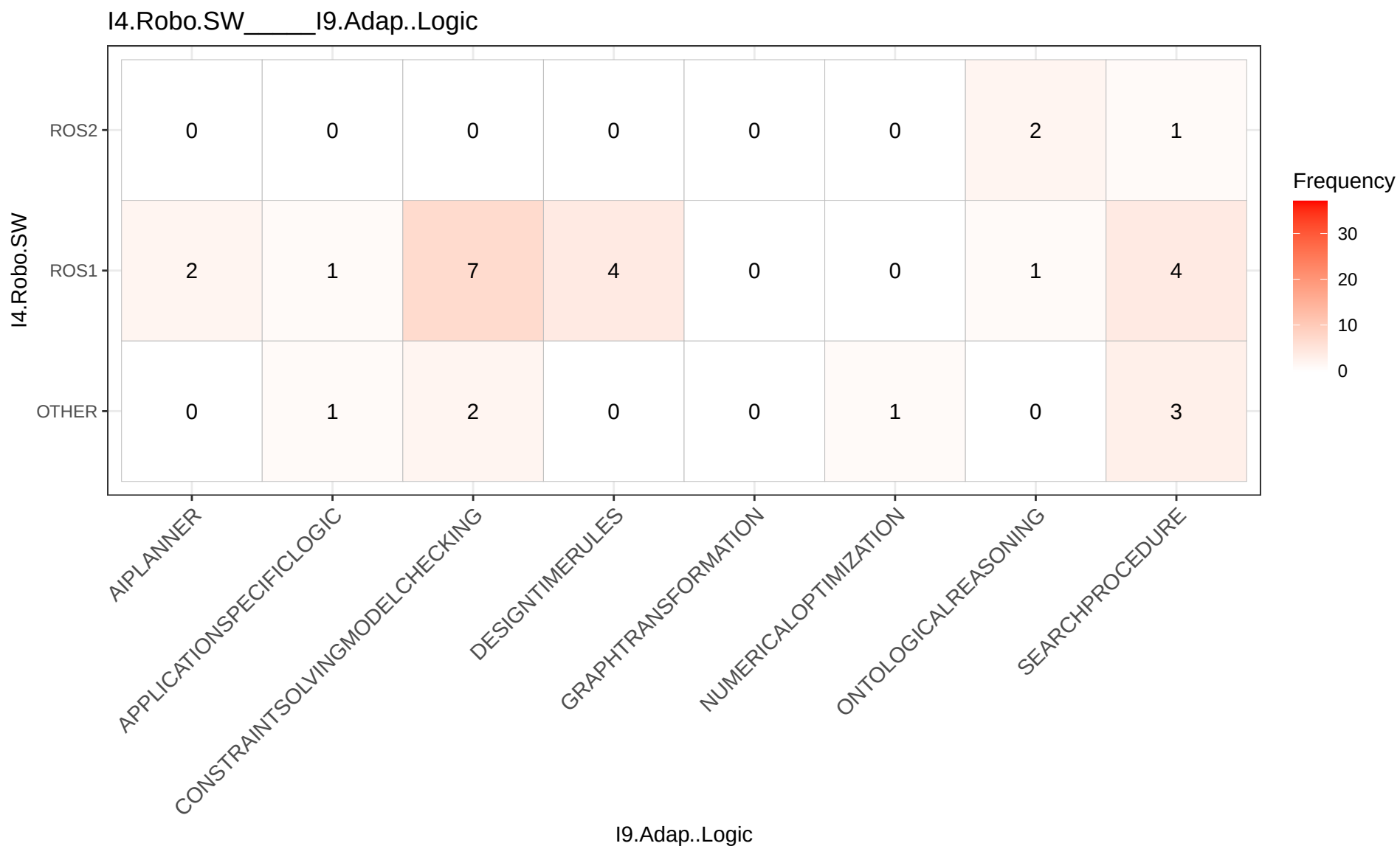
5

7

3

0





I4.Robo.SW_____I10.Monitor

I4.Robo.SW

ROS2

2

2

0

ROS1

7

13

1

OTHER

1

7

2

ENVIRONMENT

MANAGEDSYSTEM

MISSION

I10.Monitor

Frequency



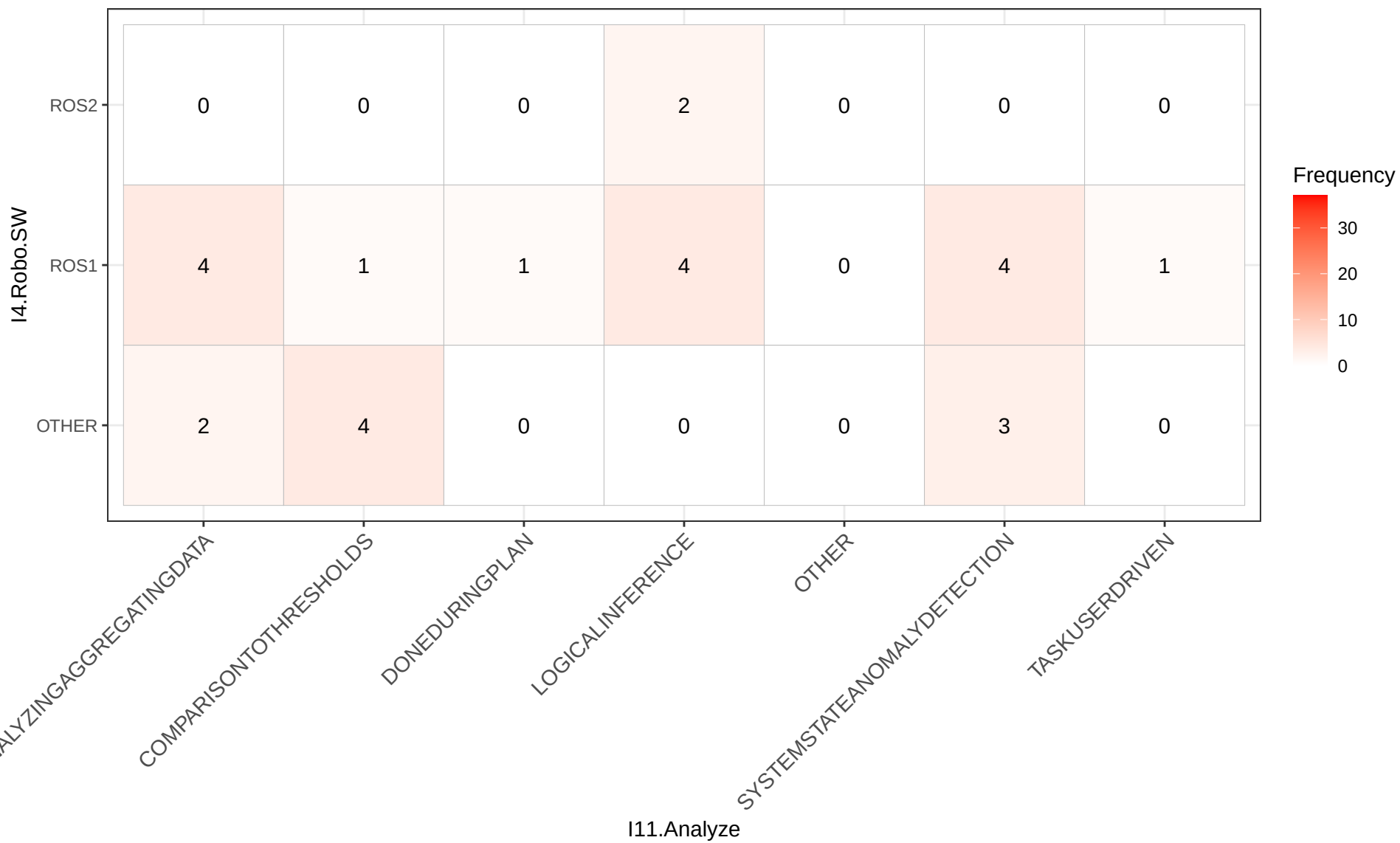
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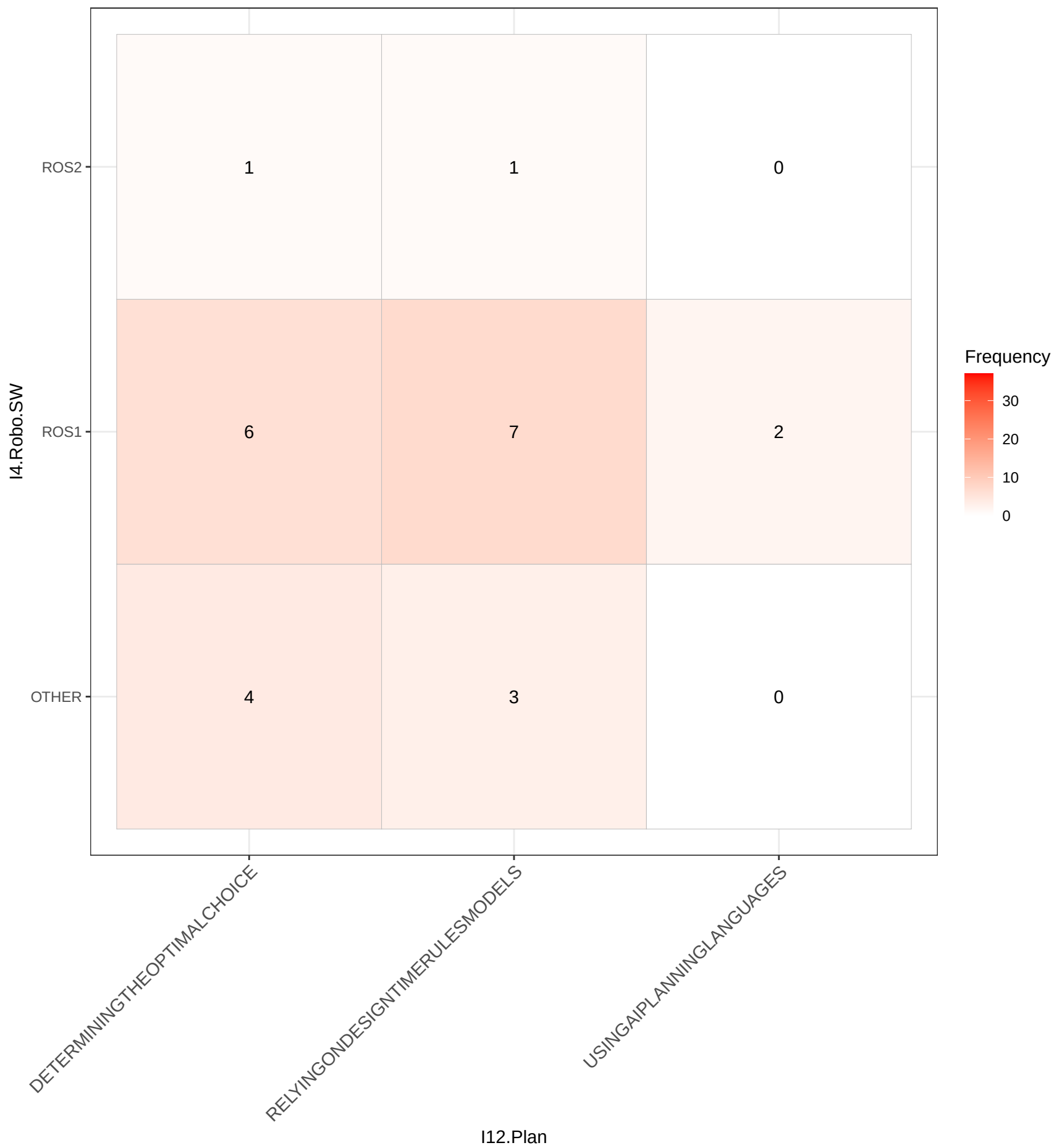
10

0

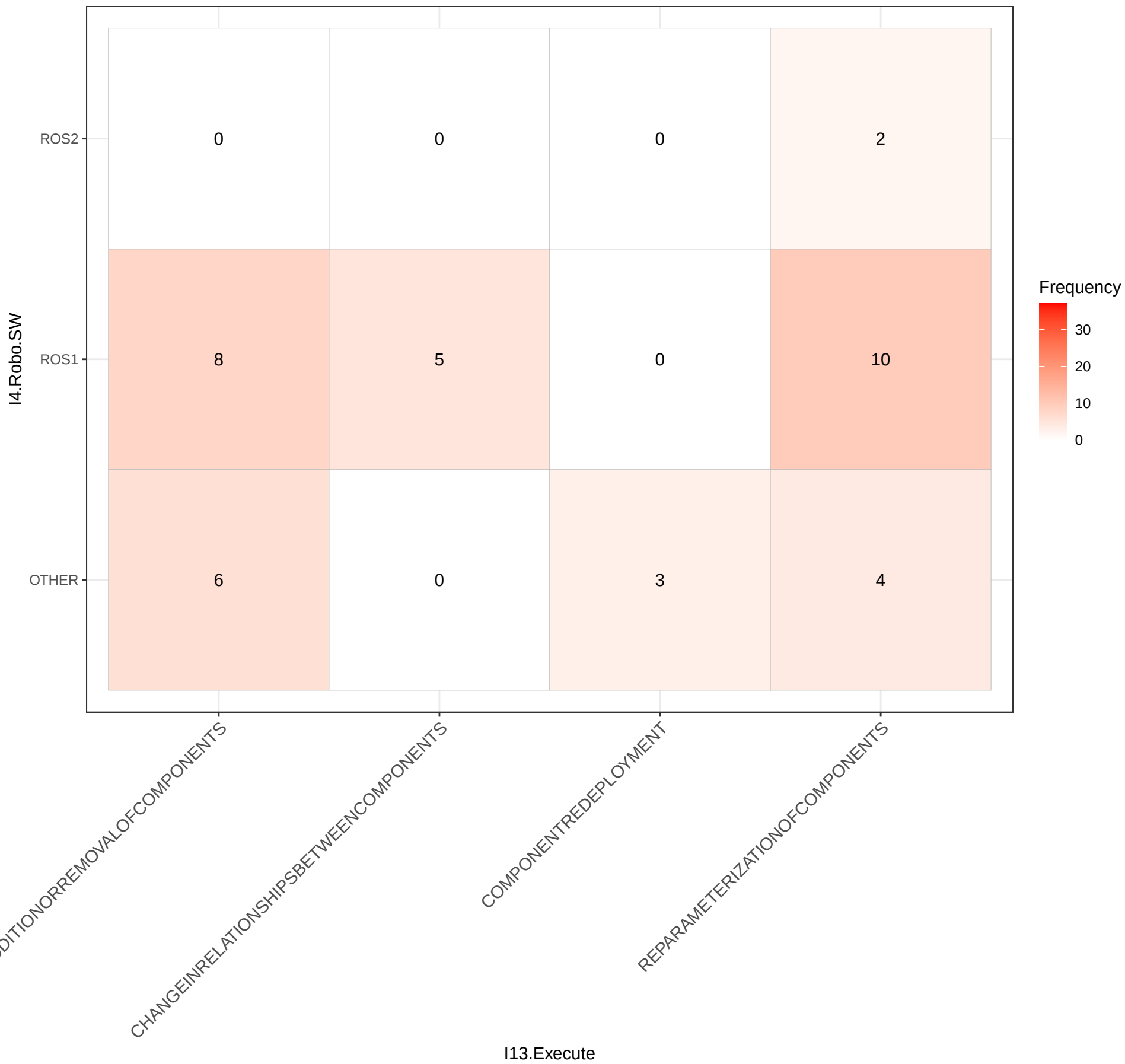
I4.Robo.SW _____ I11.Analyze

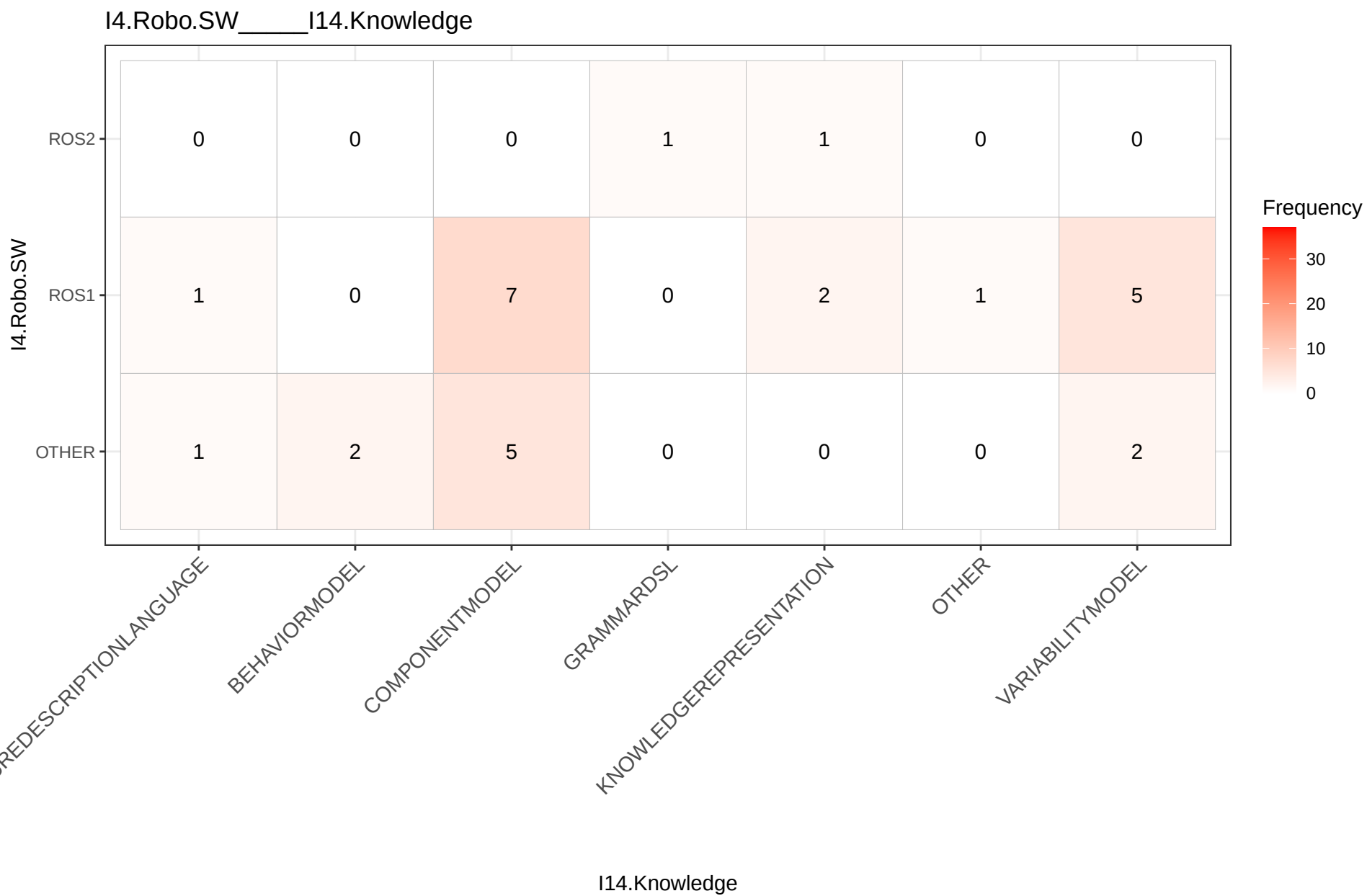


I4.Robo.SW_____I12.Plan

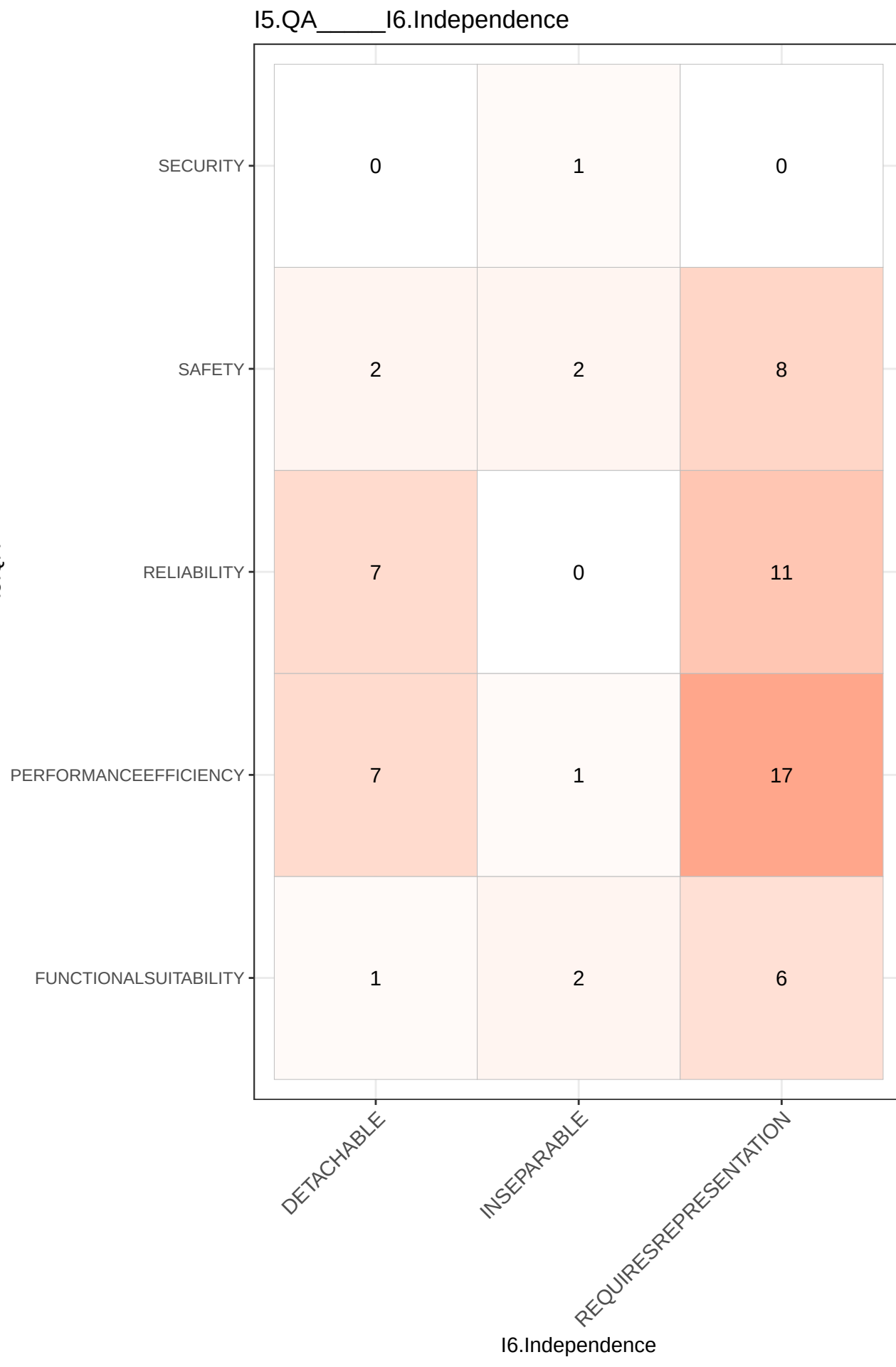


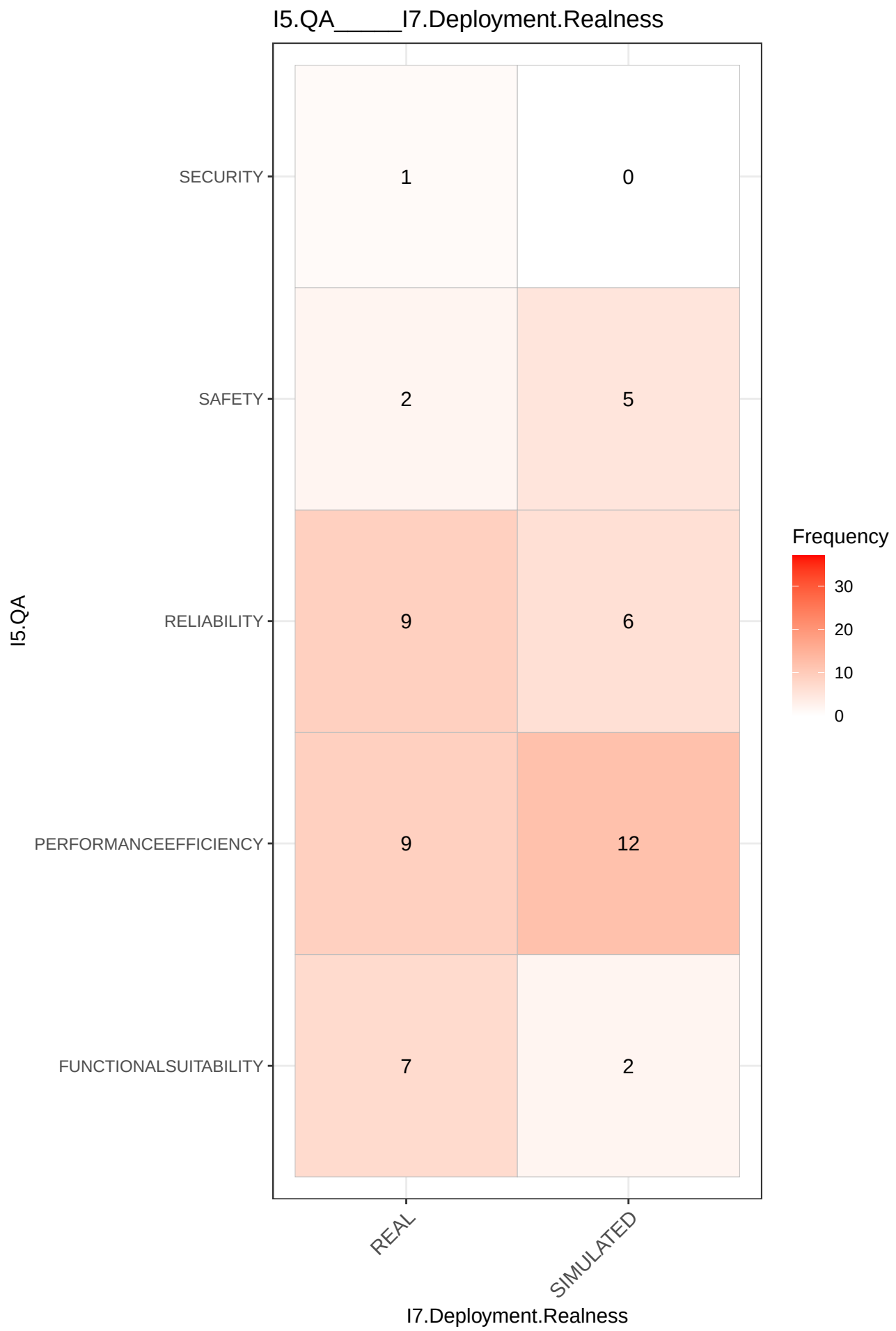
I4.Robo.SW_____I13.Execute

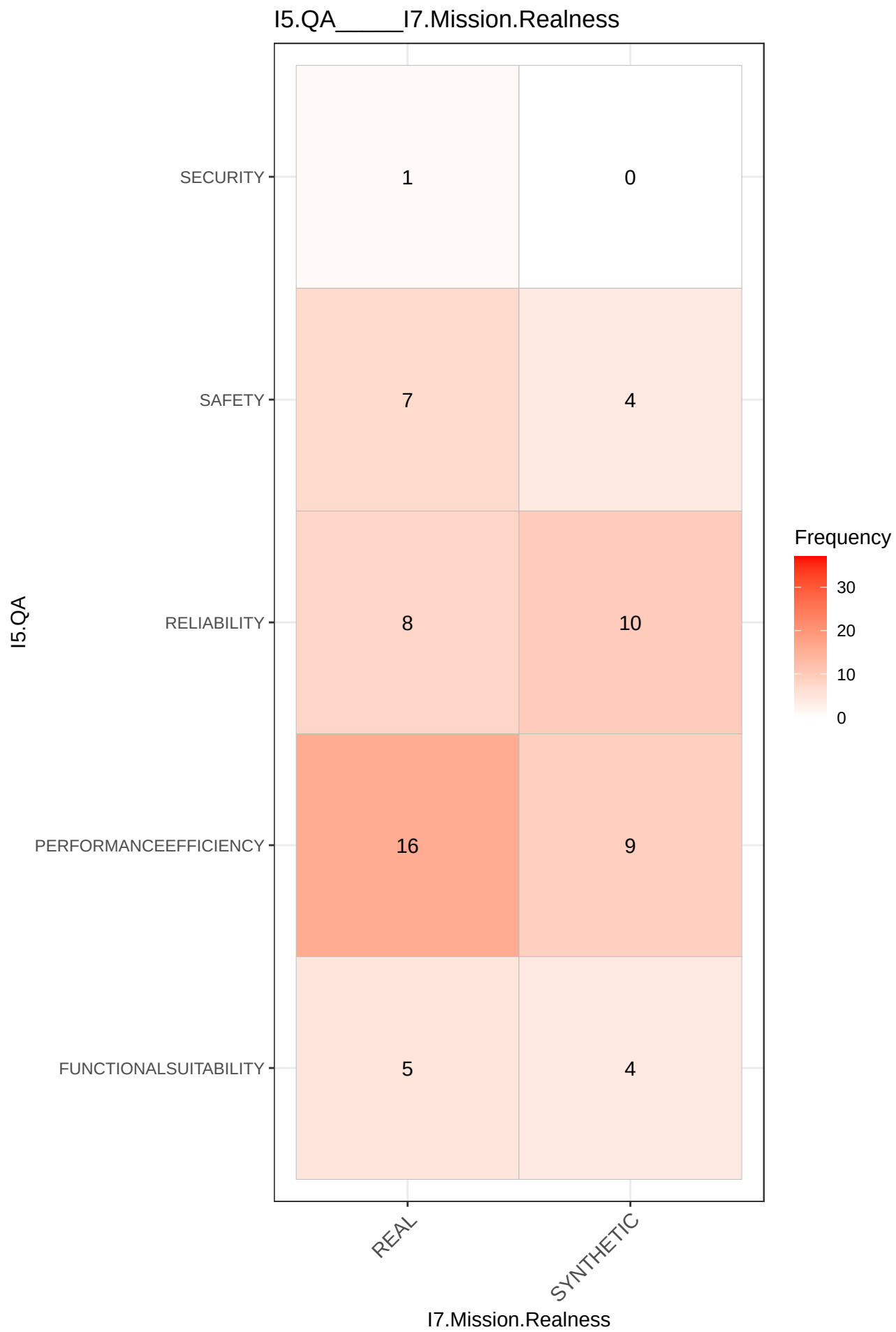


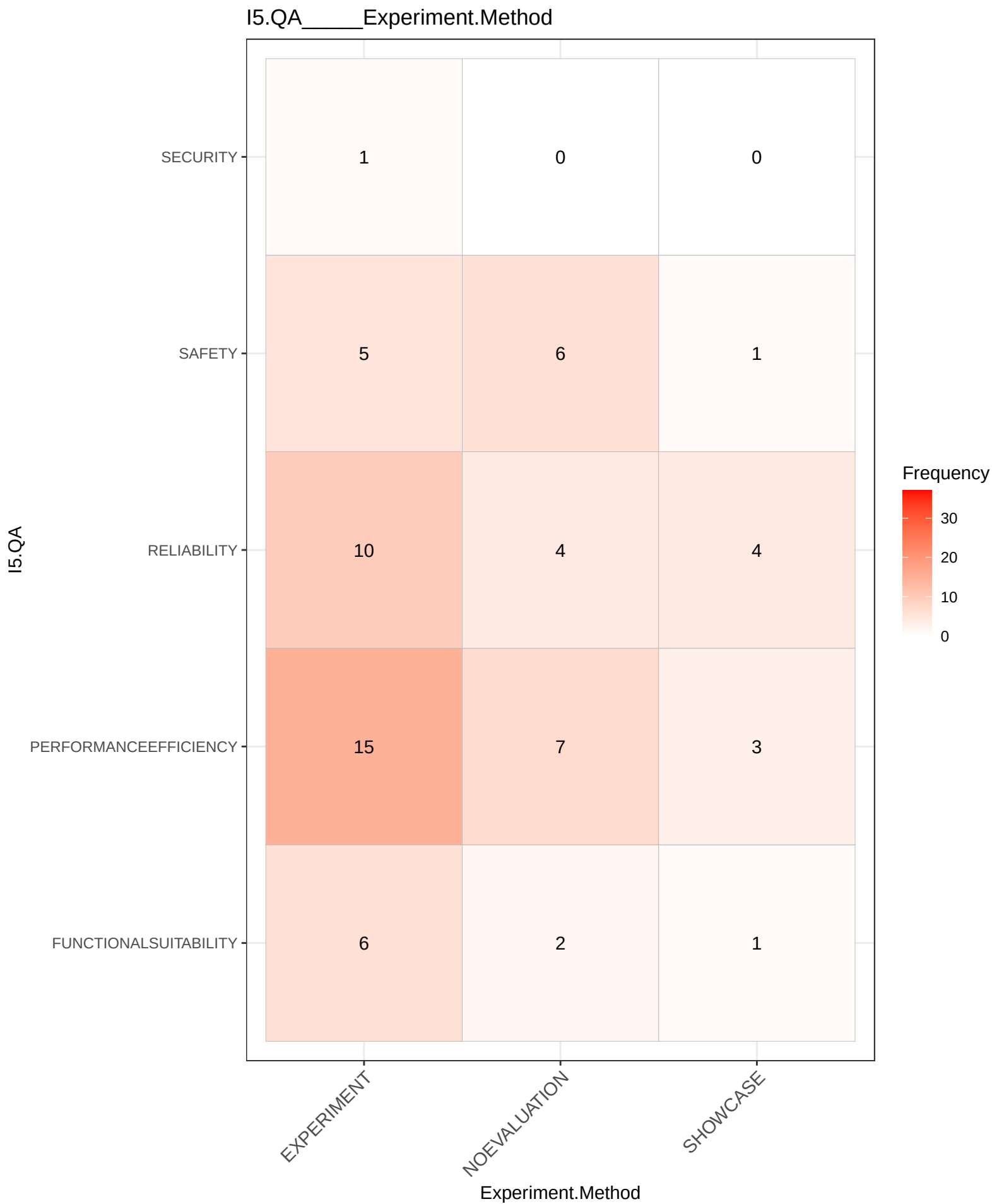


I5.QA









I5.QA

I5.QA_____I8.Evaluation

SECURITY

1

1

0

1

0

SAFETY

1

6

1

4

0

RELIABILITY

2

11

4

1

1

PERFORMANCEEFFICIENCY

2

10

7

5

2

FUNCTIONALSUITABILITY

2

4

2

1

1

Frequency



30

20

10

0

DOMAINSPECIFICPERFORMANCE

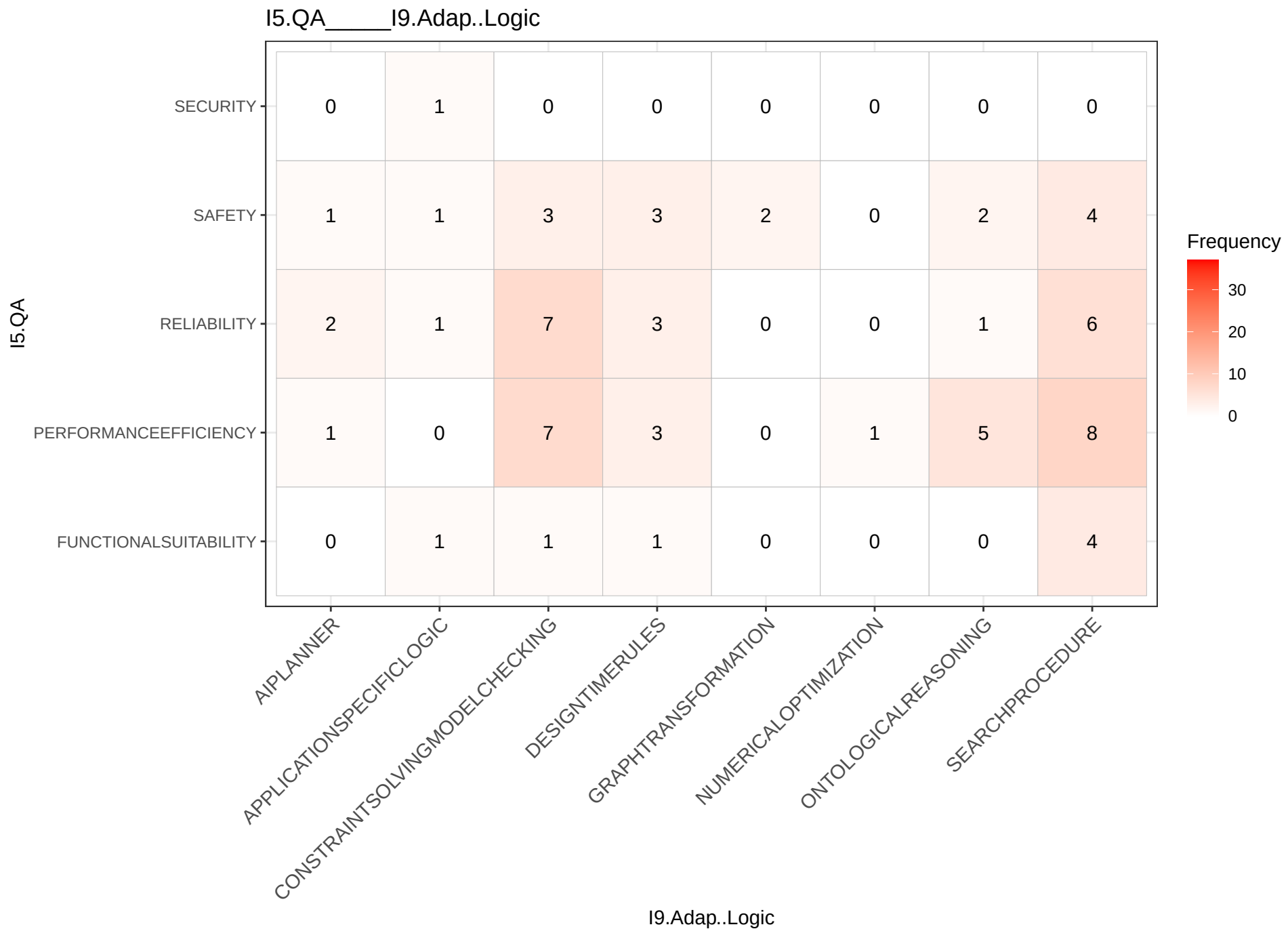
MISSIONPERFORMANCE

OVERHEADINTRODUCED

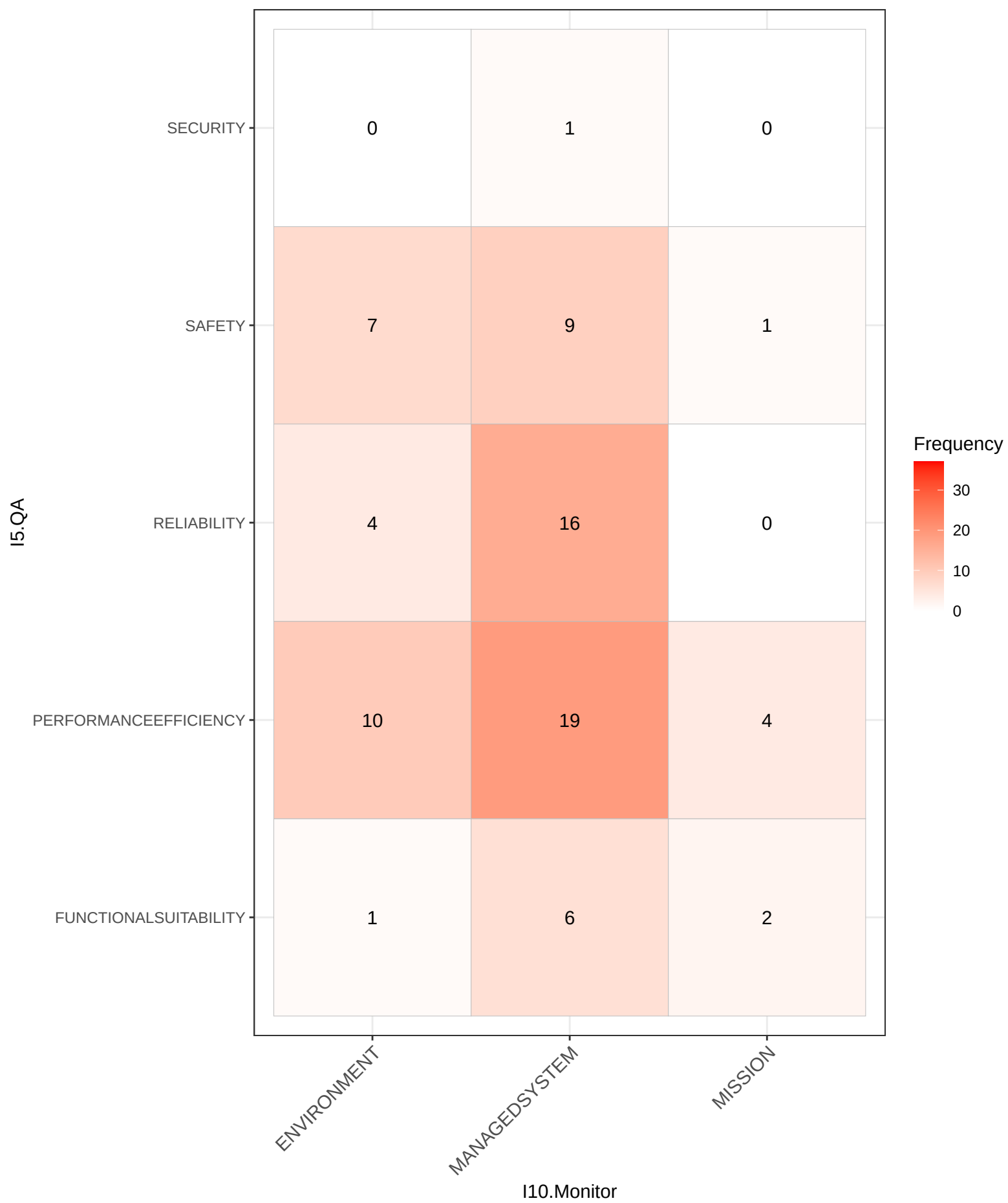
QUALITY

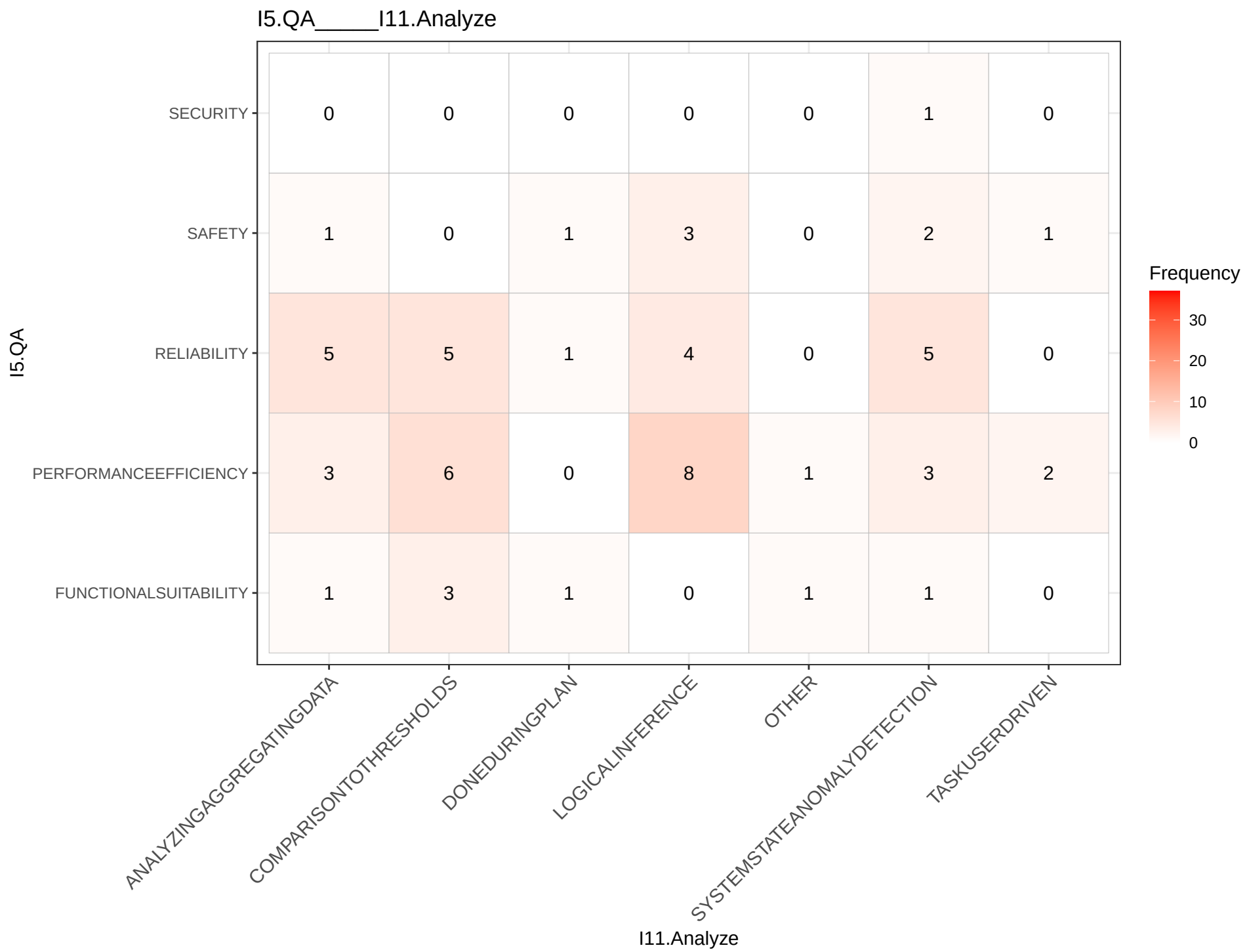
RESOURCECONSUMPTION

I8.Evaluation



I5.QA_____I10.Monitor





I5.QA

SECURITY

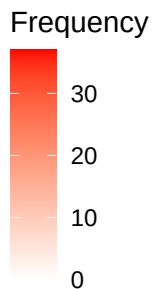
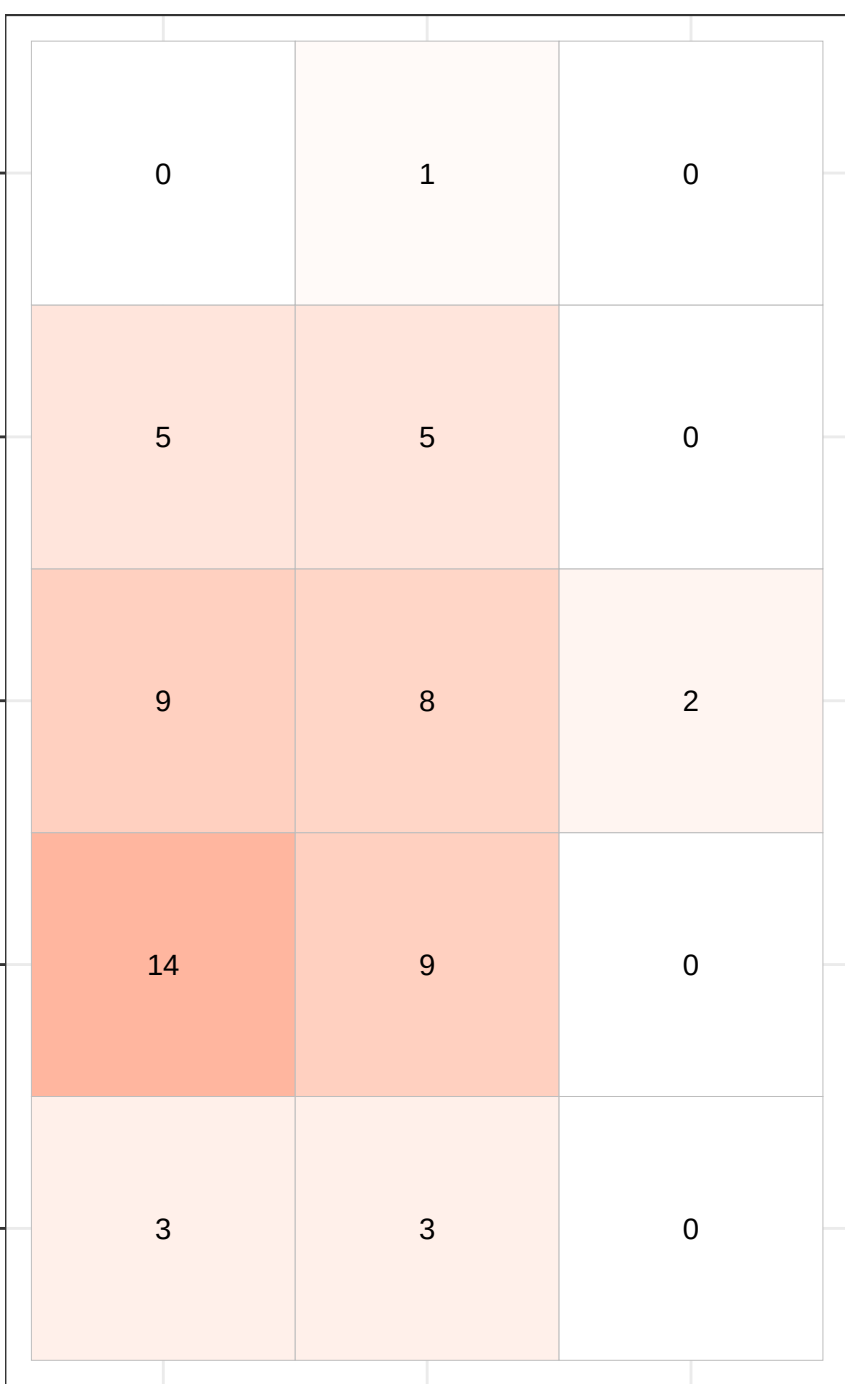
SAFETY

RELIABILITY

PERFORMANCEEFFICIENCY

FUNCTIONALSUITABILITY

I5.QA_____I12.Plan



DETERMININGTHEOPTIMALCHOICE

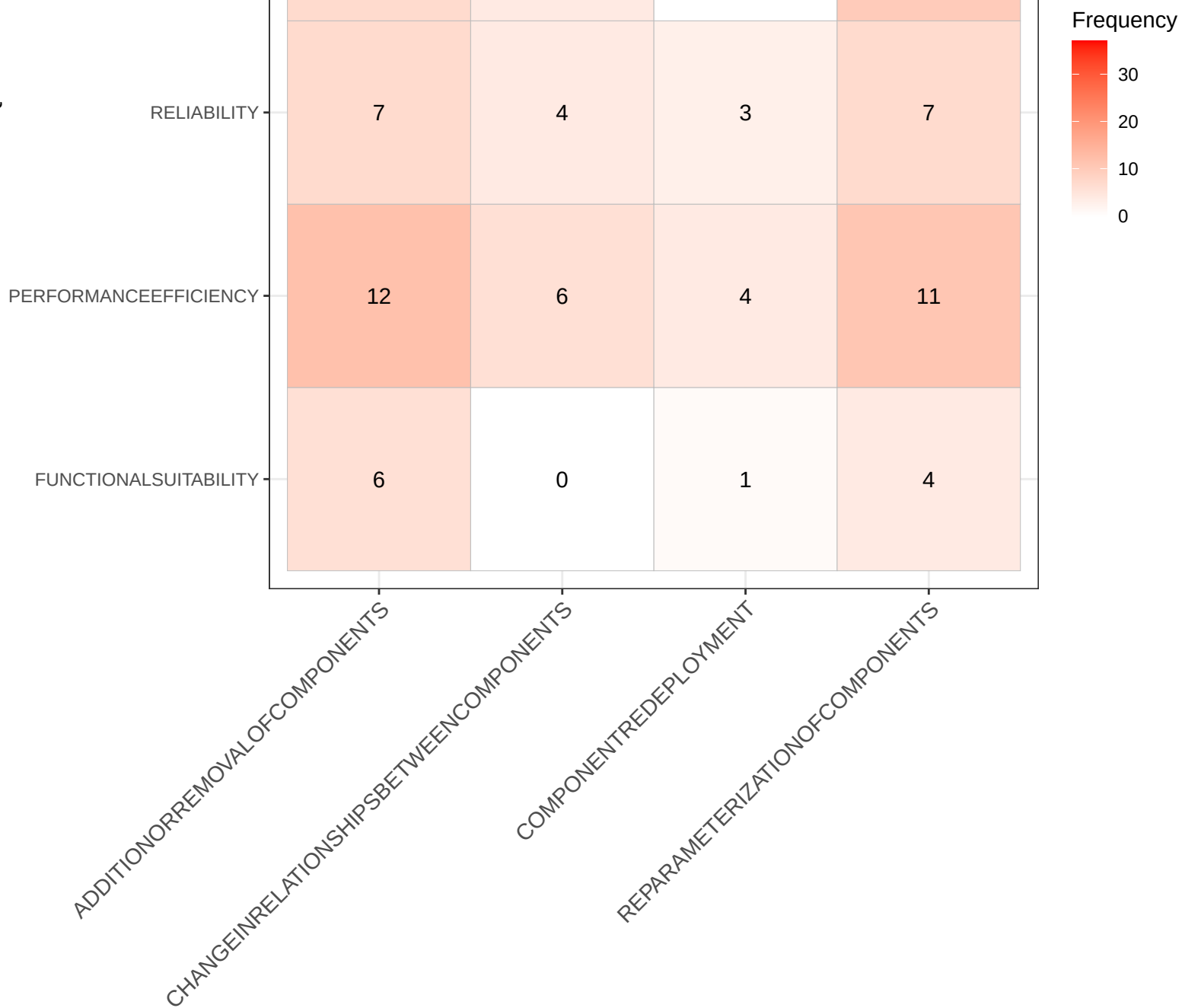
RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

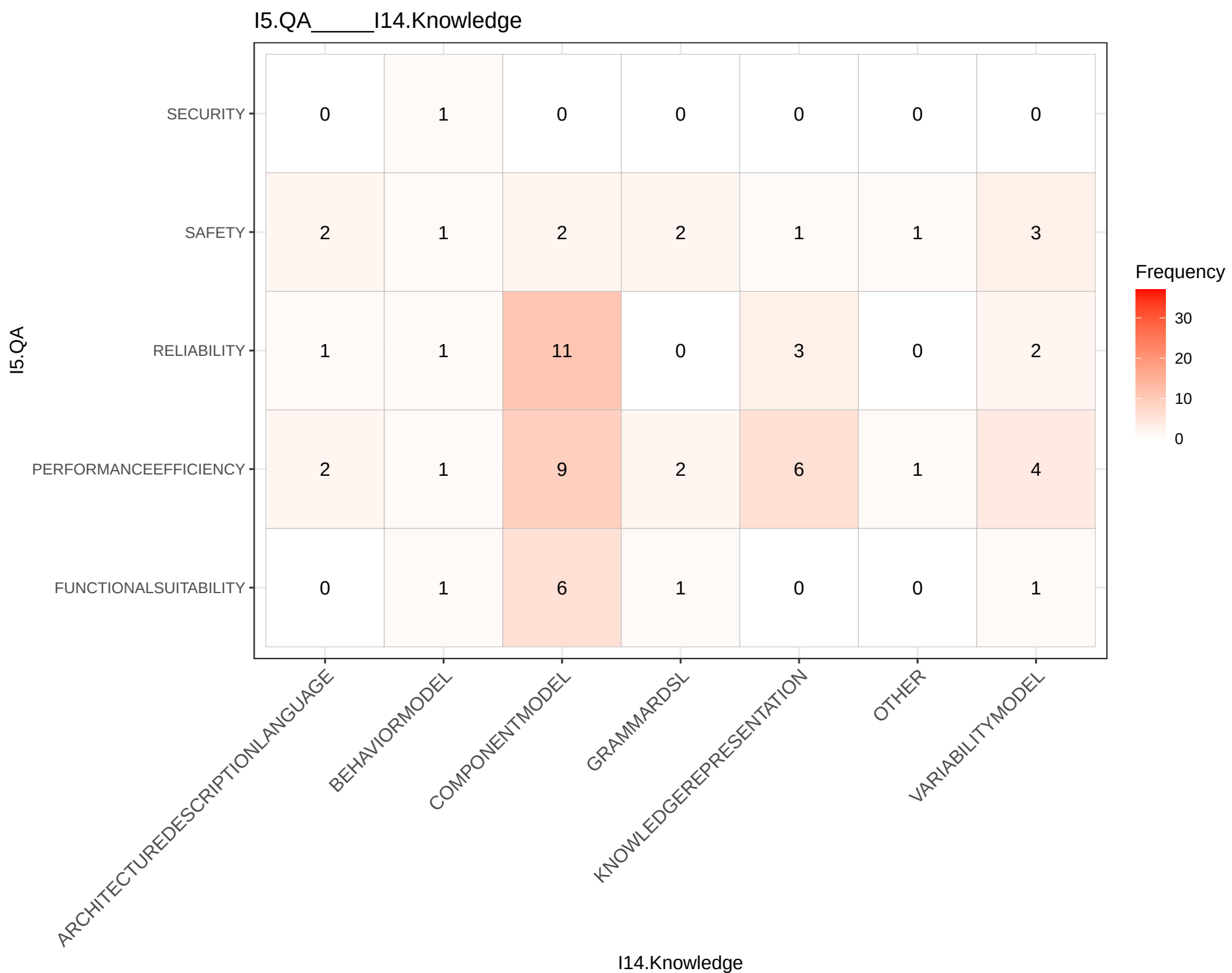
I12.Plan

I5.QA_____I13.Execute

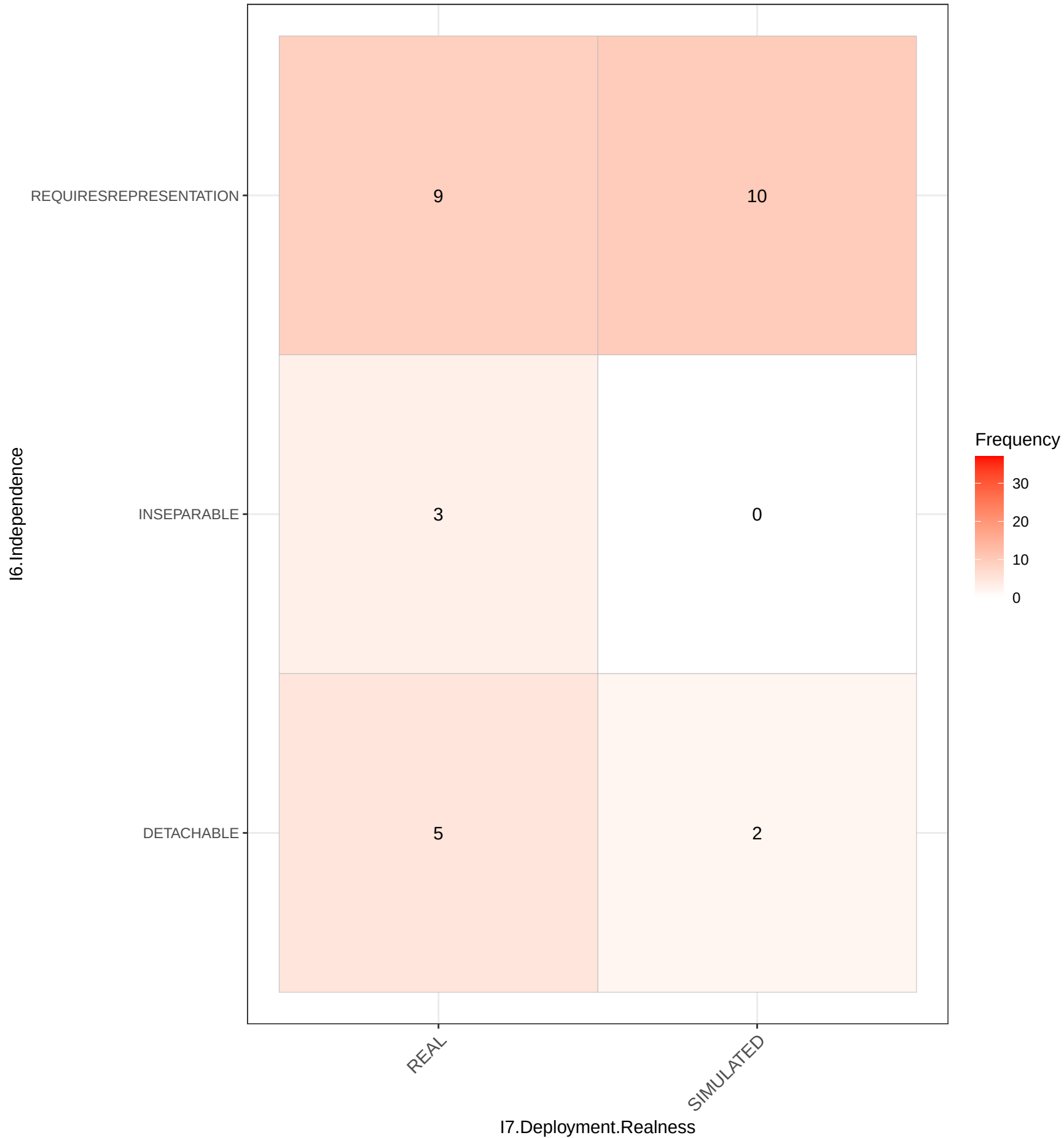
I5.QA



I13.Execute



I6.Independence_____I7.Deployment.Realness



I6.Independence_____I7.Mission.Realness

I6.Independence

REQUIRESREPRESENTATION

INSEPARABLE

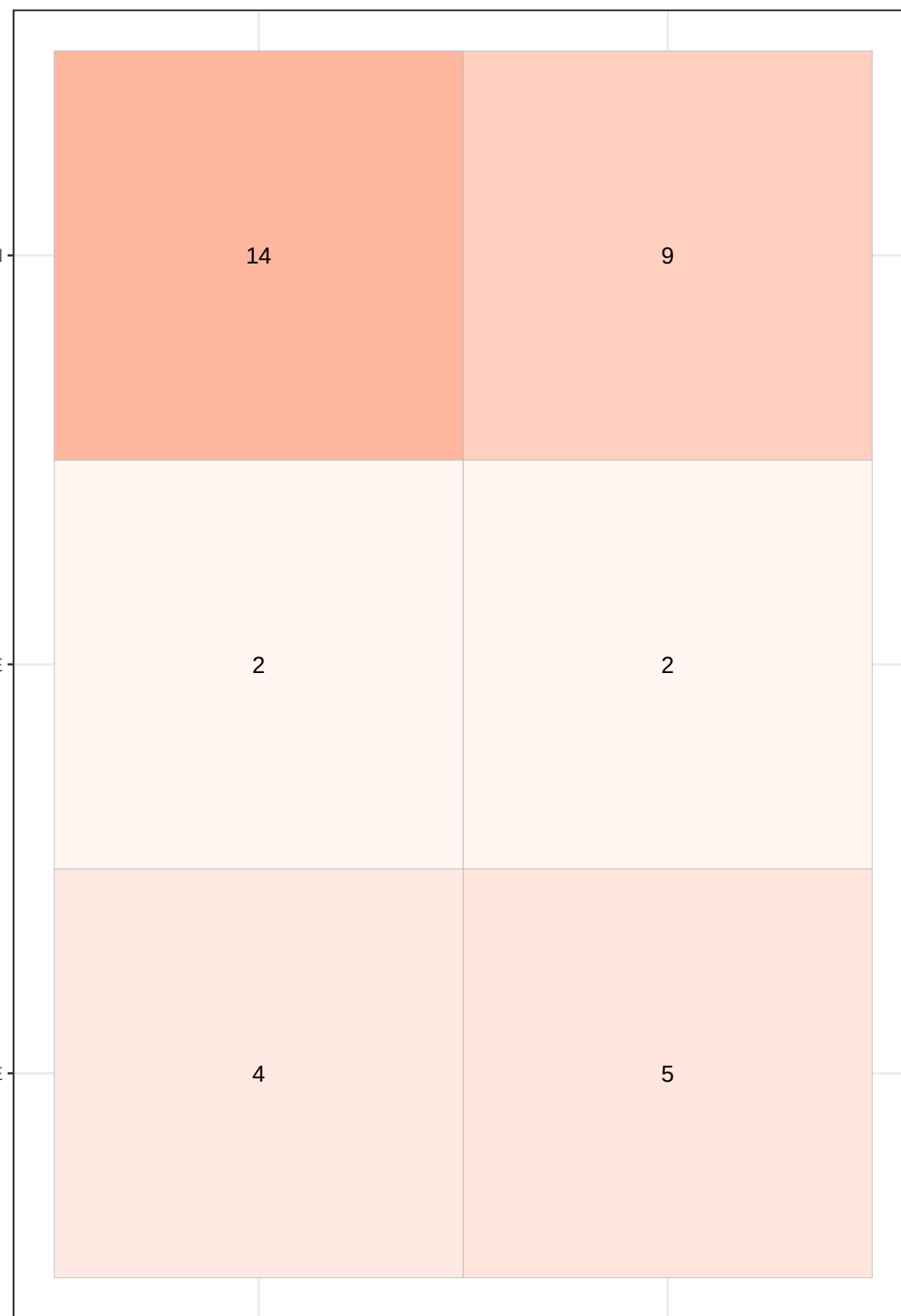
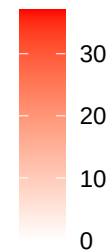
DETACHABLE

REAL

SYNTHETIC

I7.Mission.Realness

Frequency



I6.Independence

I6.Independence_____Experiment.Method

REQUIRESREPRESENTATION

INSEPARABLE

DETACHABLE

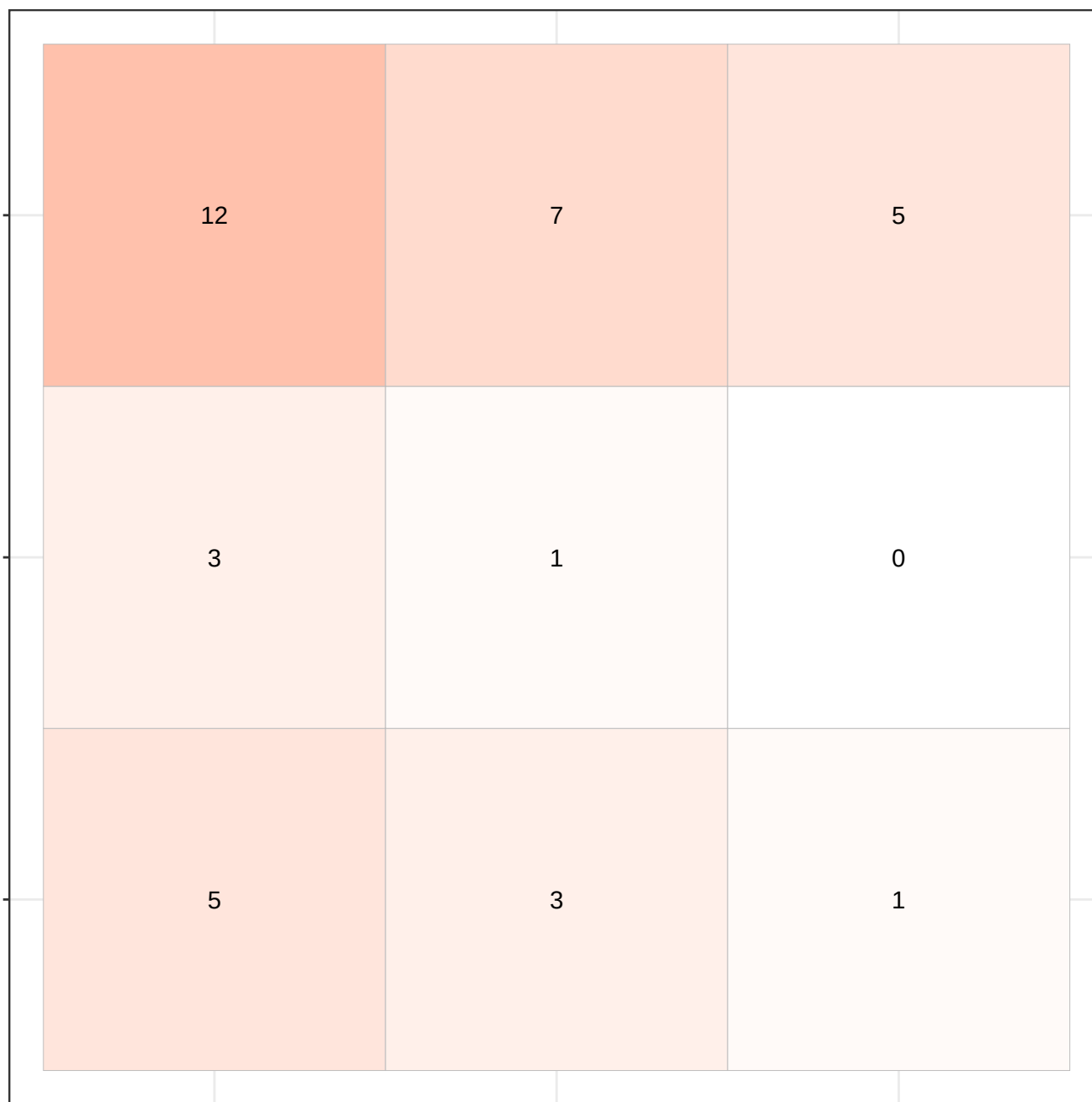
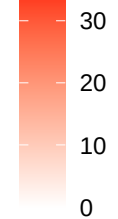
EXPERIMENT

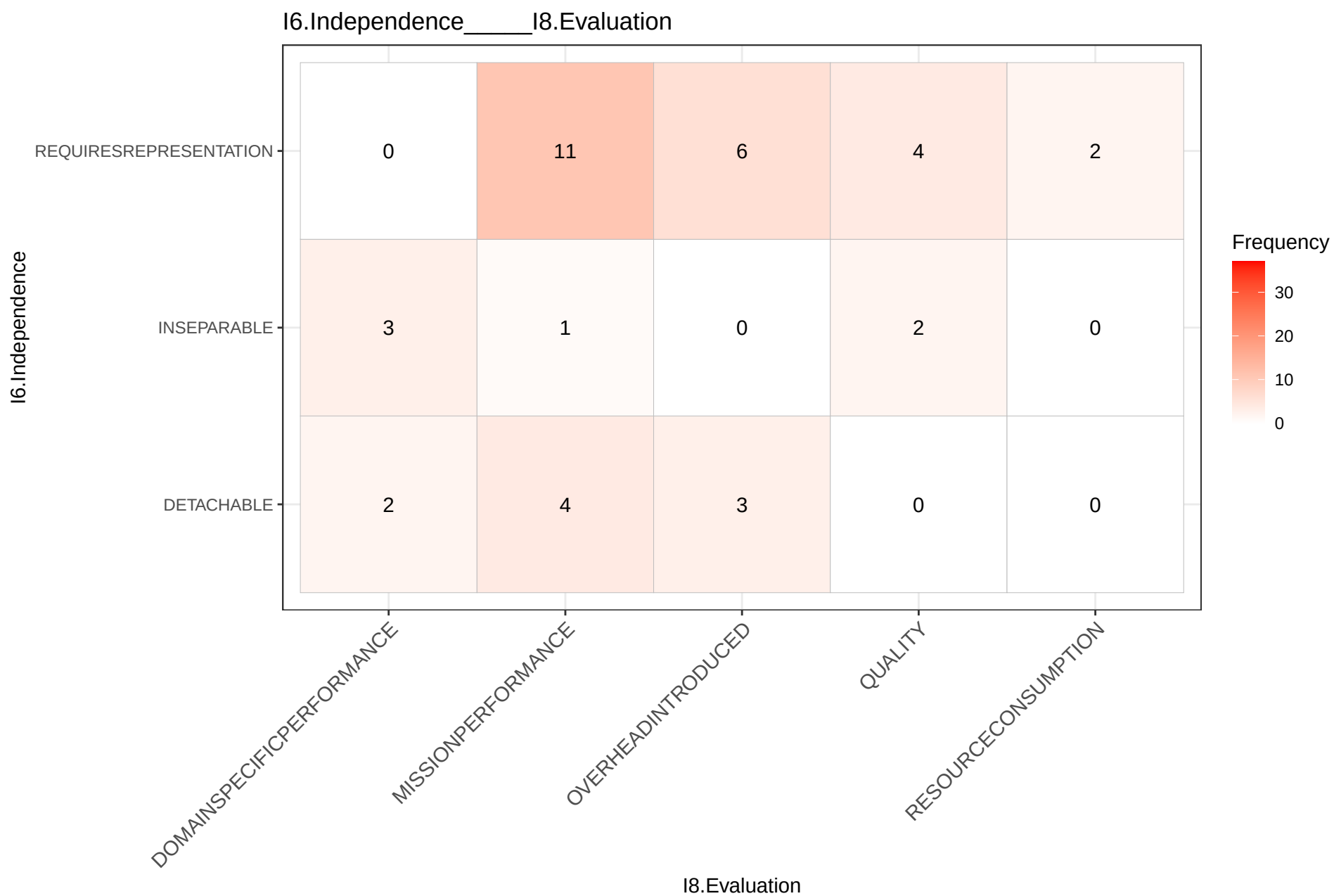
NOEVALUATION

SHOWCASE

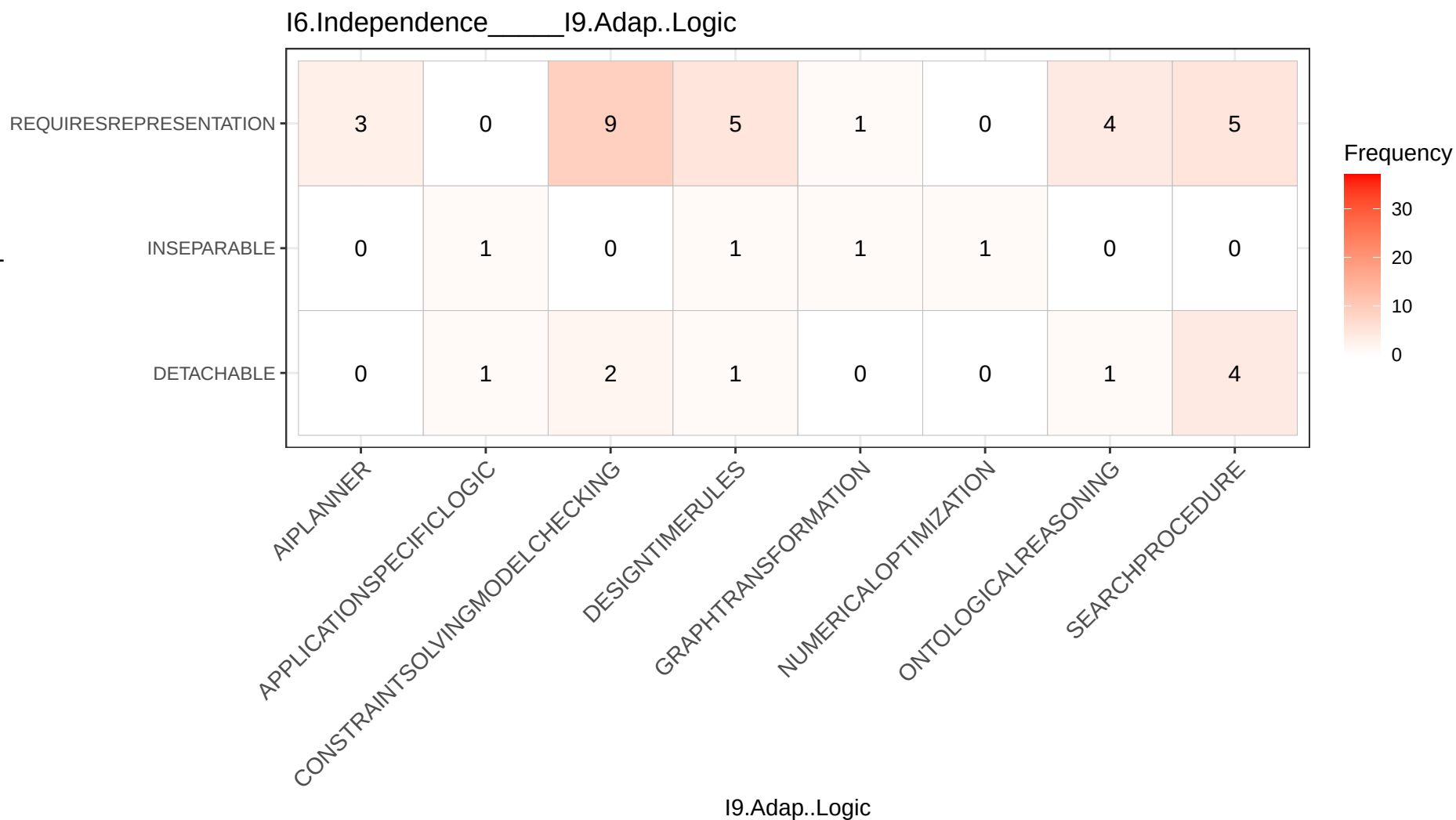
Experiment.Method

Frequency





I6.Independence



I6.Independence_____I10.Monitor

REQUIRESREPRESENTATION

11

18

3

INSEPARABLE

0

2

2

DETACHABLE

2

8

0

ENVIRONMENT

MANAGEDSYSTEM

MISSION

I10.Monitor

Frequency



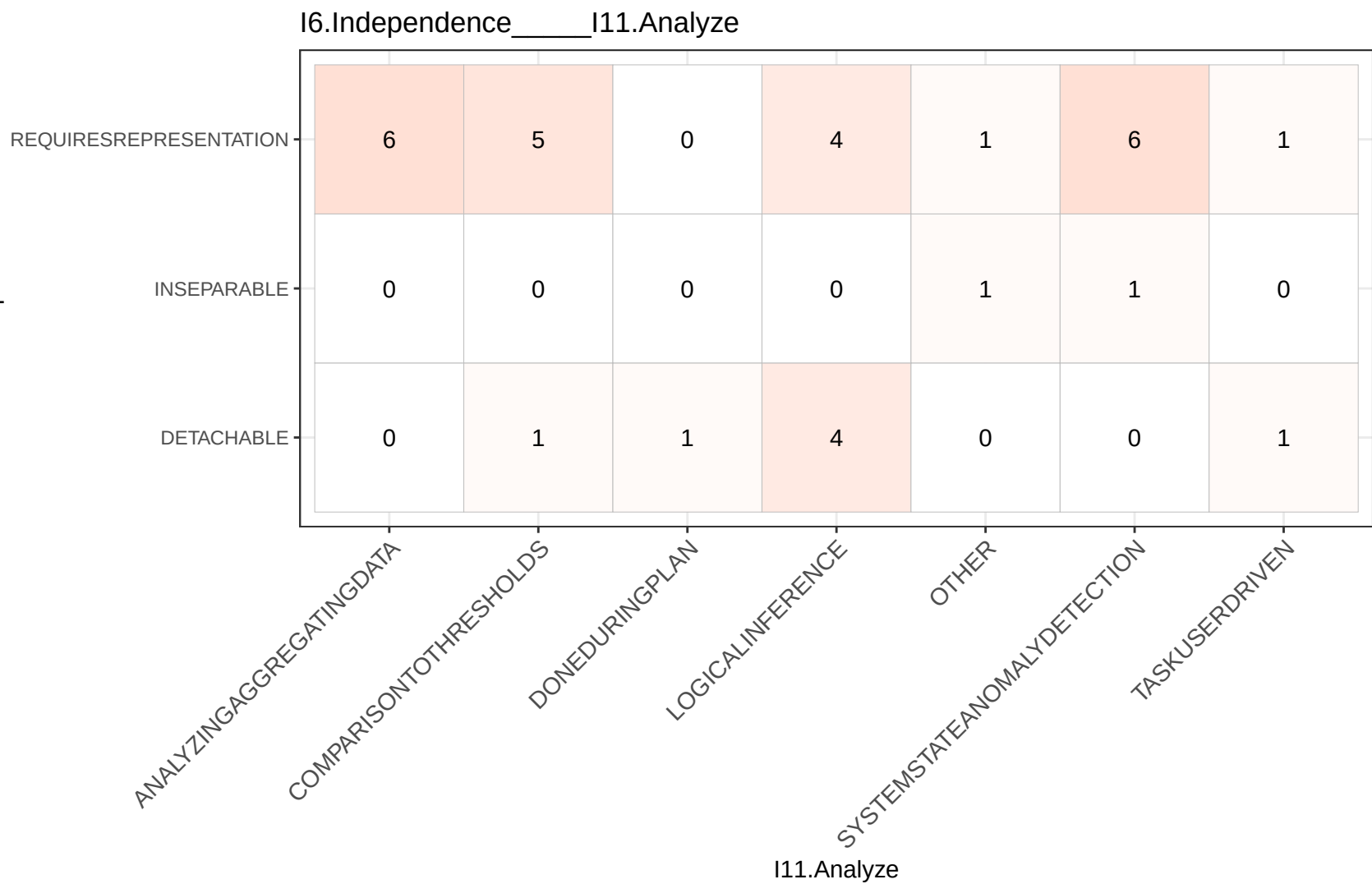
30

20

10

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I6.Independence



Frequency



I6.Independence_____I12.Plan

I6.Independence

REQUIRESREPRESENTATION

INSEPARABLE

DETACHABLE

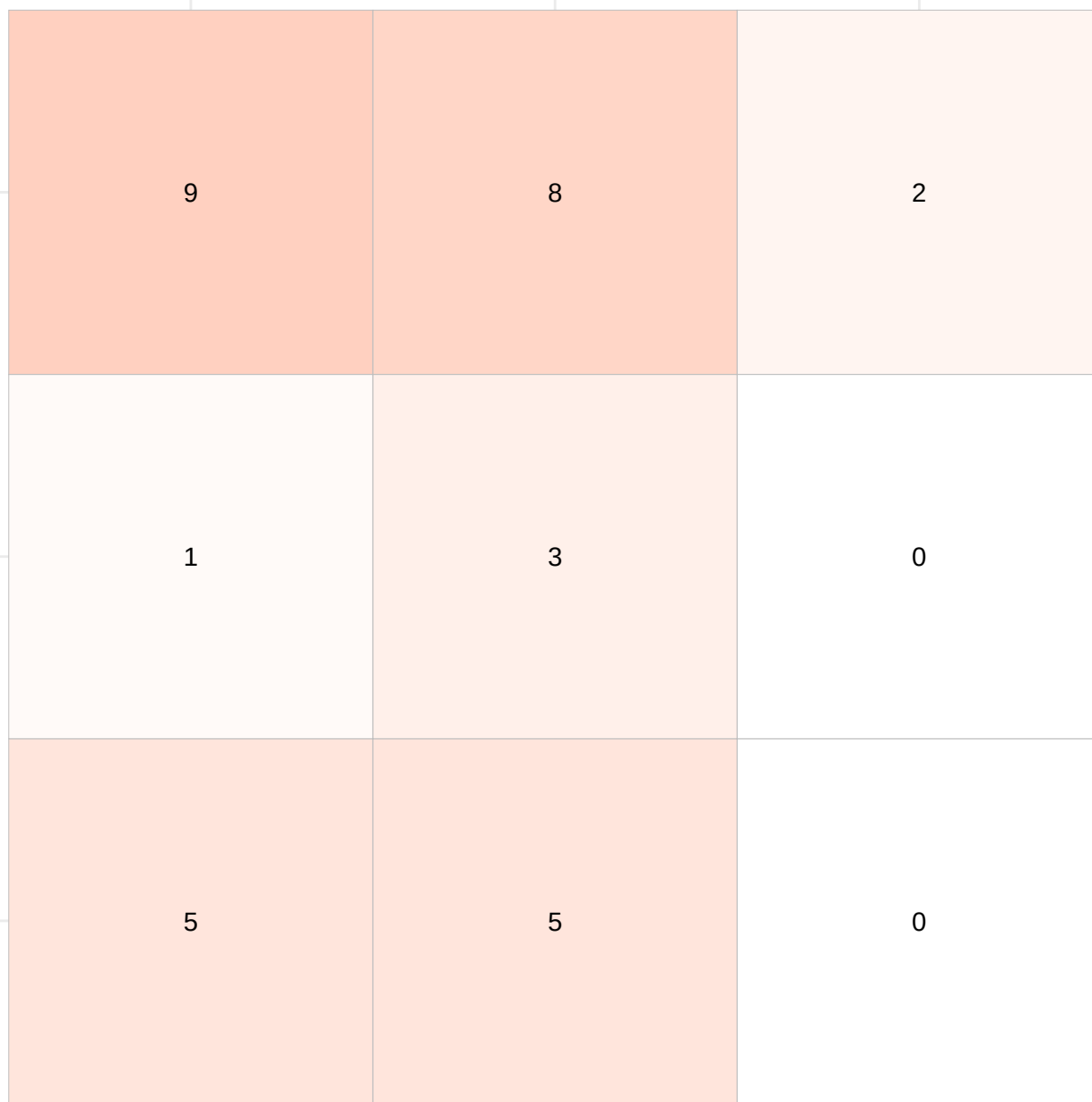
DETERMININGTHEOPTIMALCHOICE

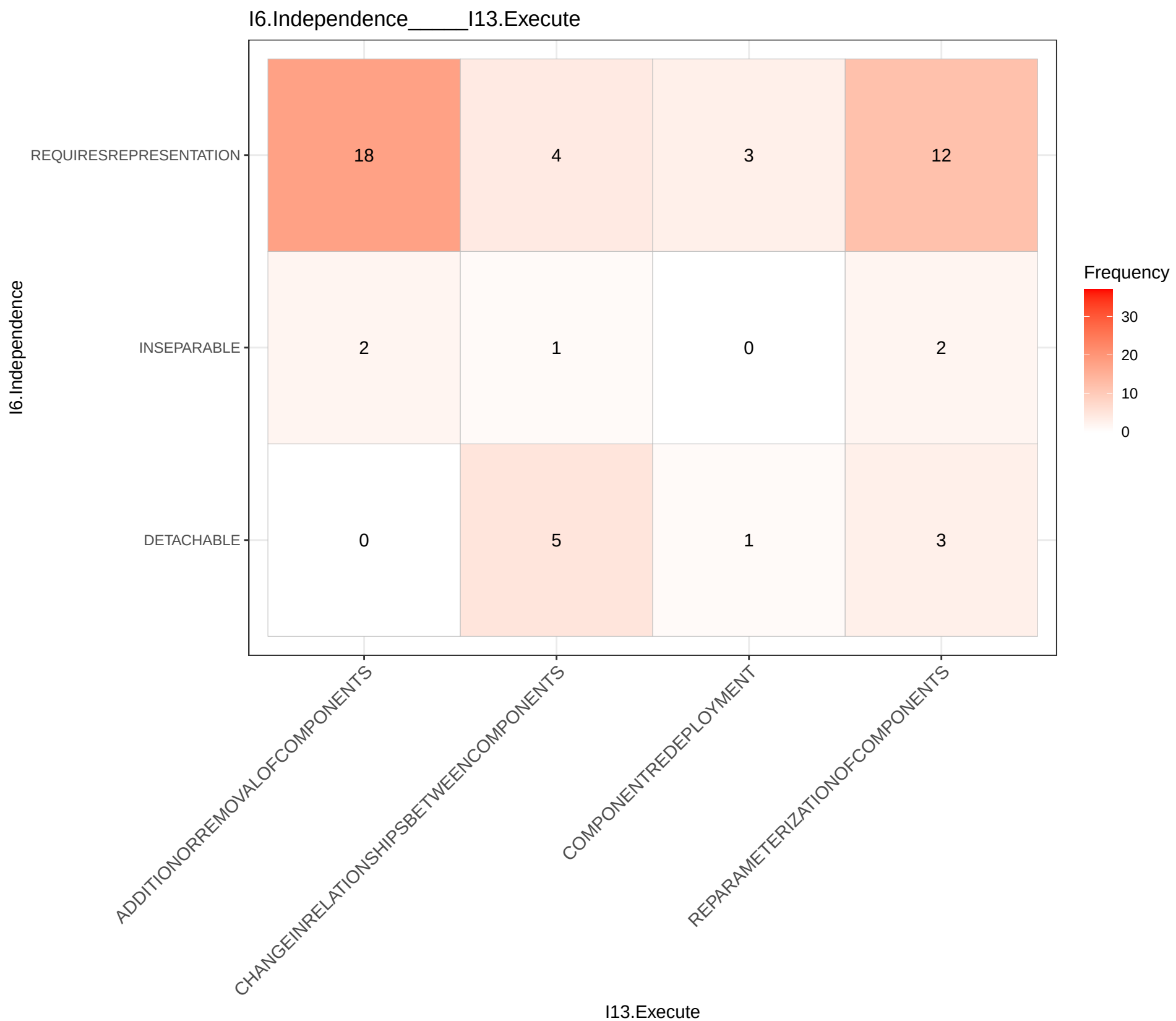
RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

I12.Plan

Frequency





I6.Independence

I6.Independence _____ I14.Knowledge

REQUIRESREPRESENTATION

INSEPARABLE

DETACHABLE

Frequency



30

20

10

0

ARCHITECTUREDESCRIPTIONLANGUAGE

BEHAVIORMODEL

COMPONENTMODEL

GRAMMARDSL

KNOWLEDGEREPRESENTATION

OTHER

VARIABILITYMODEL

I14.Knowledge

3

1

8

1

4

1

6

0

1

1

1

0

0

1

0

0

6

1

2

0

0

I7.Deployment.Realness_____I7.Mission.Realness

I7.Deployment.Realness

SIMULATED

7

5

REAL

8

9

REAL

SYNTHETIC

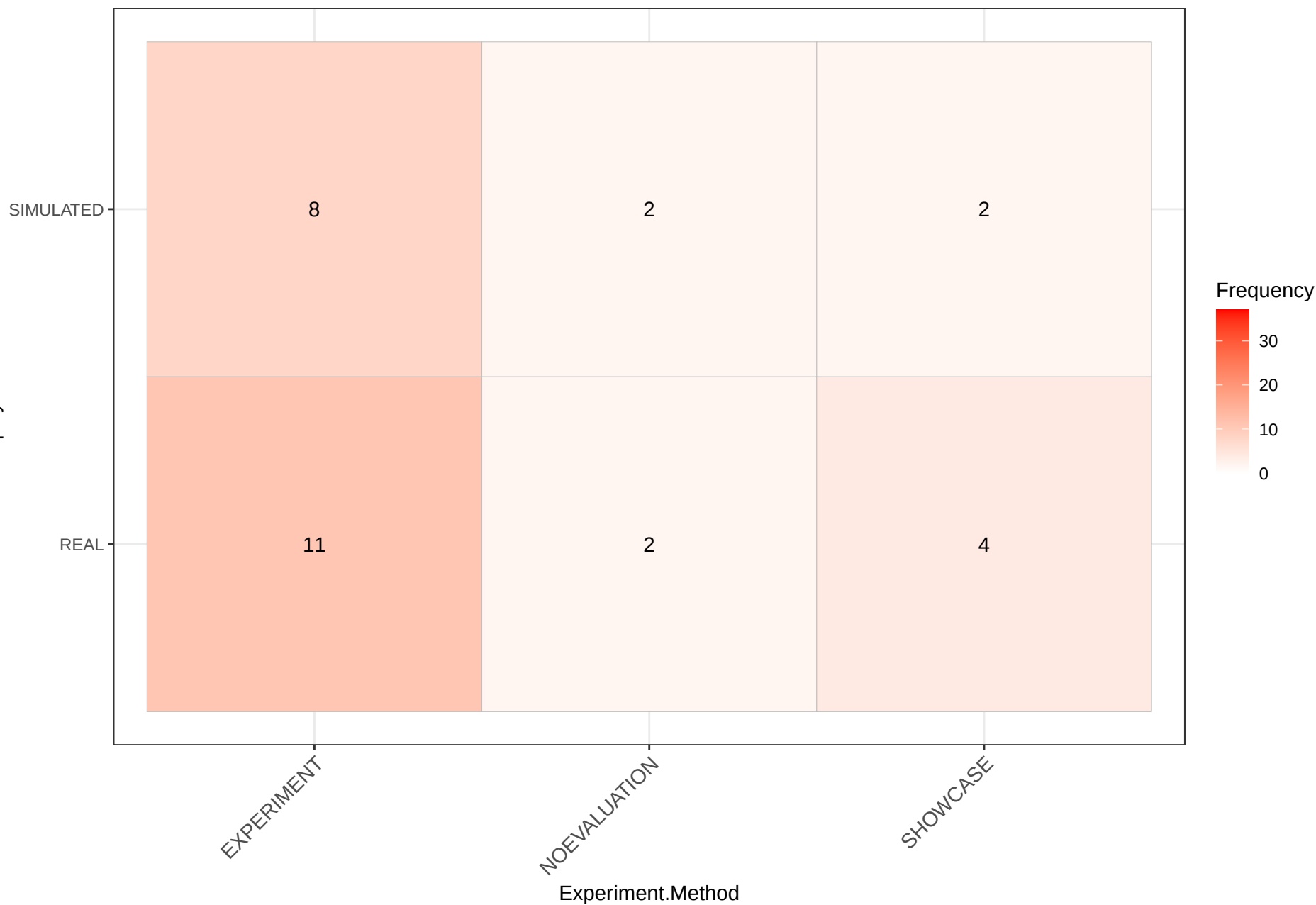
I7.Mission.Realness

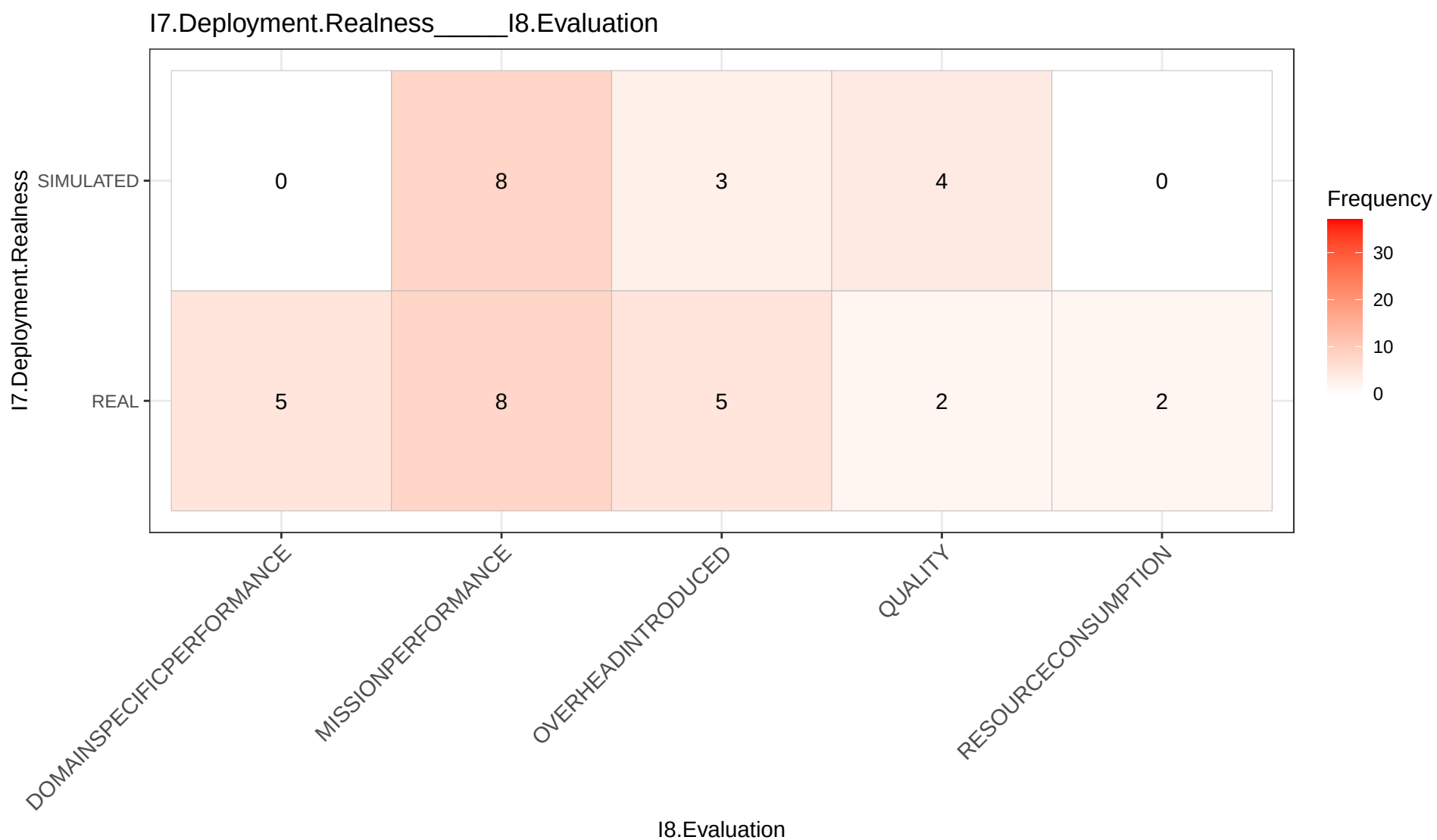
Frequency

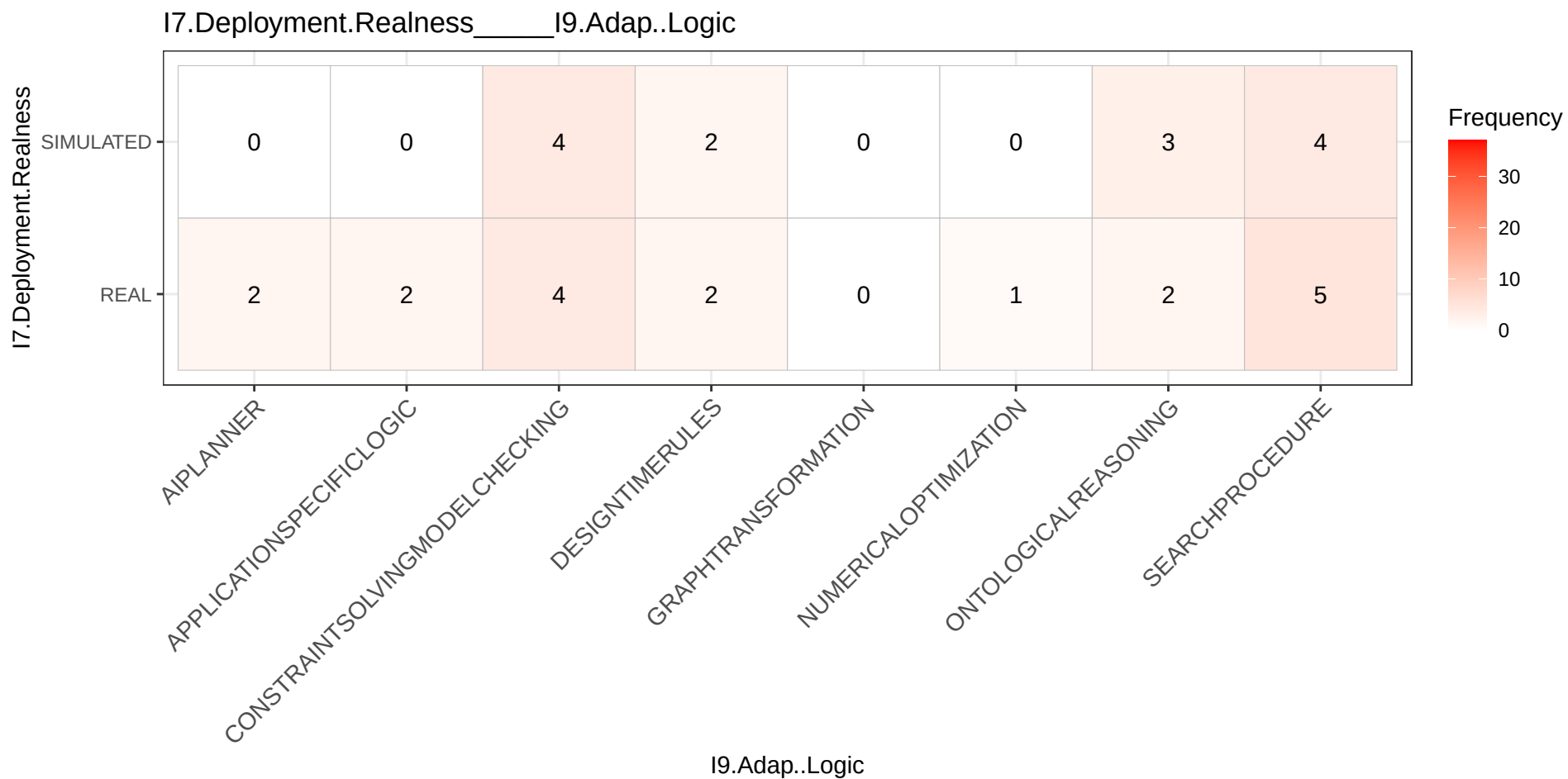


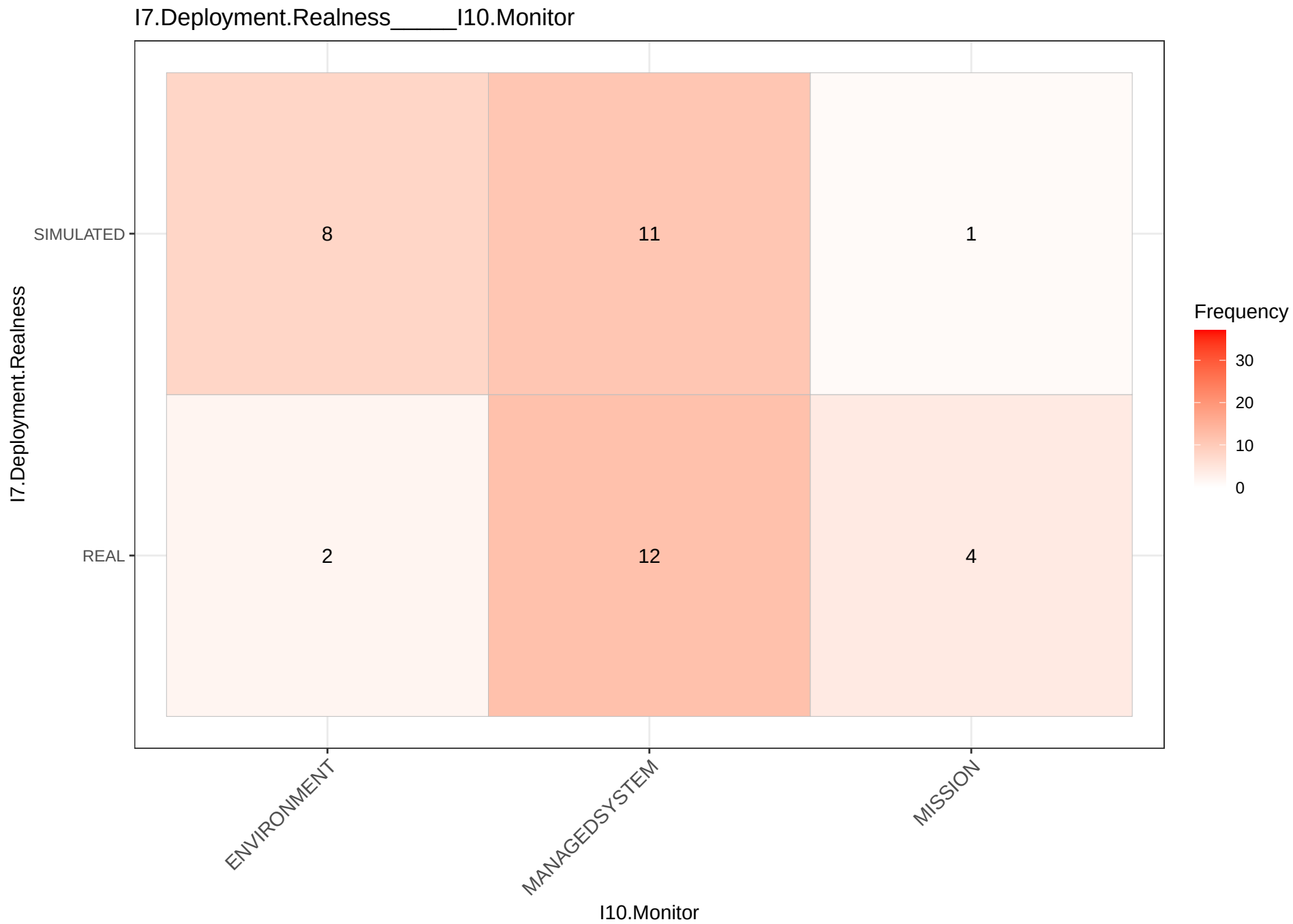
I7.Deployment.Realness_____Experiment.Method

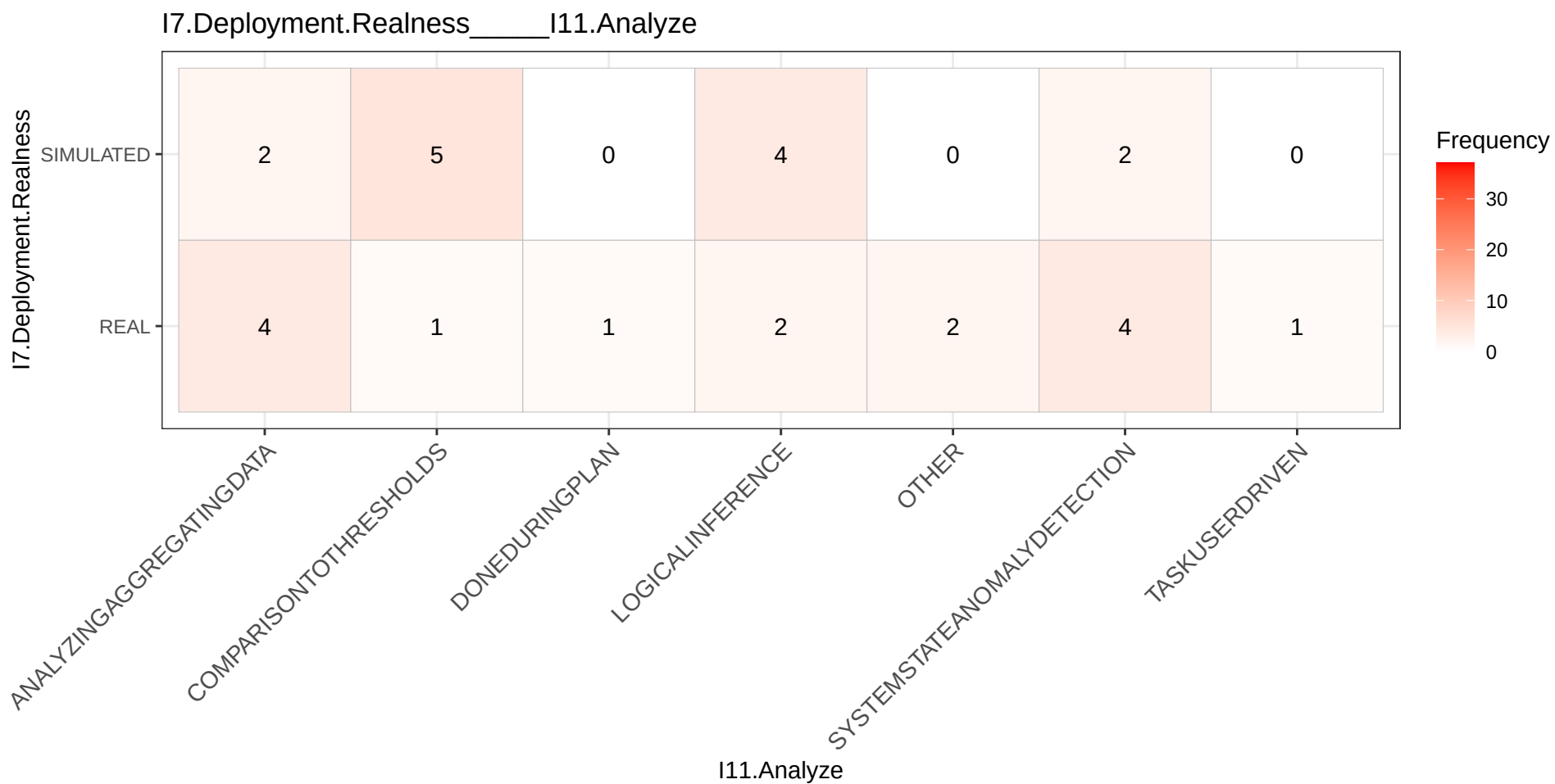
I7.Deployment.Realness











I7.Deployment.Realness_____I12.Plan

I7.Deployment.Realness

SIMULATED

6

5

0

REAL

6

6

2

DETERMININGTHEOPTIMALCHOICE

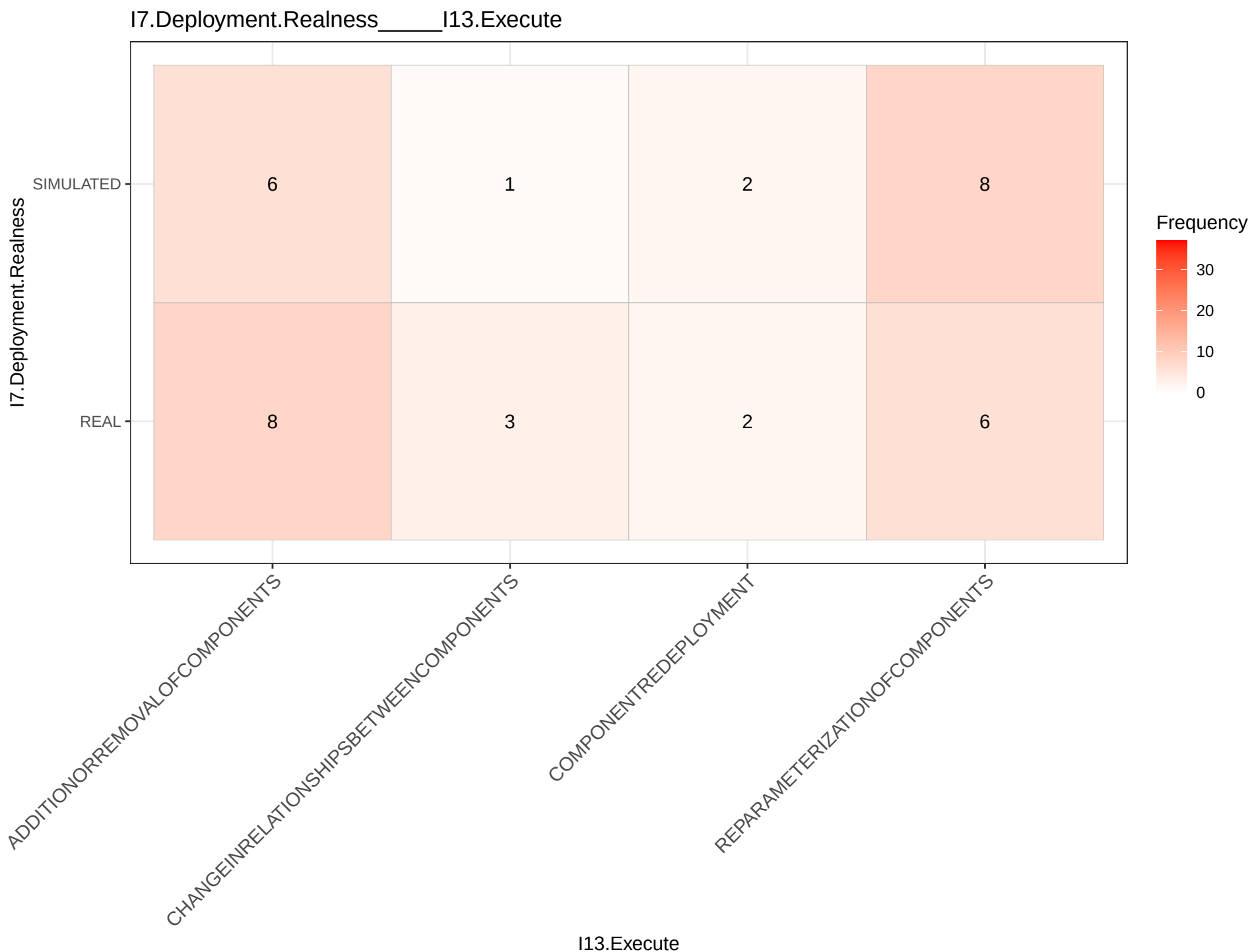
RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

I12.Plan

Frequency







I7.Mission.Realness_____Experiment.Method

I7.Mission.Realness

SYNTHETIC

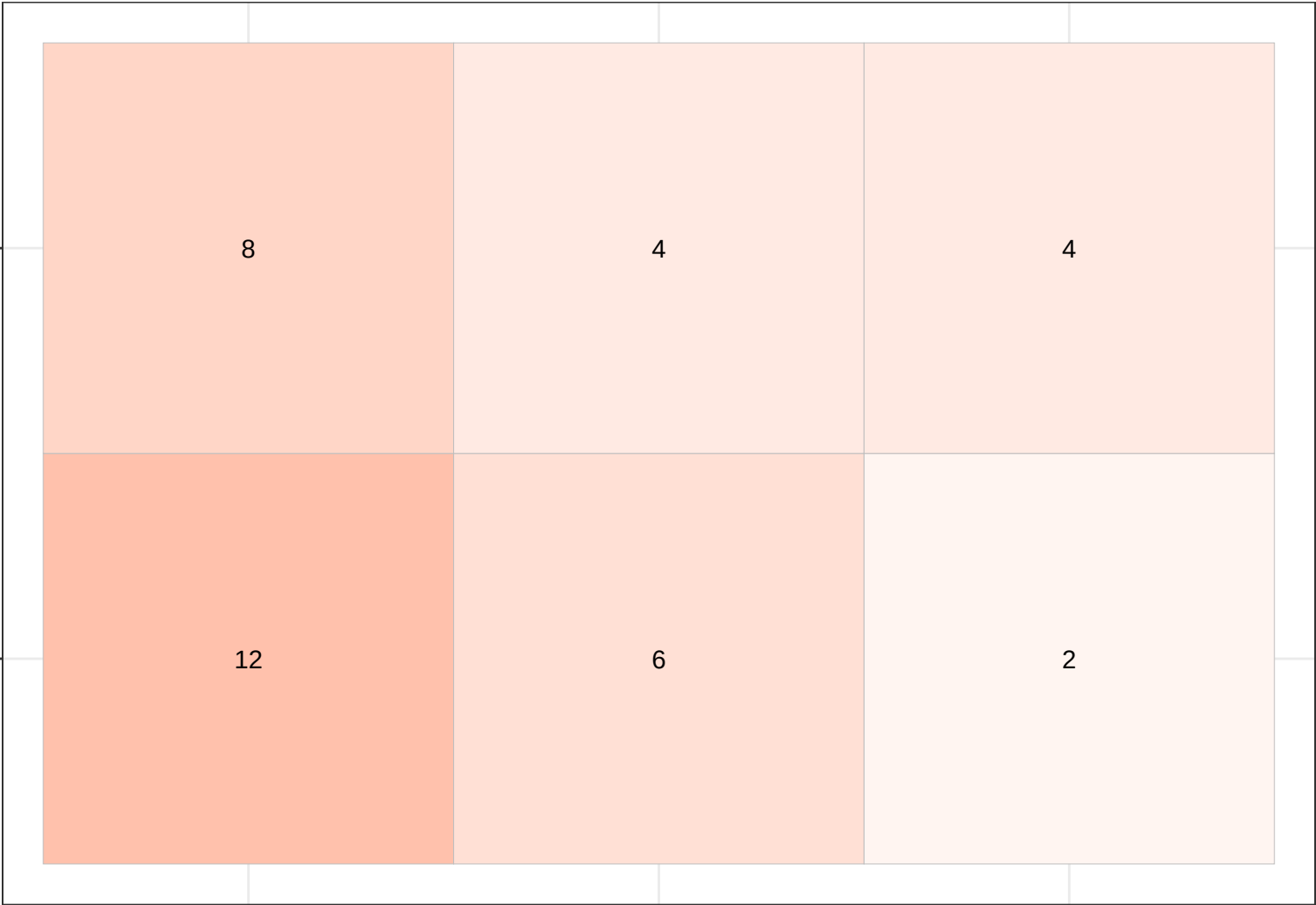
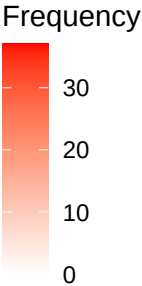
REAL

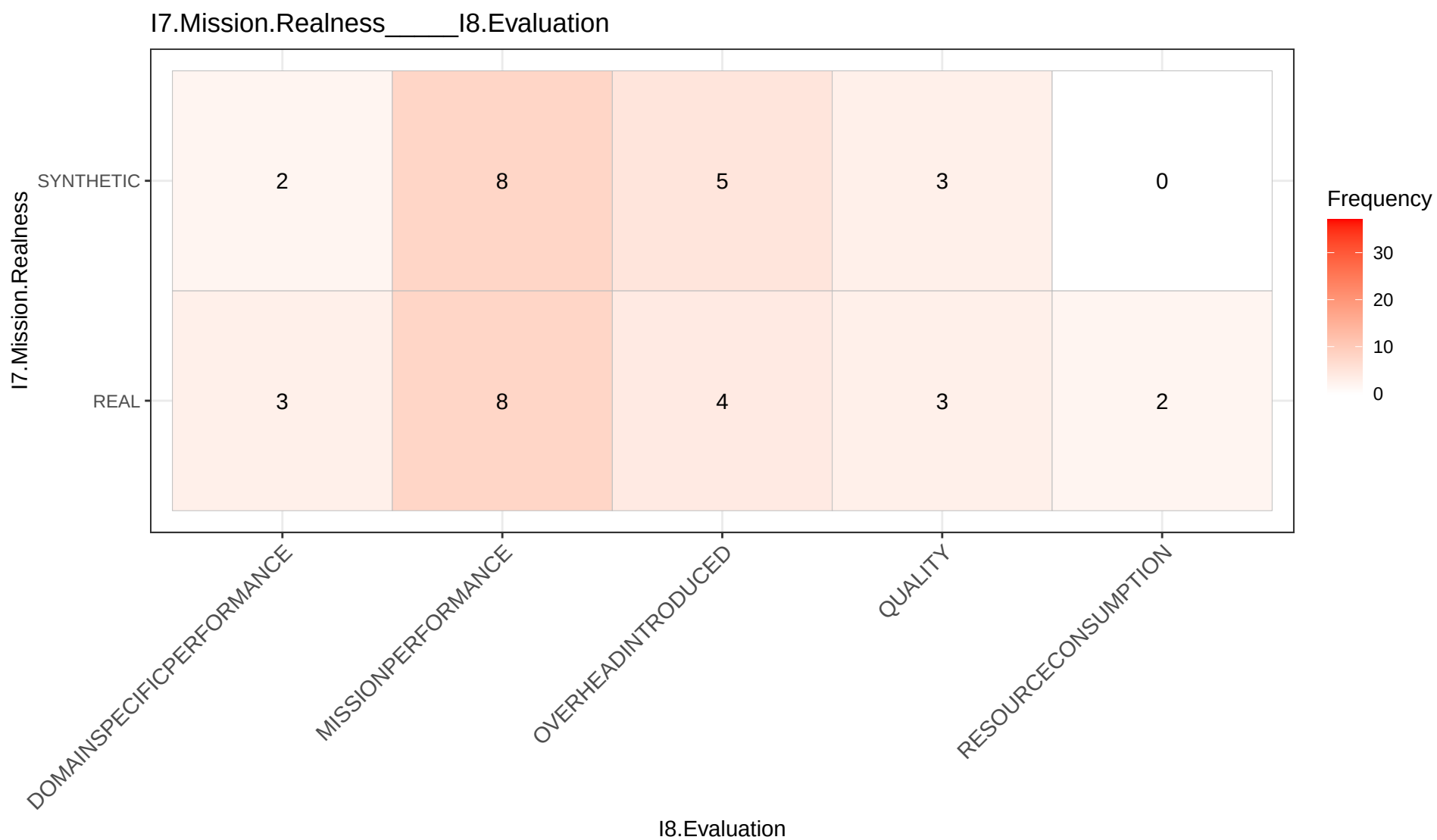
EXPERIMENT

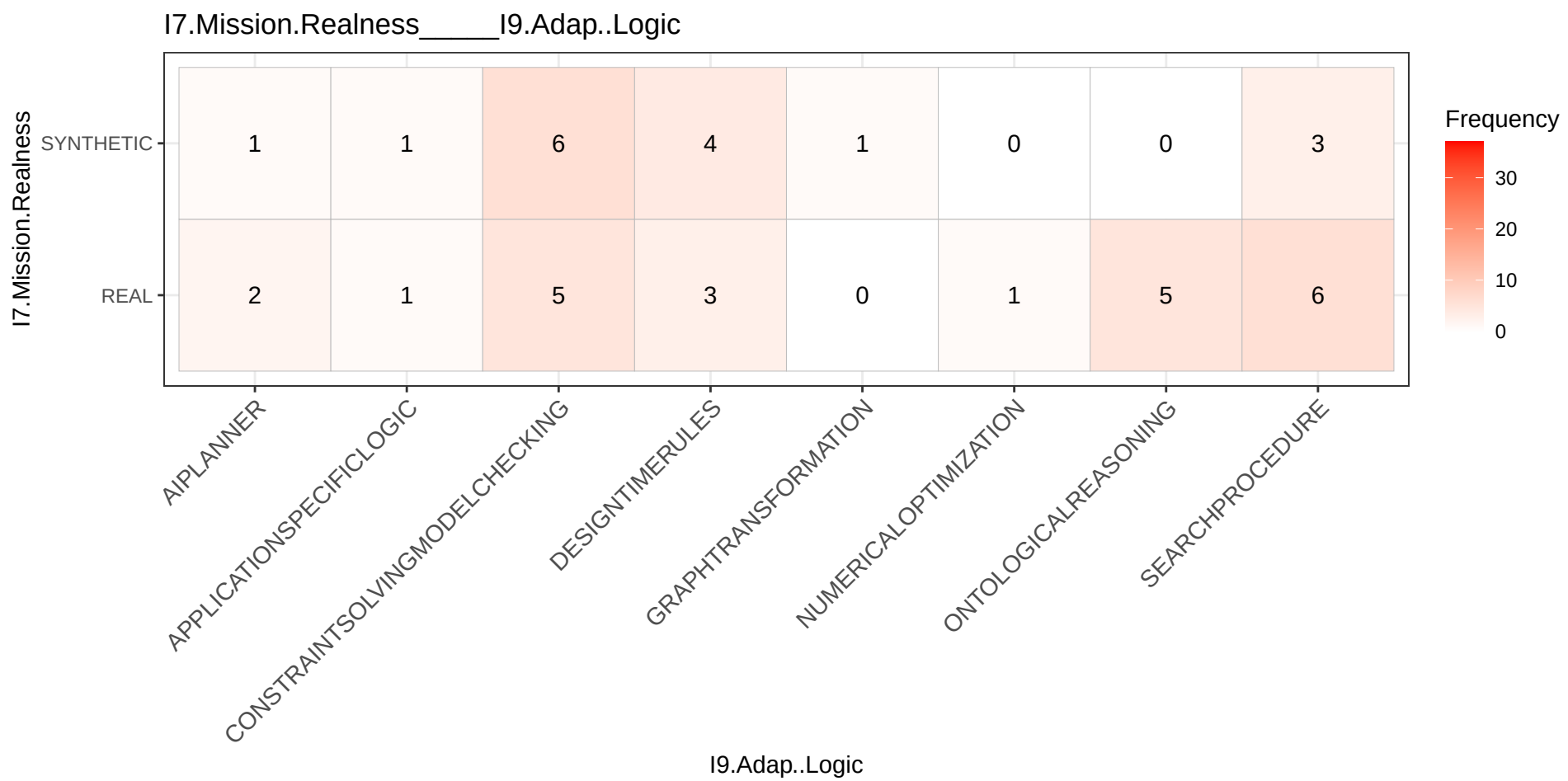
NOEVALUATION

SHOWCASE

Experiment.Method







I7.Mission.Realness

I7.Mission.Realness_____I10.Monitor

SYNTHETIC

5

13

2

REAL

8

15

3

ENVIRONMENT

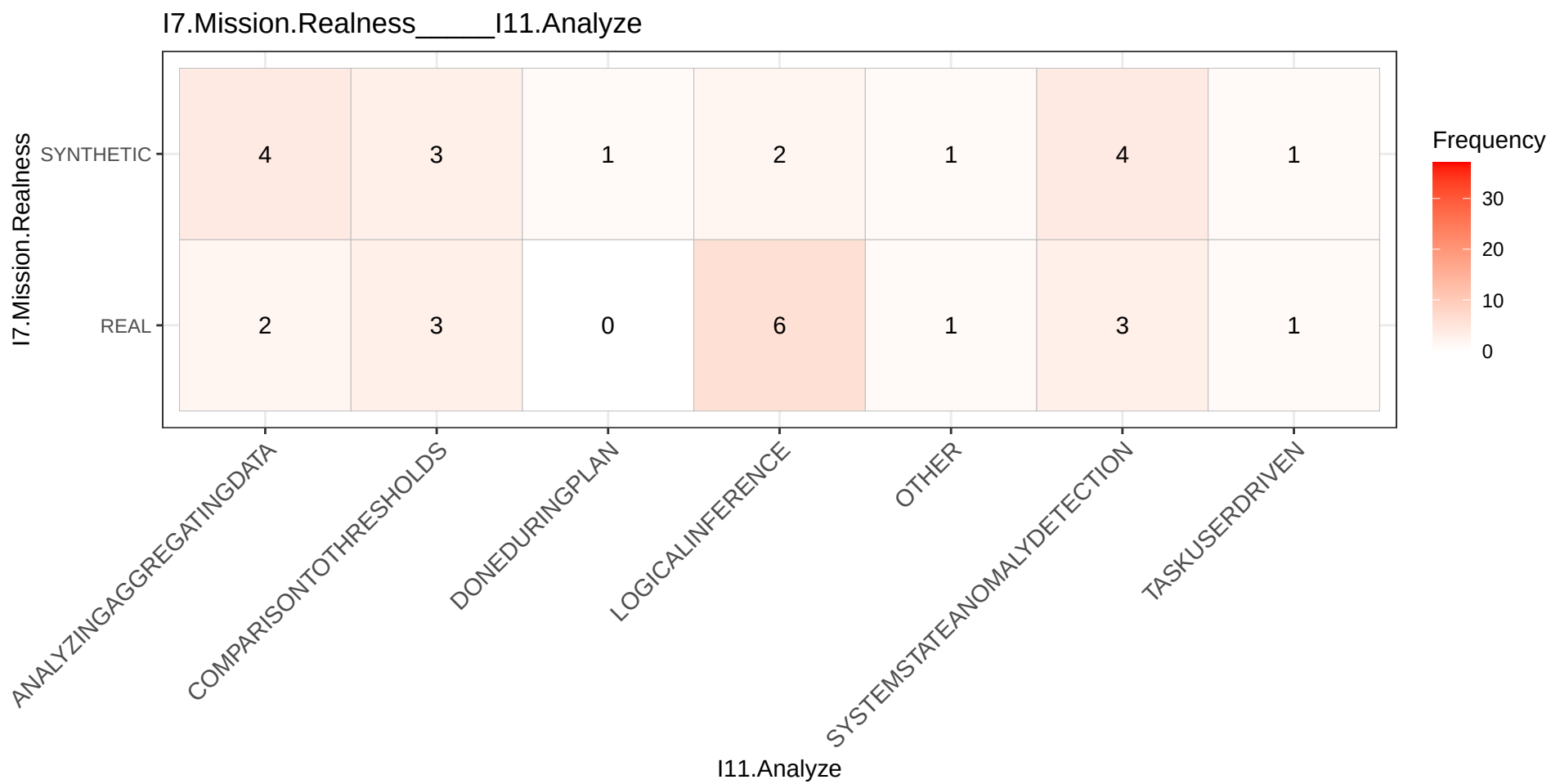
MANAGEDSYSTEM

MISSION

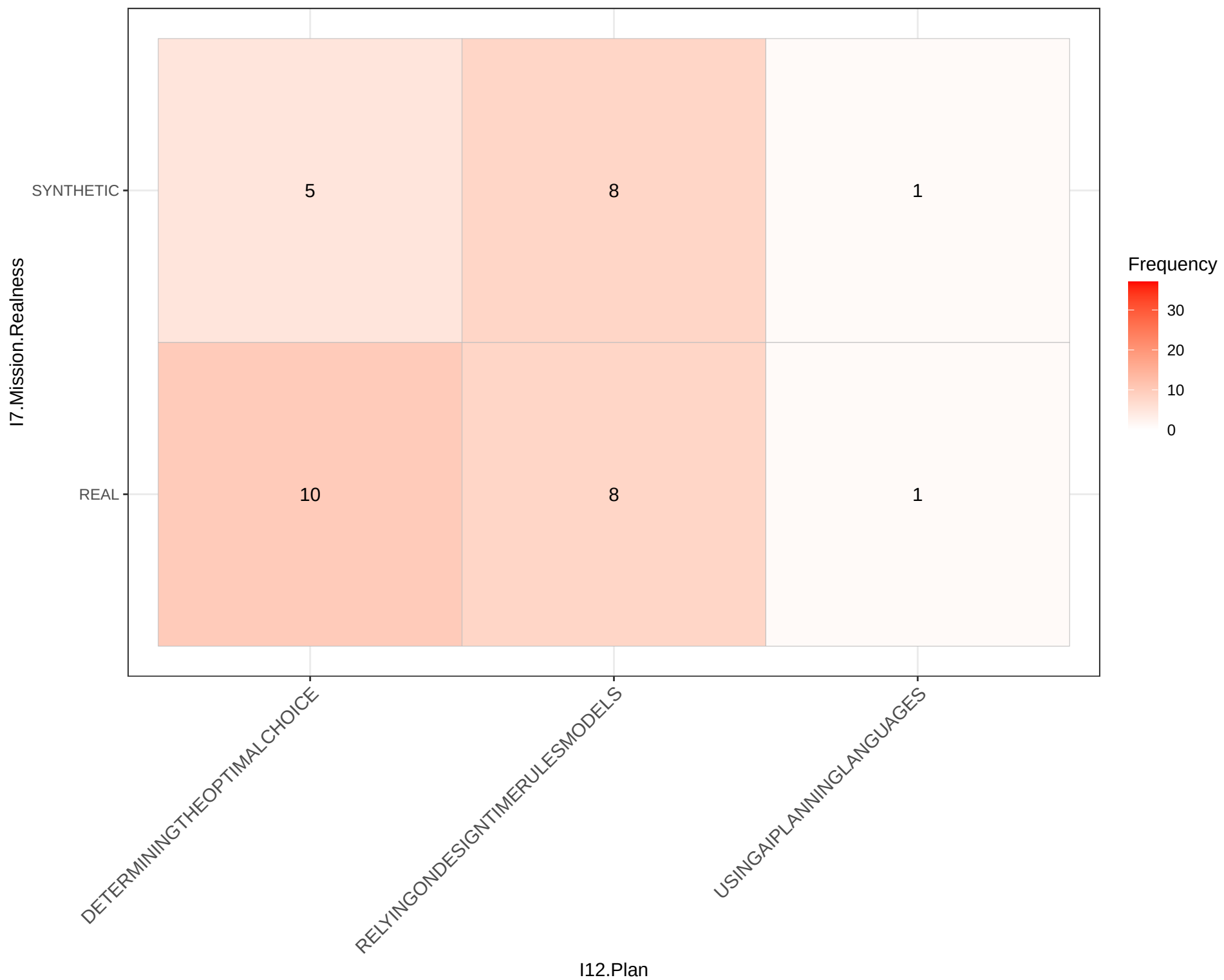
I10.Monitor

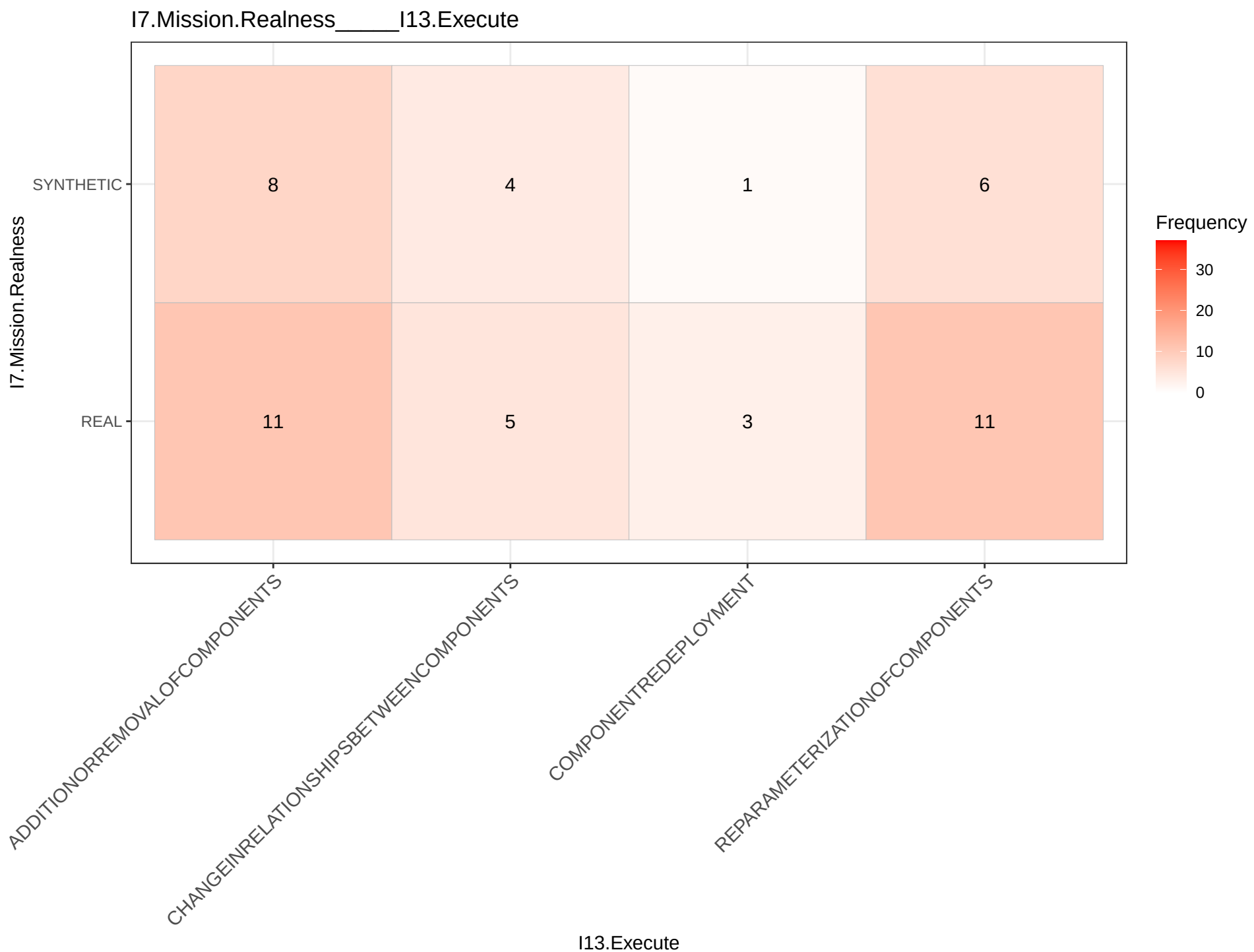
Frequency

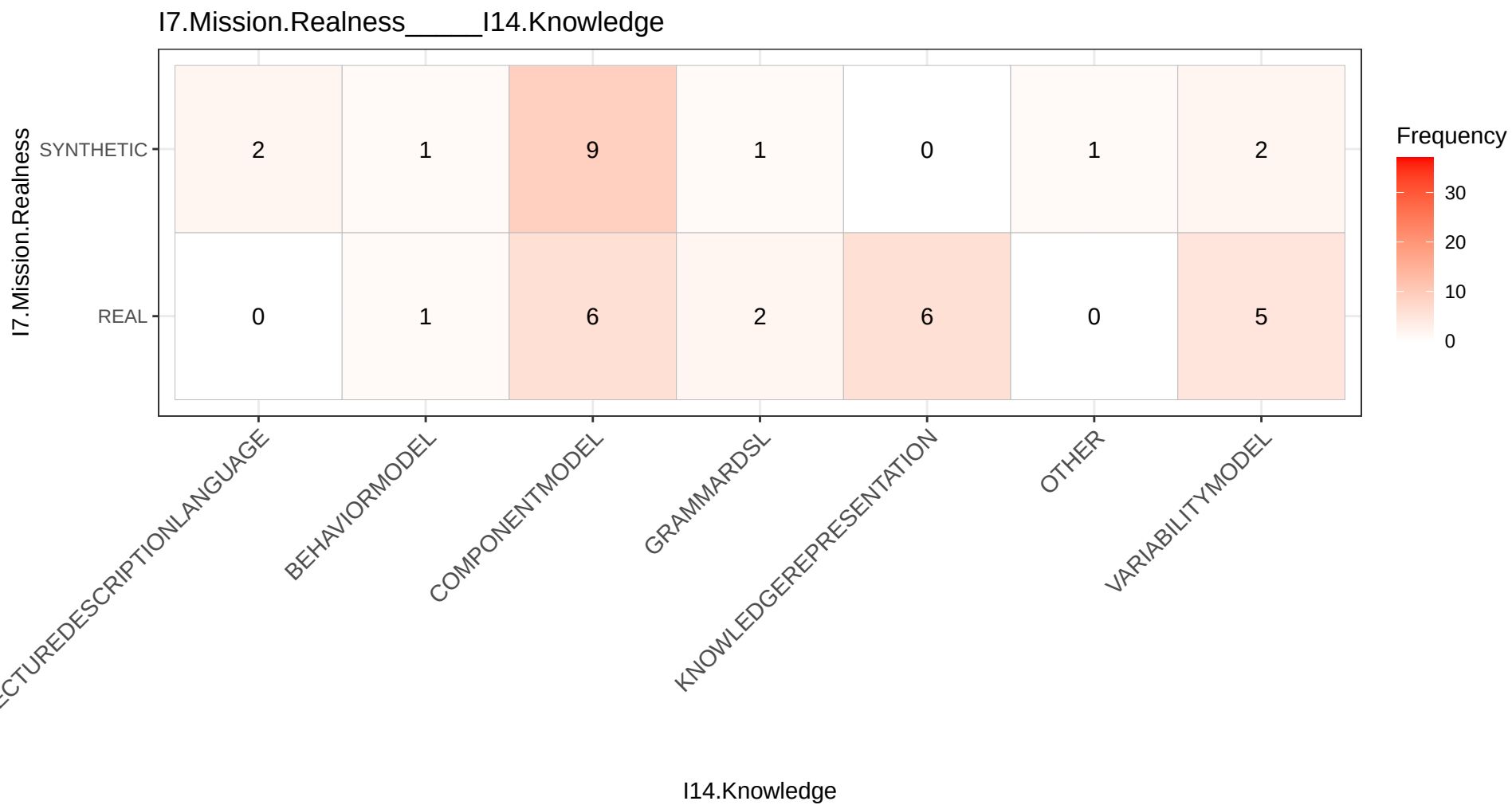




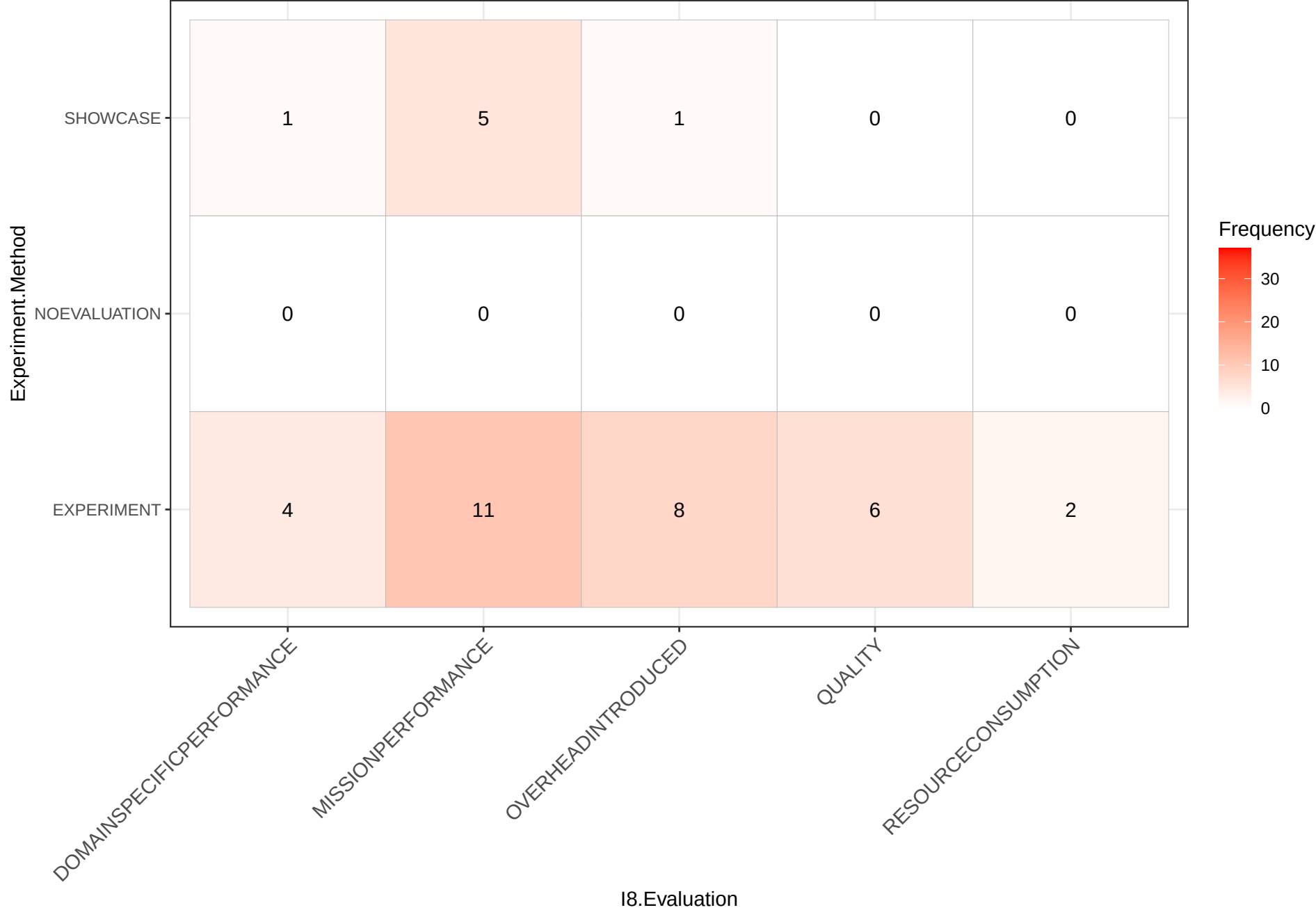
I7.Mission.Realness_____I12.Plan

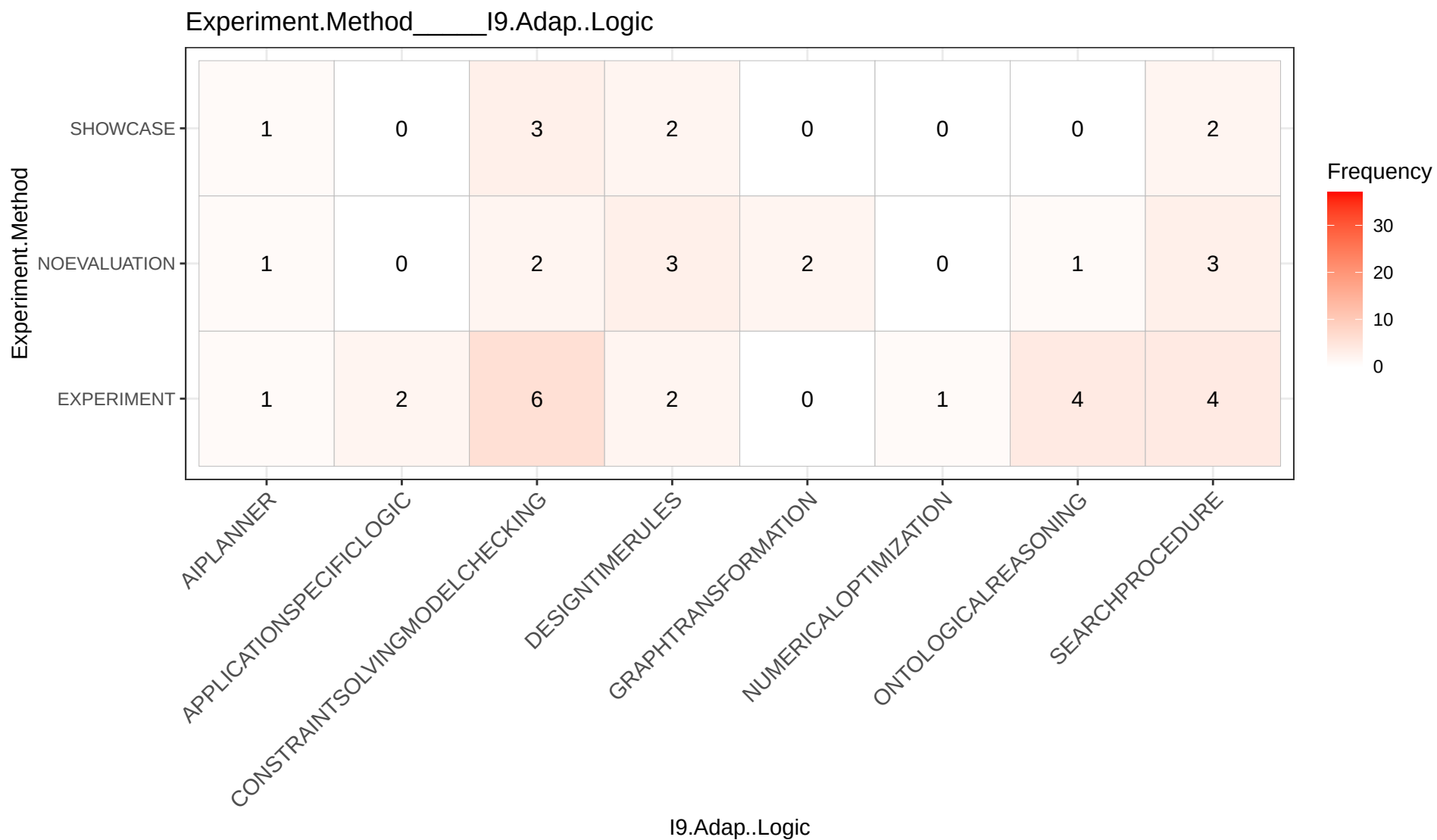




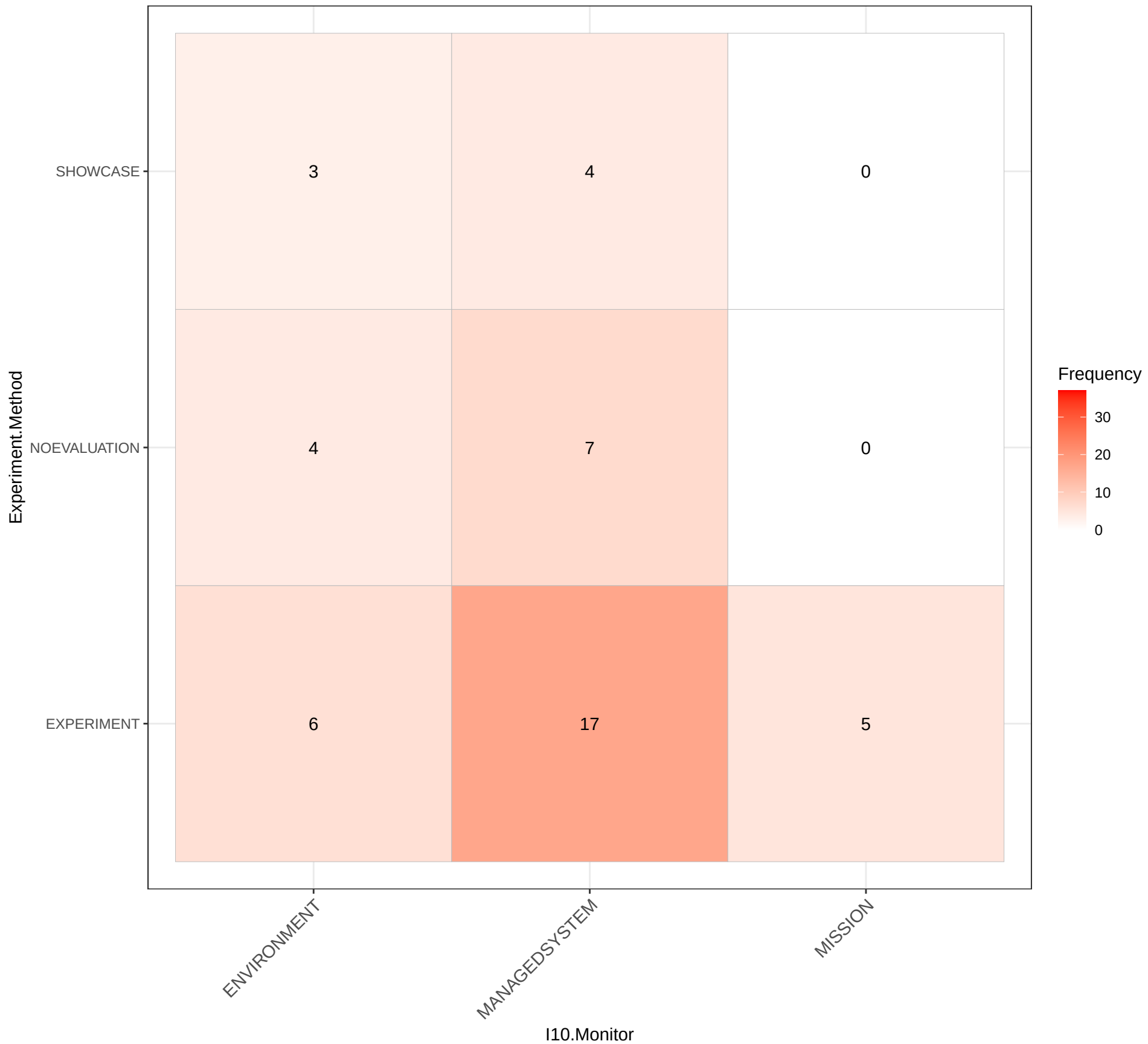


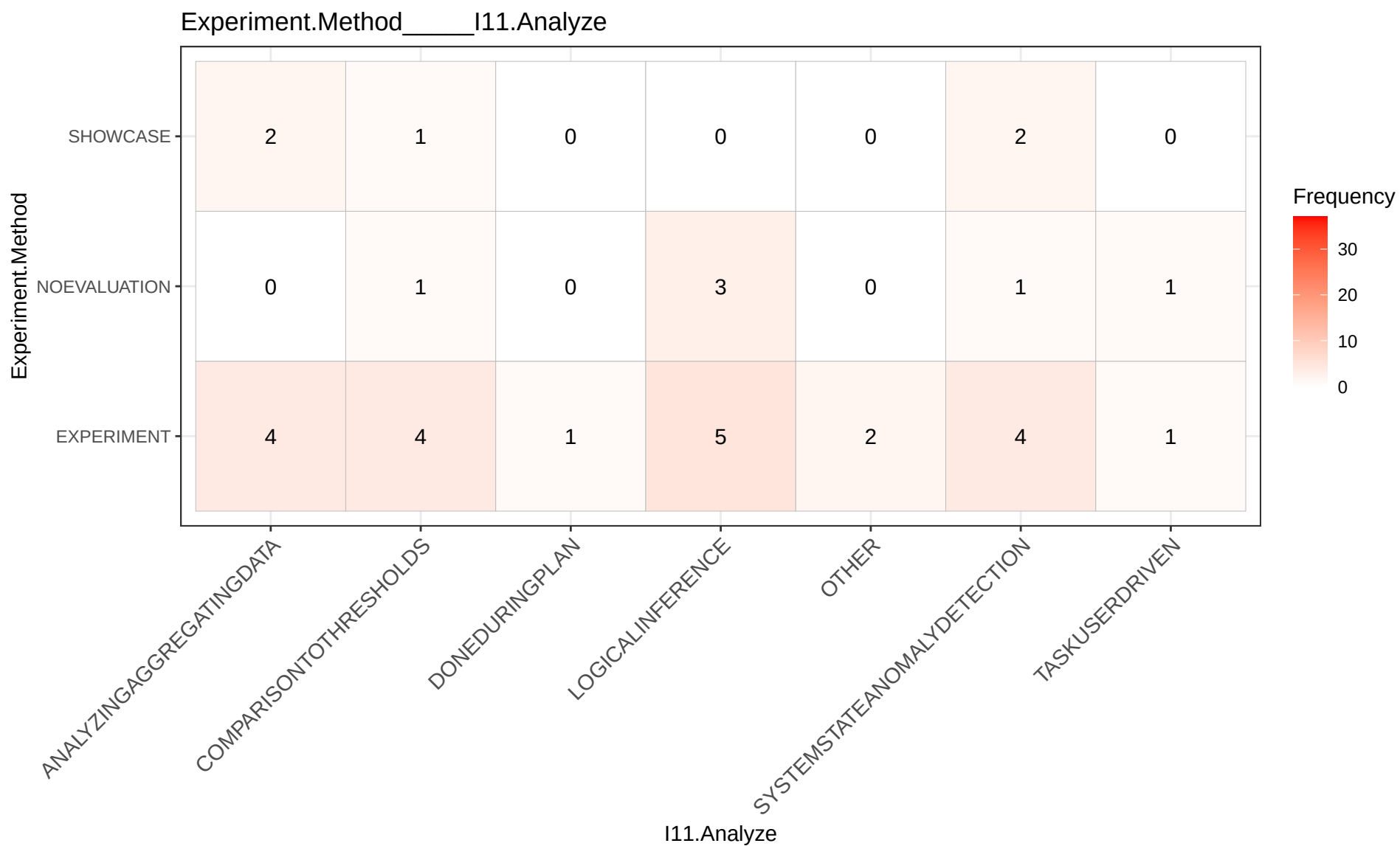
Experiment.Method_____I8.Evaluation



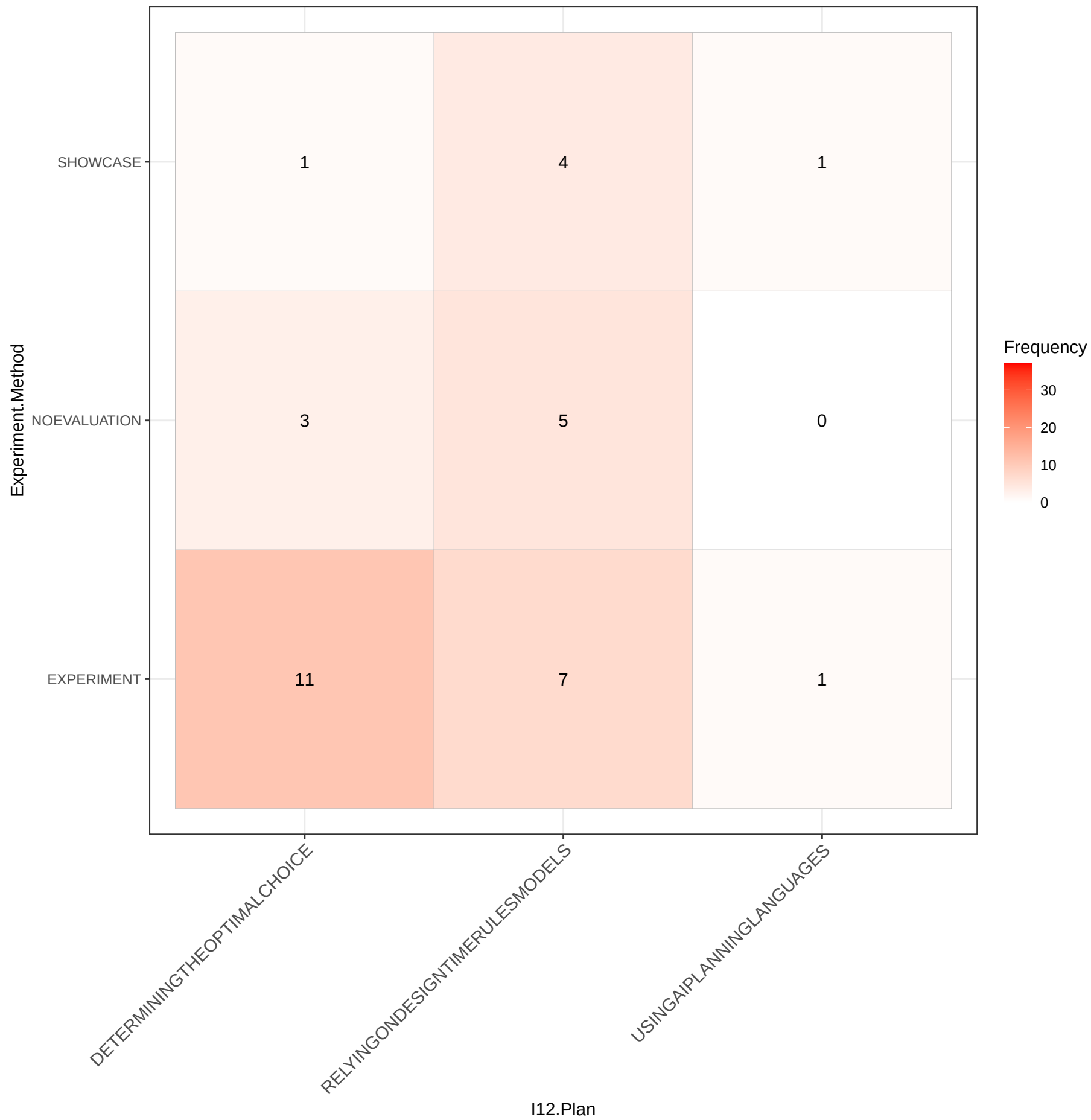


Experiment.Method_____I10.Monitor

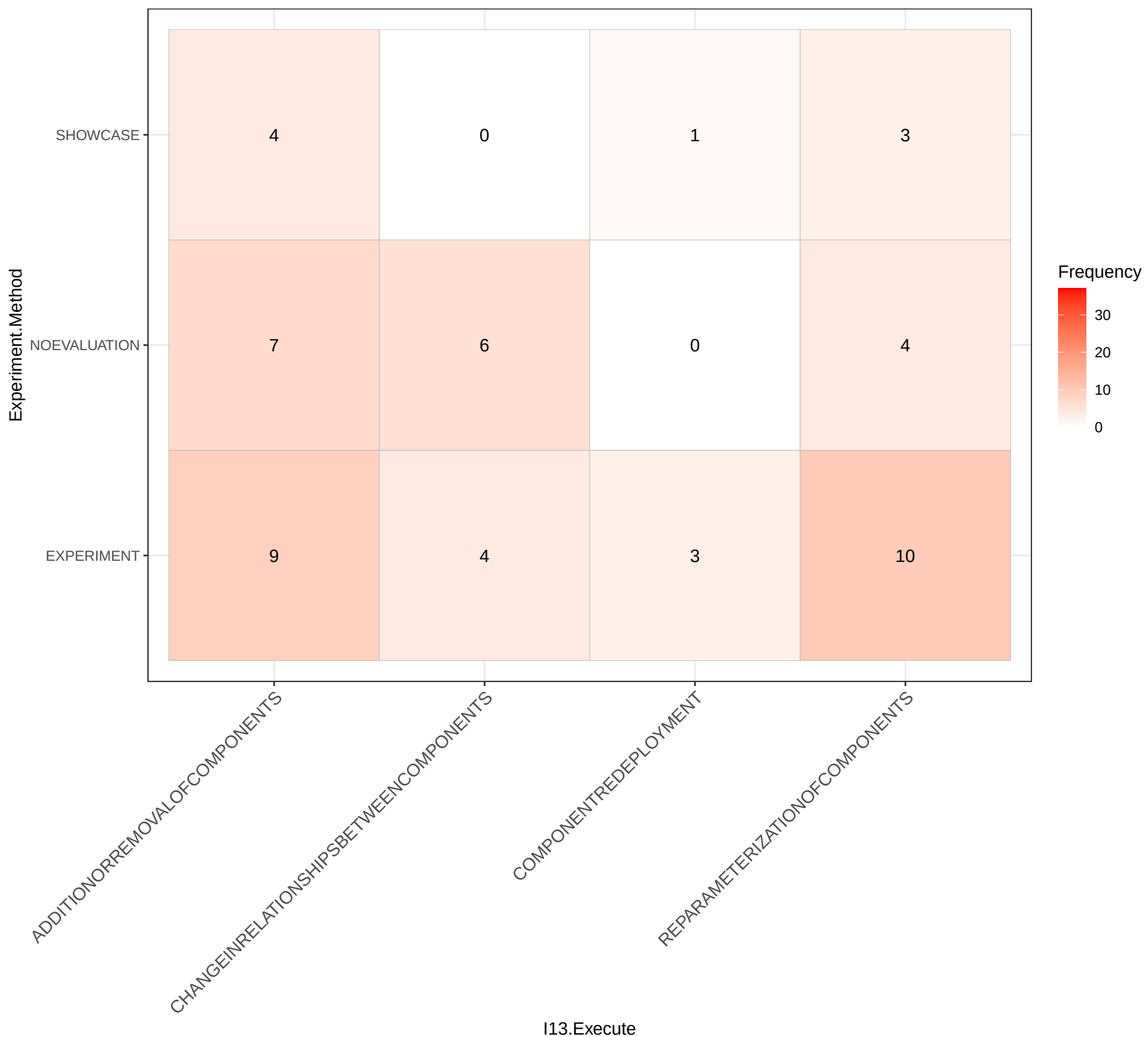


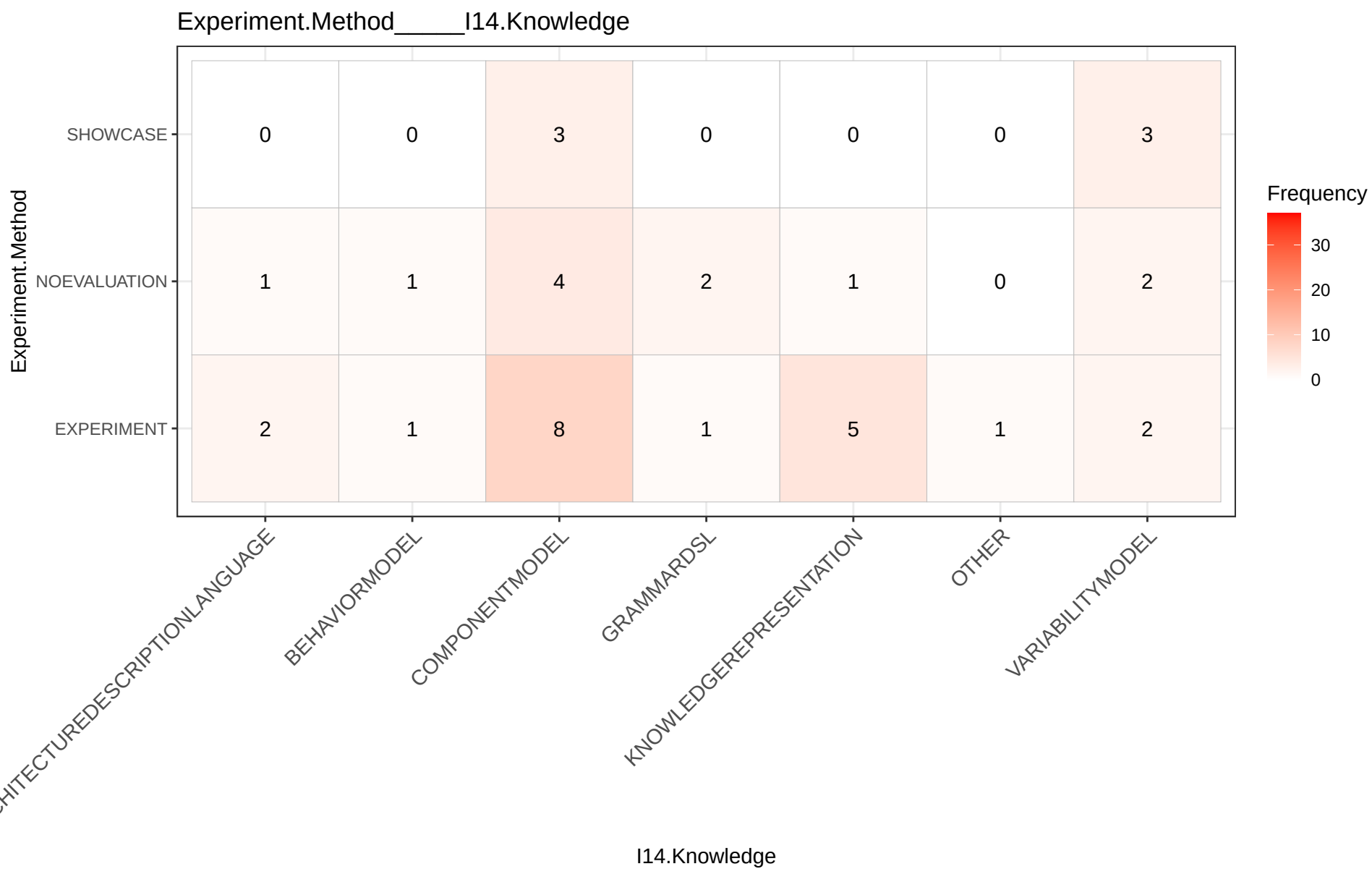


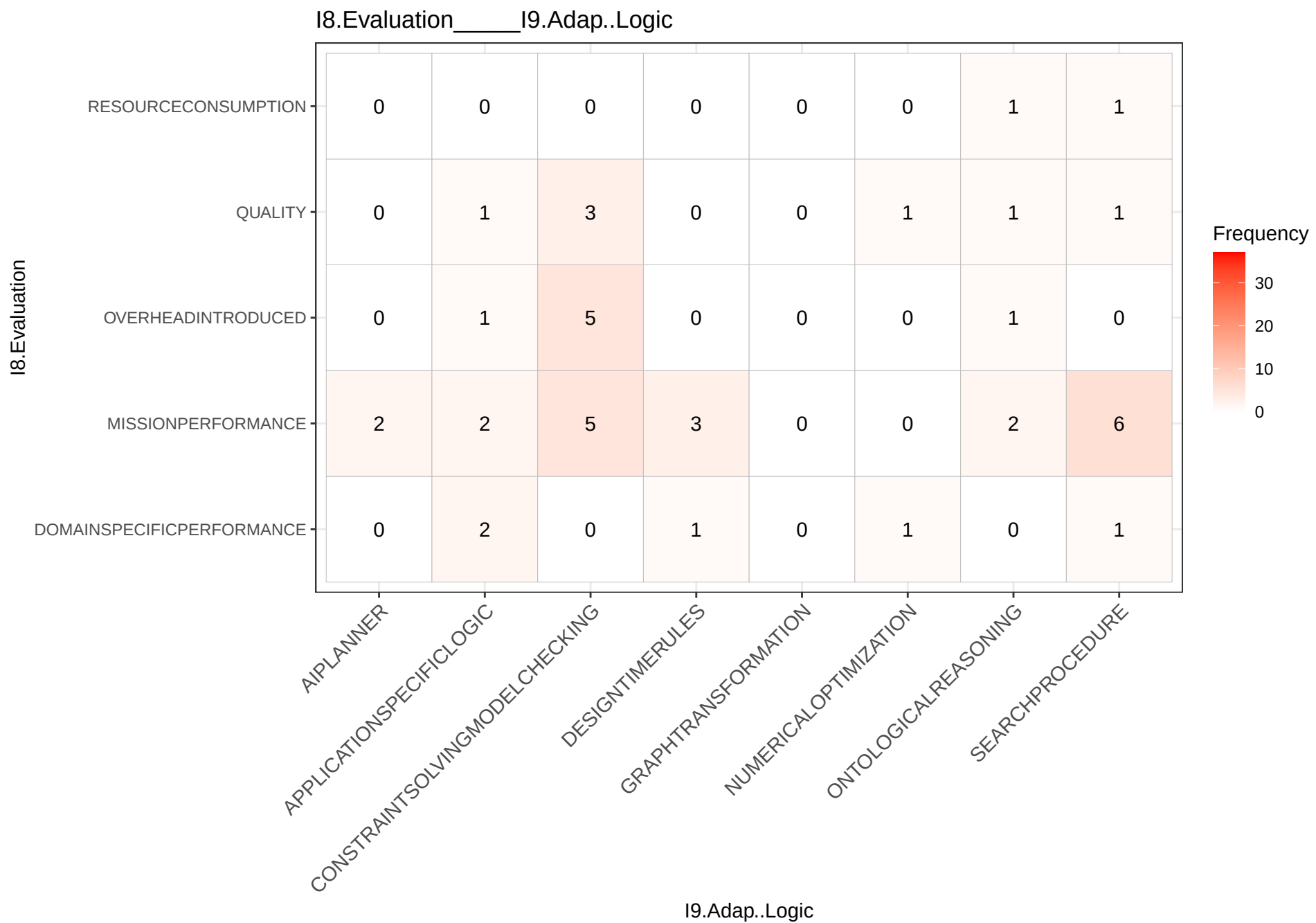
Experiment.Method_____I12.Plan



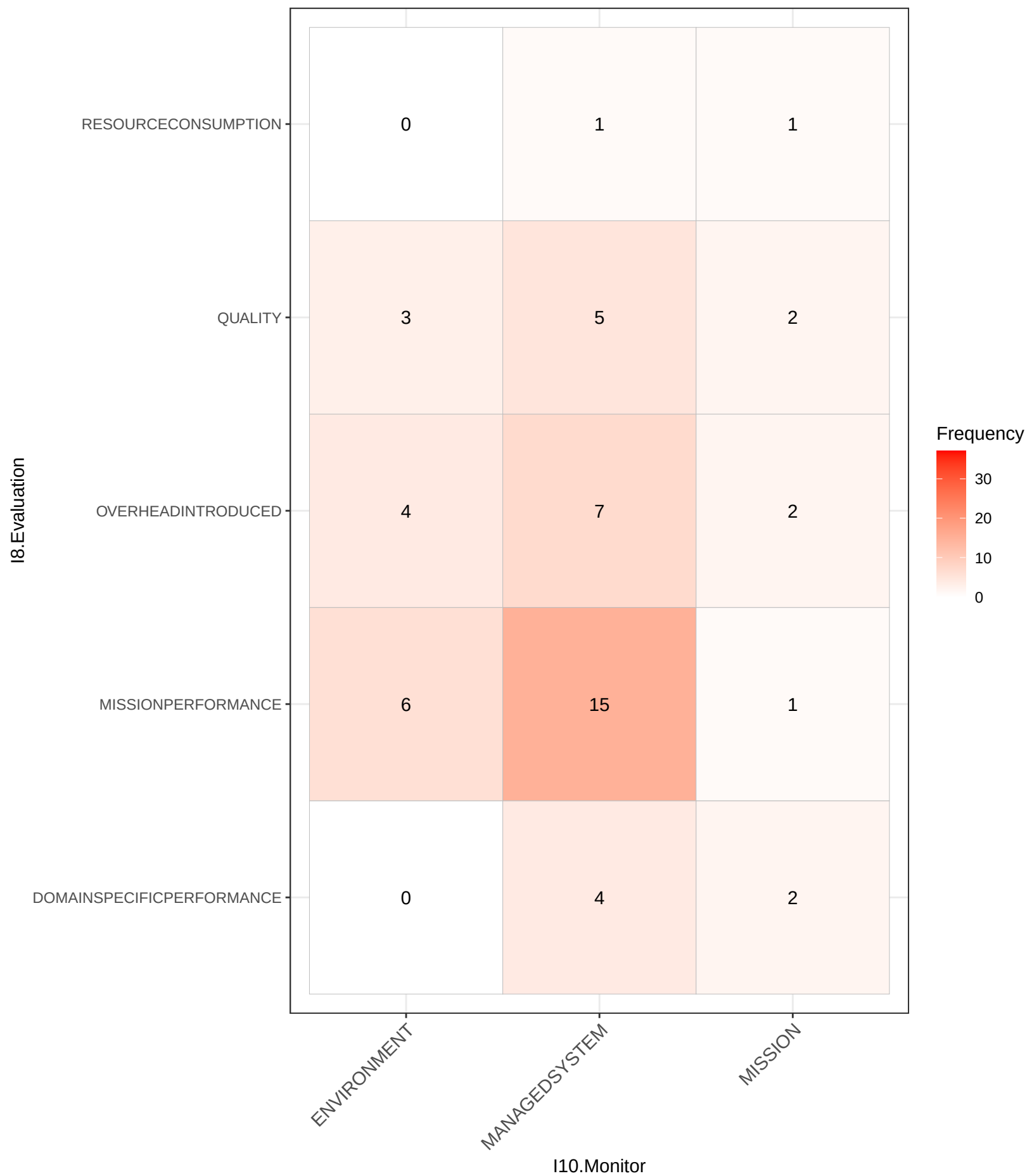
Experiment.Method____I13.Execute





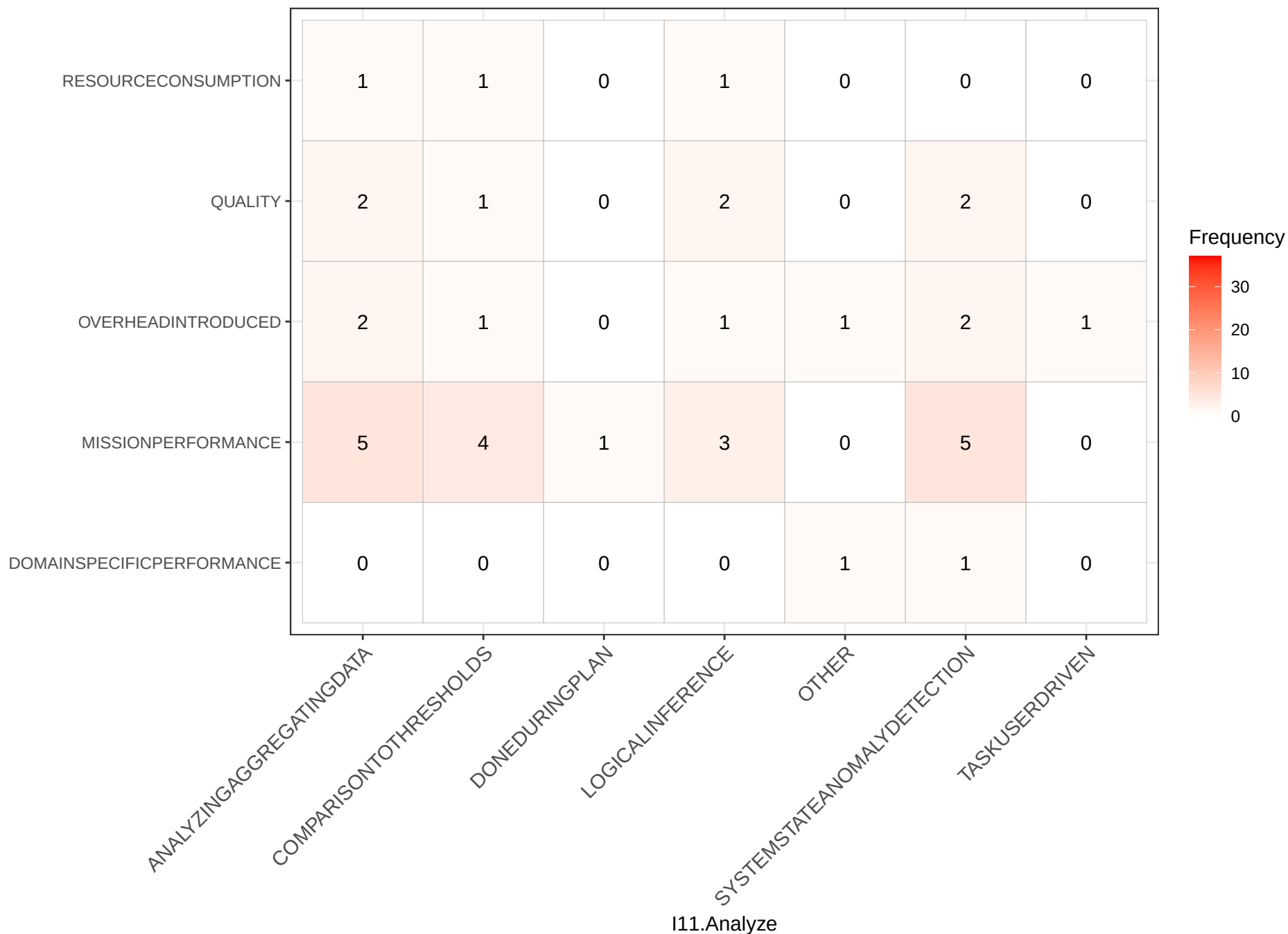


I8.Evaluation_____I10.Monitor

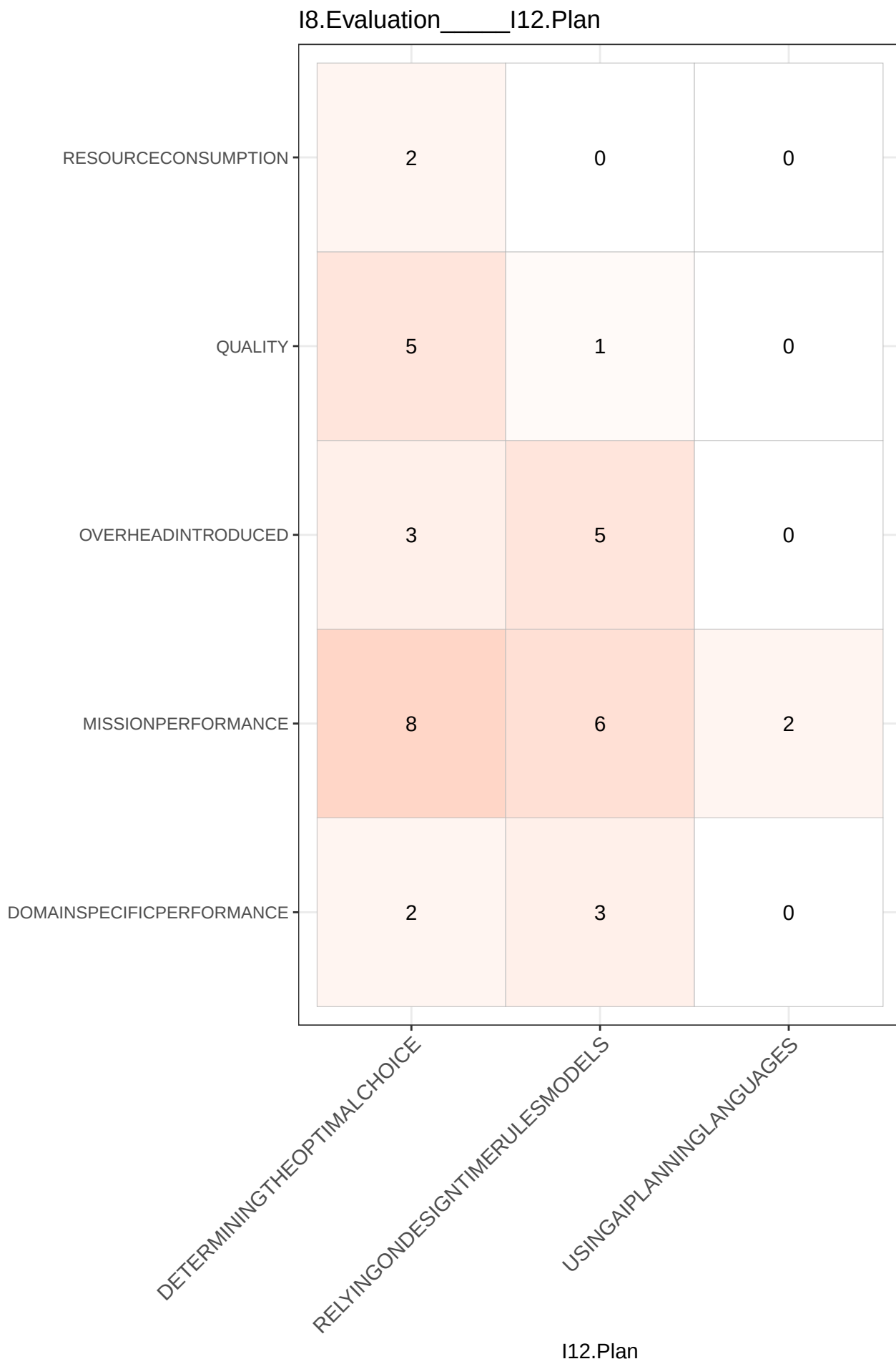


I8.Evaluation

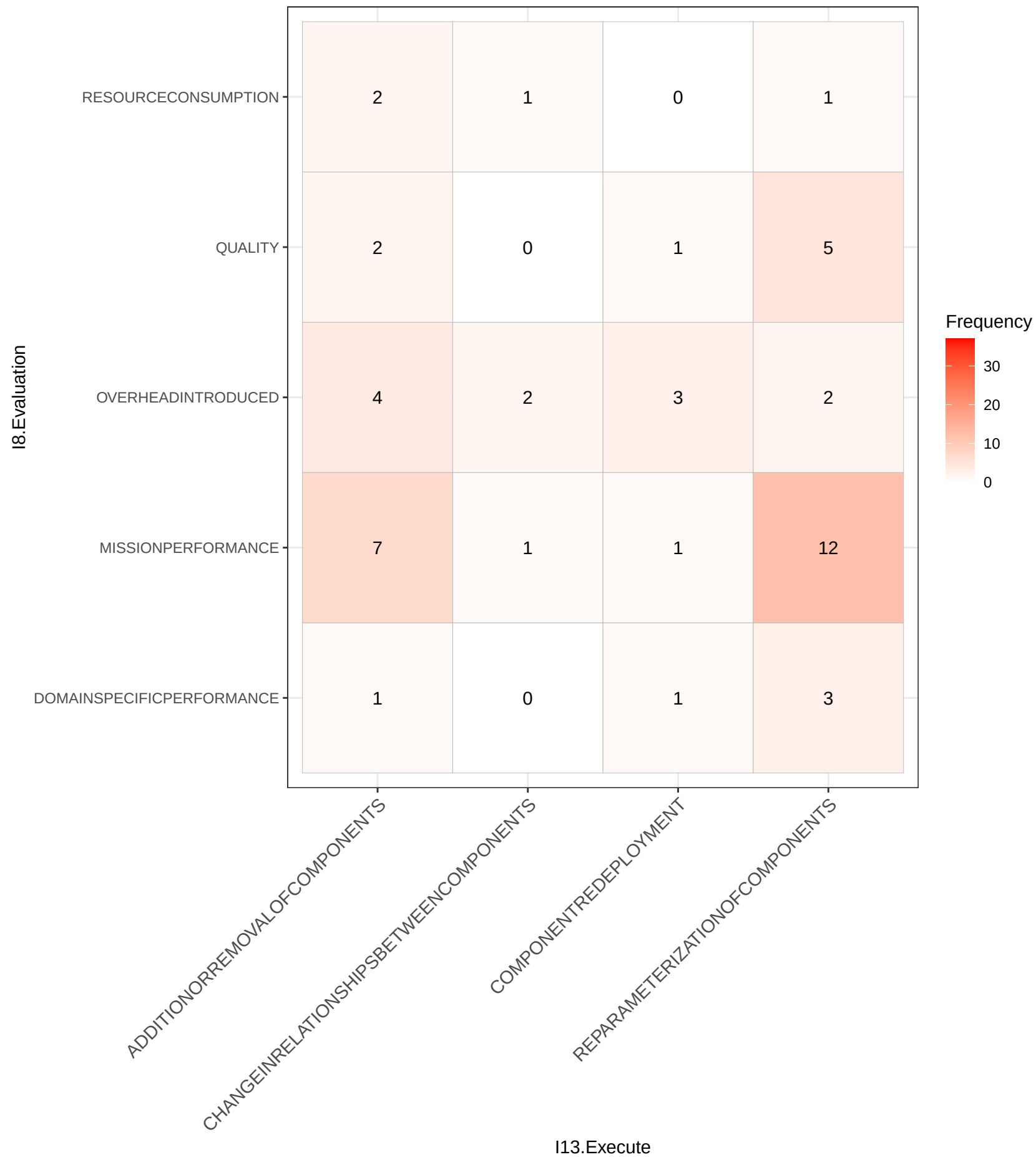
I8.Evaluation_____I11.Analyze

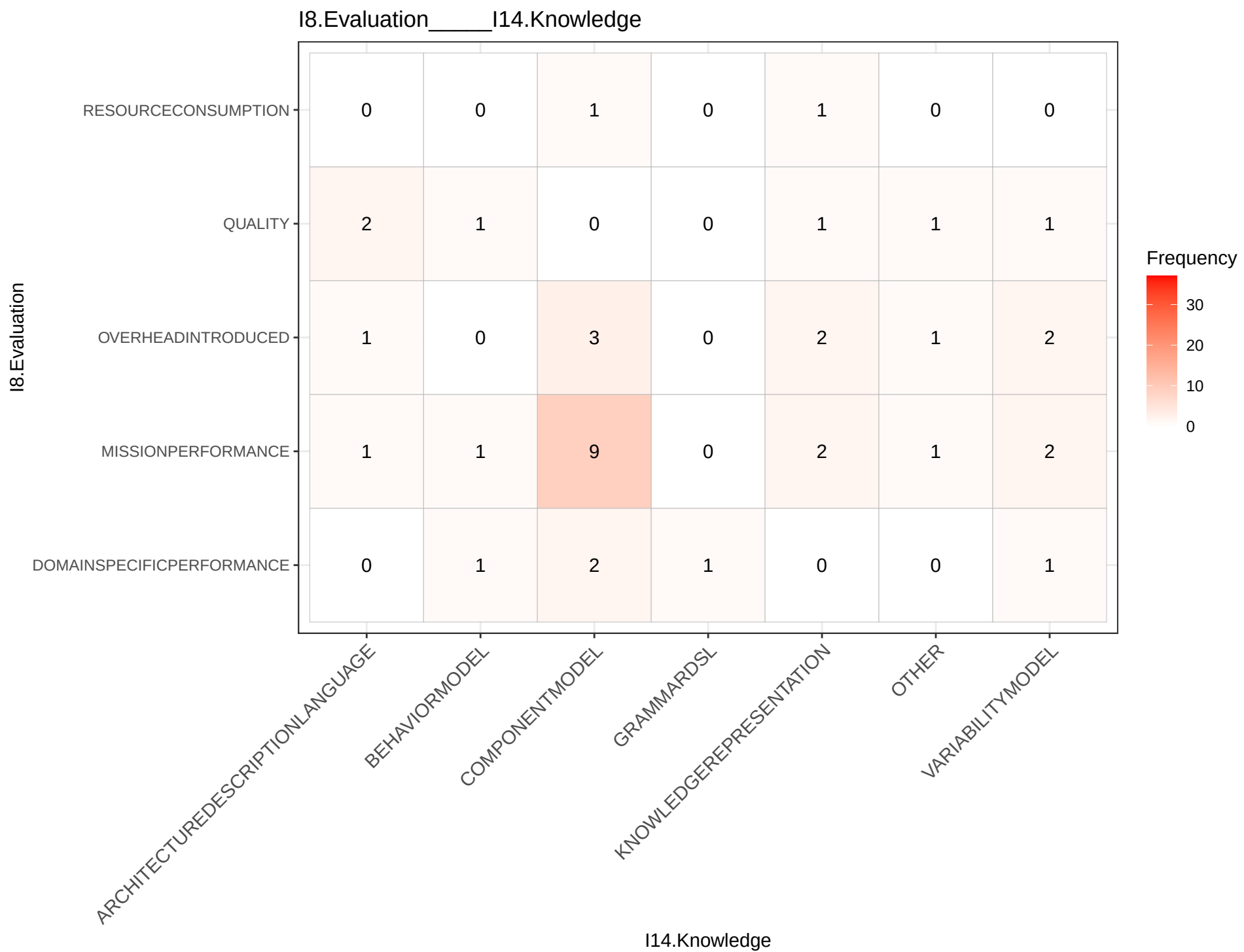


I8.Evaluation

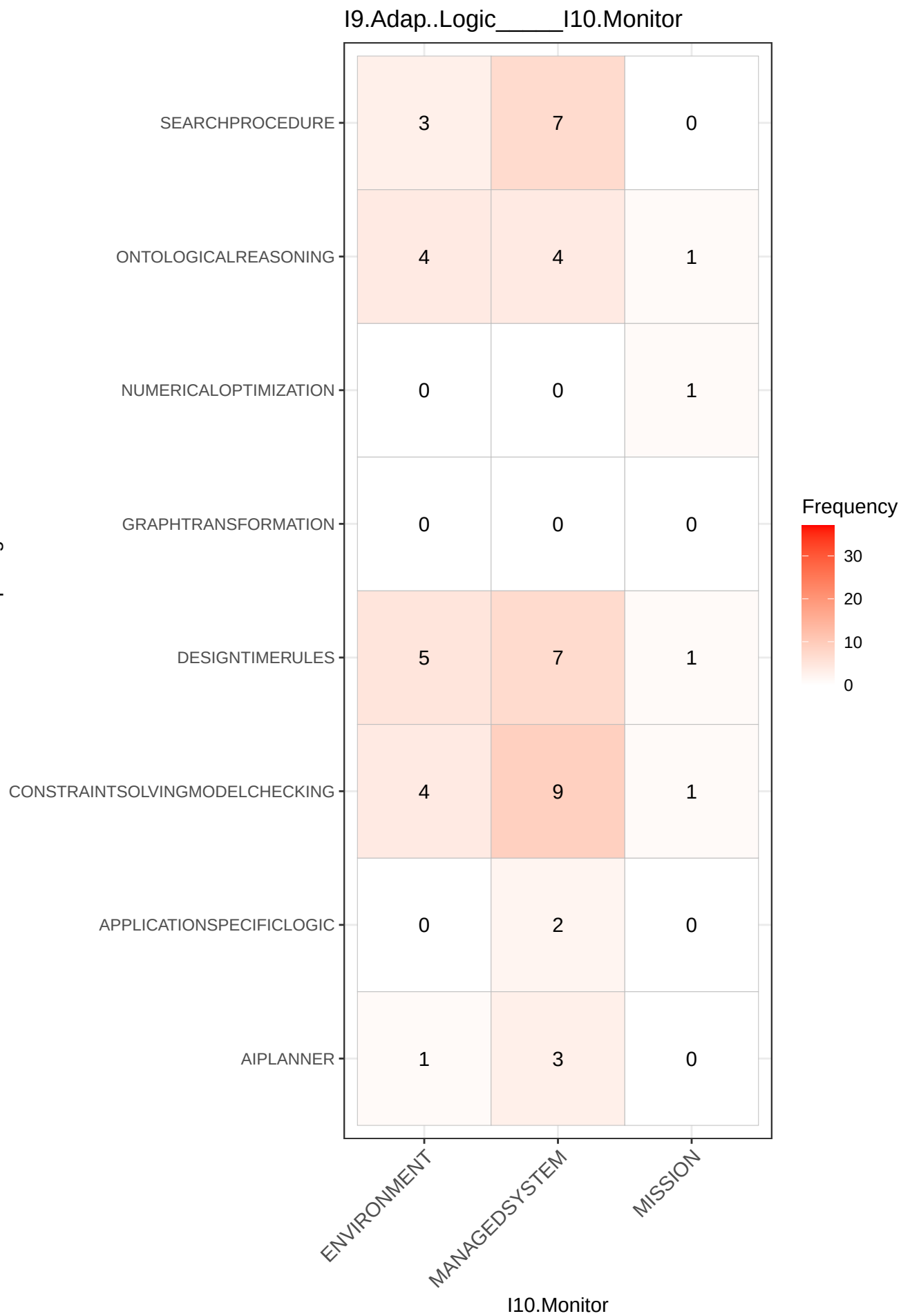


I8.Evaluation_____I13.Execute



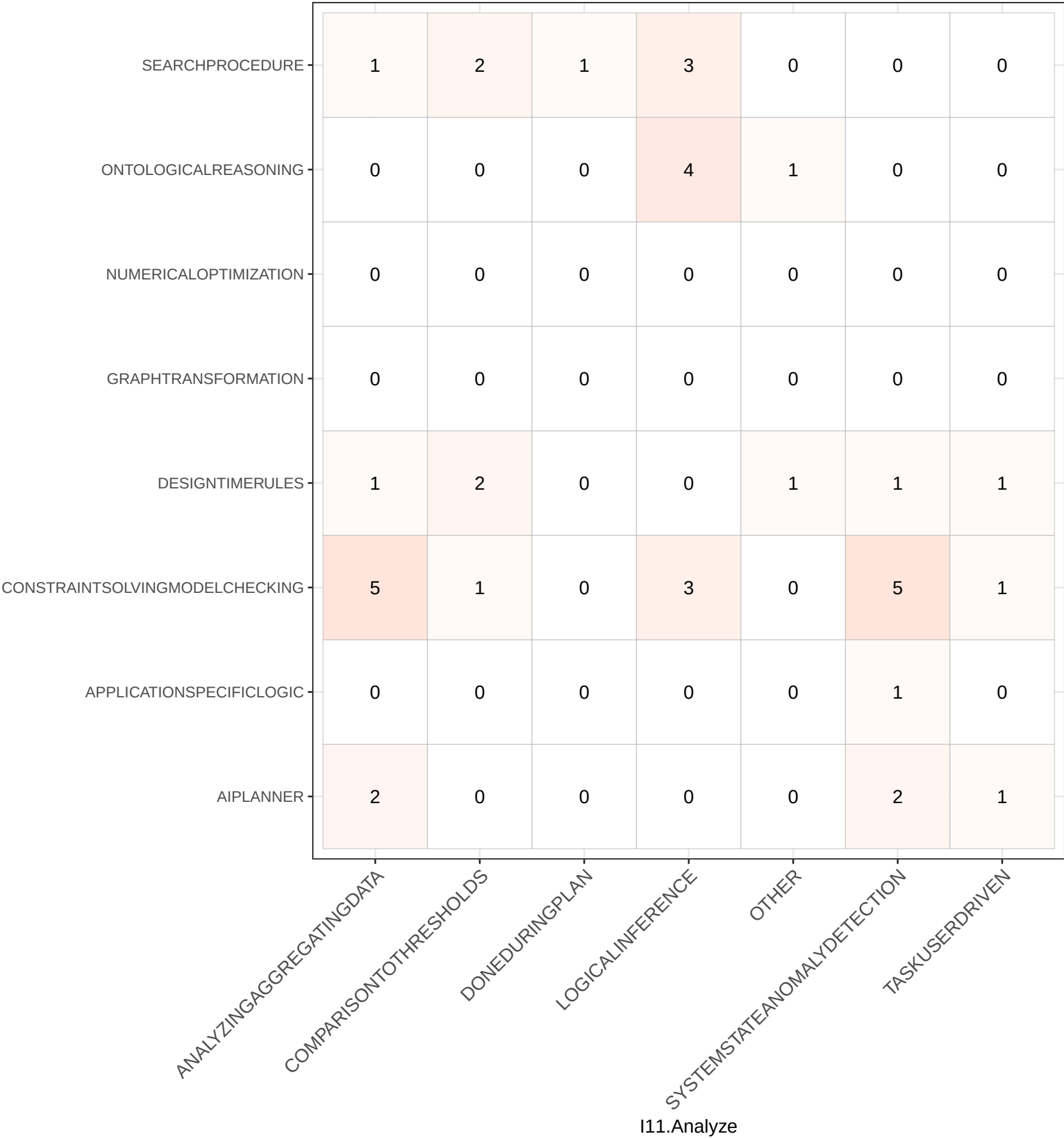


I9.Adap..Logic



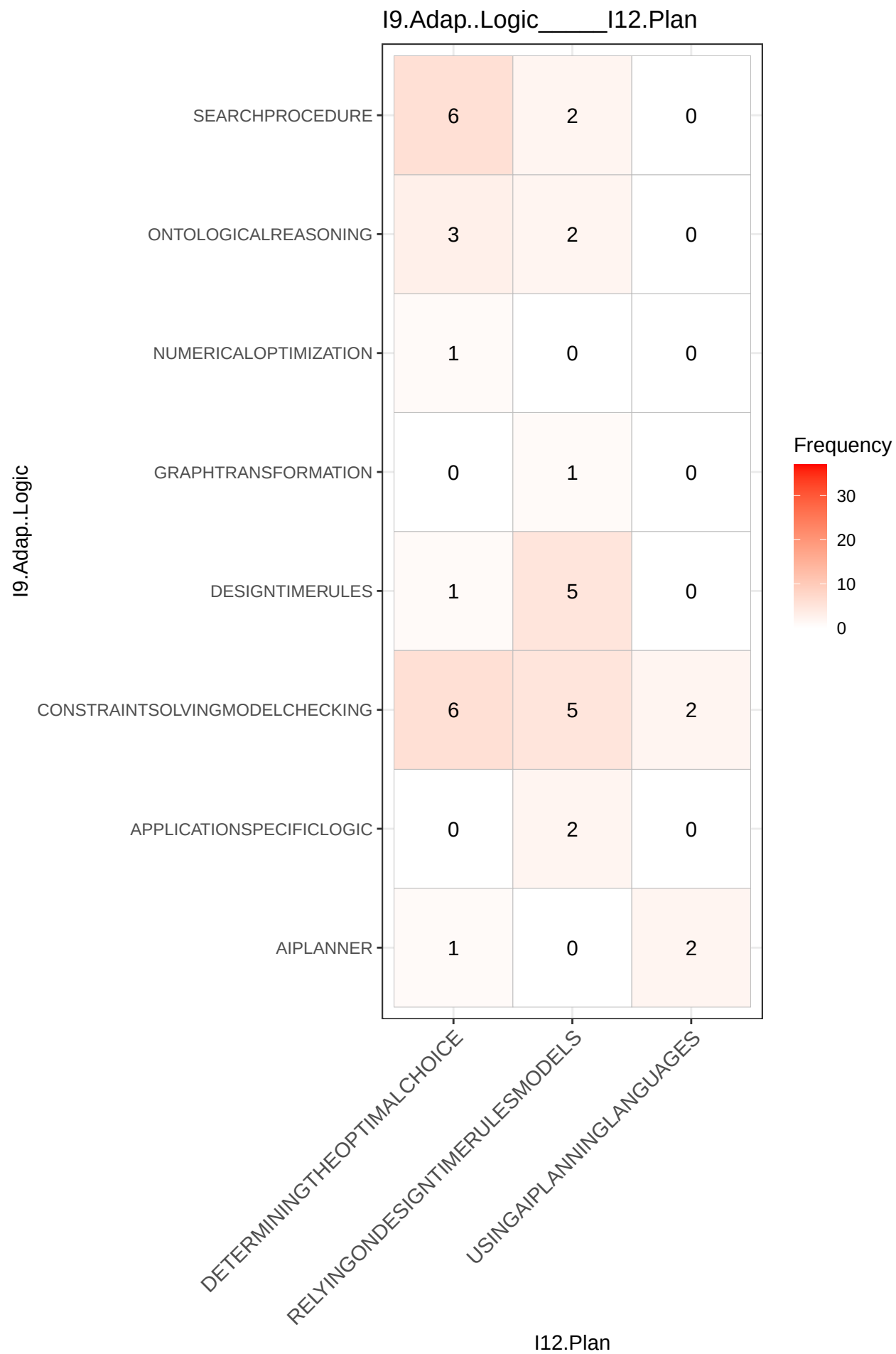
I9.Adap..Logic_____I11.Analyze

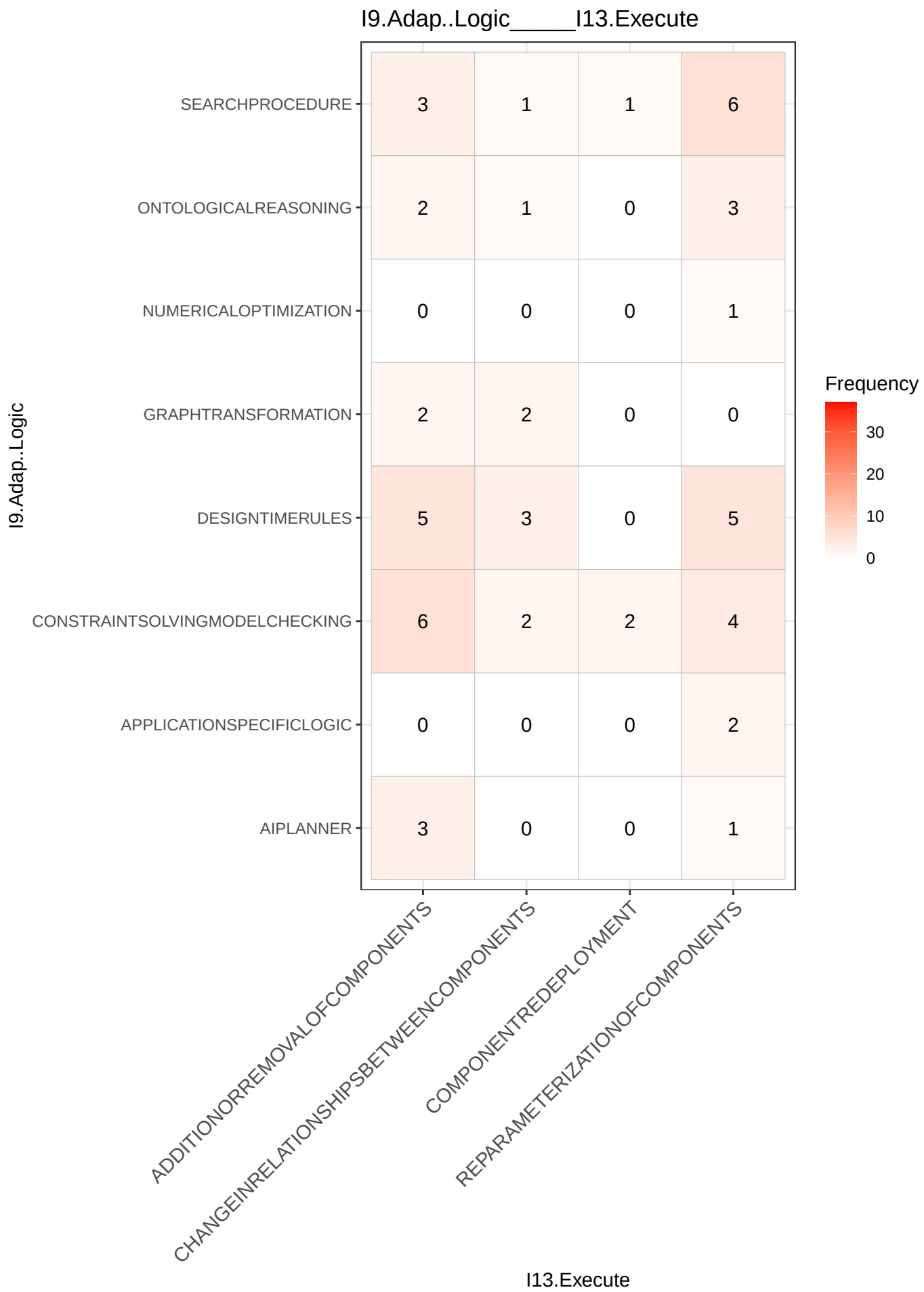
I9.Adap..Logic



Frequency

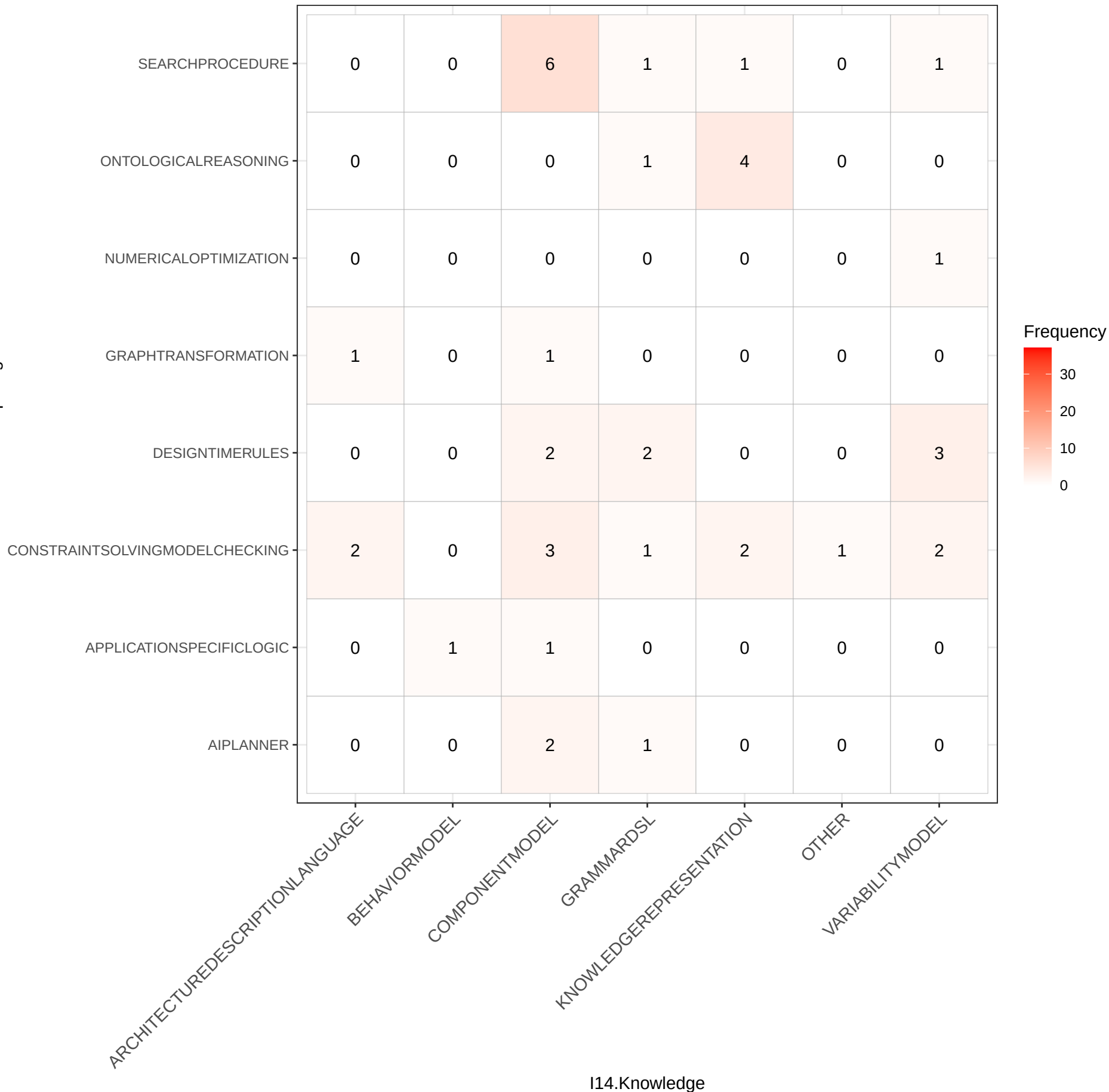




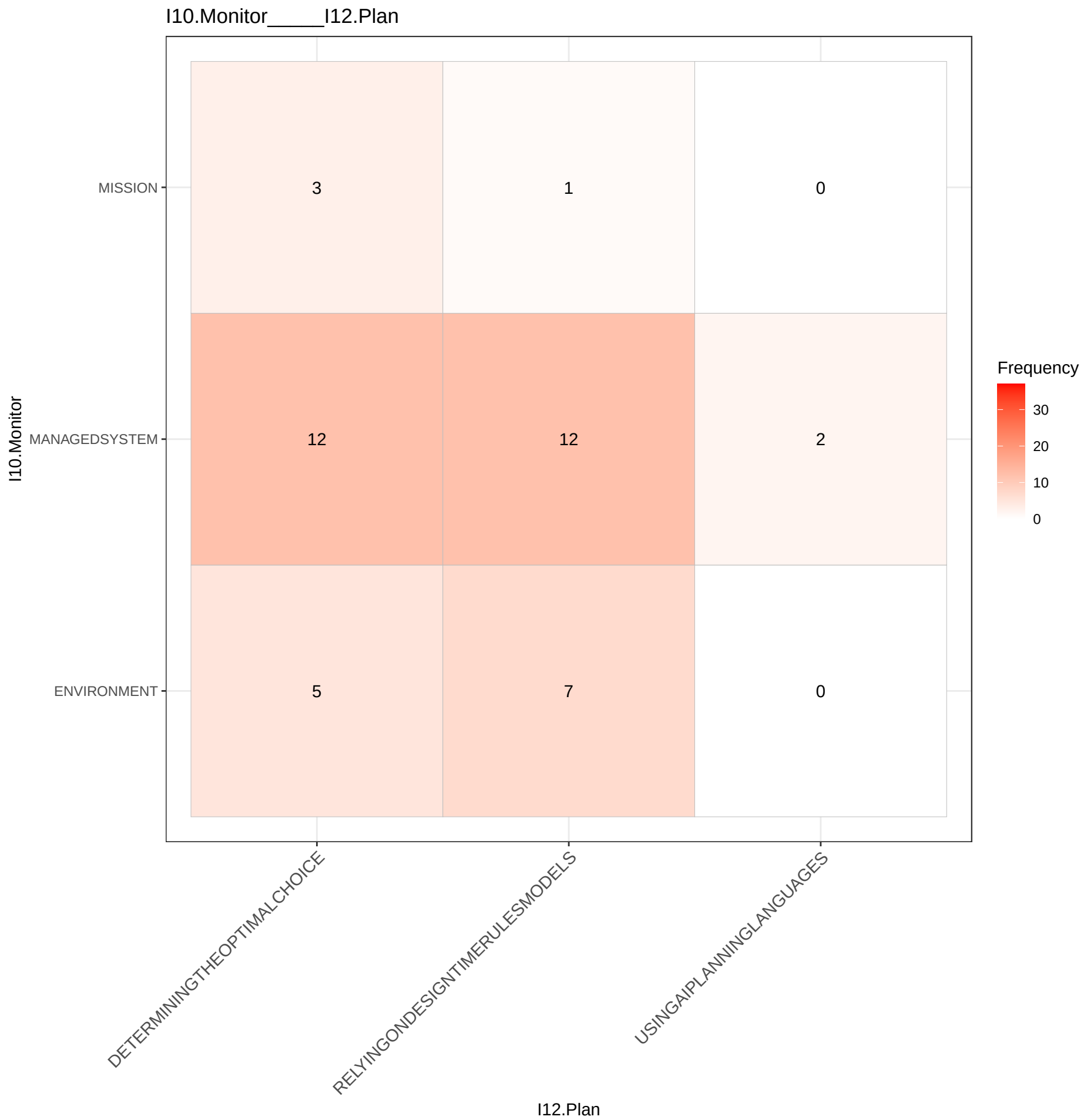


I9.Adap..Logic_____I14.Knowledge

I9.Adap..Logic

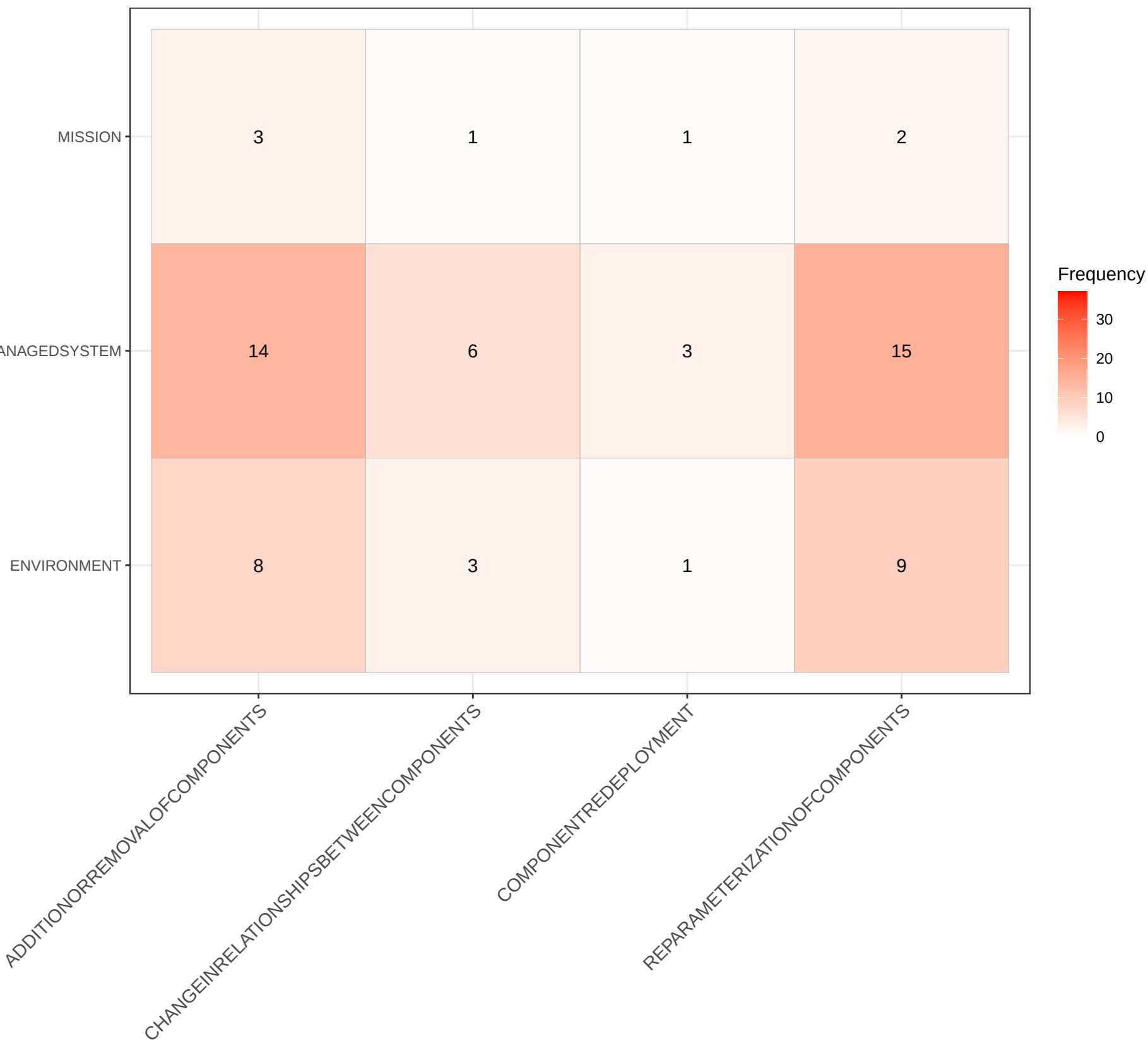




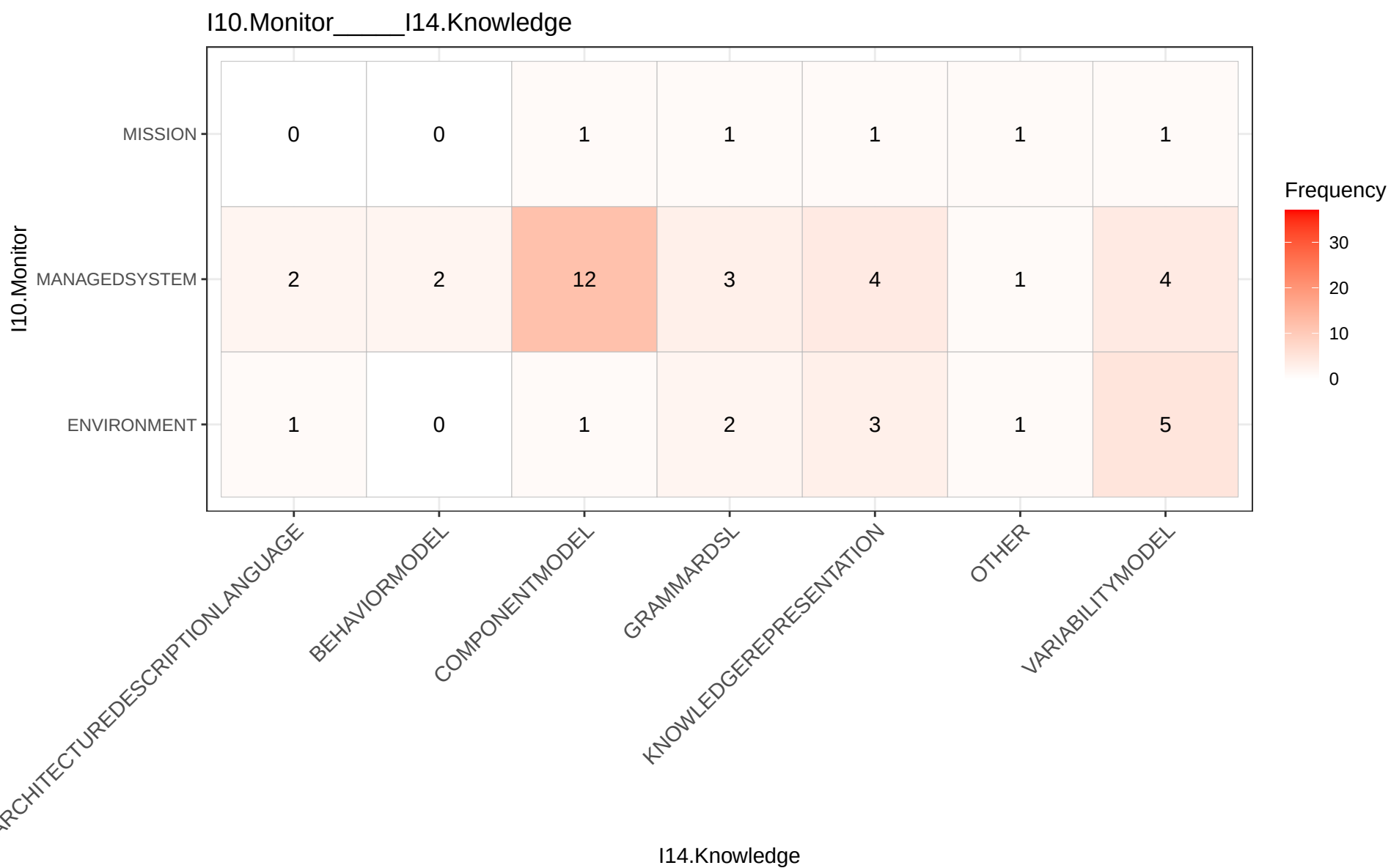


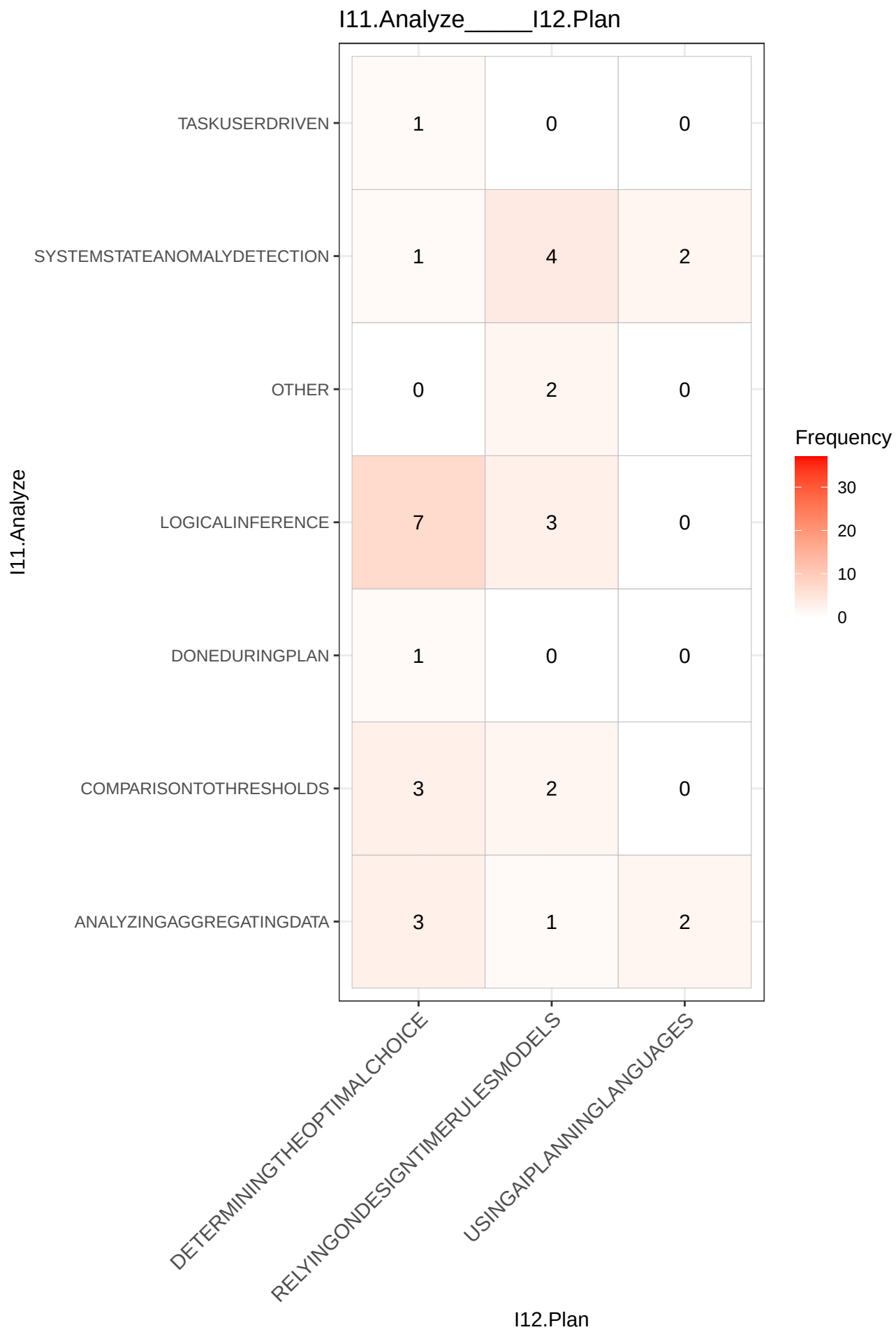
I10.Monitor

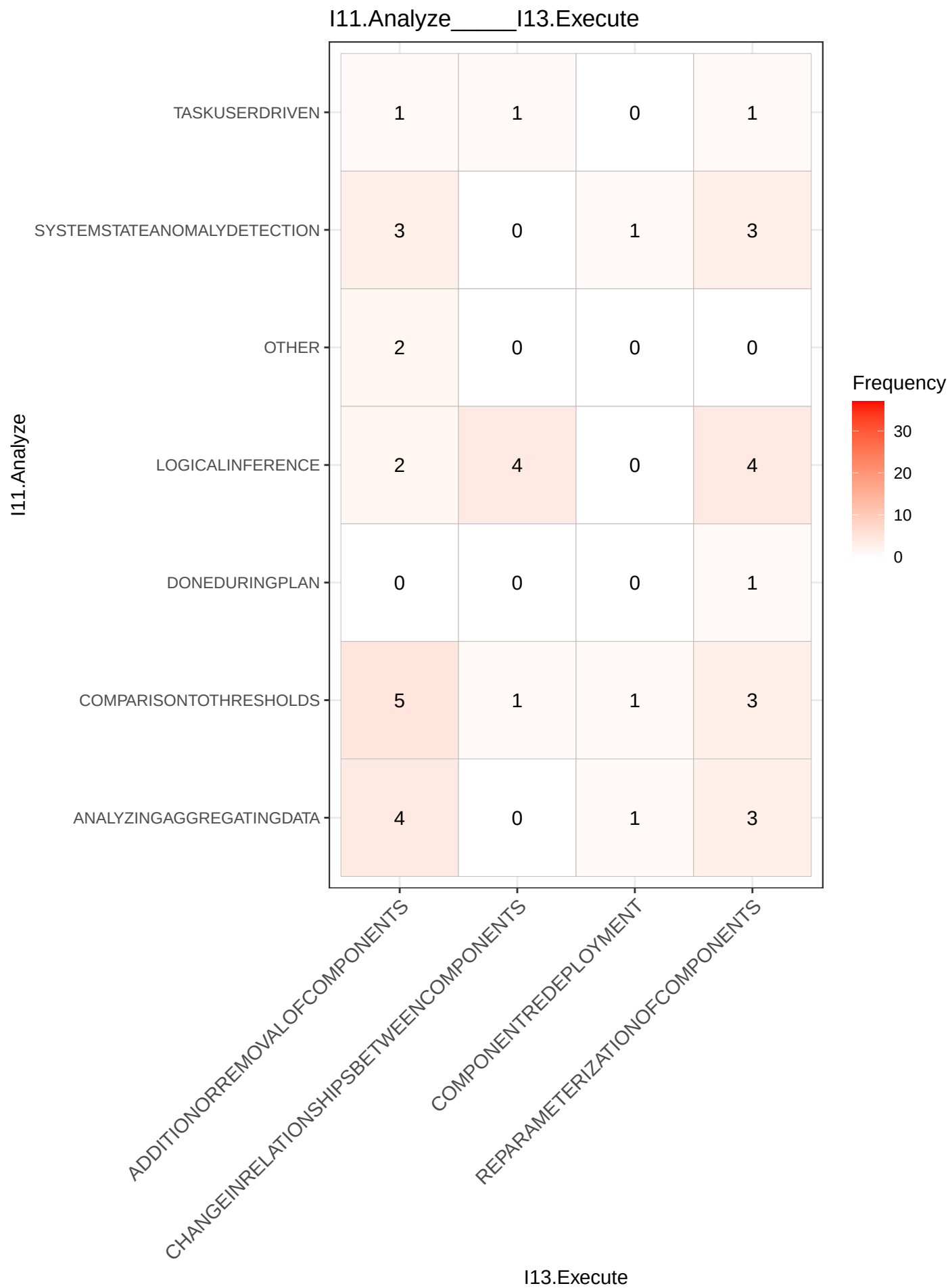
I10.Monitor_____I13.Execute



I13.Execute

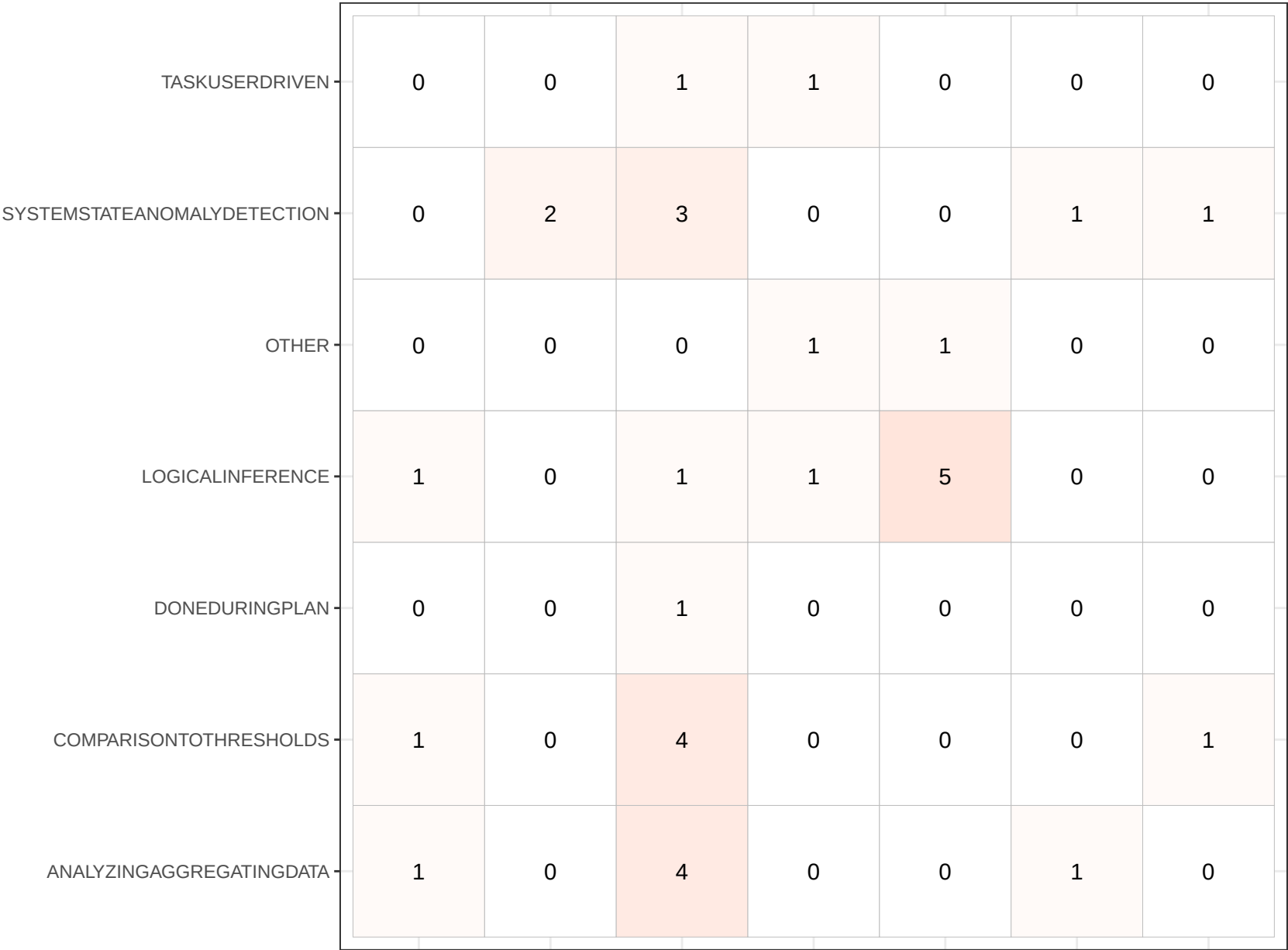






I11.Analyze_____I14.Knowledge

I11.Analyze



Frequency



I14.Knowledge

